



DeepLearning.AI

Derivatives and Optimization

Machine learning motivation

Machine Learning Motivation

Machine Learning Motivation



Machine Learning Motivation



Number of Bedrooms

Machine Learning Motivation



1



2

Number of Bedrooms

Machine Learning Motivation



1

\$150,000



2

\$250,000

Number of Bedrooms

Machine Learning Motivation

Predict the price of a house using the number of bedrooms it has



1

\$150,000



2

\$250,000

Number of Bedrooms

Machine Learning Motivation

Predict the price of a house using the number of bedrooms it has

Machine Learning Motivation

Predict the price of a house using the number of bedrooms it has

Number of Bedrooms
1
2
3
5
6
7
8
9
10

Machine Learning Motivation

Predict the price of a house using the number of bedrooms it has

Number of Bedrooms	Price of House
1	\$150,000
2	\$250,000
3	\$350,000
5	\$600,000
6	\$650,000
7	\$750,000
8	\$800,000
9	??
10	\$1,050,000

Machine Learning Motivation

Predict the price of a house using the number of bedrooms it has

9 ???

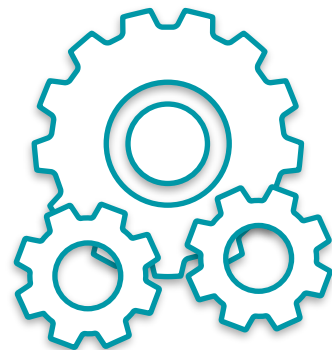
Number of Bedrooms	Price of House
1	\$150,000
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5	\$600,000
6	\$650,000
7	\$750,000
8	\$800,000
9	??
10	\$1,050,000

Machine Learning Motivation

Predict the price of a house using the number of bedrooms it has

9 ???

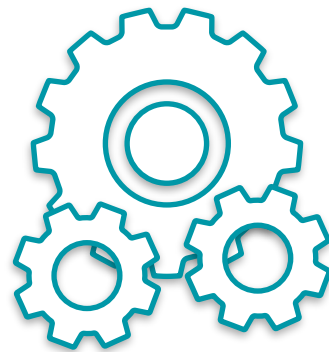
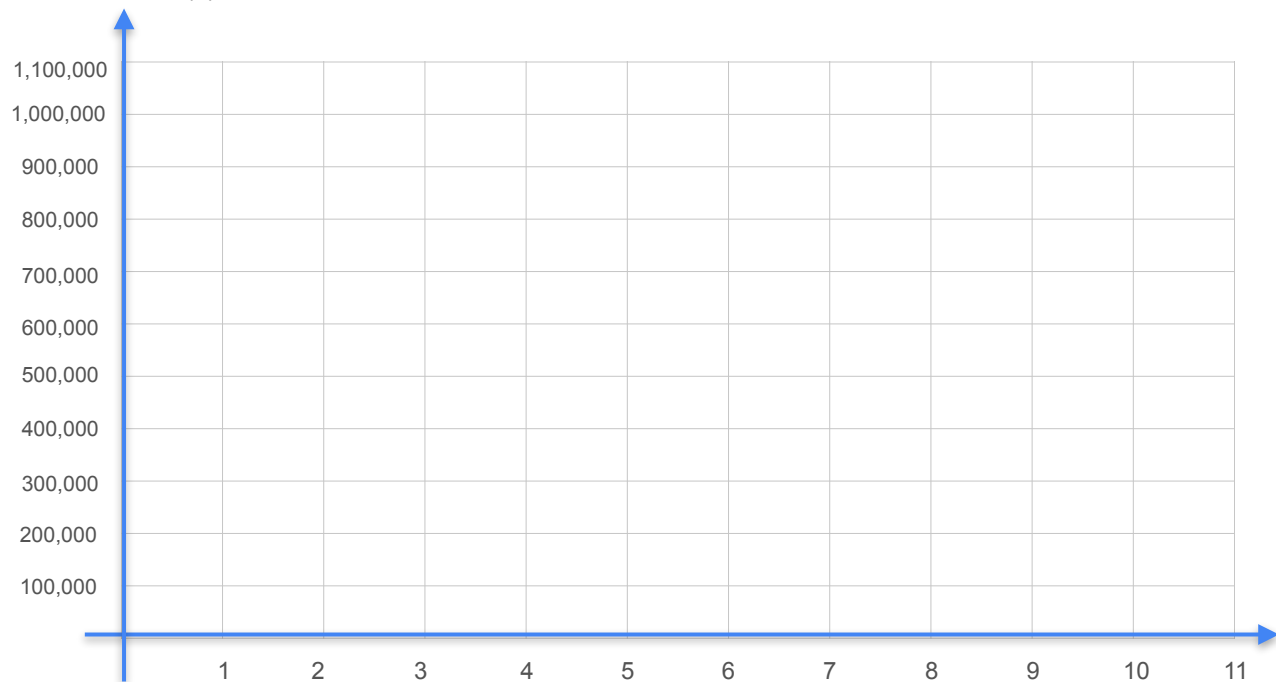
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10	\$1,050,000



Machine Learning
Model

Machine Learning Motivation

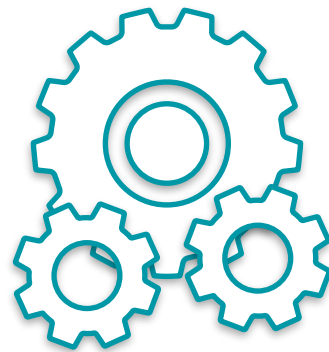
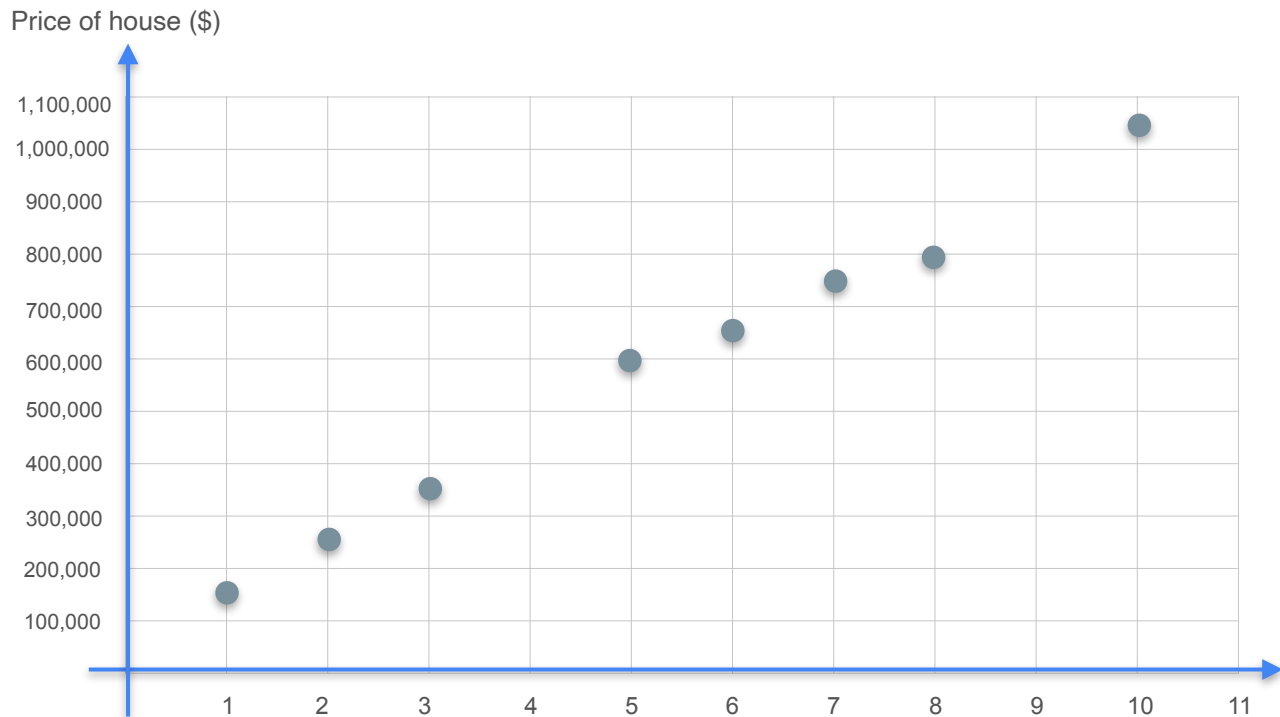
Price of house (\$)



Machine Learning
Model

*Number of
Bedrooms*

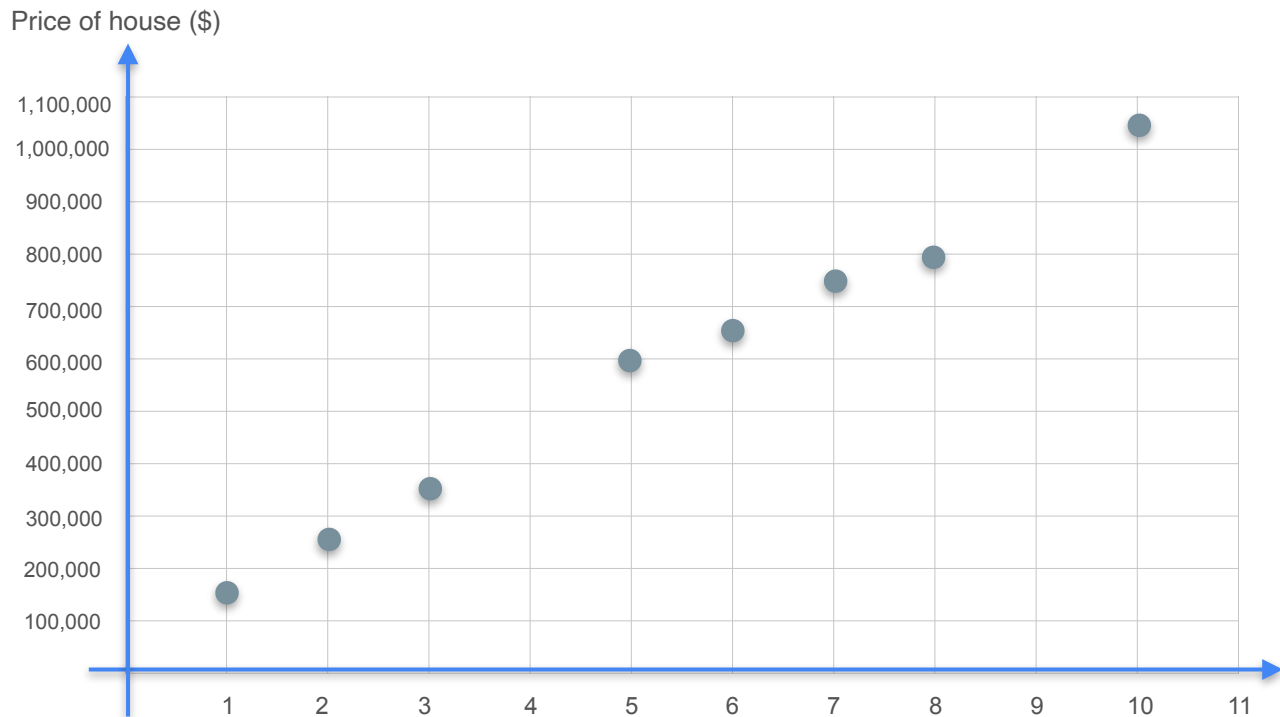
Machine Learning Motivation



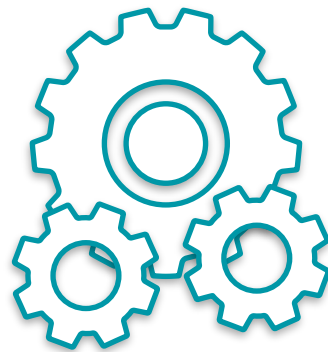
Machine Learning
Model

*Number of
Bedrooms*

Machine Learning Motivation



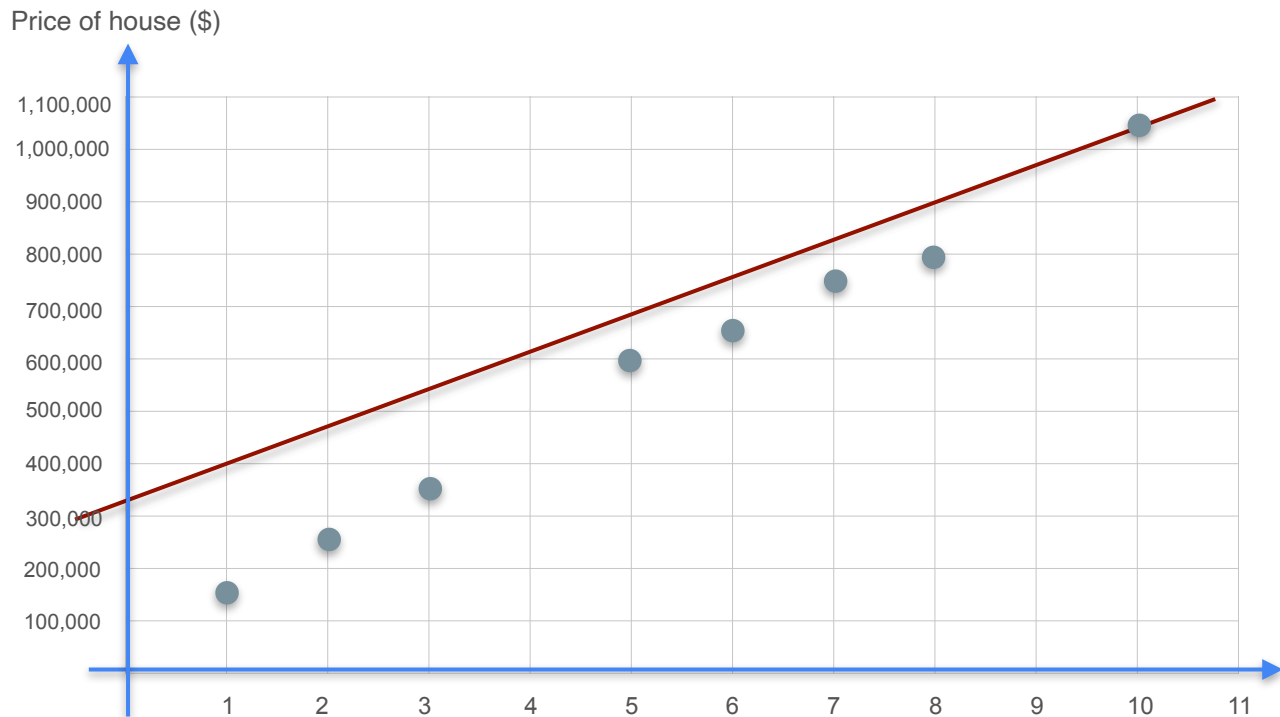
Model Training



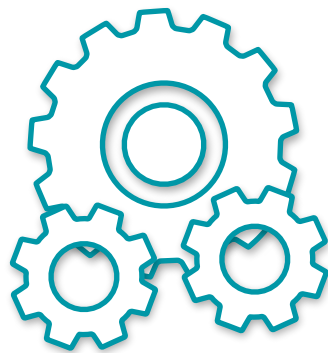
Machine Learning
Model

*Number of
Bedrooms*

Machine Learning Motivation



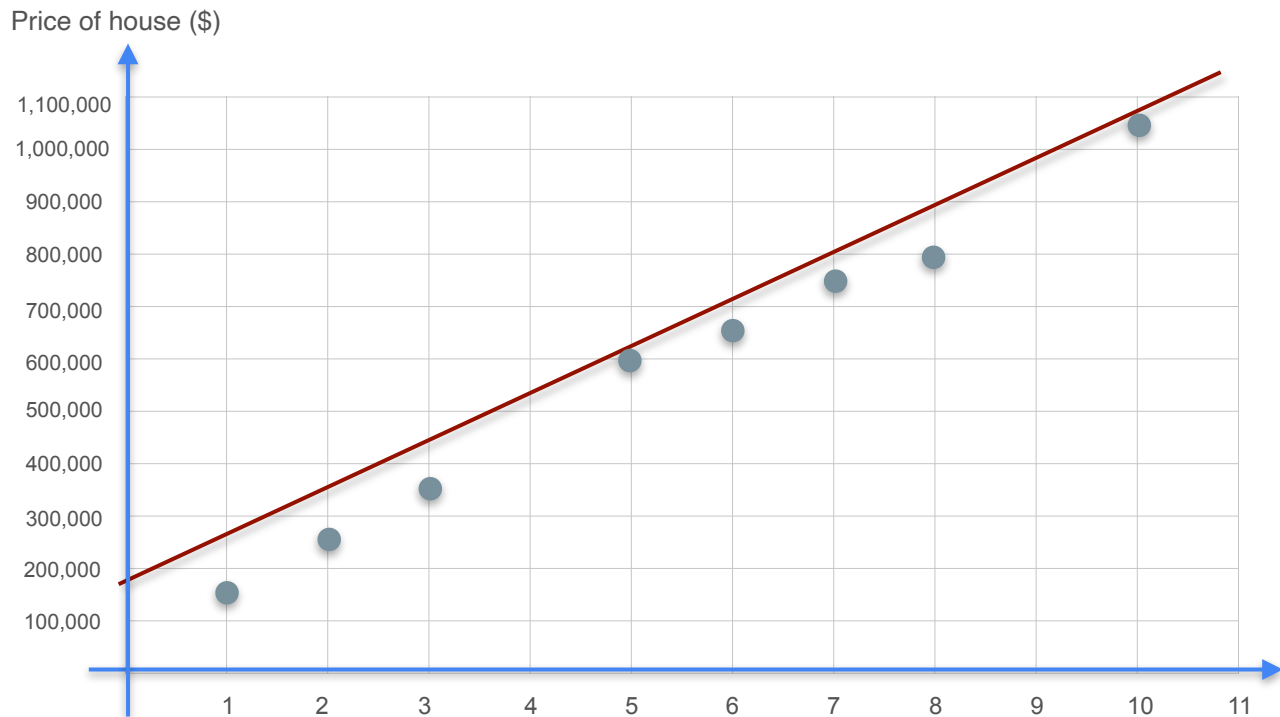
Model Training



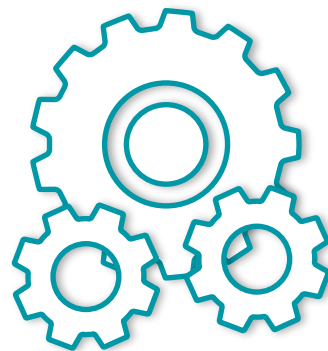
Machine Learning
Model

*Number of
Bedrooms*

Machine Learning Motivation



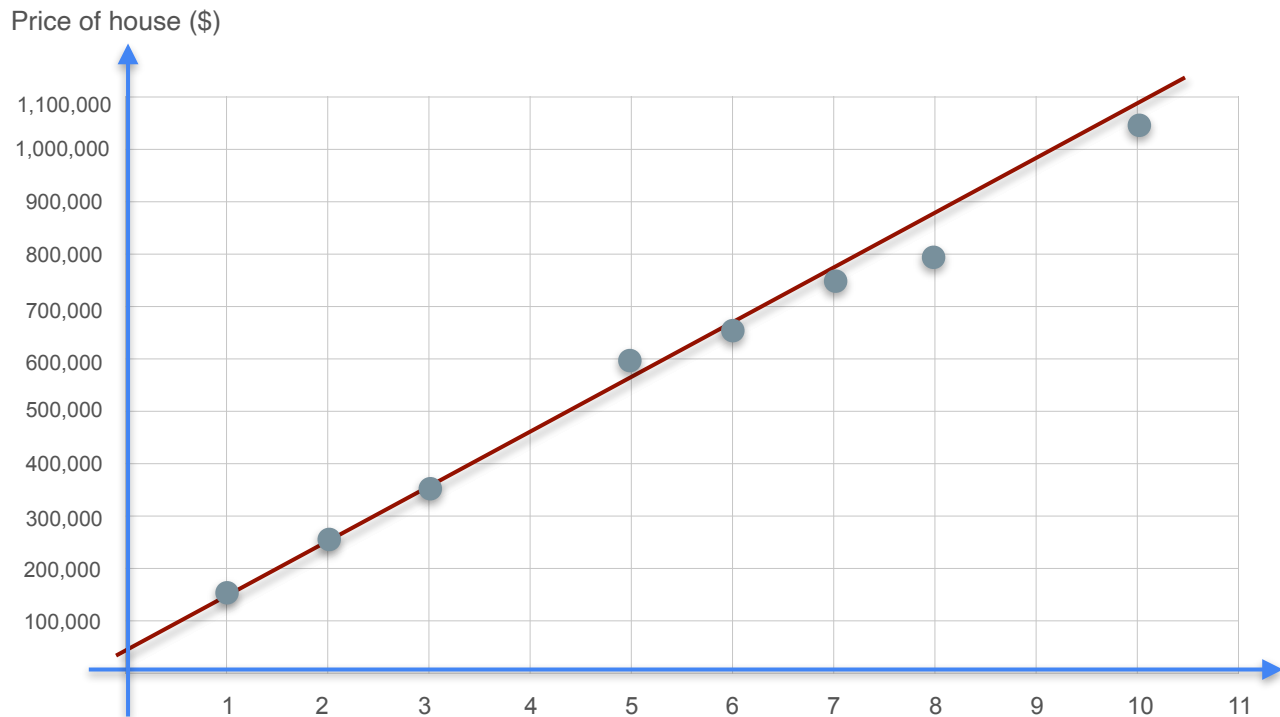
Model Training



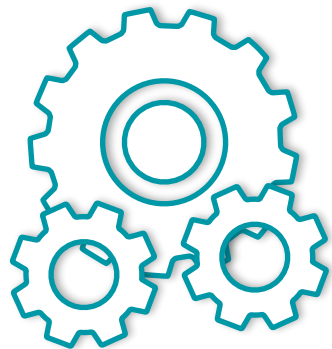
Machine Learning
Model

*Number of
Bedrooms*

Machine Learning Motivation



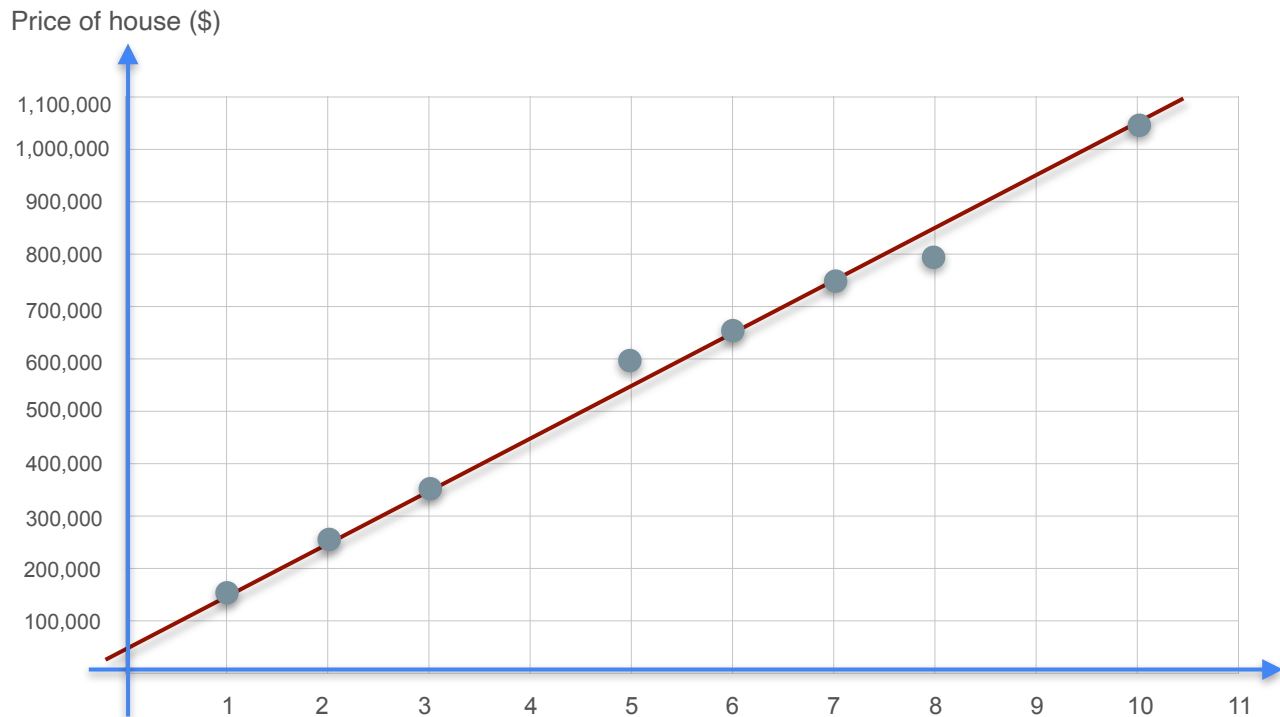
Model Training



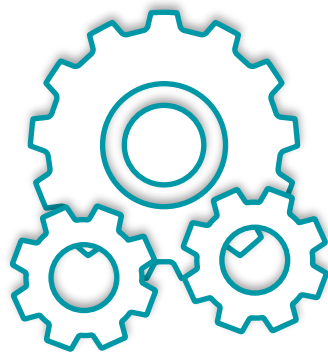
Machine Learning
Model

*Number of
Bedrooms*

Machine Learning Motivation



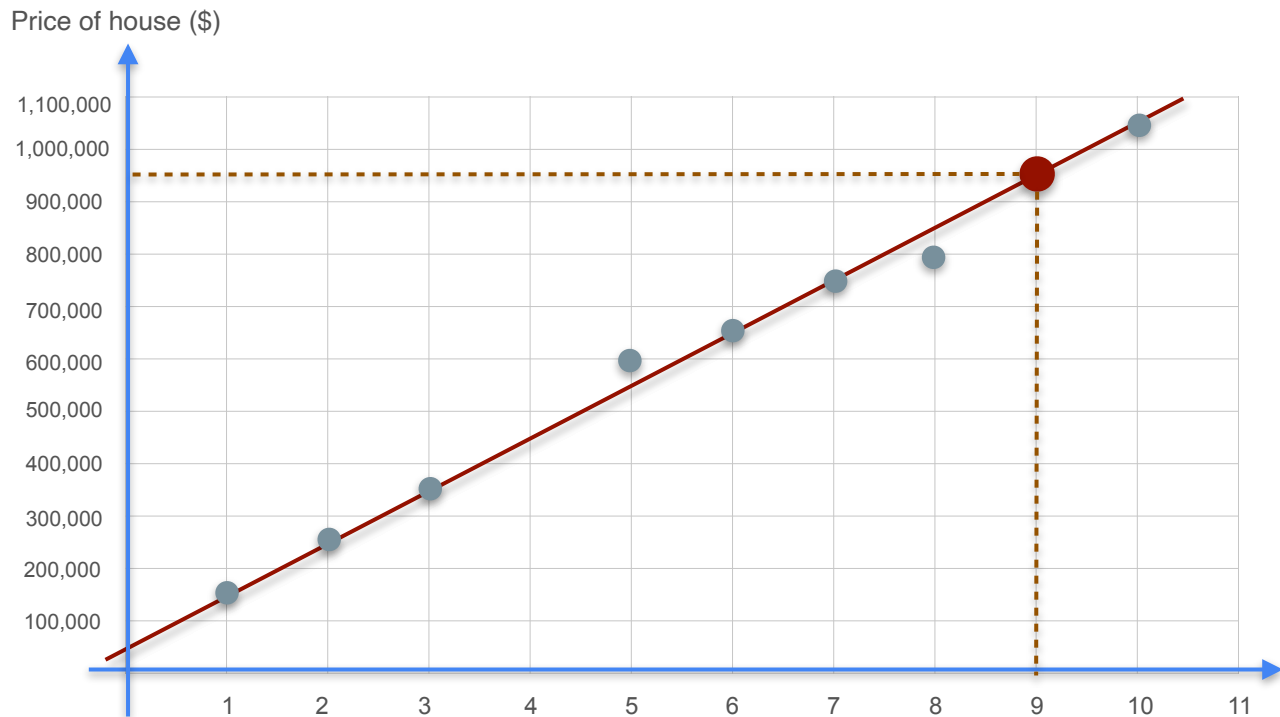
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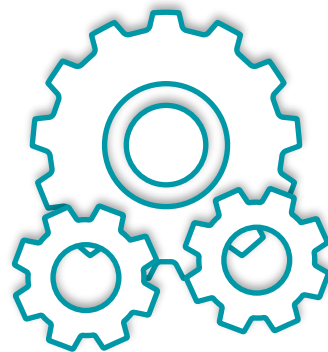
Machine Learning
Model

*Number of
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Machine Learning Motivation



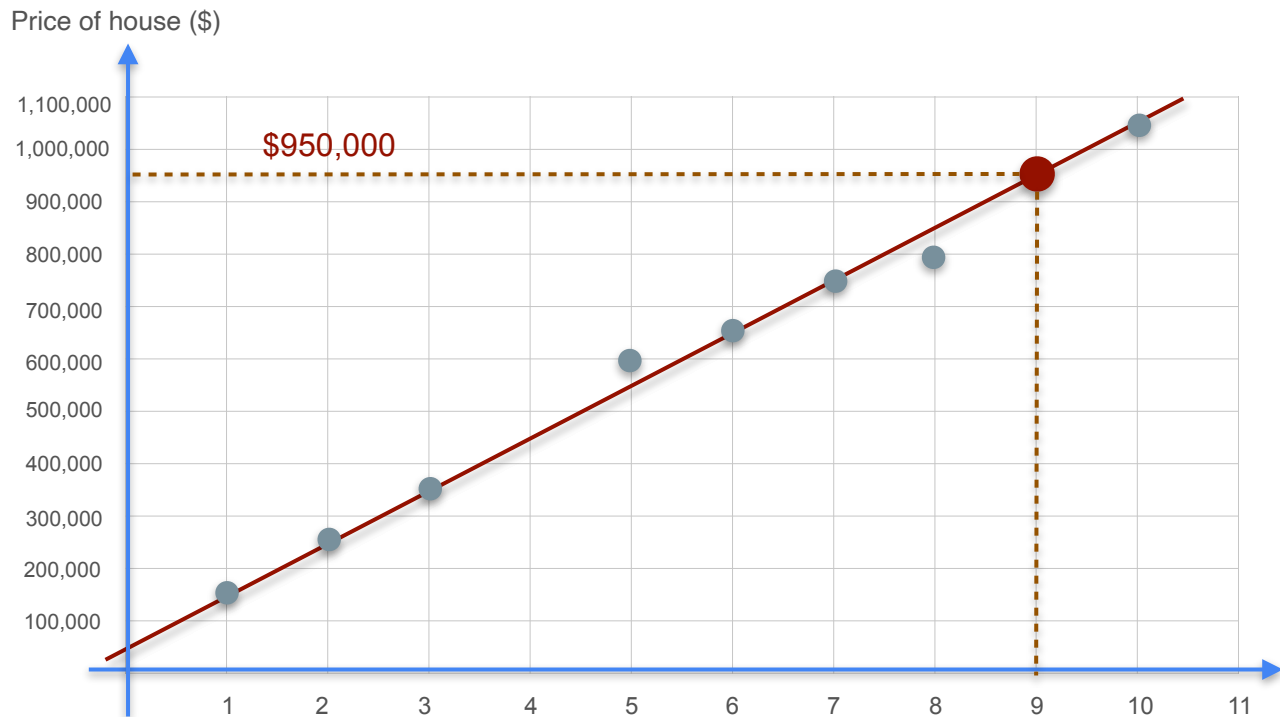
Model Training



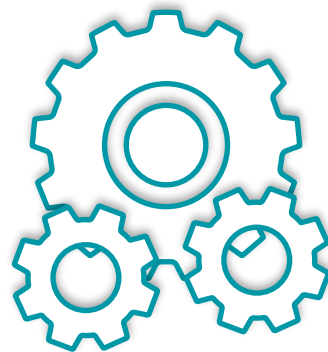
Machine Learning
Model

*Number of
Bedrooms*

Machine Learning Motivation



Model Training

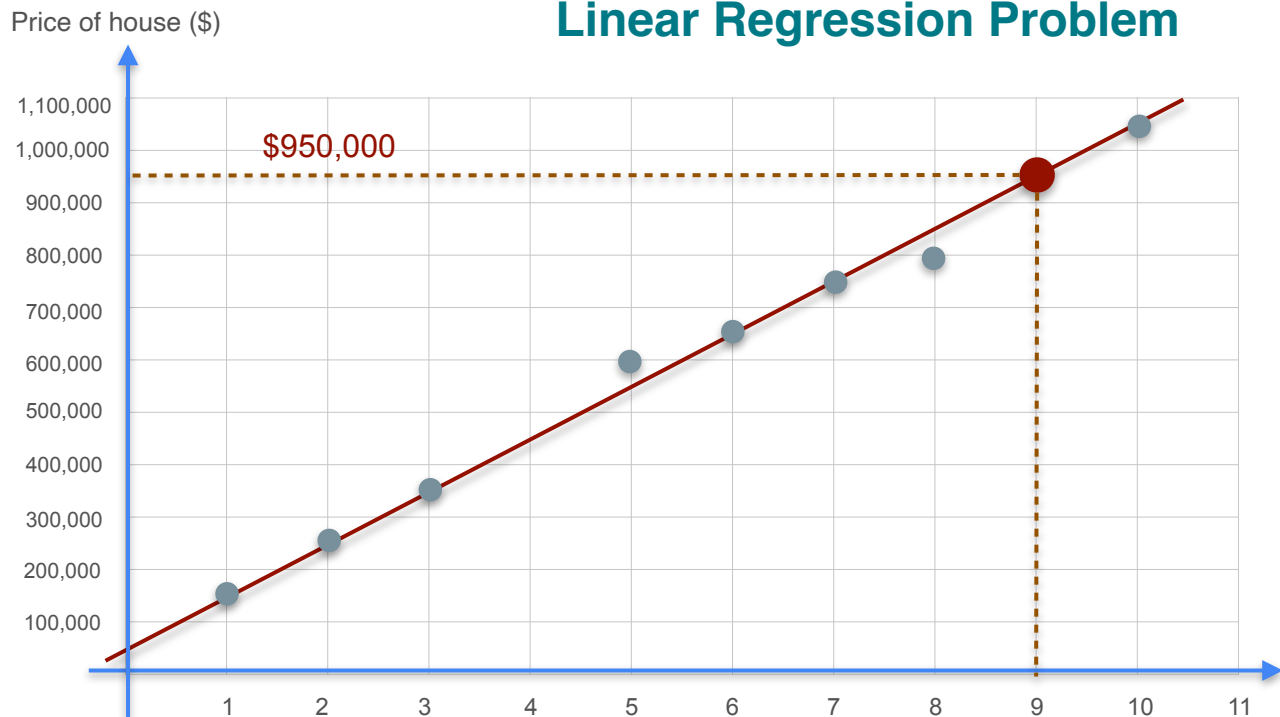


Machine Learning
Model

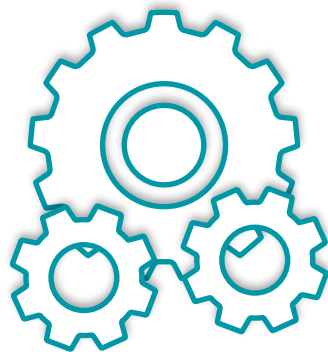
*Number of
Bedrooms*

Machine Learning Motivation

Linear Regression Problem



Model Training

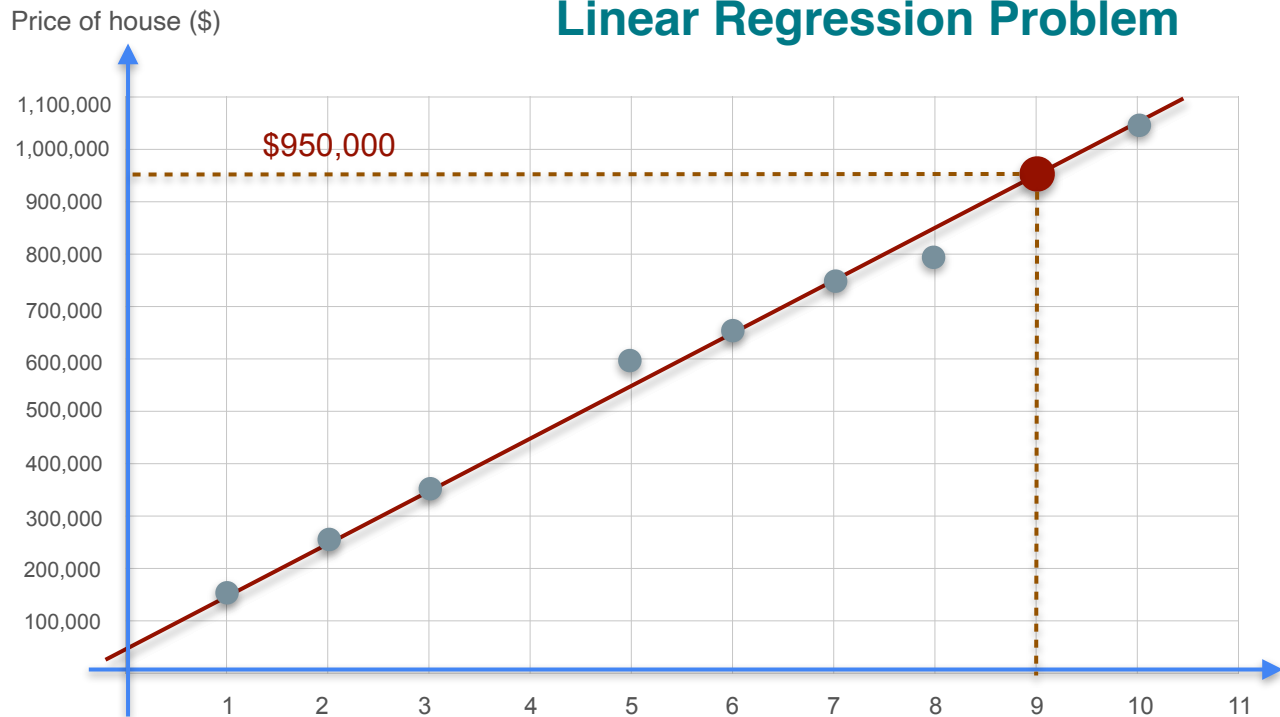


Machine Learning
Model

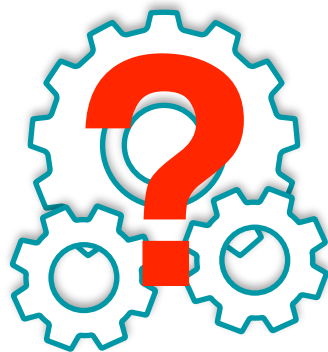
*Number of
Bedrooms*

Machine Learning Motivation

Linear Regression Problem



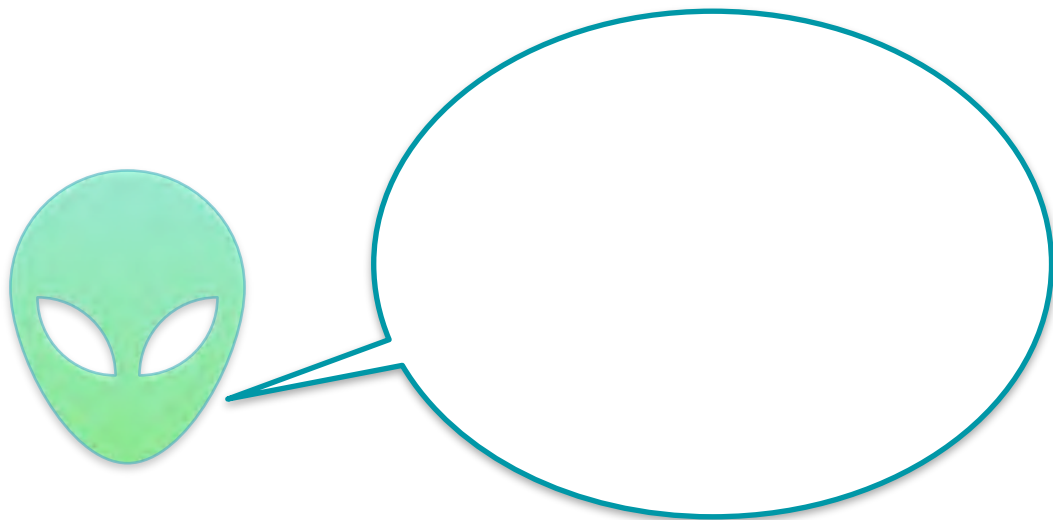
Model Training



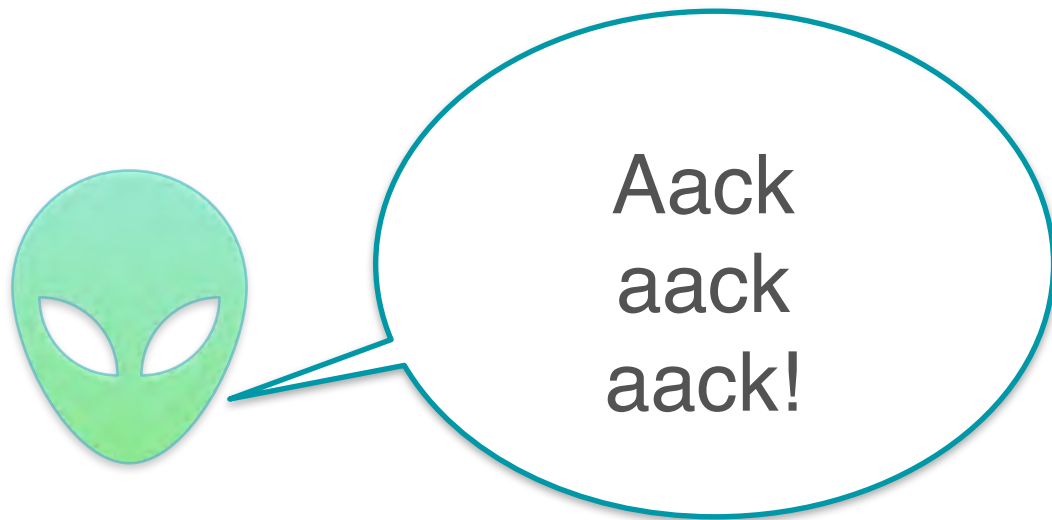
Machine Learning
Model

*Number of
Bedrooms*

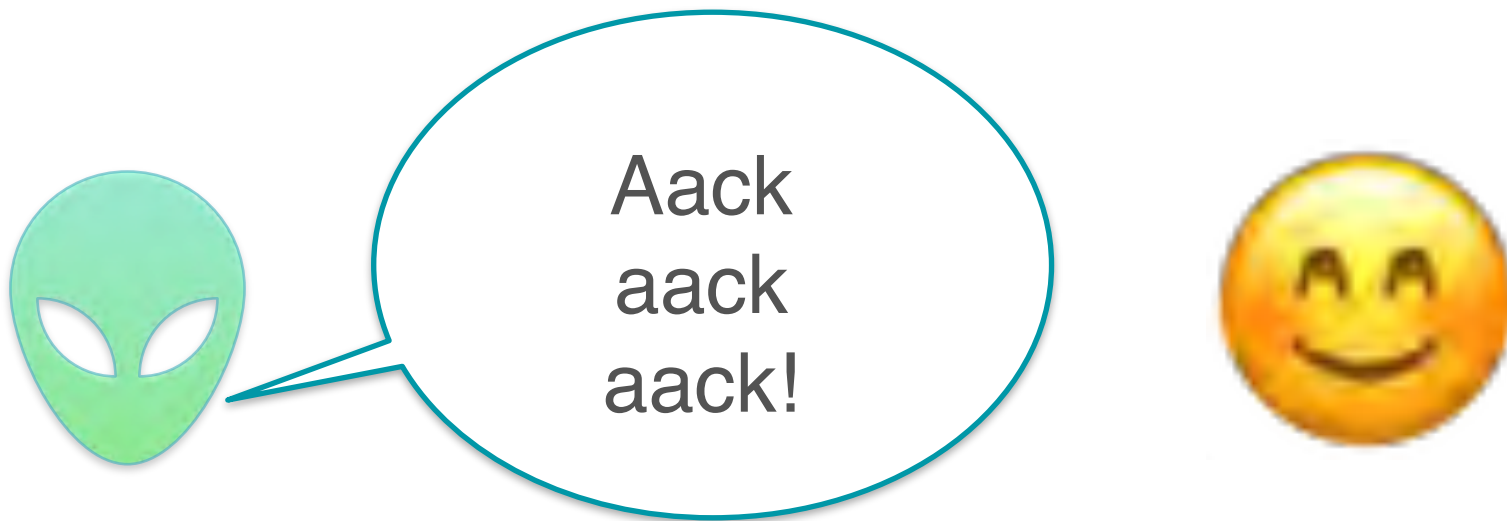
Machine Learning Motivation



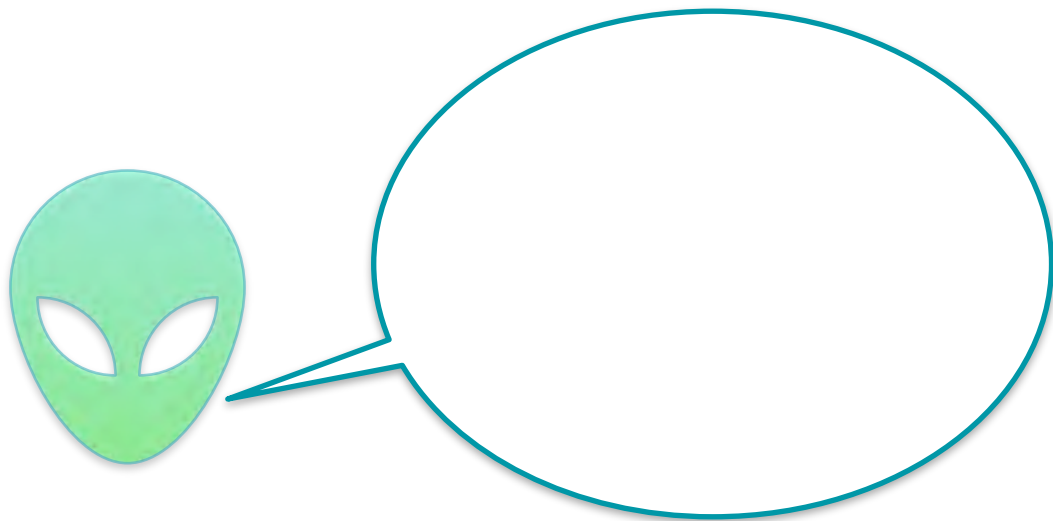
Machine Learning Motivation



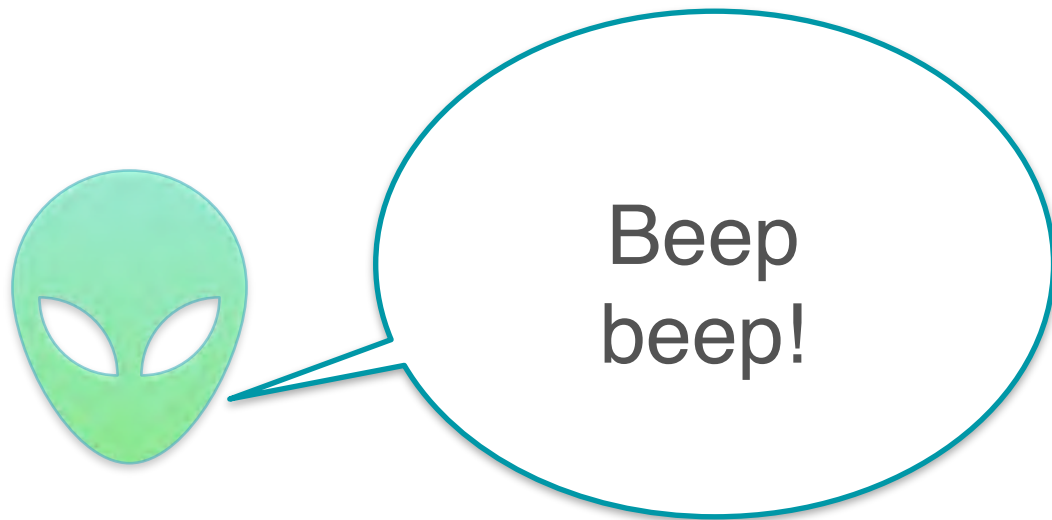
Machine Learning Motivation



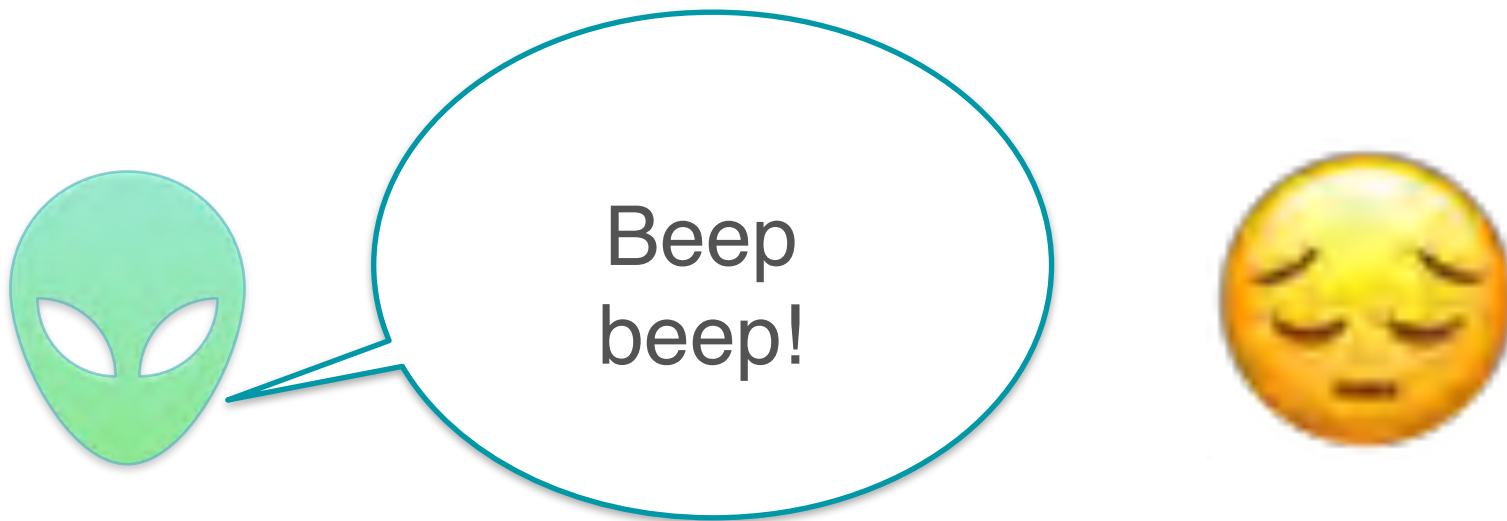
Machine Learning Motivation



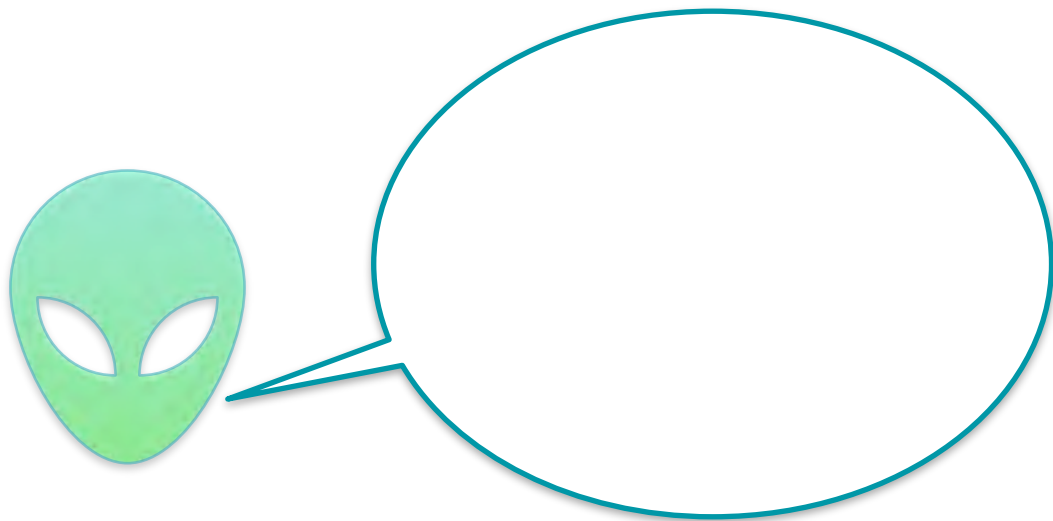
Machine Learning Motivation



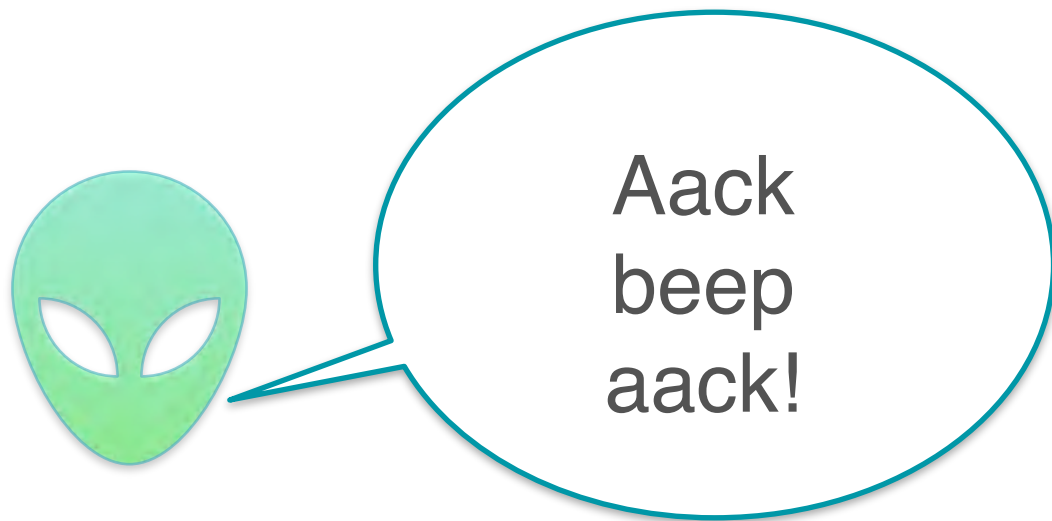
Machine Learning Motivation



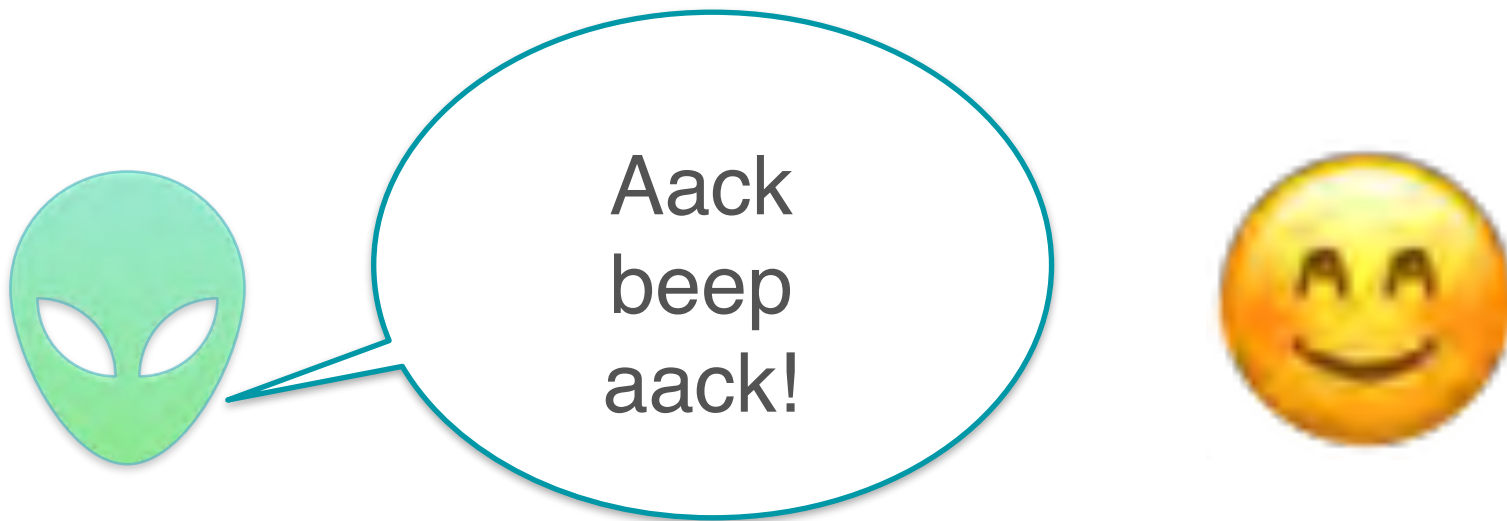
Machine Learning Motivation



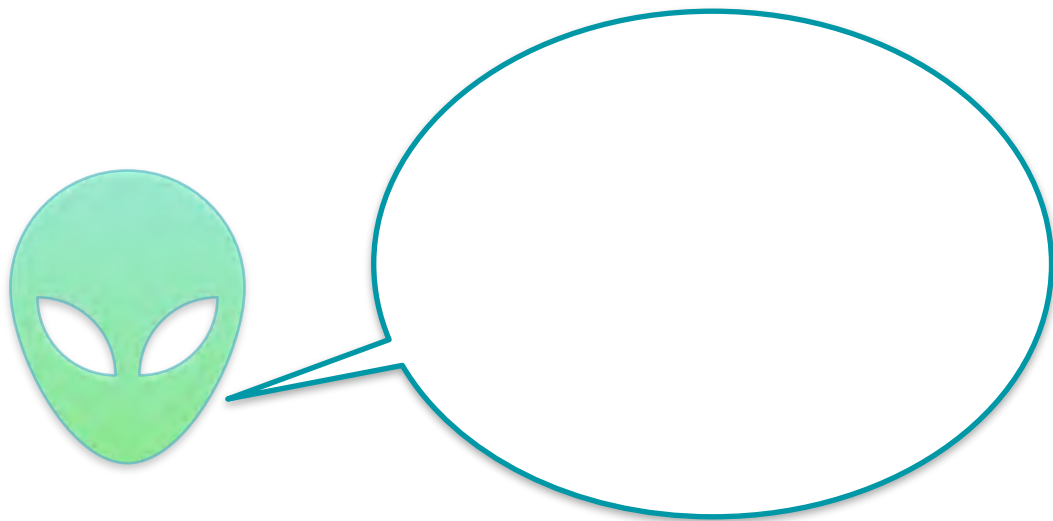
Machine Learning Motivation



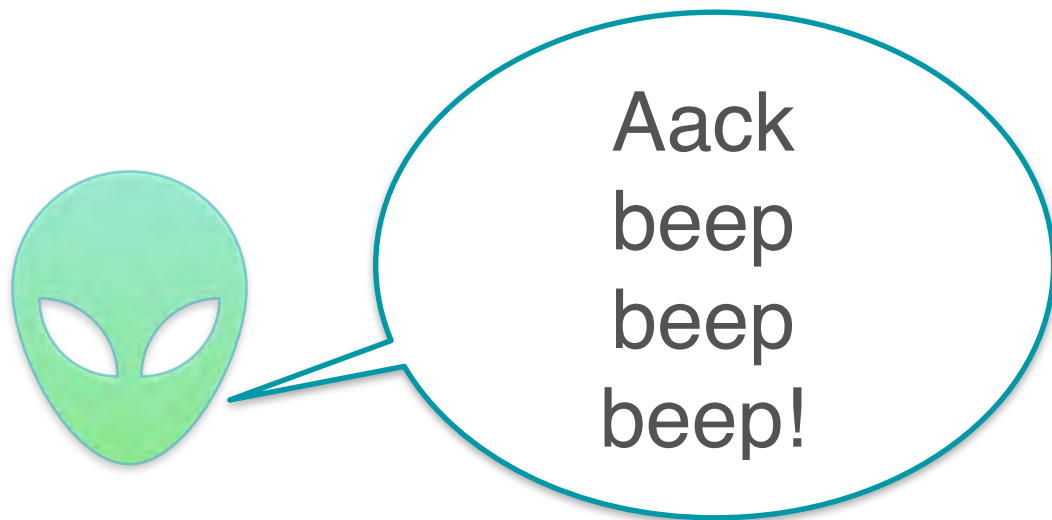
Machine Learning Motivation



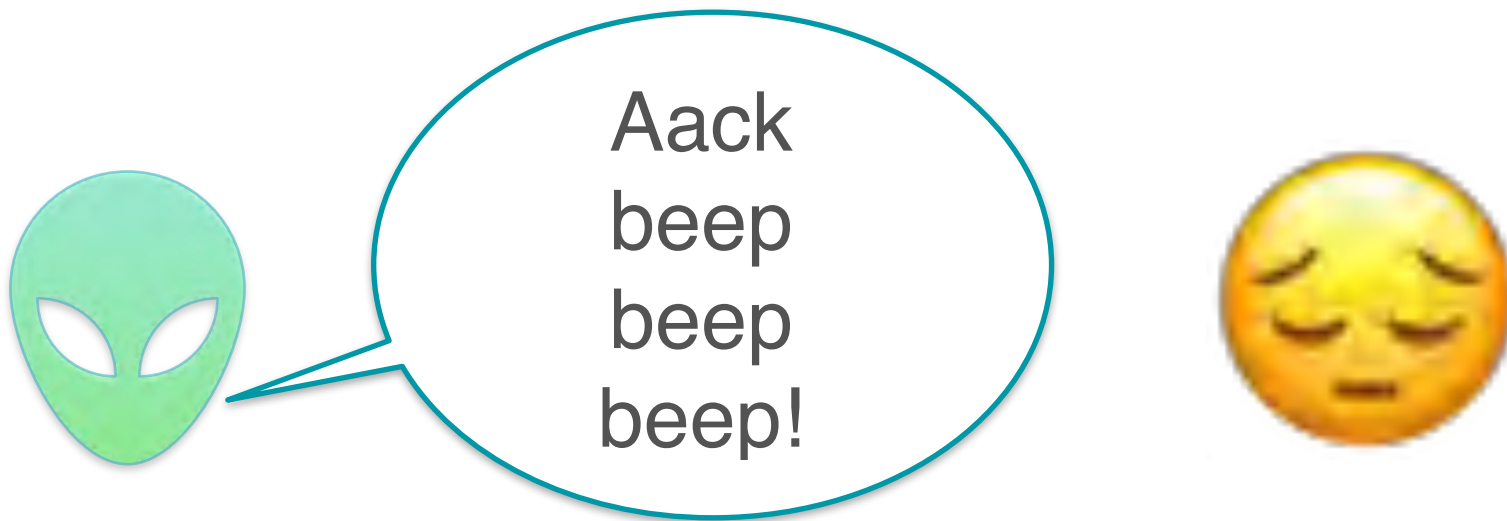
Machine Learning Motivation



Machine Learning Motivation



Machine Learning Motivation



Machine Learning Motivation



Machine Learning Motivation



Sentence

Aack aack aack!

Beep beep!

Aack beep aack!

*Aack beep beep
beep!*

Machine Learning Motivation



<i>Sentence</i>	<i>Aack</i>	<i>Beep</i>
<i>Aack aack aack!</i>	<i>3</i>	<i>0</i>
<i>Beep beep!</i>	<i>0</i>	<i>2</i>
<i>Aack beep aack!</i>	<i>2</i>	<i>1</i>
<i>Aack beep beep beep!</i>	<i>1</i>	<i>3</i>

Machine Learning Motivation



<i>Sentence</i>	<i>Aack</i>	<i>Beep</i>	<i>Mood</i>
<i>Aack aack aack!</i>	<i>3</i>	<i>0</i>	😊
<i>Beep beep!</i>	<i>0</i>	<i>2</i>	😞
<i>Aack beep aack!</i>	<i>2</i>	<i>1</i>	😊
<i>Aack beep beep beep!</i>	<i>1</i>	<i>3</i>	😞

Machine Learning Motivation



<i>Sentence</i>	<i>Aack</i>	<i>Beep</i>	<i>Mood</i>
<i>Aack aack aack!</i>	<i>3</i>	<i>0</i>	😊
<i>Beep beep!</i>	<i>0</i>	<i>2</i>	😞
<i>Aack beep aack!</i>	<i>2</i>	<i>1</i>	😊
<i>Aack beep beep beep!</i>	<i>1</i>	<i>3</i>	😞

**A classification
problem**

Machine Learning Motivation

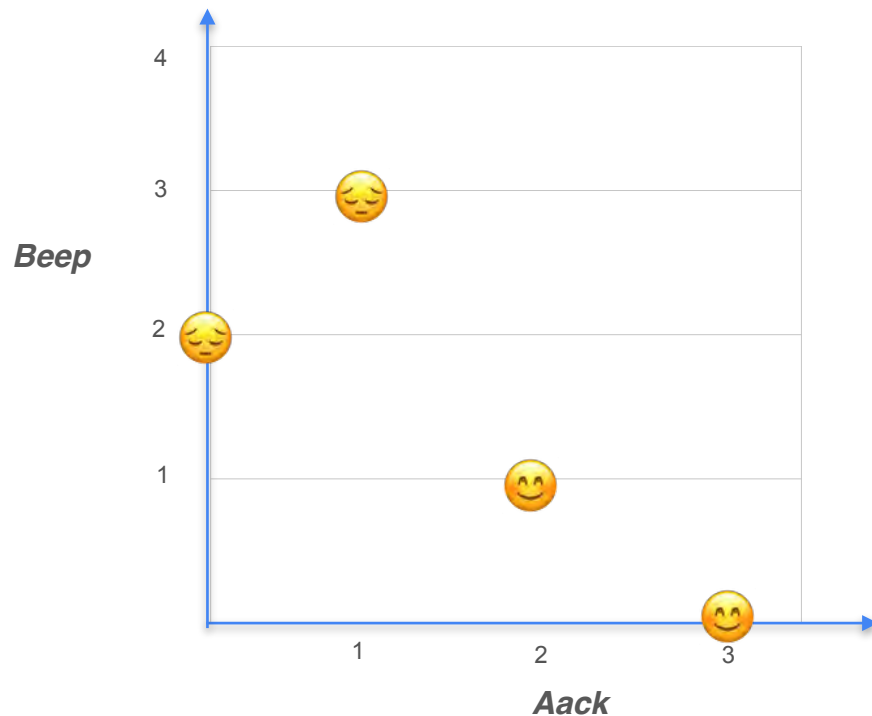


<i>Sentence</i>	<i>Aack</i>	<i>Beep</i>	<i>Mood</i>
<i>Aack aack aack!</i>	<i>3</i>	<i>0</i>	😊
<i>Beep beep!</i>	<i>0</i>	<i>2</i>	😞
<i>Aack beep aack!</i>	<i>2</i>	<i>1</i>	😊
<i>Aack beep beep beep!</i>	<i>1</i>	<i>3</i>	😞

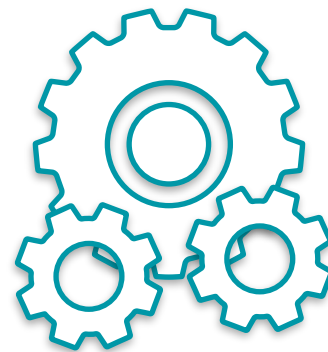
**A classification
problem**

Sentiment analysis

Machine Learning Motivation

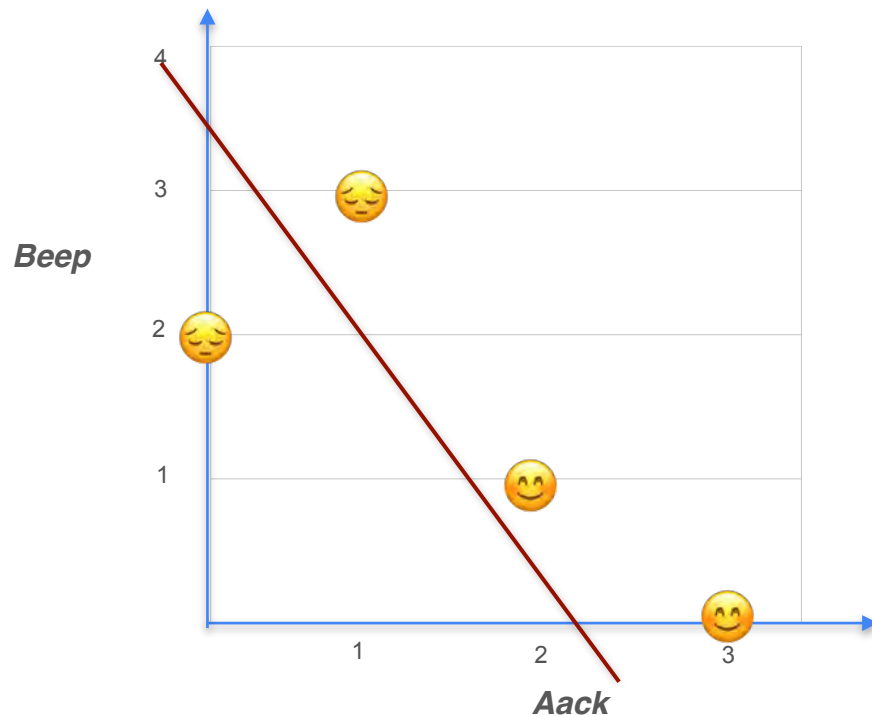


Model Training

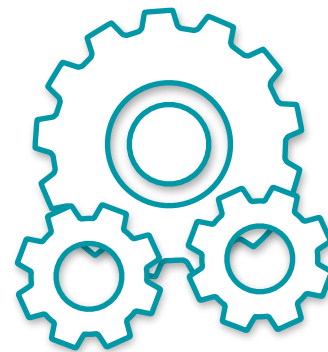


Machine Learning
Model

Machine Learning Motivation

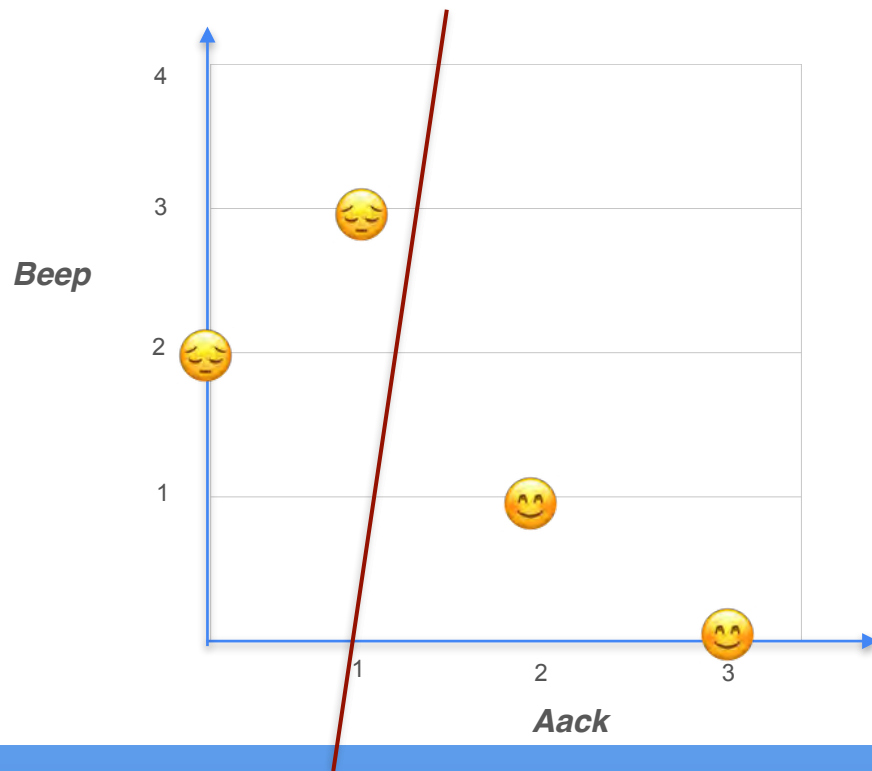


Model Training

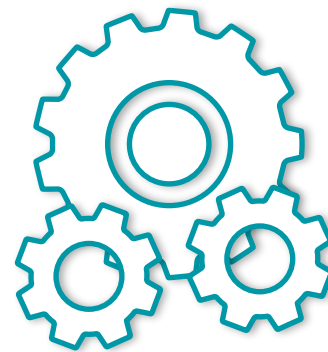


Machine Learning
Model

Machine Learning Motivation

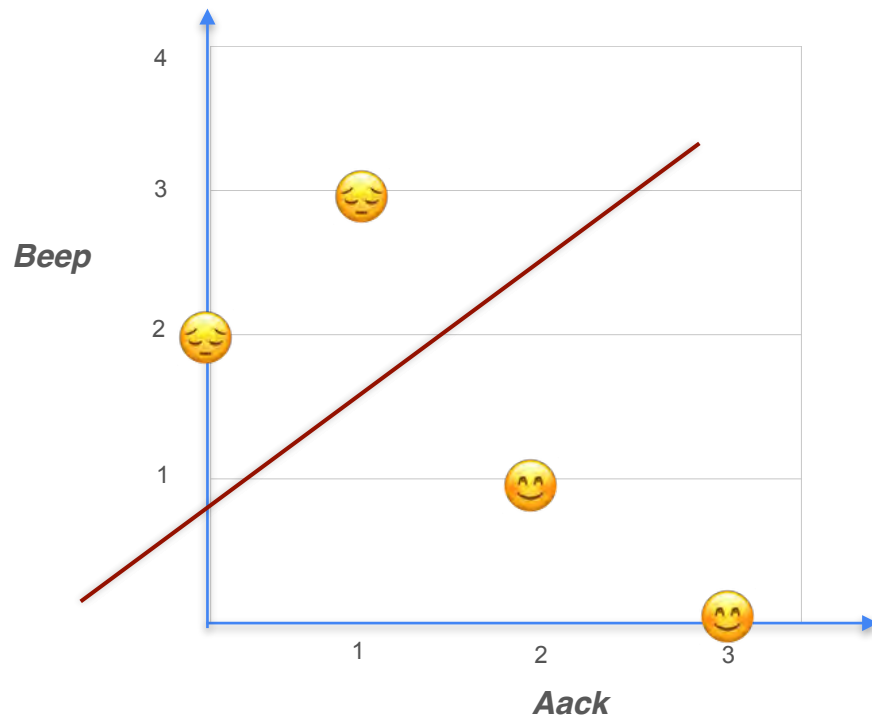


Model Training

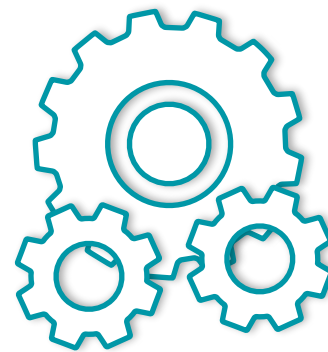


Machine Learning
Model

Machine Learning Motivation

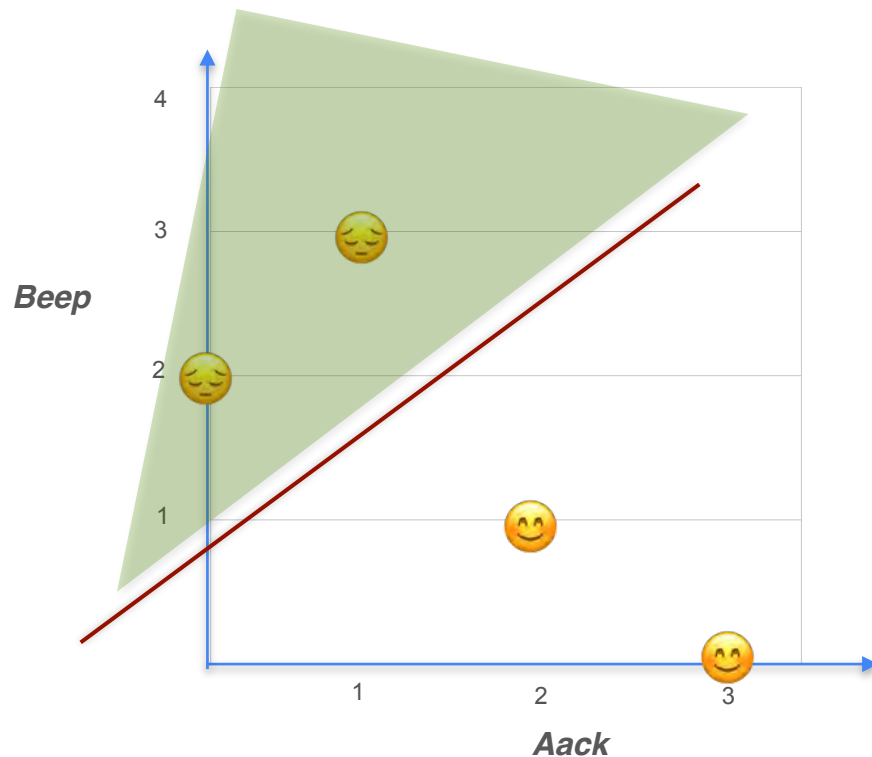


Model Training

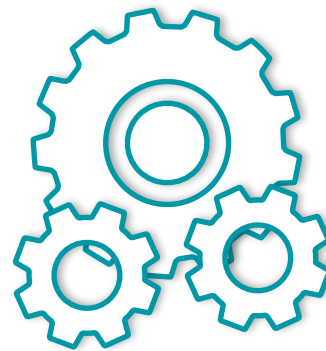


Machine Learning
Model

Machine Learning Motivation

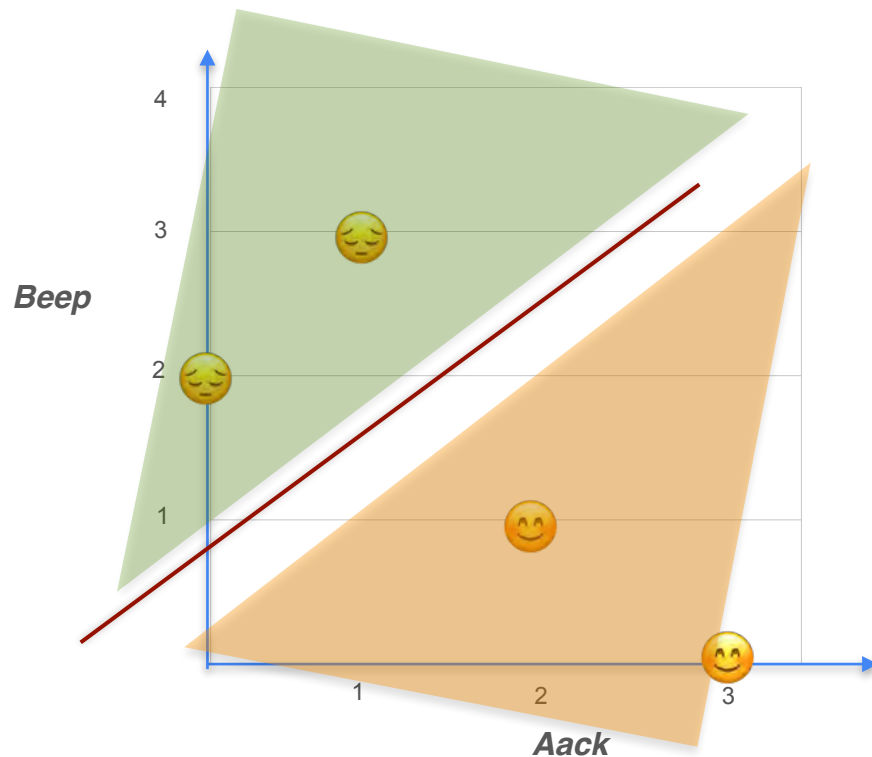


Model Training

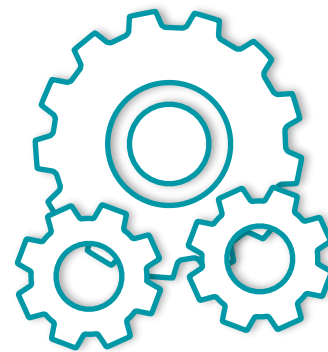


Machine Learning
Model

Machine Learning Motivation

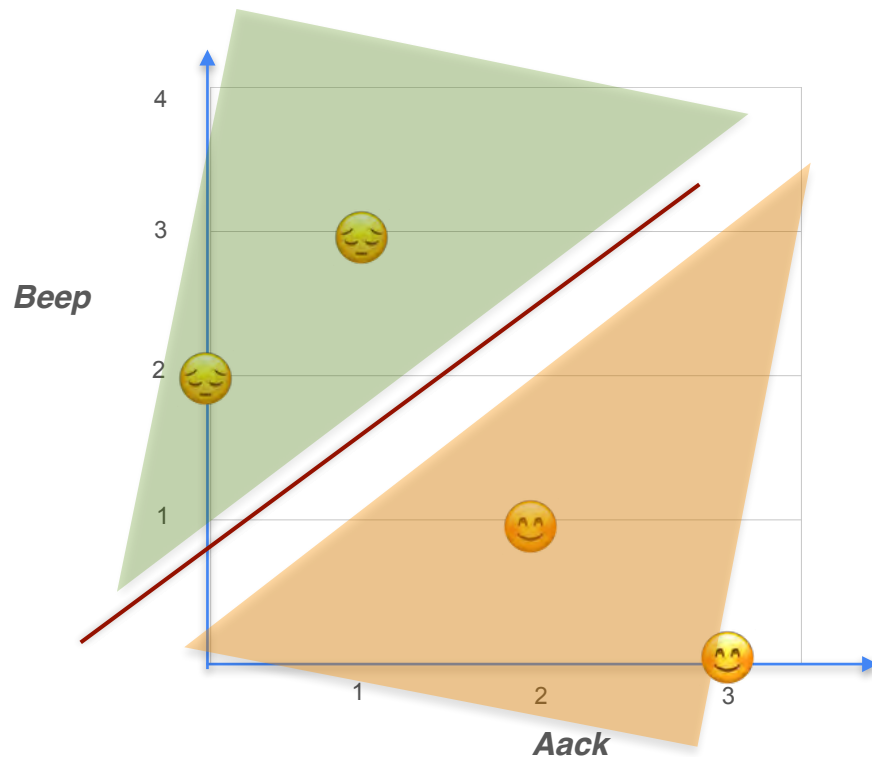


Model Training

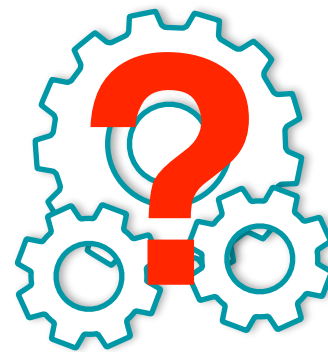


Machine Learning
Model

Machine Learning Motivation

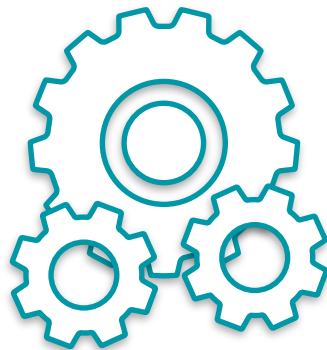


Model Training



Machine Learning
Model

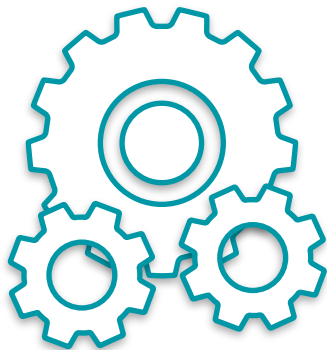
Machine Learning Motivation



Machine Learning
Model

Machine Learning Motivation

Maths concepts used in training a model



Machine Learning
Model

Machine Learning Motivation

Maths concepts used in training a model

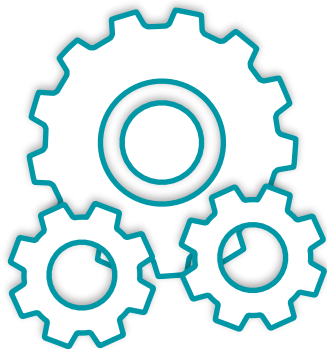
Gradients

Derivatives

Optimization

Loss and Cost functions

Gradient Descent



Machine Learning
Model

Machine Learning Motivation

Maths concepts used in training a model

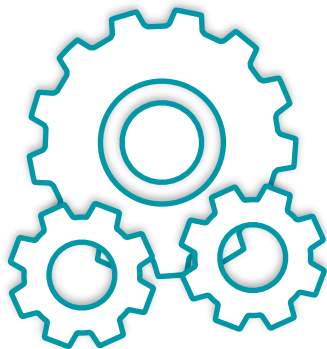
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Machine Learning
Model

Linear Regression

Machine Learning Motivation

Maths concepts used in training a model

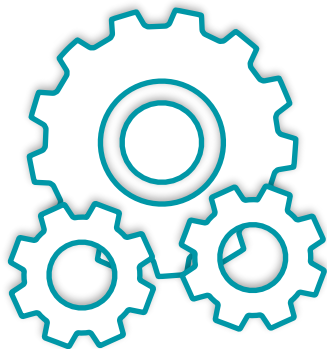
Gradients

Derivatives

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Gradient Descent



Machine Learning
Model

Linear Regression

Classification

Machine Learning Motivation

Maths concepts used in training a model

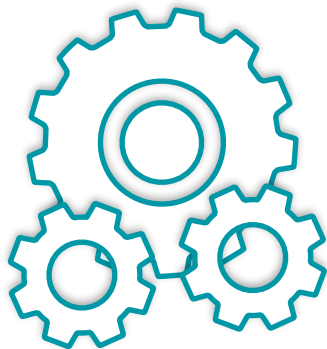
Gradients

Derivatives

Optimization

Loss and Cost functions

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Machine Learning
Model

Linear Regression

Classification

Neural Networks



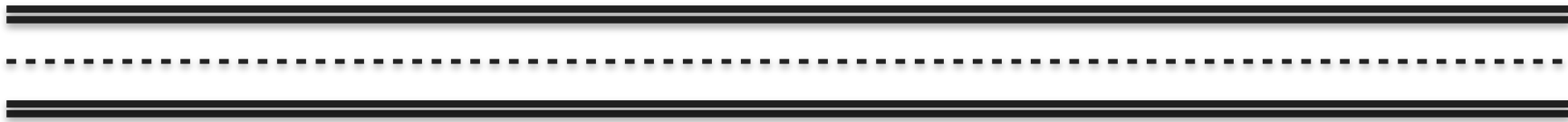
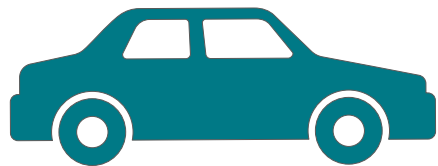
DeepLearning.AI

Derivatives and Optimization

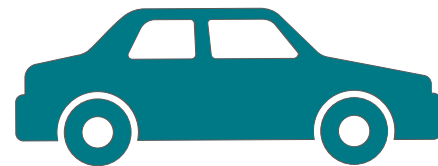
Introduction to derivatives

Introduction to Derivatives

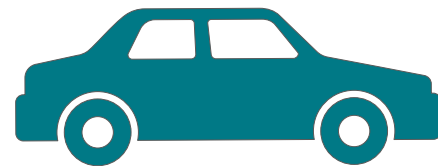
Introduction to Derivatives



Introduction to Derivatives



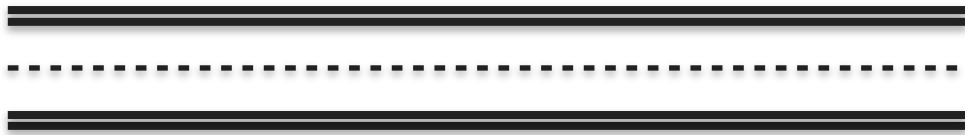
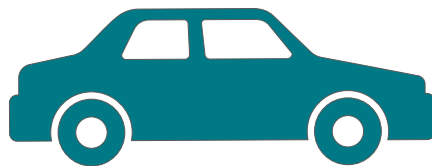
Introduction to Derivatives



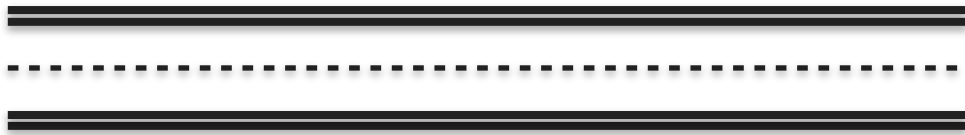
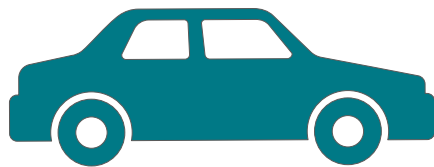
Introduction to Derivatives



Introduction to Derivatives

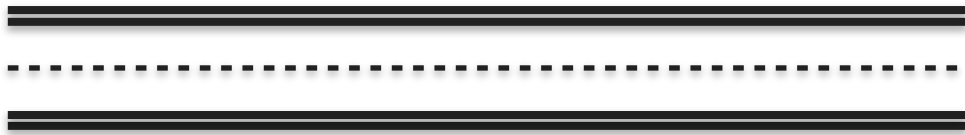
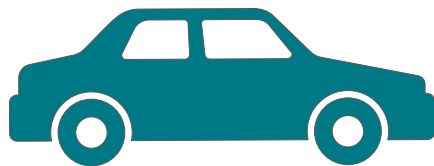


Introduction to Derivatives



Introduction to Derivatives

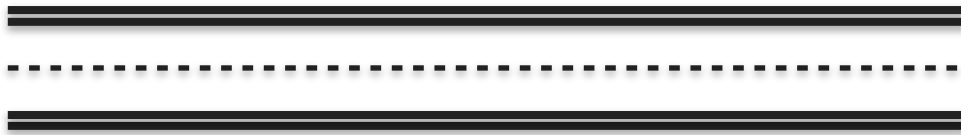
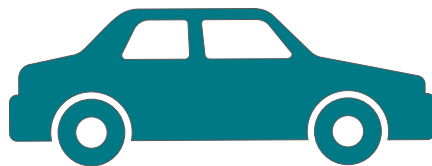
Every 5 seconds



Introduction to Derivatives

Every 5 seconds

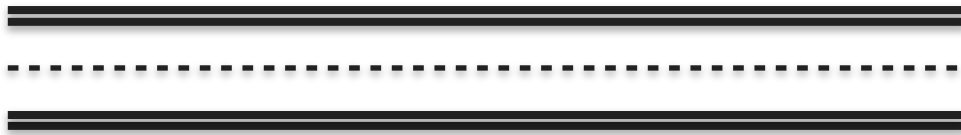
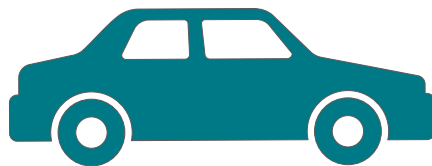
t (seconds)
0
5
10
15
20
25
30
35
40
45
50
55
60



Introduction to Derivatives

Every 5 seconds

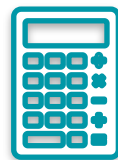
t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000



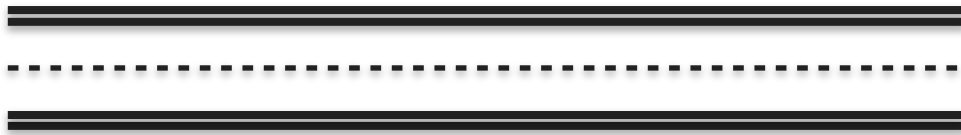
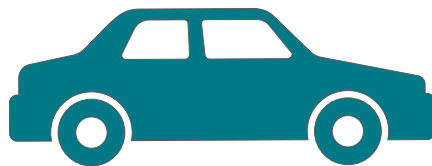
Introduction to Derivatives

Every 5 seconds

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000



Is the car moving at a constant speed?



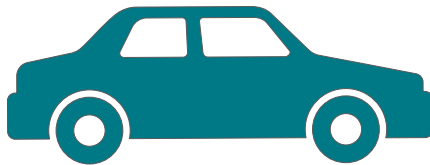
Introduction to Derivatives

Every 5 seconds

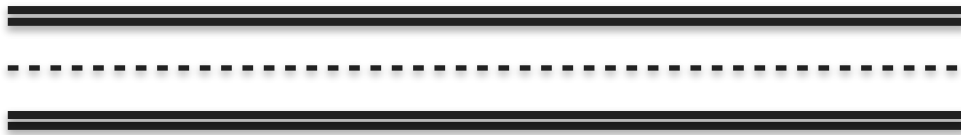
t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000



Is the car moving at a constant speed?



Hint: Look at the distance values in the table



Introduction to Derivatives

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

Introduction to Derivatives

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

Introduction to Derivatives

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

10 - 15 seconds: $202 - 122 = 80$ meters

Introduction to Derivatives

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

10 - 15 seconds: $202 - 122 = 80$ meters

Introduction to Derivatives

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

10 - 15 seconds: $202 - 122 = \mathbf{80 \text{ meters}}$

15 - 20 seconds: $265 - 202 = \mathbf{63 \text{ meters}}$

Introduction to Derivatives

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

10 - 15 seconds: $202 - 122 = 80$ meters

15 - 20 seconds: $265 - 202 = 63$ meters

Not traveling at constant speed

Introduction to Derivatives

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

Introduction to Derivatives

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

Can you use the information presented to determine your velocity at exactly ***$t = 12.5 \text{ seconds}$*** ?

Introduction to Derivatives

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

Can you use the information presented to determine your velocity at exactly ***$t = 12.5 \text{ seconds}$*** ?

Hint: velocity = distance traveled / time taken

Introduction to Derivatives

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	275
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

Can you use the information presented to determine your velocity at exactly ***$t = 12.5$ seconds***?

Introduction to Derivatives

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	275
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

Can you use the information presented to determine your velocity at exactly $t = 12.5 \text{ seconds}$?

Quiz : Slope of a Line 1

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	275
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

Quiz : Slope of a Line 1

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	275
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

What was the average velocity of the car on the interval from 10 to 15 seconds?

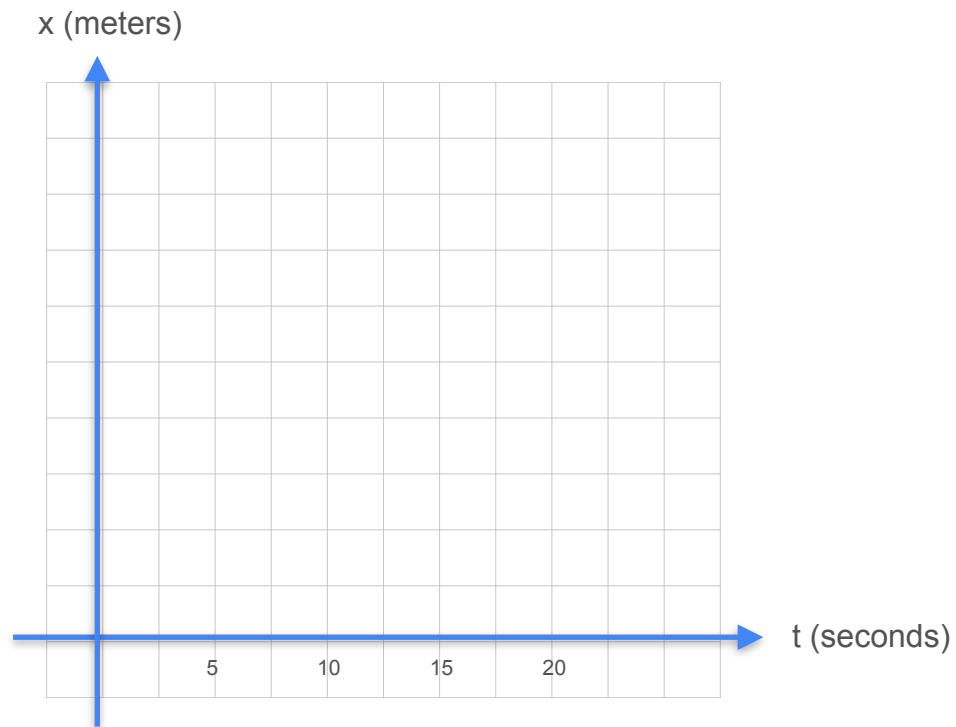
Quiz : Slope of a Line 1 - Solution

t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	275
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

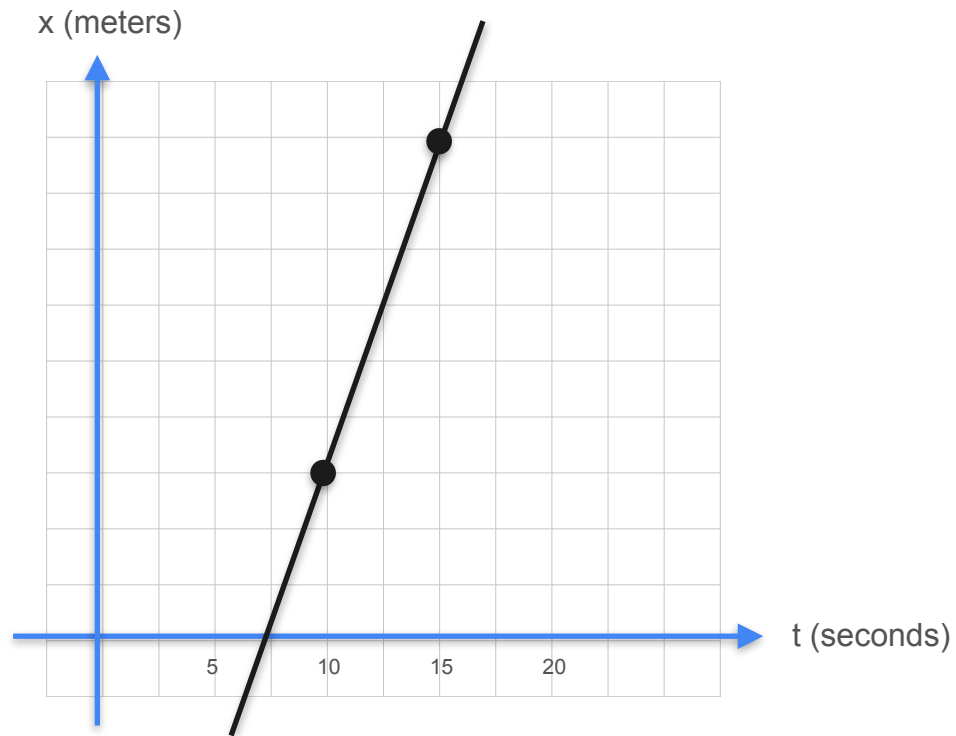
What was the average velocity of the car on the interval from 10 to 15 seconds?

Calculating the Slope

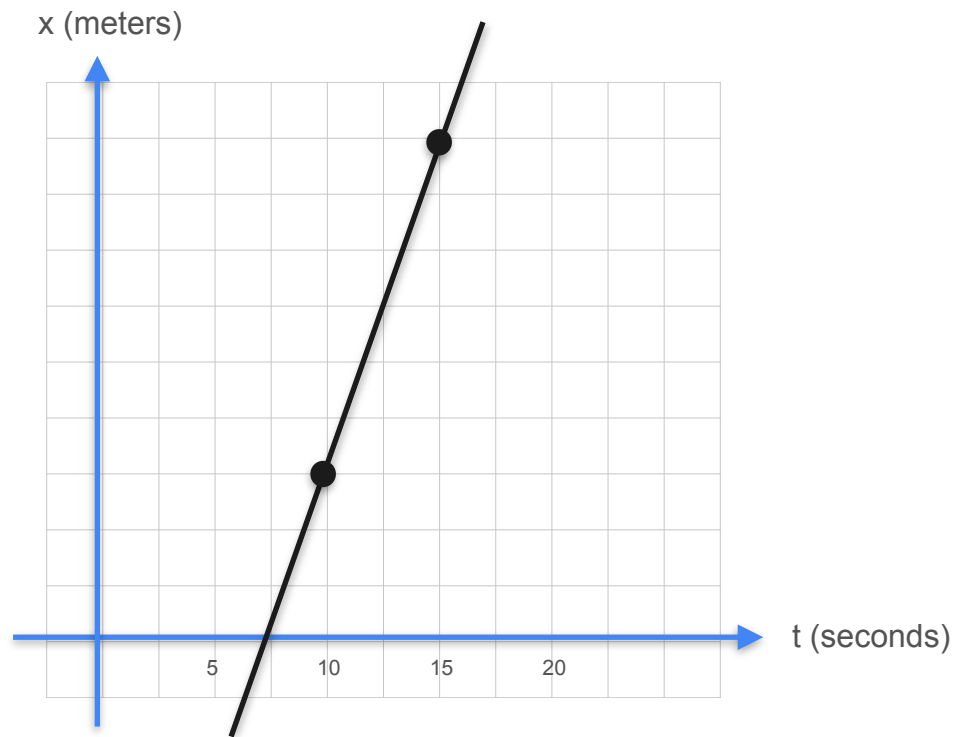
Calculating the Slope



Calculating the Slope

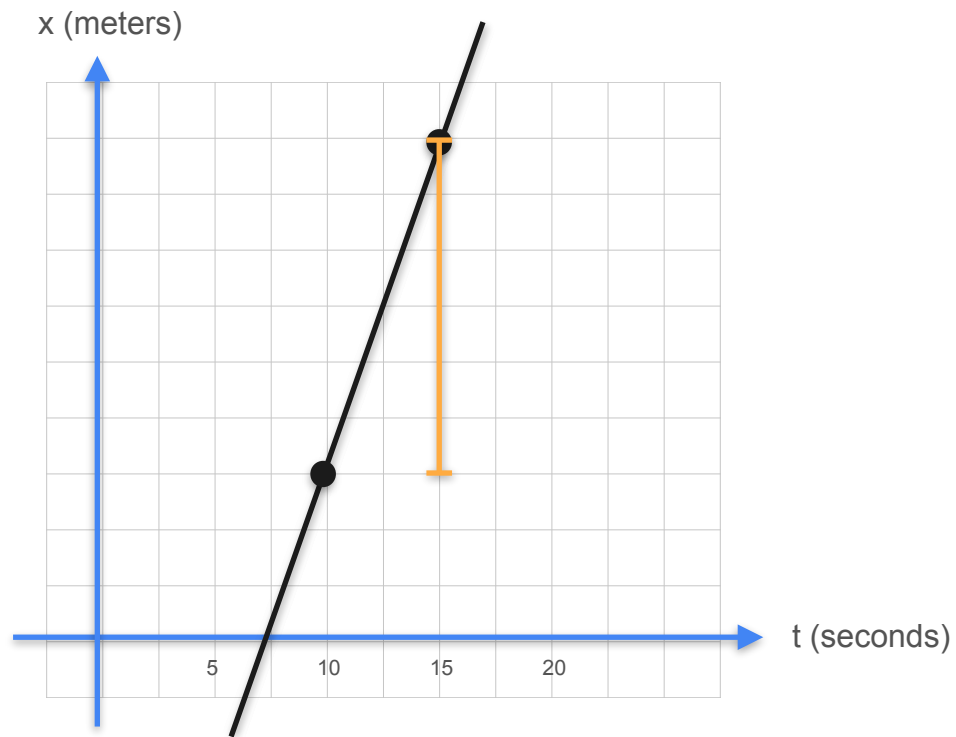


Calculating the Slope



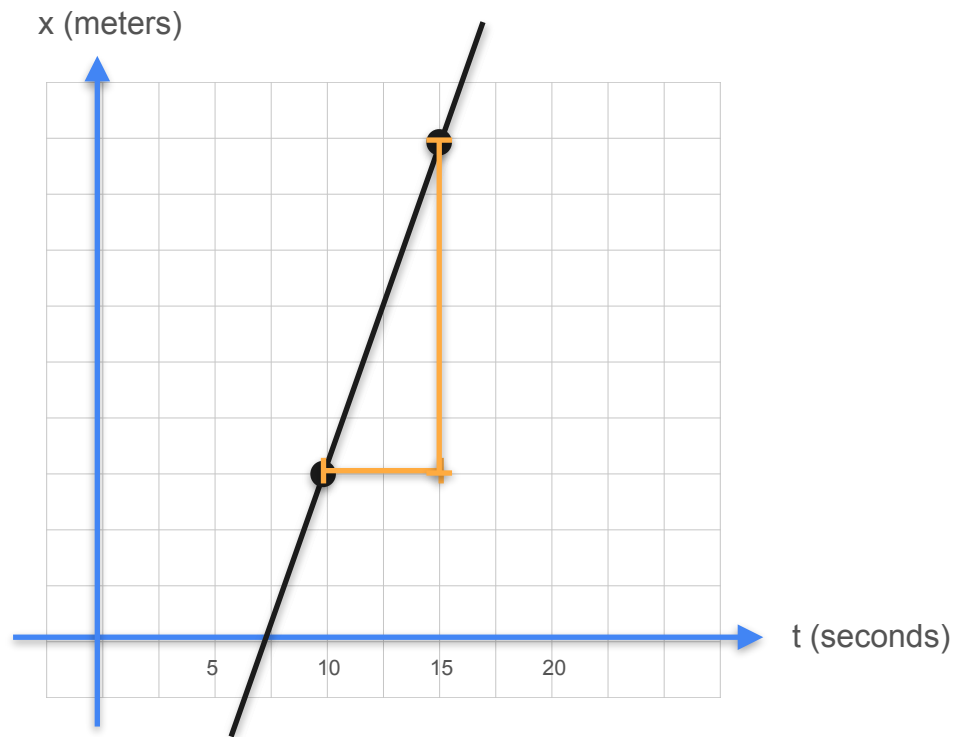
$$\text{slope} = \frac{\text{rise}}{\text{run}}$$

Calculating the Slope



$$\text{slope} = \frac{\text{rise}}{\text{run}} \quad \uparrow$$

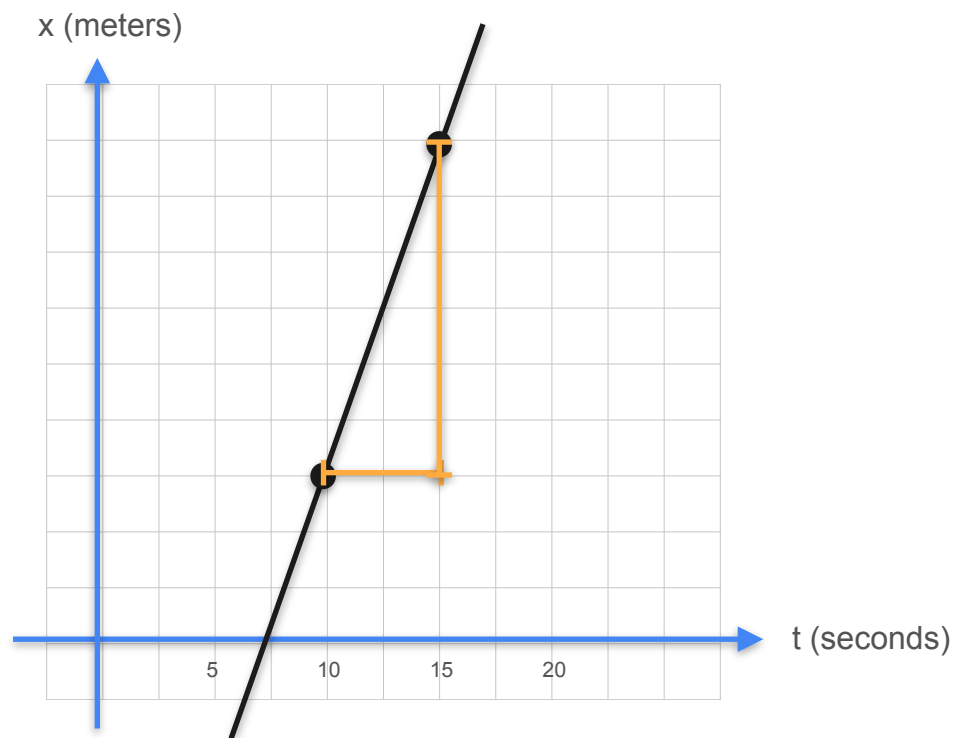
Calculating the Slope



$$\text{slope} = \frac{\text{rise}}{\text{run}}$$

↑
→

Calculating the Slope

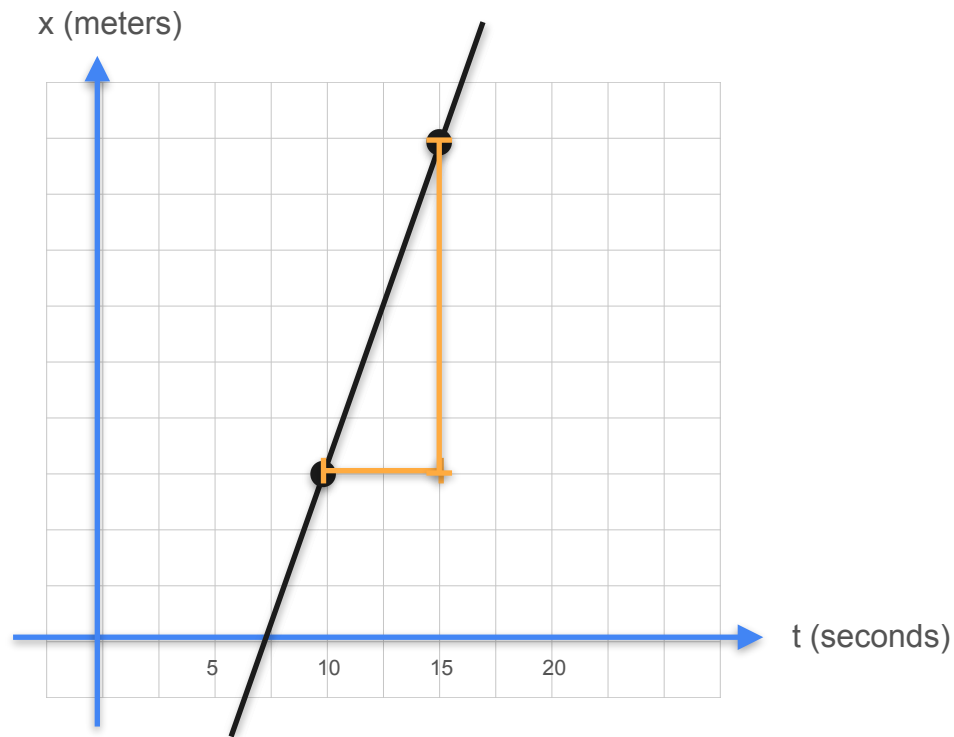


$$\text{slope} = \frac{\text{rise}}{\text{run}}$$

↑
→

$$\text{slope} = \frac{\text{change in distance}(\Delta x)}{\text{change in time}(\Delta t)}$$

Calculating the Slope



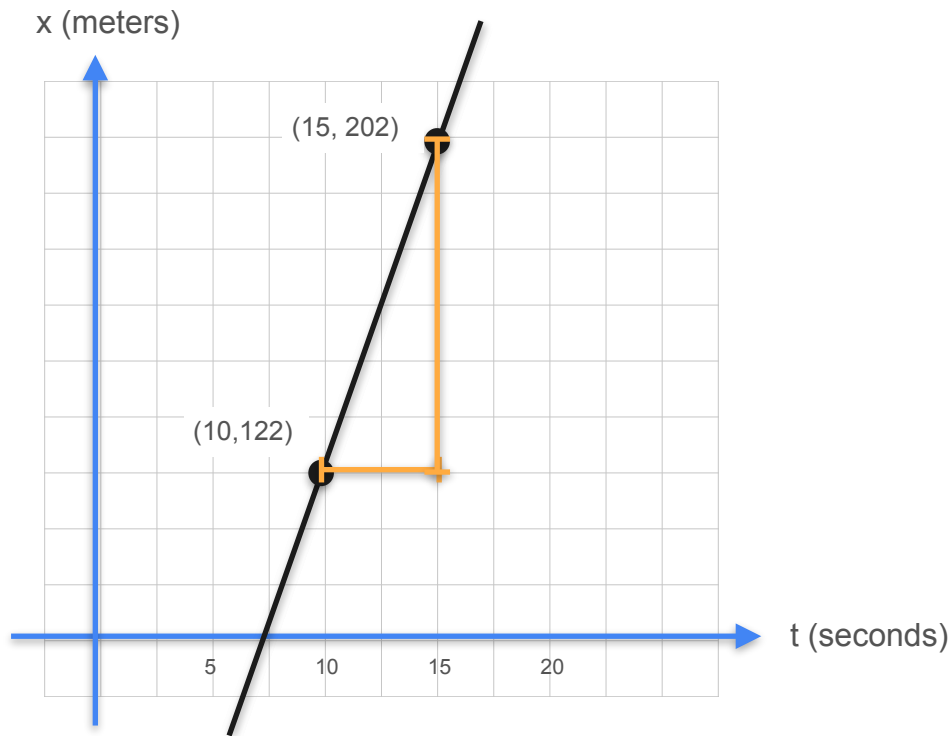
$$\text{slope} = \frac{\text{rise}}{\text{run}}$$

↑
→

$$\text{slope} = \frac{\text{change in distance}(\Delta x)}{\text{change in time}(\Delta t)}$$

$$\text{slope} = \frac{x(15) - x(10)}{t(15) - t(10)}$$

Calculating the Slope



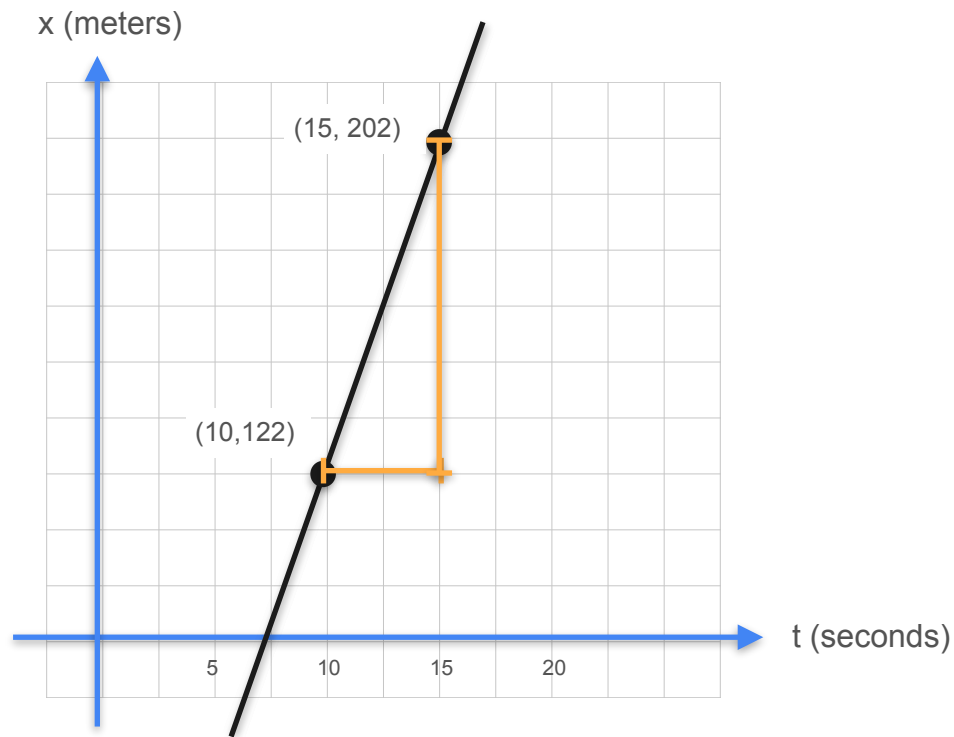
$$\text{slope} = \frac{\text{rise}}{\text{run}}$$

↑
→

$$\text{slope} = \frac{\text{change in distance}(\Delta x)}{\text{change in time}(\Delta t)}$$

$$\text{slope} = \frac{x(15) - x(10)}{t(15) - t(10)}$$

Calculating the Slope



$$\text{slope} = \frac{\text{rise}}{\text{run}}$$

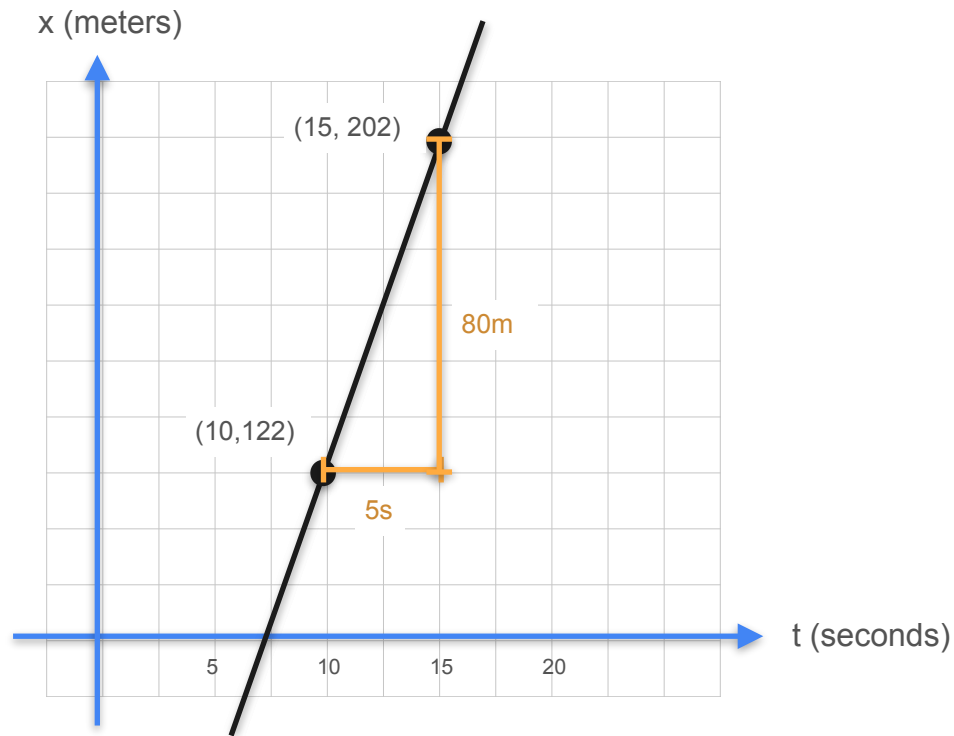
↑
→

$$\text{slope} = \frac{\text{change in distance}(\Delta x)}{\text{change in time}(\Delta t)}$$

$$\text{slope} = \frac{x(15) - x(10)}{t(15) - t(10)}$$

$$\text{slope} = \frac{202\text{m} - 122\text{m}}{15\text{s} - 10\text{s}}$$

Calculating the Slope



$$\text{slope} = \frac{\text{rise}}{\text{run}}$$

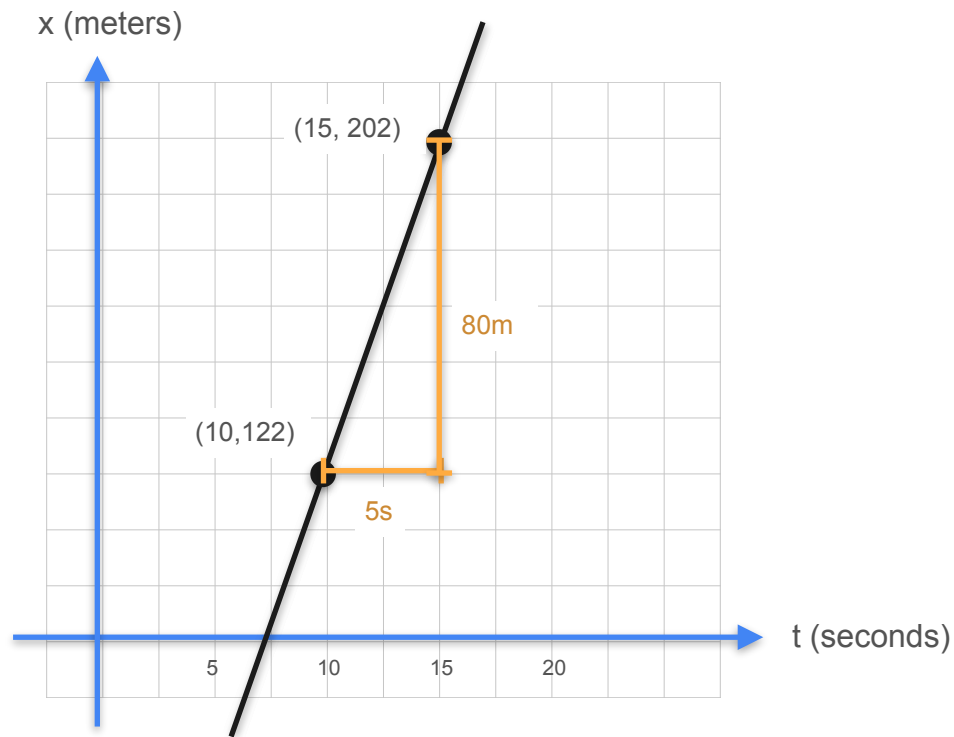
↑
→

$$\text{slope} = \frac{\text{change in distance}(\Delta x)}{\text{change in time}(\Delta t)}$$

$$\text{slope} = \frac{x(15) - x(10)}{t(15) - t(10)}$$

$$\text{slope} = \frac{202m - 122m}{15s - 10s}$$

Calculating the Slope



$$\text{slope} = \frac{\text{rise}}{\text{run}}$$

↑
→

$$\text{slope} = \frac{\text{change in distance}(\Delta x)}{\text{change in time}(\Delta t)}$$

$$\text{slope} = \frac{x(15) - x(10)}{t(15) - t(10)}$$

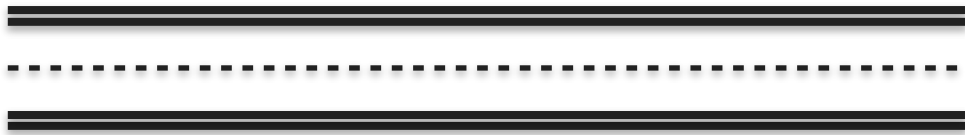
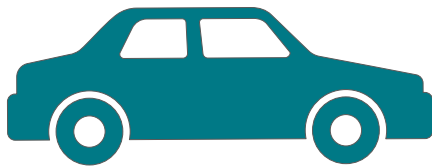
$$\text{slope} = \frac{202m - 122m}{15s - 10s}$$

$$\text{slope} = \frac{80m}{5s} = 16m/s$$

Introduction to Derivatives



Can you find a good estimate of the velocity of the car at time $t = 12.5$ seconds using this data?

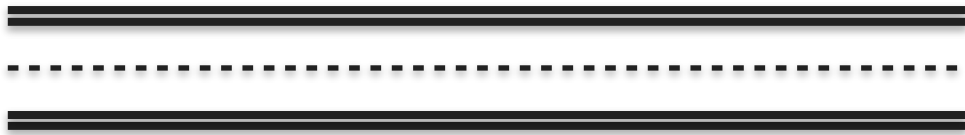
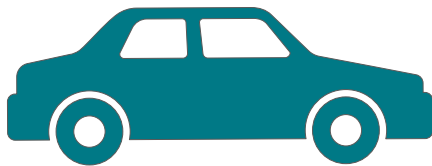


Introduction to Derivatives

Every second



Can you find a good estimate of the velocity of the car at time $t = 12.5$ seconds using this data?



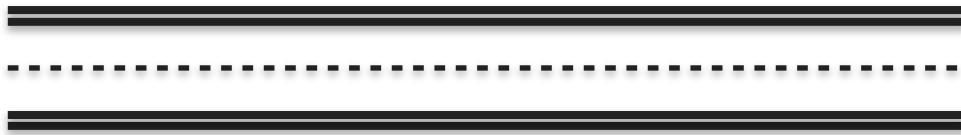
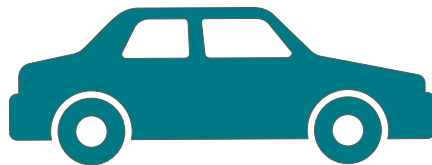
Introduction to Derivatives

Every second

t (s)
10
11
12
13
14
15
16
17
18
19
20



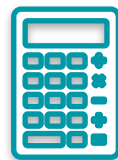
Can you find a good estimate of the velocity of the car at time $t = 12.5$ seconds using this data?



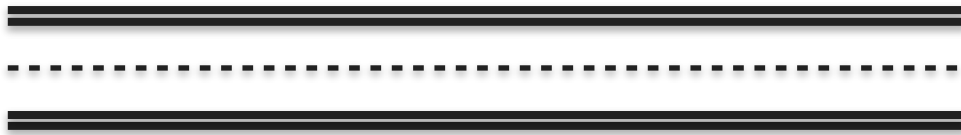
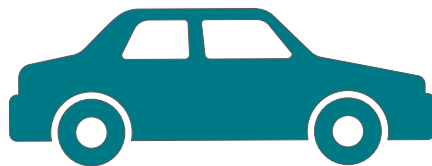
Introduction to Derivatives

Every second

t (s)	x(m)
10	122
11	138
12	155
13	170
14	186
15	202
16	218
17	234
18	250
19	265
20	265



Can you find a good estimate of the velocity of the car at time $t = 12.5$ seconds using this data?



Quiz : Slope of a Line 2

t (s)	x (m)
10	122
11	138
12	155
13	170
14	186
15	202
16	218
17	234
18	250
19	265
20	265

Can you find a good estimate of the velocity of the car at time $t = 12.5$ seconds using this data?

Quiz : Slope of a Line 2

t (s)	x (m)
10	122
11	138
12	155
13	170
14	186
15	202
16	218
17	234
18	250
19	265
20	265

Can you find a good estimate of the velocity of the car at time $t = 12.5$ seconds using this data?

$$slope = \frac{\Delta x}{\Delta t}$$

Quiz : Slope of a Line 2

t (s)	x (m)
10	122
11	138
12	155
13	170
14	186
15	202
16	218
17	234
18	250
19	265
20	265

Can you find a good estimate of the velocity of the car at time $t = 12.5$ seconds using this data?

$$slope = \frac{\Delta x}{\Delta t}$$

$$slope = \frac{x(13) - x(12)}{13s - 12s}$$

Quiz : Slope of a Line 2

t (s)	x (m)
10	122
11	138
12	155
13	170
14	186
15	202
16	218
17	234
18	250
19	265
20	265

Can you find a good estimate of the velocity of the car at time $t = 12.5$ seconds using this data?

$$slope = \frac{\Delta x}{\Delta t}$$

$$slope = \frac{x(13) - x(12)}{13s - 12s}$$

$$slope = \frac{170m - 155m}{1s}$$

Quiz : Slope of a Line 2

t (s)	x (m)
10	122
11	138
12	155
13	170
14	186
15	202
16	218
17	234
18	250
19	265
20	265

Can you find a good estimate of the velocity of the car at time $t = 12.5$ seconds using this data?

$$slope = \frac{\Delta x}{\Delta t}$$

$$slope = \frac{x(13) - x(12)}{13s - 12s}$$

$$slope = \frac{170m - 155m}{1s}$$

$$slope = 15m/s$$



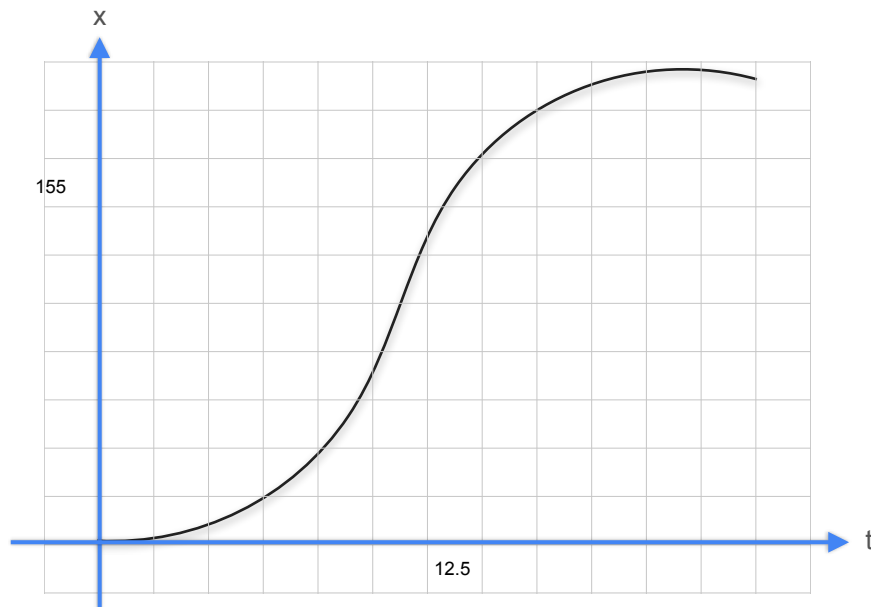
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Derivatives and Optimization

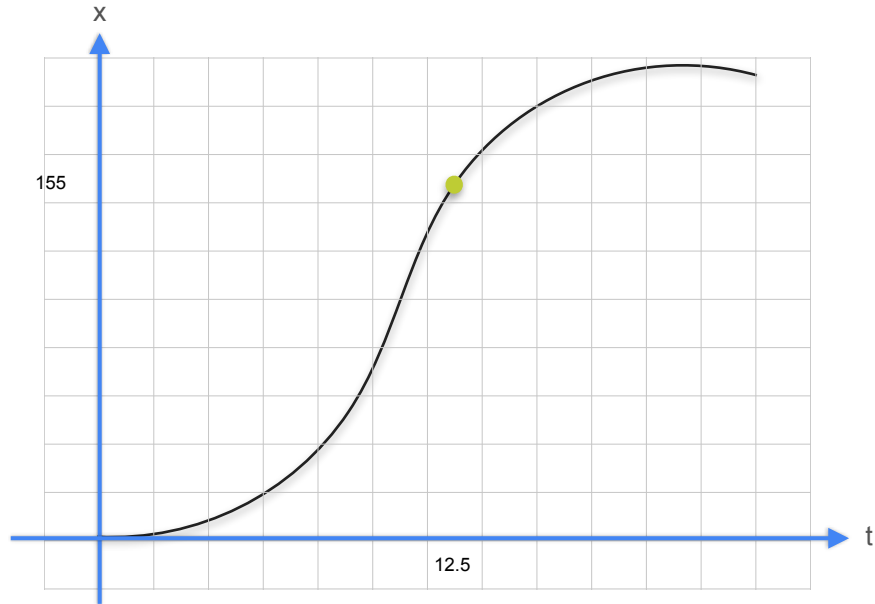
Derivatives and tangents

The derivative

The derivative

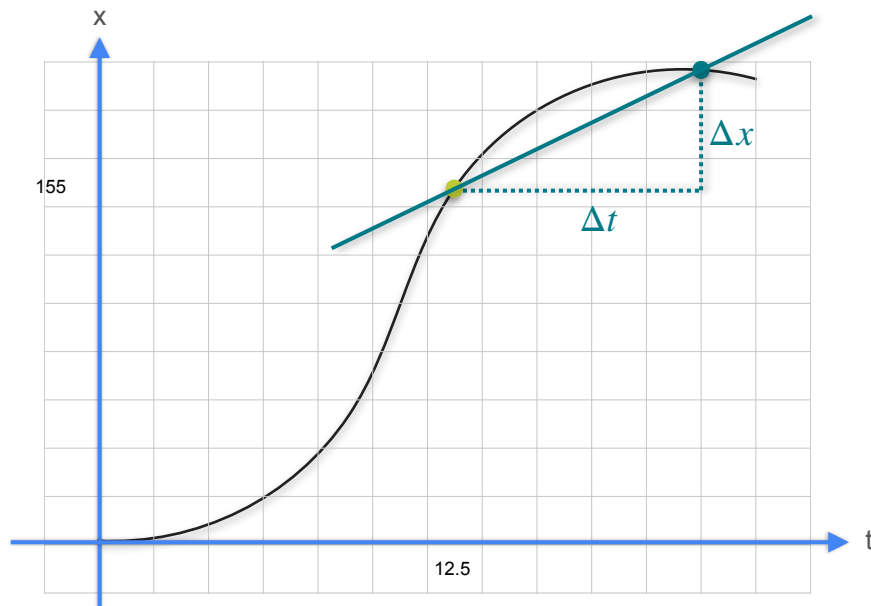


The derivative



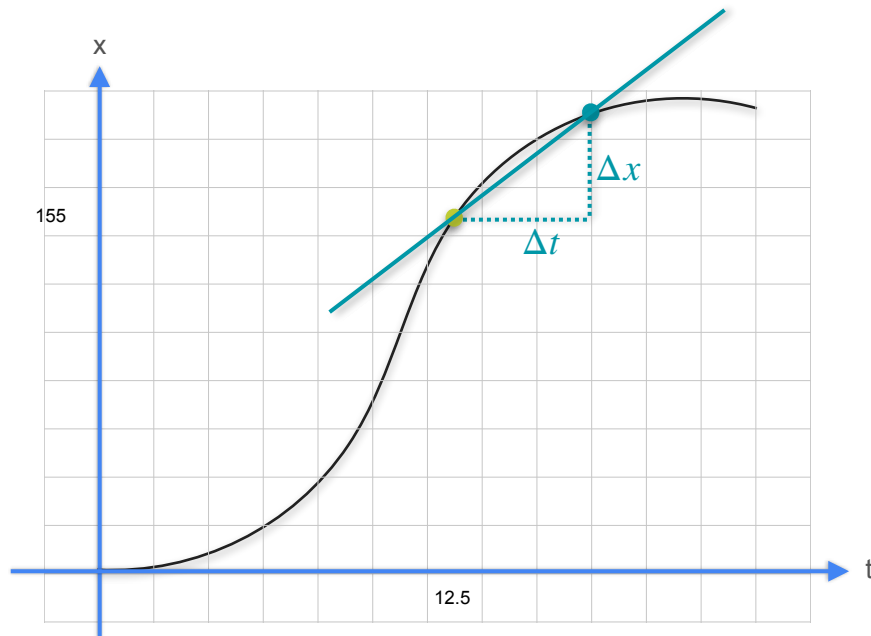
The derivative

$$\frac{\Delta x}{\Delta t}$$



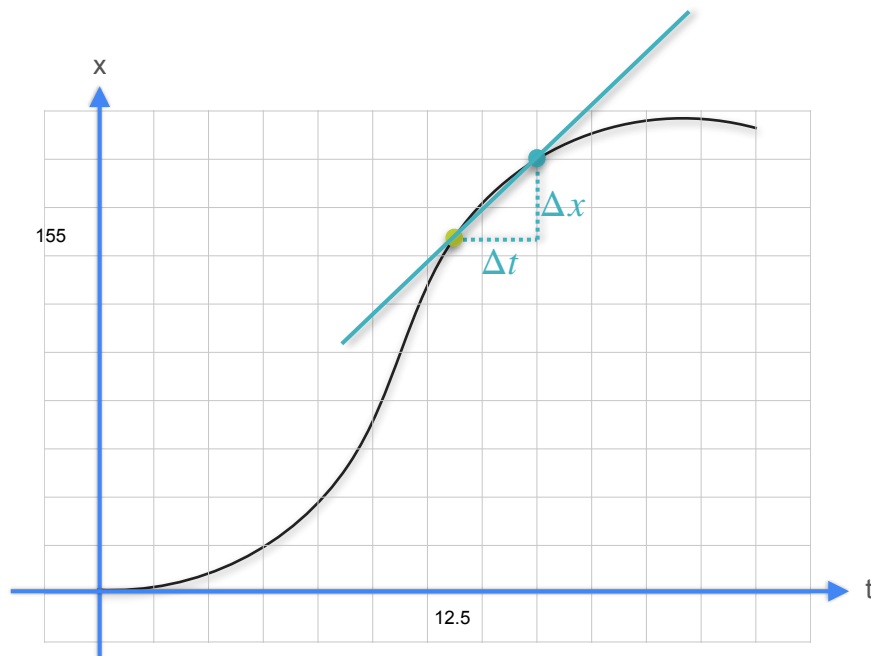
The derivative

$$\frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t}$$



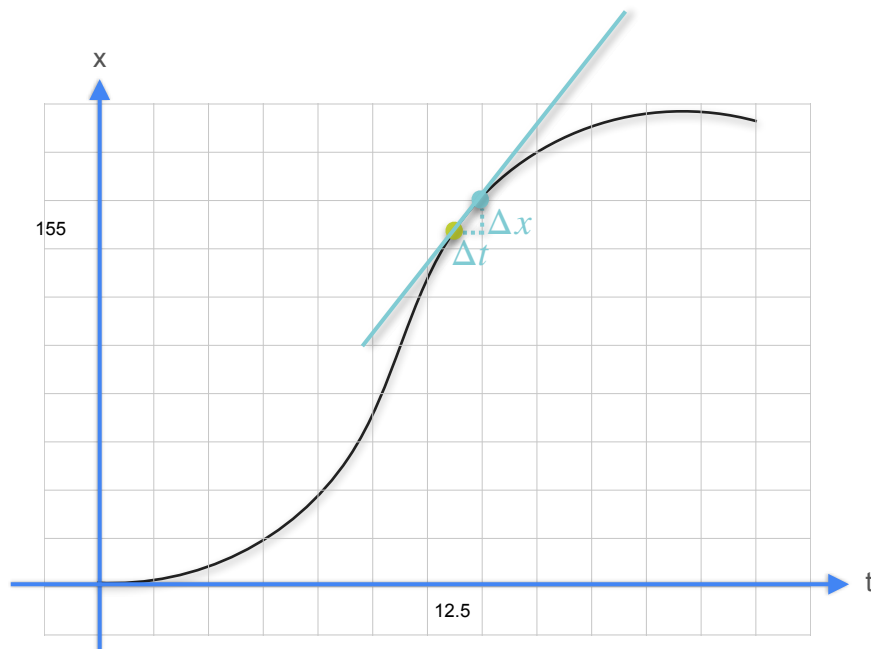
The derivative

$$\frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t}$$



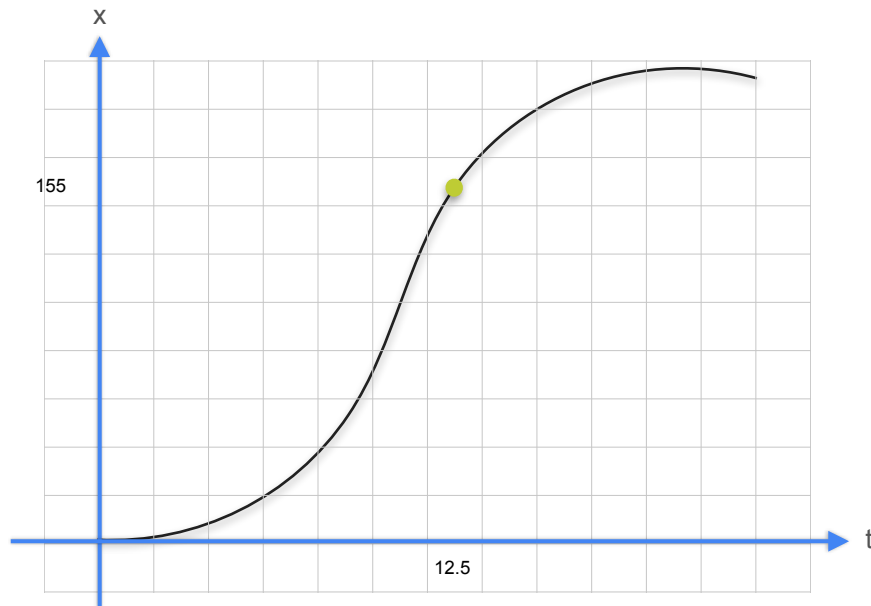
The derivative

$$\frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t}$$



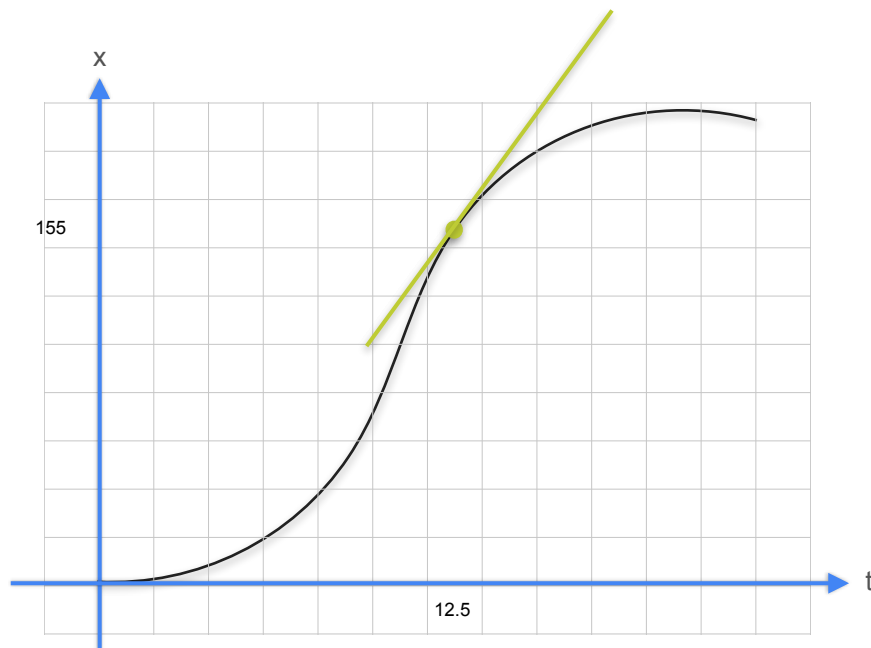
The derivative

$$\frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \longrightarrow \quad \frac{dx}{dt}$$



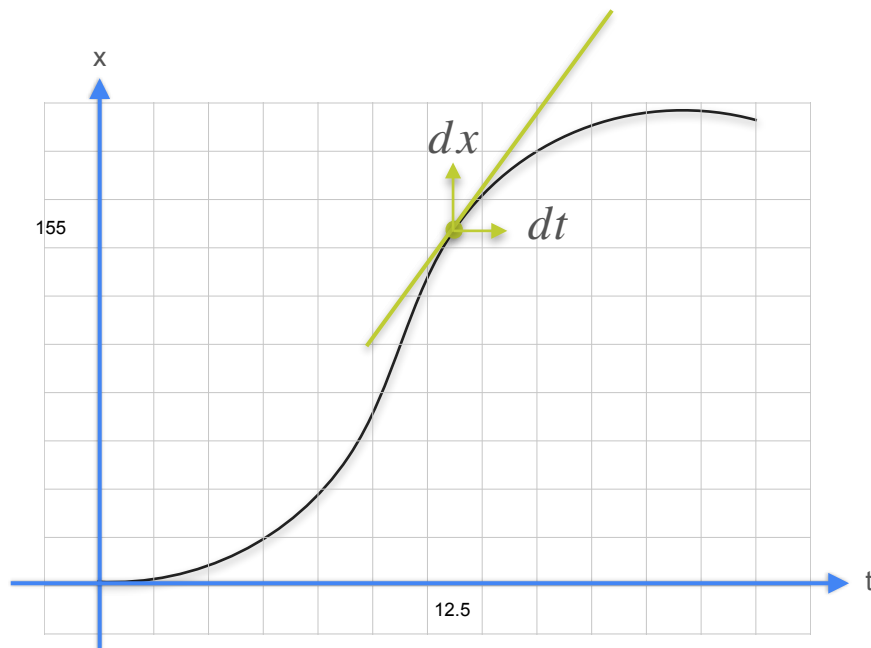
The derivative

$$\frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \longrightarrow \quad \frac{dx}{dt}$$



The derivative

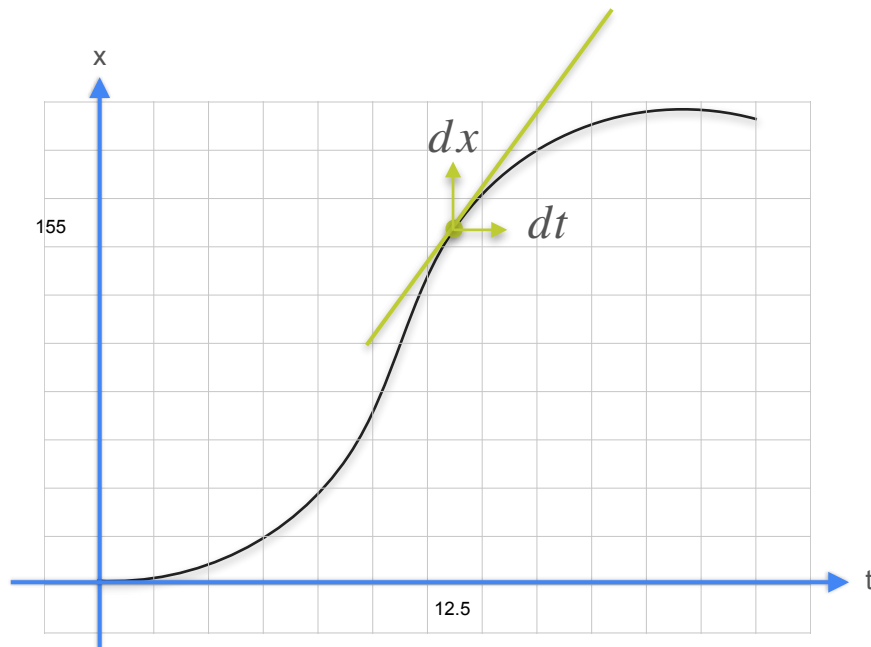
$$\frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \longrightarrow \quad \frac{dx}{dt}$$



The derivative

$$\frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \longrightarrow \quad \frac{dx}{dt}$$

Derivative





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Derivatives and Optimization

Slopes, maxima, and minima

Zero Slope

t (s)	x (m)
19	265
20	265

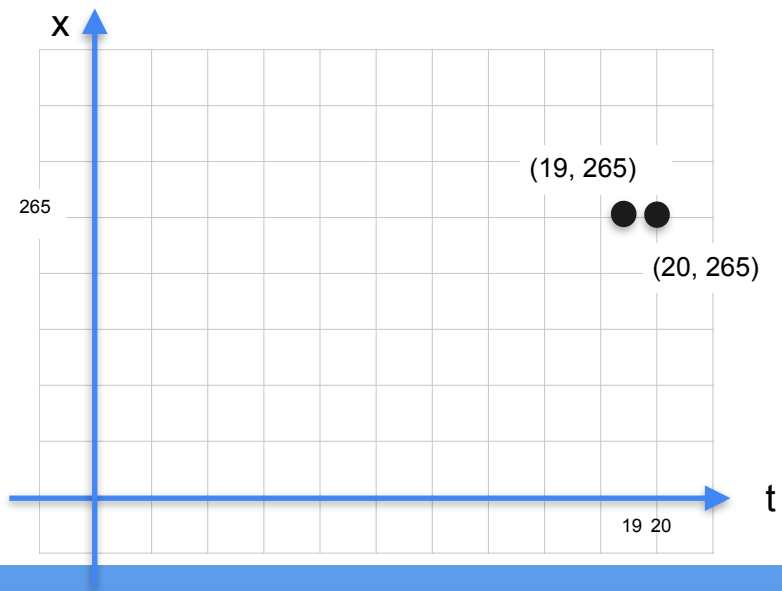
Zero Slope

t (s)	x (m)
19	265
20	265



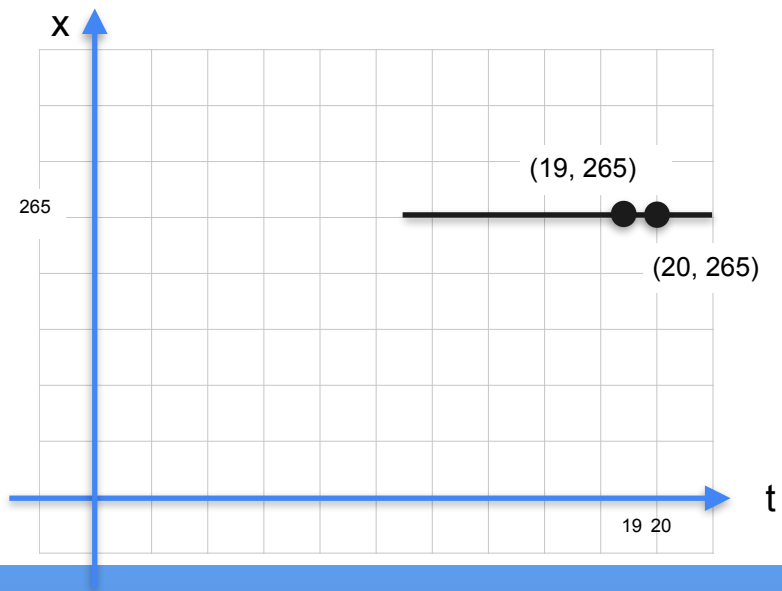
Zero Slope

t (s)	x (m)
19	265
20	265



Zero Slope

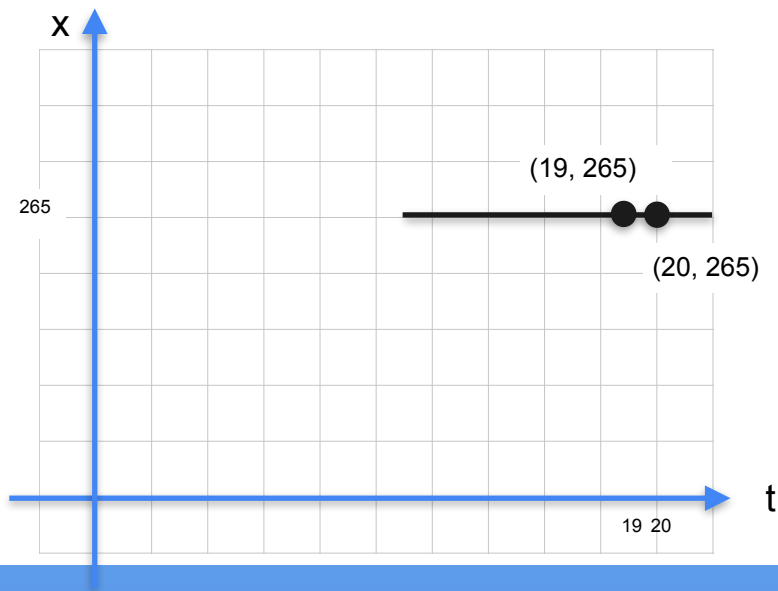
t (s)	x (m)
19	265
20	265



Zero Slope

t (s)	x (m)
19	265
20	265

$$\text{slope} = \frac{\Delta x}{\Delta t}$$

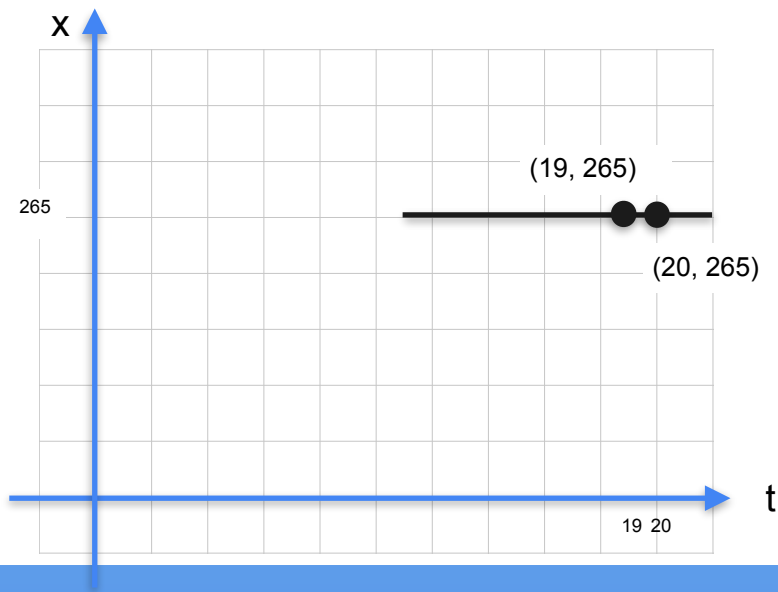


Zero Slope

t (s)	x (m)
19	265
20	265

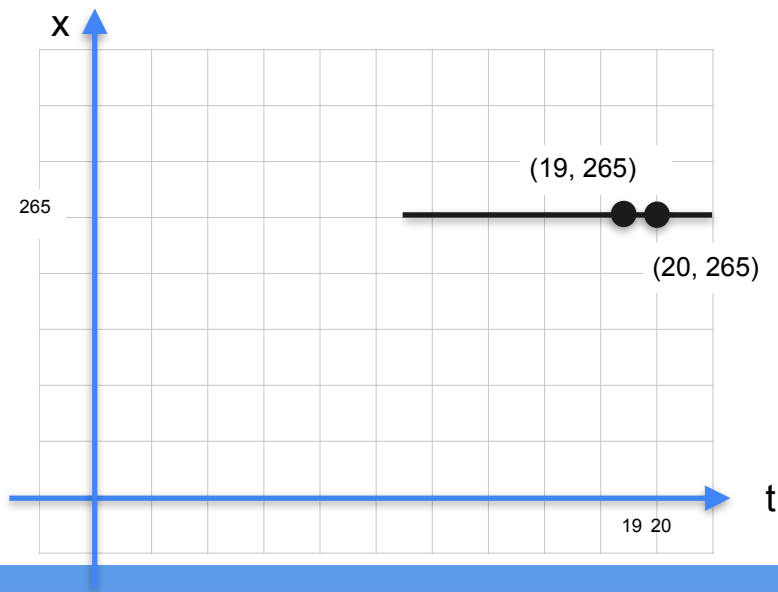
$$\text{slope} = \frac{\Delta x}{\Delta t}$$

$$\text{slope} = \frac{x(20) - x(19)}{20 - 19}$$



Zero Slope

t (s)	x (m)
19	265
20	265



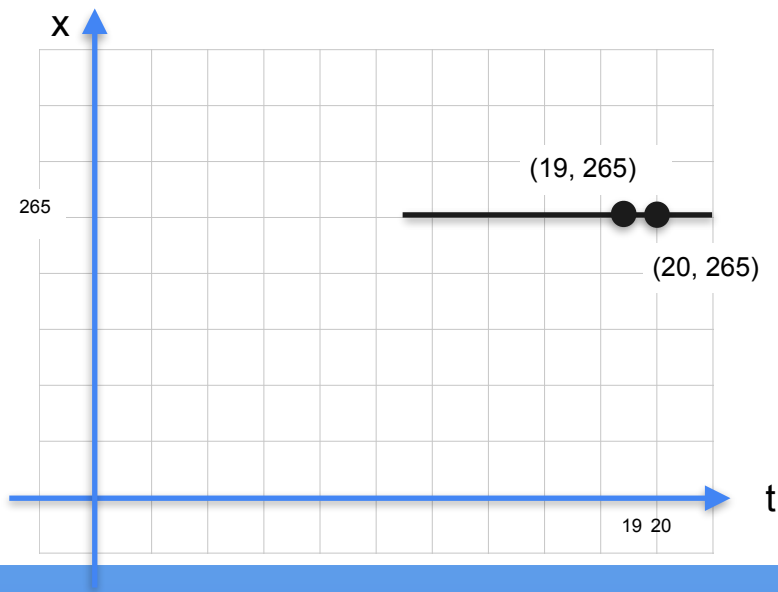
$$\text{slope} = \frac{\Delta x}{\Delta t}$$

$$\text{slope} = \frac{x(20) - x(19)}{20 - 19}$$

$$\text{slope} = \frac{265\text{m} - 265\text{m}}{1\text{s}}$$

Zero Slope

t (s)	x (m)
19	265
20	265



$$\text{slope} = \frac{\Delta x}{\Delta t}$$

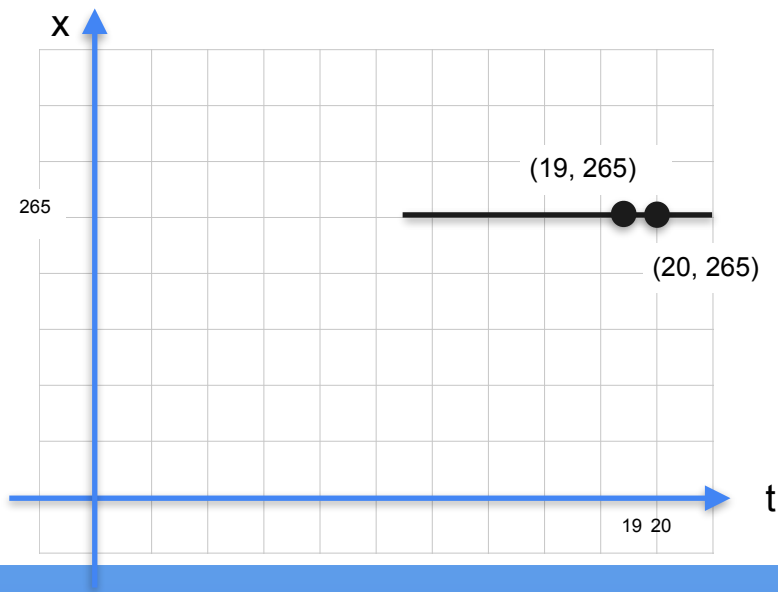
$$\text{slope} = \frac{x(20) - x(19)}{20 - 19}$$

$$\text{slope} = \frac{265\text{m} - 265\text{m}}{1\text{s}}$$

$$\text{slope} = \frac{0}{1}$$

Zero Slope

t (s)	x (m)
19	265
20	265



$$\text{slope} = \frac{\Delta x}{\Delta t}$$

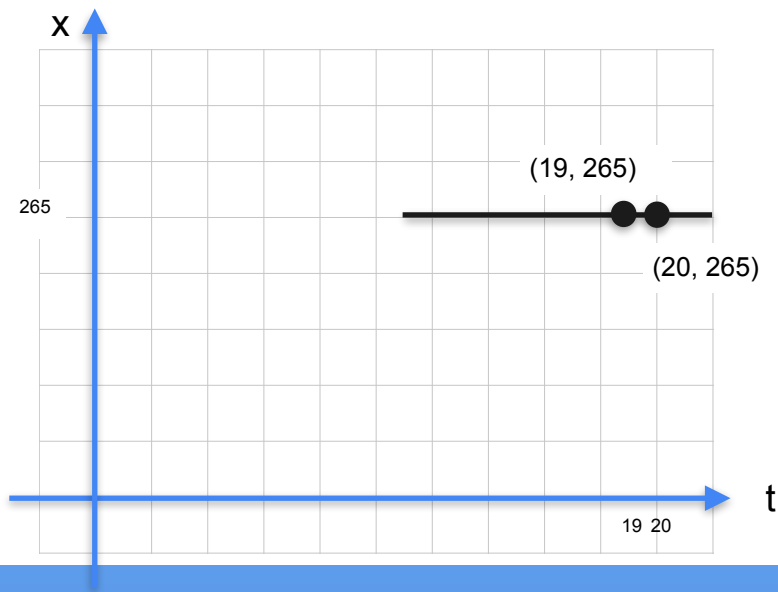
$$\text{slope} = \frac{x(20) - x(19)}{20 - 19}$$

$$\text{slope} = \frac{265\text{m} - 265\text{m}}{1\text{s}}$$

$$\text{slope} = \frac{0}{1} \quad \leftarrow \text{No rise in distance}$$

Zero Slope

t (s)	x (m)
19	265
20	265



$$\text{slope} = \frac{\Delta x}{\Delta t}$$

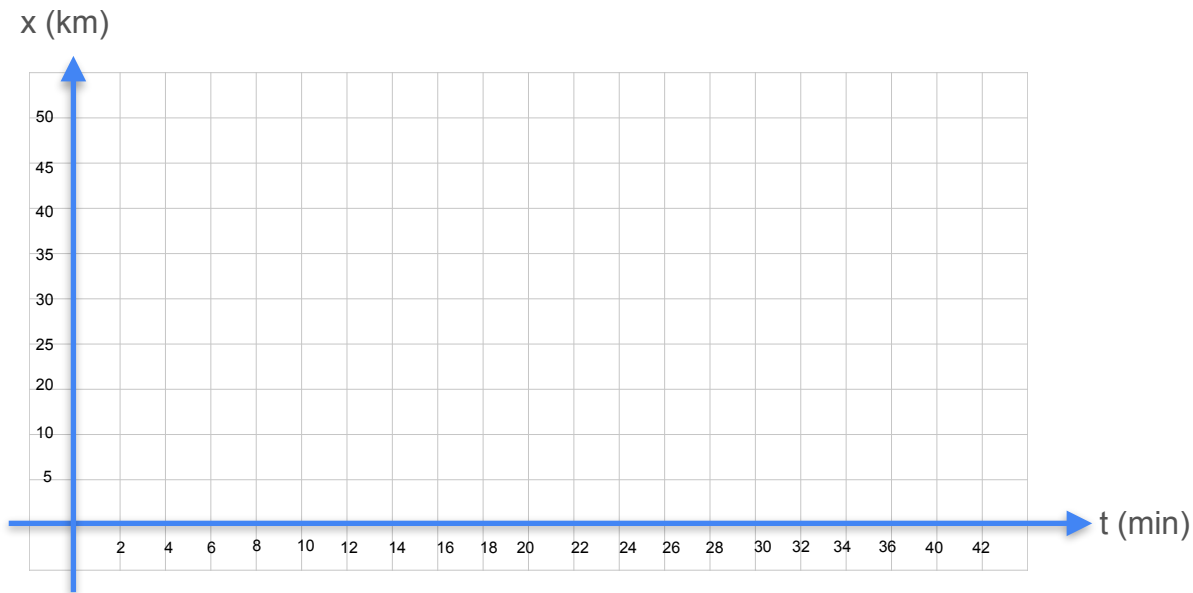
$$\text{slope} = \frac{x(20) - x(19)}{20 - 19}$$

$$\text{slope} = \frac{265\text{m} - 265\text{m}}{1\text{s}}$$

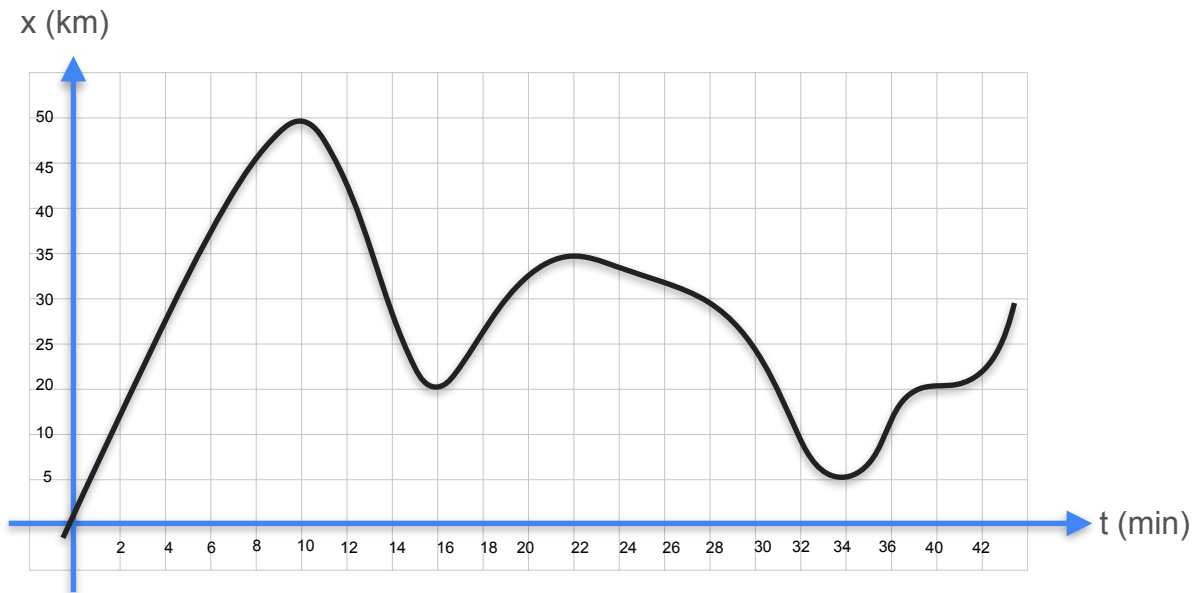
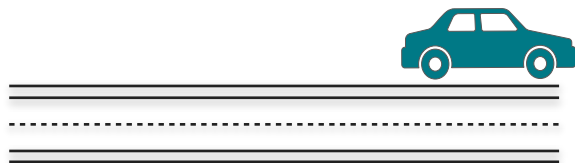
$$\text{slope} = \frac{0}{1} \quad \leftarrow \text{No rise in distance}$$

$$\text{slope} = 0 \text{ m/s}$$

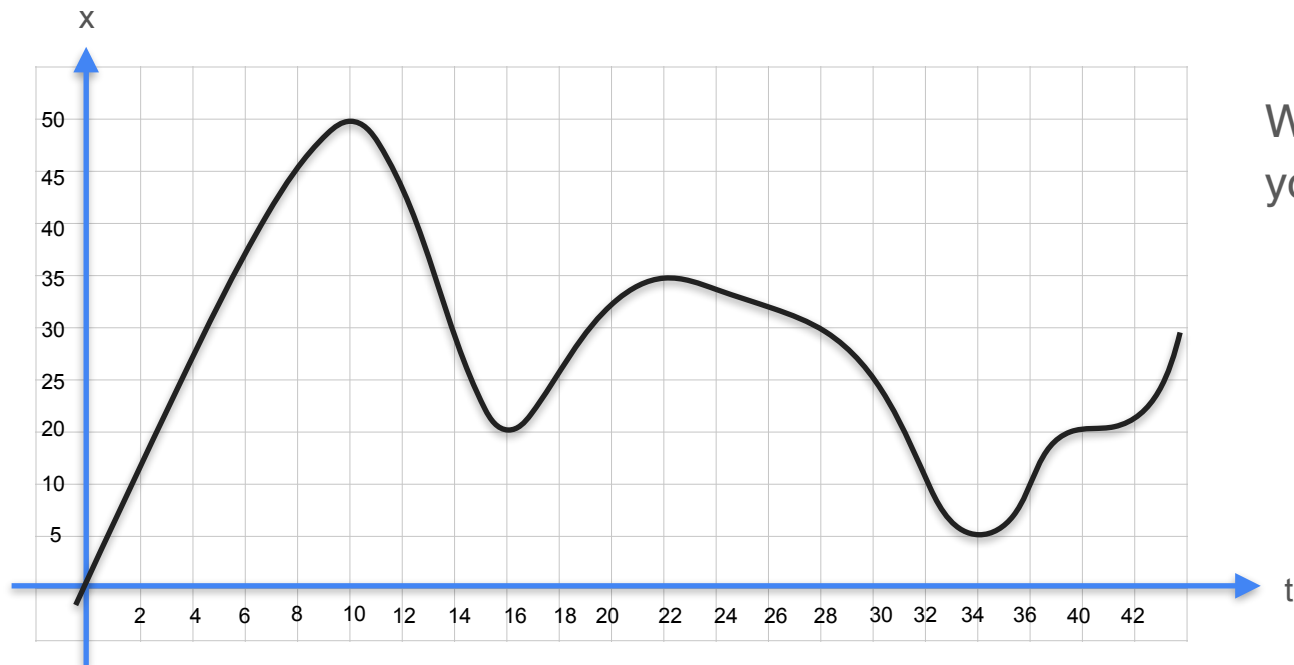
Minima and Maxima



Minima and Maxima

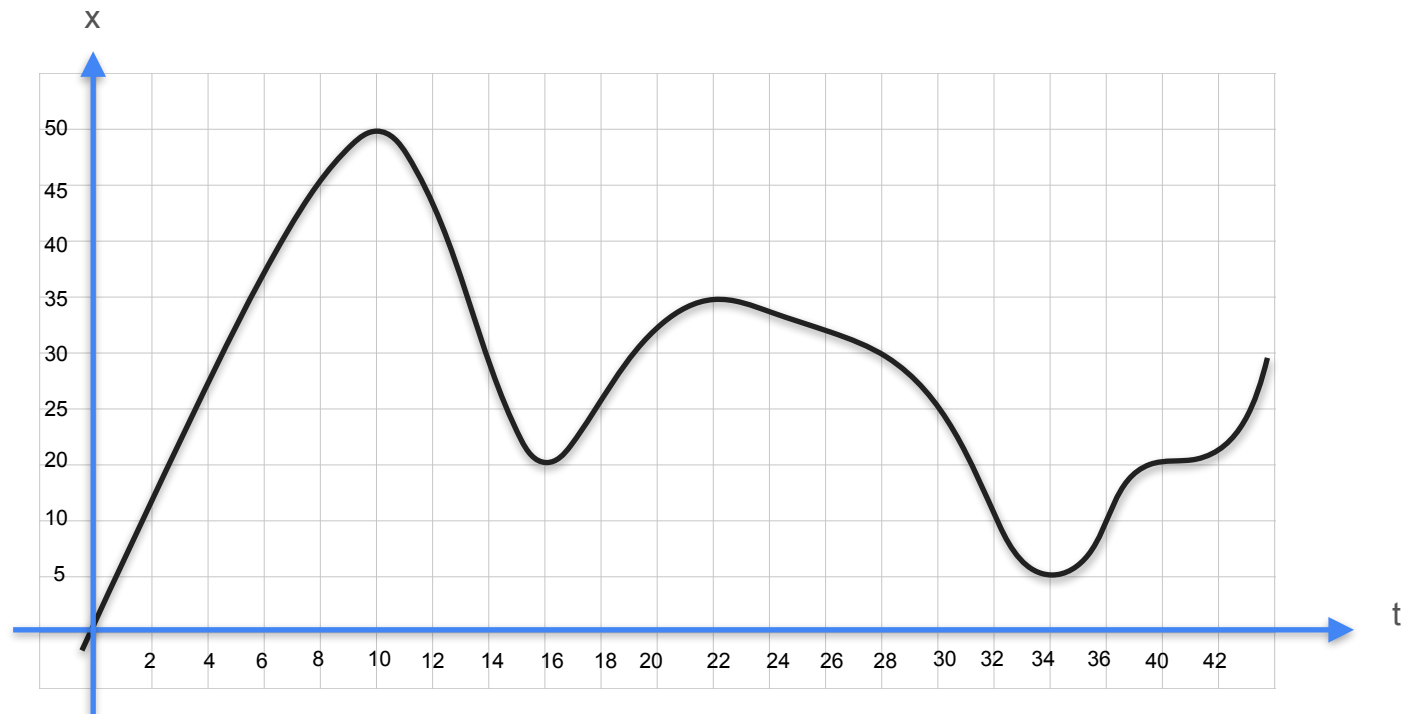


Quiz : Minima and Maxima 1

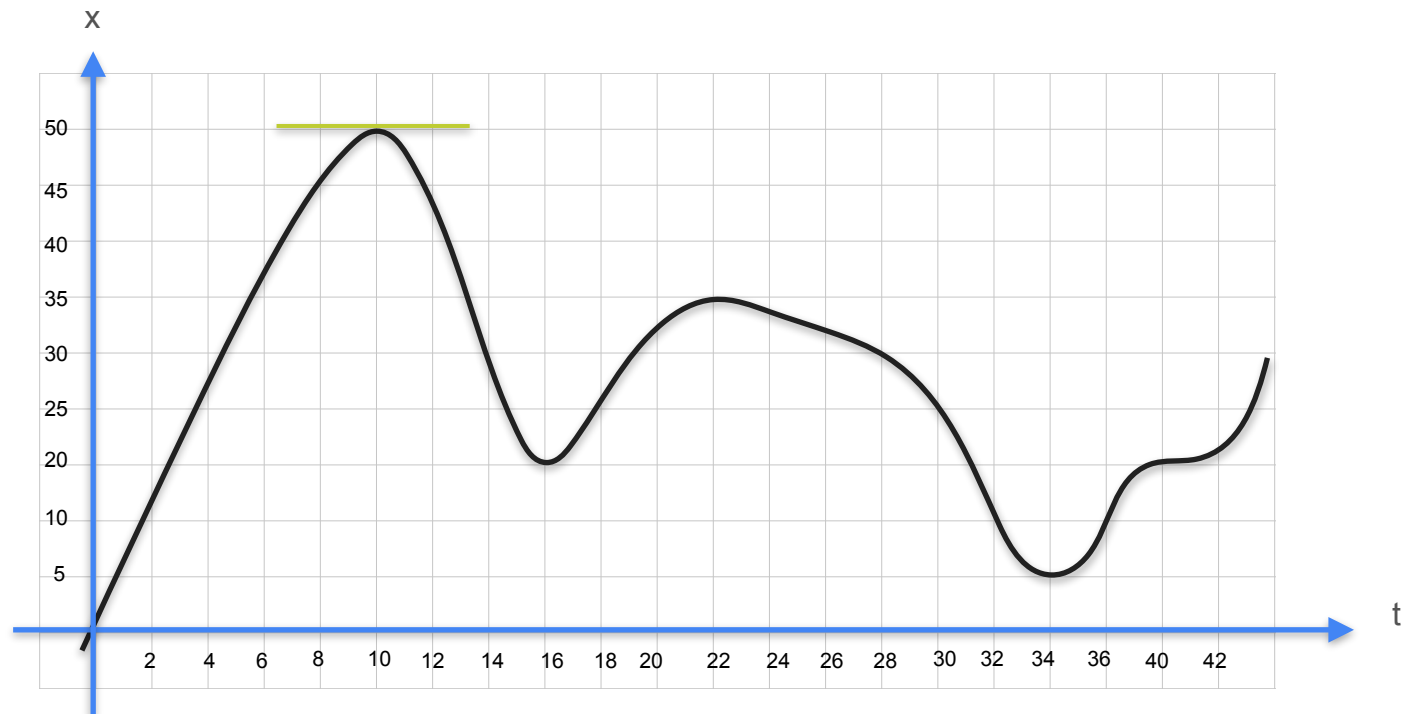


Where was the velocity of your car zero?

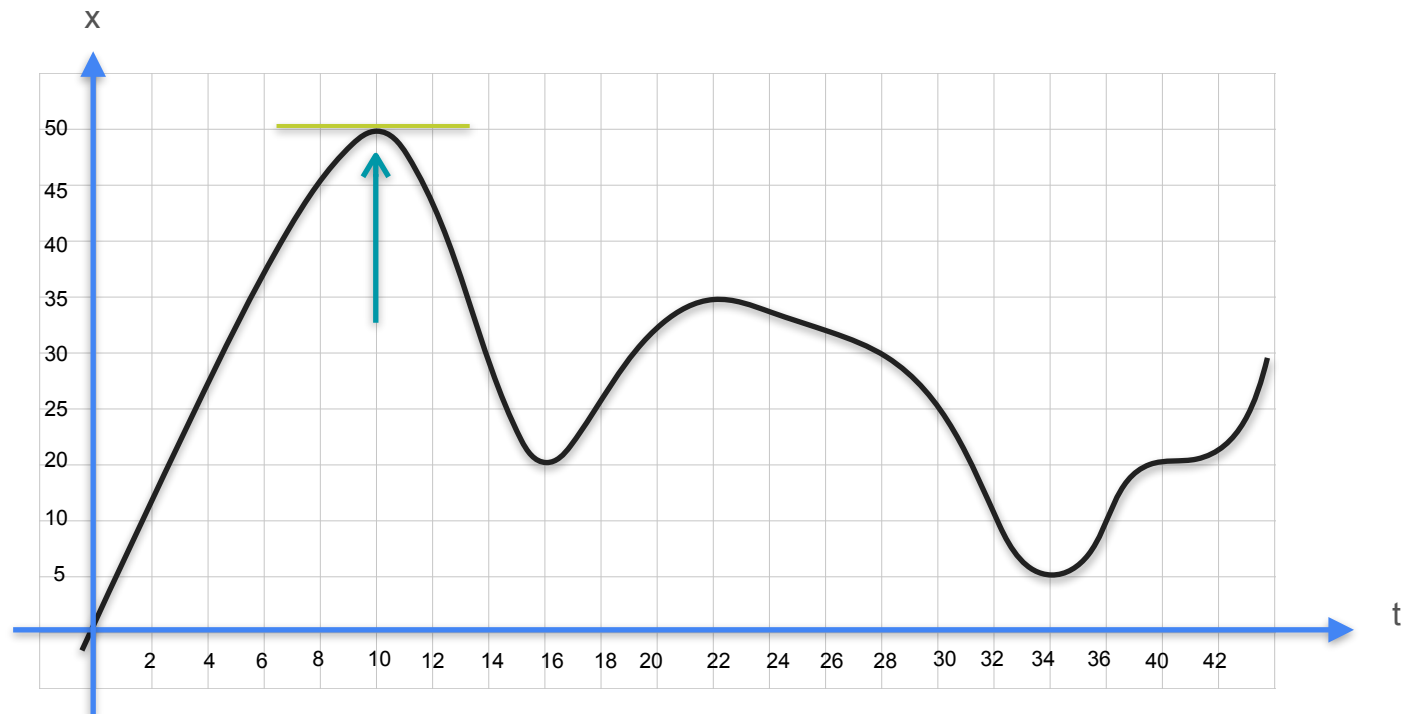
Quiz : Minima and Maxima 1 - Solution



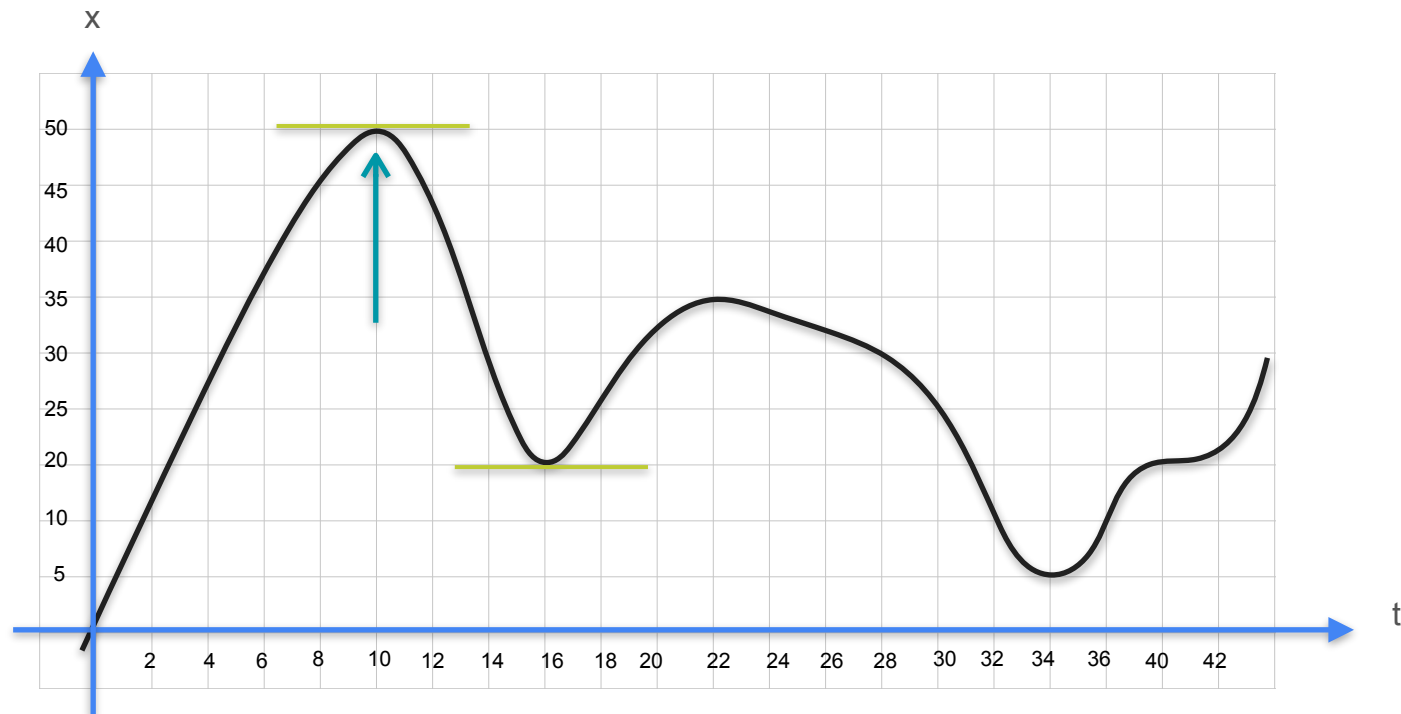
Quiz : Minima and Maxima 1 - Solution



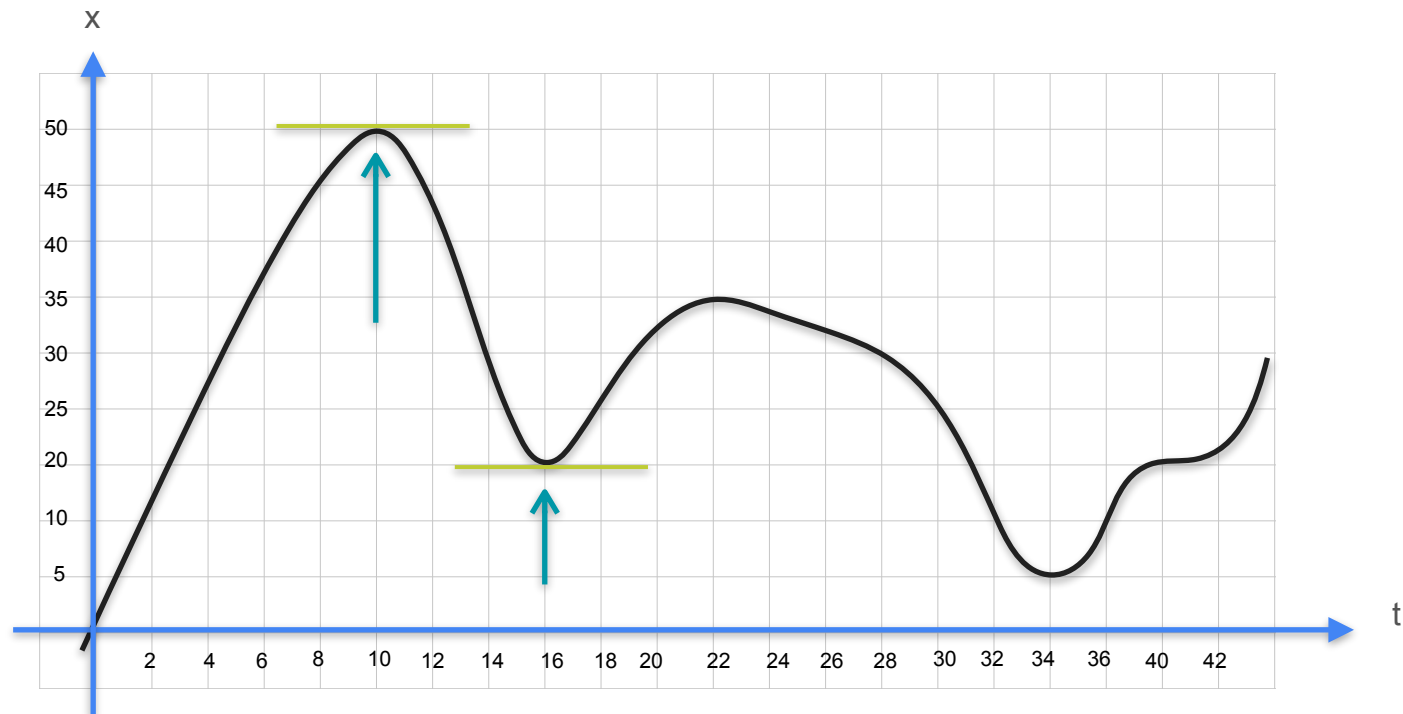
Quiz : Minima and Maxima 1 - Solution



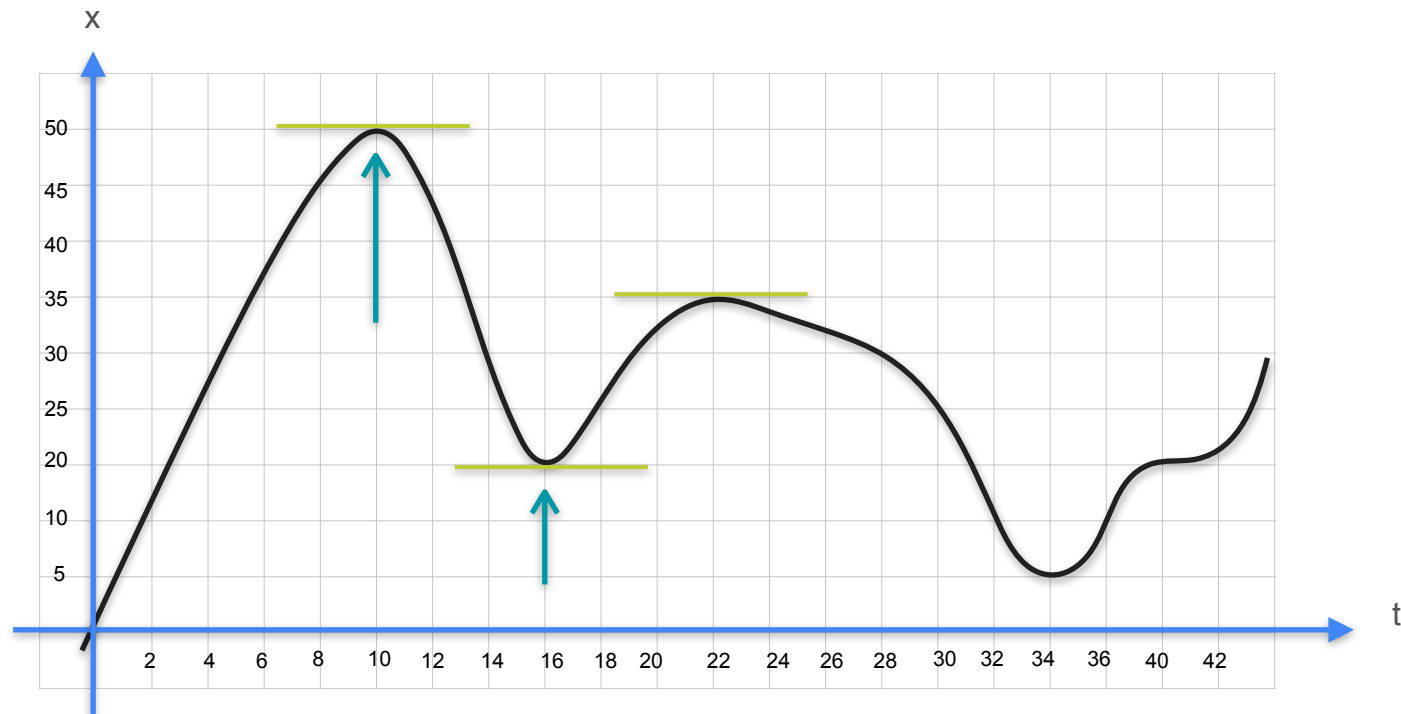
Quiz : Minima and Maxima 1 - Solution



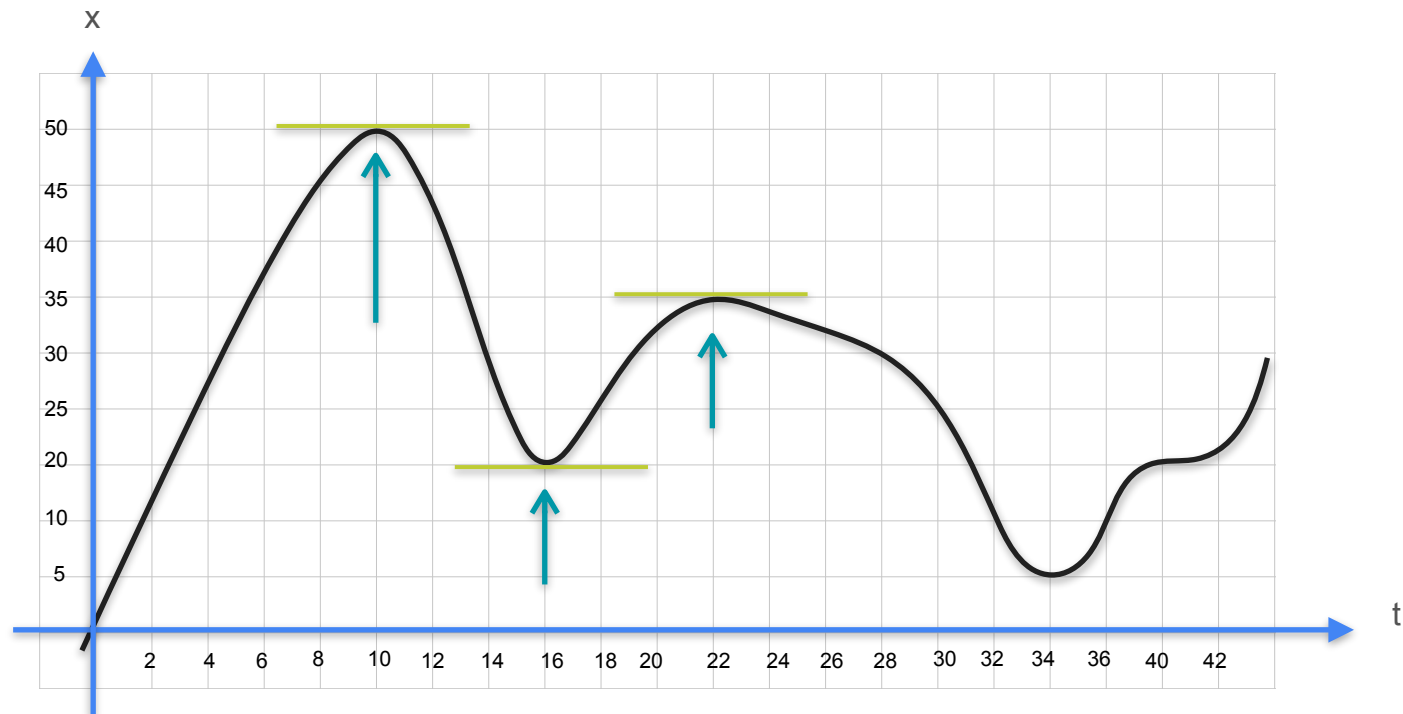
Quiz : Minima and Maxima 1 - Solution



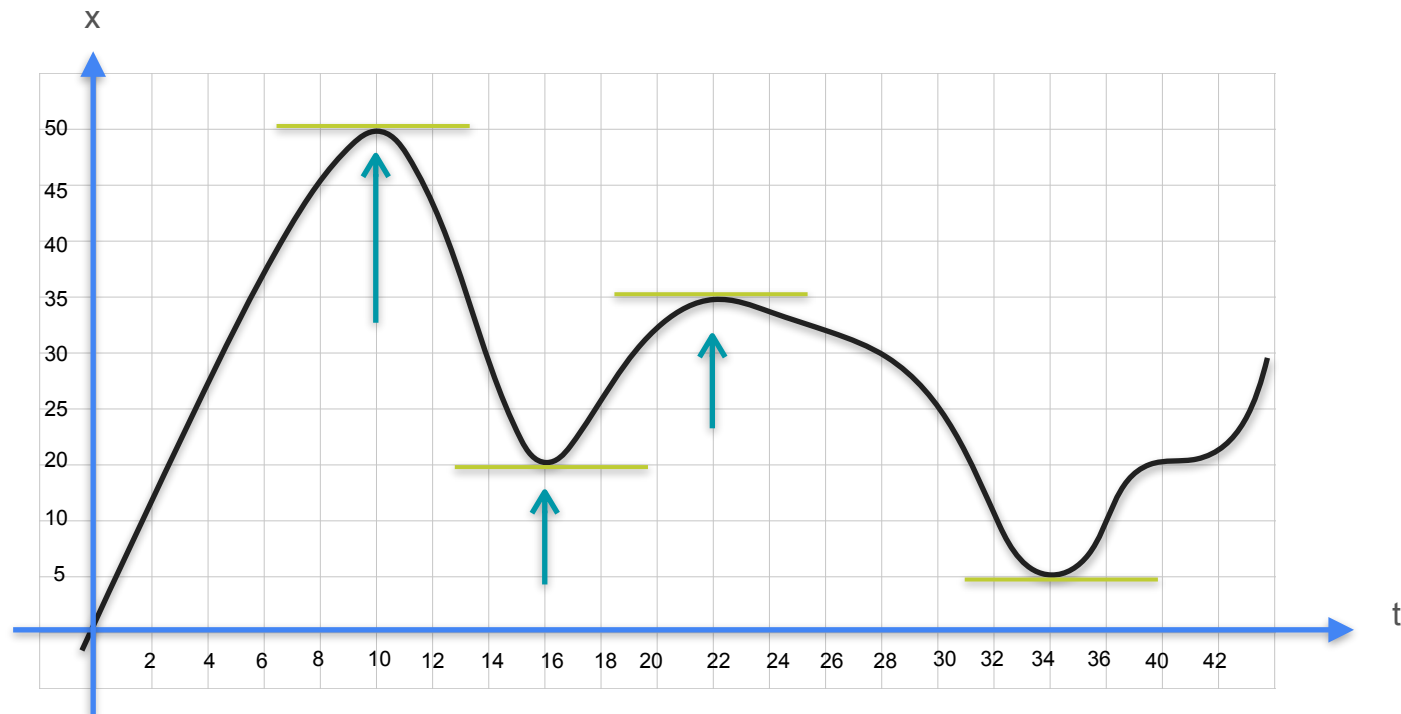
Quiz : Minima and Maxima 1 - Solution



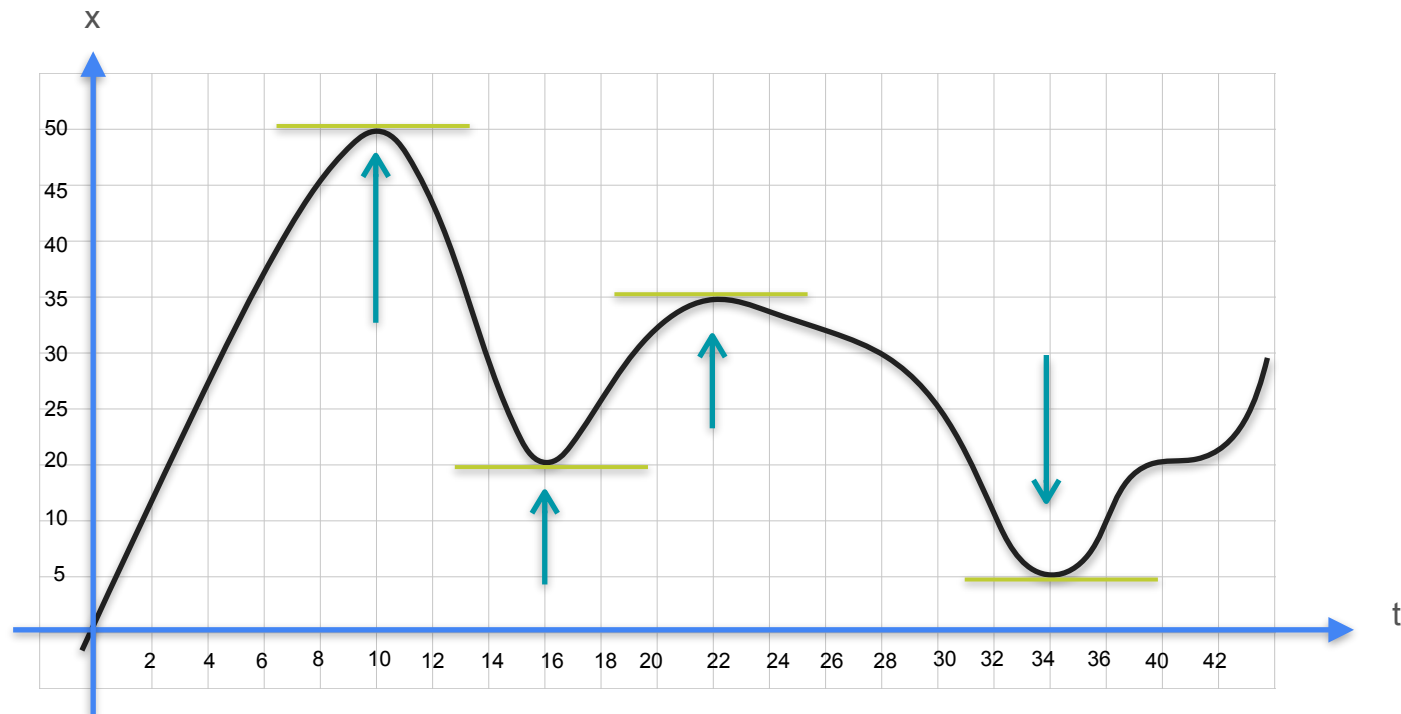
Quiz : Minima and Maxima 1 - Solution



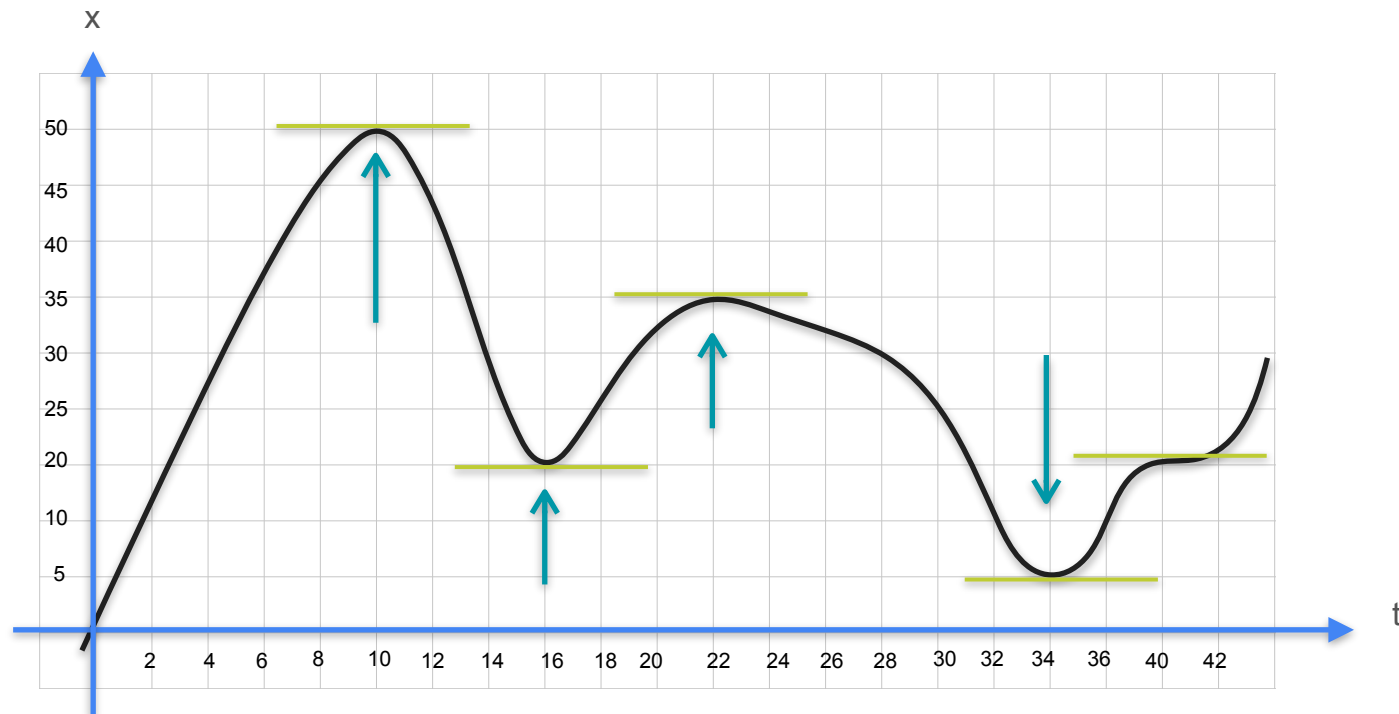
Quiz : Minima and Maxima 1 - Solution



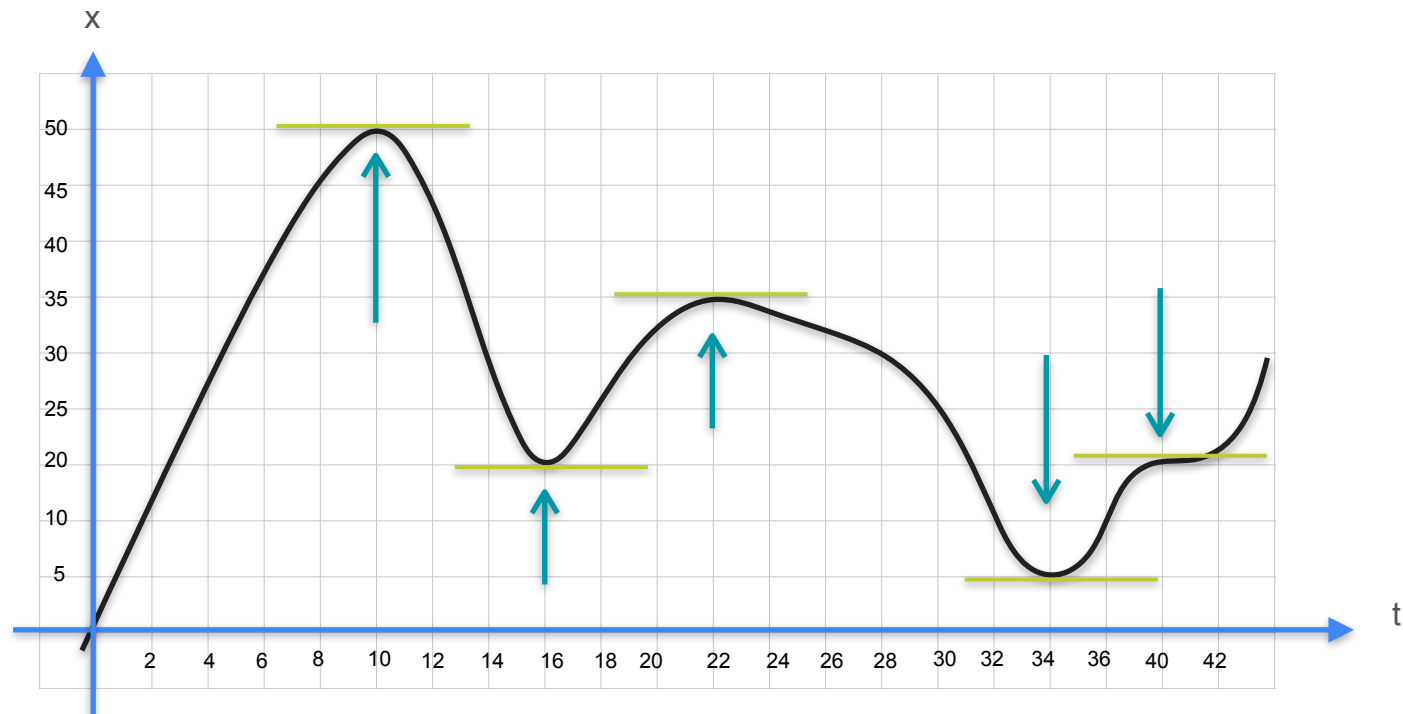
Quiz : Minima and Maxima 1 - Solution



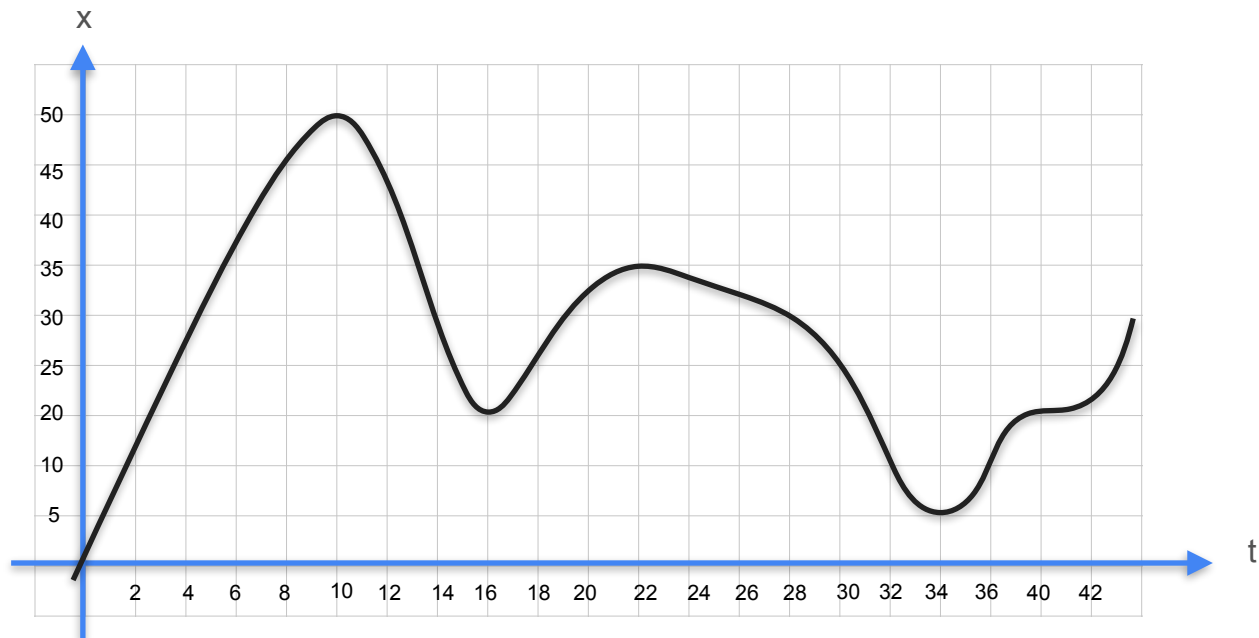
Quiz : Minima and Maxima 1 - Solution



Quiz : Minima and Maxima 1 - Solution

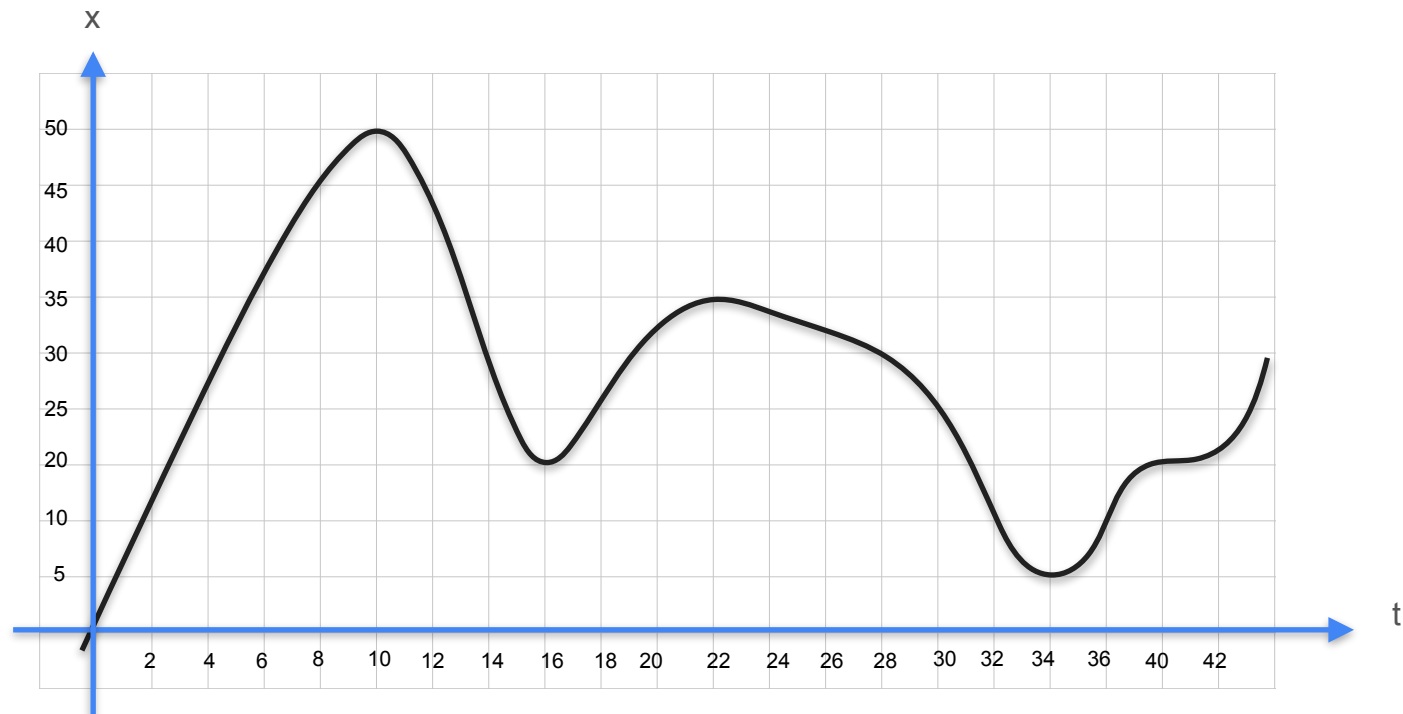


Quiz : Concept of Derivatives 3

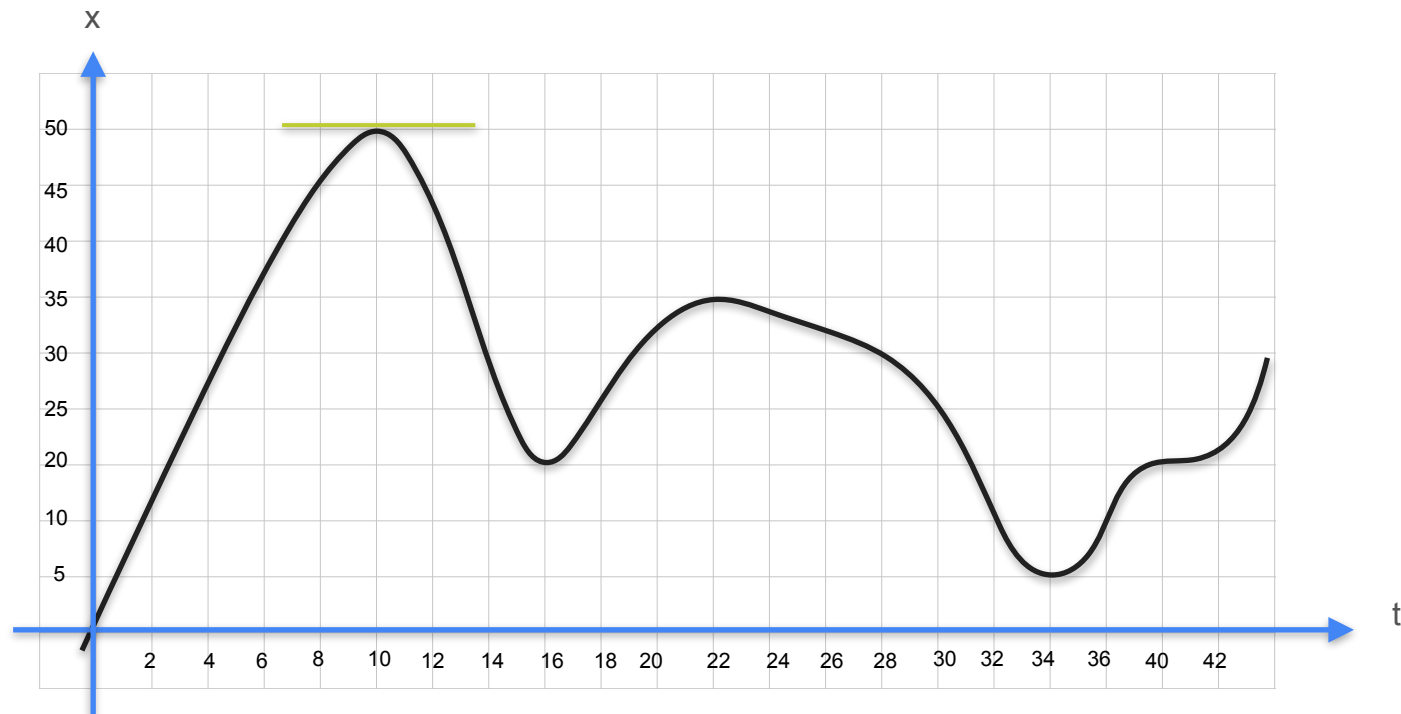


At what time was the car farthest from its starting point?

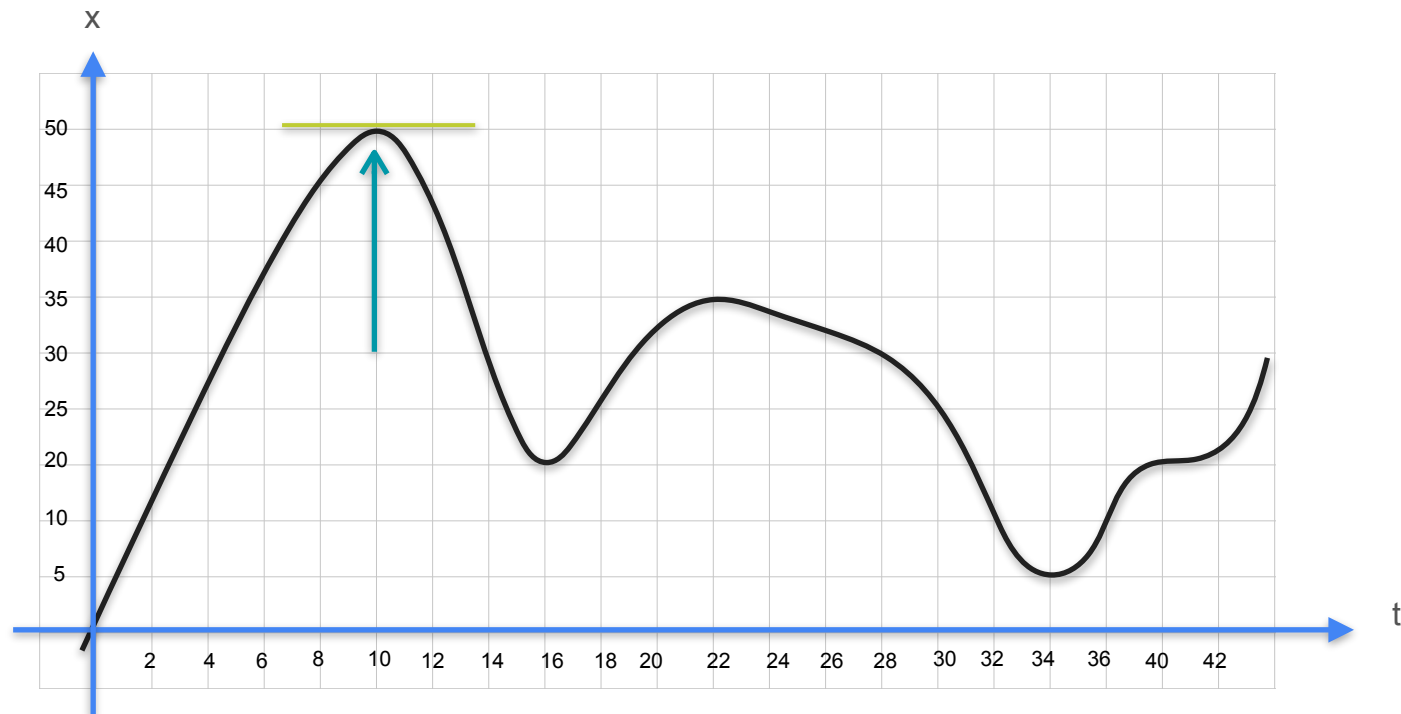
Quiz : Maxima & Minima - Solution



Quiz : Maxima & Minima - Solution



Quiz : Maxima & Minima - Solution



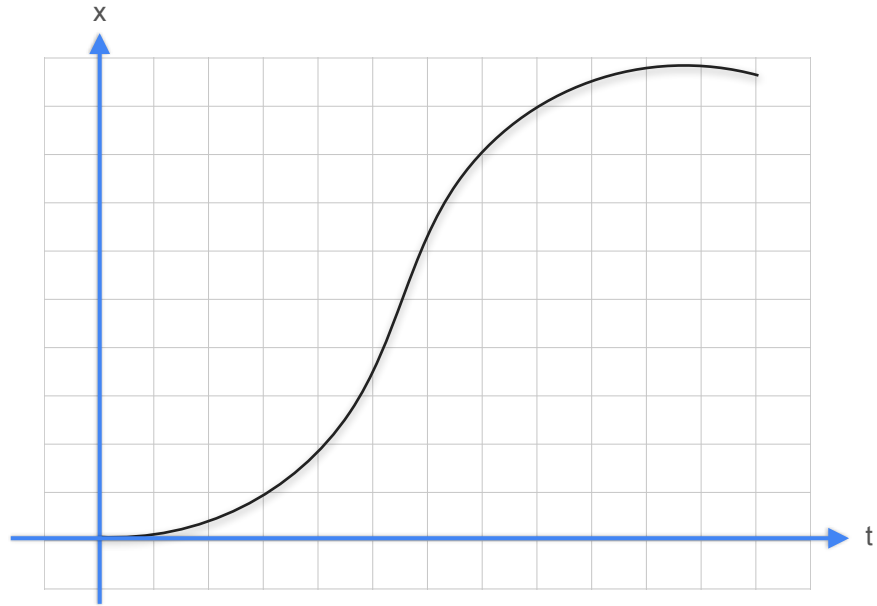


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Derivatives and Optimization

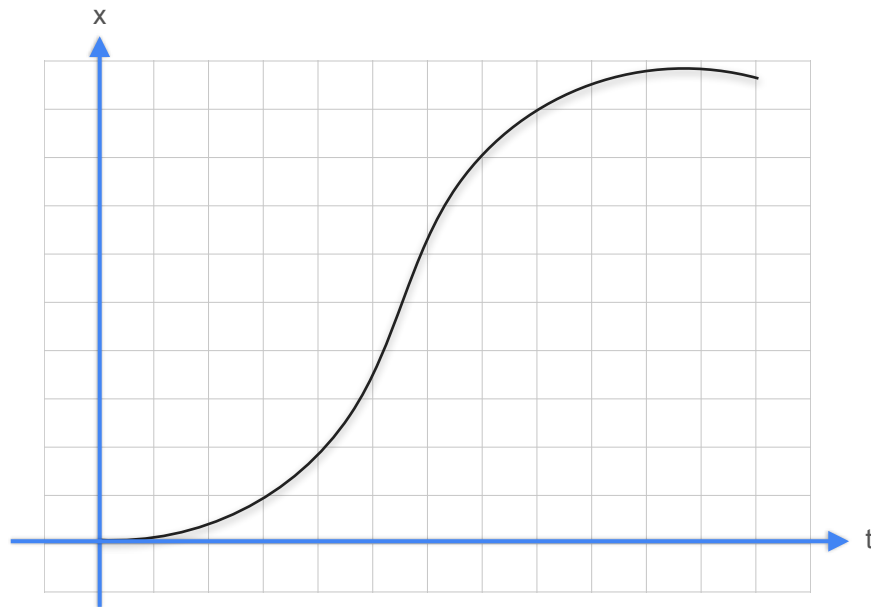
Derivative notation

Derivatives



Derivatives

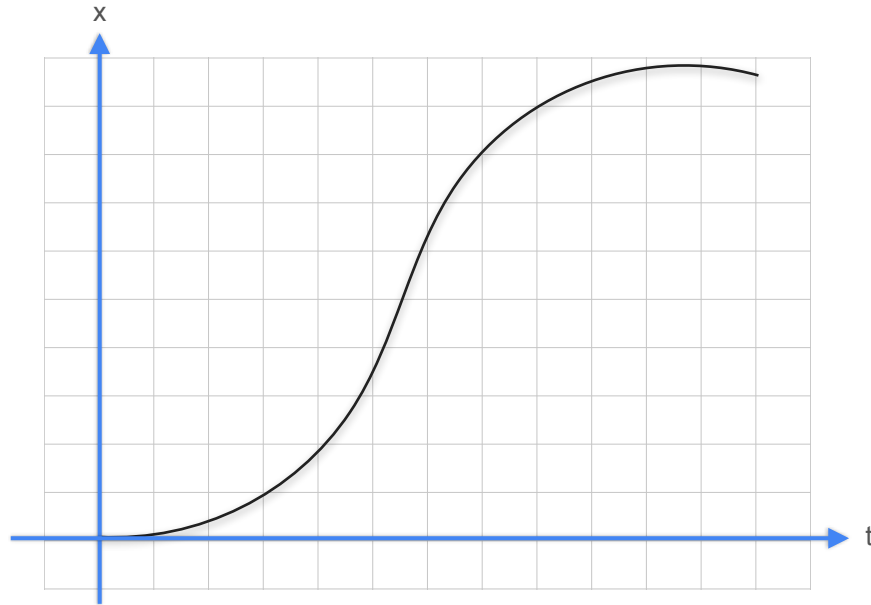
$$\text{slope} = \frac{\text{change in distance}}{\text{change in time}}$$



Derivatives

$$\text{slope} = \frac{\text{change in distance}}{\text{change in time}}$$

$$\text{slope} = \frac{\Delta x}{\Delta t}$$

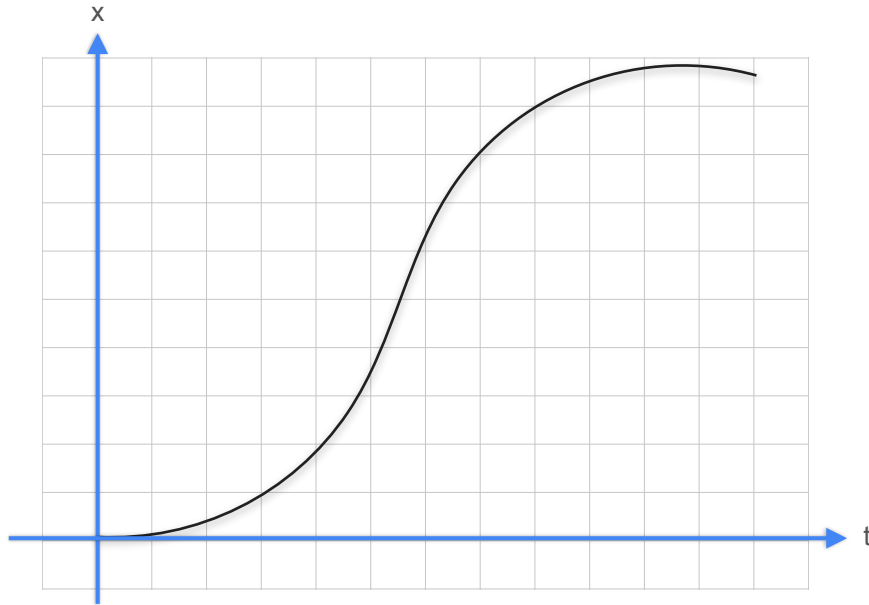


Derivatives

$$\text{slope} = \frac{\text{change in distance}}{\text{change in time}}$$

$$\text{slope} = \frac{\Delta x}{\Delta t}$$

$$\text{slope at a point} = \frac{dx}{dt}$$

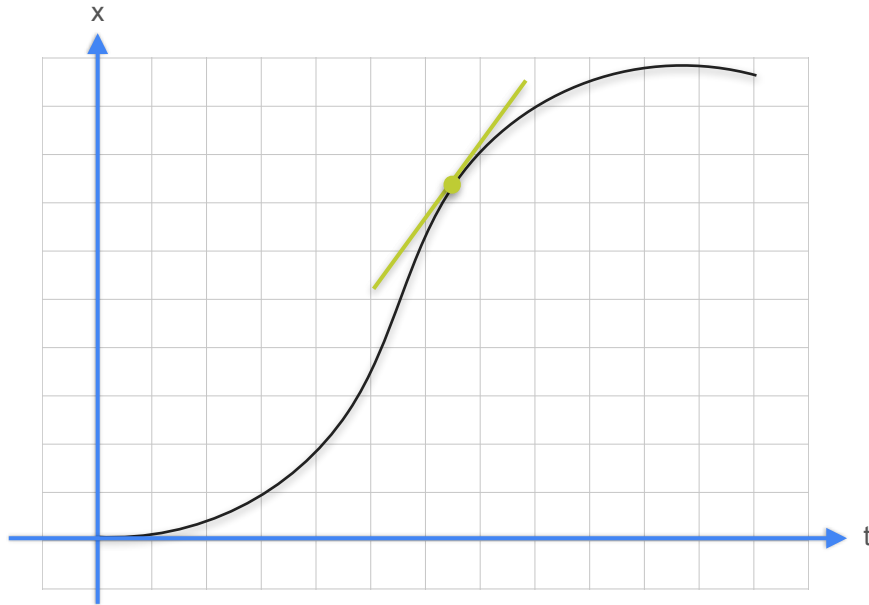


Derivatives

$$\text{slope} = \frac{\text{change in distance}}{\text{change in time}}$$

$$\text{slope} = \frac{\Delta x}{\Delta t}$$

$$\text{slope at a point} = \frac{dx}{dt}$$

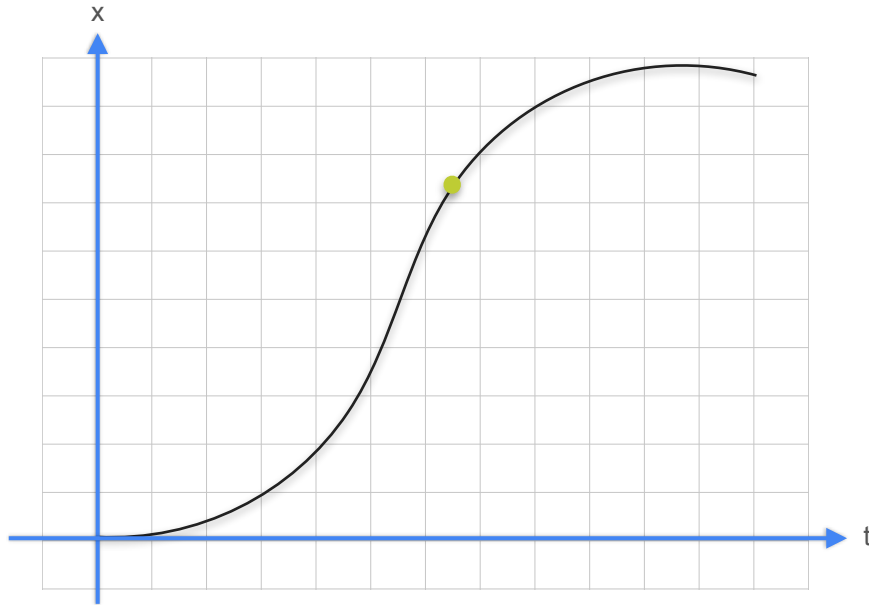


Derivatives

$$\text{slope} = \frac{\text{change in distance}}{\text{change in time}}$$

$$\text{slope} = \frac{\Delta x}{\Delta t}$$

$$\text{slope at a point} = \frac{dx}{dt}$$

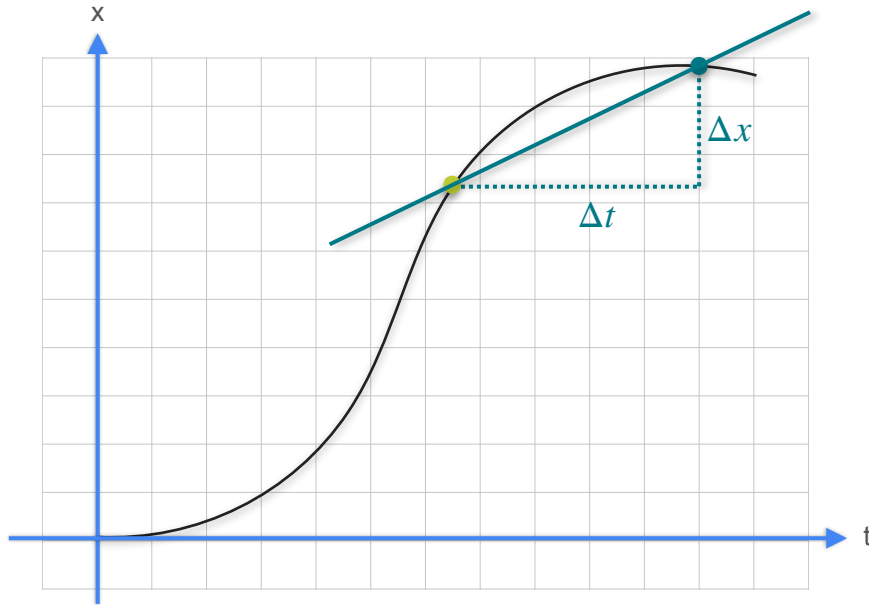


Derivatives

$$\text{slope} = \frac{\text{change in distance}}{\text{change in time}}$$

$$\text{slope} = \frac{\Delta x}{\Delta t}$$

$$\text{slope at a point} = \frac{dx}{dt}$$

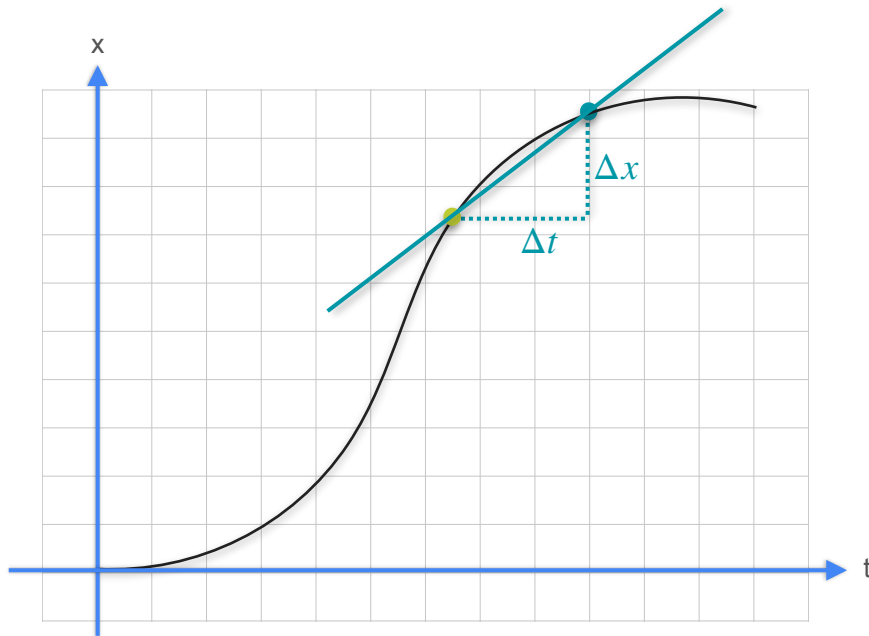


Derivatives

$$\text{slope} = \frac{\text{change in distance}}{\text{change in time}}$$

$$\text{slope} = \frac{\Delta x}{\Delta t}$$

$$\text{slope at a point} = \frac{dx}{dt}$$

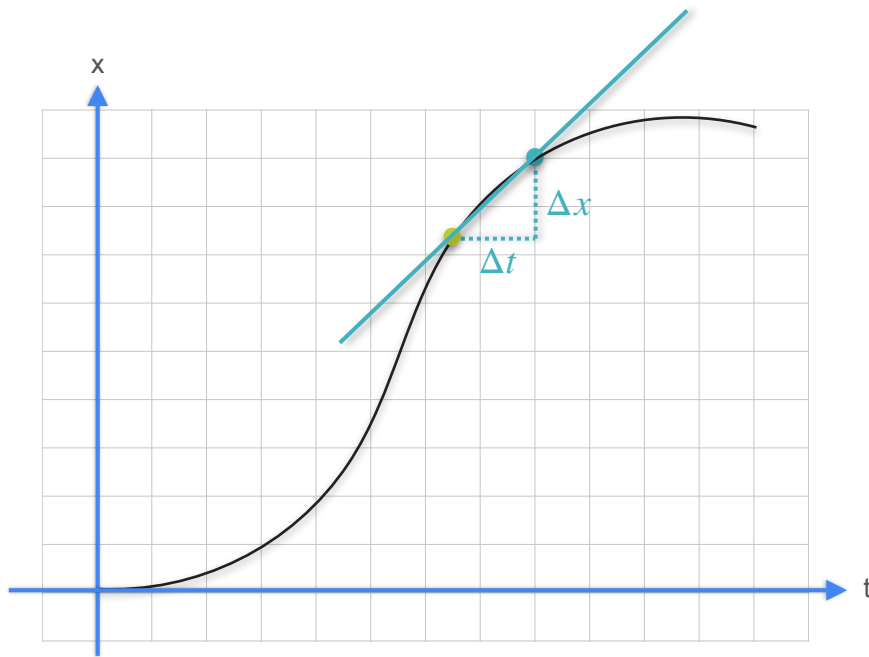


Derivatives

$$\text{slope} = \frac{\text{change in distance}}{\text{change in time}}$$

$$\text{slope} = \frac{\Delta x}{\Delta t}$$

$$\text{slope at a point} = \frac{dx}{dt}$$

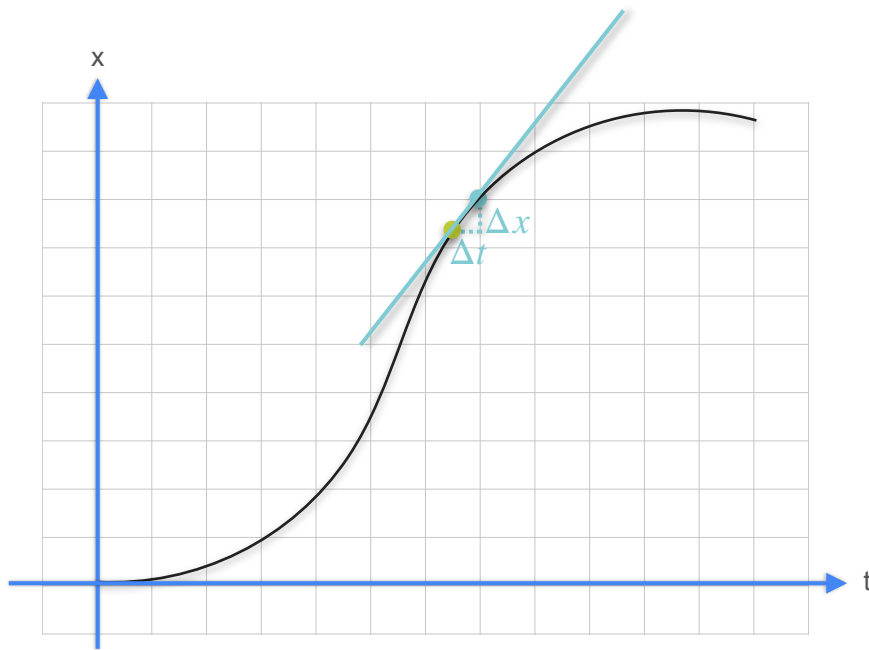


Derivatives

$$\text{slope} = \frac{\text{change in distance}}{\text{change in time}}$$

$$\text{slope} = \frac{\Delta x}{\Delta t}$$

$$\text{slope at a point} = \frac{dx}{dt}$$

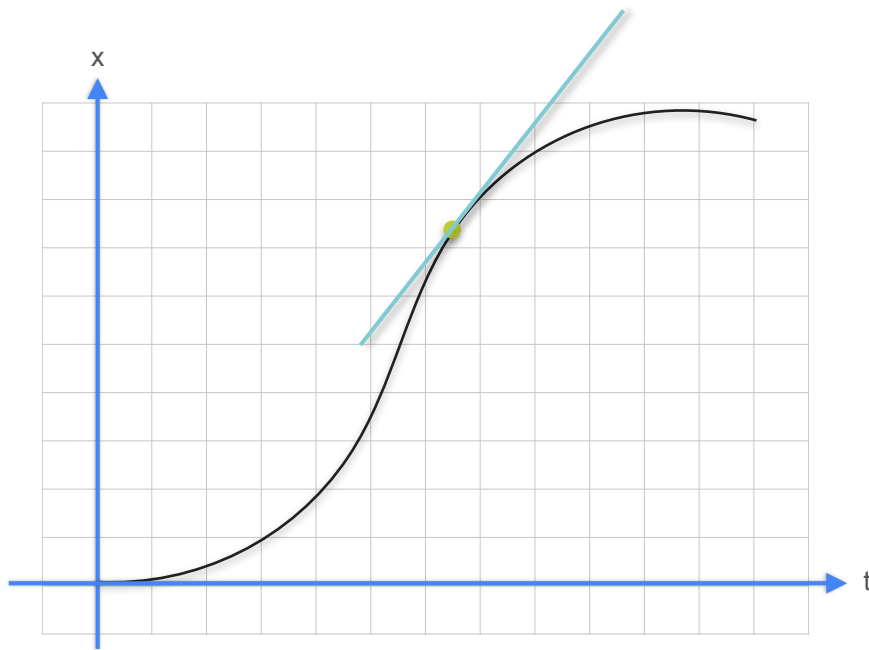


Derivatives

$$\text{slope} = \frac{\text{change in distance}}{\text{change in time}}$$

$$\text{slope} = \frac{\Delta x}{\Delta t}$$

$$\text{slope at a point} = \frac{dx}{dt}$$

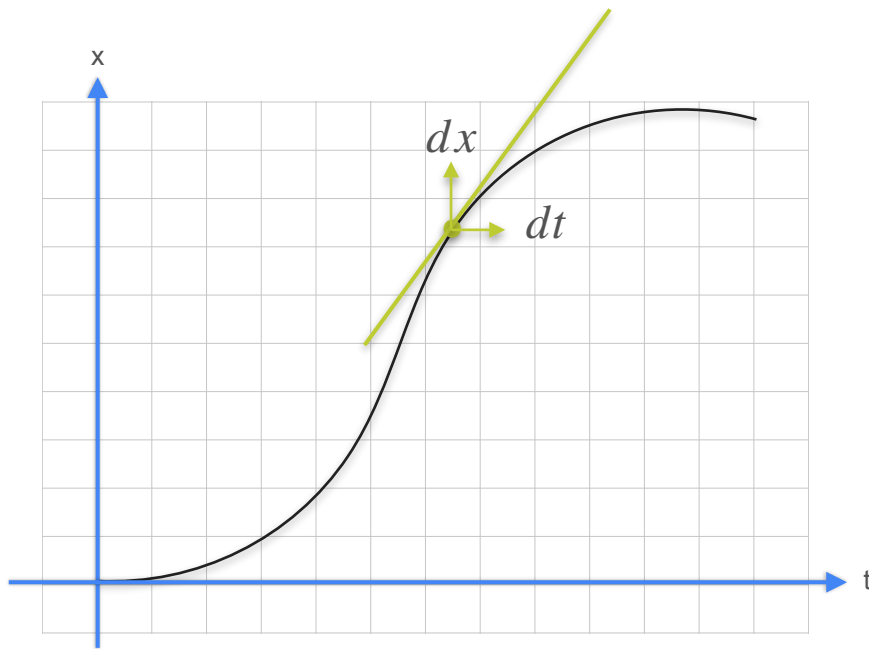


Derivatives

$$\text{slope} = \frac{\text{change in distance}}{\text{change in time}}$$

$$\text{slope} = \frac{\Delta x}{\Delta t}$$

$$\text{slope at a point} = \frac{dx}{dt}$$

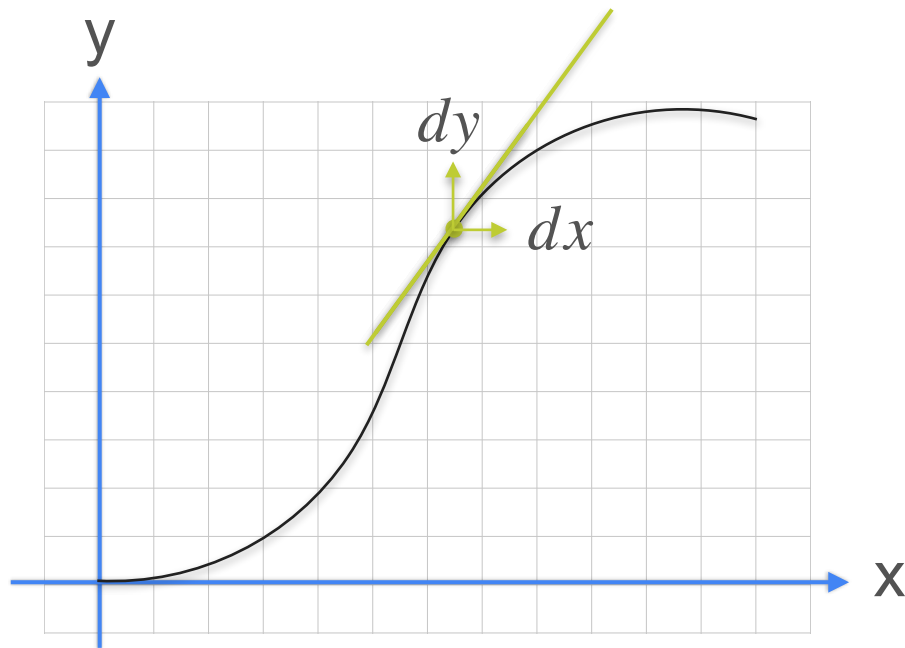


Derivatives

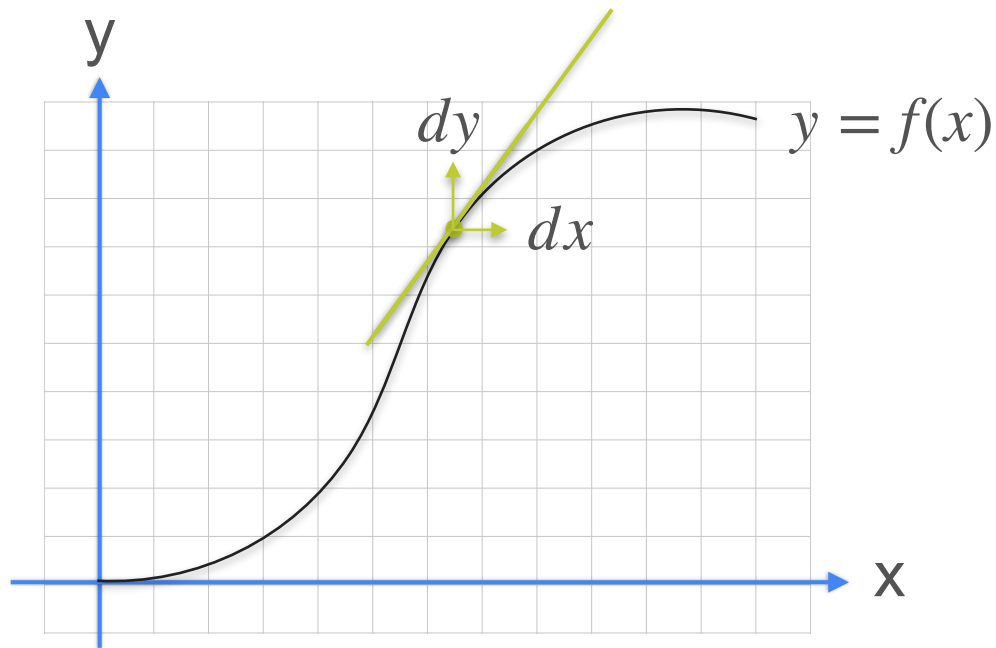
$$\text{slope} = \frac{\text{vertical change}}{\text{horizontal change}}$$

$$\text{slope} = \frac{\Delta y}{\Delta x}$$

$$\text{slope at a point} = \frac{dy}{dx}$$

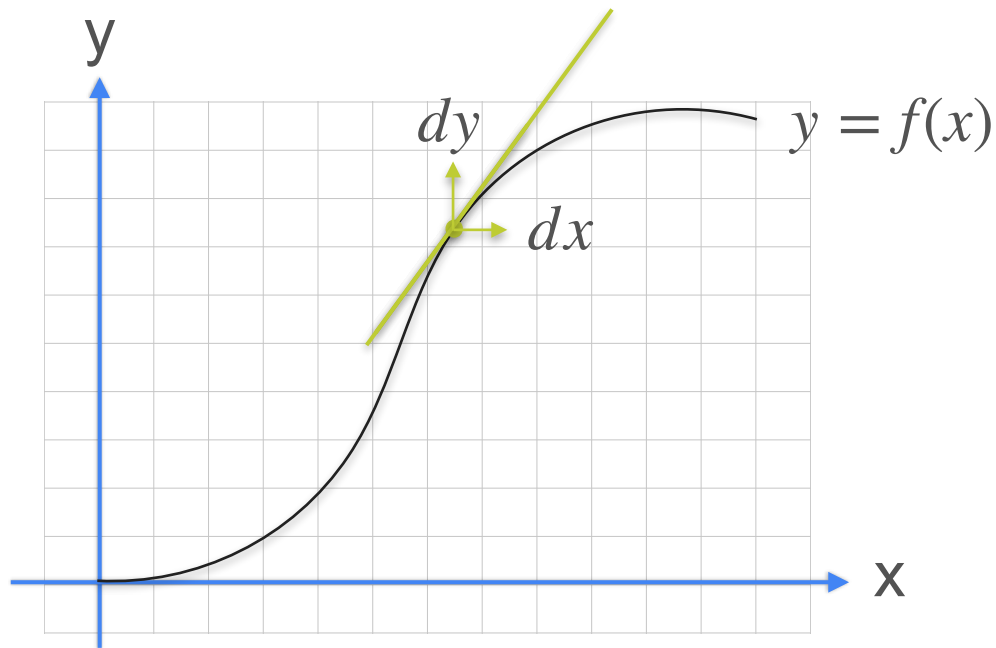


Derivatives



Derivatives

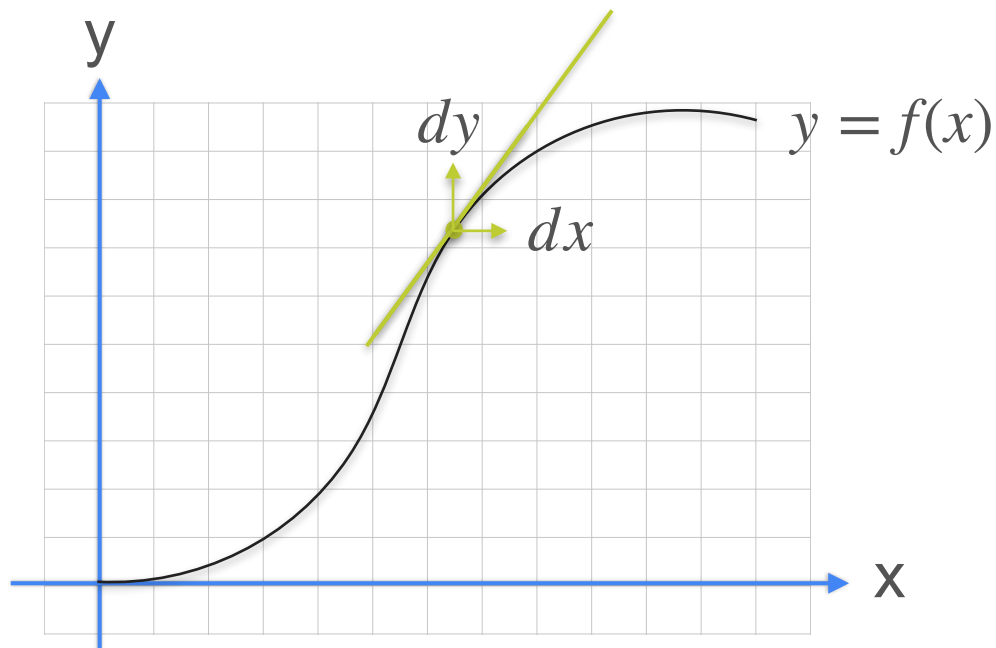
$$y = f(x)$$



Derivatives

$$y = f(x)$$

Derivative of f is expressed as:

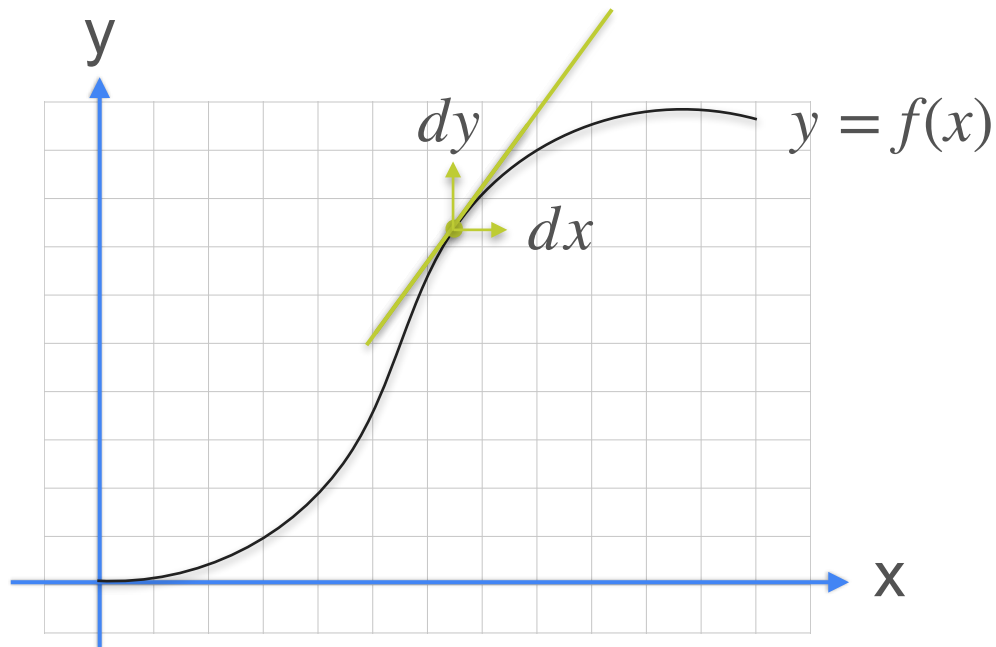


Derivatives

$$y = f(x)$$

Derivative of f is expressed as:

$$\frac{d}{dx}f(x)$$

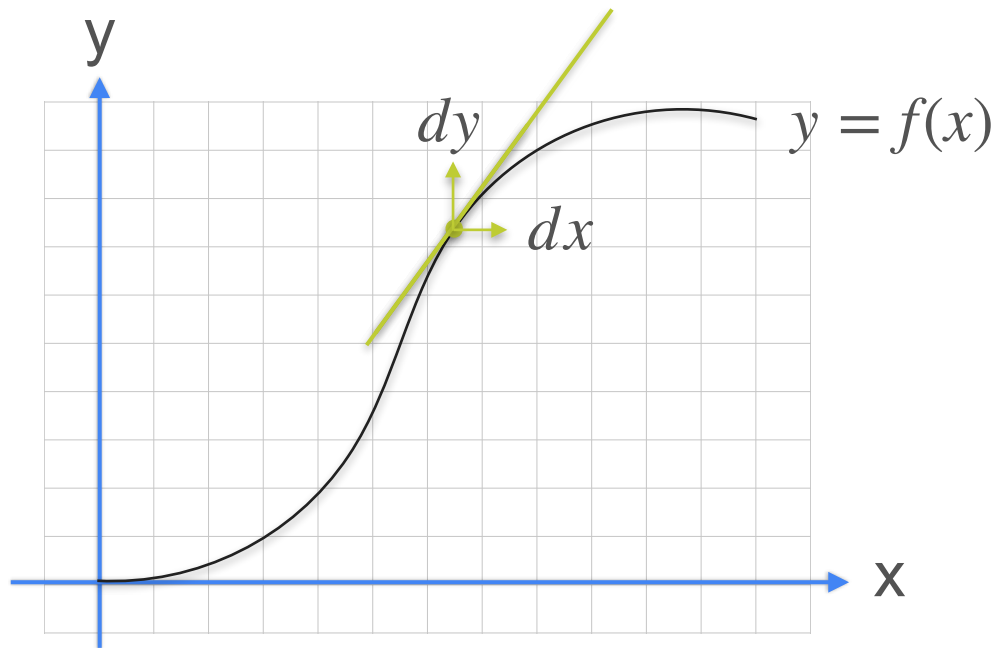


Derivatives

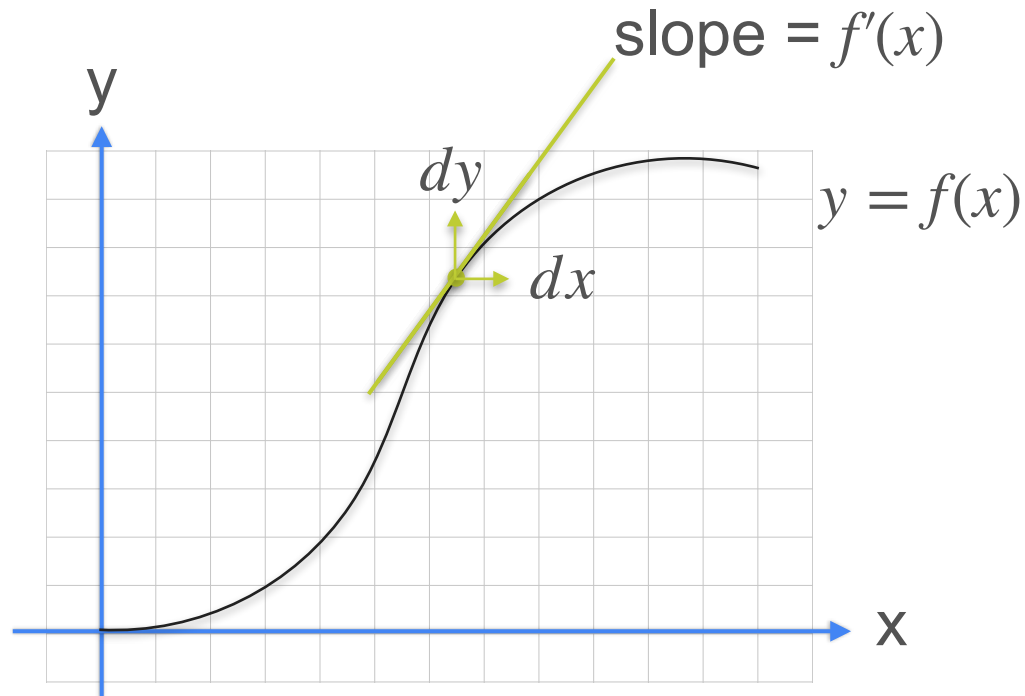
$$y = f(x)$$

Derivative of f is expressed as:

$$\frac{d}{dx}f(x) = \frac{dy}{dx}$$

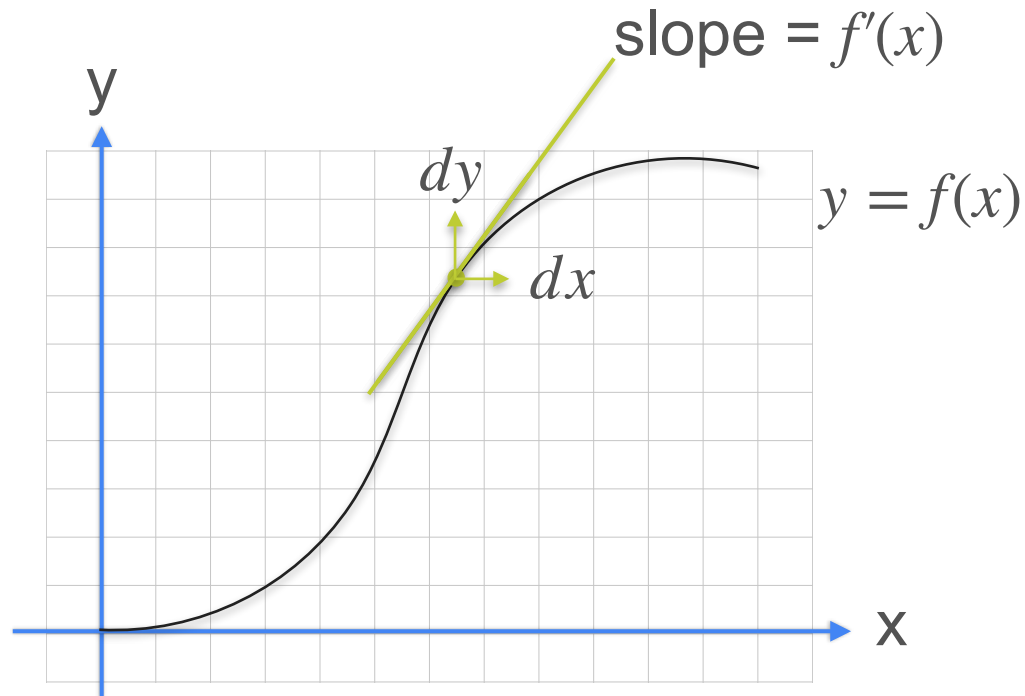


Derivatives: Lagrange's and Leibniz's Notation



Derivatives: Lagrange's and Leibniz's Notation

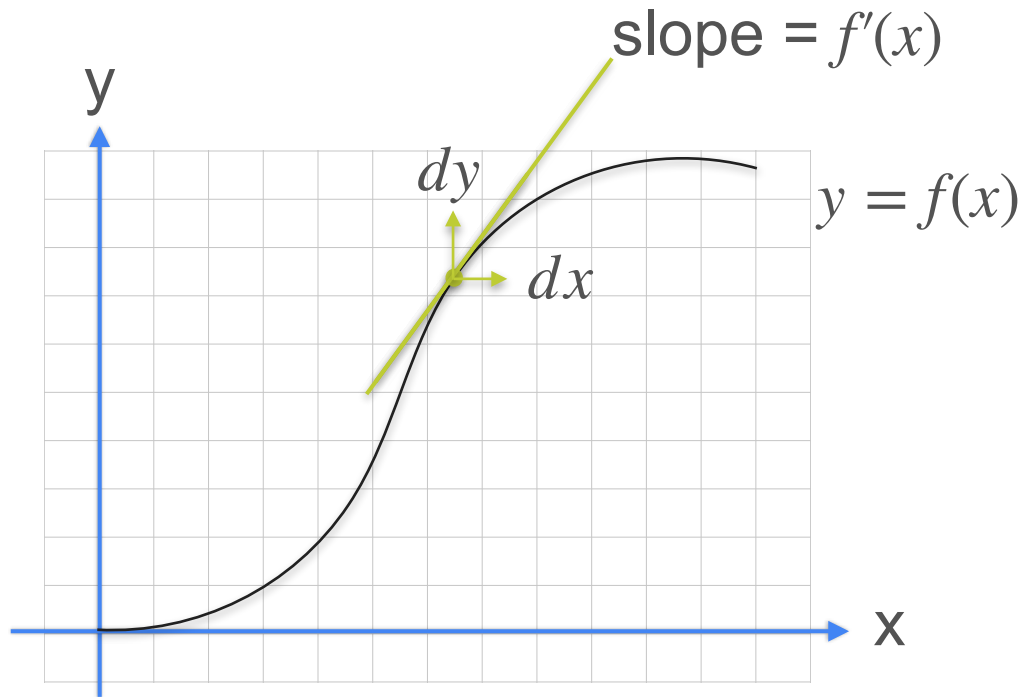
$$y = f(x)$$



Derivatives: Lagrange's and Leibniz's Notation

$$y = f(x)$$

Derivative of f is expressed as:

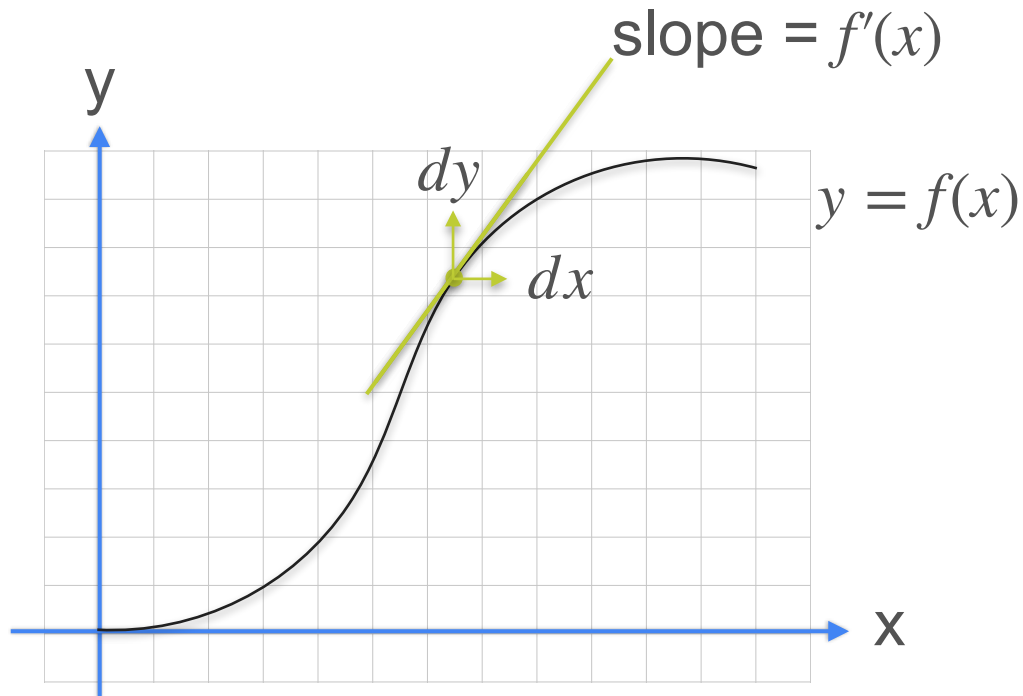


Derivatives: Lagrange's and Leibniz's Notation

$$y = f(x)$$

Derivative of f is expressed as:

$$f'(x)$$

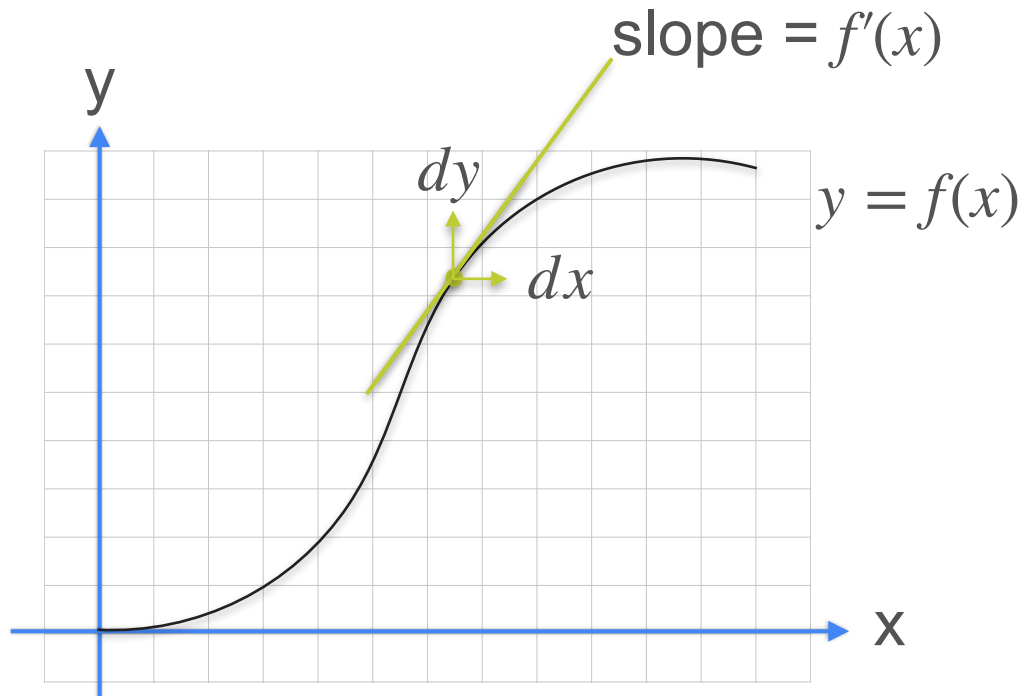


Derivatives: Lagrange's and Leibniz's Notation

$$y = f(x)$$

Derivative of f is expressed as:

$$f'(x) \quad \text{Lagrange's notation}$$



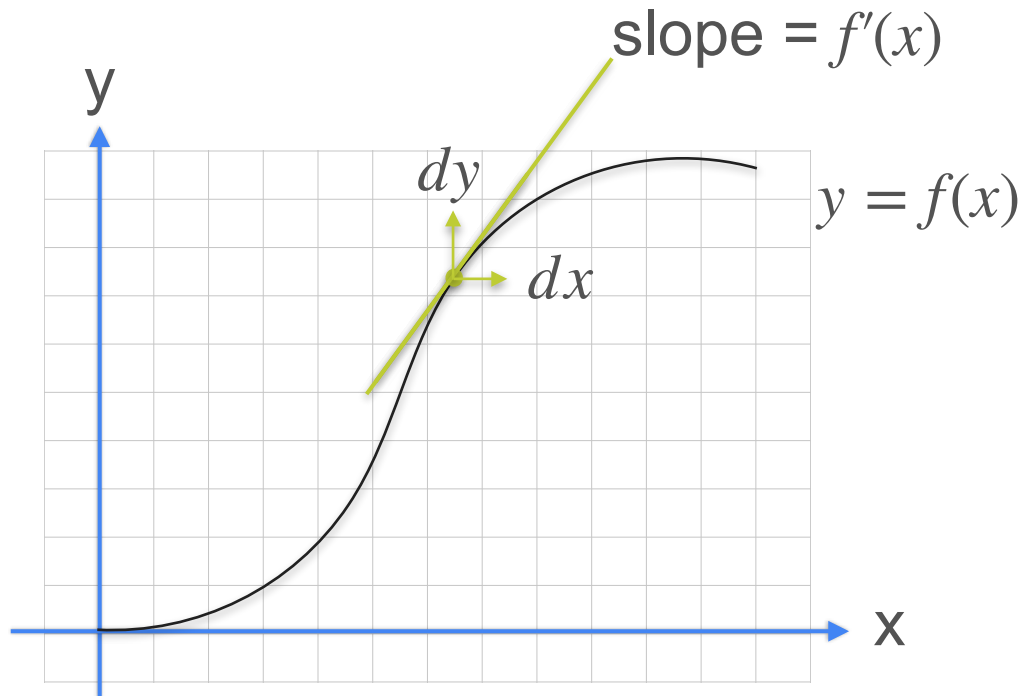
Derivatives: Lagrange's and Leibniz's Notation

$$y = f(x)$$

Derivative of f is expressed as:

$$f'(x) \quad \text{Lagrange's notation}$$

$$\frac{dy}{dx}$$



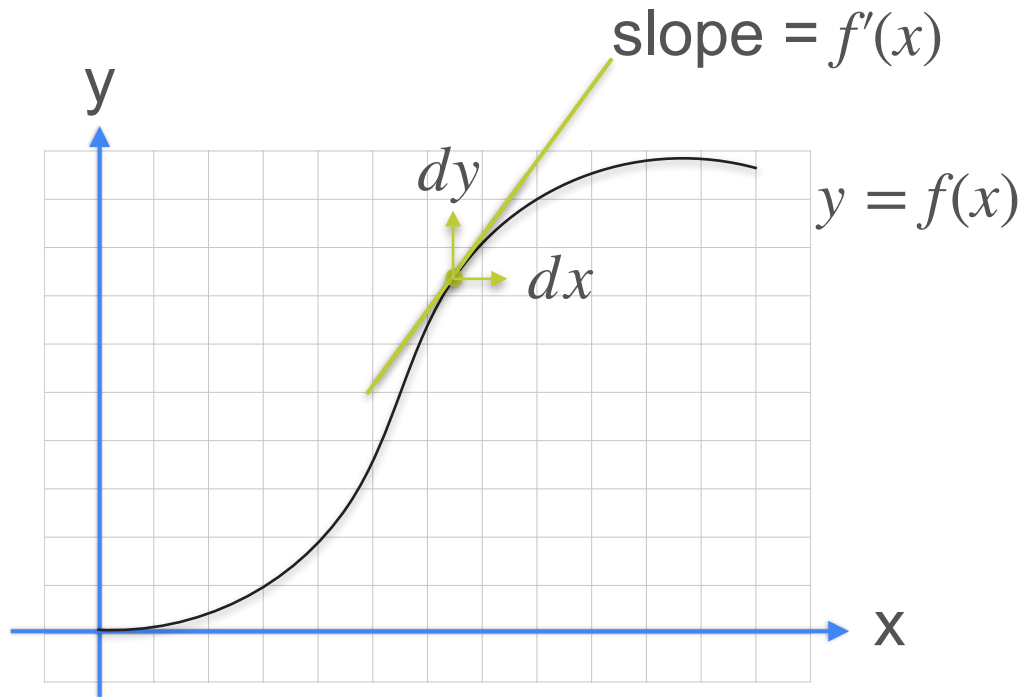
Derivatives: Lagrange's and Leibniz's Notation

$$y = f(x)$$

Derivative of f is expressed as:

$f'(x)$ **Lagrange's notation**

$$\frac{dy}{dx} = \frac{d}{dx} f(x)$$



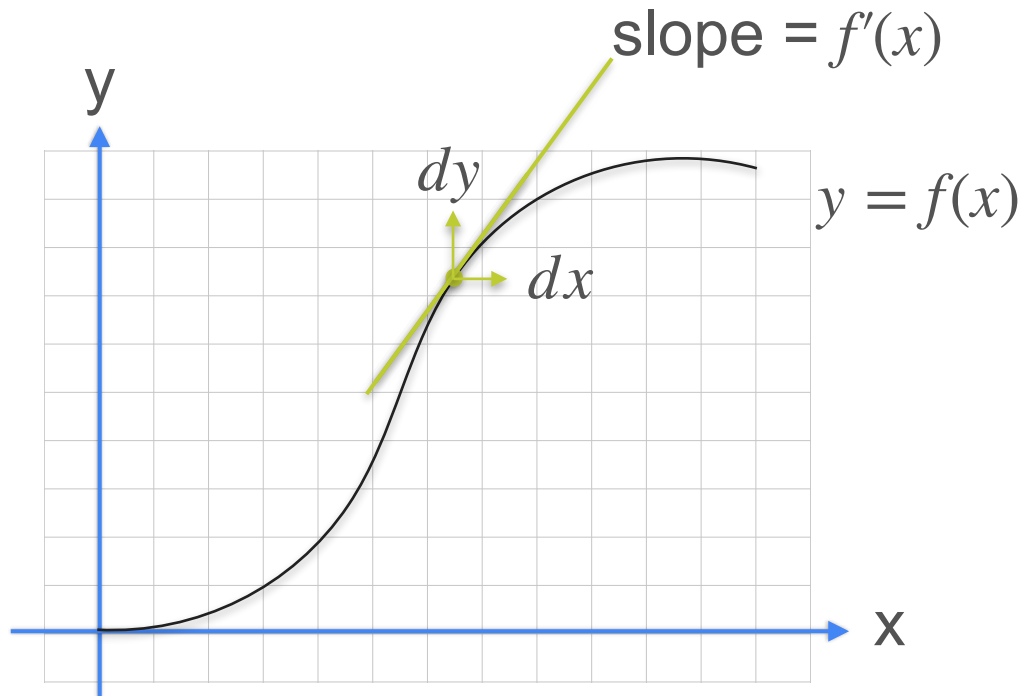
Derivatives: Lagrange's and Leibniz's Notation

$$y = f(x)$$

Derivative of f is expressed as:

$$f'(x) \quad \text{Lagrange's notation}$$

$$\frac{dy}{dx} = \frac{d}{dx} f(x) \quad \text{Leibniz's notation}$$





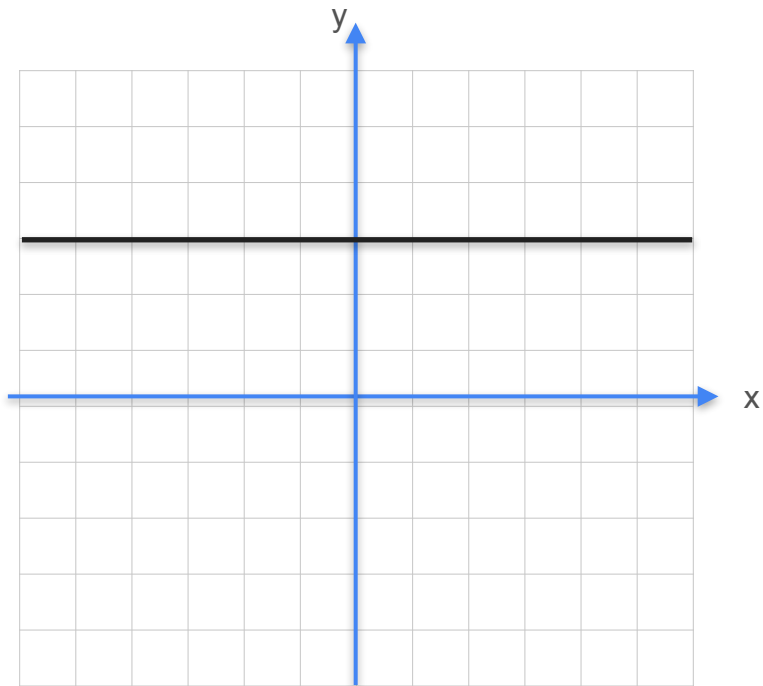
DeepLearning.AI

Derivatives and Optimization

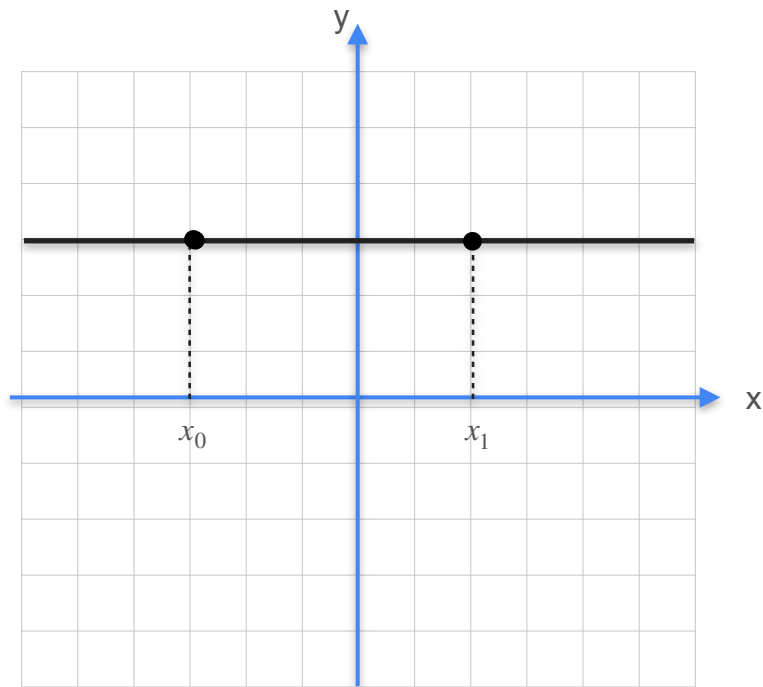
**Some common derivatives:
Lines**

Derivative of a Constant

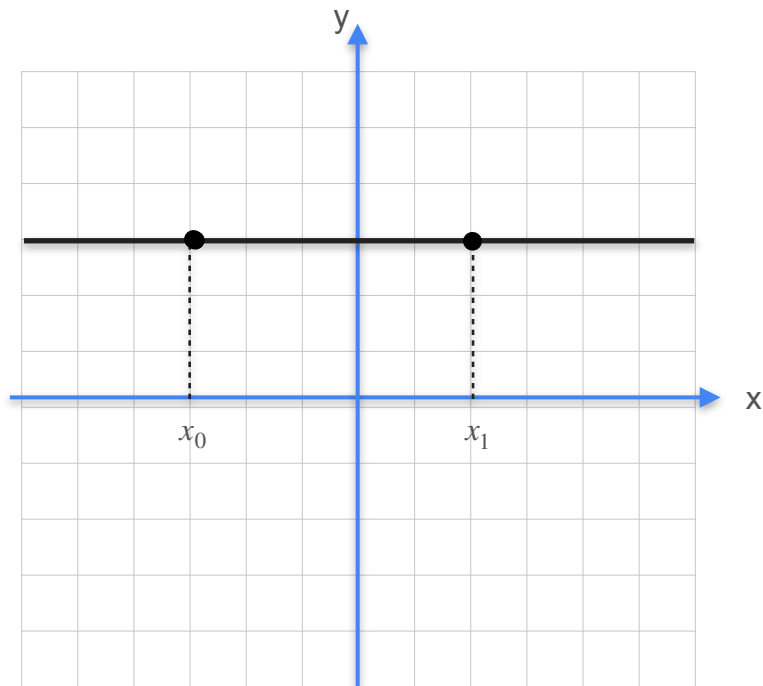
Derivative of a Constant



Derivative of a Constant

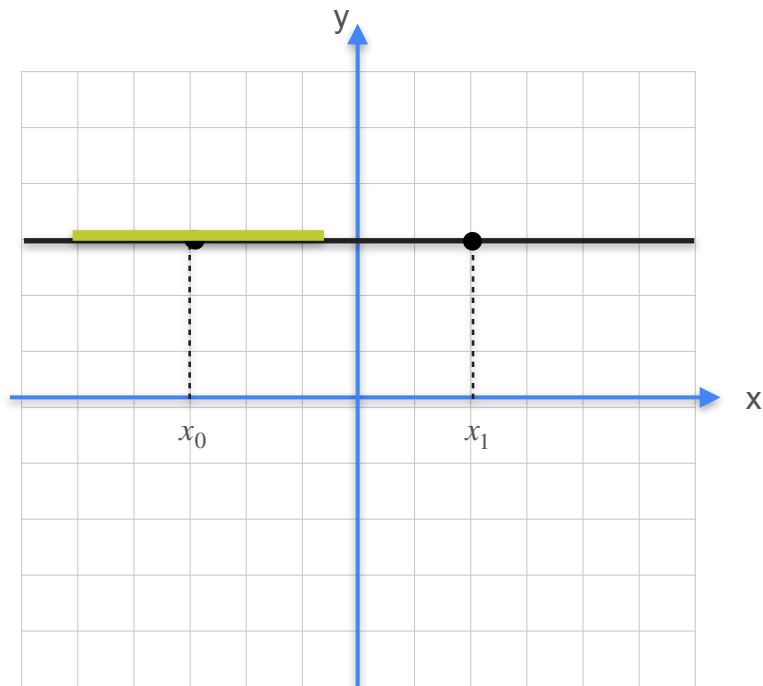


Derivative of a Constant



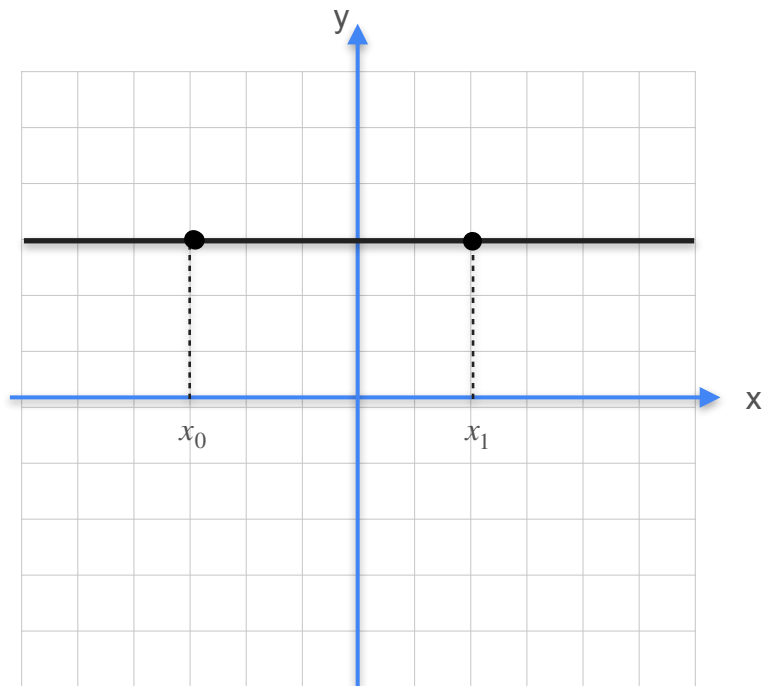
$$y = f(x) = c$$

Derivative of a Constant



$$y = f(x) = c$$

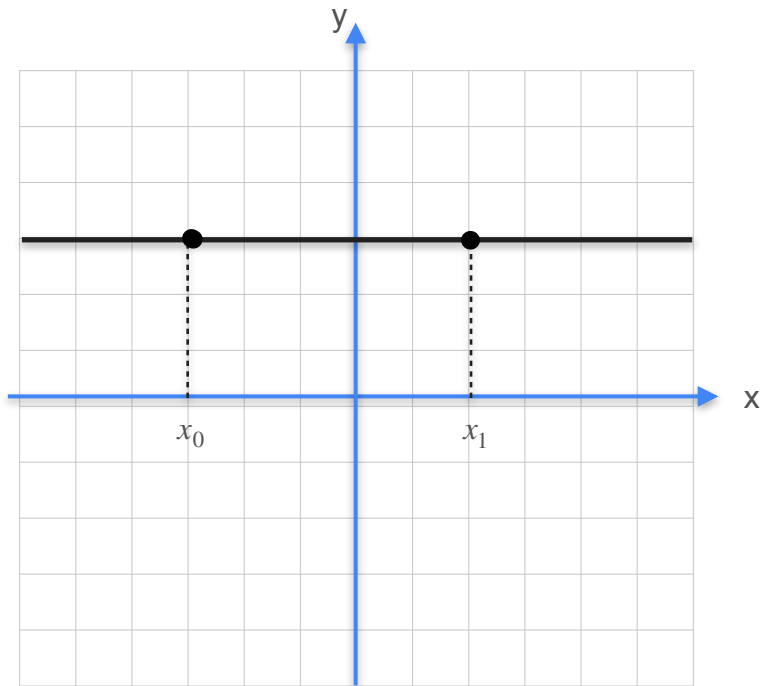
Derivative of a Constant



$$y = f(x) = c$$

Slope?

Derivative of a Constant

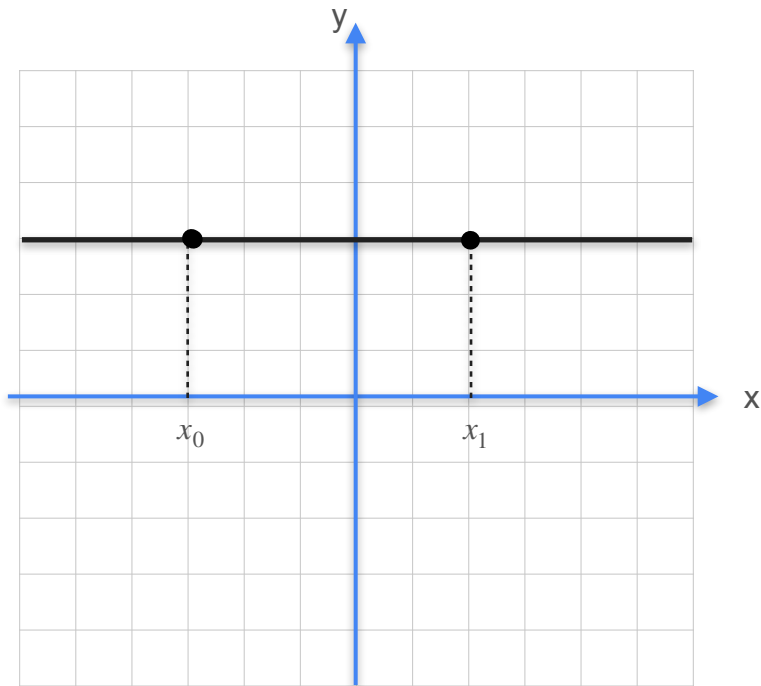


$$y = f(x) = c$$

Slope?

$$\frac{\Delta y}{\Delta x}$$

Derivative of a Constant

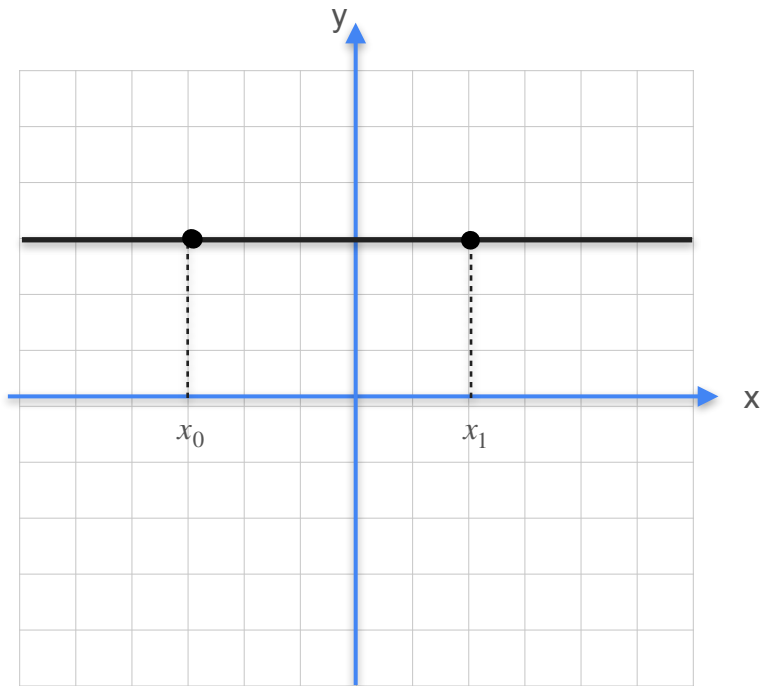


$$y = f(x) = c$$

Slope?

$$\frac{\Delta y}{\Delta x} = \frac{c - c}{x_1 - x_0} = 0$$

Derivative of a Constant



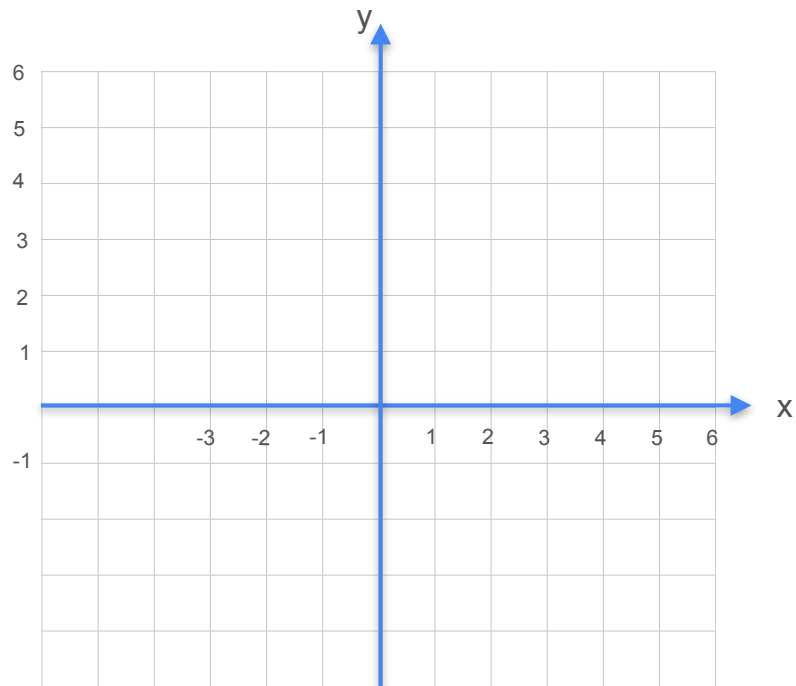
$$y = f(x) = c$$

Slope?

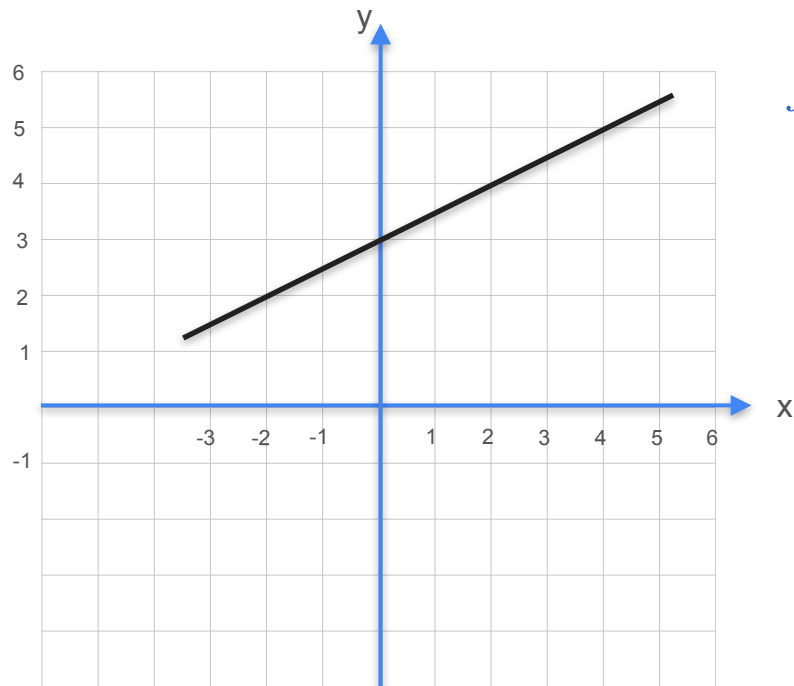
$$\frac{\Delta y}{\Delta x} = \frac{c - c}{x_1 - x_0} = 0 \rightarrow f'(x) = 0$$

Derivative of a Line

Derivative of a Line

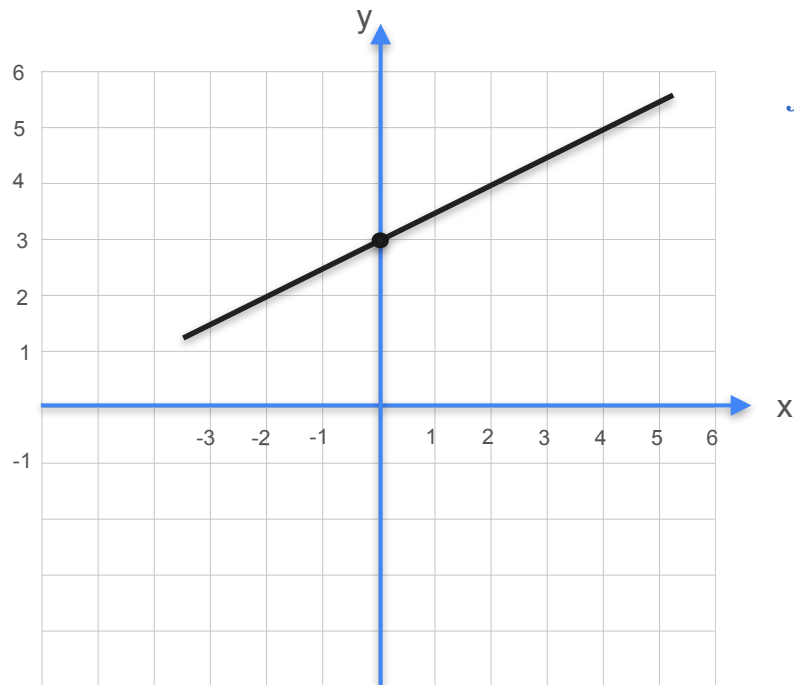


Derivative of a Line



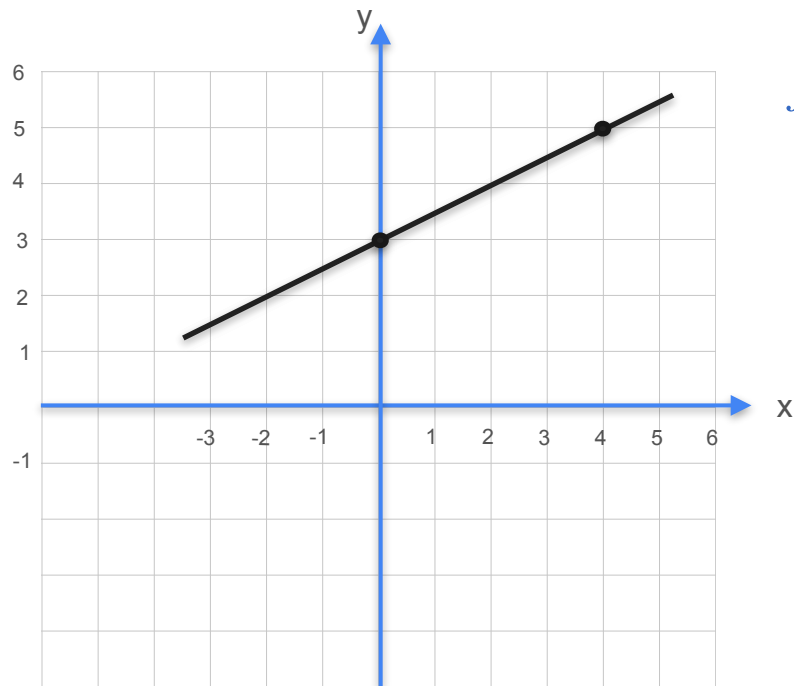
$$f(x) = ax + b$$

Derivative of a Line



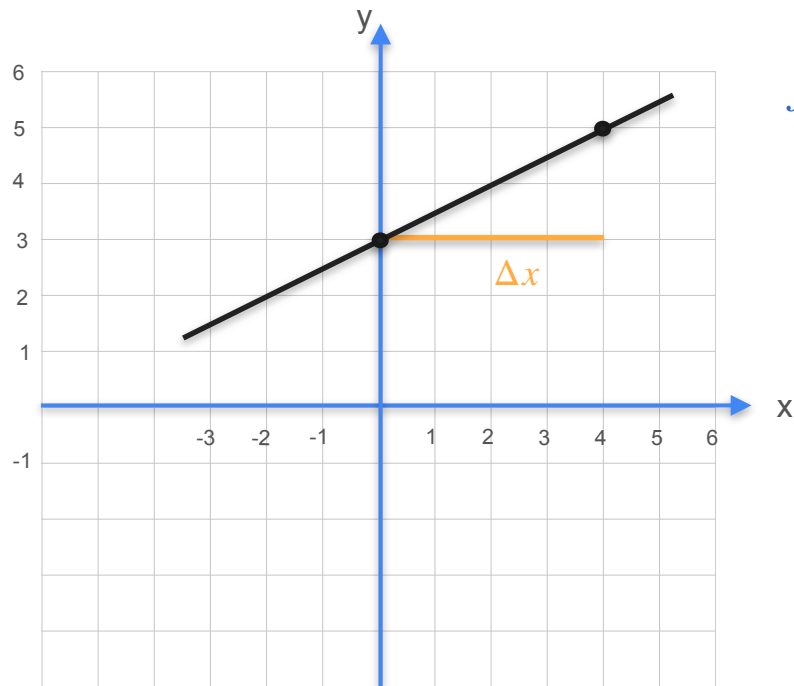
$$f(x) = ax + b$$

Derivative of a Line



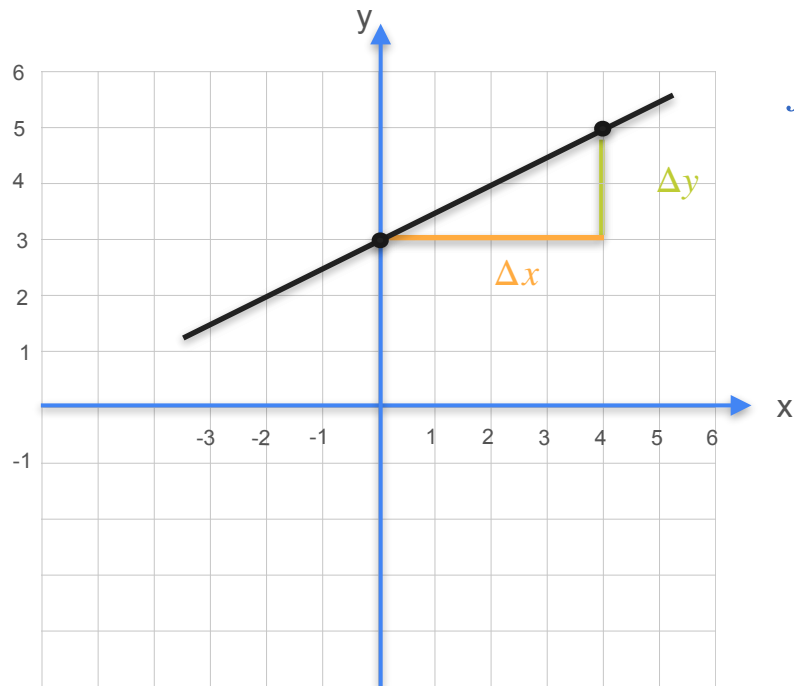
$$f(x) = ax + b$$

Derivative of a Line



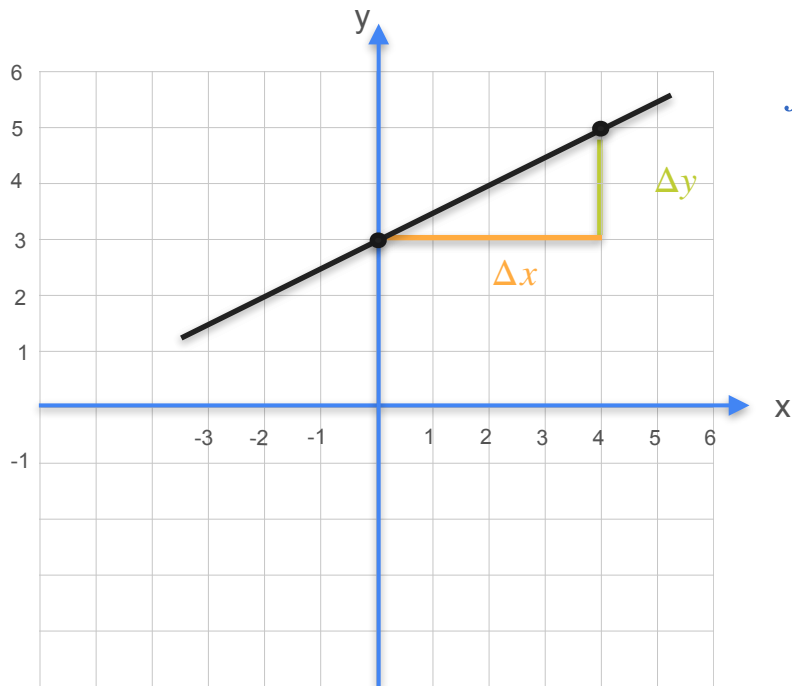
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Derivative of a Line



$$f(x) = ax + b$$

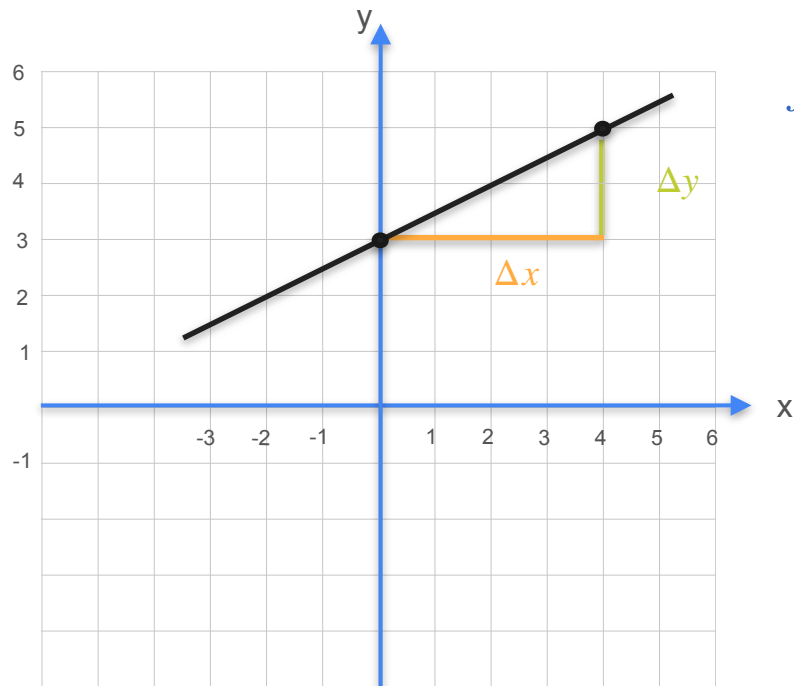
Derivative of a Line



$$f(x) = ax + b$$

$$\frac{\Delta y}{\Delta x}$$

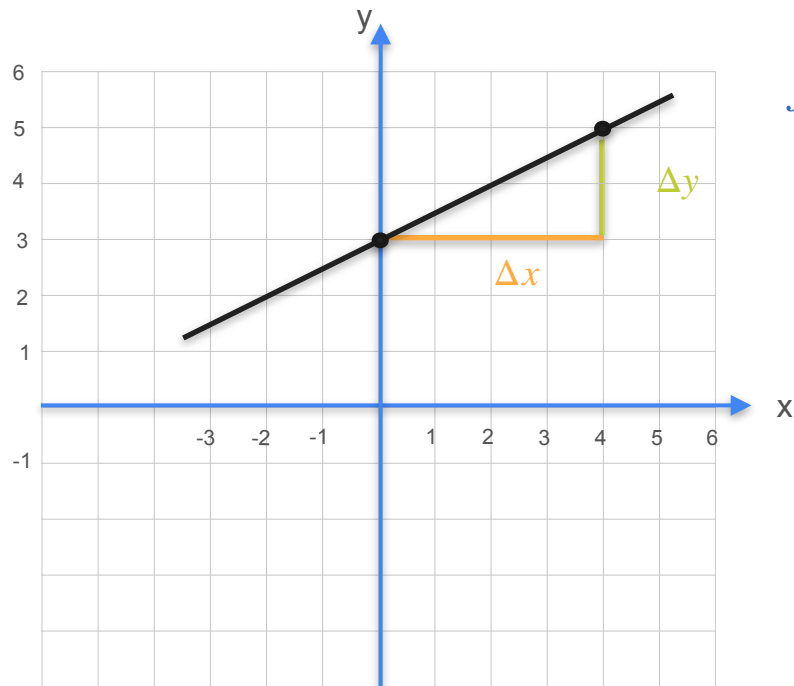
Derivative of a Line



$$f(x) = ax + b$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}}$$

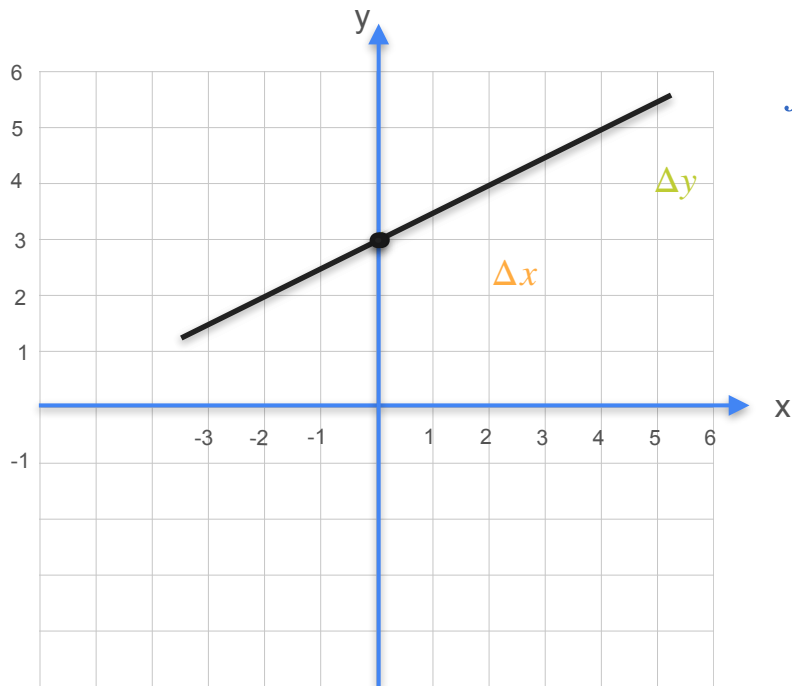
Derivative of a Line



$$f(x) = ax + b$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

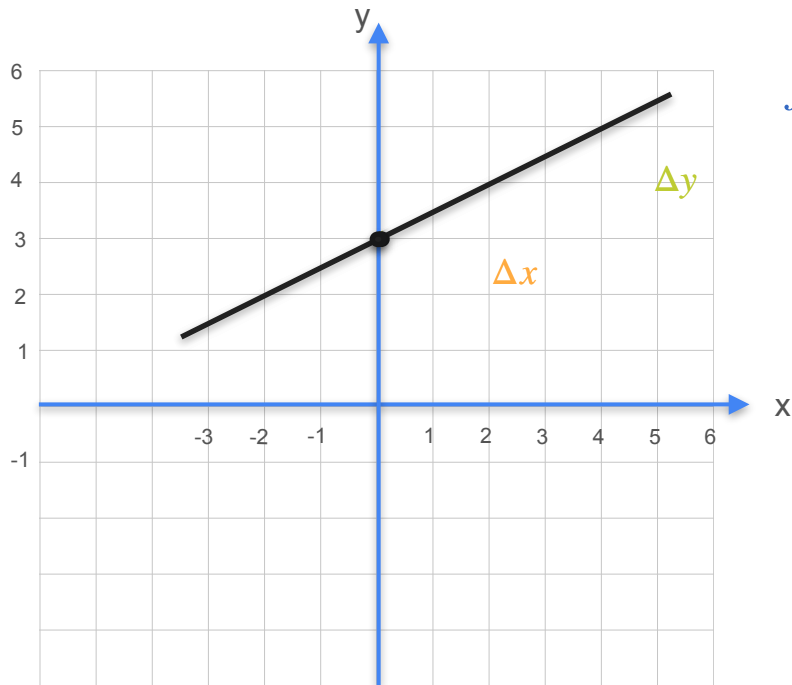
Derivative of a Line



$$f(x) = ax + b$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

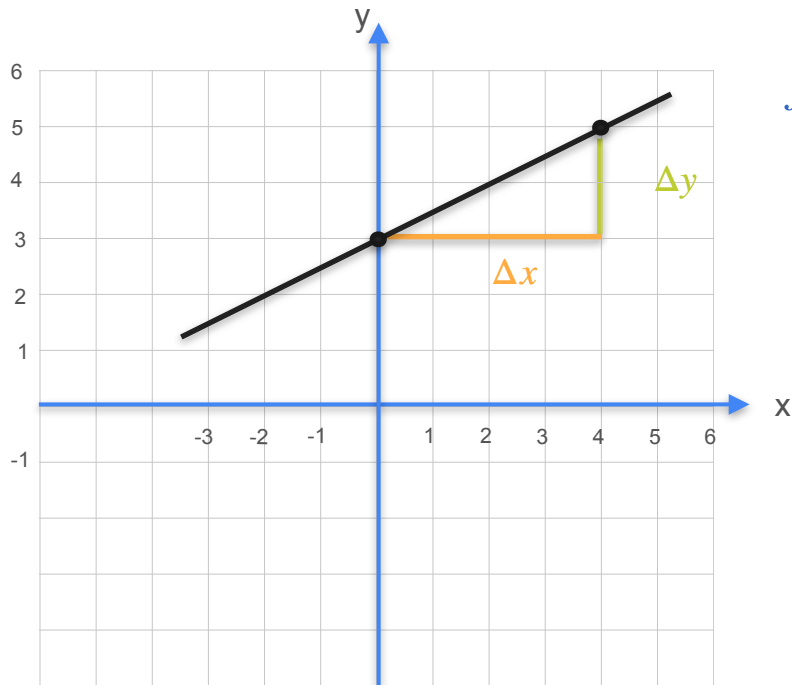
Derivative of a Line



$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

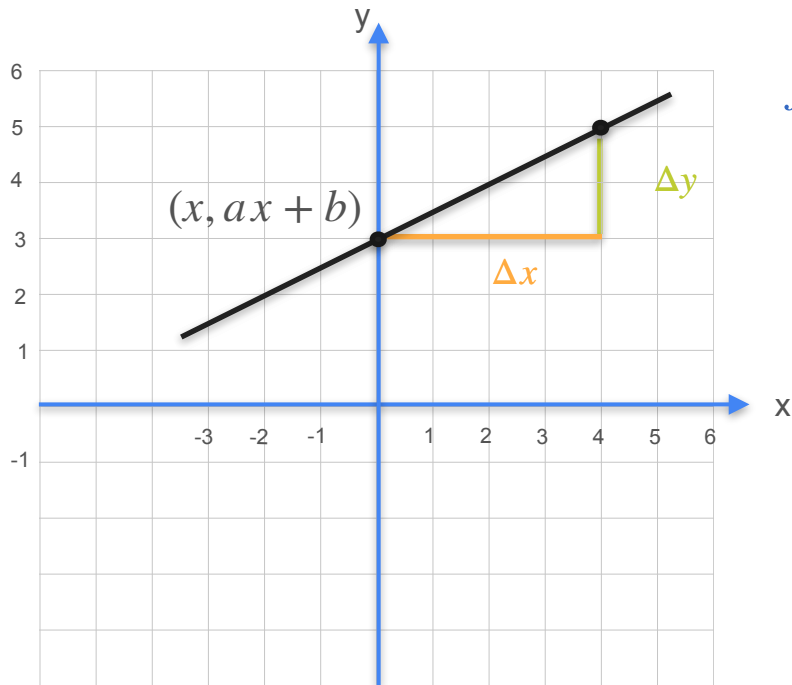
Derivative of a Line



$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

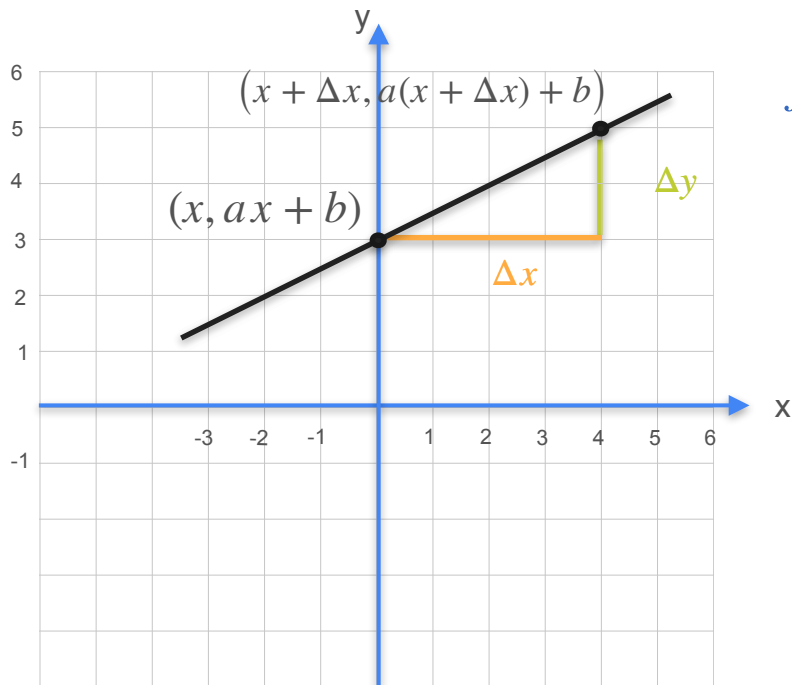
Derivative of a Line



$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

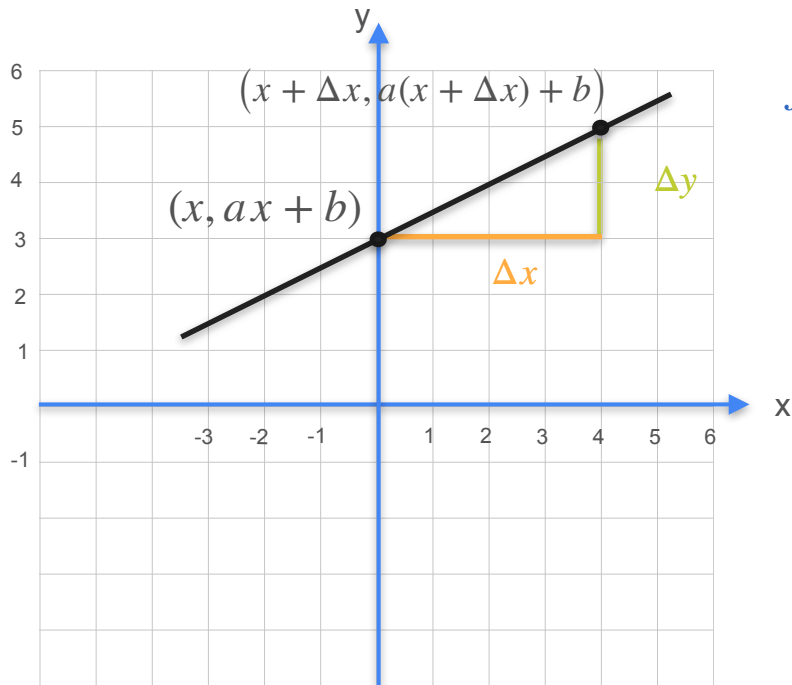
Derivative of a Line



$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

Derivative of a Line

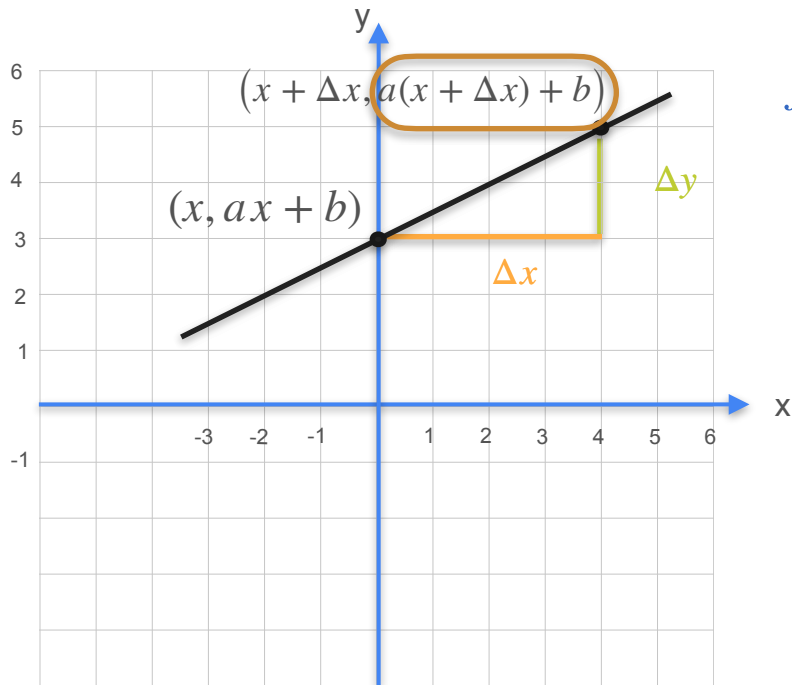


$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

$$\frac{\Delta y}{\Delta x}$$

Derivative of a Line

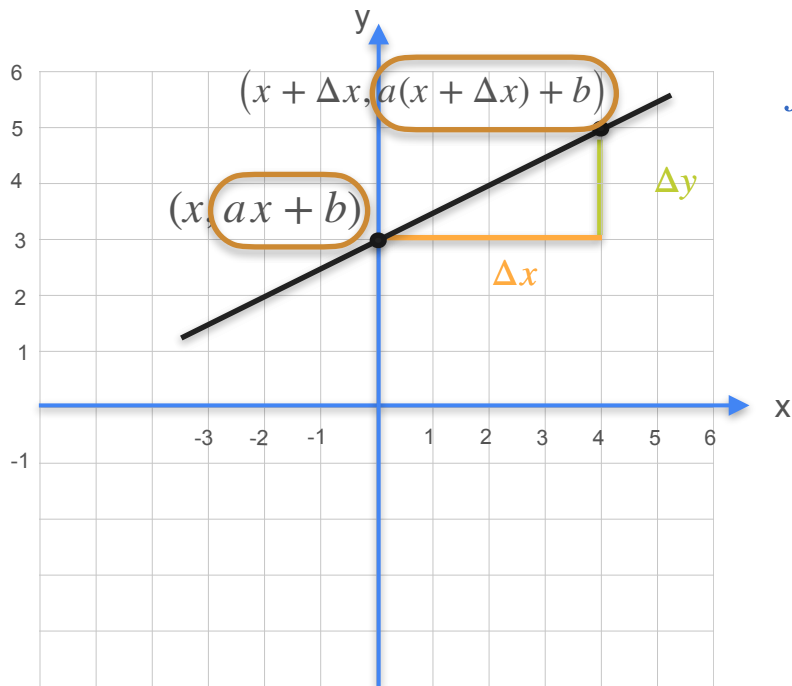


$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

$$\frac{\Delta y}{\Delta x}$$

Derivative of a Line

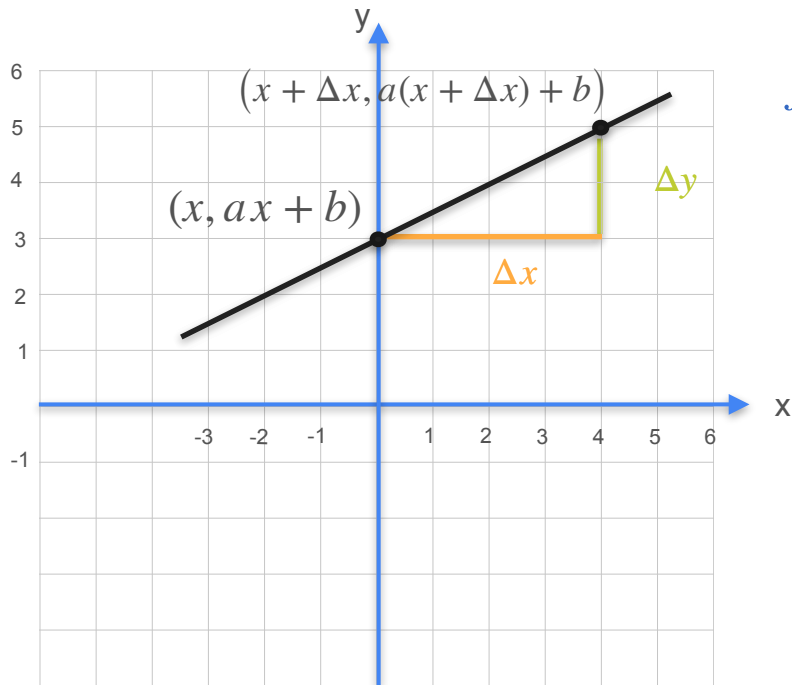


$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

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Derivative of a Line

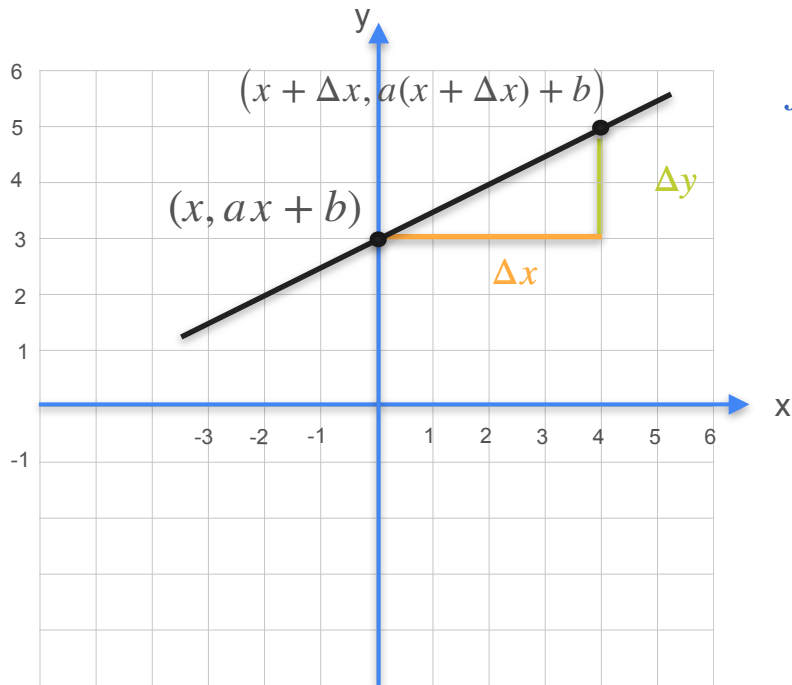


$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

$$\frac{\Delta y}{\Delta x} = \frac{a(x + \Delta x) + b - (ax + b)}{\Delta x}$$

Derivative of a Line

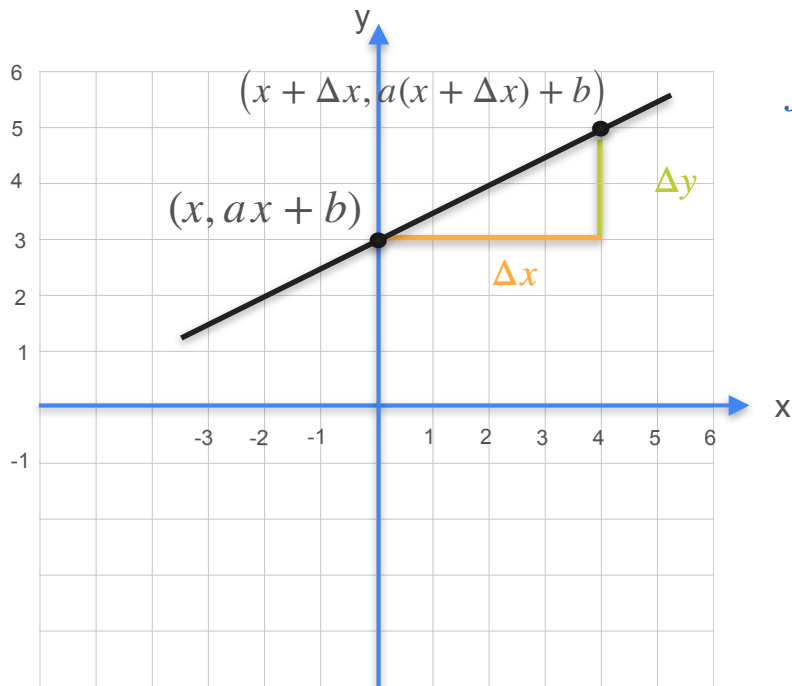


$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

$$\frac{\Delta y}{\Delta x} = \frac{a\cancel{(x + \Delta x)} + b - (\cancel{a}x + b)}{\Delta x}$$

Derivative of a Line

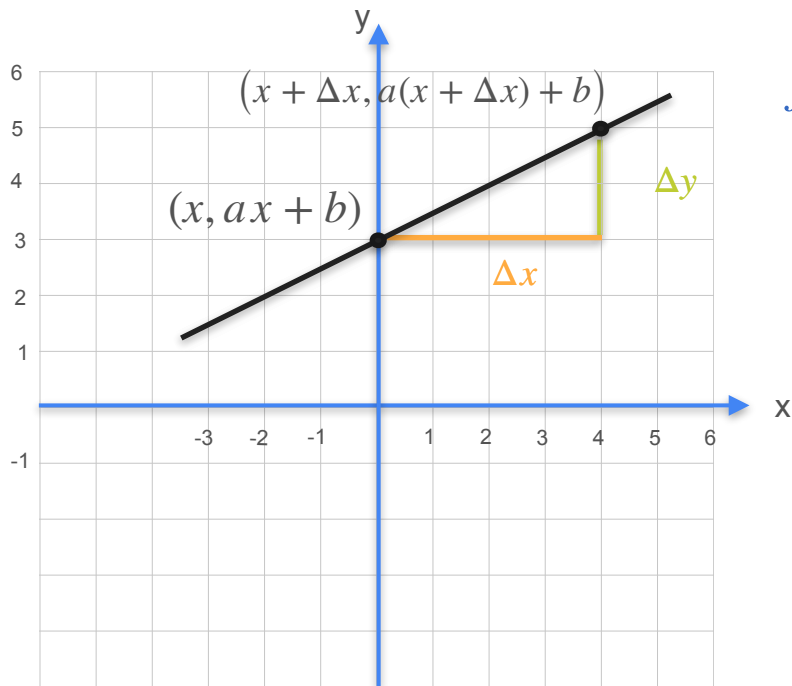


$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

$$\frac{\Delta y}{\Delta x} = \frac{a\cancel{(x + \Delta x)} + \cancel{b} - (\cancel{a}x + \cancel{b})}{\Delta x}$$

Derivative of a Line

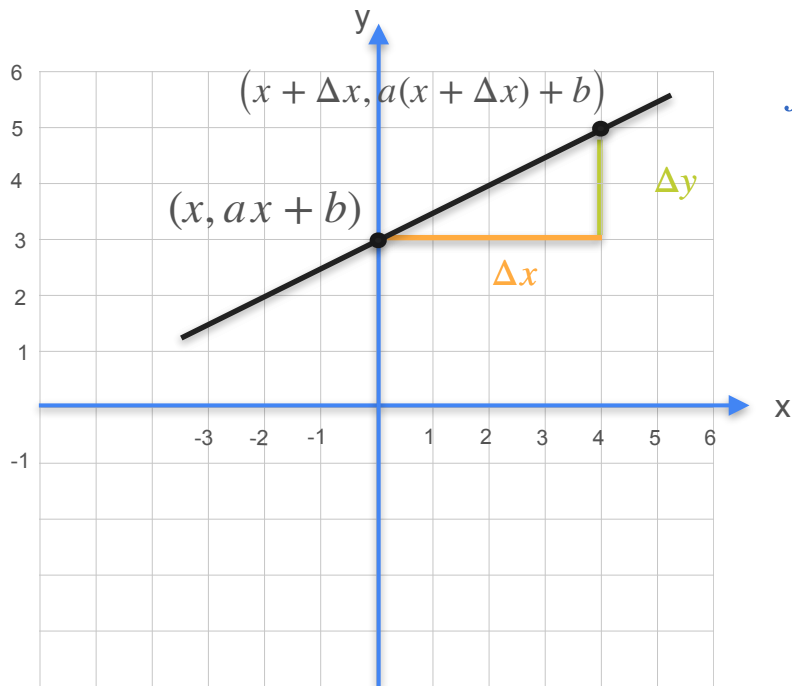


$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

$$\begin{aligned} \frac{\Delta y}{\Delta x} &= \frac{a(x + \Delta x) + b - (ax + b)}{\Delta x} \\ &= a \frac{\Delta x}{\Delta x} \end{aligned}$$

Derivative of a Line

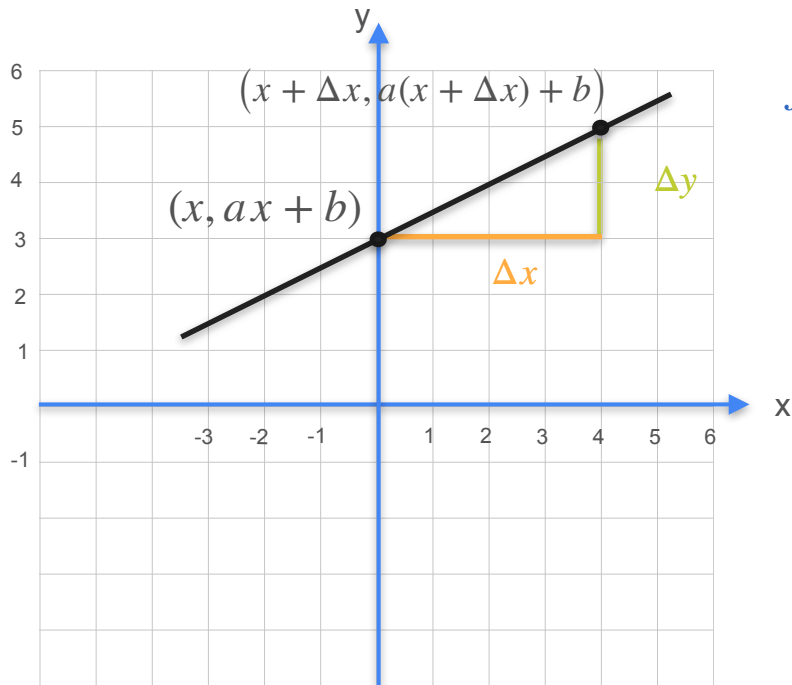


$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

$$\begin{aligned} \frac{\Delta y}{\Delta x} &= \frac{a(x + \Delta x) + b - (ax + b)}{\Delta x} \\ &= a \frac{\cancel{\Delta x}}{\cancel{\Delta x}} \end{aligned}$$

Derivative of a Line

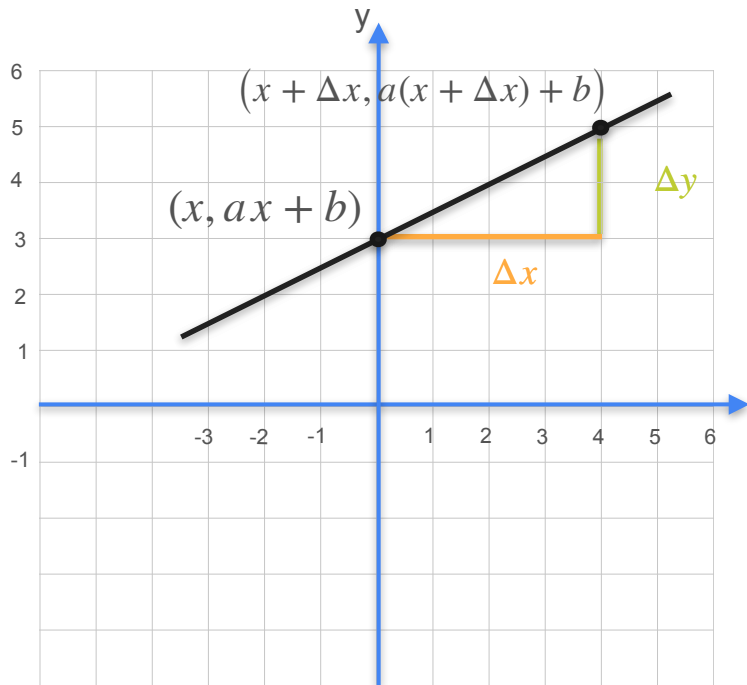


$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

$$\begin{aligned} \frac{\Delta y}{\Delta x} &= \frac{a\cancel{(x + \Delta x)} + \cancel{b} - (\cancel{a}x + \cancel{b})}{\Delta x} \\ &= a \frac{\cancel{\Delta x}}{\cancel{\Delta x}} = a \end{aligned}$$

Derivative of a Line



$$f(x) = ax + b \quad \Rightarrow \quad f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

$$\begin{aligned} \frac{\Delta y}{\Delta x} &= \frac{a(x + \Delta x) + b - (ax + b)}{\Delta x} \\ &= a \frac{\cancel{\Delta x}}{\cancel{\Delta x}} = a \end{aligned}$$



DeepLearning.AI

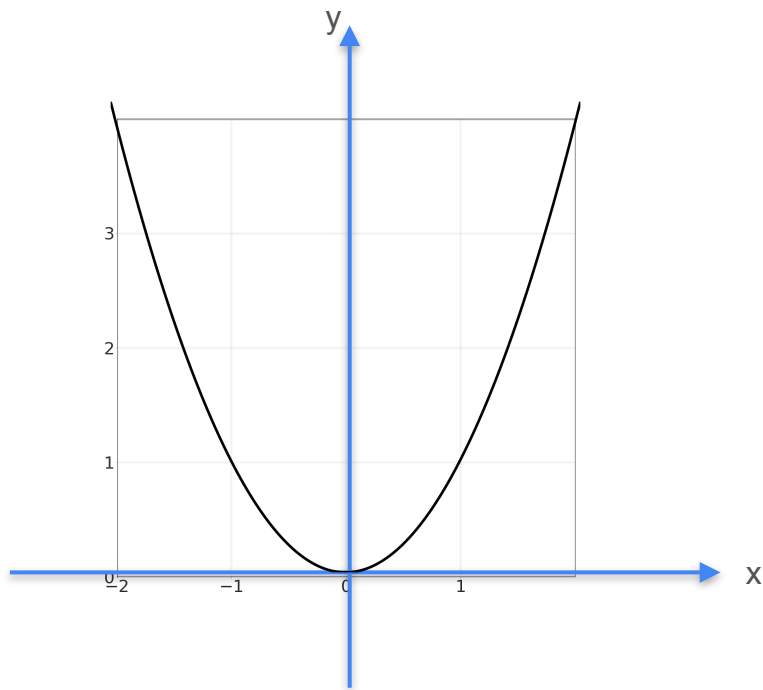
Derivatives and Optimization

**Some common derivatives:
Quadratics**

Derivative of Quadratic Functions

Derivative of Quadratic Functions

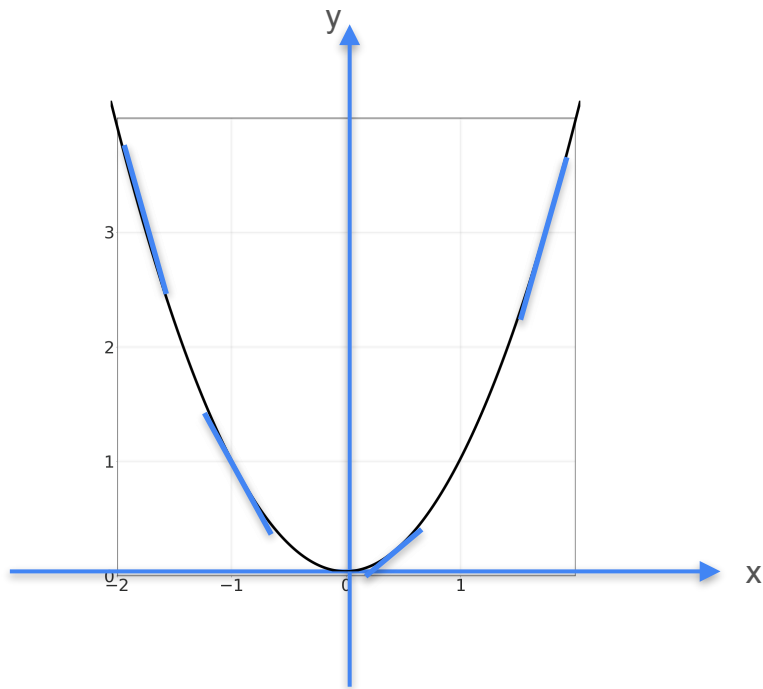
Quadratics: $y = f(x) = x^2$



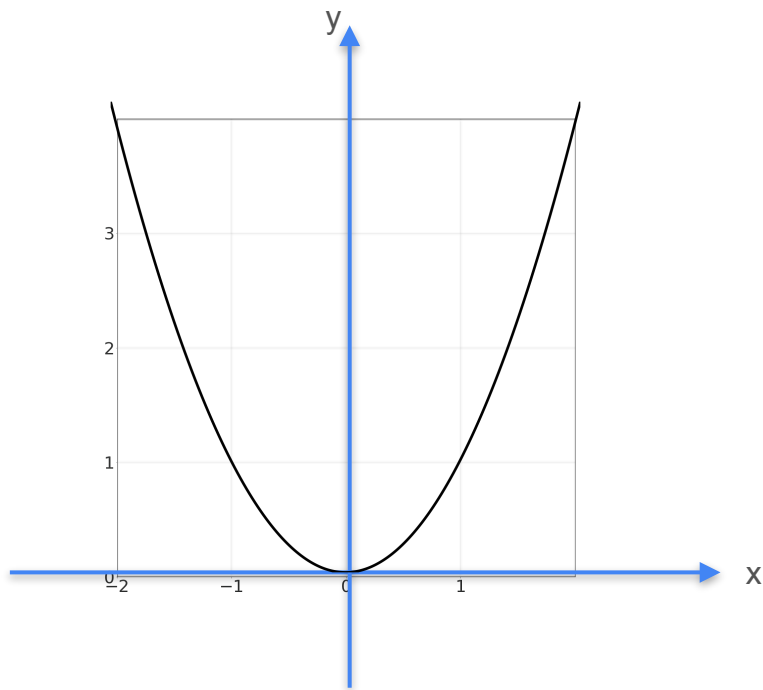
Derivative of Quadratic Functions

Quadratics: $y = f(x) = x^2$

Slope: $\frac{\Delta f}{\Delta x}$



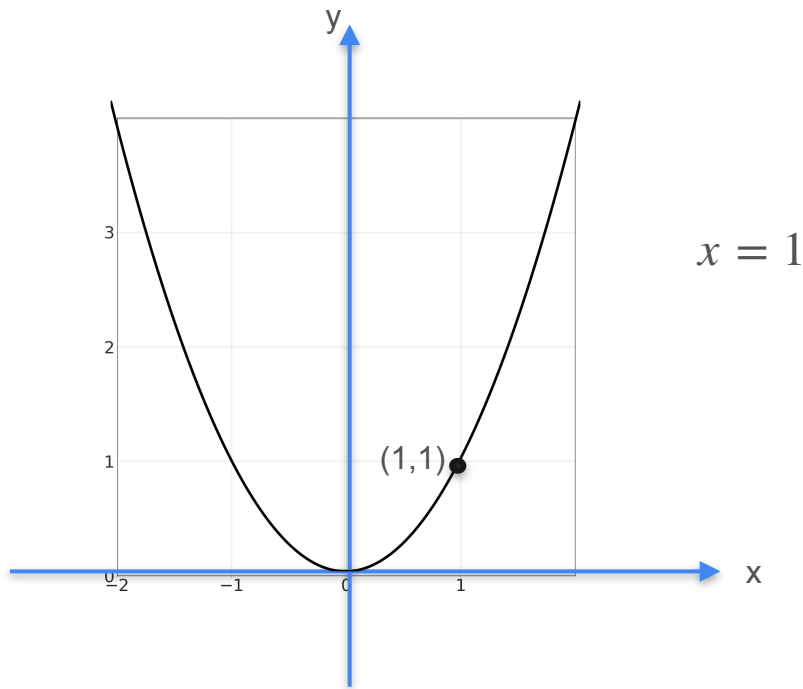
Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

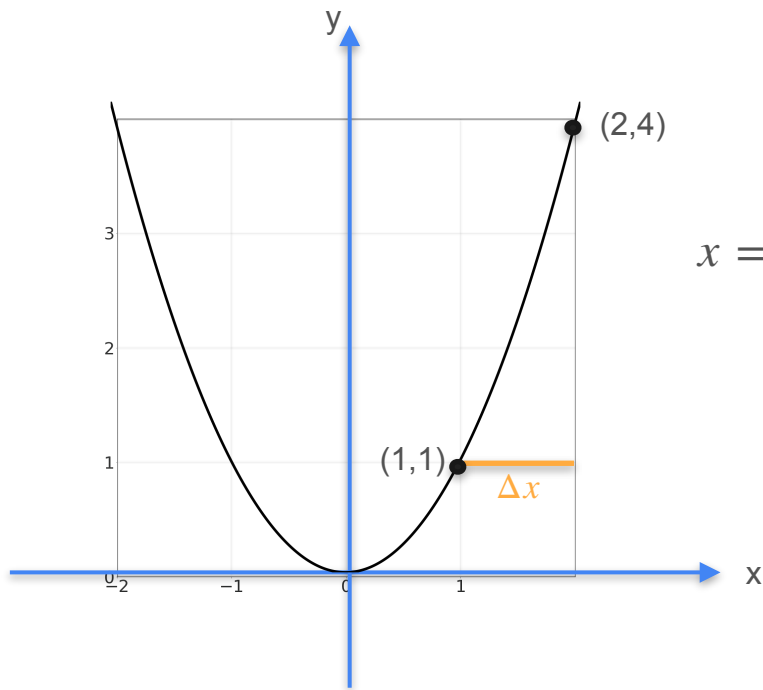
Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

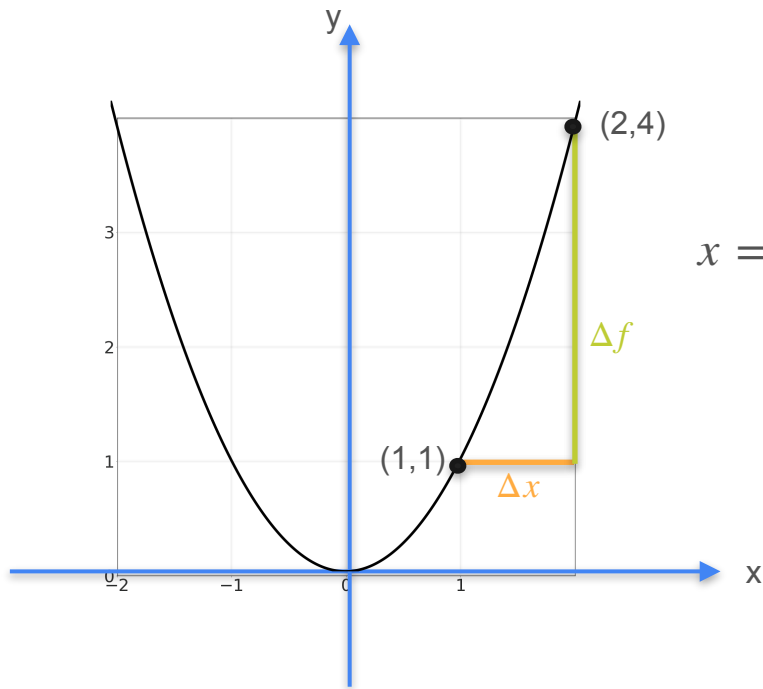
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

$x = 1$

Δx

1.0

Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

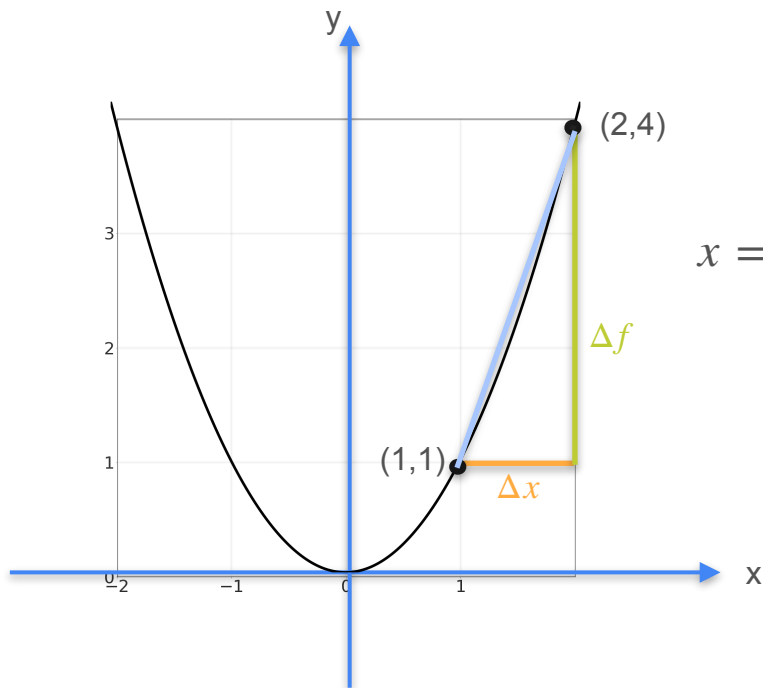
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

$x = 1$

Δx	1.0
Δf	3

$$(1 + 1)^2 - 1^2 = 4 - 1$$

Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

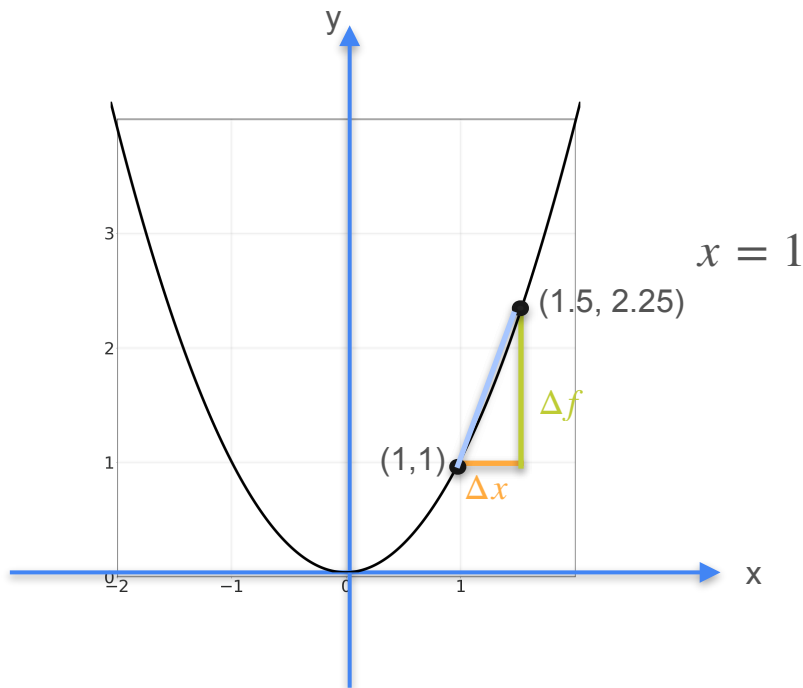
$x = 1$

Δx	1.0
Δf	3
Slope	3

$$(1 + 1)^2 - 1^2 = 4 - 1$$

$$\frac{3}{1}$$

Derivative of Quadratic Functions

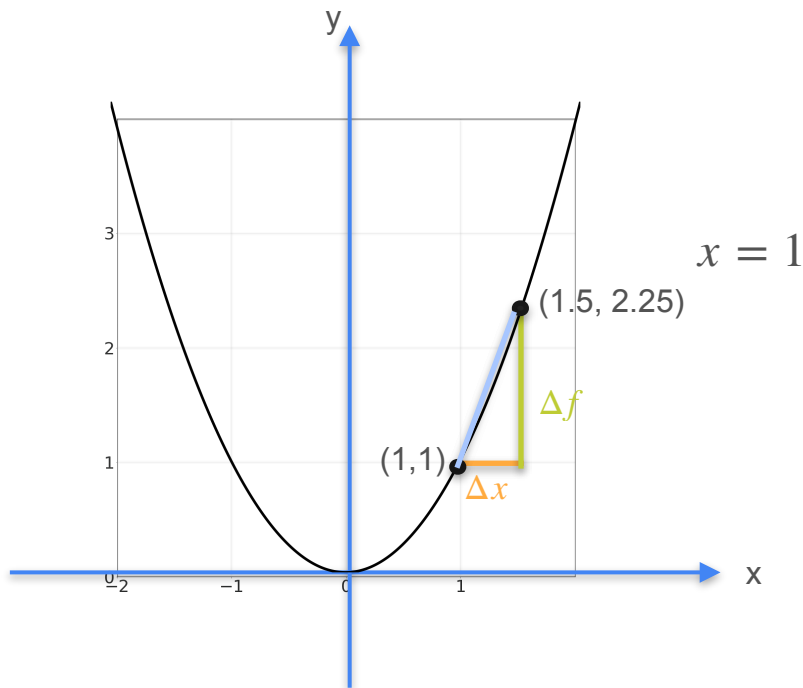


Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

Δx	1.0	1/2
Δf	3	
Slope	3	

Derivative of Quadratic Functions



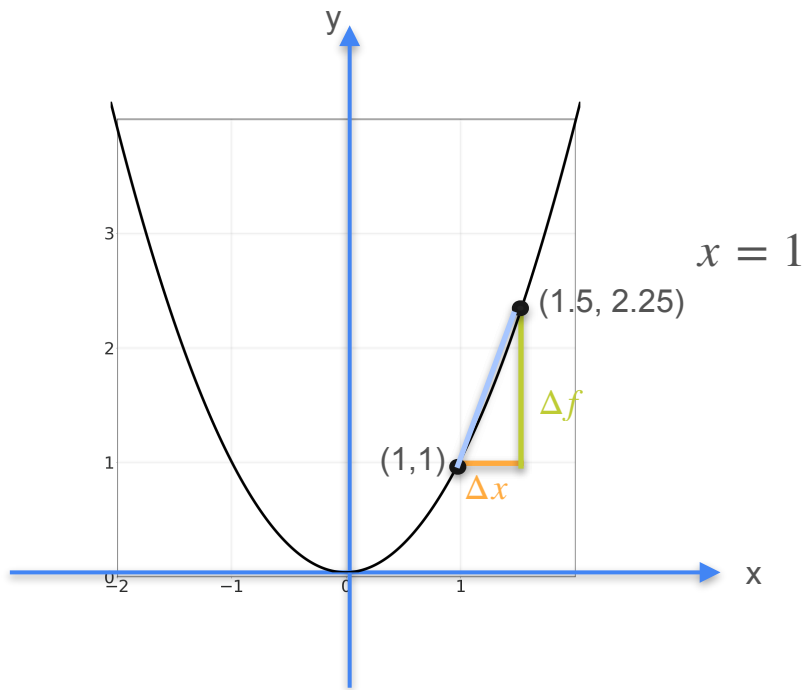
Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

Δx	1.0	1/2
Δf	3	1.25
Slope	3	

$$(1 + 0.5)^2 - 1^2 = 2.25 - 1$$

Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

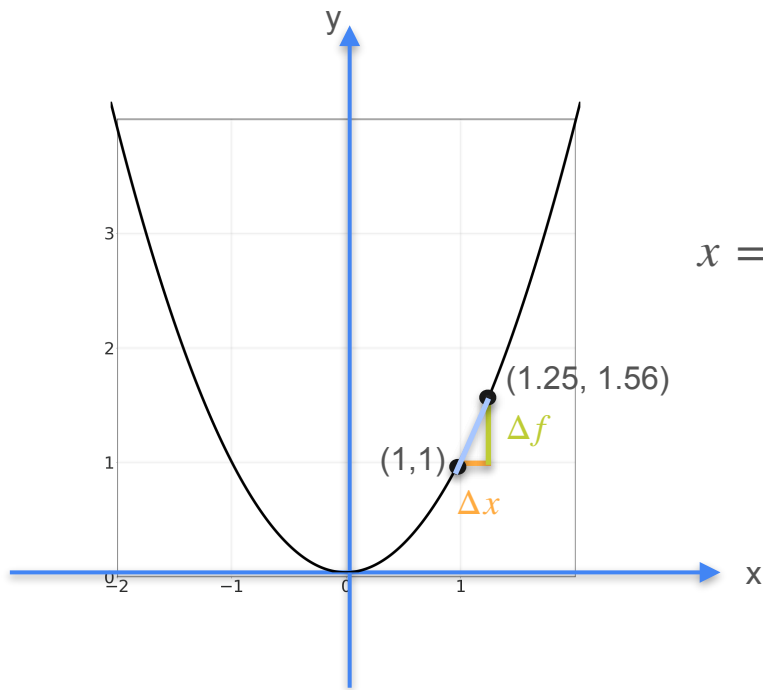
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

Δx	1.0	1/2
Δf	3	1.25
Slope	3	2.5

$$(1 + 0.5)^2 - 1^2 = 2.25 - 1$$

$$\frac{1.25}{0.5}$$

Derivative of Quadratic Functions



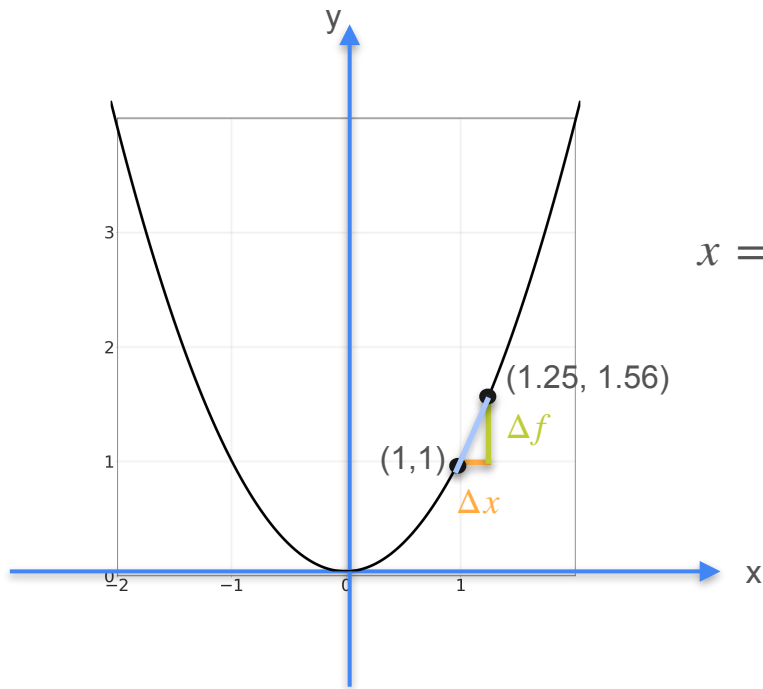
$x = 1$

Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

Δx	1.0	1/2	1/4
Δf	3	1.25	
Slope	3	2.5	

Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

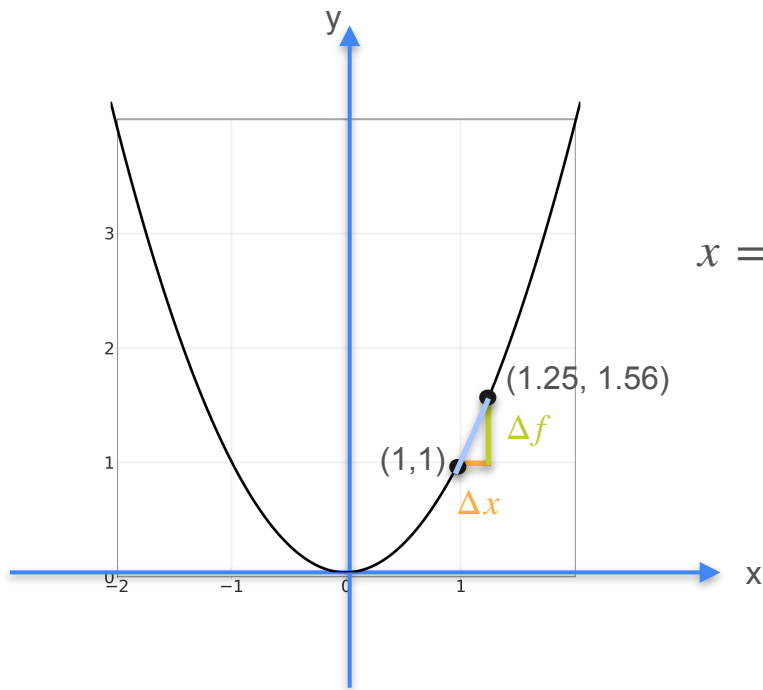
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

$x = 1$

Δx	1.0	1/2	1/4
Δf	3	1.25	0.562
Slope	3	2.5	

$$(1 + 0.25)^2 - 1^2 = 1.56 - 1$$

Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

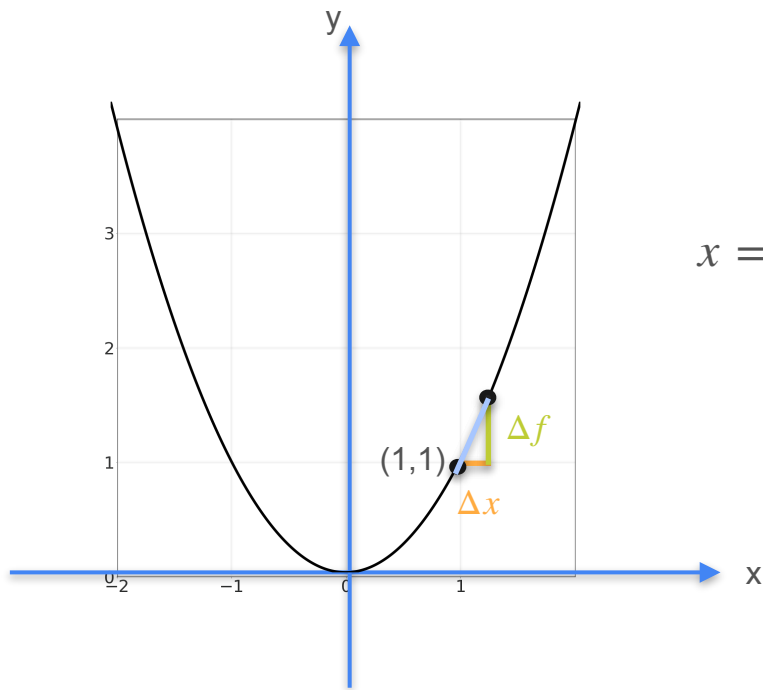
$x = 1$

Δx	1.0	1/2	1/4
Δf	3	1.25	0.562
Slope	3	2.5	2.25

$$(1 + 0.25)^2 - 1^2 = 1.56 - 1$$

$$\frac{0.56}{0.25}$$

Derivative of Quadratic Functions

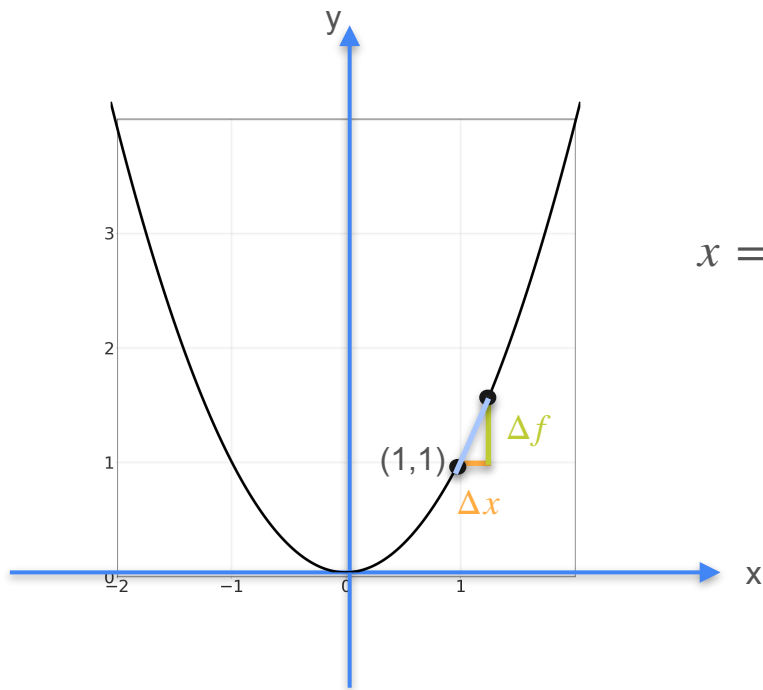


Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

Δx	1.0	1/2	1/4
Δf	3	1.25	0.562
Slope	3	2.5	2.25

Derivative of Quadratic Functions



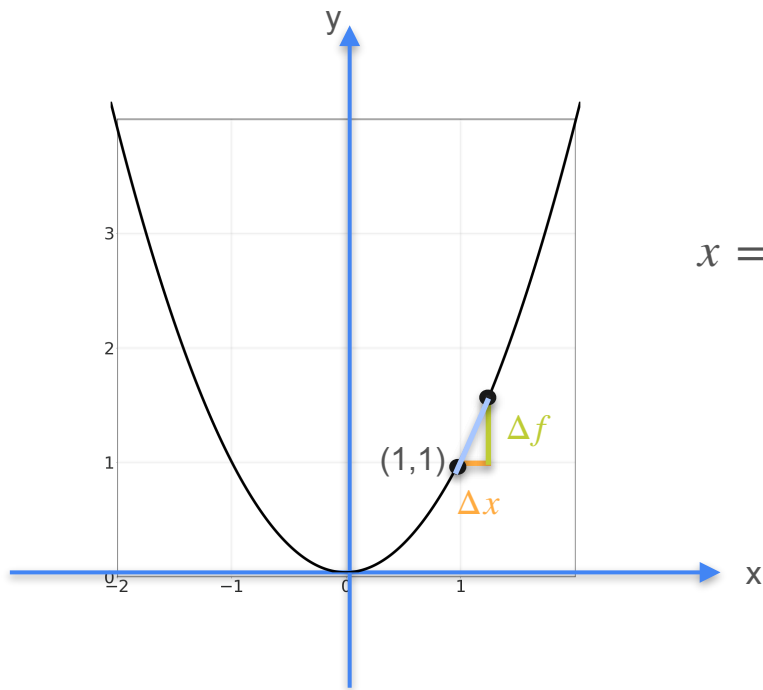
Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

$x = 1$

Δx	1.0	1/2	1/4	1/8	1/16
Δf	3	1.25	0.562	0.265	
Slope	3	2.5	2.25	2.125	

Derivative of Quadratic Functions



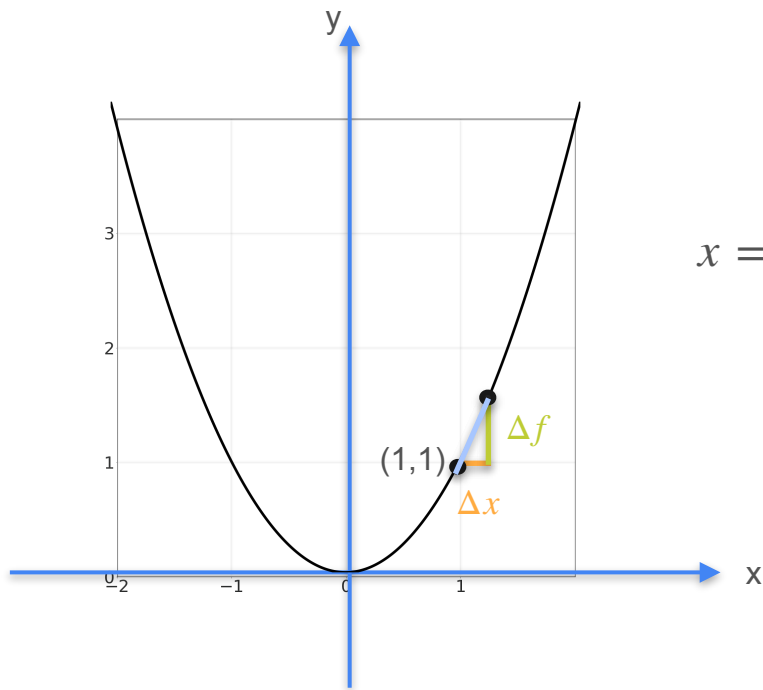
$x = 1$

Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8	1/16
Δf	3	1.25	0.562	0.265	0.128
Slope	3	2.5	2.25	2.125	

Derivative of Quadratic Functions

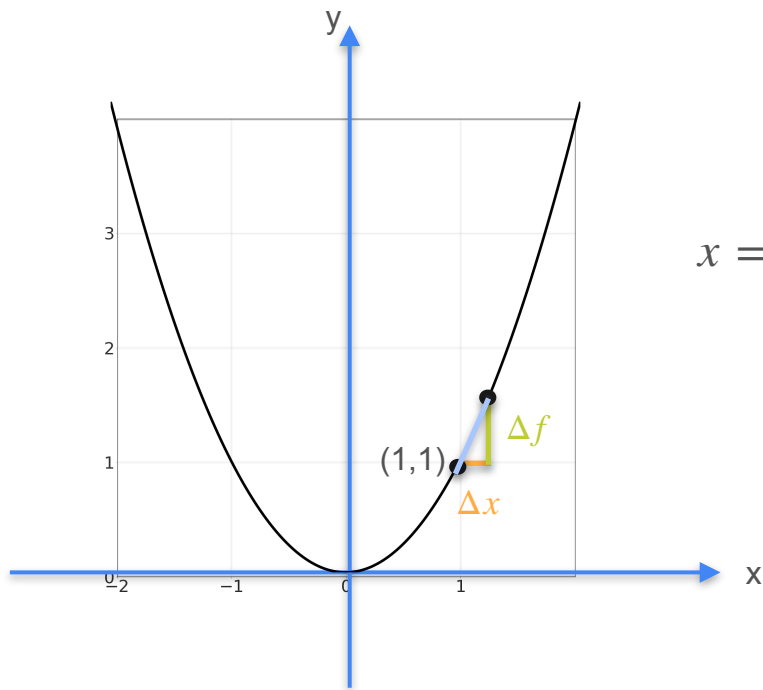


Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8	1/16
Δf	3	1.25	0.562	0.265	0.128
Slope	3	2.5	2.25	2.125	2.065

Derivative of Quadratic Functions



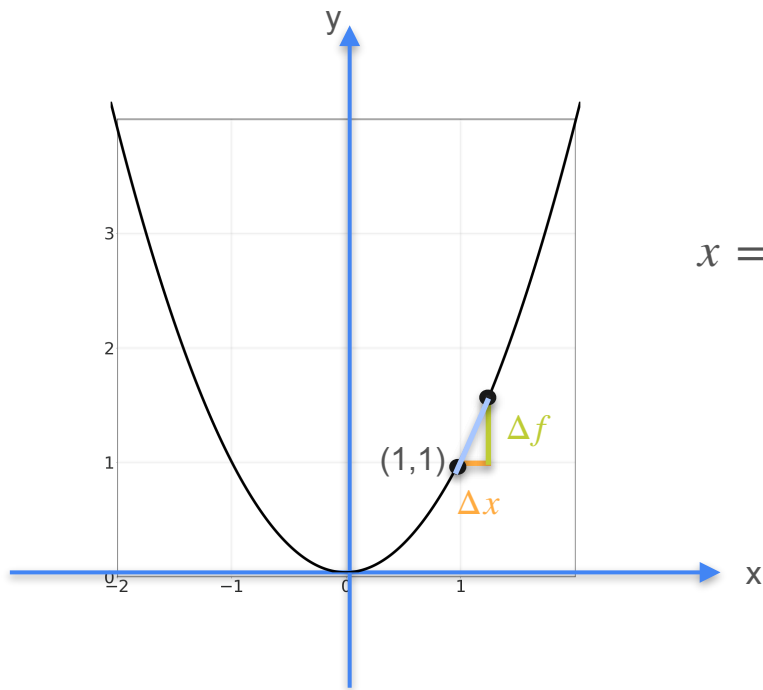
Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

$x = 1$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	3	1.25	0.562	0.265	0.128	
Slope	3	2.5	2.25	2.125	2.065	

Derivative of Quadratic Functions



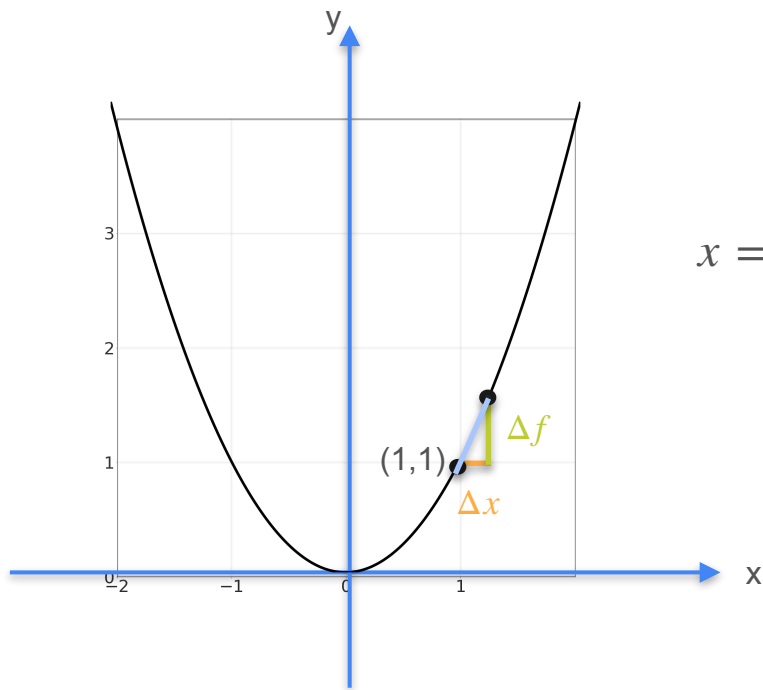
$x = 1$

Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	3	1.25	0.562	0.265	0.128	0.002
Slope	3	2.5	2.25	2.125	2.065	

Derivative of Quadratic Functions



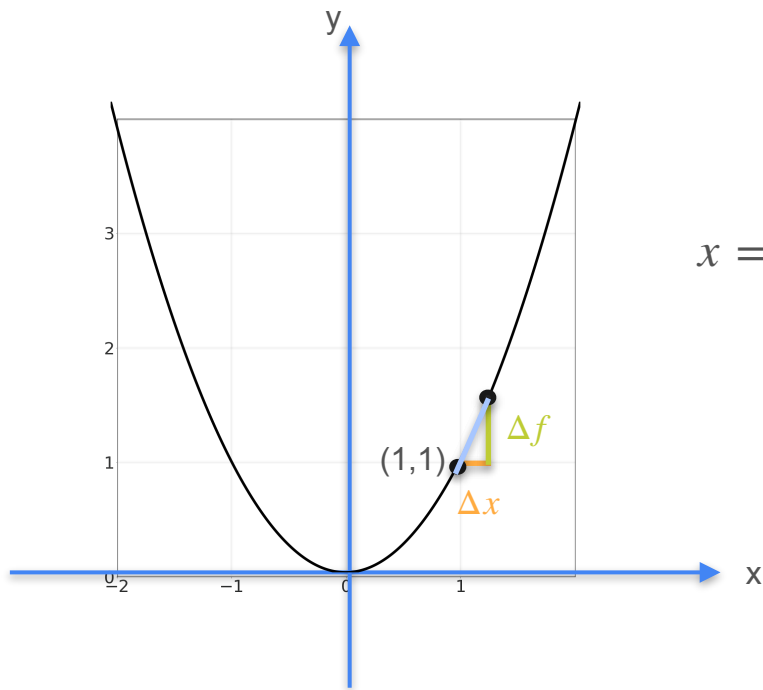
$x = 1$

Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	3	1.25	0.562	0.265	0.128	0.002
Slope	3	2.5	2.25	2.125	2.065	2.001

Derivative of Quadratic Functions



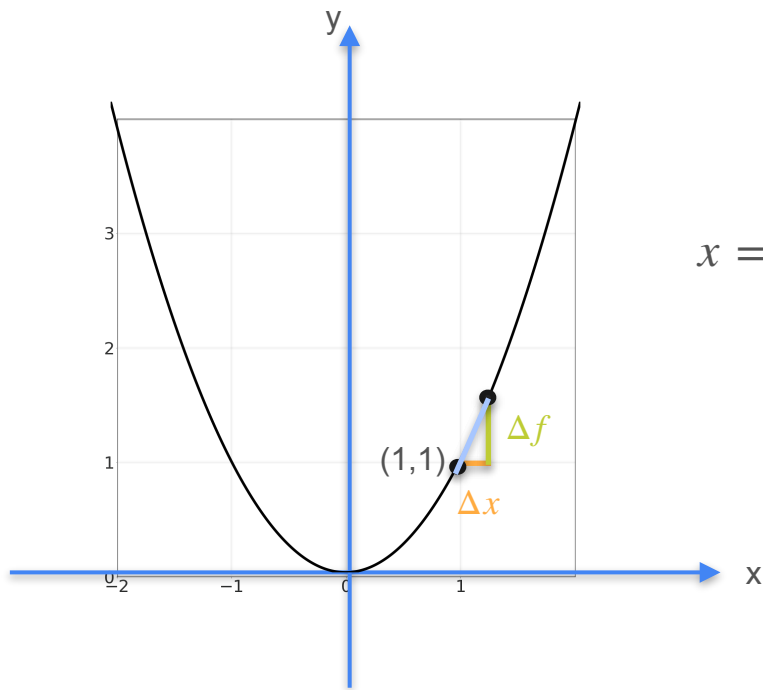
Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	3	1.25	0.562	0.265	0.128	0.002
Slope	3	2.5	2.25	2.125	2.065	2.001

$$f'(1) = \frac{d}{dx} f(1) = 2$$

Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

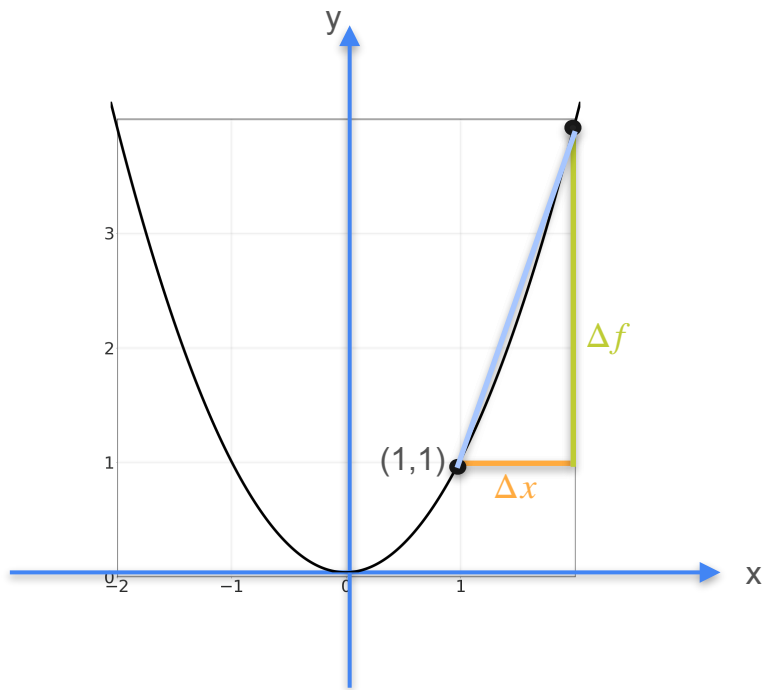
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

$x = 1$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	3	1.25	0.562	0.265	0.128	0.002
Slope	3	2.5	2.25	2.125	2.065	2.001

$$f'(1) = \frac{d}{dx} f(1) = 2 = 2 \times 1$$

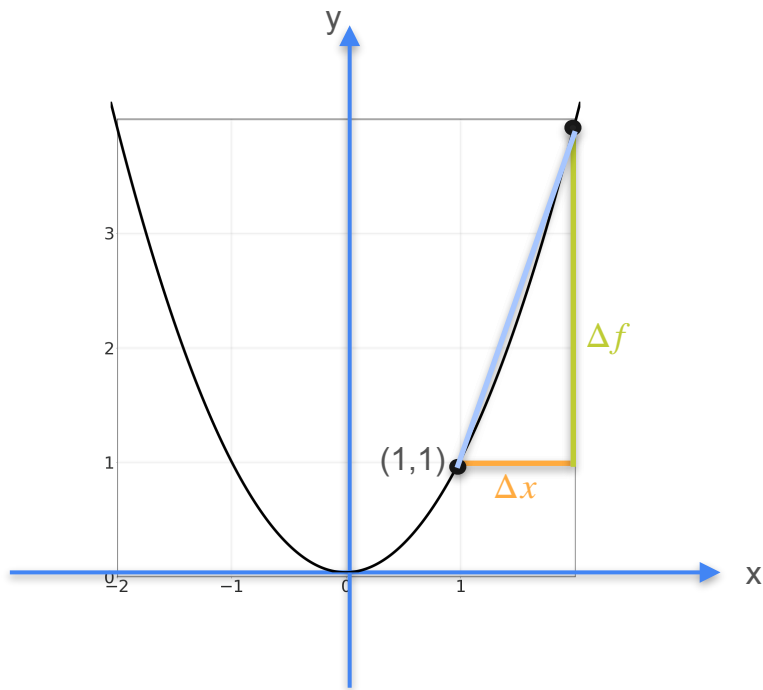
Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Derivative of Quadratic Functions

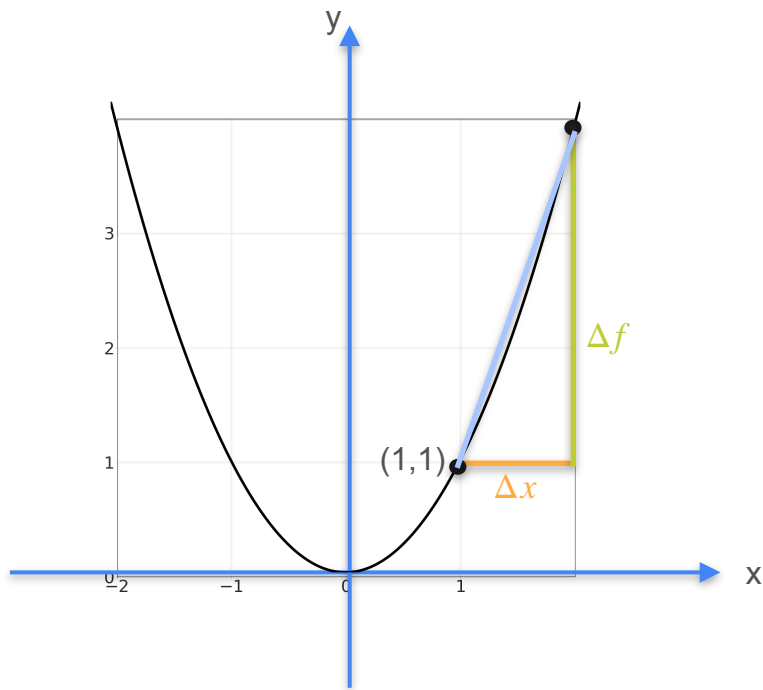


Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\frac{\Delta f}{\Delta x}$$

Derivative of Quadratic Functions

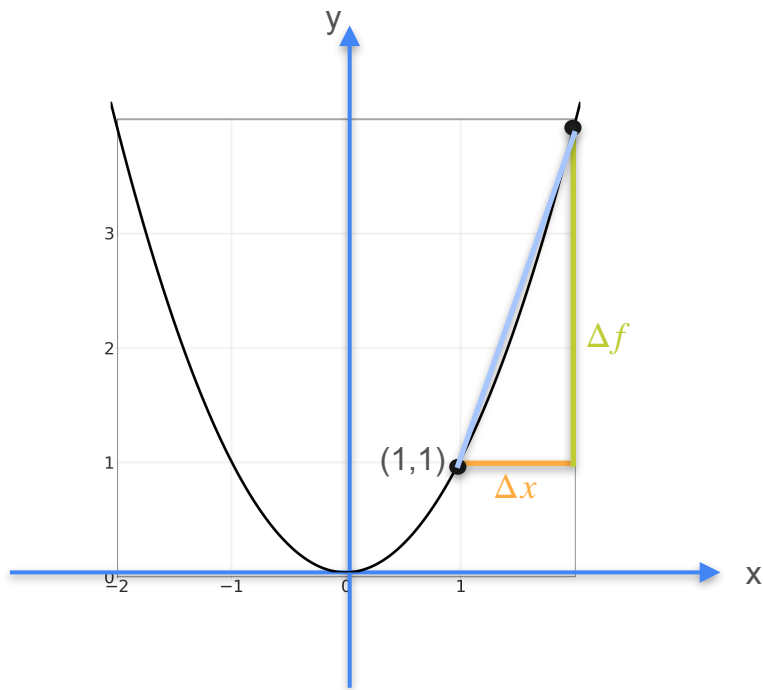


Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - x^2}{\Delta x}$$

Derivative of Quadratic Functions



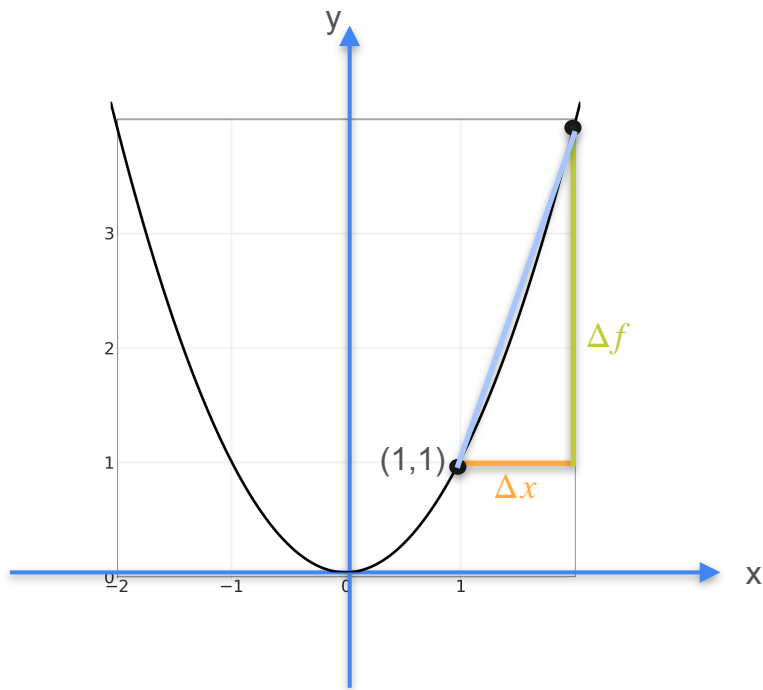
Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - x^2}{\Delta x}$$

$$= \frac{x^2 + 2x\Delta x + (\Delta x)^2 - x^2}{\Delta x}$$

Derivative of Quadratic Functions



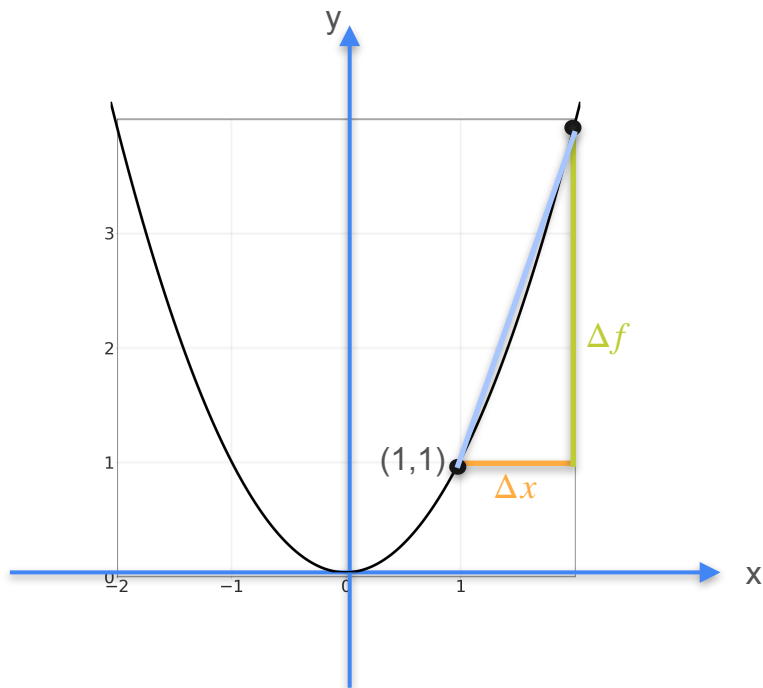
Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - x^2}{\Delta x}$$

$$= \frac{\cancel{x^2} + 2x\Delta x + (\Delta x)^2 - \cancel{x^2}}{\Delta x}$$

Derivative of Quadratic Functions



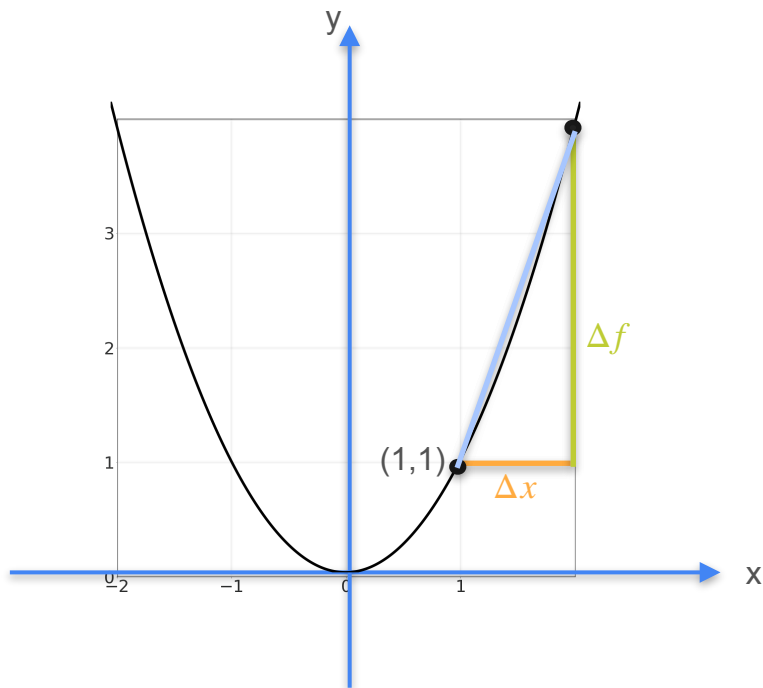
Quadratics: $y = f(x) = x^2$

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$$\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - x^2}{\Delta x}$$

$$= \frac{\cancel{x^2} + 2x\cancel{\Delta x} + (\cancel{\Delta x})^2 - \cancel{x^2}}{\cancel{\Delta x}}$$

Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

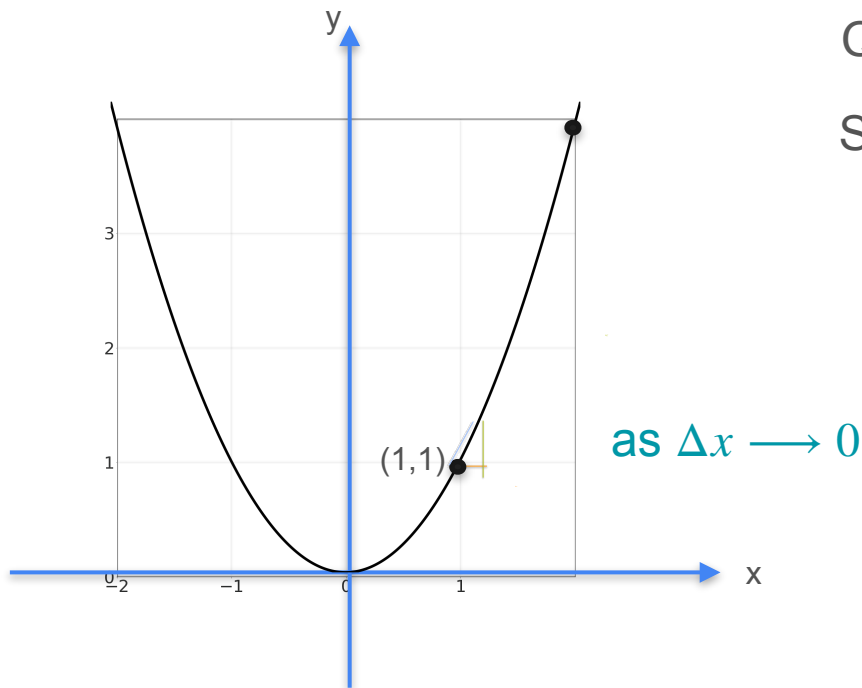
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - x^2}{\Delta x}$$

$$= \frac{\cancel{x^2} + 2x\cancel{\Delta x} + (\cancel{\Delta x})^2 - \cancel{x^2}}{\cancel{\Delta x}}$$

$$= 2x + \Delta x$$

Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

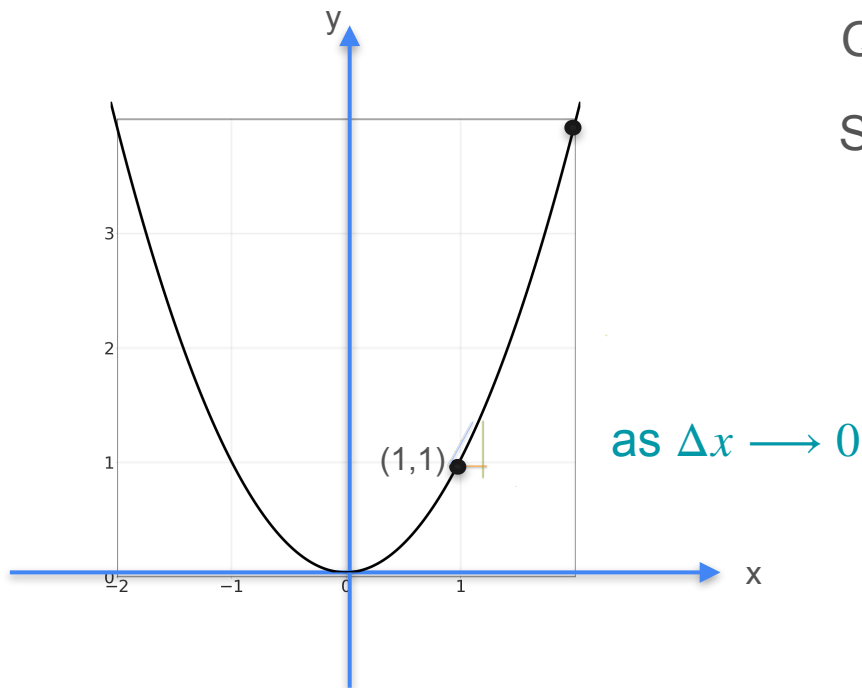
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - x^2}{\Delta x}$$

$$= \frac{\cancel{x^2} + 2x\cancel{\Delta x} + (\cancel{\Delta x})^2 - \cancel{x^2}}{\cancel{\Delta x}}$$

$$= 2x + \Delta x$$

Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

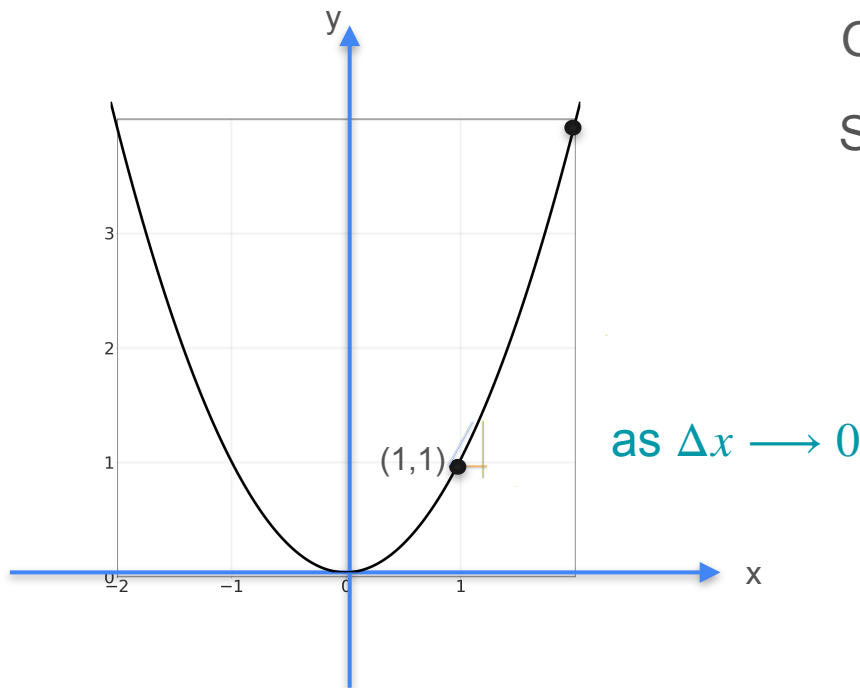
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^2 - x^2}{\Delta x}$$

$$= \frac{\cancel{x^2} + 2x\cancel{\Delta x} + (\cancel{\Delta x})^2 - \cancel{x^2}}{\cancel{\Delta x}}$$

$$= 2x + \boxed{\Delta x}^0$$

Derivative of Quadratic Functions

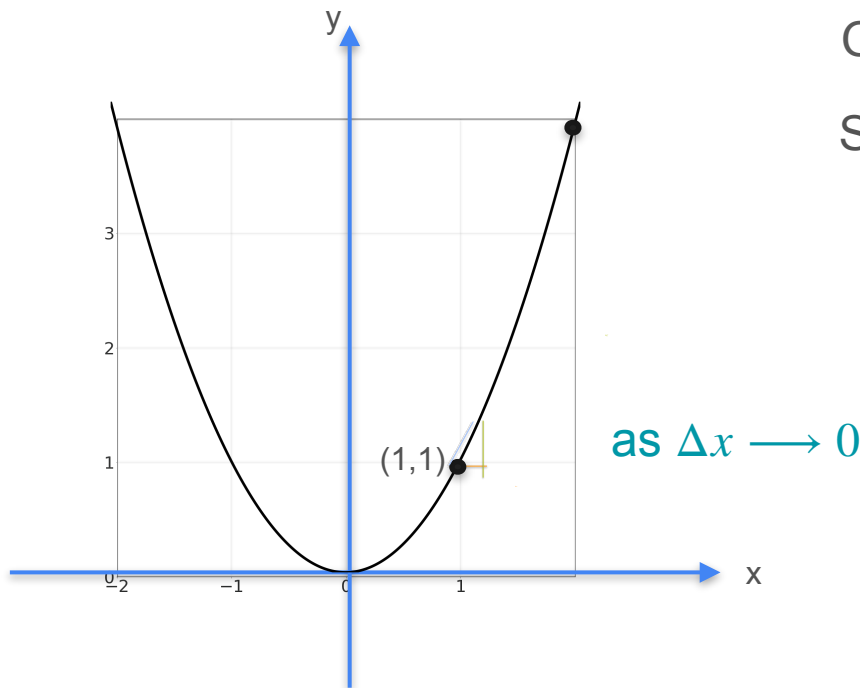


Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\begin{aligned} \boxed{\frac{\Delta f}{\Delta x}} &= \frac{(x + \Delta x)^2 - x^2}{\Delta x} \\ &= \frac{\cancel{x^2} + 2x\cancel{\Delta x} + (\cancel{\Delta x})^2 - \cancel{x^2}}{\cancel{\Delta x}} \\ &= 2x + \boxed{\Delta x}^0 \end{aligned}$$

Derivative of Quadratic Functions

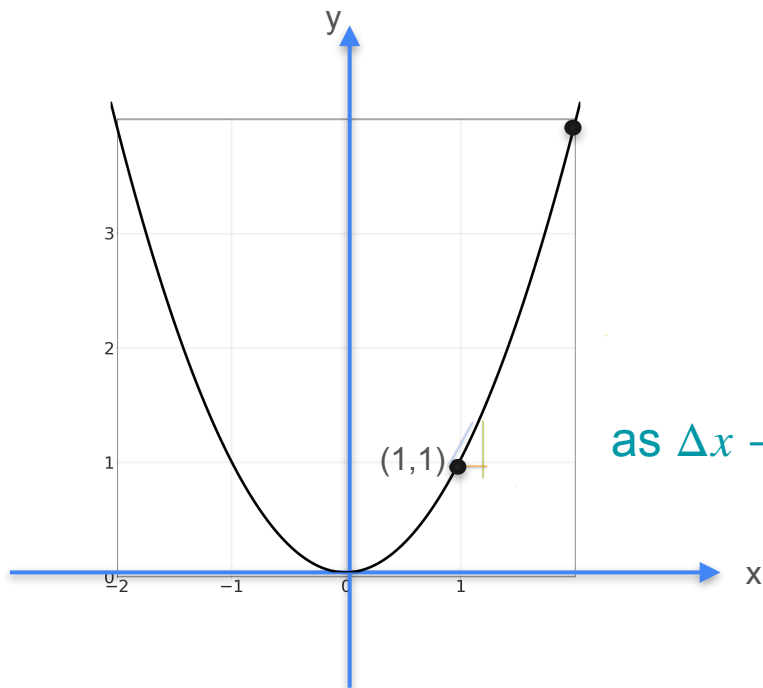


Quadratics: $y = f(x) = x^2$

Slope: $\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$$\begin{aligned} \frac{df}{dx} \quad \boxed{\frac{\Delta f}{\Delta x}} &= \frac{(x + \Delta x)^2 - x^2}{\Delta x} \\ &= \frac{\cancel{x^2} + 2x\cancel{\Delta x} + (\cancel{\Delta x})^2 - \cancel{x^2}}{\cancel{\Delta x}} \\ &= 2x + \boxed{\Delta x}^0 \end{aligned}$$

Derivative of Quadratic Functions



Quadratics: $y = f(x) = x^2$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\begin{aligned} \frac{df}{dx} \quad \boxed{\frac{\Delta f}{\Delta x}} &= \frac{(x + \Delta x)^2 - x^2}{\Delta x} \\ &= \frac{\cancel{x^2} + 2x\cancel{\Delta x} + (\cancel{\Delta x})^2 - \cancel{x^2}}{\cancel{\Delta x}} \\ &= 2x + \boxed{\Delta x}^0 \end{aligned}$$

$$f(x) = x^2 \Rightarrow f'(x) = 2x$$



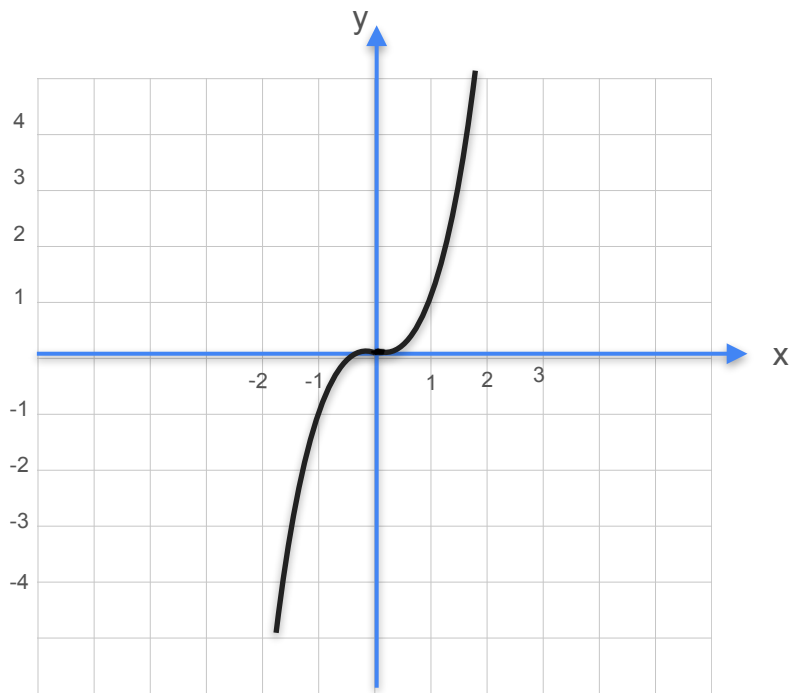
DeepLearning.AI

Derivatives and Optimization

**Some common derivatives:
Higher degree polynomials**

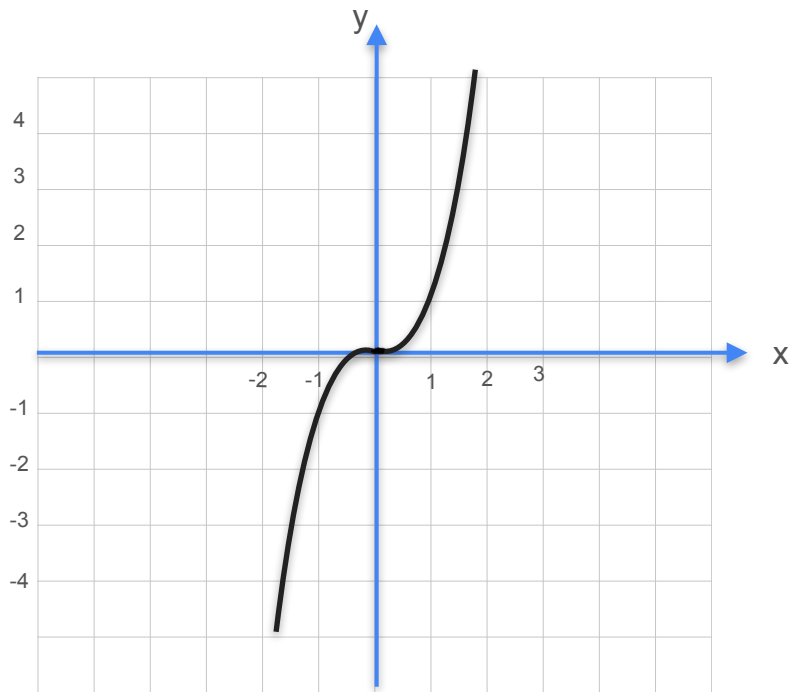
Derivative of Cubic Functions

Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

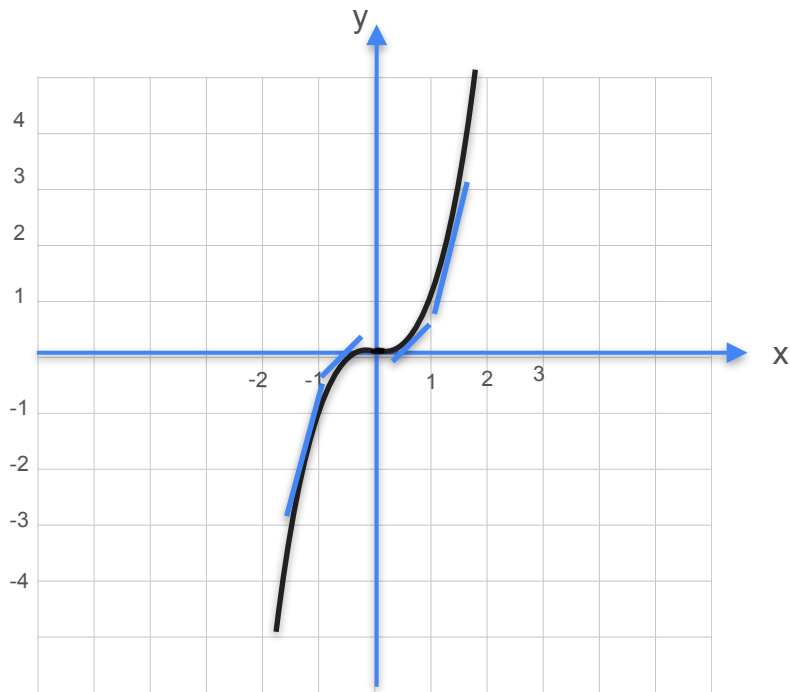
Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x}$

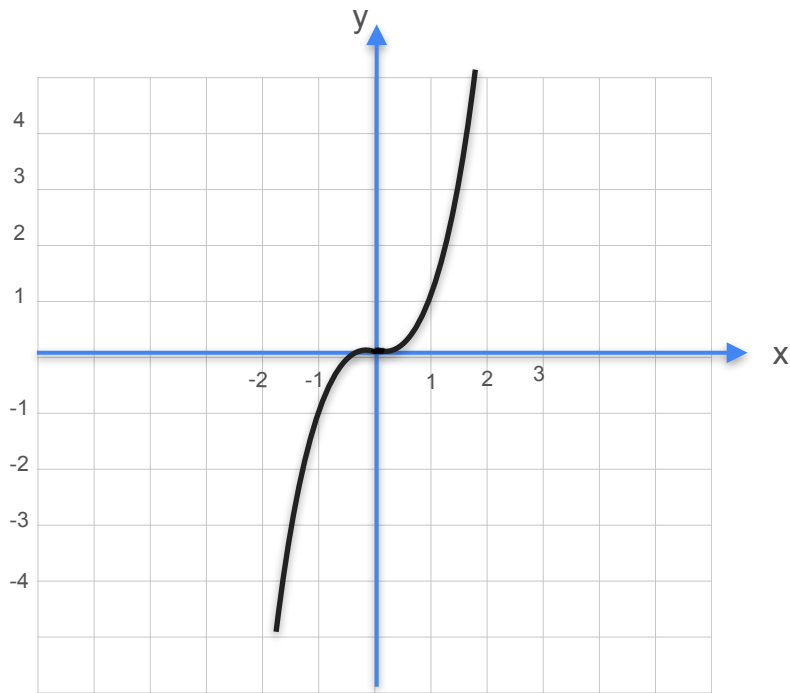
Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x}$

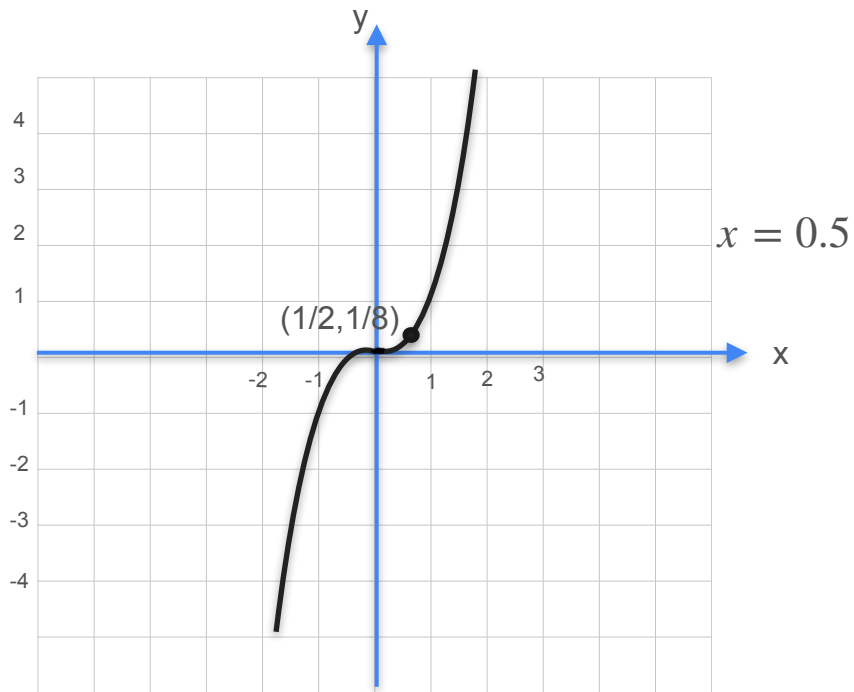
Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

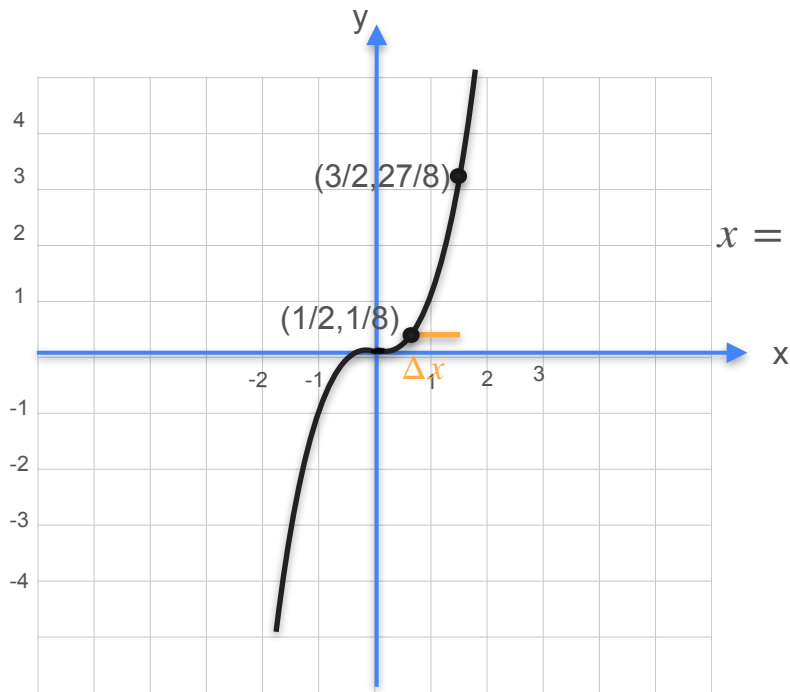
Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$

Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

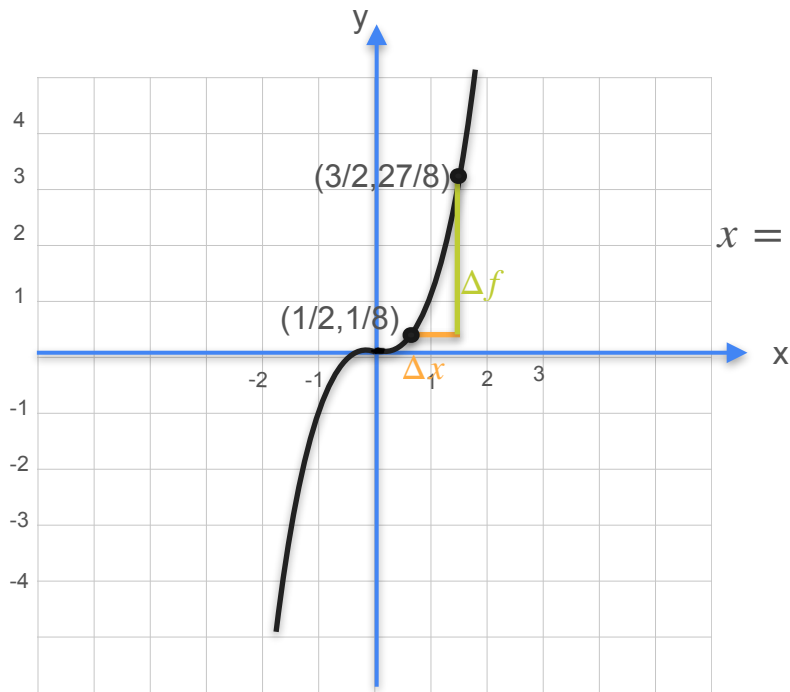
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

$x = 0.5$

Δx

1.0

Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

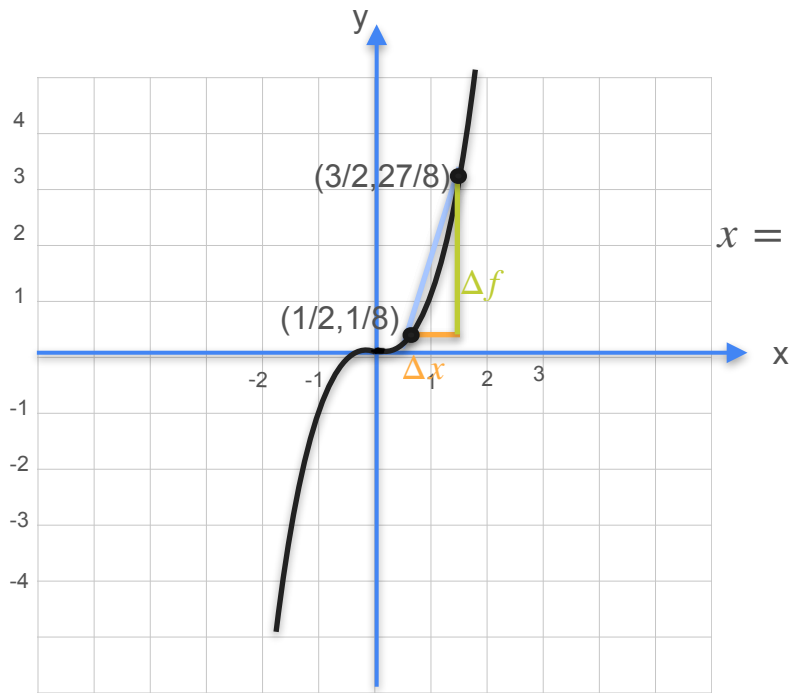
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

$x = 0.5$

Δx	1.0
Δf	3.25

$$\left(\frac{1}{2} + 1\right)^3 - \left(\frac{1}{2}\right)^3 = \frac{27}{8} - \frac{1}{8}$$

Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

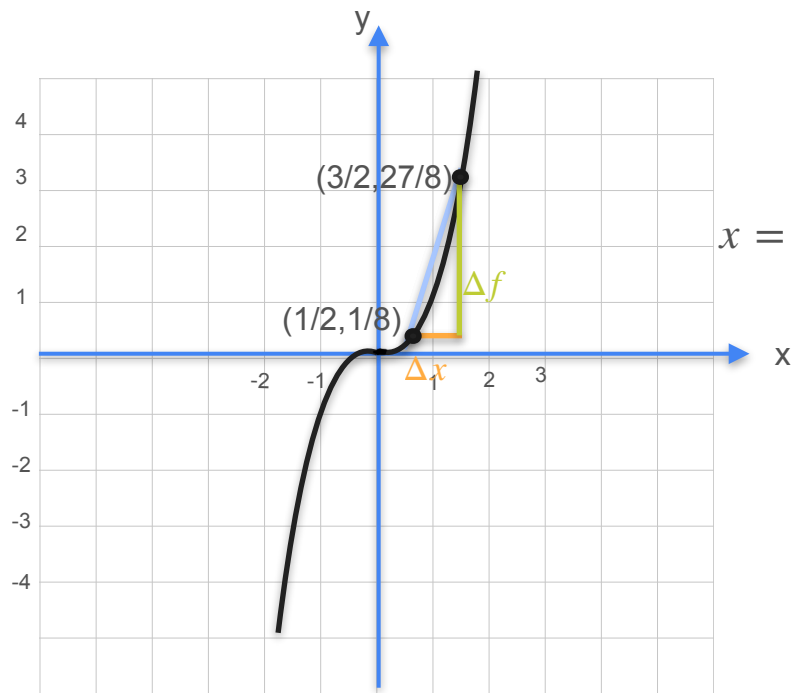
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

$x = 0.5$

Δx	1.0
Δf	3.25
Slope	

$$\left(\frac{1}{2} + 1\right)^3 - \left(\frac{1}{2}\right)^3 = \frac{27}{8} - \frac{1}{8}$$

Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

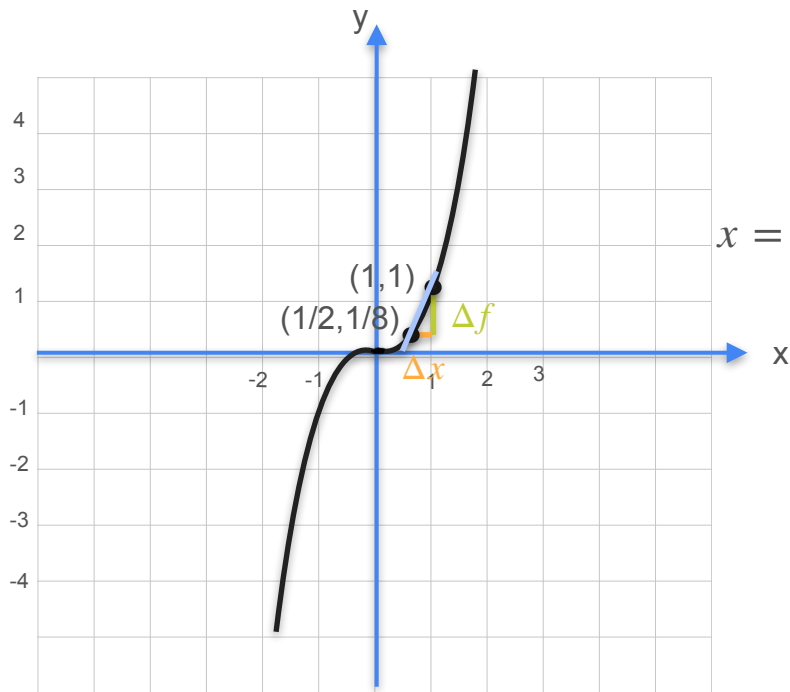
$x = 0.5$

Δx	1.0
Δf	3.25
Slope	3.25

$$\left(\frac{1}{2} + 1\right)^3 - \left(\frac{1}{2}\right)^3 = \frac{27}{8} - \frac{1}{8}$$

$$\frac{3.25}{1}$$

Derivative of Cubic Functions



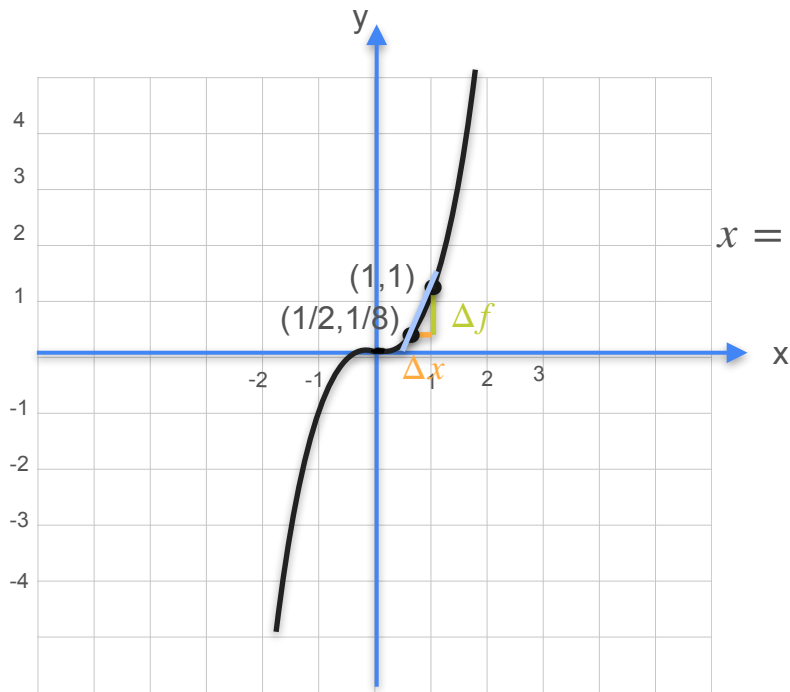
Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

$x = 0.5$

Δx	1.0
Δf	3.25
Slope	3.25

Derivative of Cubic Functions



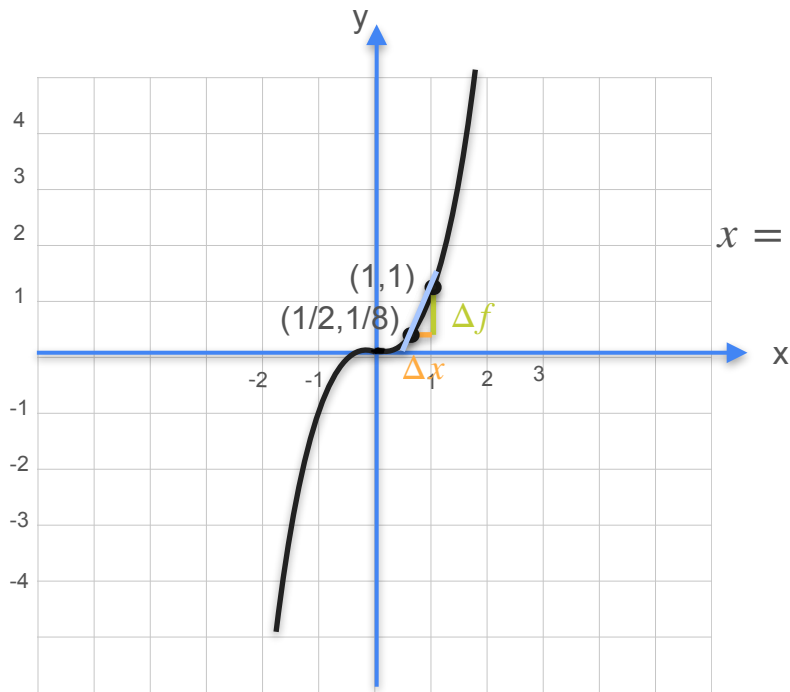
Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

$x = 0.5$

Δx	1.0	1/2
Δf	3.25	
Slope	3.25	

Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

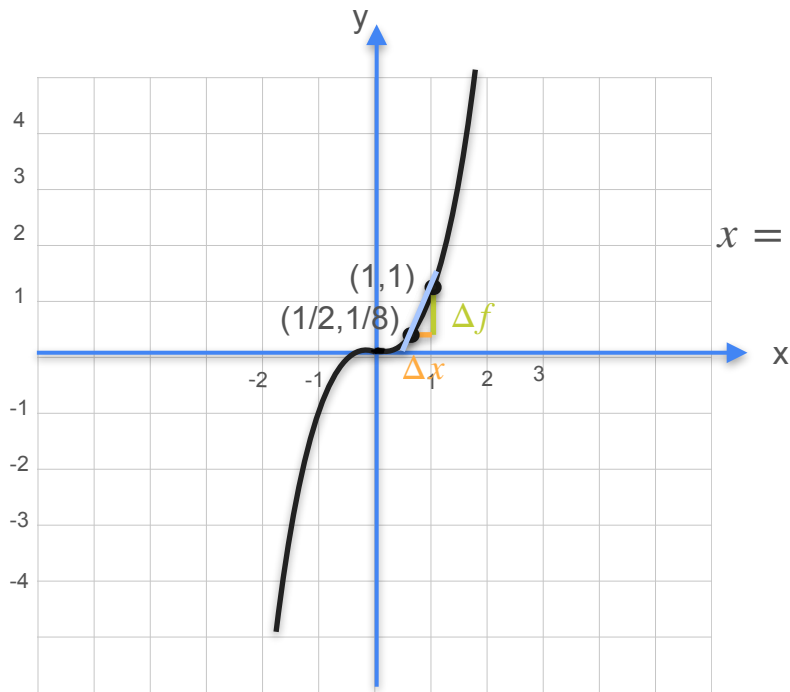
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

$x = 0.5$

Δx	1.0	1/2
Δf	3.25	0.86
Slope	3.25	

$$\left(\frac{1}{2} + \frac{1}{2}\right)^3 - \left(\frac{1}{2}\right)^3 = 1 - \frac{1}{8}$$

Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$

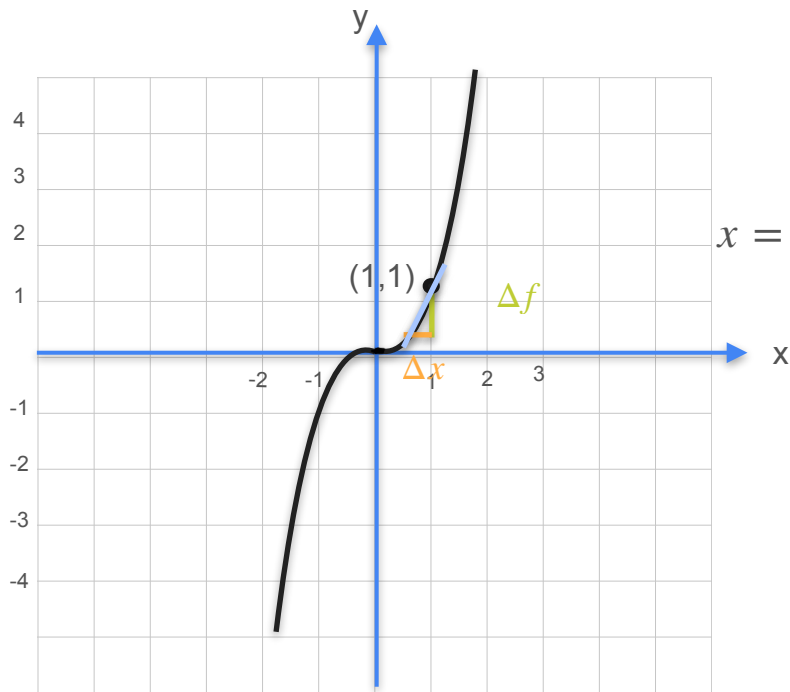
$x = 0.5$

Δx	1.0	1/2
Δf	3.25	0.86
Slope	3.25	1.75

$$\left(\frac{1}{2} + \frac{1}{2}\right)^3 - \left(\frac{1}{2}\right)^3 = 1 - \frac{1}{8}$$

$$\frac{0.86}{0.5}$$

Derivative of Cubic Functions

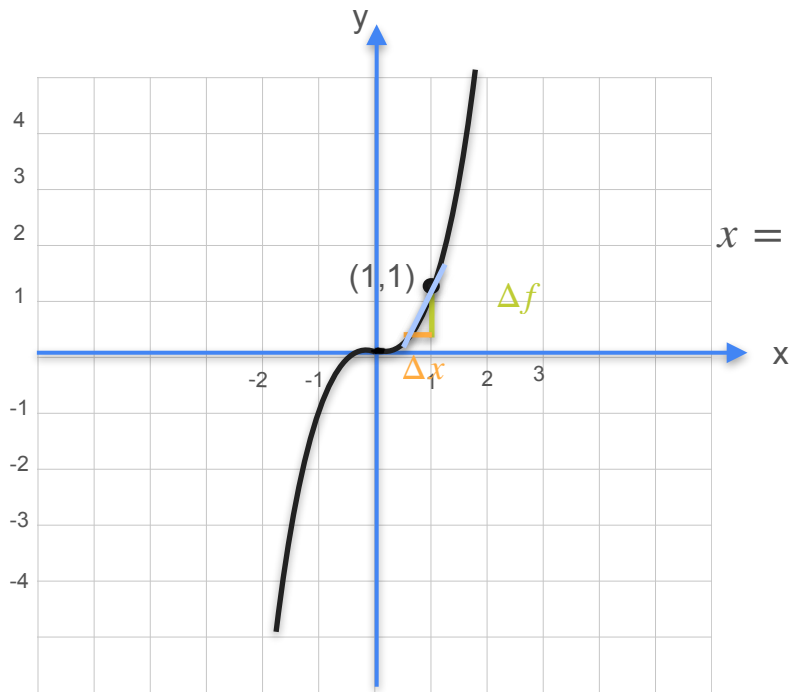


Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

Δx	1.0	1/2
Δf	3.25	0.86
Slope	3.25	1.75

Derivative of Cubic Functions



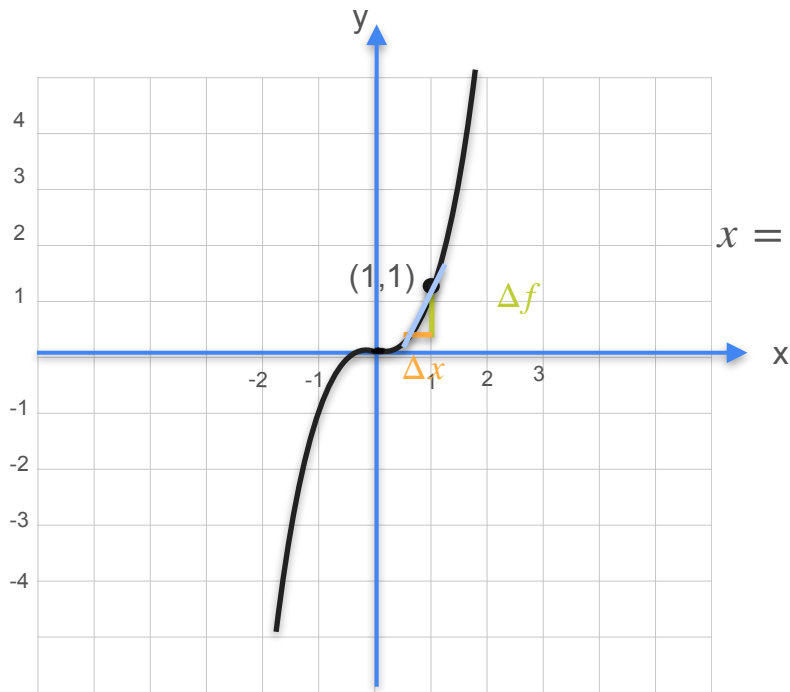
Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

$x = 0.5$

Δx	1.0	1/2	1/4
Δf	3.25	0.86	
Slope	3.25	1.75	

Derivative of Cubic Functions



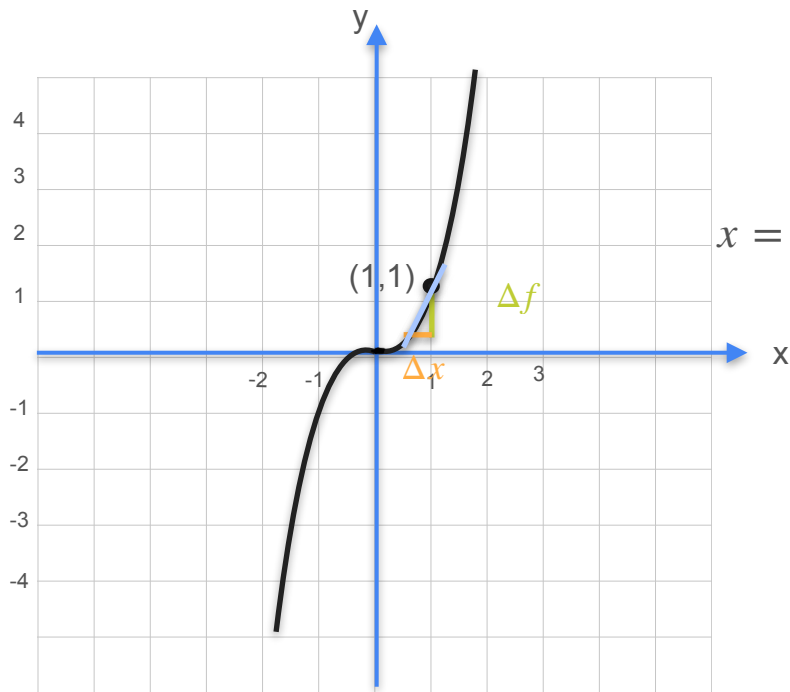
Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

$x = 0.5$

Δx	1.0	1/2	1/4
Δf	3.25	0.86	0.30
Slope	3.25	1.75	

Derivative of Cubic Functions



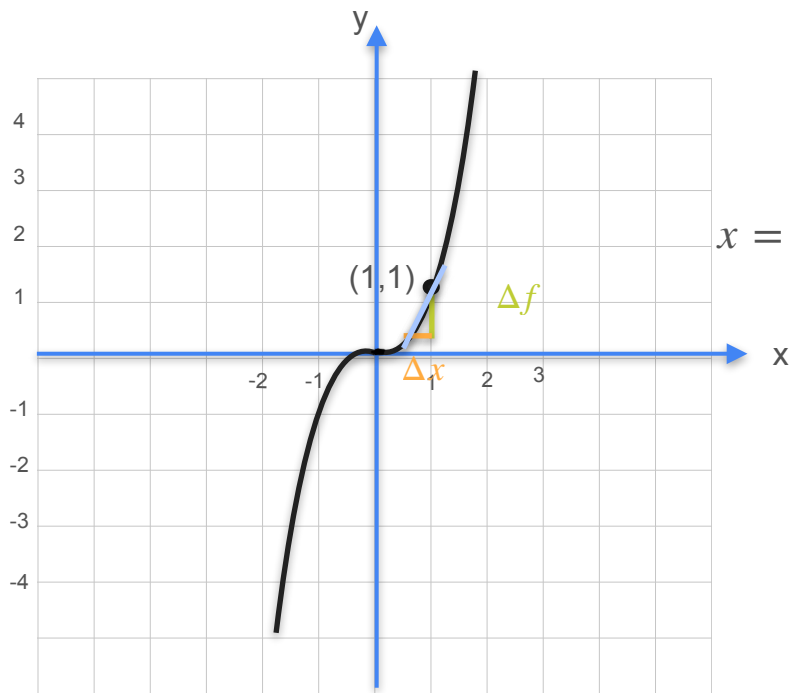
Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

$x = 0.5$

Δx	1.0	1/2	1/4
Δf	3.25	0.86	0.30
Slope	3.25	1.75	1.188

Derivative of Cubic Functions



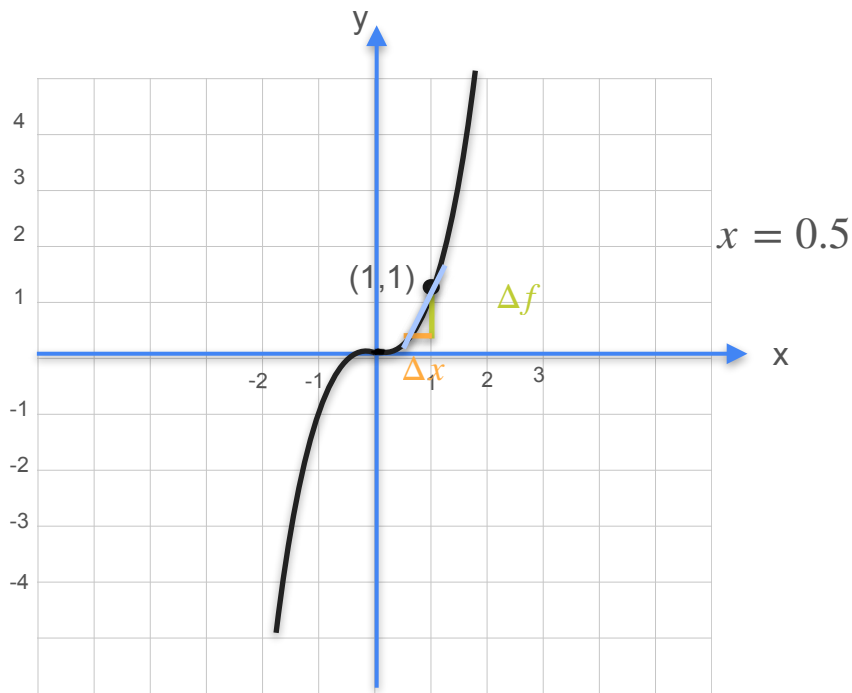
Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

$x = 0.5$

Δx	1.0	1/2	1/4	1/8
Δf	3.25	0.86	0.30	
Slope	3.25	1.75	1.188	

Derivative of Cubic Functions

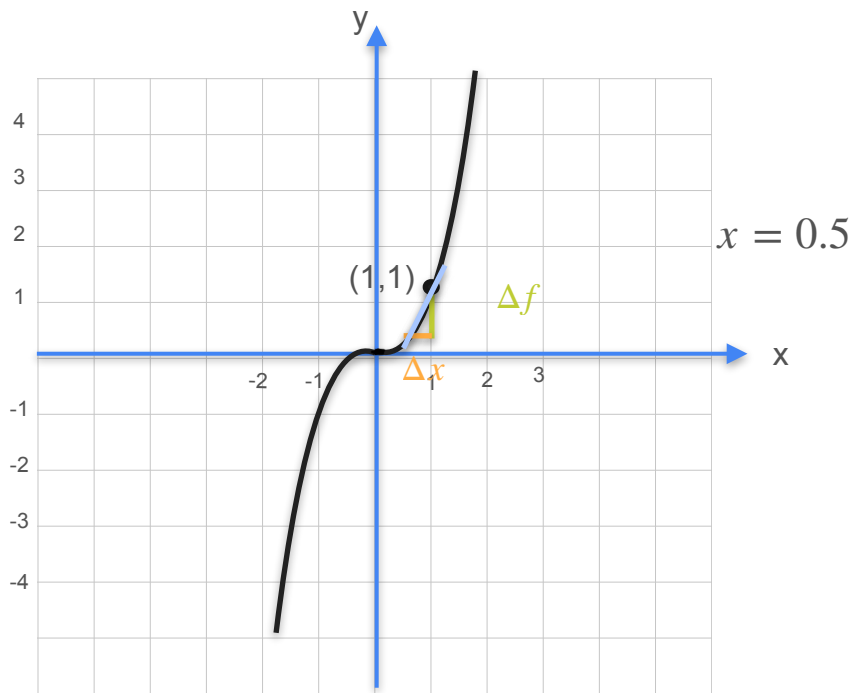


Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8
Δf	3.25	0.86	0.30	0.12
Slope	3.25	1.75	1.188	

Derivative of Cubic Functions

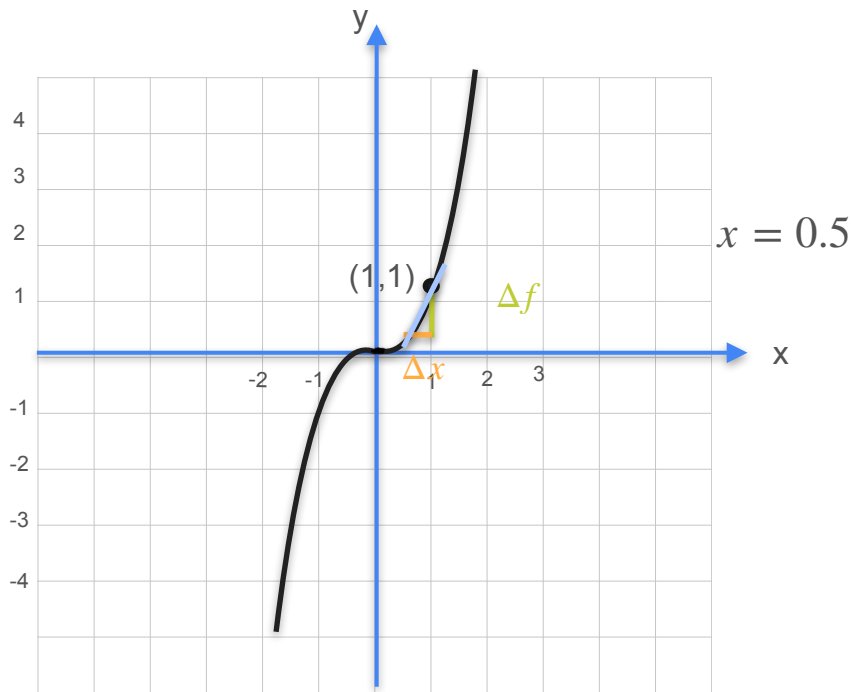


Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$

Δx	1.0	1/2	1/4	1/8
Δf	3.25	0.86	0.30	0.12
Slope	3.25	1.75	1.188	0.95

Derivative of Cubic Functions

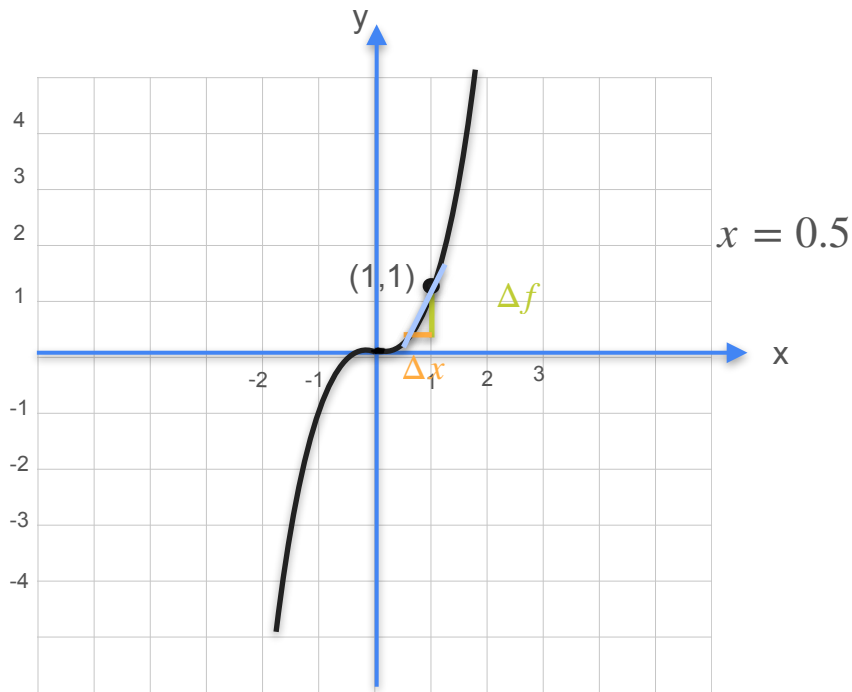


Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$

Δx	1.0	1/2	1/4	1/8	1/16
Δf	3.25	0.86	0.30	0.12	
Slope	3.25	1.75	1.188	0.95	

Derivative of Cubic Functions

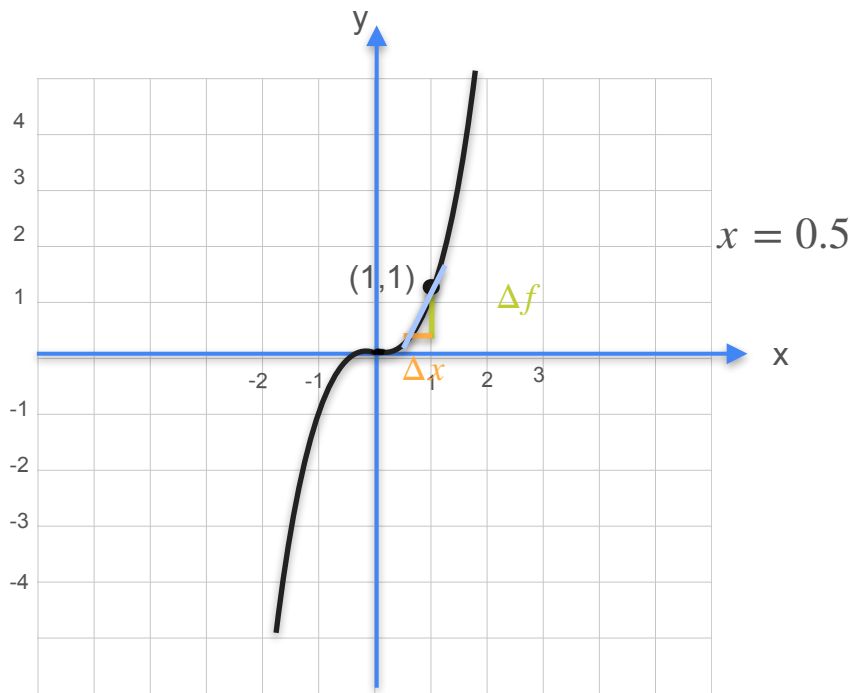


Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$

Δx	1.0	1/2	1/4	1/8	1/16
Δf	3.25	0.86	0.30	0.12	0.05
Slope	3.25	1.75	1.188	0.95	

Derivative of Cubic Functions

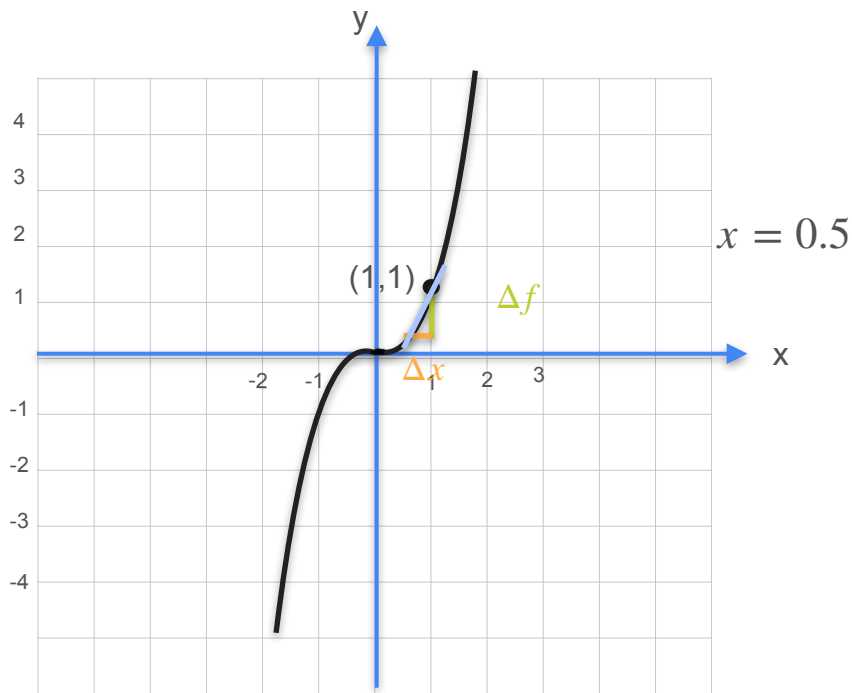


Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8	1/16
Δf	3.25	0.86	0.30	0.12	0.05
Slope	3.25	1.75	1.188	0.95	0.85

Derivative of Cubic Functions

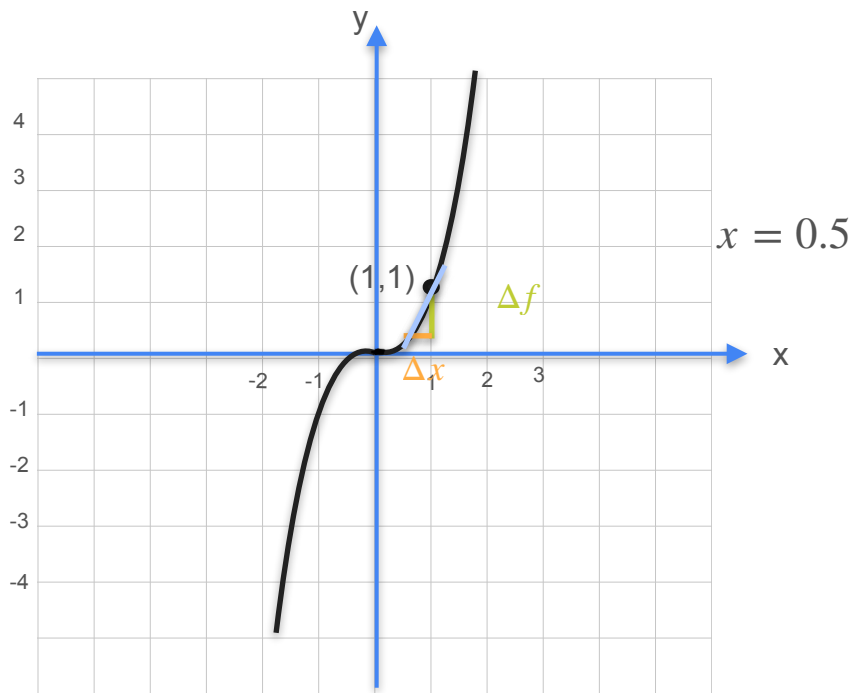


Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	3.25	0.86	0.30	0.12	0.05	
Slope	3.25	1.75	1.188	0.95	0.85	

Derivative of Cubic Functions

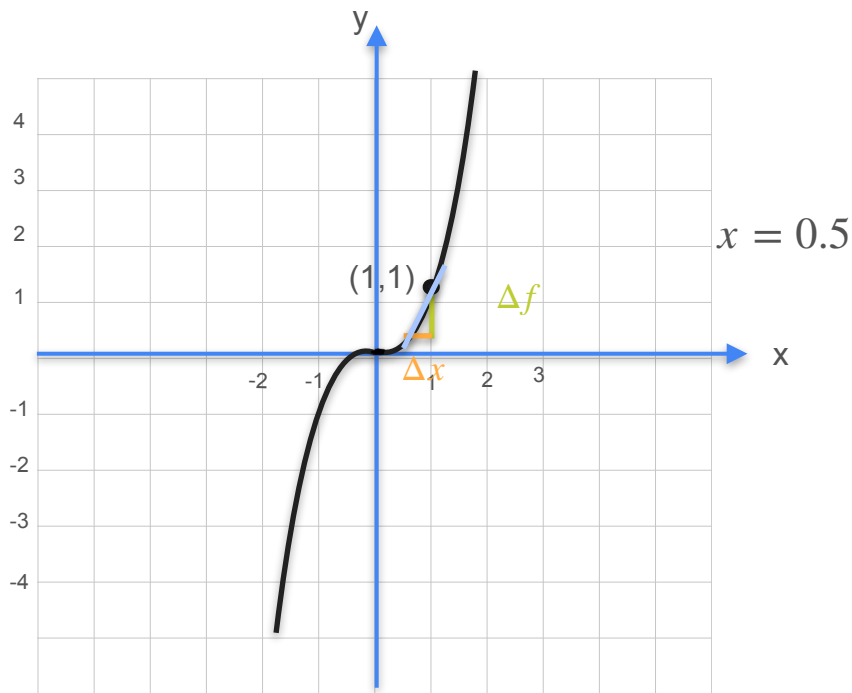


Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	3.25	0.86	0.30	0.12	0.05	0.0008
Slope	3.25	1.75	1.188	0.95	0.85	

Derivative of Cubic Functions

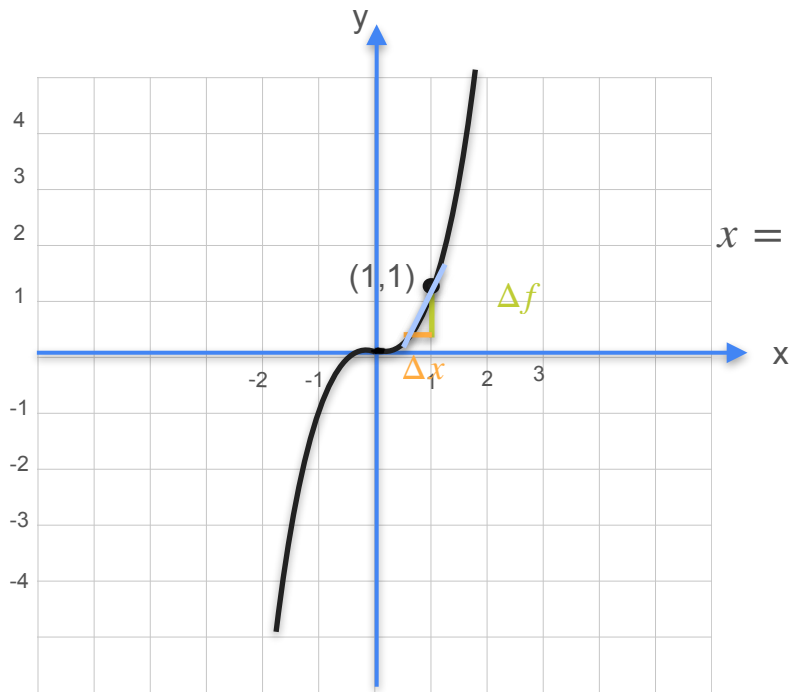


Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	3.25	0.86	0.30	0.12	0.05	0.0008
Slope	3.25	1.75	1.188	0.95	0.85	0.752

Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

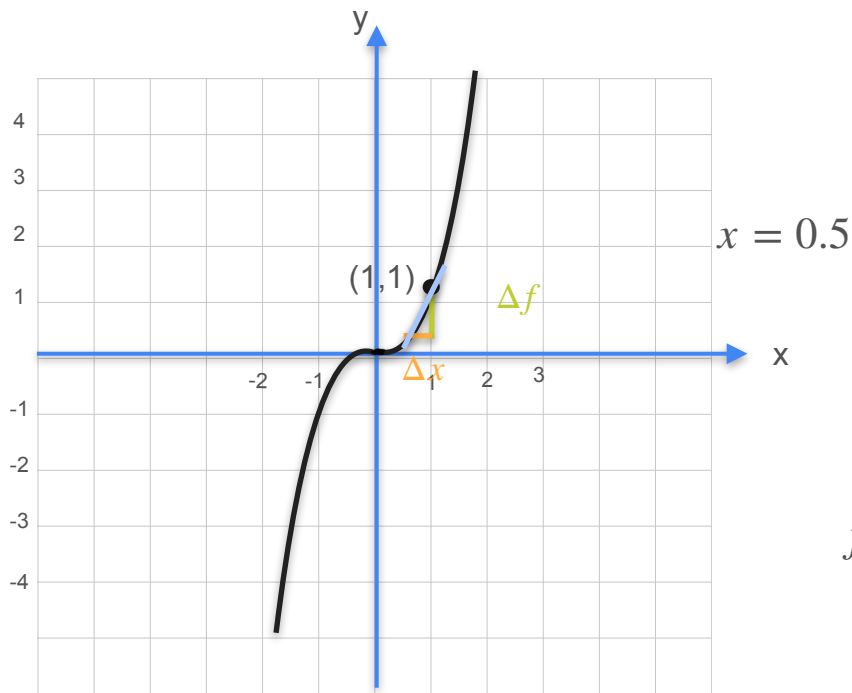
Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$

$x = 0.5$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	3.25	0.86	0.30	0.12	0.05	0.0008
Slope	3.25	1.75	1.188	0.95	0.85	0.752

$$f'(0.5) = \frac{d}{dx} f(0.5) = 0.75 = 3 \times 0.5^2$$

Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

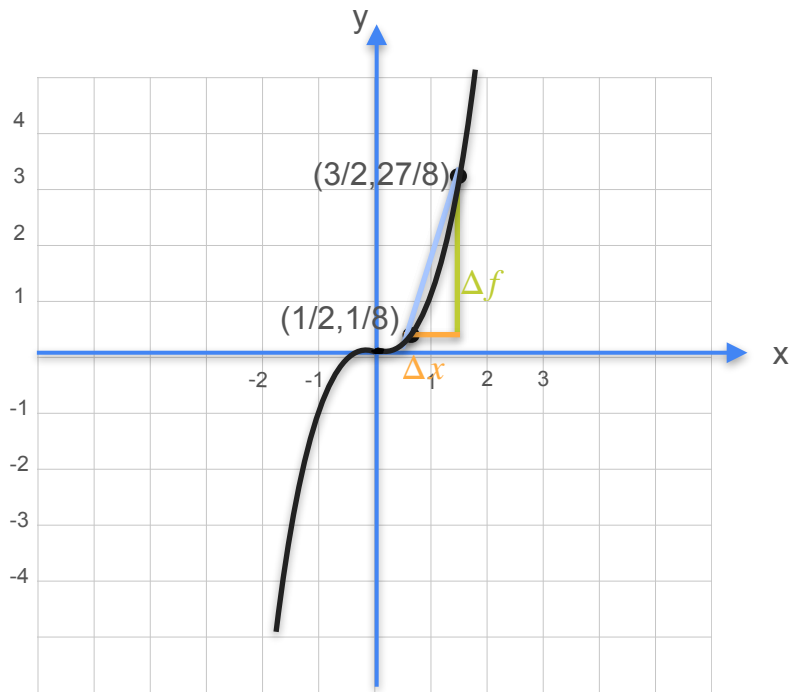
Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - (x)^3}{\Delta x}$

$x = 0.5$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	3.25	0.86	0.30	0.12	0.05	0.0008
Slope	3.25	1.75	1.188	0.95	0.85	0.752

$$f'(0.5) = \frac{d}{dx} f(0.5) = 0.75 = 3 \times 0.5^2 = 3 \times 0.5^2$$

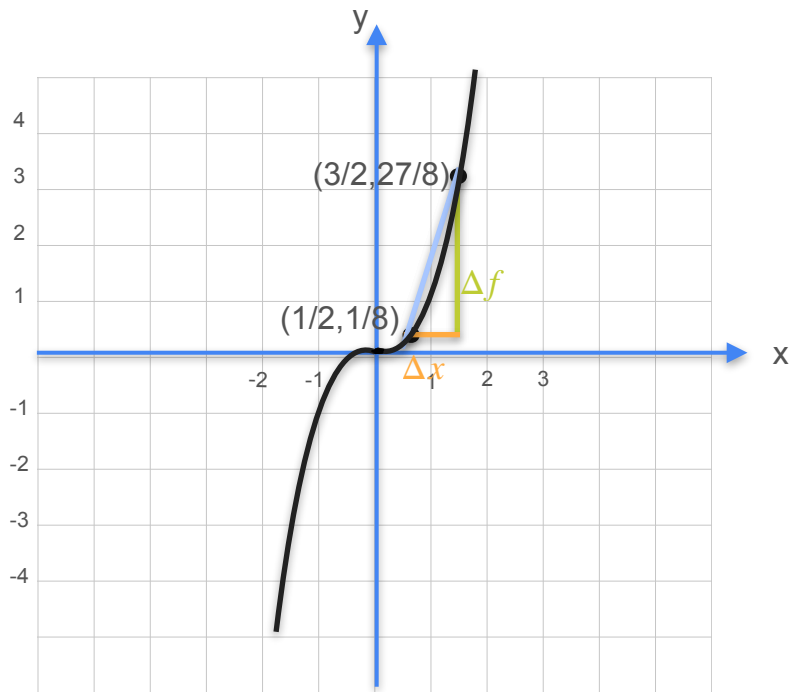
Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$

Derivative of Cubic Functions

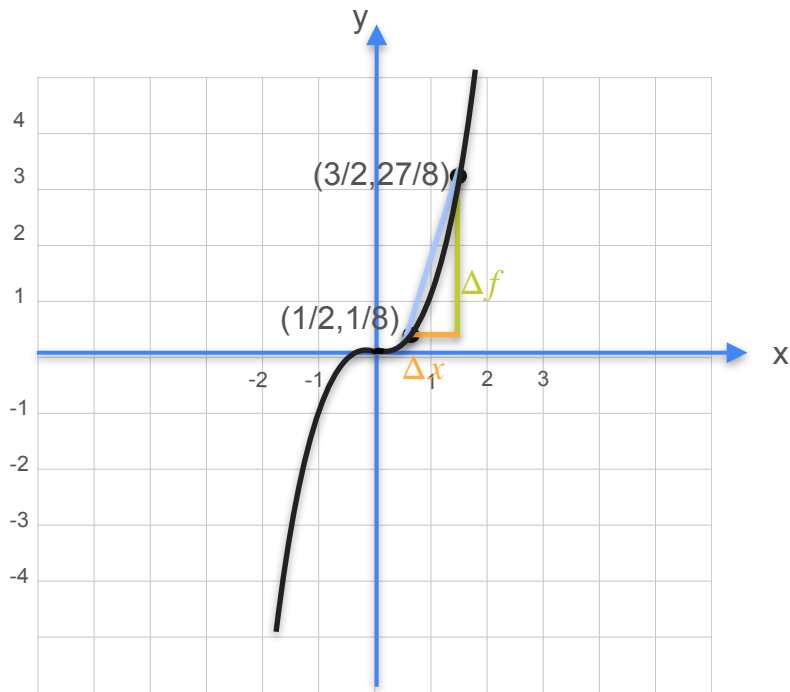


Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$$\frac{\Delta f}{\Delta x}$$

Derivative of Cubic Functions

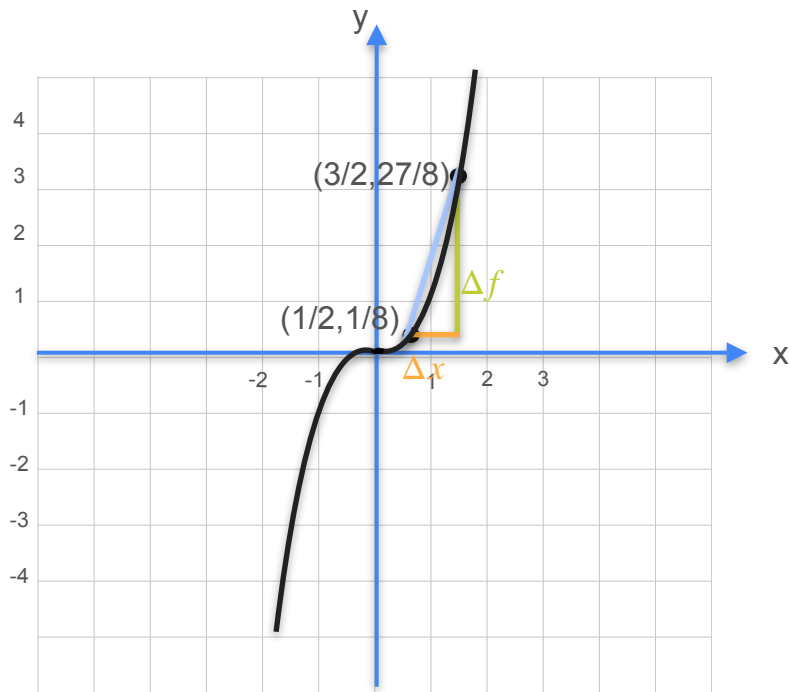


Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - x^3}{\Delta x}$$

Derivative of Cubic Functions

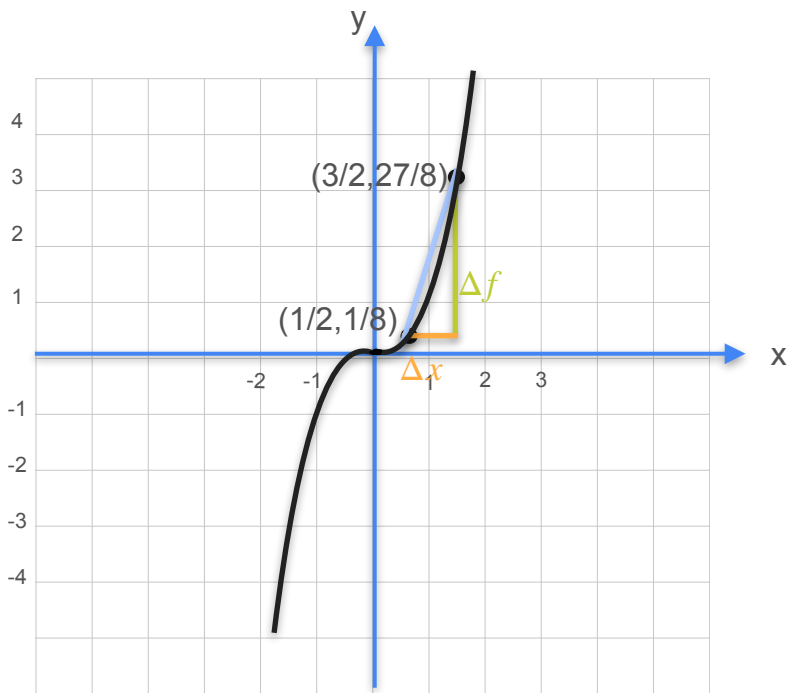


Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - x^3}{\Delta x}$$

Derivative of Cubic Functions

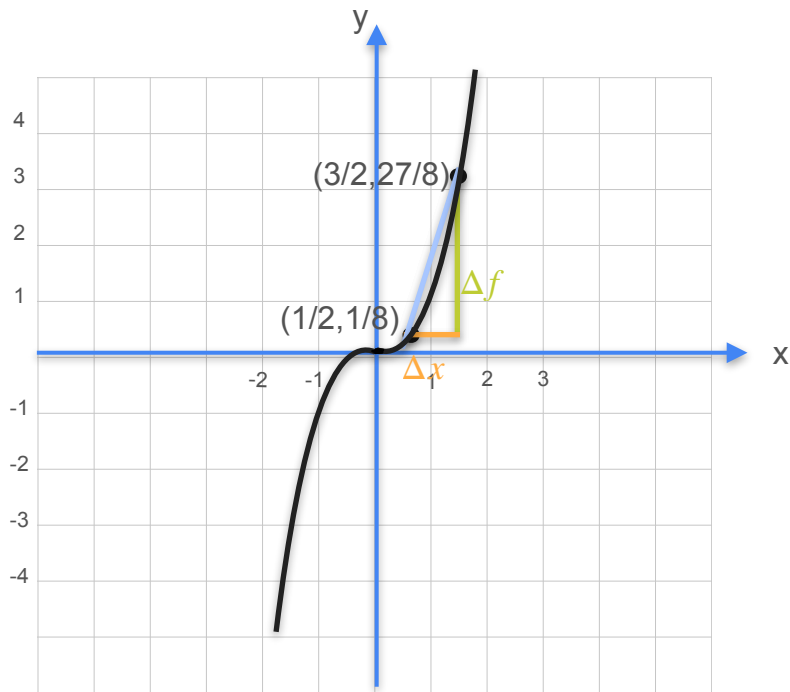


Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$$\begin{aligned}\frac{\Delta f}{\Delta x} &= \frac{(x + \Delta x)^3 - x^3}{\Delta x} \\ &= \frac{x^3 + 3x(\Delta x)^2 + 3x^2\Delta x + (\Delta x)^3 - x^3}{\Delta x}\end{aligned}$$

Derivative of Cubic Functions



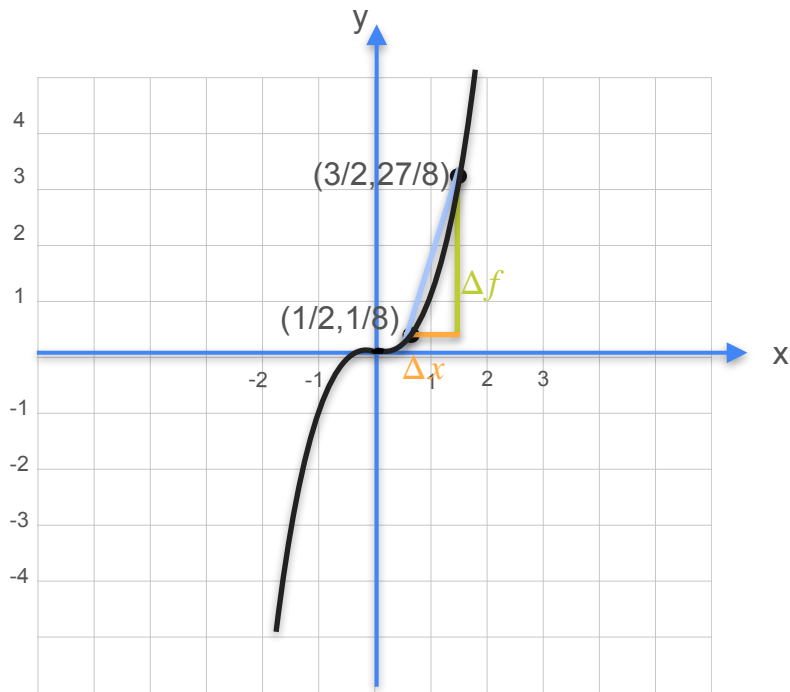
Cubic: $y = f(x) = x^3$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - x^3}{\Delta x}$$

$$= \frac{\cancel{x^3} + 3x(\Delta x)^2 + 3x^2\Delta x + (\Delta x)^3 - \cancel{x^3}}{\Delta x}$$

Derivative of Cubic Functions

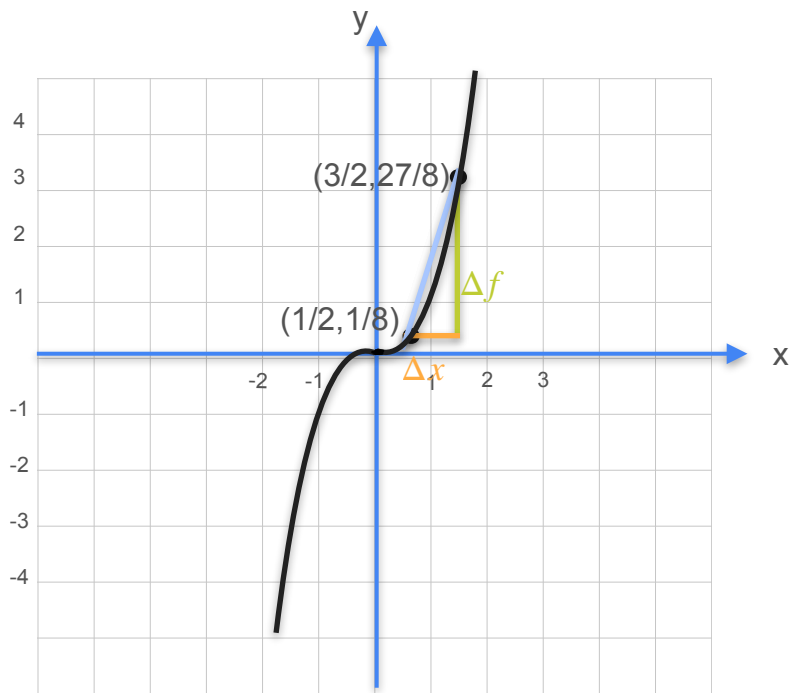


Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$$\begin{aligned} \frac{\Delta f}{\Delta x} &= \frac{(x + \Delta x)^3 - x^3}{\Delta x} \\ &= \frac{\cancel{x^3} + 3x(\cancel{\Delta x})^2 + 3x^2\cancel{\Delta x} + (\cancel{\Delta x})^3 - \cancel{x^3}}{\cancel{\Delta x}} \end{aligned}$$

Derivative of Cubic Functions

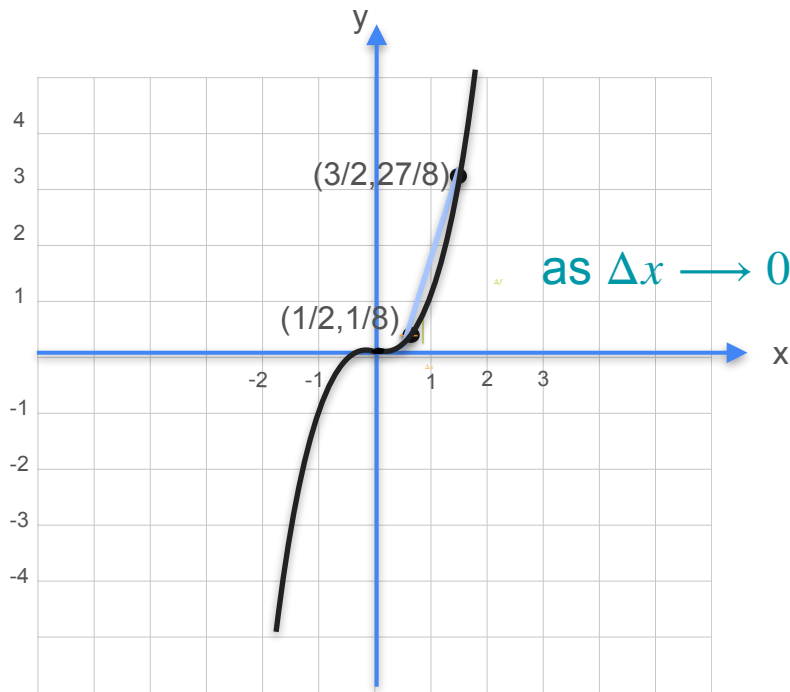


Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$$\begin{aligned} \frac{\Delta f}{\Delta x} &= \frac{(x + \Delta x)^3 - x^3}{\Delta x} \\ &= \frac{\cancel{x^3} + 3x(\cancel{\Delta x})^2 + 3x^2\cancel{\Delta x} + (\cancel{\Delta x})^3 - \cancel{x^3}}{\cancel{\Delta x}} \\ &= 3x\Delta x + 3x^2 + \Delta x^2 \end{aligned}$$

Derivative of Cubic Functions

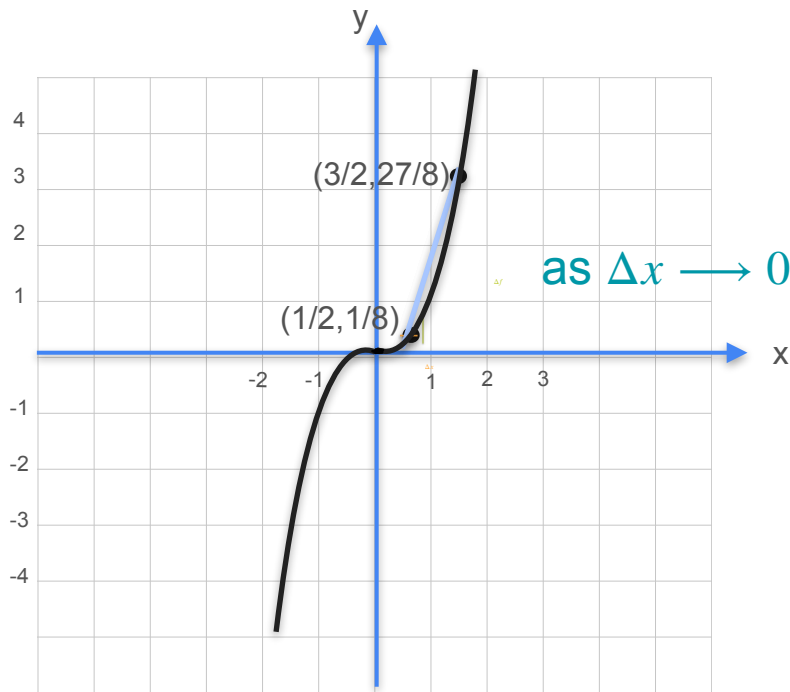


Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$$\begin{aligned}
 \frac{\Delta f}{\Delta x} &= \frac{(x + \Delta x)^3 - x^3}{\Delta x} \\
 &= \frac{x^3 + 3x(\Delta x)^2 + 3x^2\Delta x + (\Delta x)^3 - x^3}{\Delta x} \\
 &= 3x\Delta x + 3x^2 + \Delta x^2
 \end{aligned}$$

Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$$\frac{df}{dx}$$

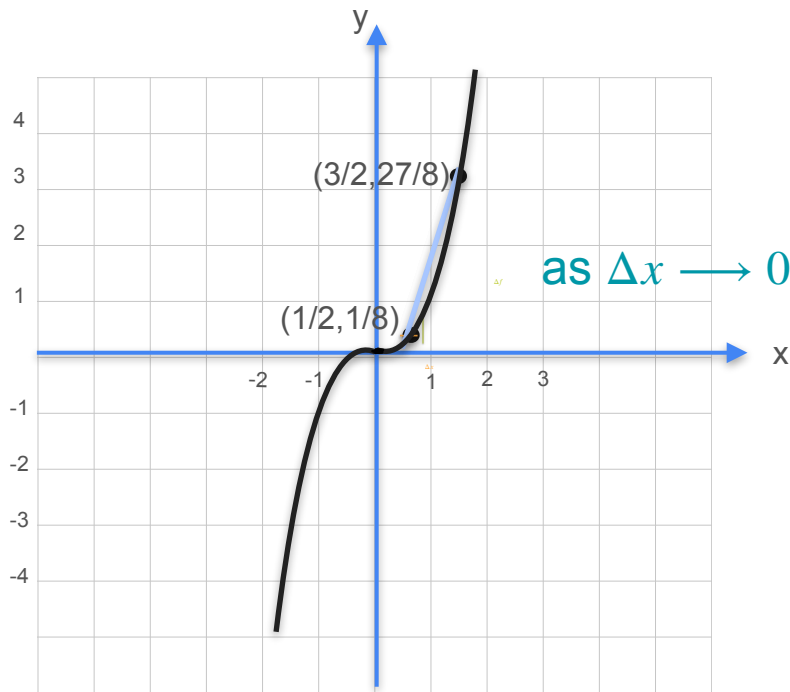
$$\frac{\Delta f}{\Delta x}$$

$$= \frac{(x + \Delta x)^3 - x^3}{\Delta x}$$

$$= \frac{x^3 + 3x(\Delta x)^2 + 3x^2\Delta x + (\Delta x)^3 - x^3}{\Delta x}$$

$$= 3x\Delta x + 3x^2 + \Delta x^2$$

Derivative of Cubic Functions



Cubic: $y = f(x) = x^3$

Slope: $\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$$\begin{aligned} \frac{df}{dx} &= \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^3 - x^3}{\Delta x} \\ &= \frac{x^3 + 3x(\Delta x)^2 + 3x^2\Delta x + (\Delta x)^3 - x^3}{\Delta x} \\ &= 3x\Delta x + 3x^2 + \Delta x^2 \end{aligned}$$

$f(x) = x^3 \Rightarrow f'(x) = 3x^2$



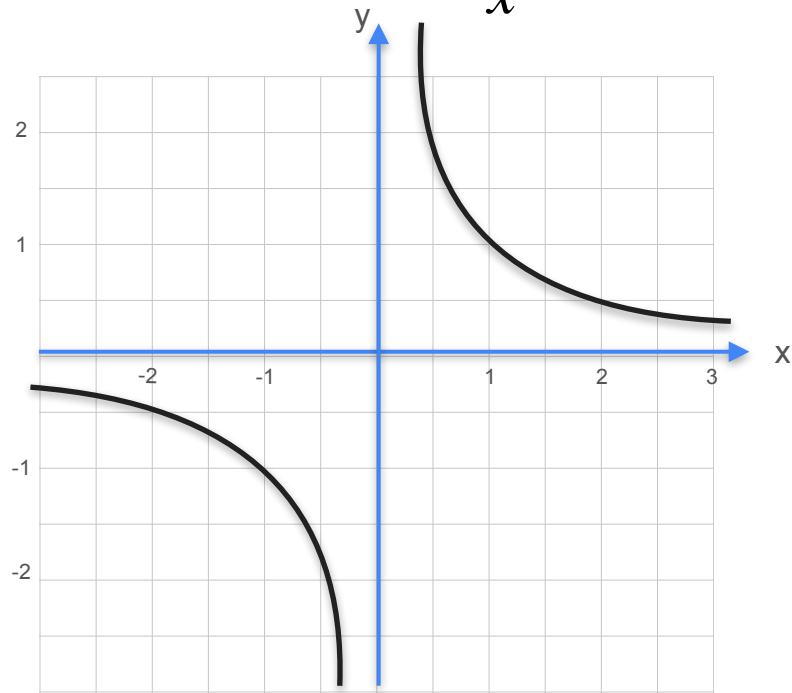
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Derivatives and Optimization

**Some common derivatives:
Other power functions**

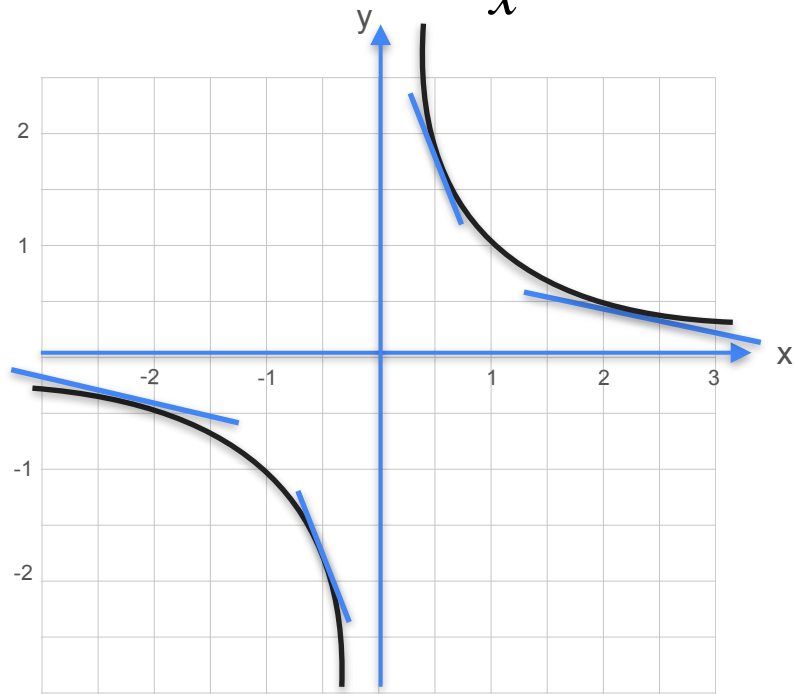
Derivative of $\frac{1}{x}$

Derivative of $\frac{1}{x}$



$$y = f(x) = x^{-1} = \frac{1}{x}$$

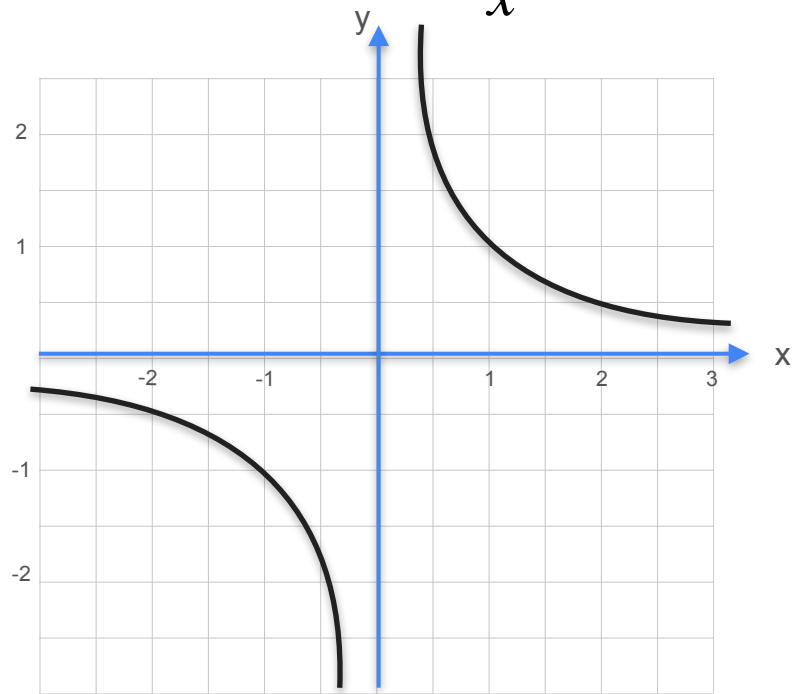
Derivative of $\frac{1}{x}$



$$y = f(x) = x^{-1} = \frac{1}{x}$$

Slope: $\frac{\Delta f}{\Delta x}$

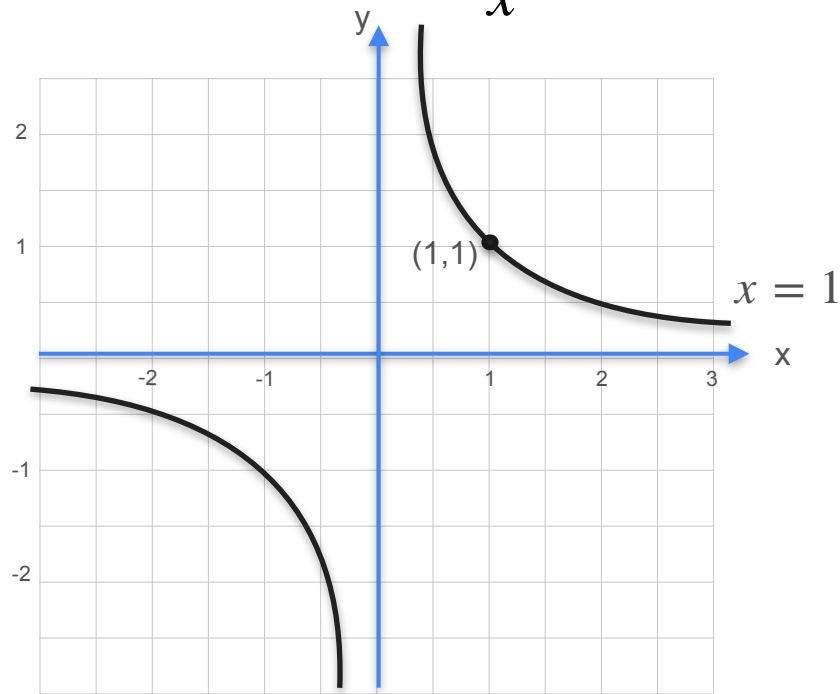
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Slope: $y = f(x) = x^{-1} = \frac{1}{x}$

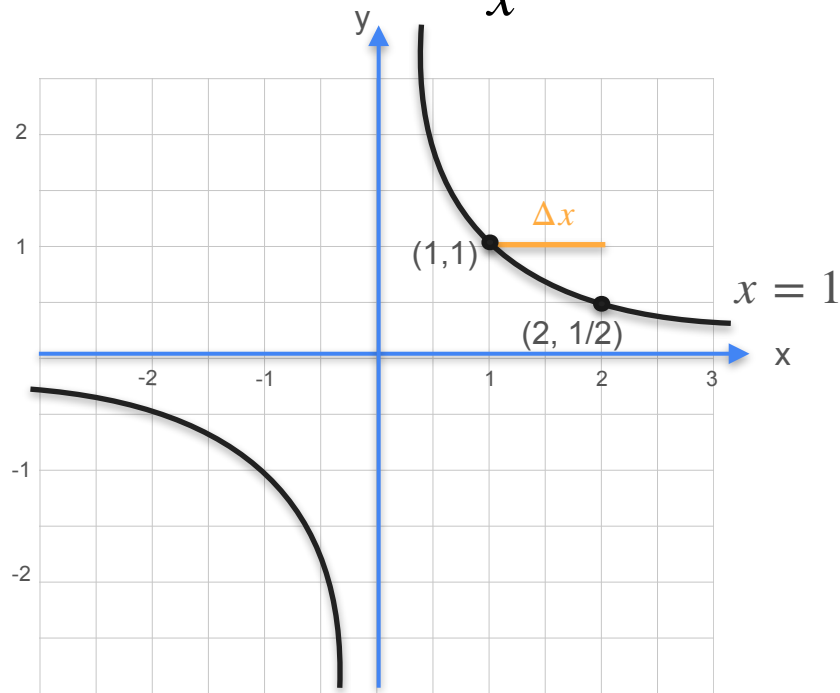
$$\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^{-1} - (x)^{-1}}{\Delta x}$$

Derivative of $\frac{1}{x}$



$$y = f(x) = x^{-1} = \frac{1}{x}$$
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Derivative of $\frac{1}{x}$

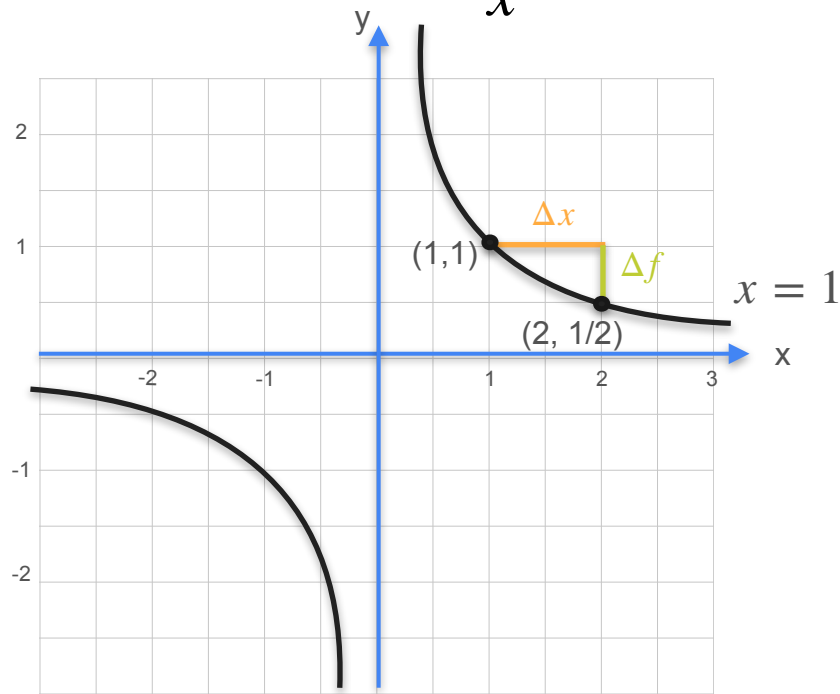


$$y = f(x) = x^{-1} = \frac{1}{x}$$
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^{-1} - (x)^{-1}}{\Delta x}$$

Δx

1.0

Derivative of $\frac{1}{x}$

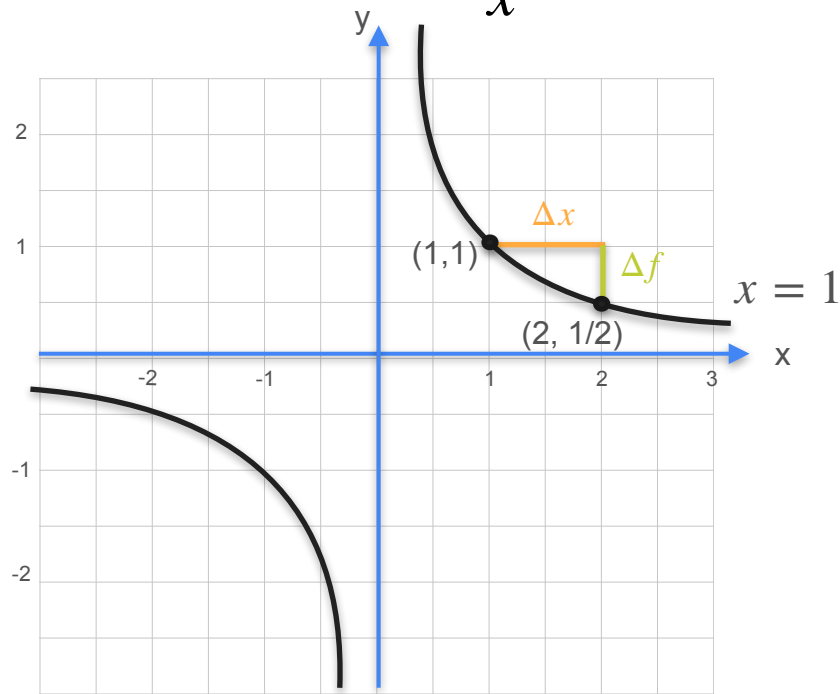


Slope: $y = f(x) = x^{-1} = \frac{1}{x}$

$$\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^{-1} - (x)^{-1}}{\Delta x}$$

Δx	1.0
Δf	

Derivative of $\frac{1}{x}$



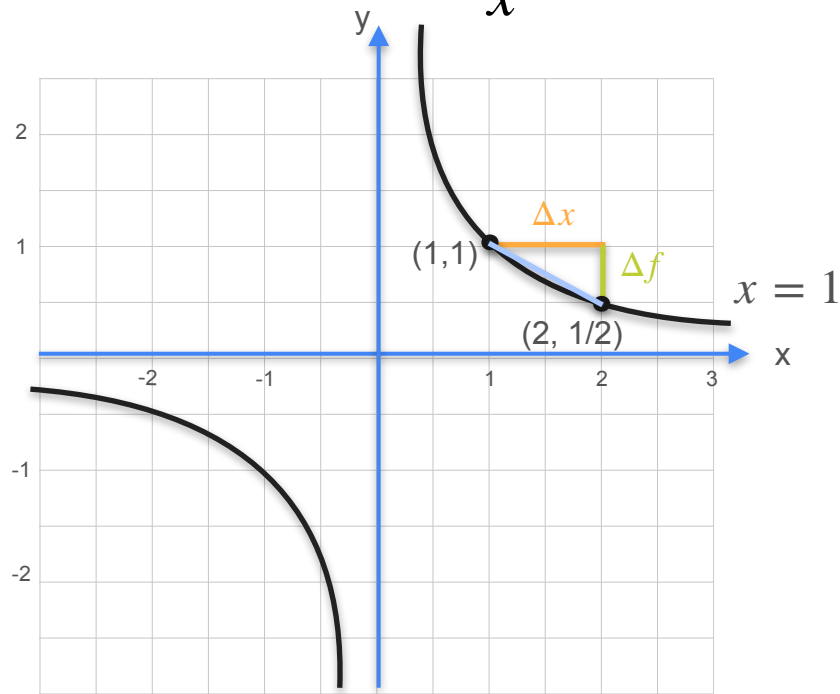
$$y = f(x) = x^{-1} = \frac{1}{x}$$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^{-1} - (x)^{-1}}{\Delta x}$$

Δx	1.0
Δf	-0.5

$$(1 + 1)^{-1} - 1^{-1} = \frac{1}{2} - 1$$

Derivative of $\frac{1}{x}$



$$y = f(x) = x^{-1} = \frac{1}{x}$$

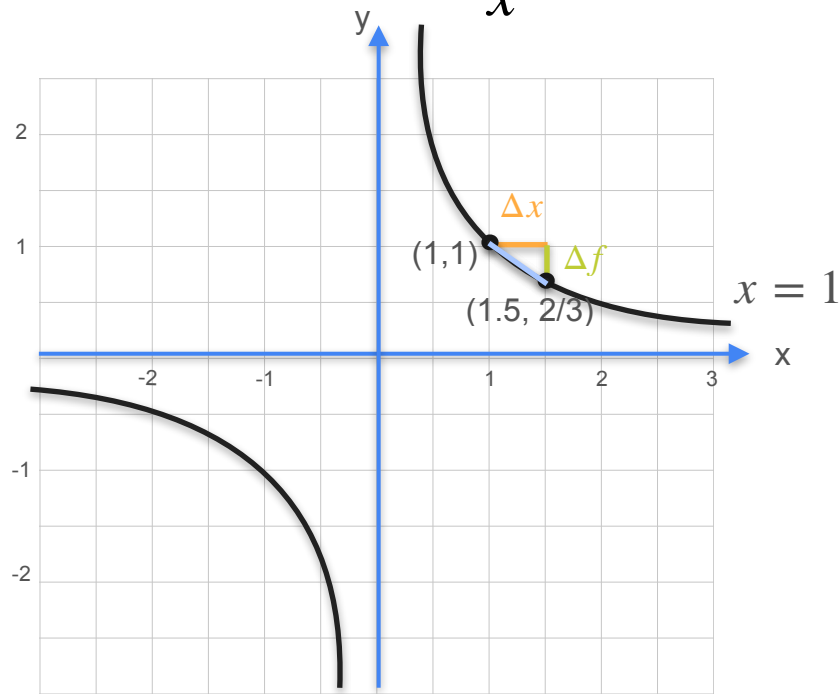
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^{-1} - (x)^{-1}}{\Delta x}$$

Δx	1.0
Δf	-0.5
Slope	-0.5

$$(1 + 1)^{-1} - 1^{-1} = \frac{1}{2} - 1$$

$$\frac{-0.5}{1}$$

Derivative of $\frac{1}{x}$

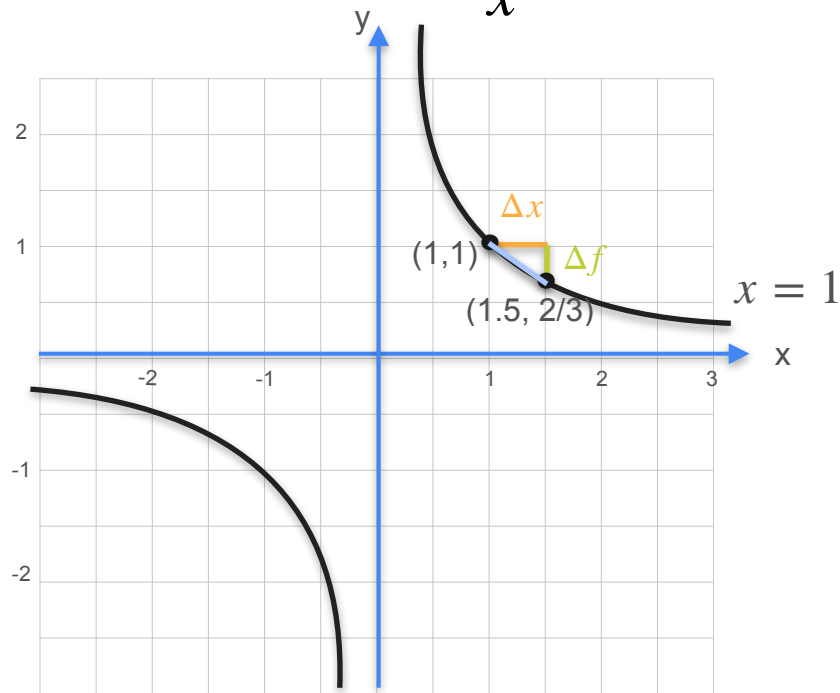


Inverse: $y = f(x) = x^{-1} = \frac{1}{x}$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^{-1} - (x)^{-1}}{\Delta x}$

Δx	1.0
Δf	-0.5
Slope	-0.5

Derivative of $\frac{1}{x}$

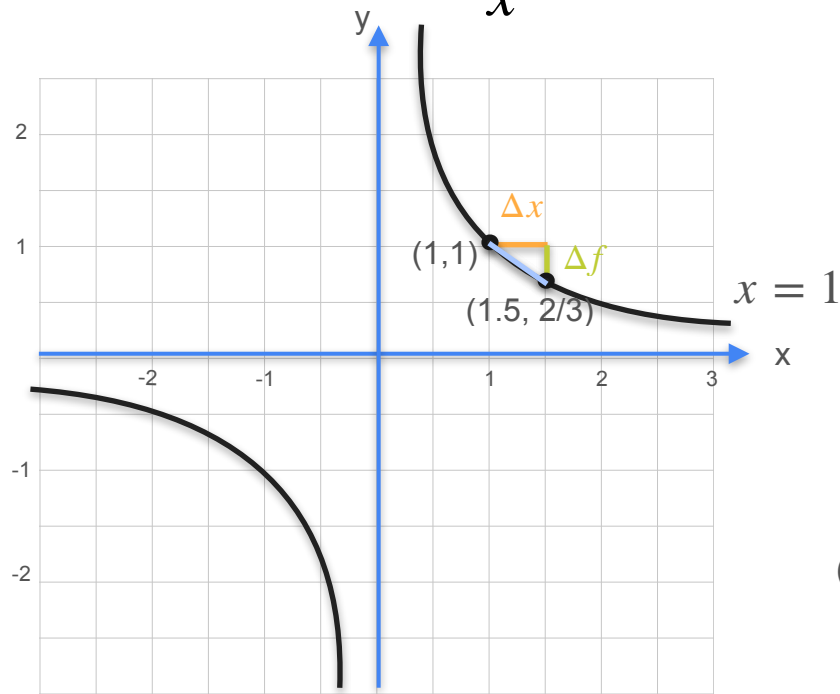


Inverse: $y = f(x) = x^{-1} = \frac{1}{x}$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^{-1} - (x)^{-1}}{\Delta x}$

Δx	1.0	1/2
Δf	-0.5	
Slope	-0.5	

Derivative of $\frac{1}{x}$



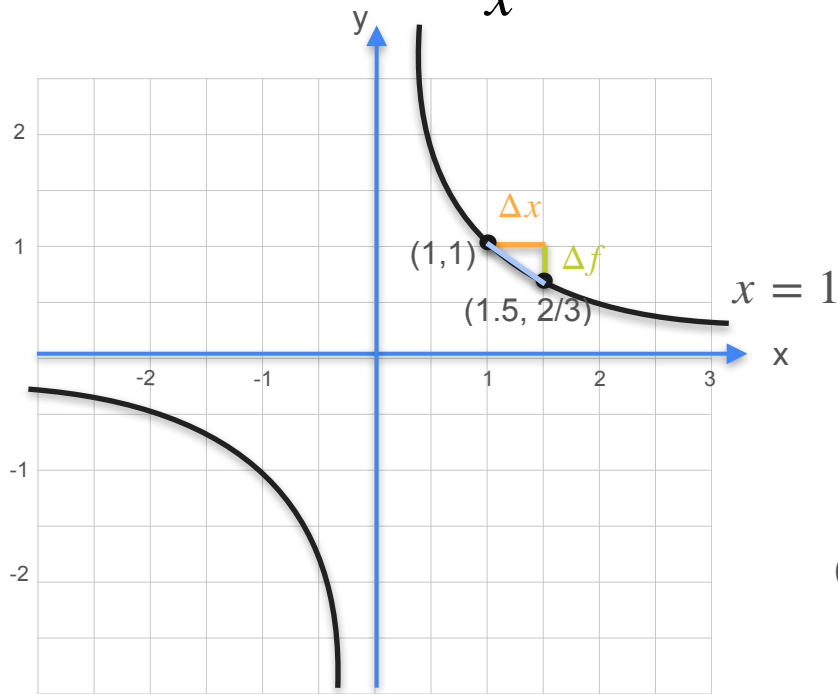
Inverse: $y = f(x) = x^{-1} = \frac{1}{x}$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^{-1} - (x)^{-1}}{\Delta x}$

Δx	1.0	1/2
Δf	-0.5	-0.33
Slope	-0.5	

$$(1 + 0.5)^{-1} - 1^{-1} = \frac{2}{3} - 1$$

Derivative of $\frac{1}{x}$



Inverse: $y = f(x) = x^{-1} = \frac{1}{x}$

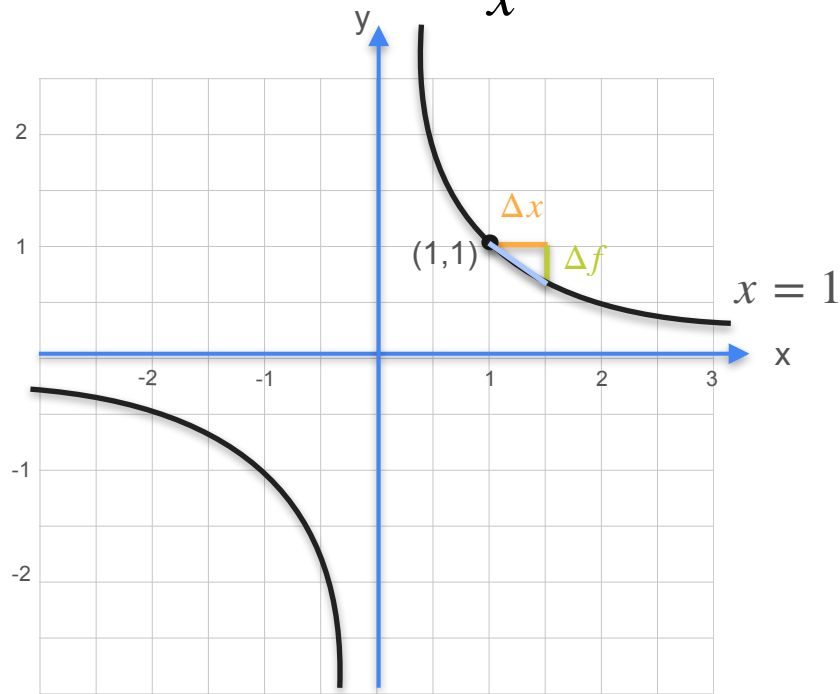
Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^{-1} - (x)^{-1}}{\Delta x}$

Δx	1.0	1/2
Δf	-0.5	-0.33
Slope	-0.5	-0.67

$$(1 + 0.5)^{-1} - 1^{-1} = \frac{2}{3} - 1$$

$$\frac{0.33}{0.5}$$

Derivative of $\frac{1}{x}$

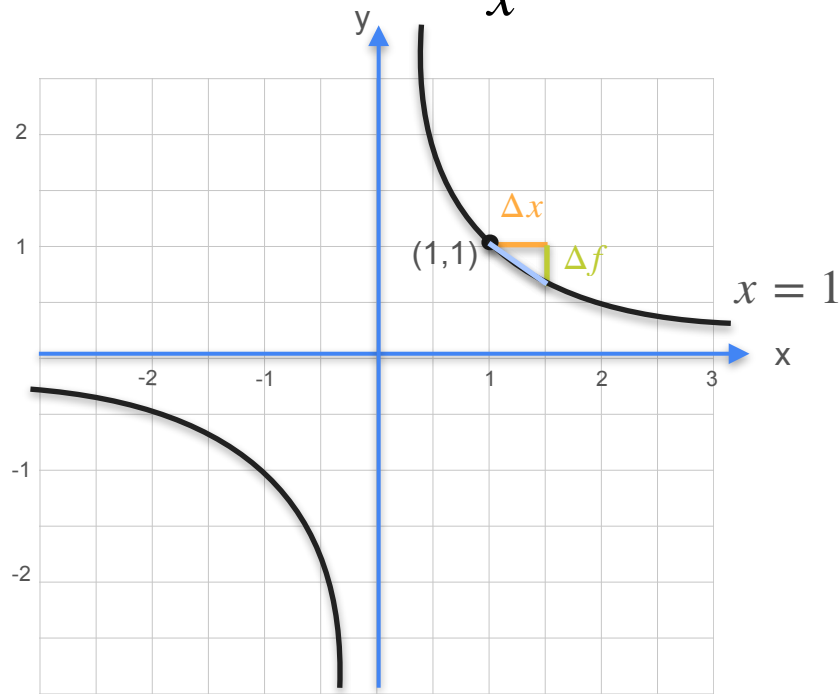


Inverse: $y = f(x) = x^{-1} = \frac{1}{x}$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^{-1} - (x)^{-1}}{\Delta x}$

Δx	1.0	1/2
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Derivative of $\frac{1}{x}$

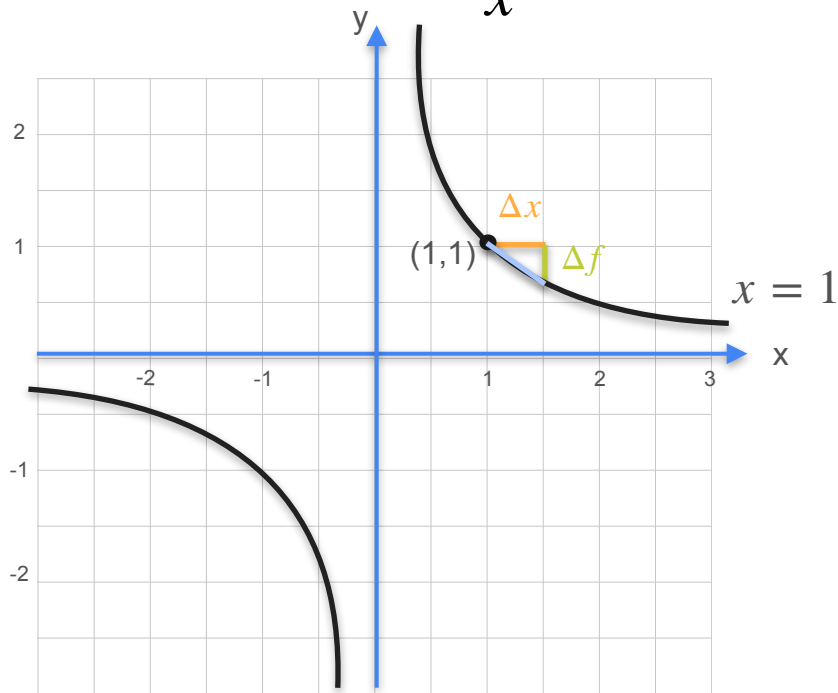


Inverse: $y = f(x) = x^{-1} = \frac{1}{x}$

Slope: $\frac{\Delta f}{\Delta x} = \frac{(x + \Delta x)^{-1} - (x)^{-1}}{\Delta x}$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	-0.5	-0.33	-0.2	-0.11	-0.06	-0.001
Slope	-0.5	-0.67	-0.8	-0.89	-0.94	-0.999

Derivative of $\frac{1}{x}$



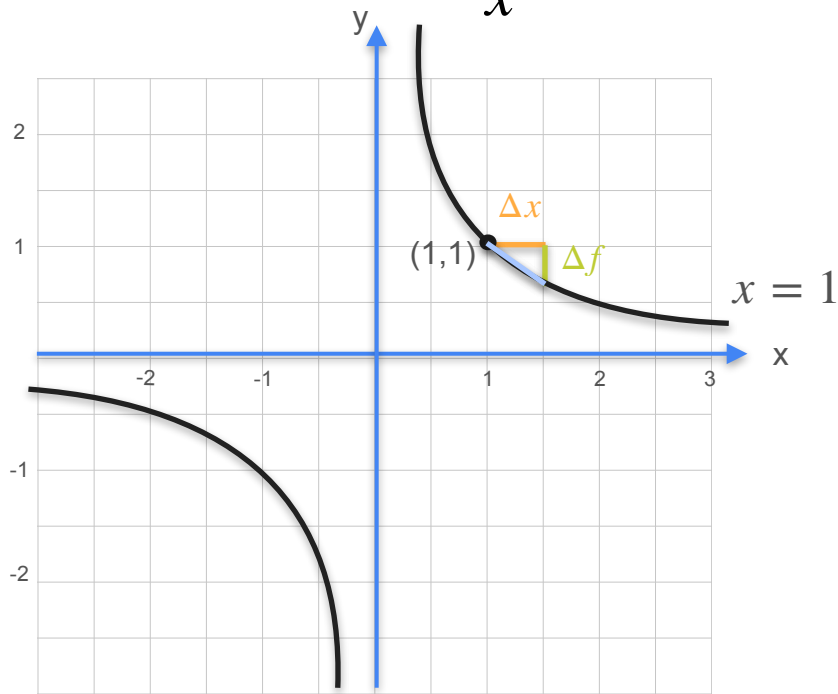
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Δx	1.0	1/2	1/4	1/8	1/16	1/1000
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Slope	-0.5	-0.67	-0.8	-0.89	-0.94	-0.999

$$f'(1) = \frac{d}{dx} f(1) = -1$$

Derivative of $\frac{1}{x}$



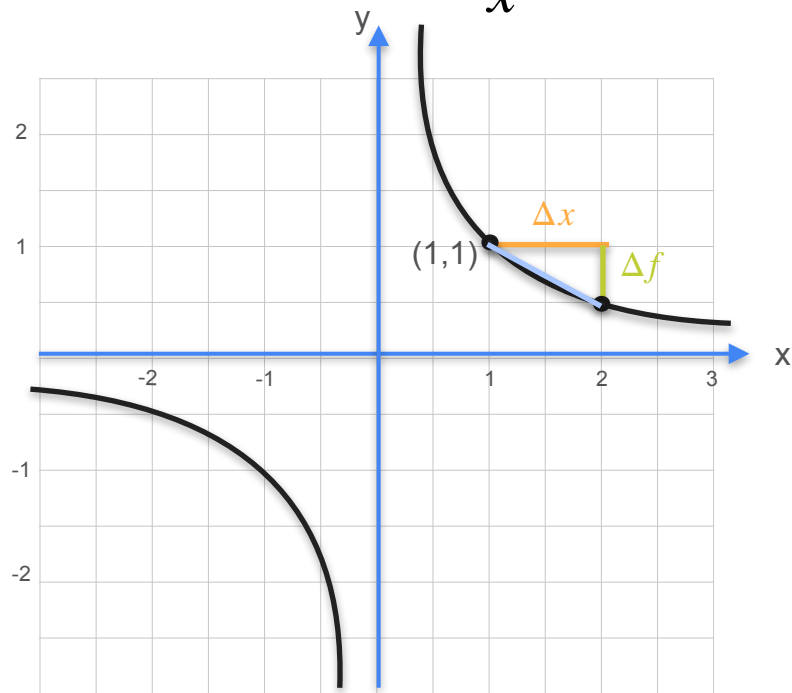
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Δf	-0.5	-0.33	-0.2	-0.11	-0.06	-0.001
Slope	-0.5	-0.67	-0.8	-0.89	-0.94	-0.999

$$f'(1) = \frac{d}{dx} f(1) = -1 = -1 \times 1^2$$

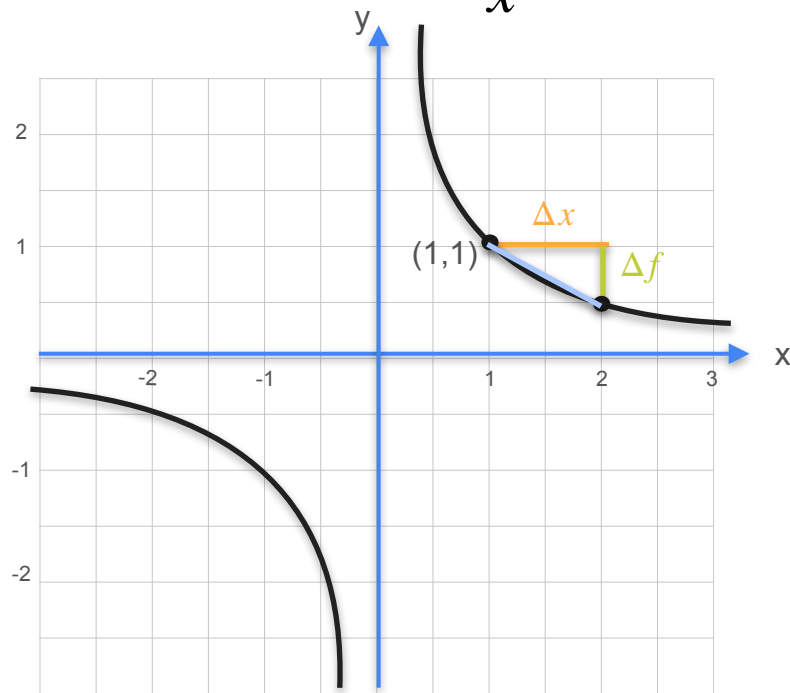
Derivative of $\frac{1}{x}$



Inverse: $y = f(x) = x^{-1} = \frac{1}{x}$

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Derivative of $\frac{1}{x}$

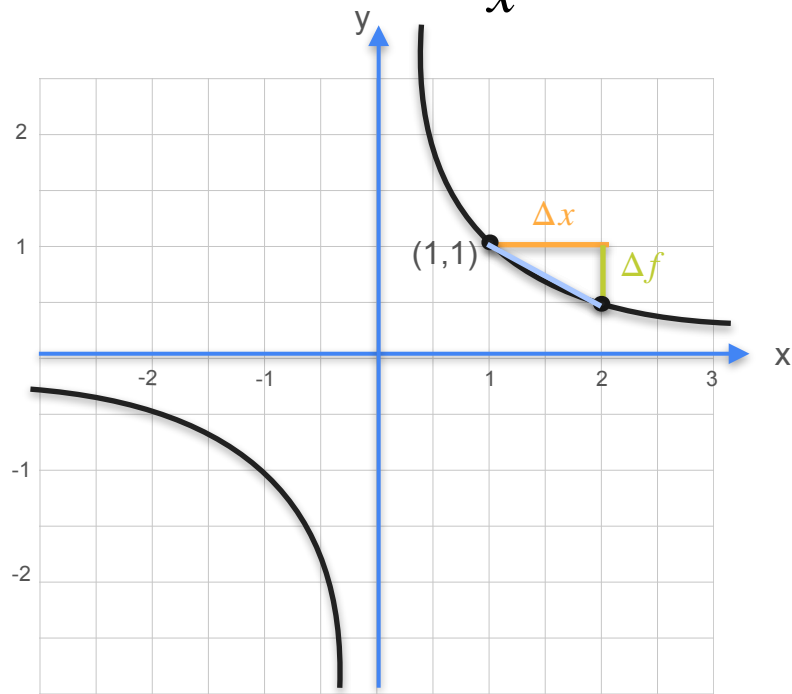


Inverse: $y = f(x) = x^{-1} = \frac{1}{x}$

Slope: $\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$\frac{\Delta f}{\Delta x}$

Derivative of $\frac{1}{x}$

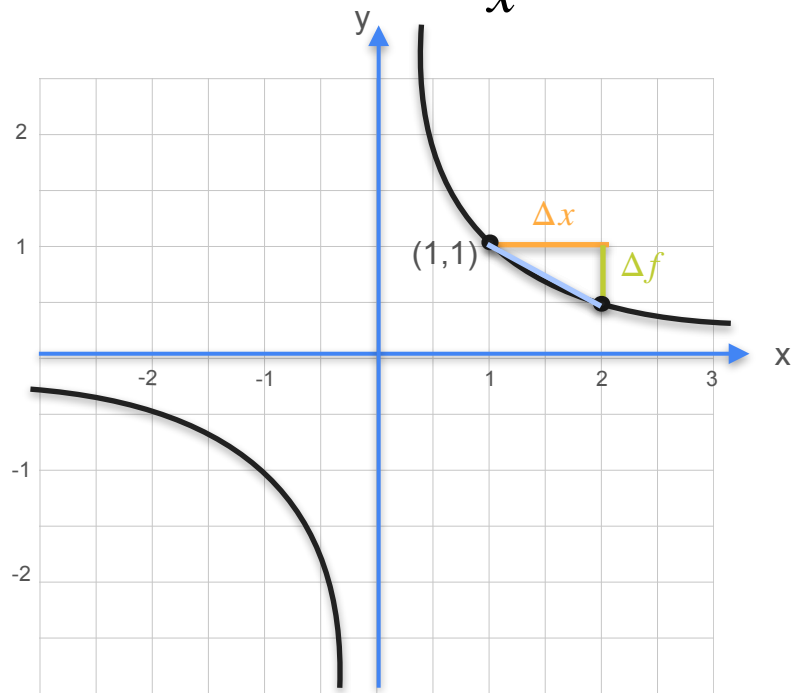


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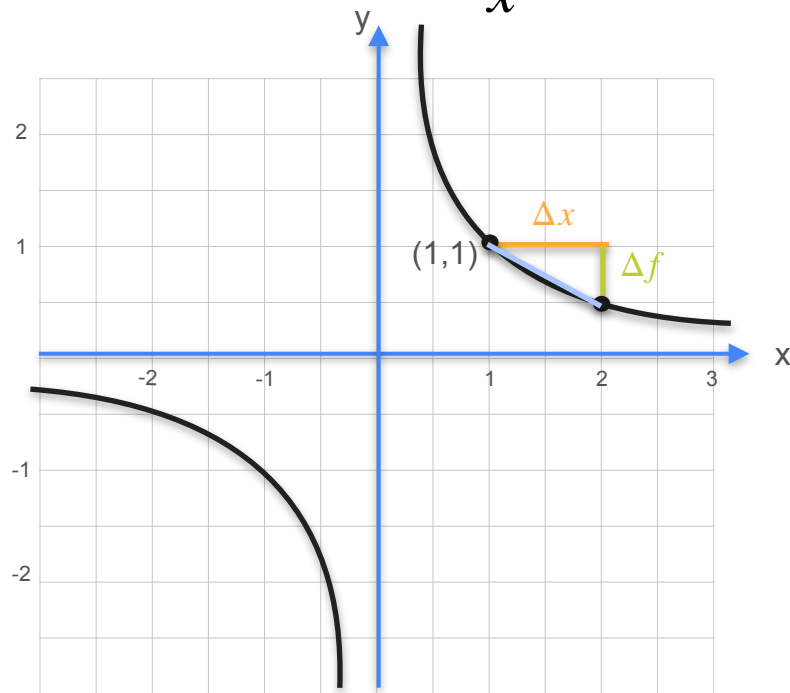
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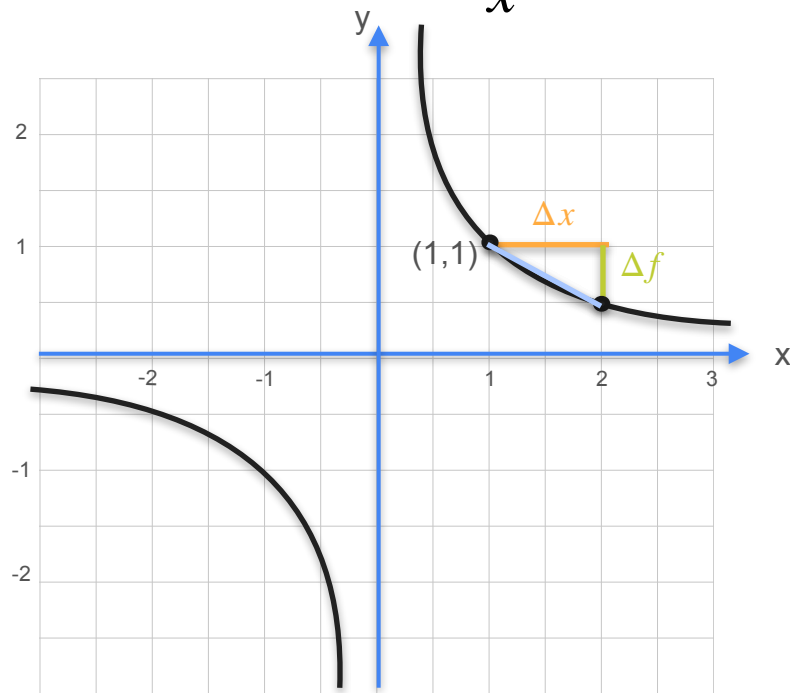
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Derivative of $\frac{1}{x}$

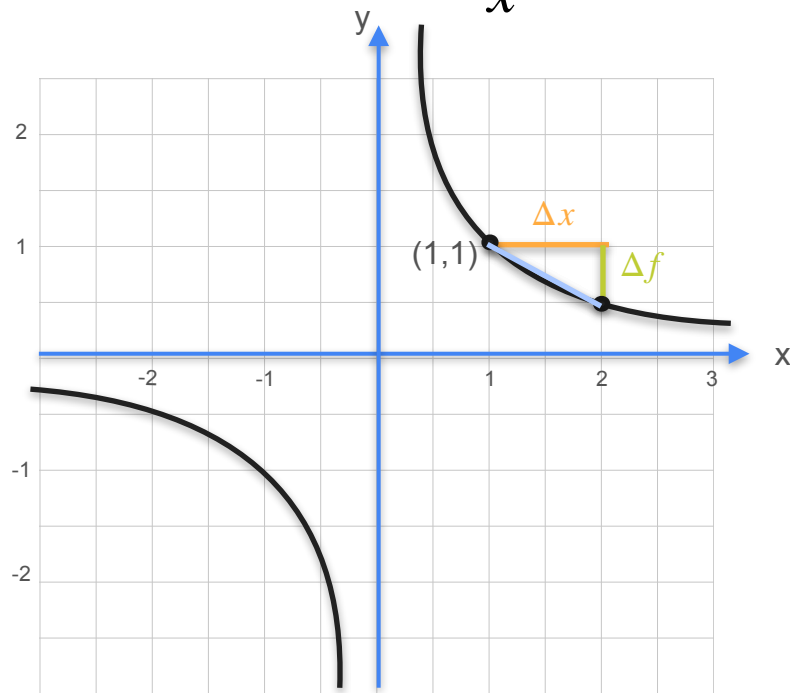


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Derivative of $\frac{1}{x}$



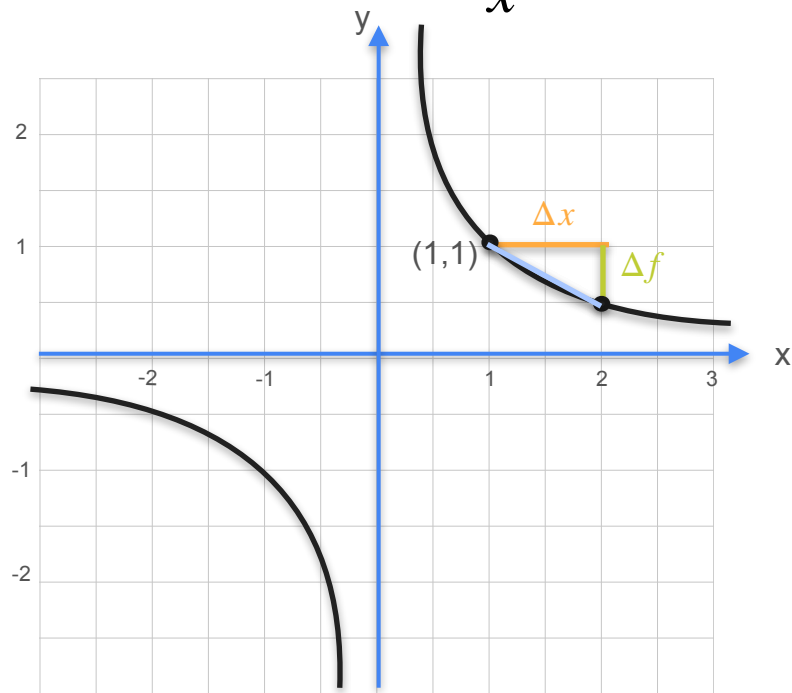
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Derivative of $\frac{1}{x}$

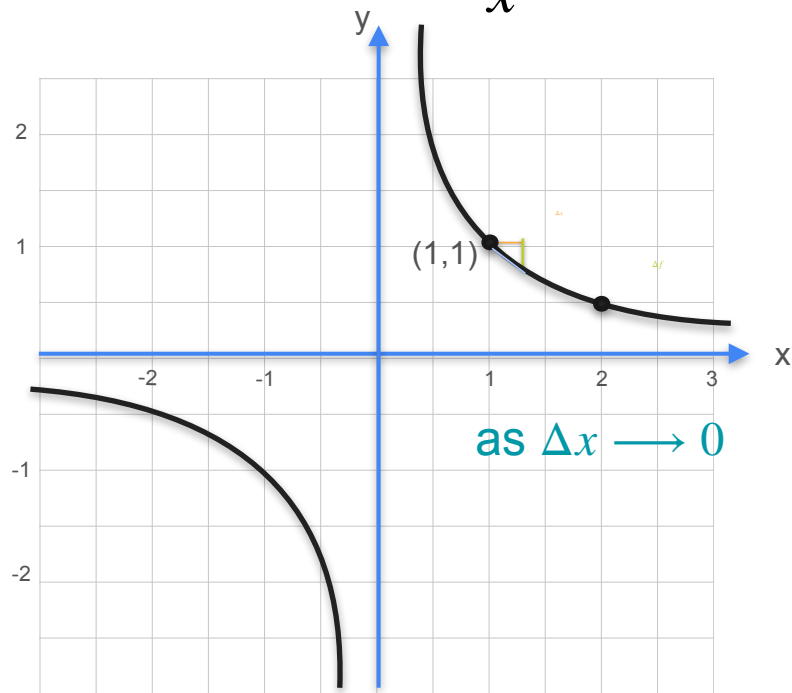


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$$\begin{aligned} \frac{\Delta f}{\Delta x} &= \frac{(x + \Delta x)^{-1} - x^{-1}}{\Delta x} \\ &= \frac{\frac{1}{x + \Delta x} - \frac{1}{x}}{\Delta x} = \frac{\cancel{x} - \cancel{(x + \Delta x)}}{(x + \Delta x)x \cancel{\Delta x}} \\ &= -\frac{1}{x^2 + x\Delta x} \end{aligned}$$

Derivative of $\frac{1}{x}$

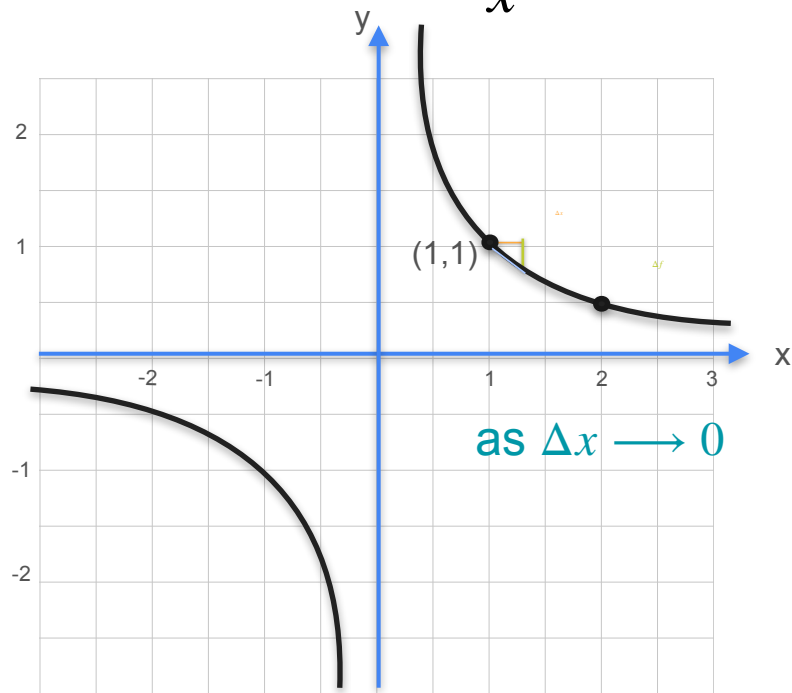


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Derivative of $\frac{1}{x}$



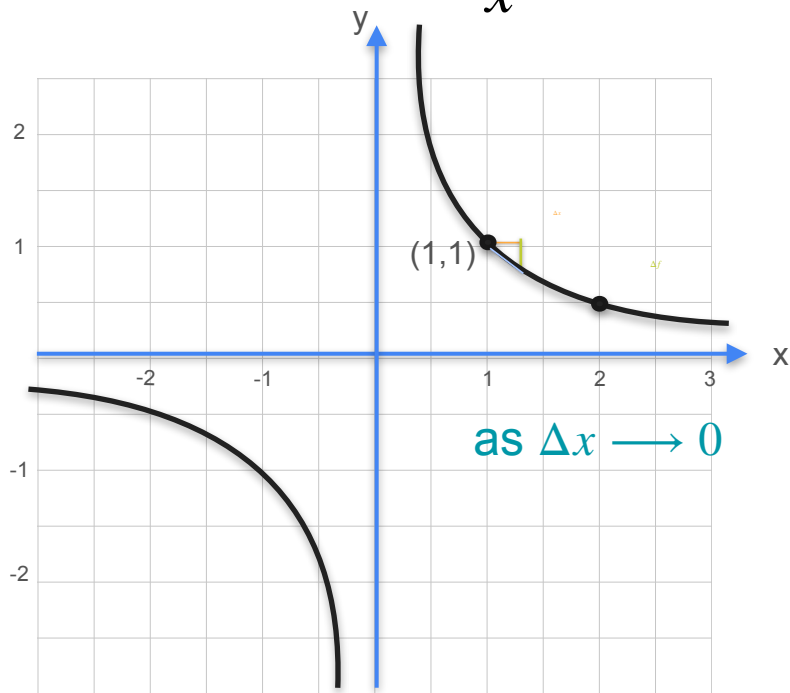
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As $\Delta x \rightarrow 0$, the term $x\Delta x$ in the denominator approaches 0, and the derivative becomes $-\frac{1}{x^2}$.

Derivative of $\frac{1}{x}$



Inverse: $y = f(x) = x^{-1} = \frac{1}{x}$

Slope: $\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$

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$f(x) = x^{-1} \Rightarrow f'(x) = -x^{-2}$

Derivative of Power Functions

Derivative of Power Functions

$$f(x) = x^2$$

$$f(x) = x^3$$

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Derivative of Power Functions

$$f(x) = x^2$$

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$$f(x) = x^{-1}$$

$$f'(x) = 2x^1$$

$$f'(x) = 3x^2$$

$$f'(x) = (-1)x^{-2}$$

Derivative of Power Functions

Can you spot the pattern?

$$f(x) = x^2$$

$$f(x) = x^3$$

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Derivative of Power Functions

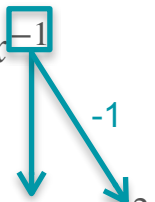
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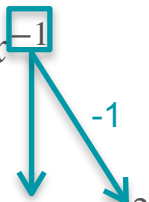
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$$f'(x) = (-1)x^{-2}$$

$$f(x) = x^n \quad \Rightarrow \quad f'(x) = \frac{d}{dx} f(x) = nx^{n-1}$$



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Derivatives and Optimization

The inverse function and its derivative

Inverse Function

What's an inverse?

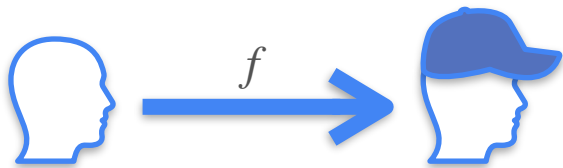
Inverse Function

What's an inverse?



Inverse Function

What's an inverse?



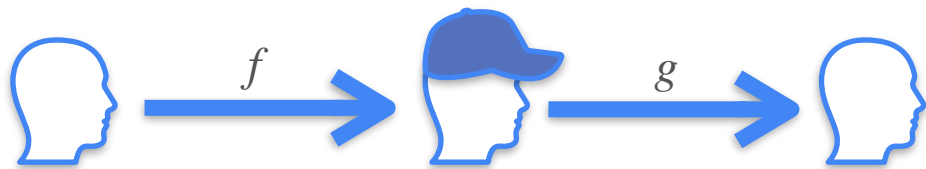
Inverse Function

What's an inverse?



Inverse Function

What's an inverse?



Inverse Function

What's an inverse?



x

Inverse Function

What's an inverse?



$$x \xrightarrow{f} x^2$$

Inverse Function

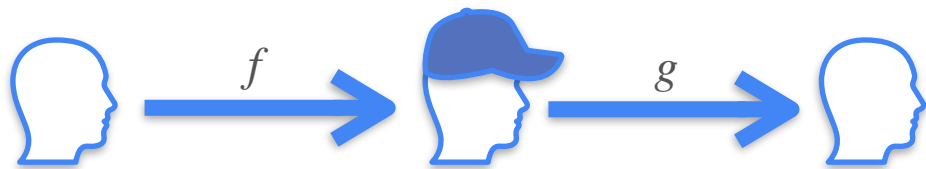
What's an inverse?



$$x \xrightarrow{f} x^2 \xrightarrow{g} x$$

Inverse Function

What's an inverse?



$g(x)$ and $f(x)$ are
inverses

$$x \xrightarrow{f} x^2 \xrightarrow{g} x$$

Inverse Function

What's an inverse?



$$x \xrightarrow{f} x^2 \xrightarrow{g} x$$

$g(x)$ and $f(x)$ are
inverses

$$g(x) = f^{-1}(x)$$

Inverse Function

What's an inverse?



$$x \xrightarrow{f} x^2 \xrightarrow{g} x$$

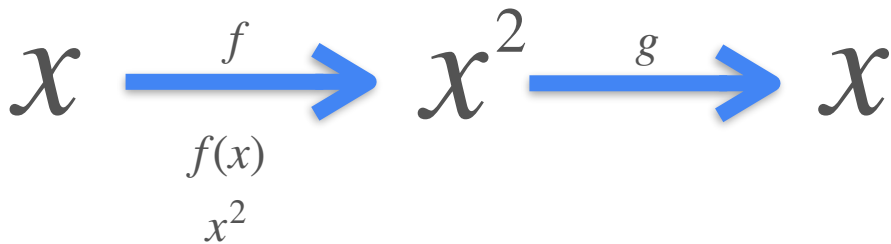
$g(x)$ and $f(x)$ are
inverses

$$g(x) = f^{-1}(x)$$

$$g(f(x)) = x$$

Inverse Function

What's an inverse?



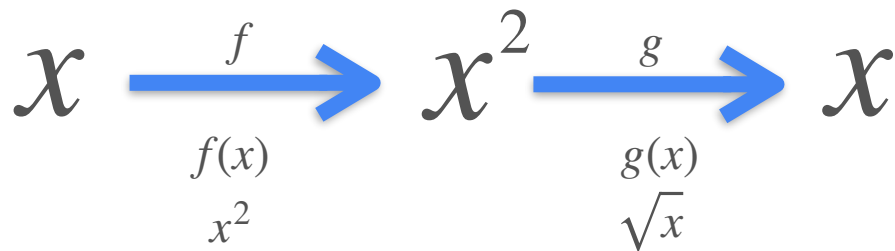
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inverses

$$g(x) = f^{-1}(x)$$

$$g(f(x)) = x$$

Inverse Function

What's an inverse?



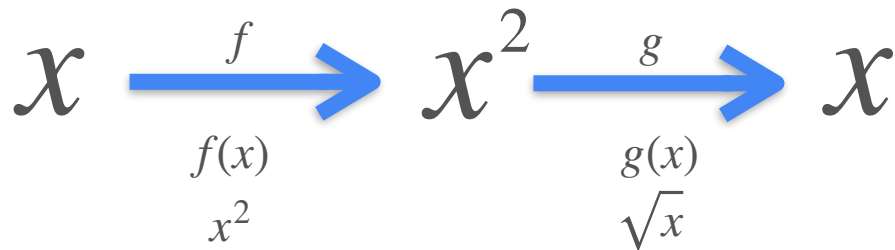
$g(x)$ and $f(x)$ are
inverses

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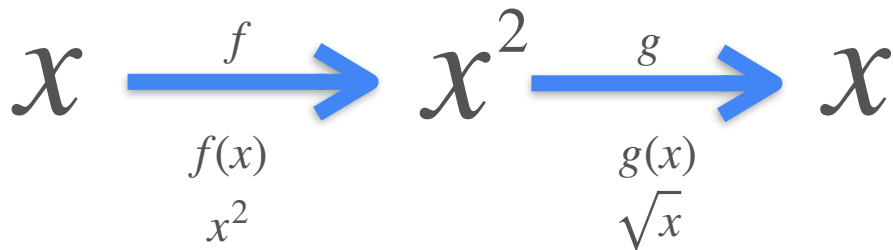
$$g(x) = f^{-1}(x)$$

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$$\sqrt{x^2}$$

Inverse Function

What's an inverse?



$g(x)$ and $f(x)$ are
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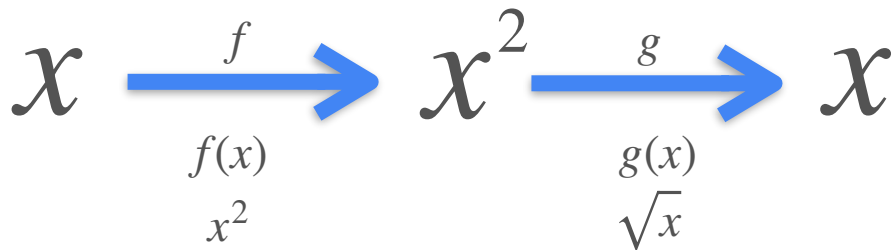
$$g(x) = f^{-1}(x)$$

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Inverse Function

What's an inverse?



$g(x)$ and $f(x)$ are
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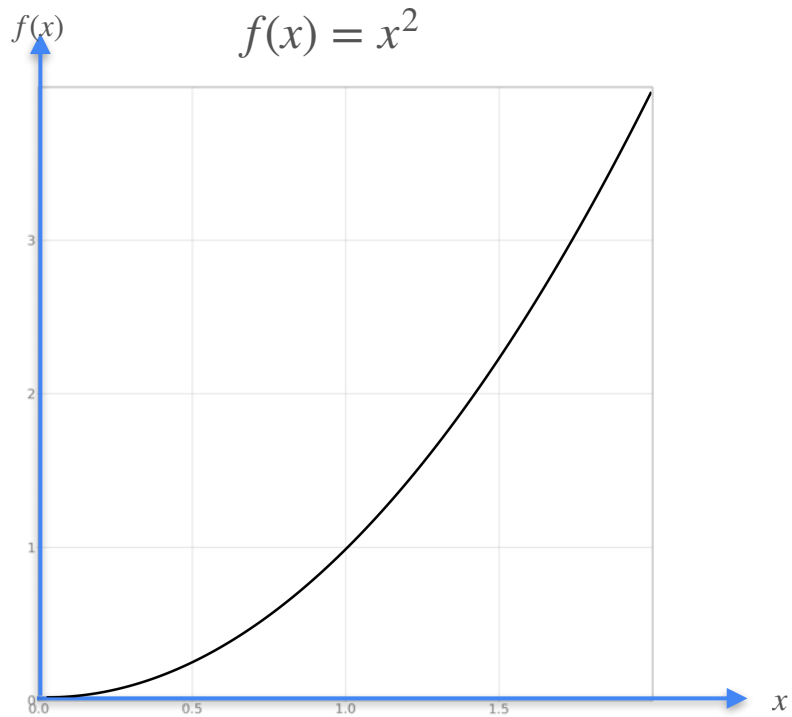
$$g(x) = f^{-1}(x)$$

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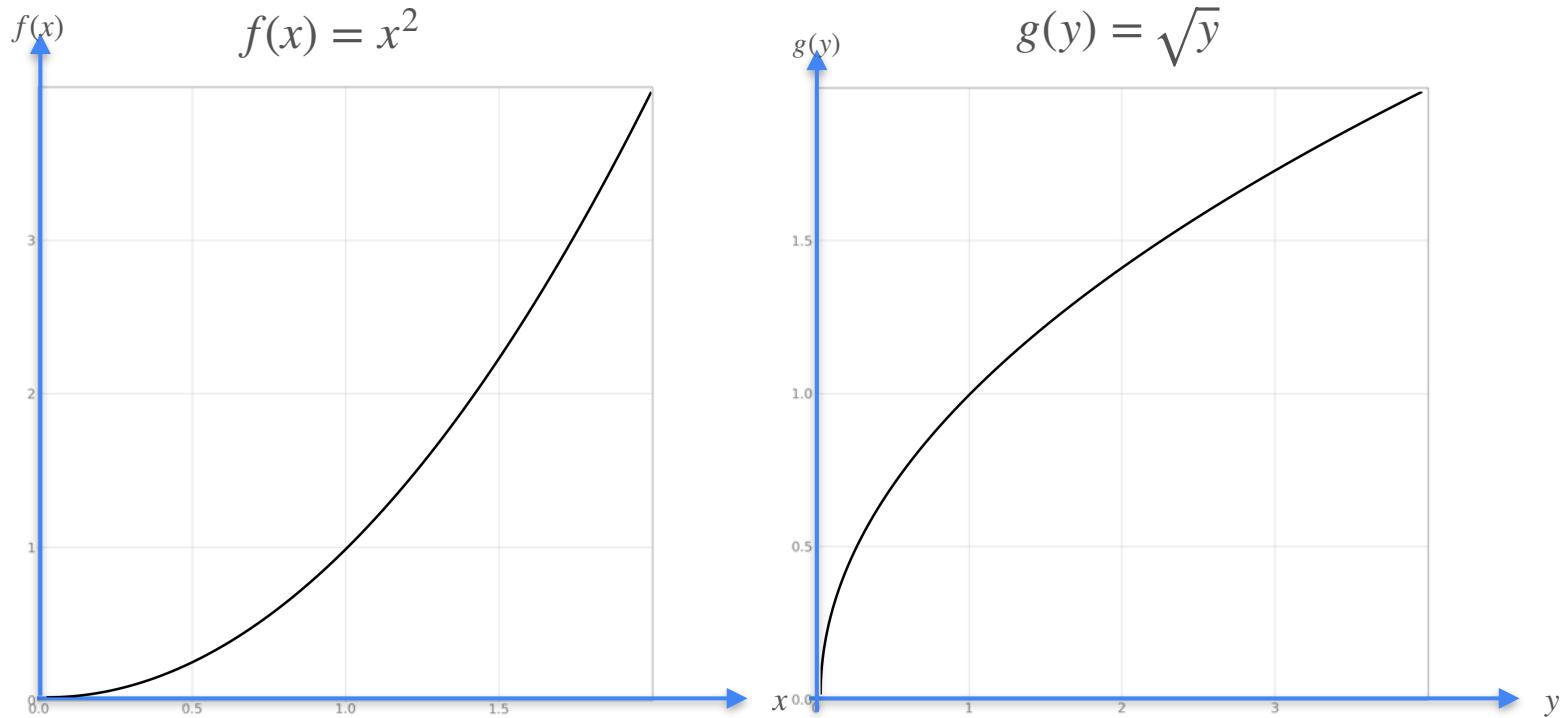
$$\sqrt{x^2} = x \quad \text{for } x > 0$$

Derivative of the Inverse

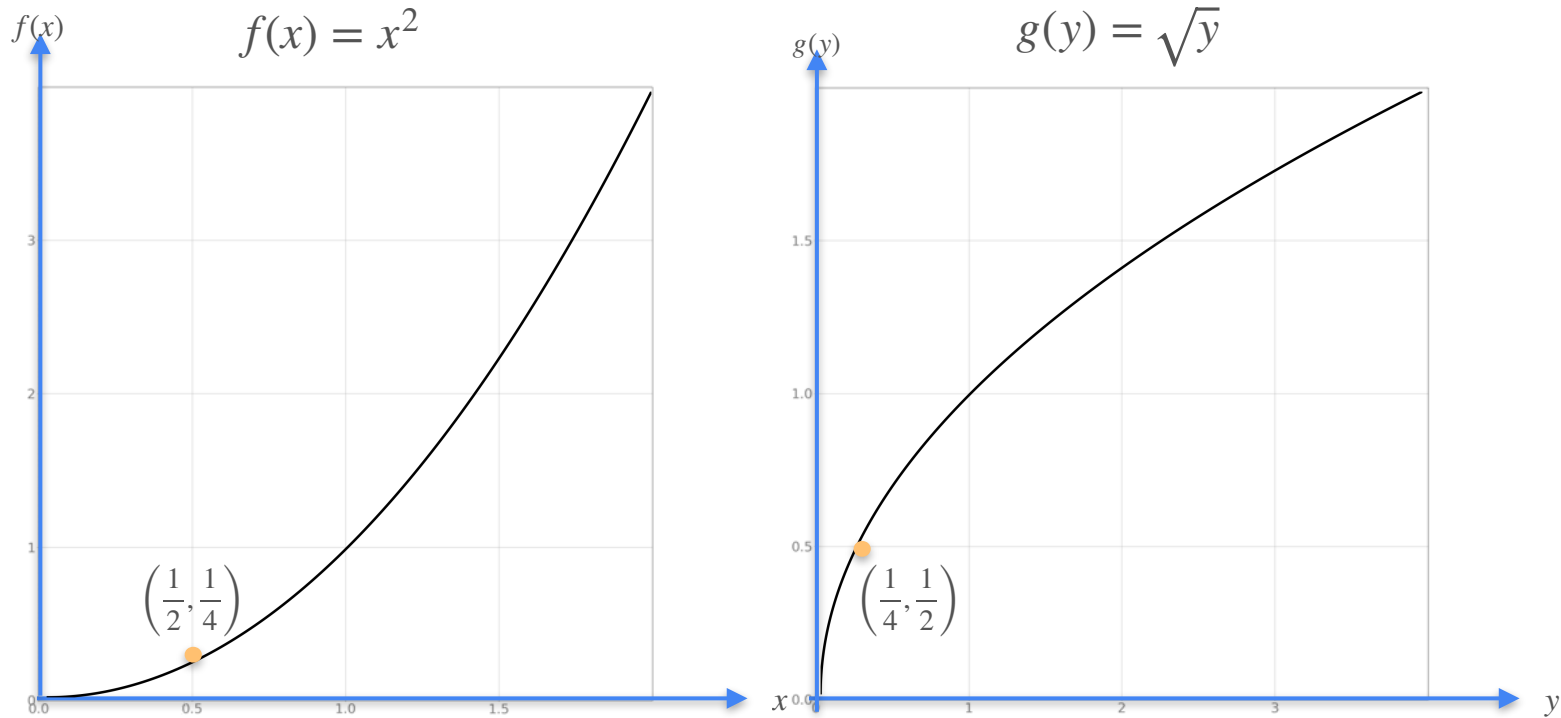
Derivative of the Inverse



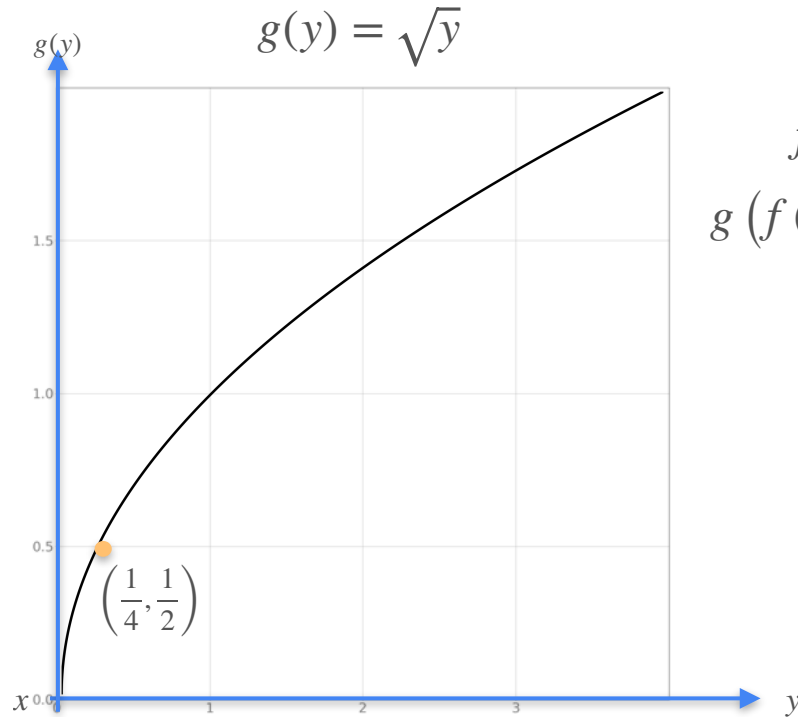
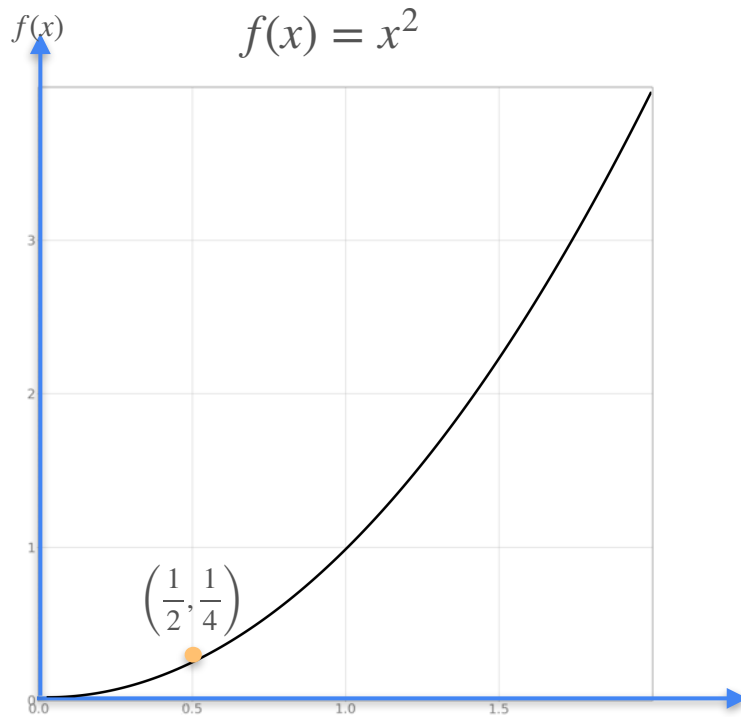
Derivative of the Inverse



Derivative of the Inverse

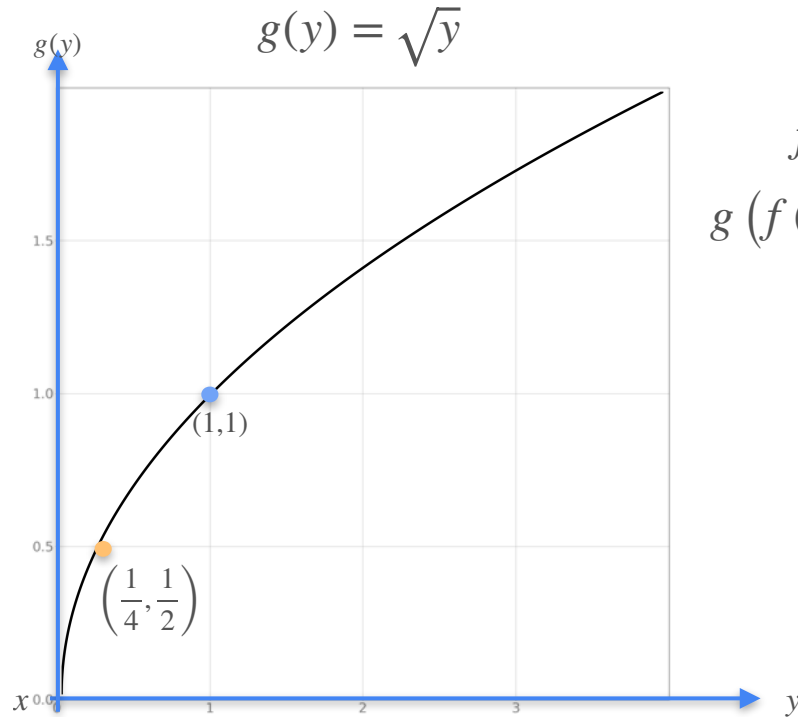
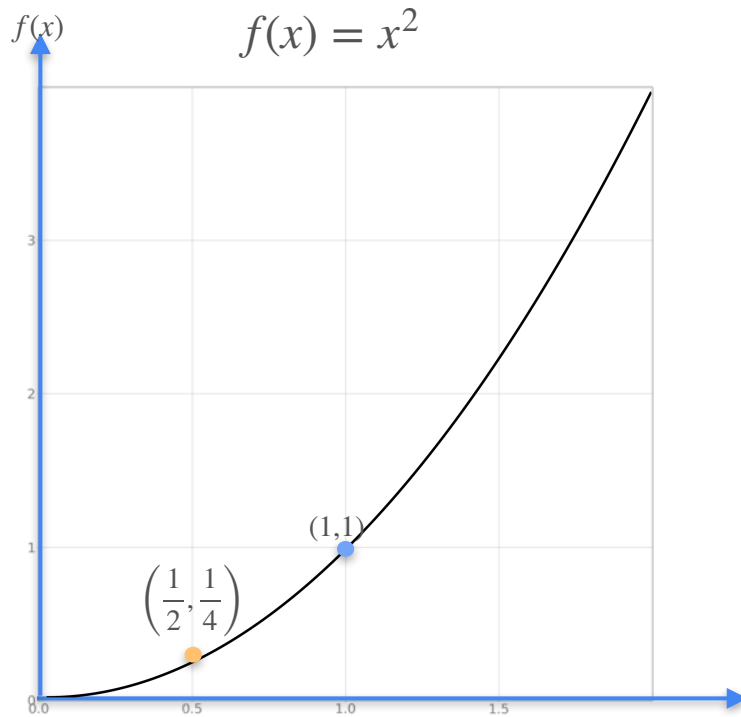


Derivative of the Inverse



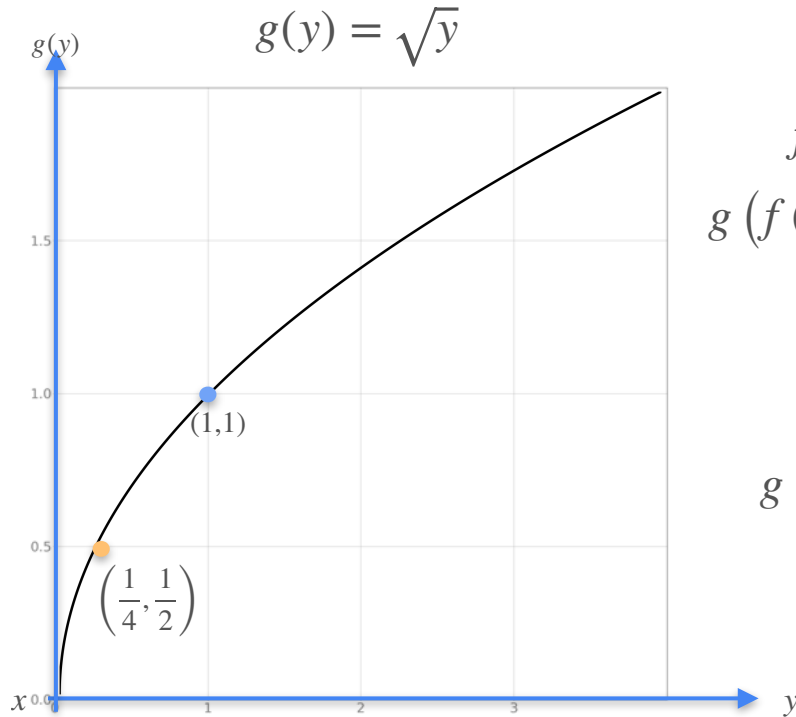
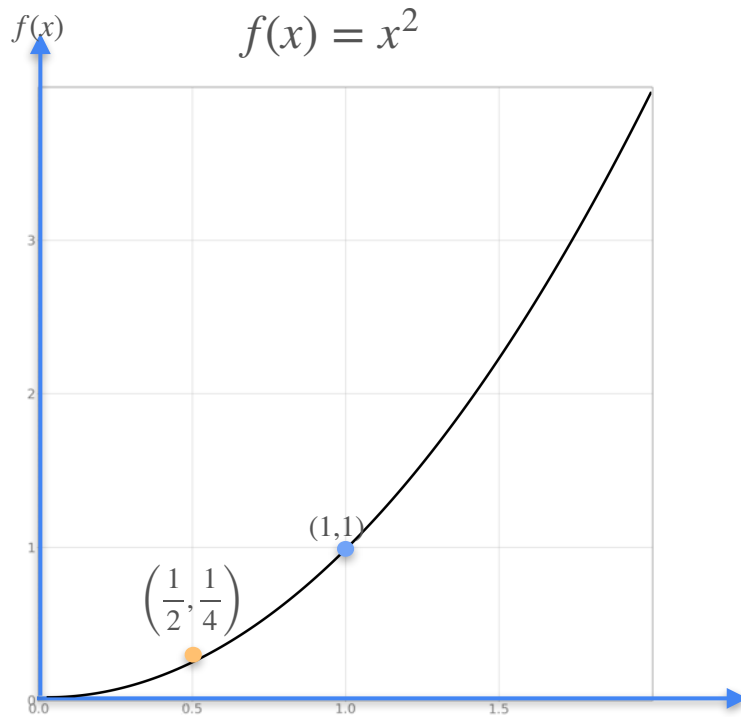
$$\begin{aligned} f(1/2) &= 1/4 \\ g(f(1/2)) &= g(1/4) \\ &= 1/2 \end{aligned}$$

Derivative of the Inverse



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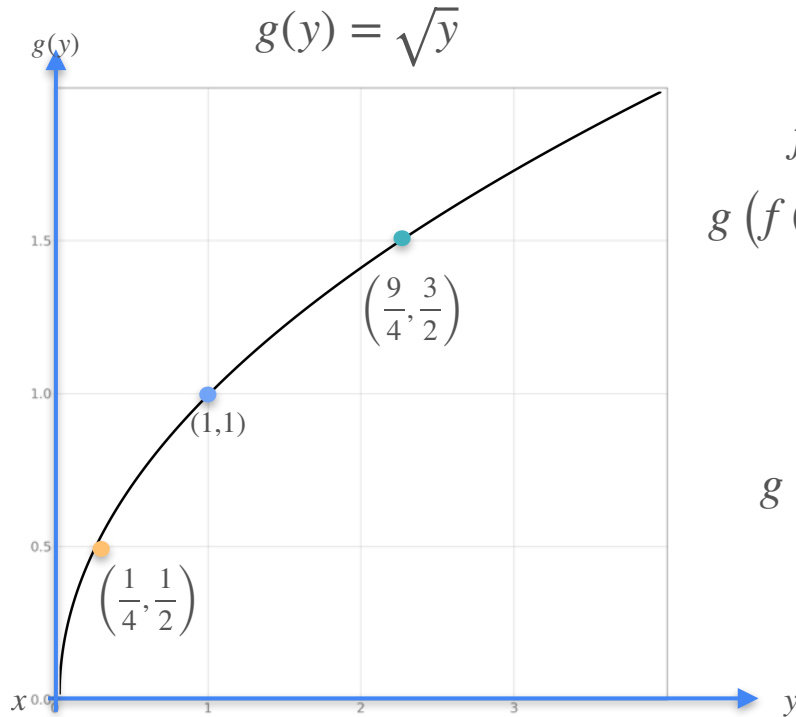
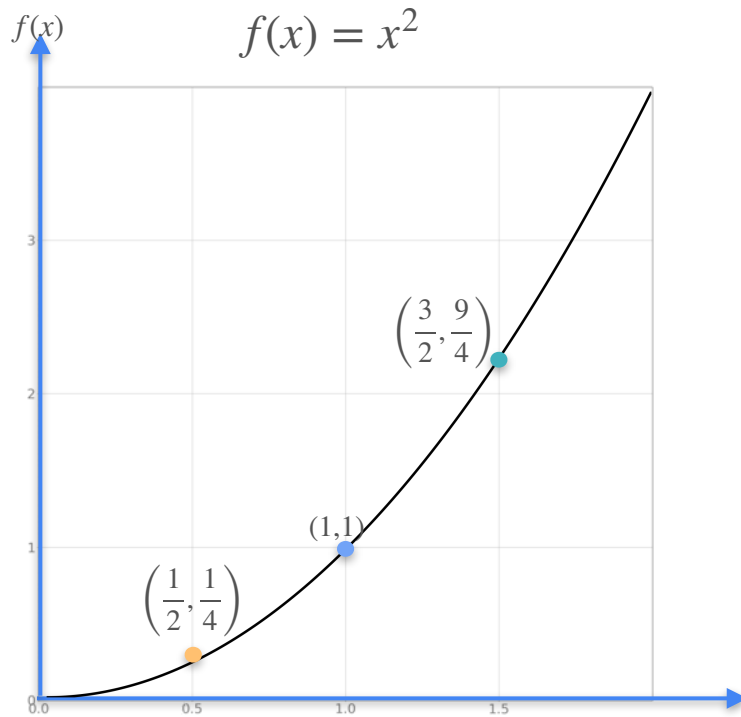
Derivative of the Inverse



$$\begin{aligned} f(1/2) &= 1/4 \\ g(f(1/2)) &= g(1/4) \\ &= 1/2 \end{aligned}$$

$$\begin{aligned} f(1) &= 1 \\ g(f(1)) &= g(1) \\ &= 1 \end{aligned}$$

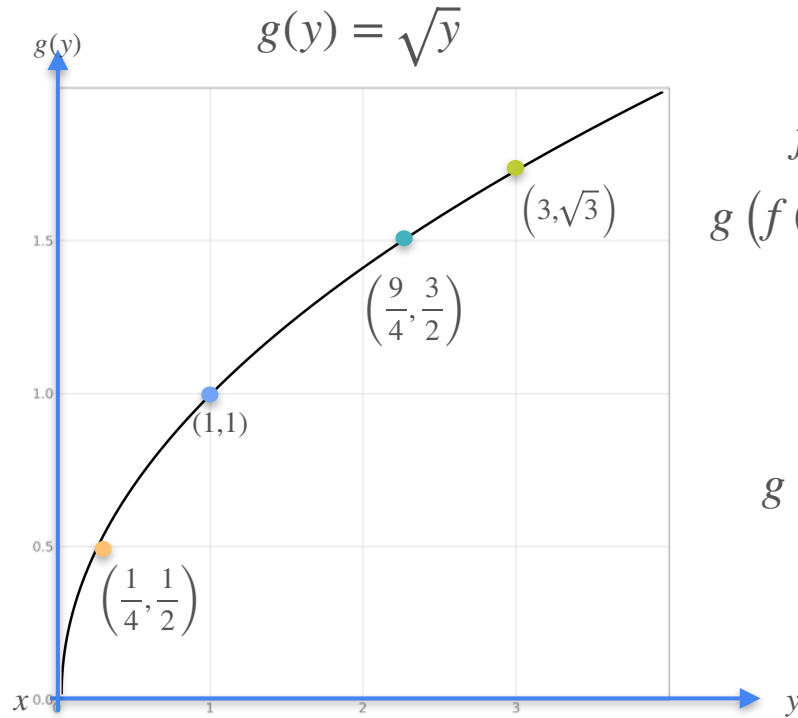
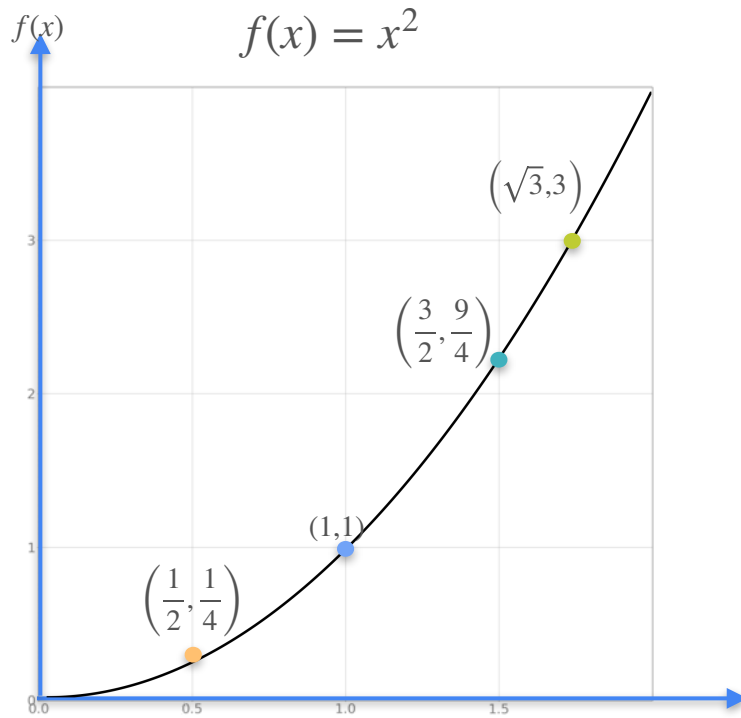
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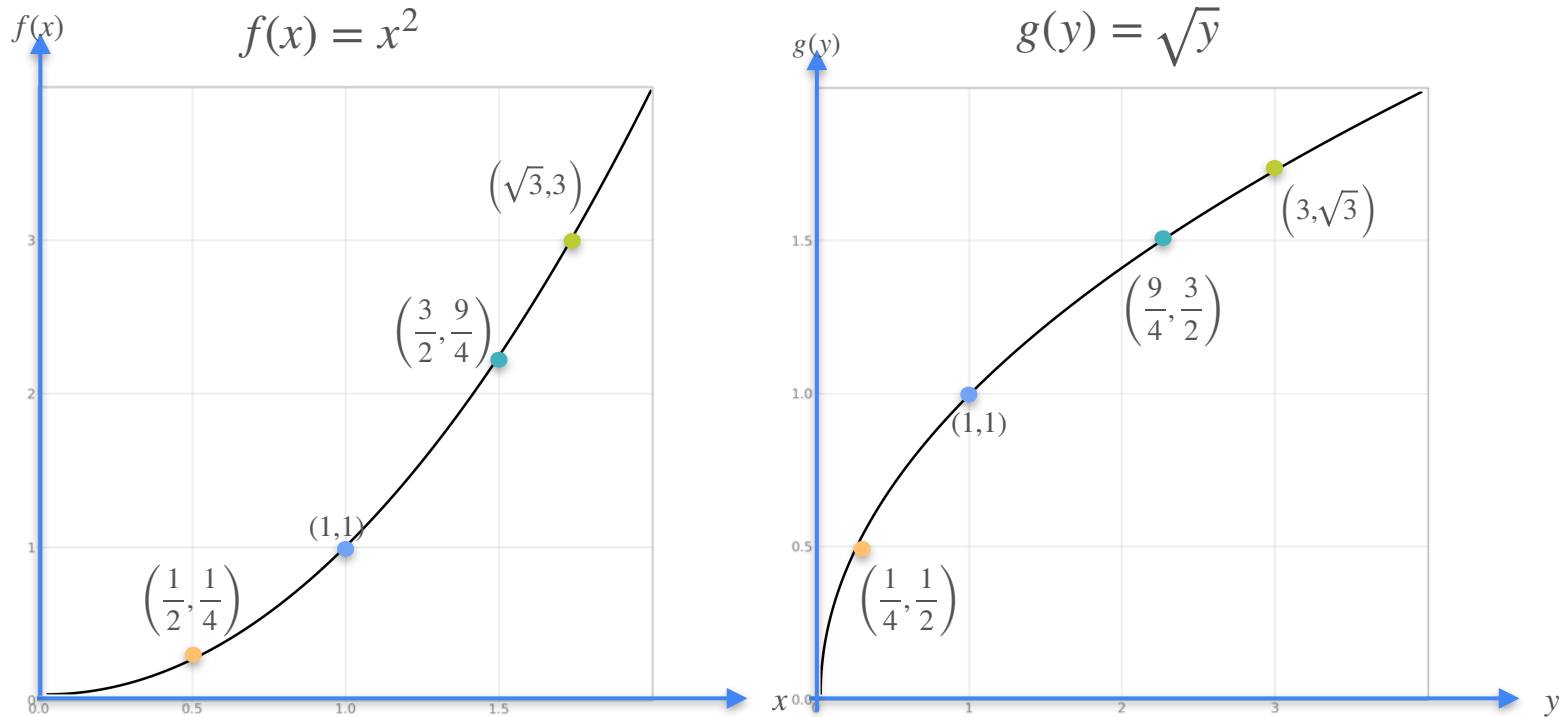
Derivative of the Inverse



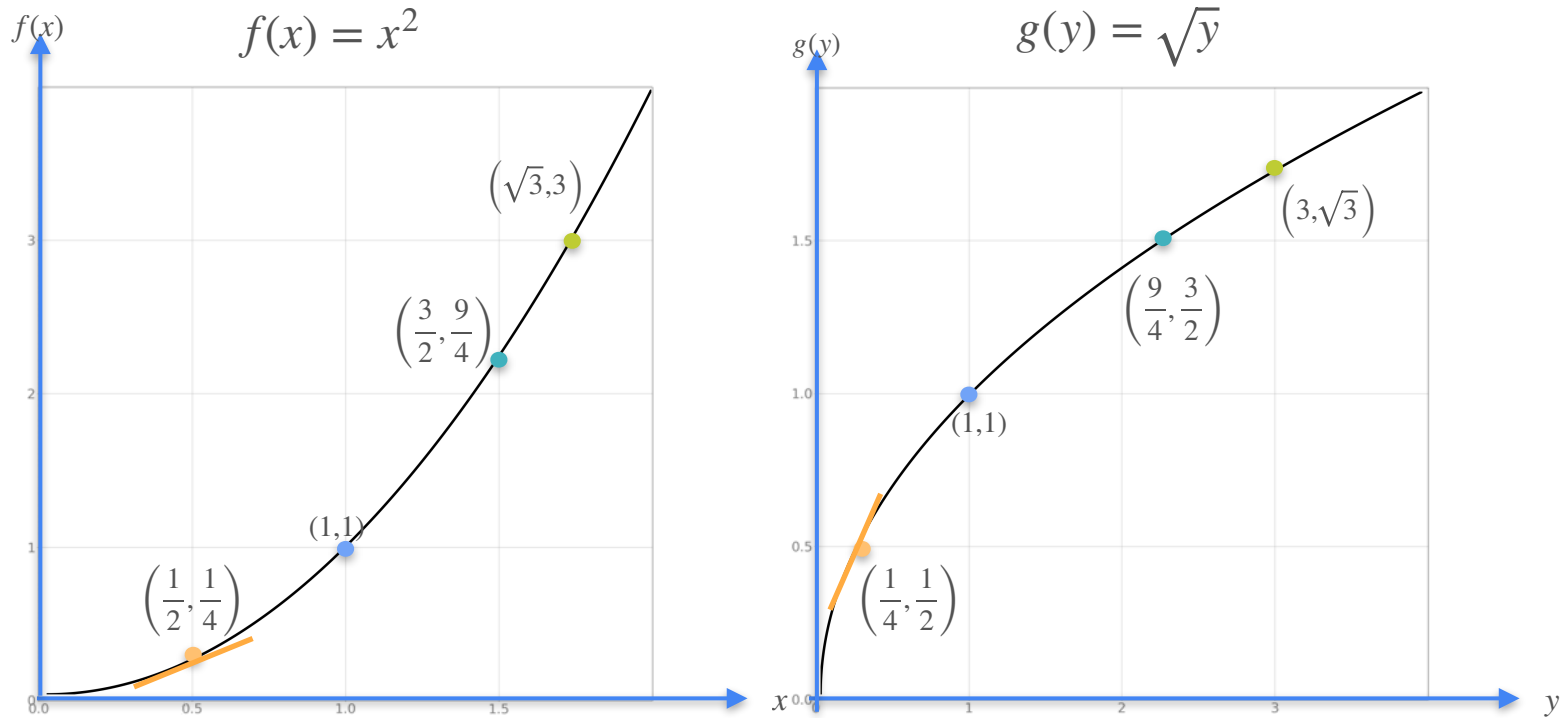
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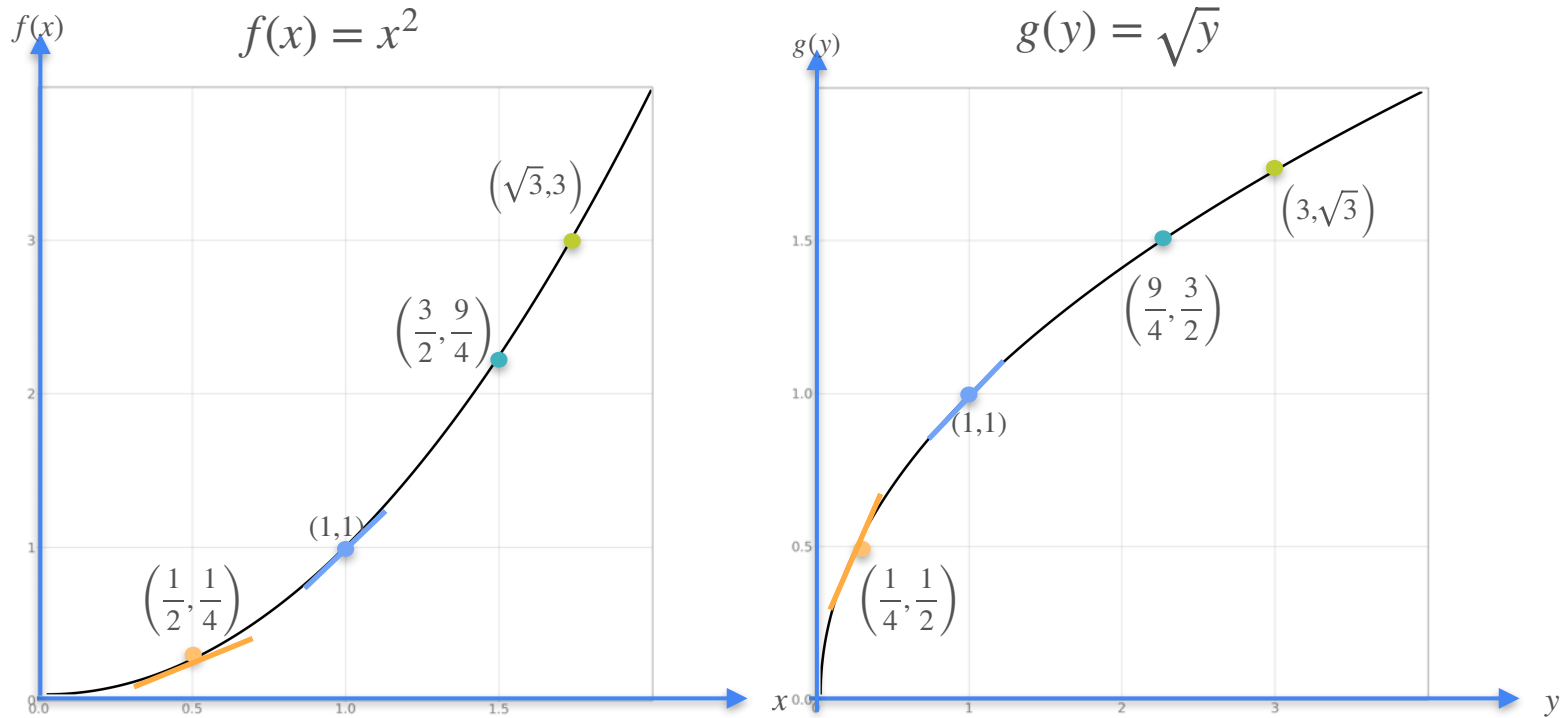
Derivative of the Inverse



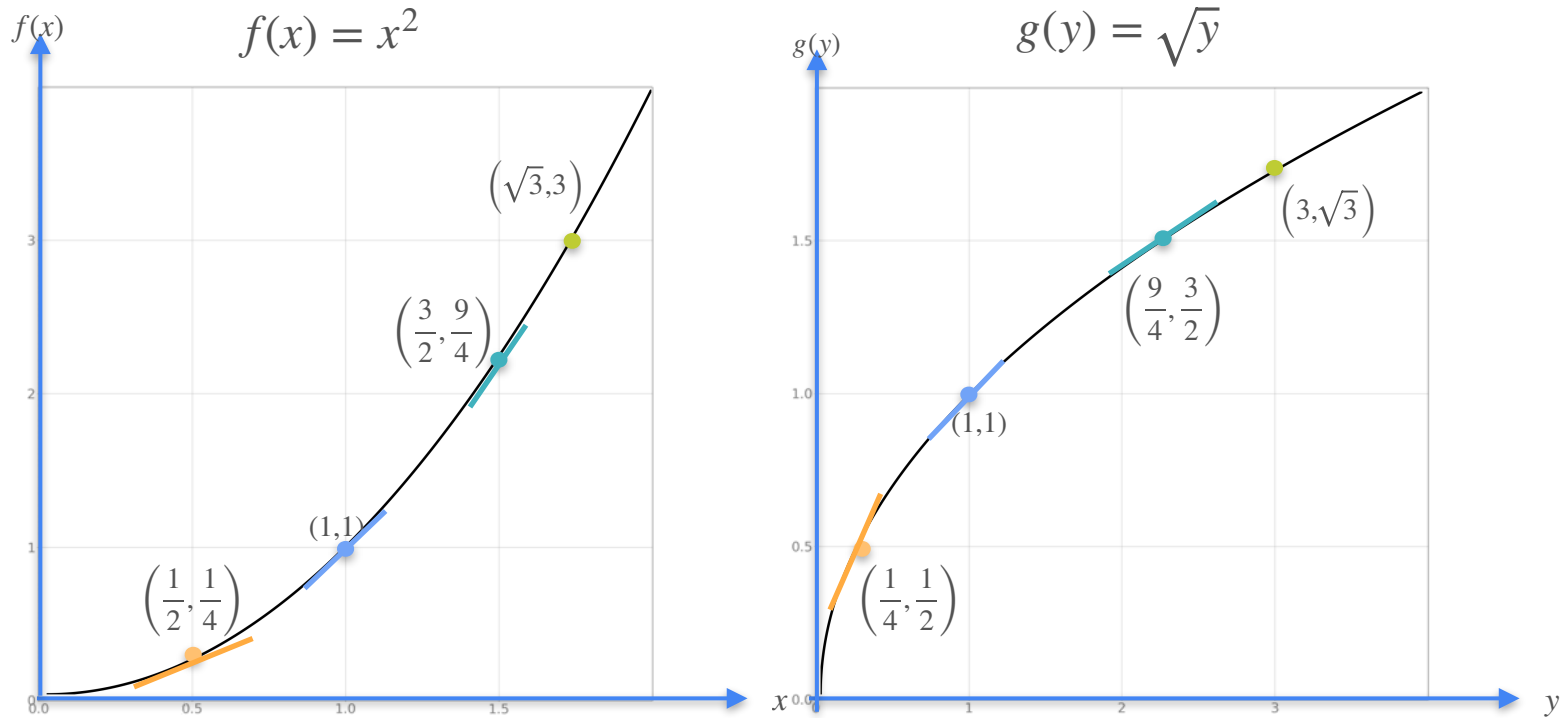
Derivative of the Inverse



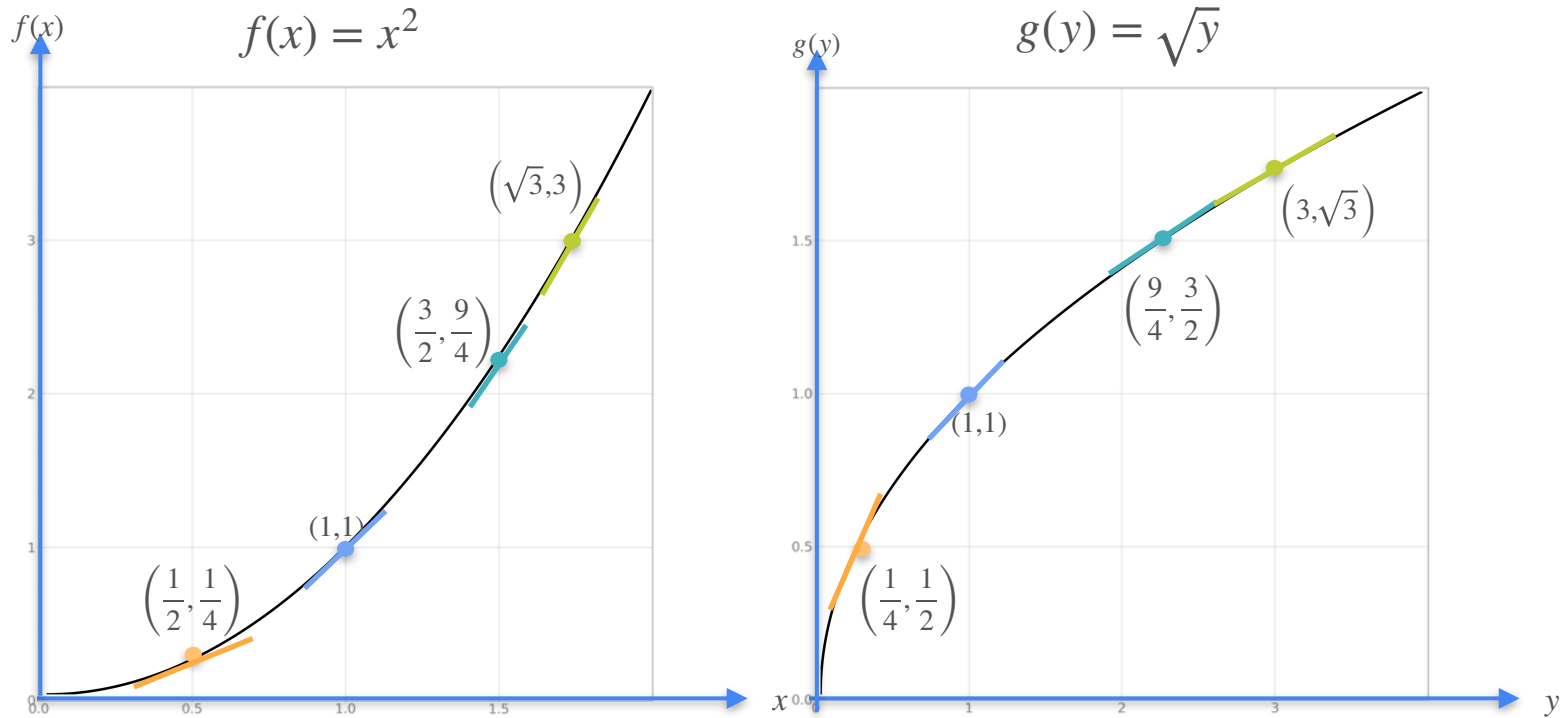
Derivative of the Inverse



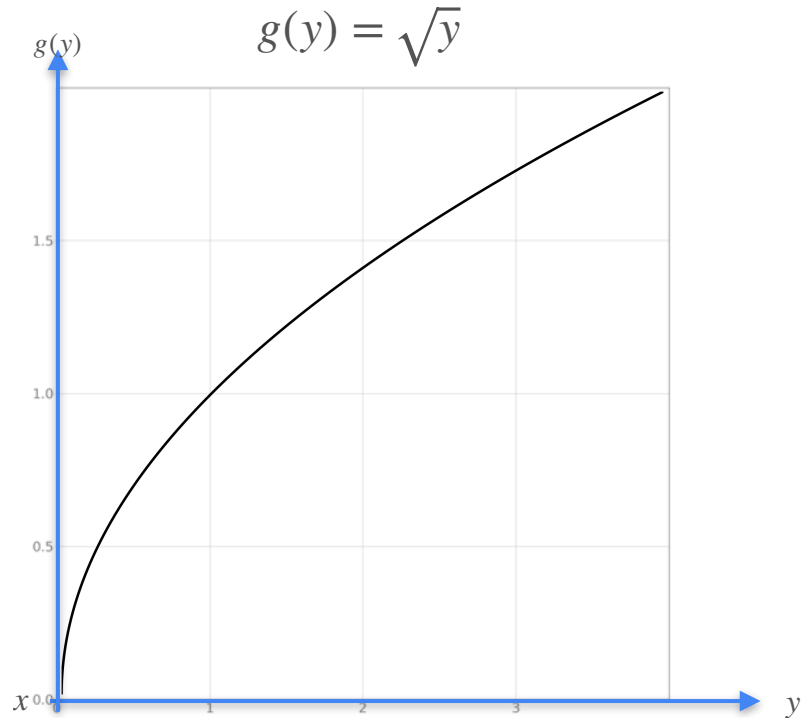
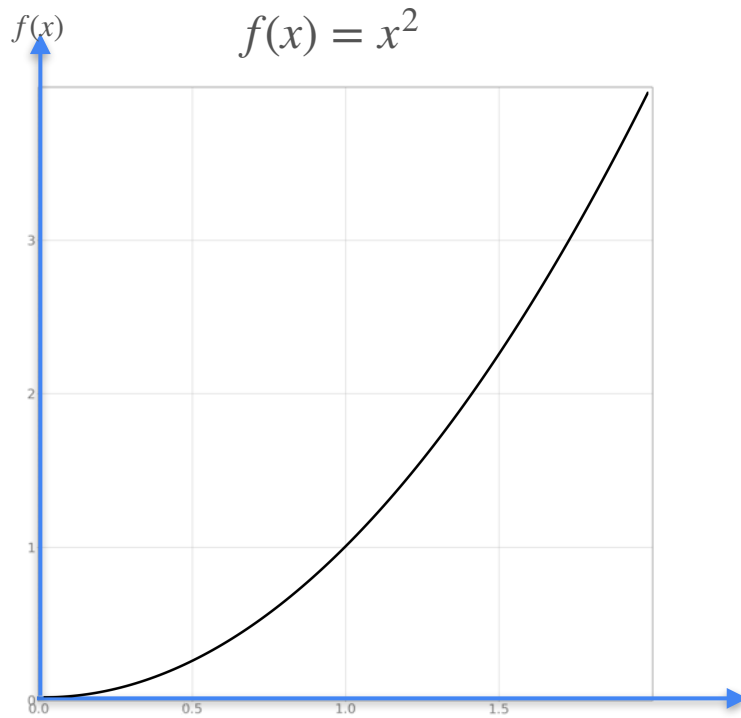
Derivative of the Inverse



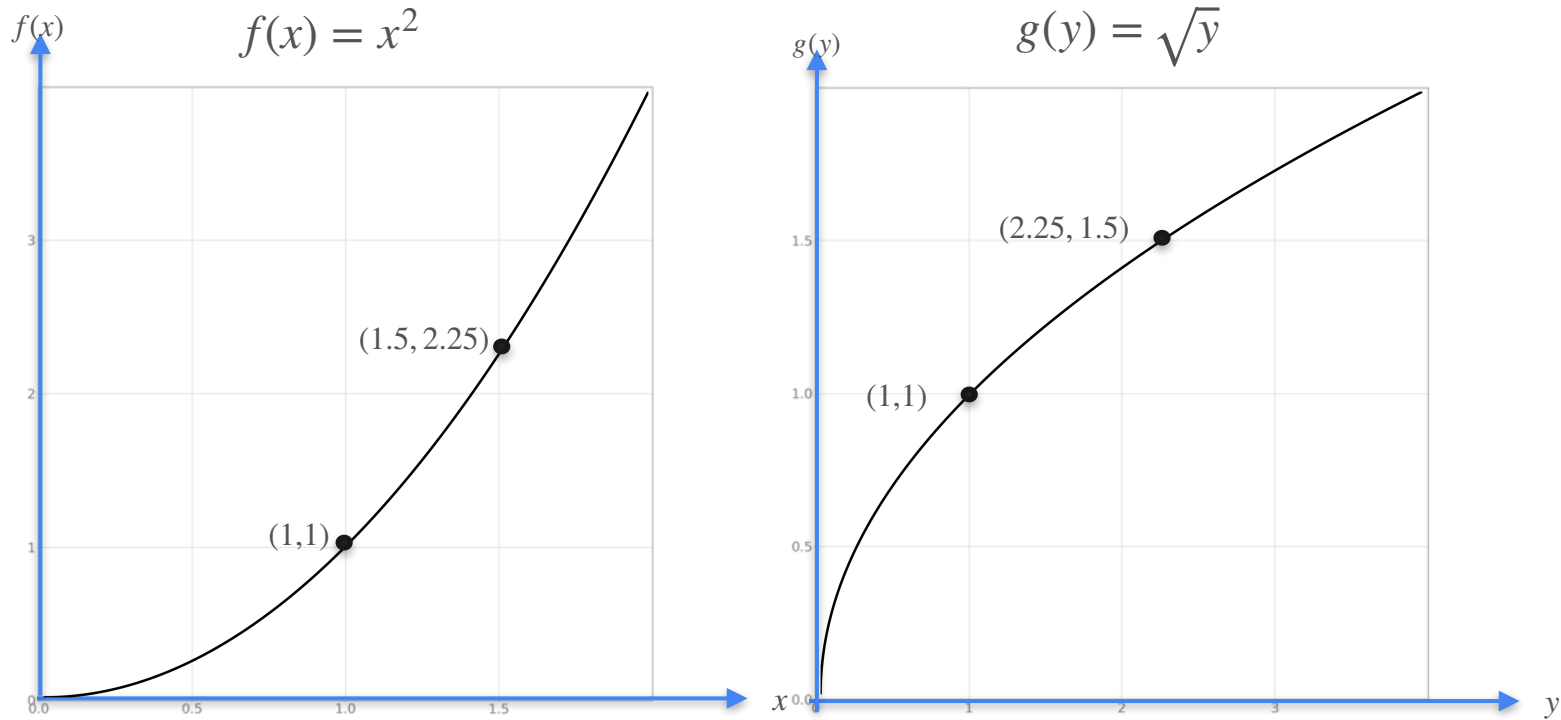
Derivative of the Inverse



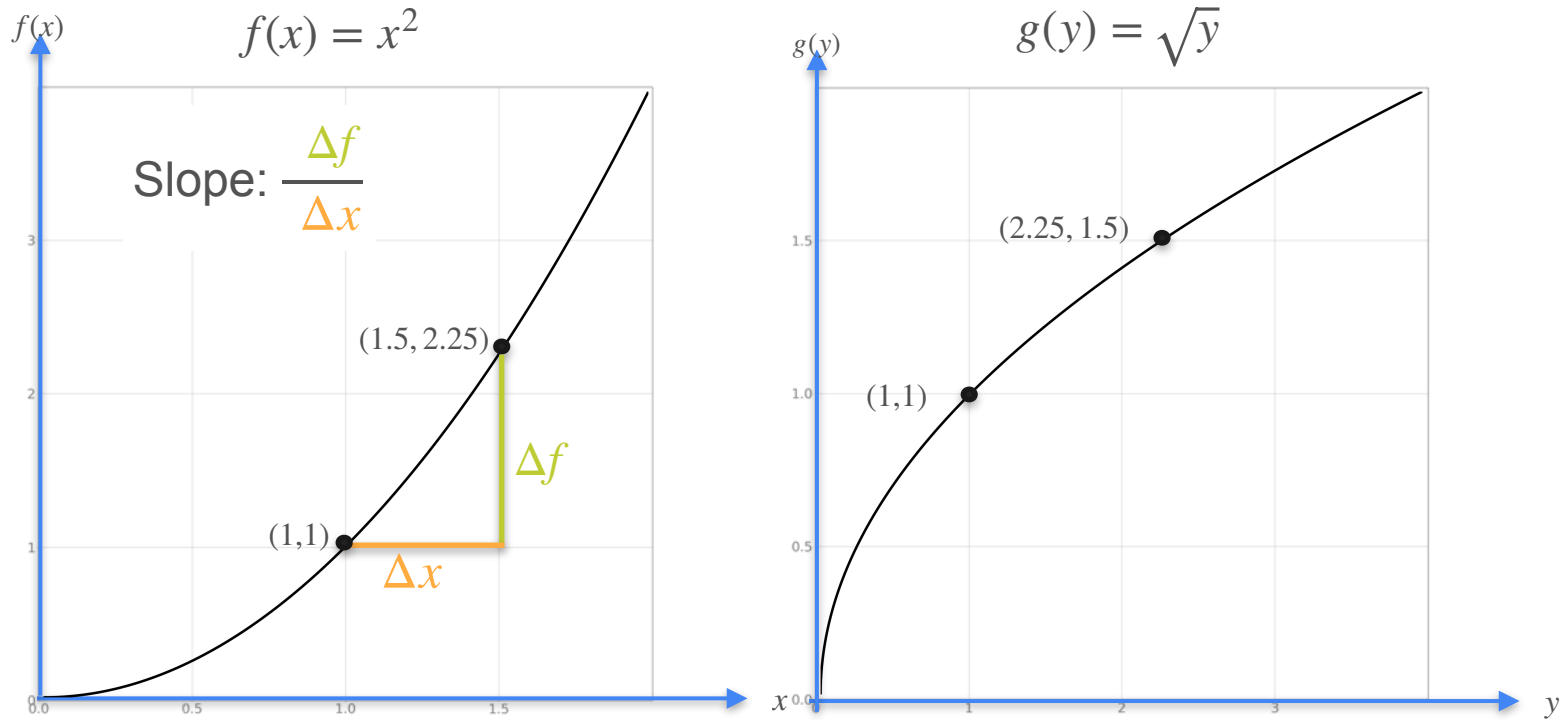
Derivative of the Inverse



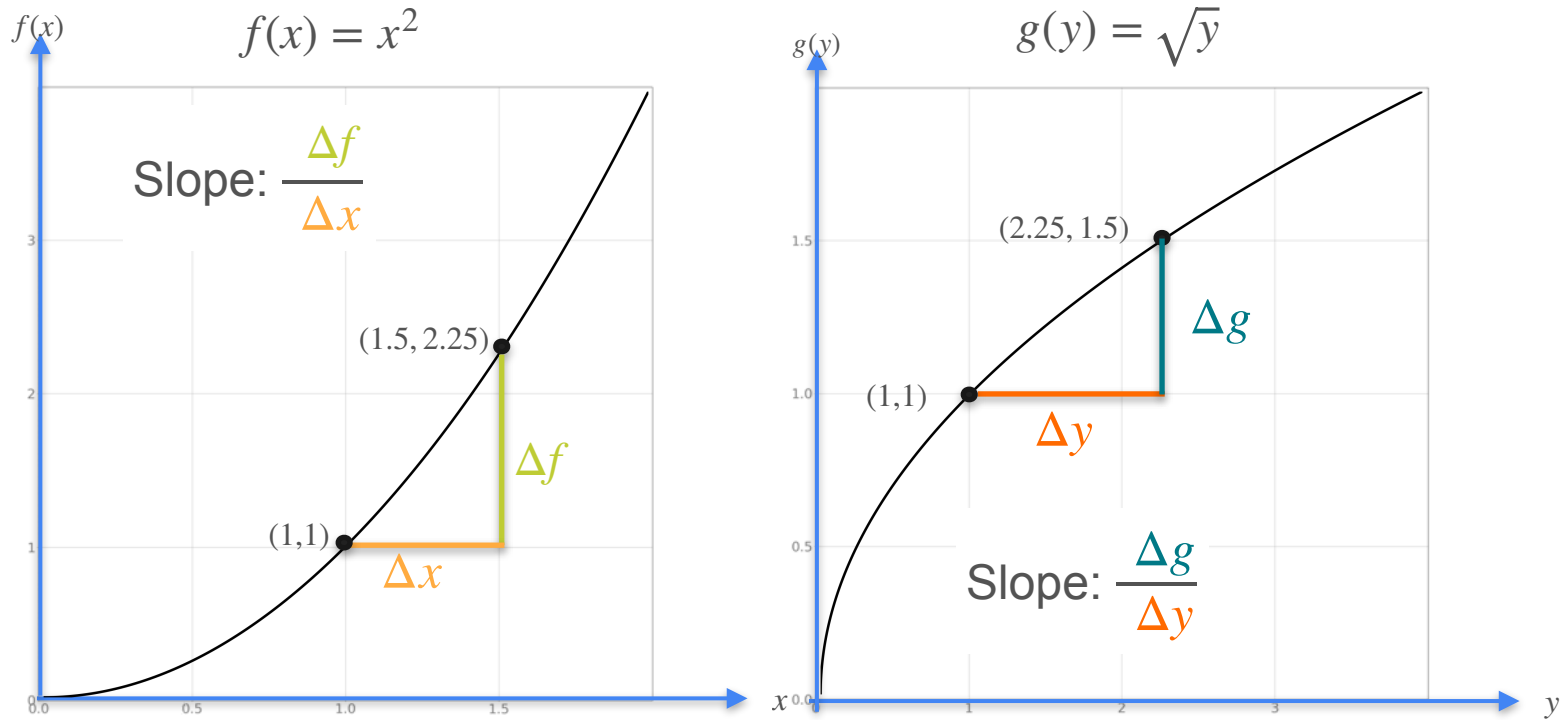
Derivative of the Inverse



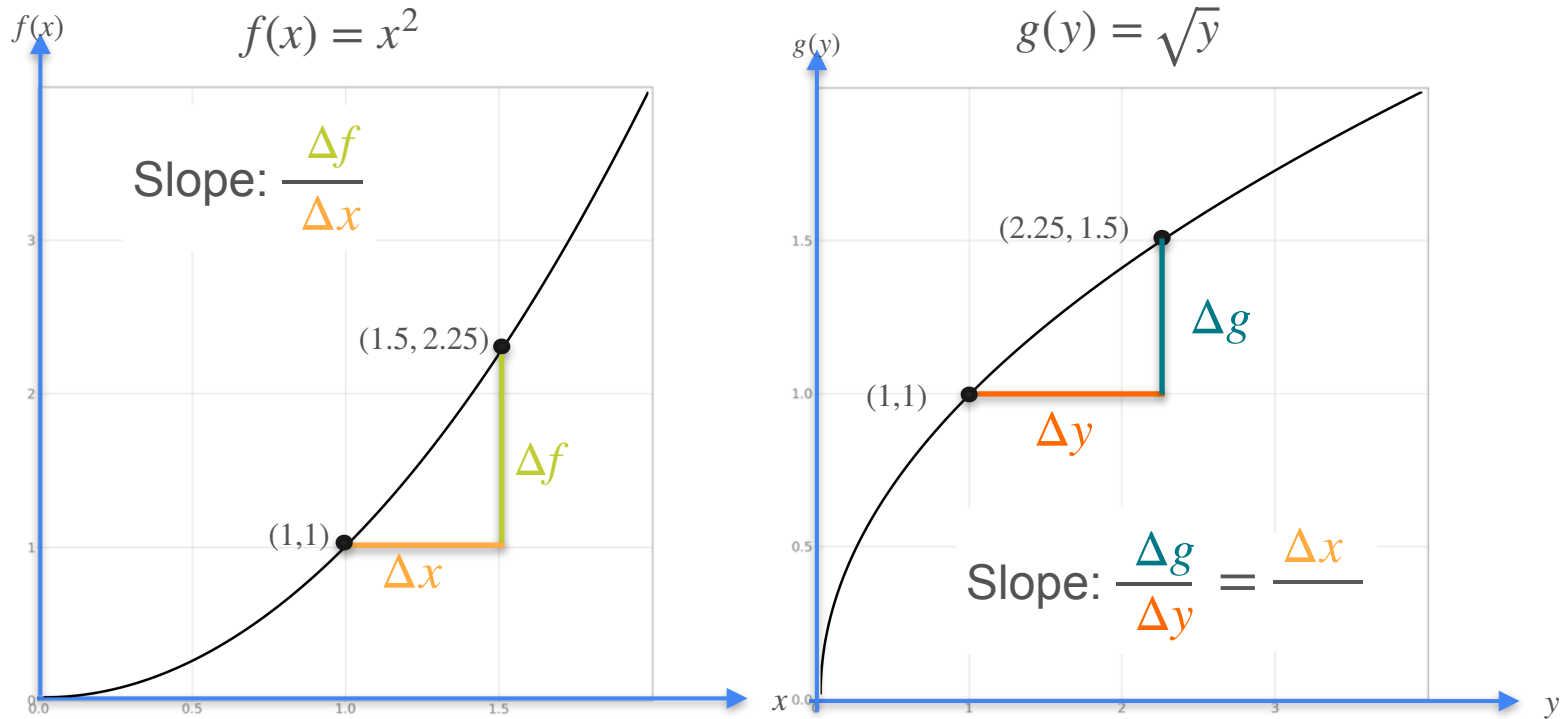
Derivative of the Inverse



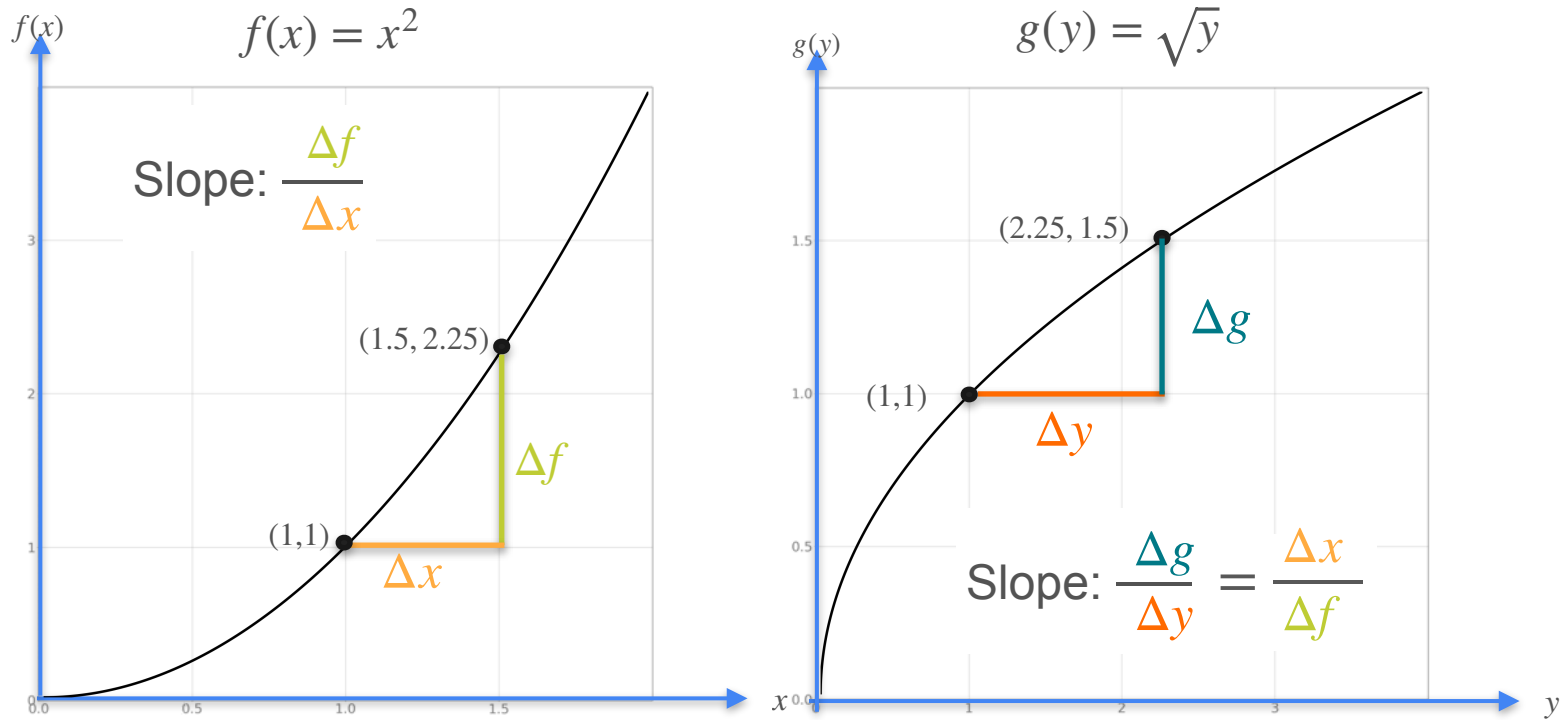
Derivative of the Inverse



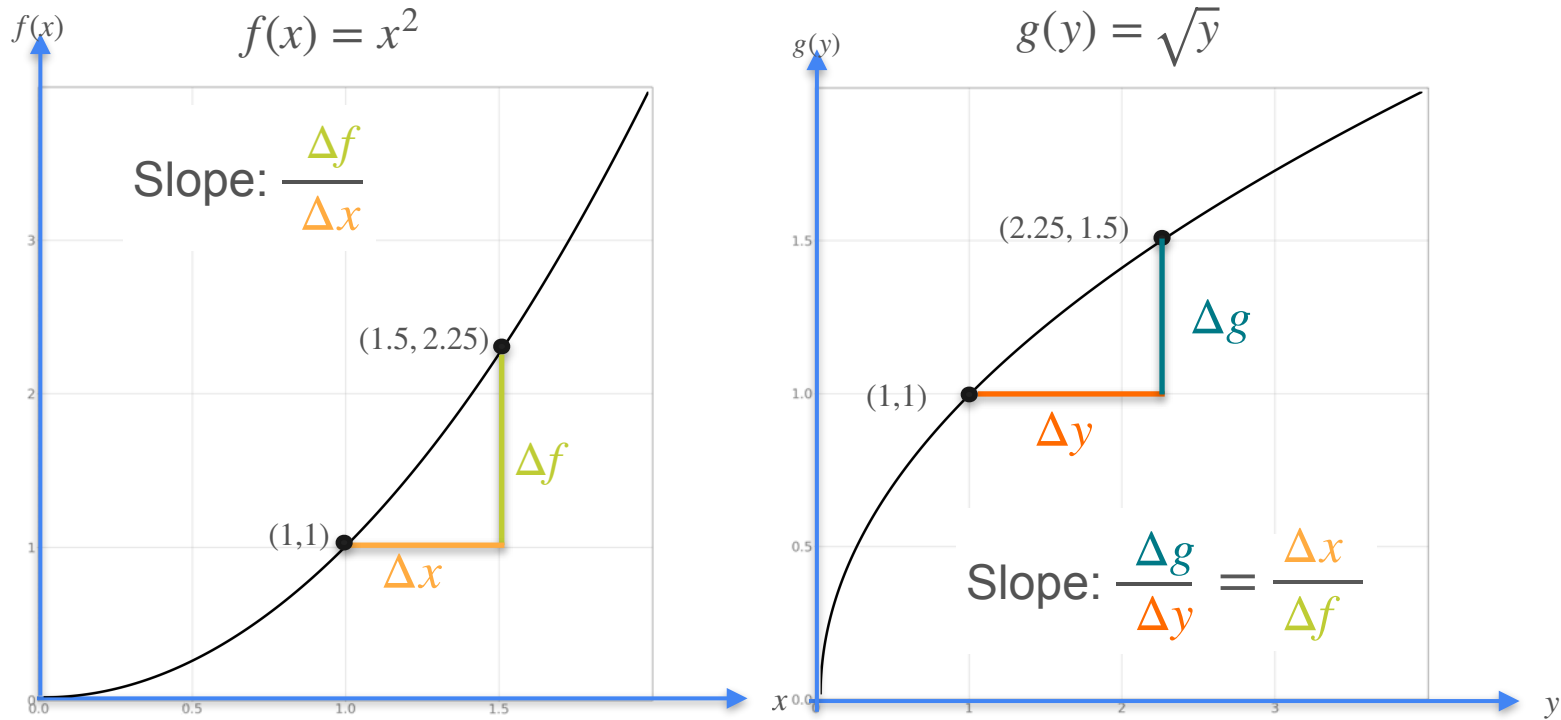
Derivative of the Inverse



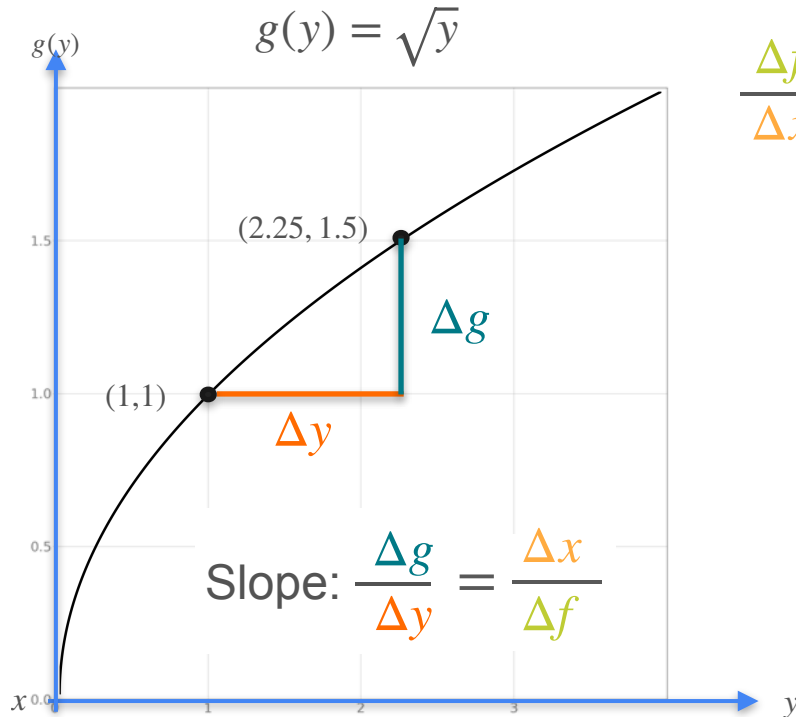
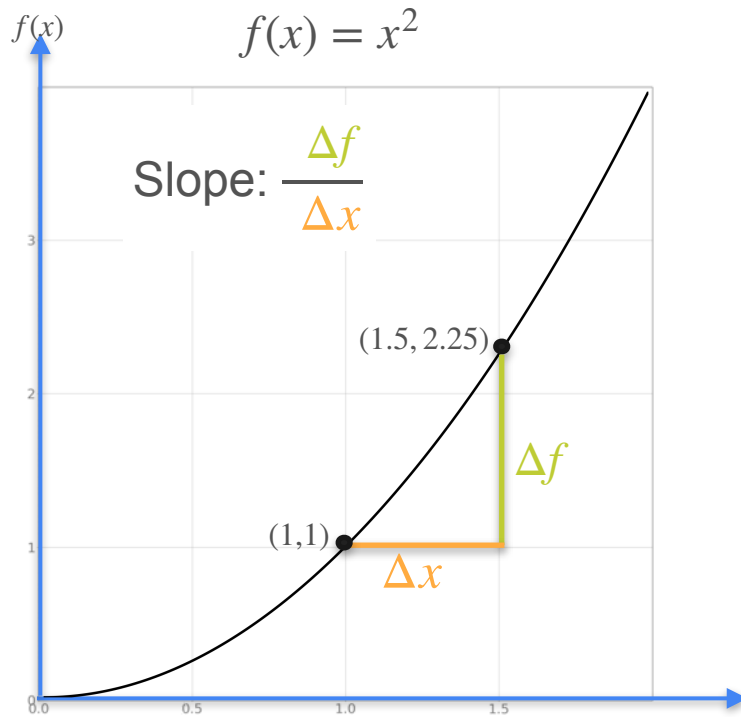
Derivative of the Inverse



Derivative of the Inverse

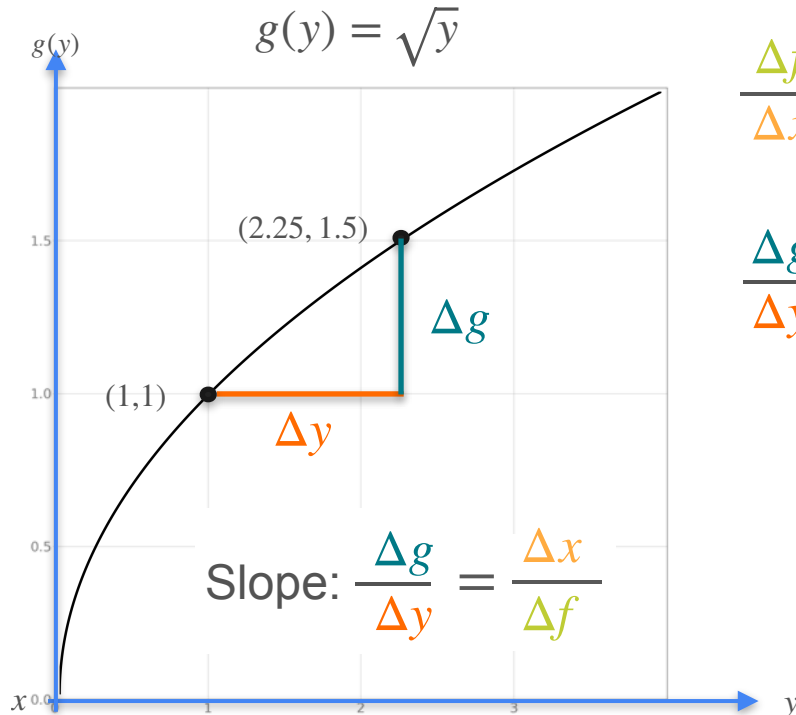
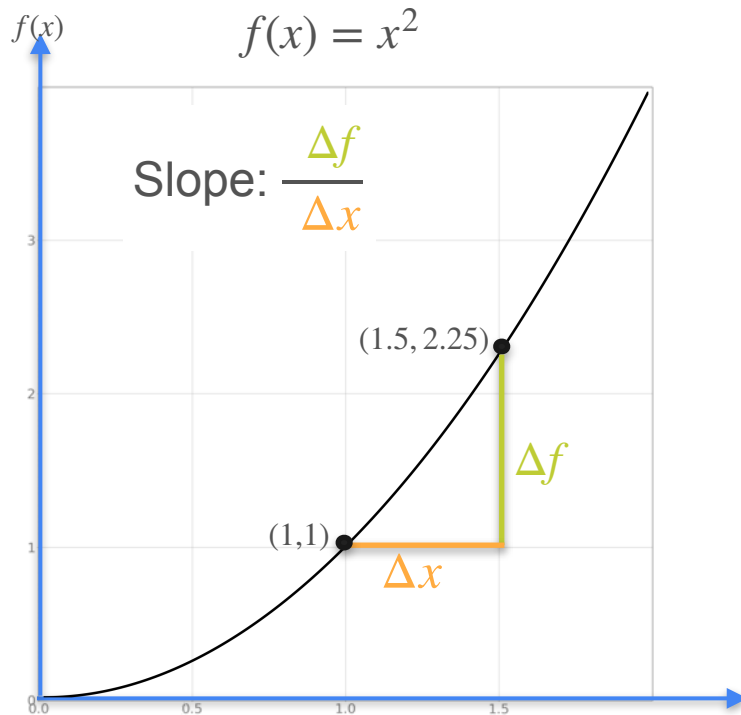


Derivative of the Inverse



$$\frac{\Delta f}{\Delta x} = f'(x)$$

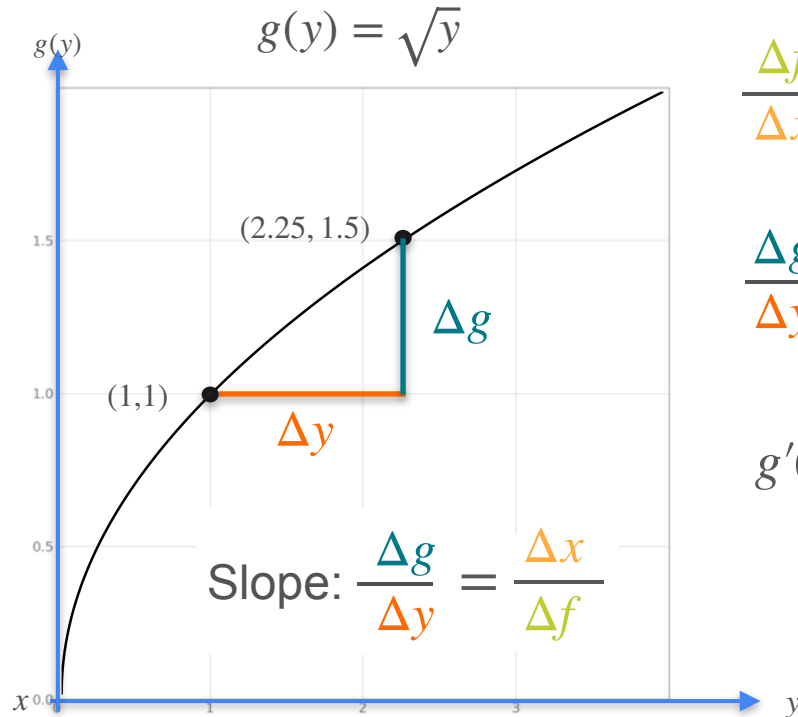
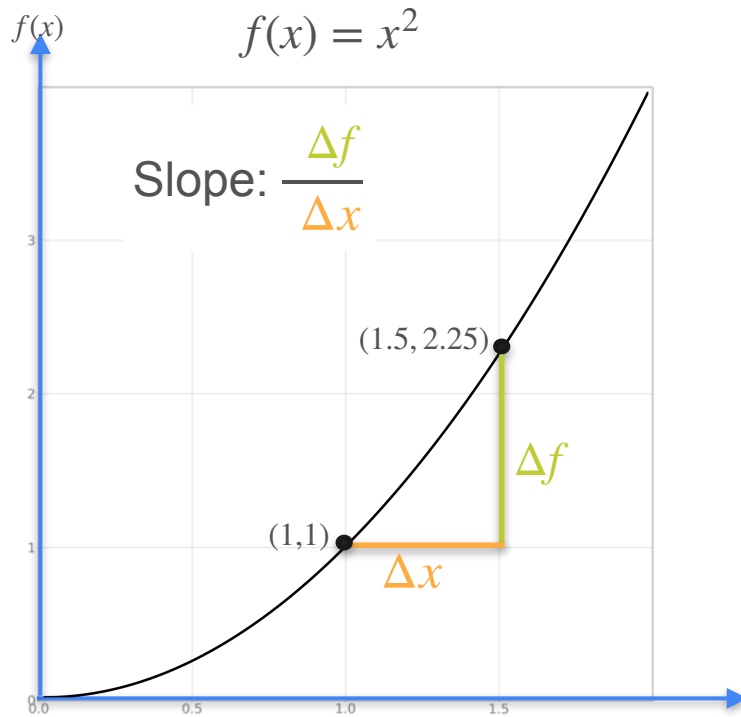
Derivative of the Inverse



$$\frac{\Delta f}{\Delta x} = f'(x)$$

$$\frac{\Delta g}{\Delta y} = g'(y)$$

Derivative of the Inverse

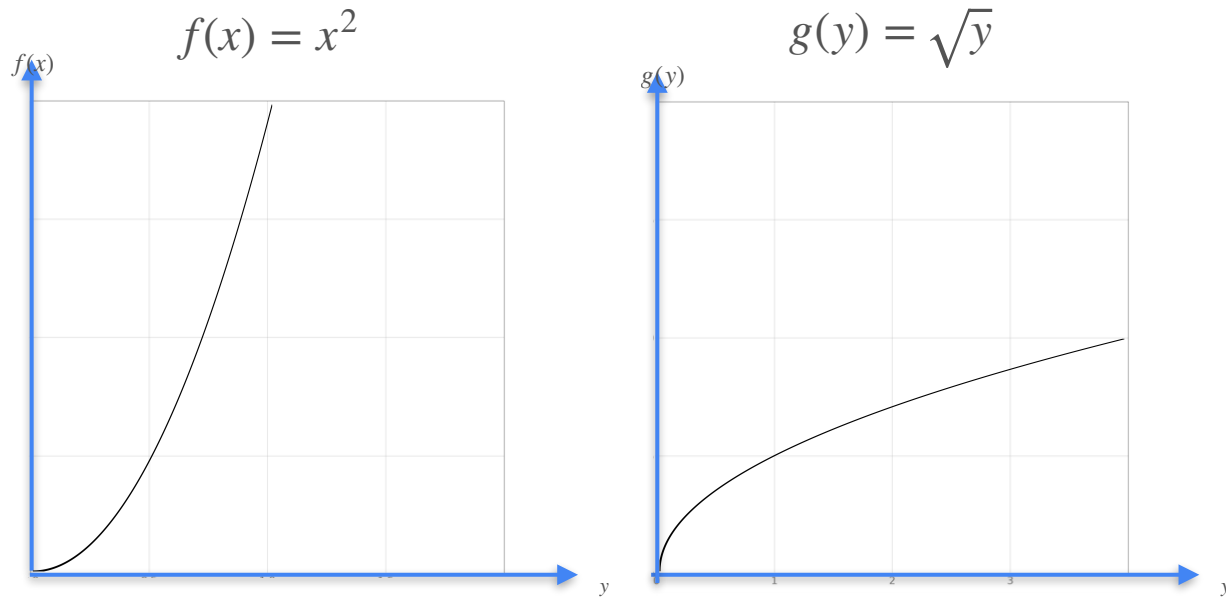


$$\frac{\Delta f}{\Delta x} = f'(x)$$

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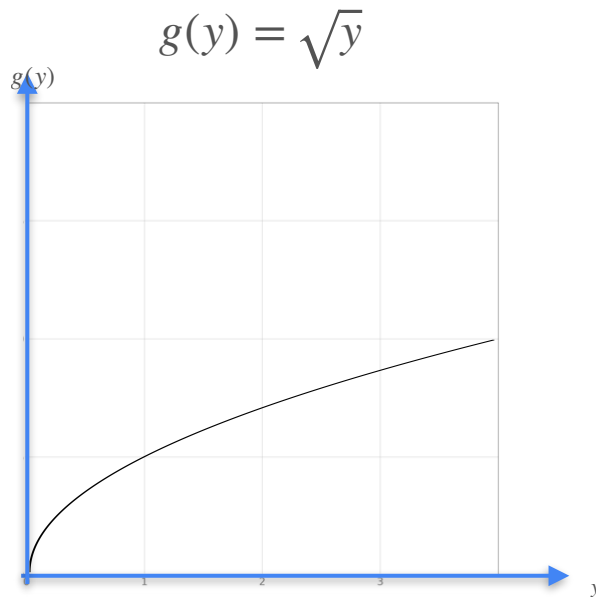
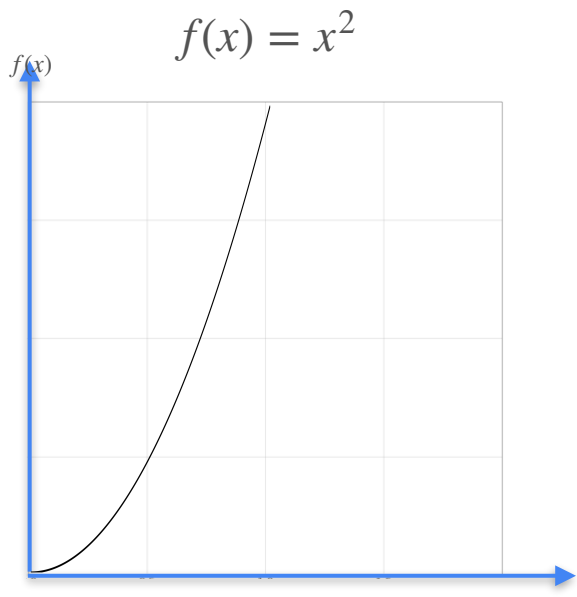
$$g'(y) = \frac{1}{f'(x)}$$

Derivative of the Inverse



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Derivative of the Inverse



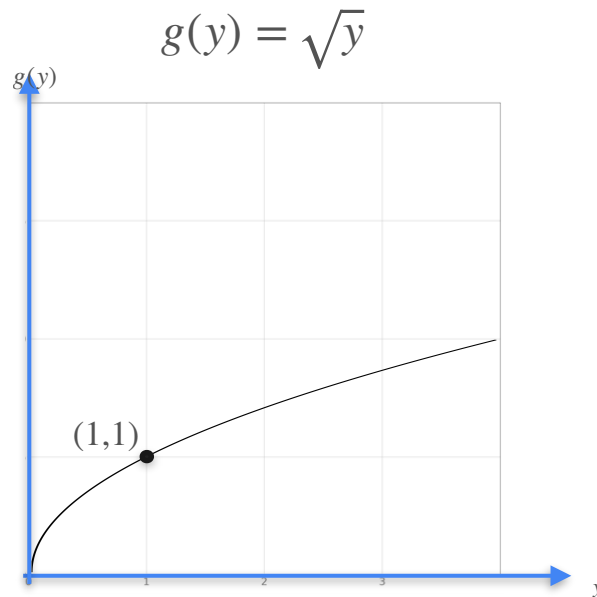
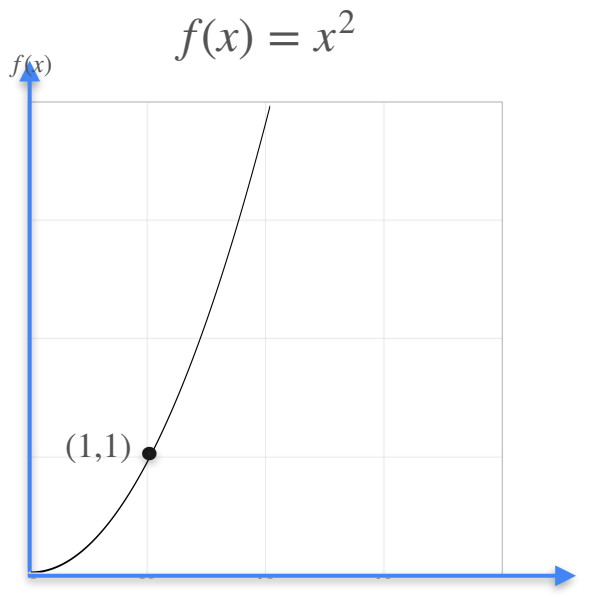
$$g'(y) = \frac{1}{f'(x)}$$

at the point (1,1)

$$f(1) = 1$$

$$g(1) = 1$$

Derivative of the Inverse



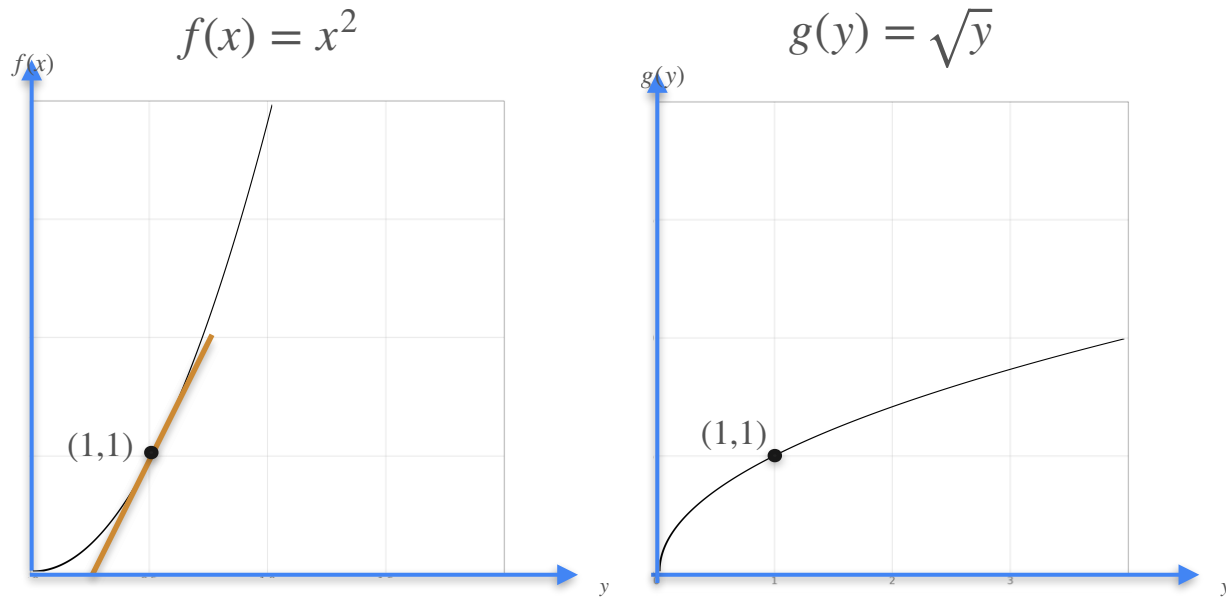
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Derivative of the Inverse



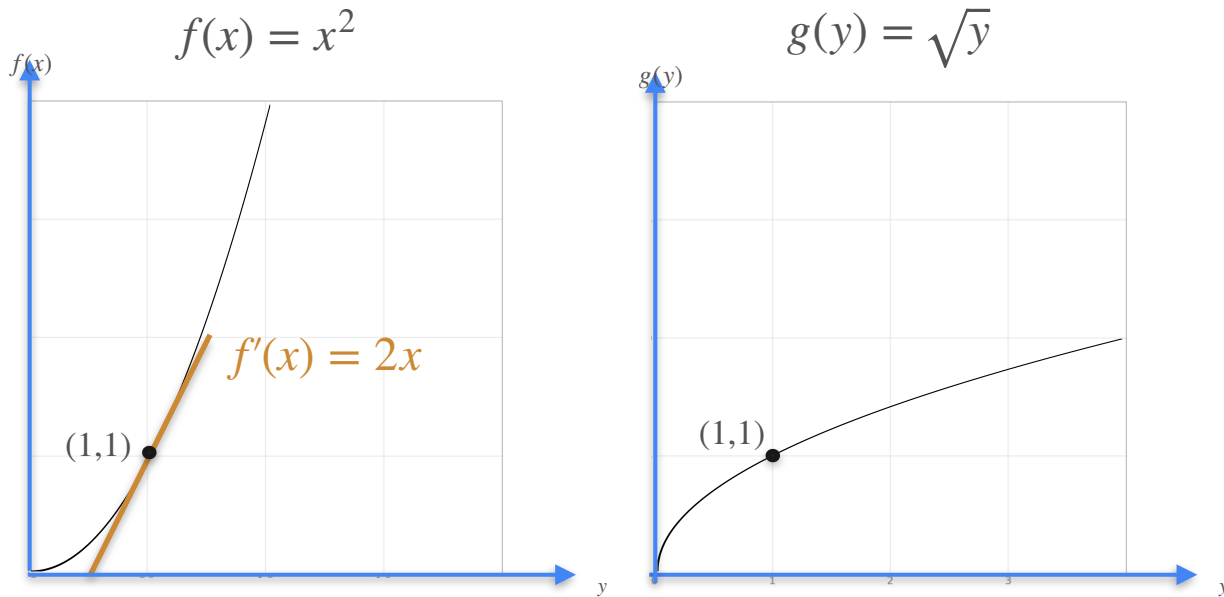
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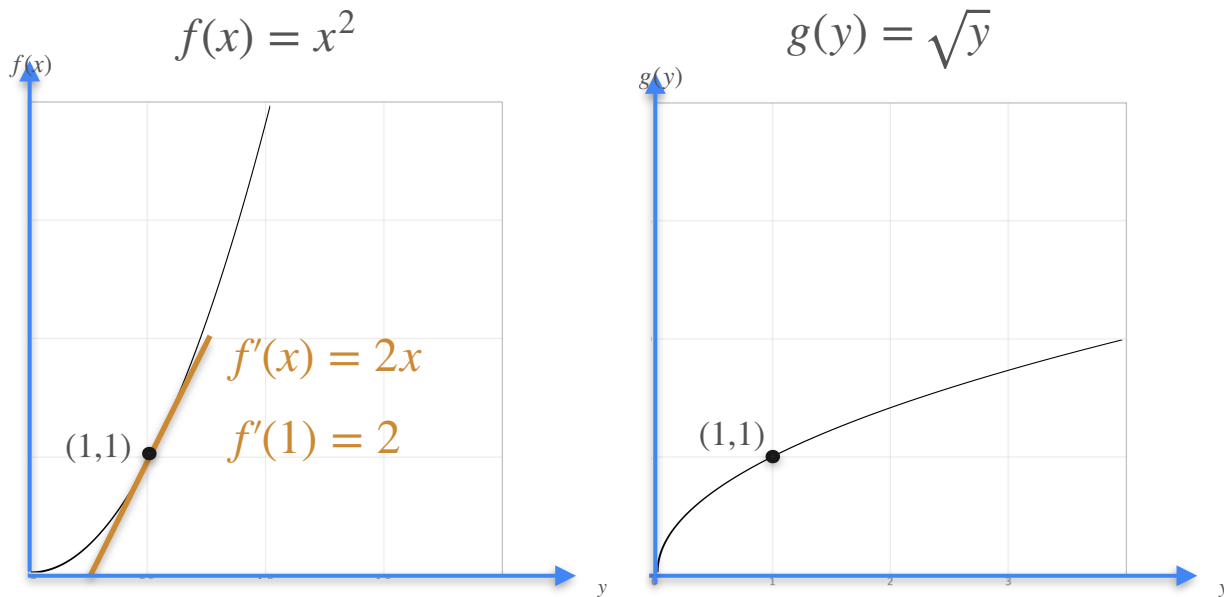
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Derivative of the Inverse



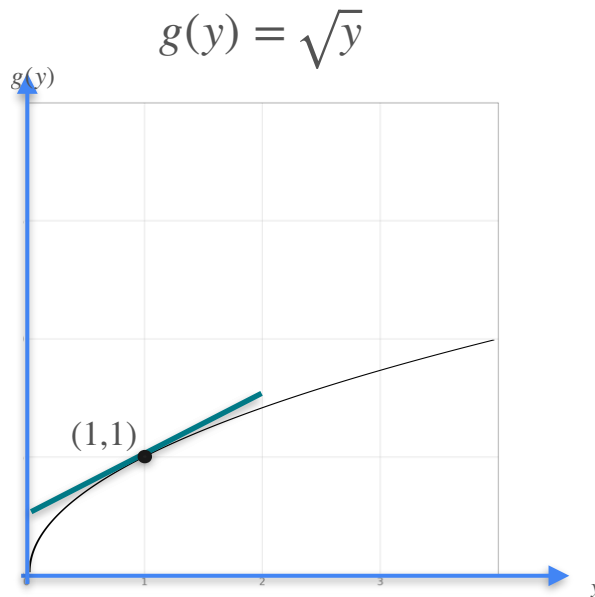
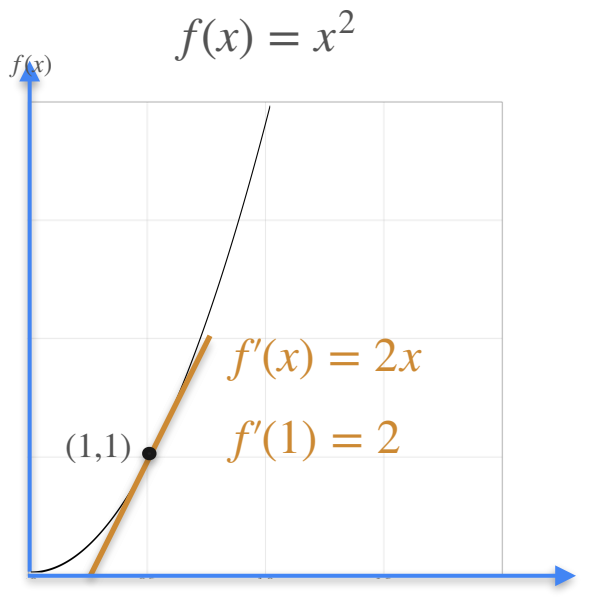
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Derivative of the Inverse



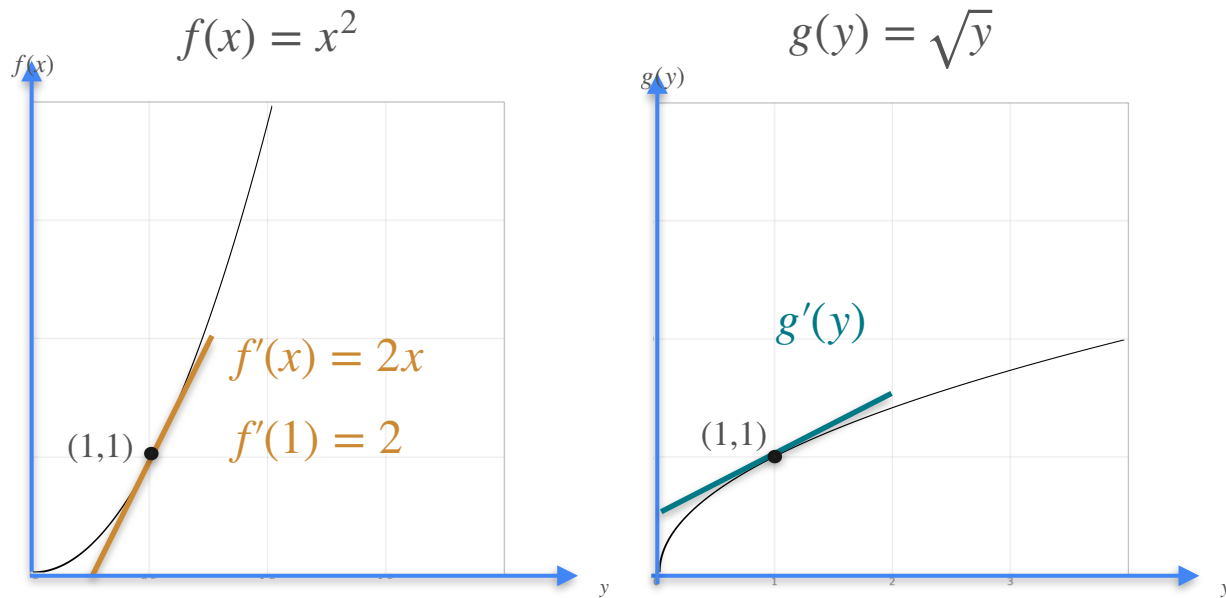
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Derivative of the Inverse



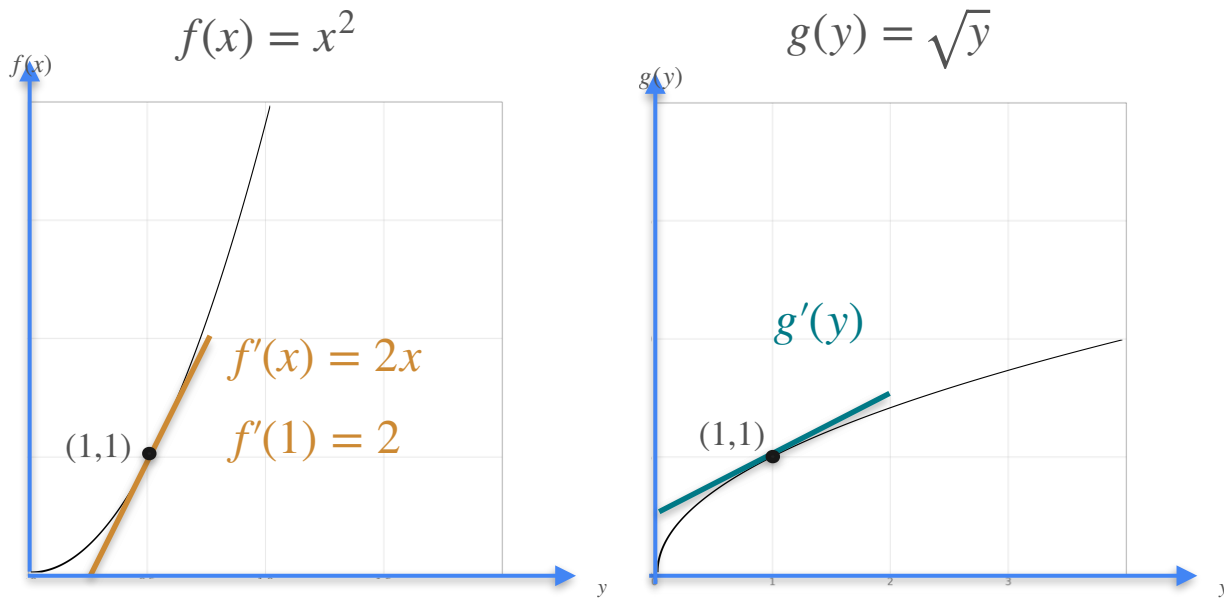
$$g'(y) = \frac{1}{f'(x)}$$

at the point $(1,1)$

$$f(1) = 1$$

$$g(1) = 1$$

Derivative of the Inverse



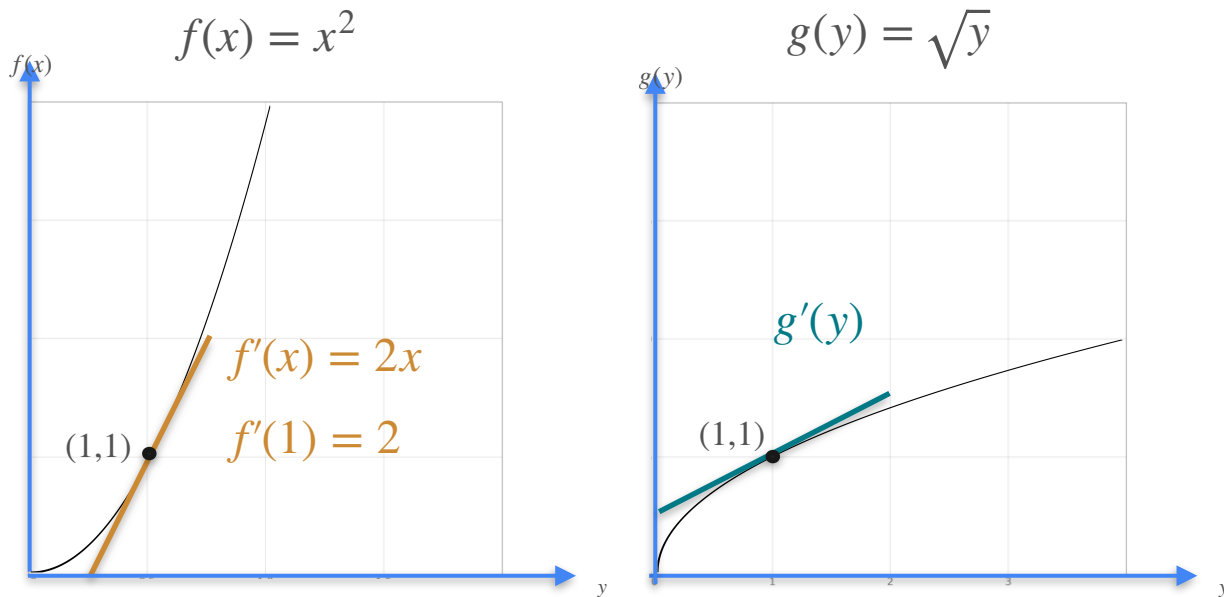
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at the point $(1,1)$

$$f(1) = 1 \qquad g(1) = 1$$

$$g'(1) = \frac{1}{f'(1)}$$

Derivative of the Inverse



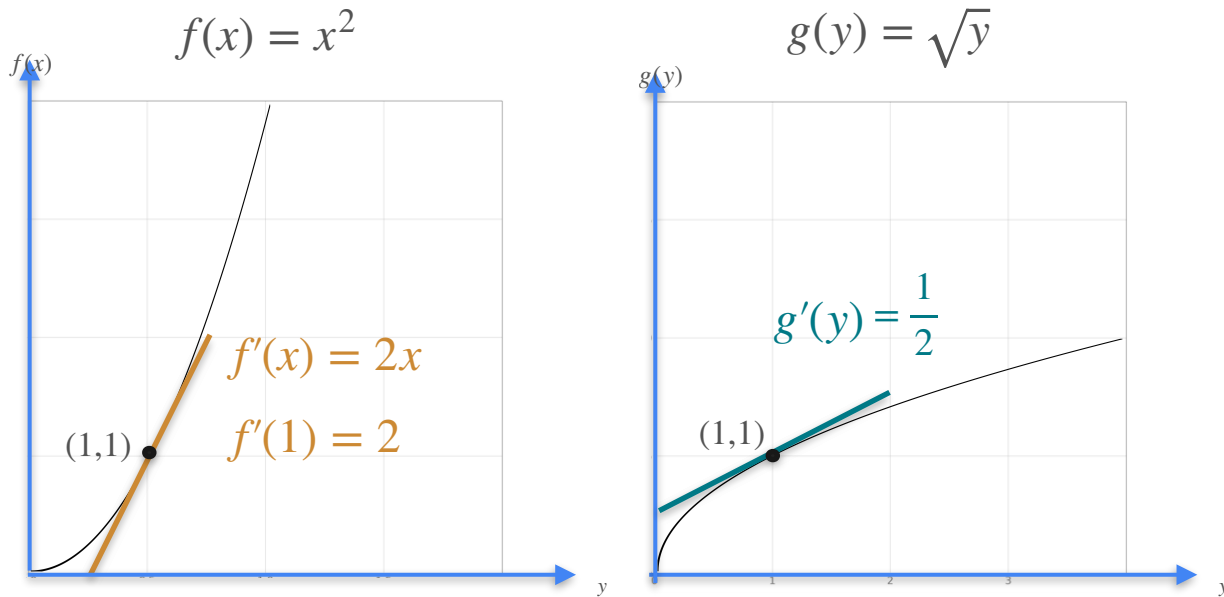
$$g'(y) = \frac{1}{f'(x)}$$

at the point $(1,1)$

$$f(1) = 1 \qquad g(1) = 1$$

$$g'(1) = \frac{1}{f'(1)} = \frac{1}{2}$$

Derivative of the Inverse



$$g'(y) = \frac{1}{f'(x)}$$

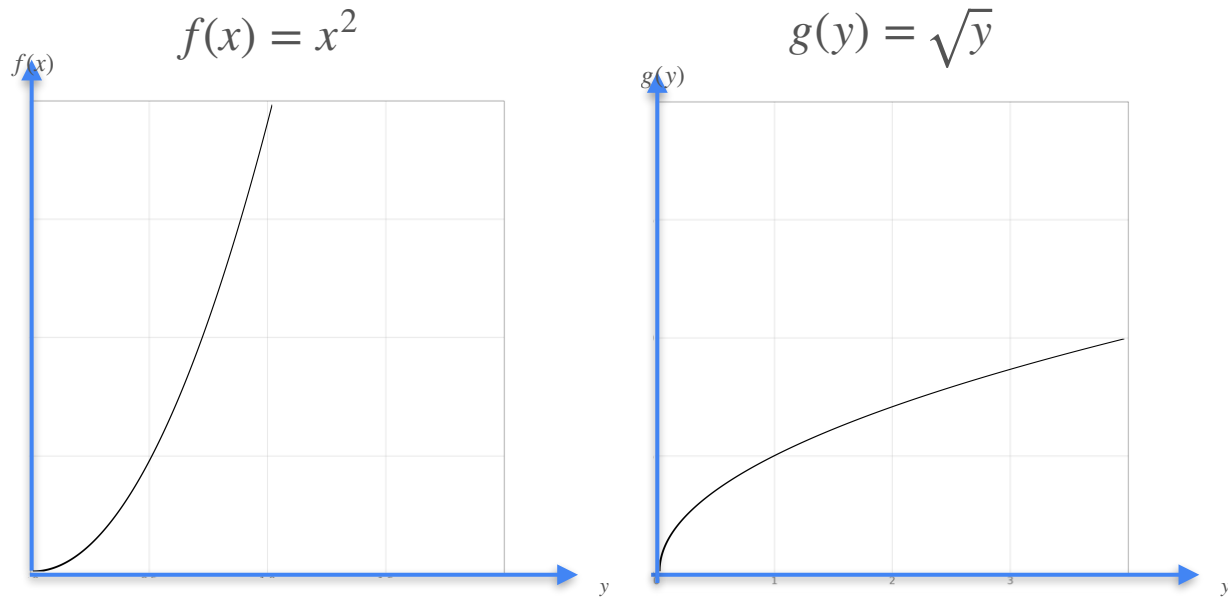
at the point $(1,1)$

$$f(1) = 1$$

$$g(1) = 1$$

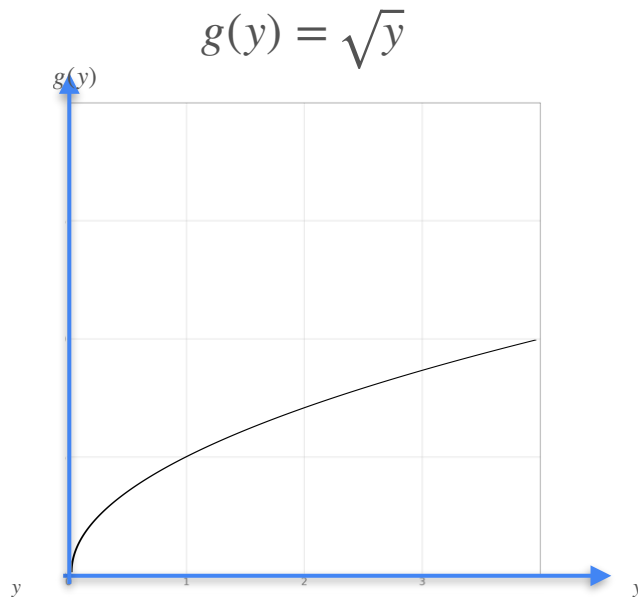
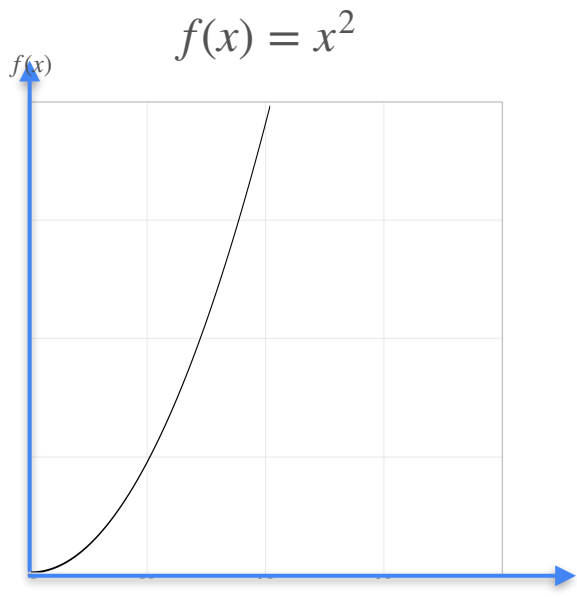
$$g'(1) = \frac{1}{f'(1)}$$

Derivative of the Inverse



$$g'(y) = \frac{1}{f'(x)}$$

Derivative of the Inverse



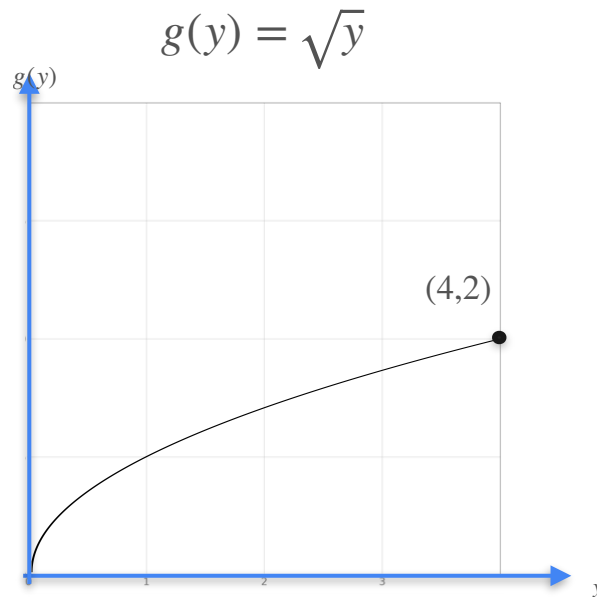
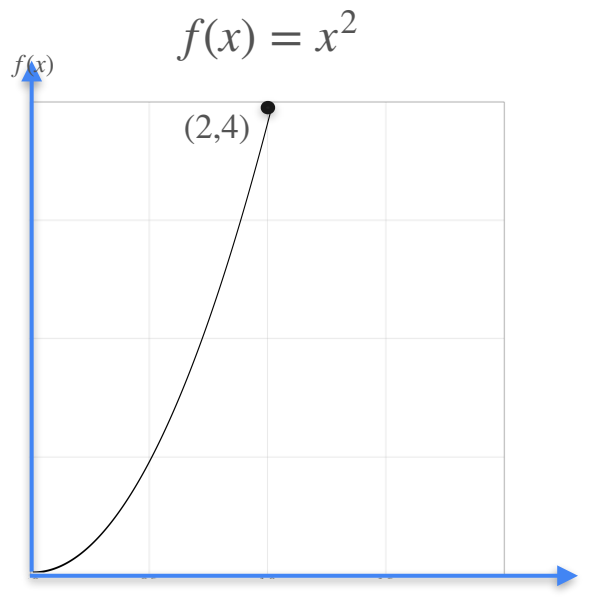
$$g'(y) = \frac{1}{f'(x)}$$

at the point (2,4)

$$f(2) = 4$$

$$g(4) = 2$$

Derivative of the Inverse



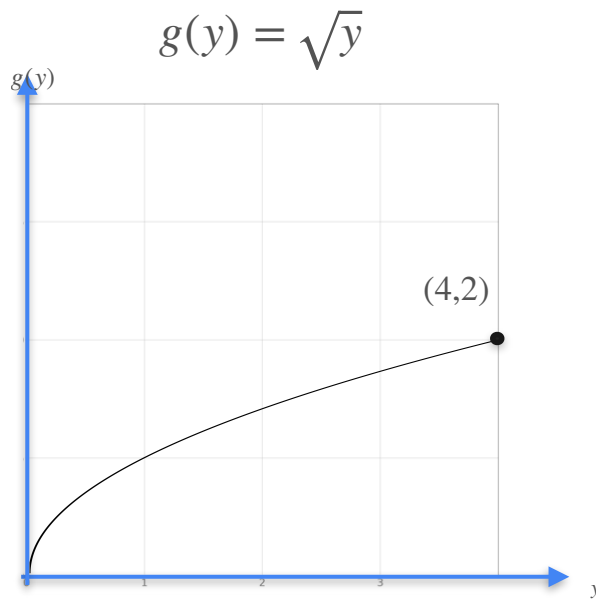
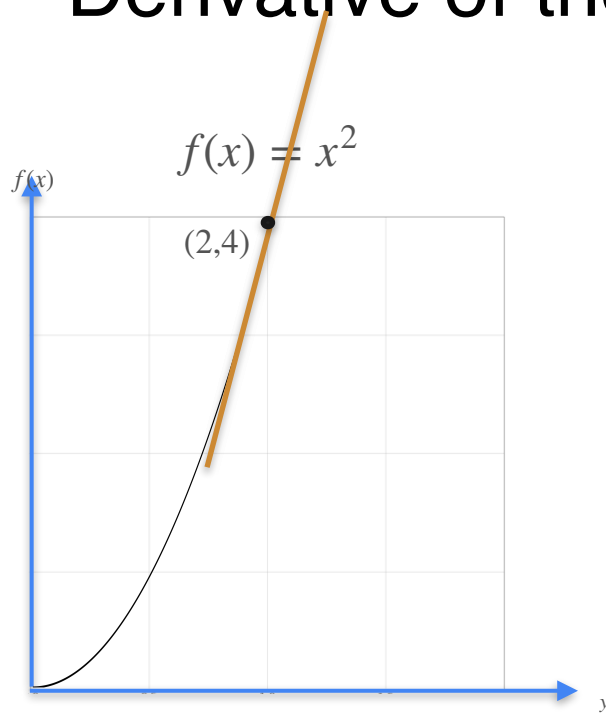
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Derivative of the Inverse



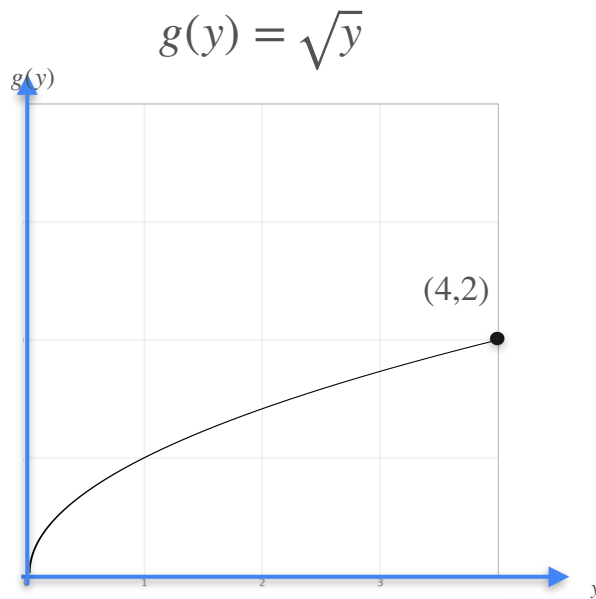
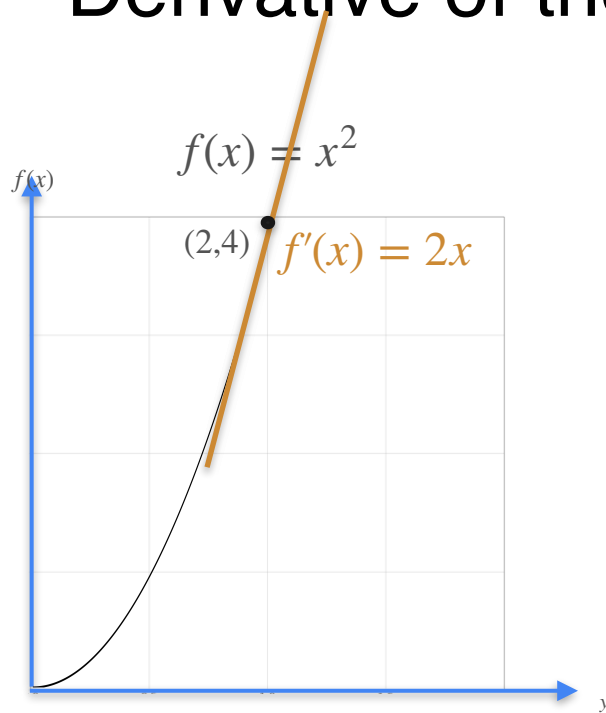
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$$g(4) = 2$$

Derivative of the Inverse



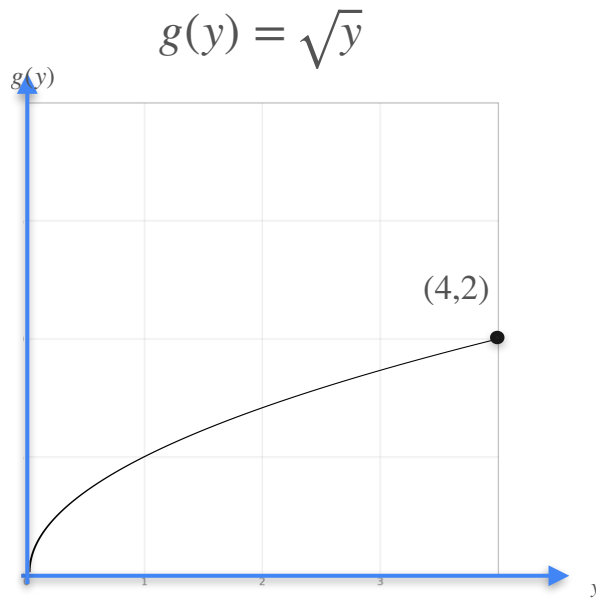
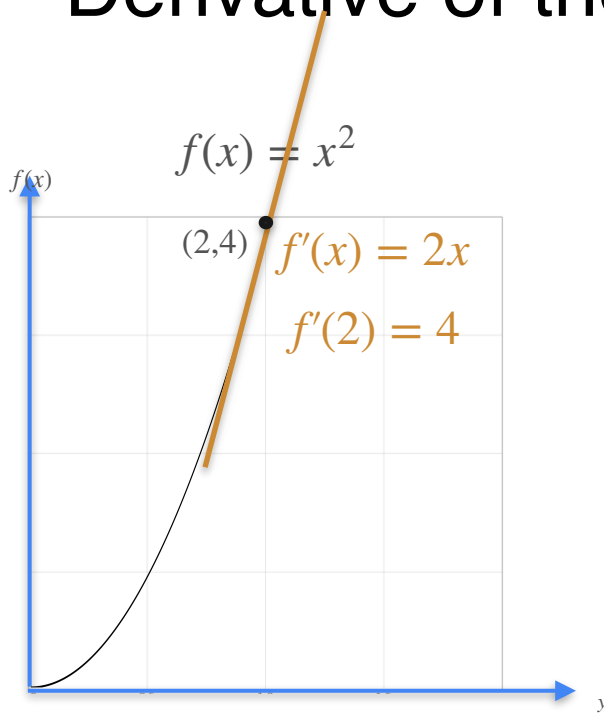
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Derivative of the Inverse



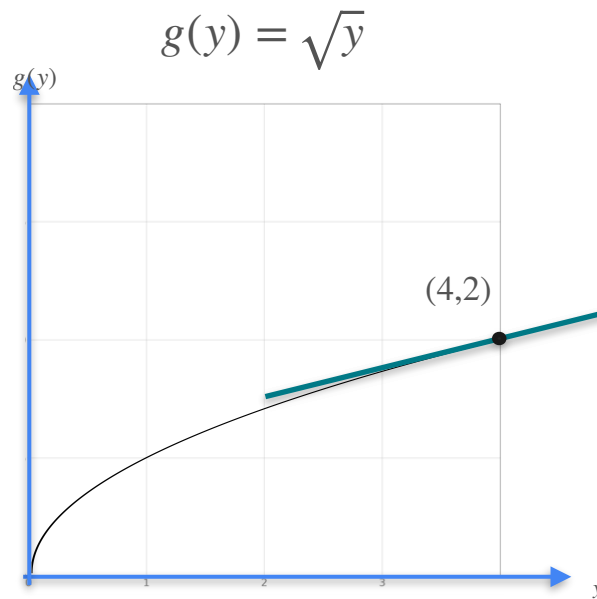
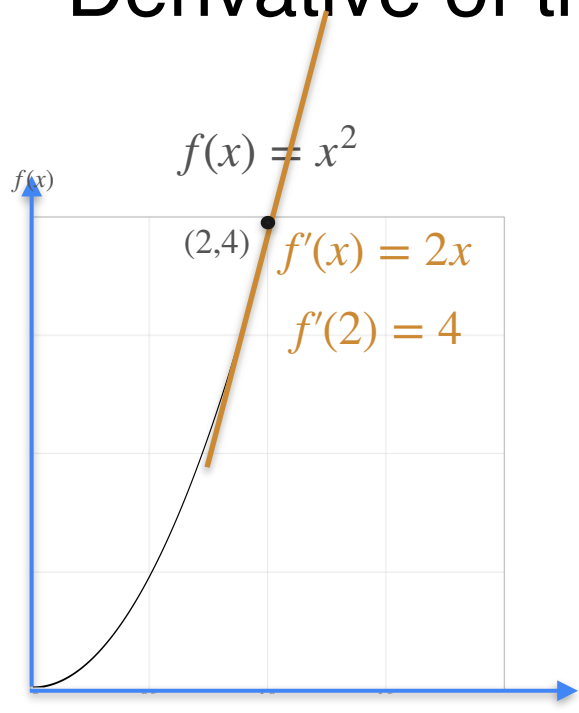
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Derivative of the Inverse



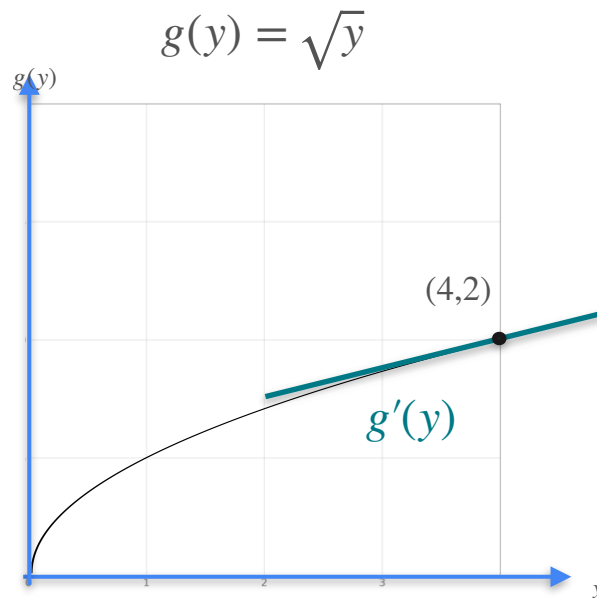
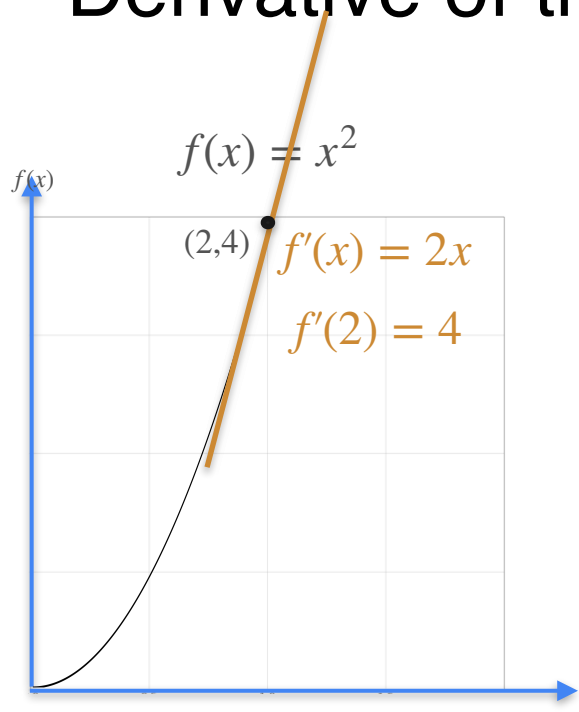
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Derivative of the Inverse



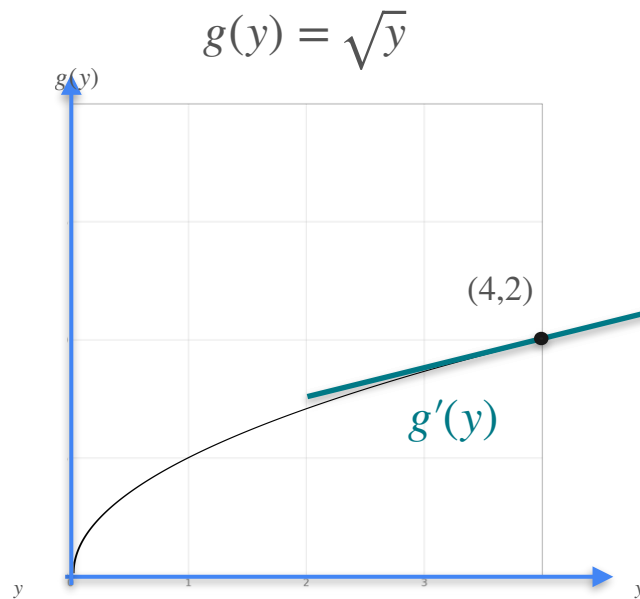
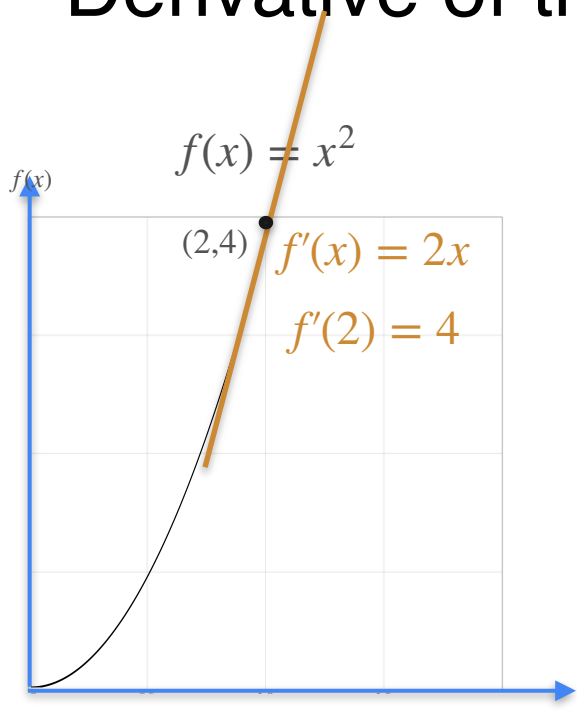
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Derivative of the Inverse



$$g'(y) = \frac{1}{f'(x)}$$

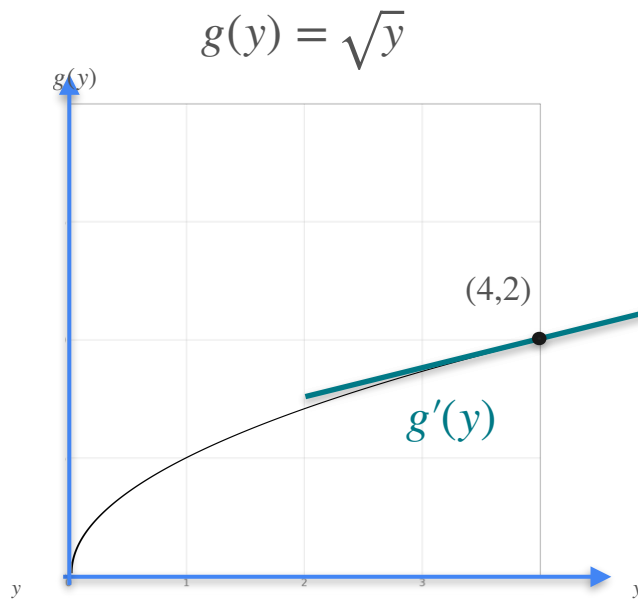
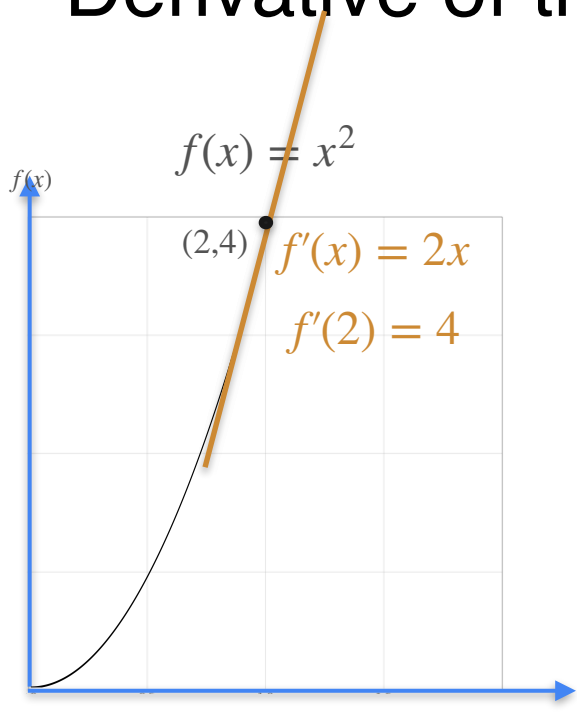
at the point $(2,4)$

$$f(2) = 4$$

$$g(4) = 2$$

$$g'(4) = \frac{1}{f'(2)}$$

Derivative of the Inverse



$$g'(y) = \frac{1}{f'(x)}$$

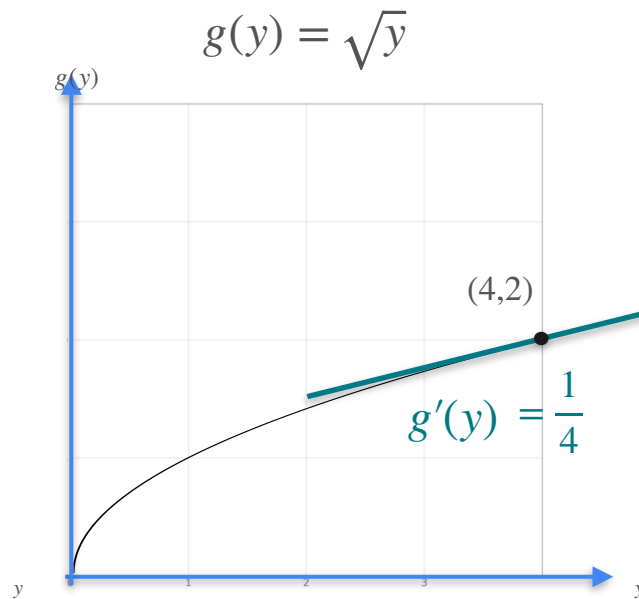
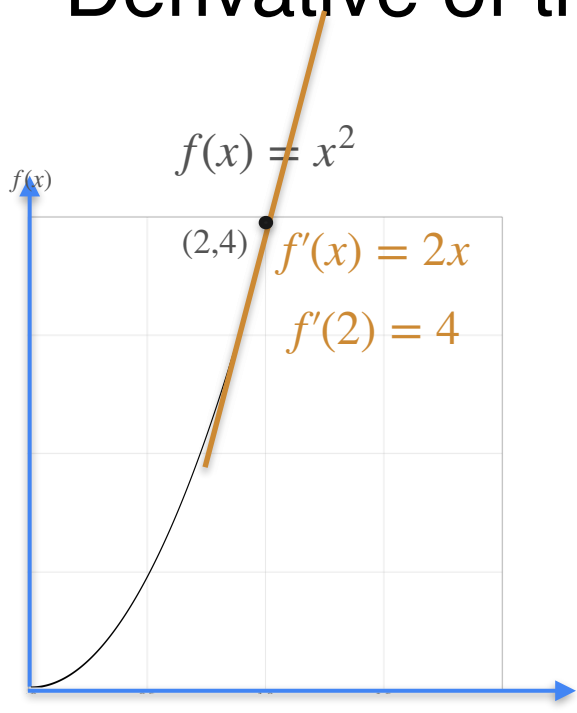
at the point $(2,4)$

$$f(2) = 4$$

$$g(4) = 2$$

$$g'(4) = \frac{1}{f'(2)} = \frac{1}{4}$$

Derivative of the Inverse



$$g'(y) = \frac{1}{f'(x)}$$

at the point $(2,4)$

$$f(2) = 4$$

$$g(4) = 2$$

$$g'(4) = \frac{1}{f'(2)}$$



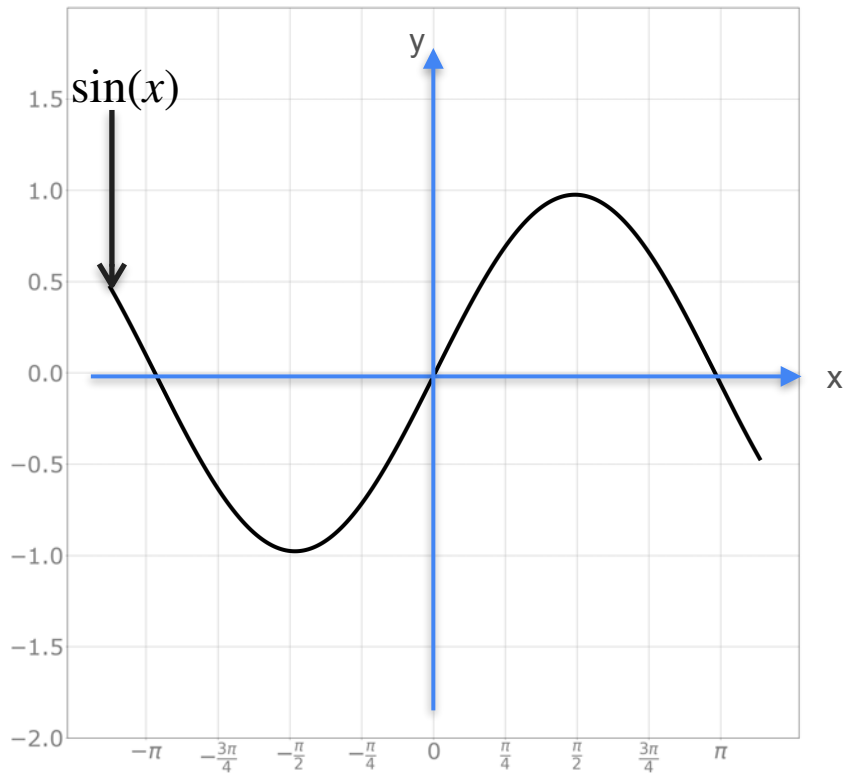
DeepLearning.AI

Derivatives and Optimization

Derivative of trigonometric functions

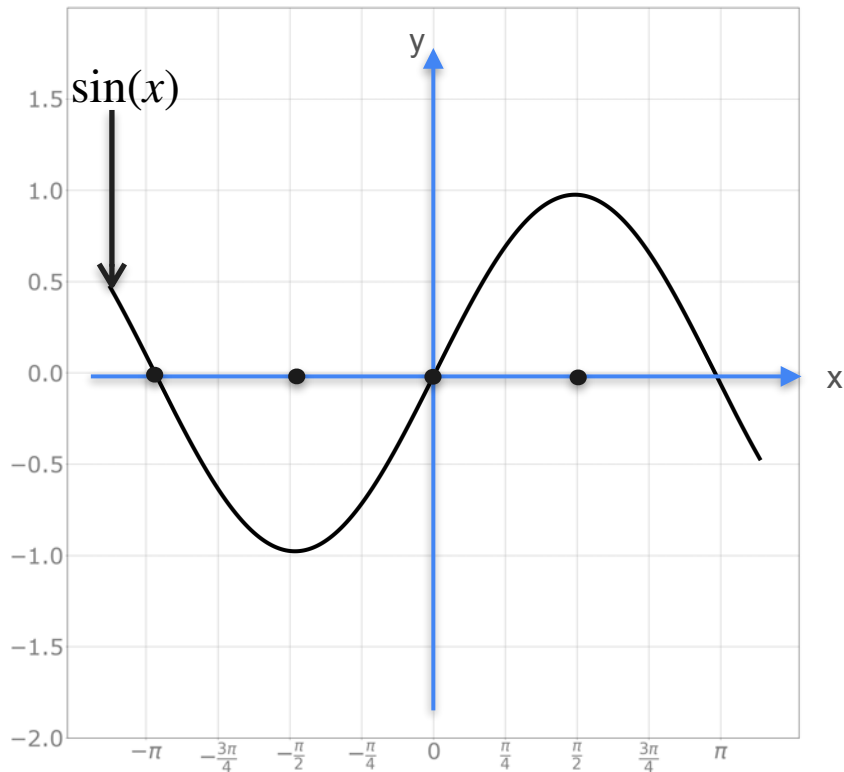
Derivative of Trigonometric Functions

Derivative of Trigonometric Functions



Sine $y = f(x) = \sin(x)$

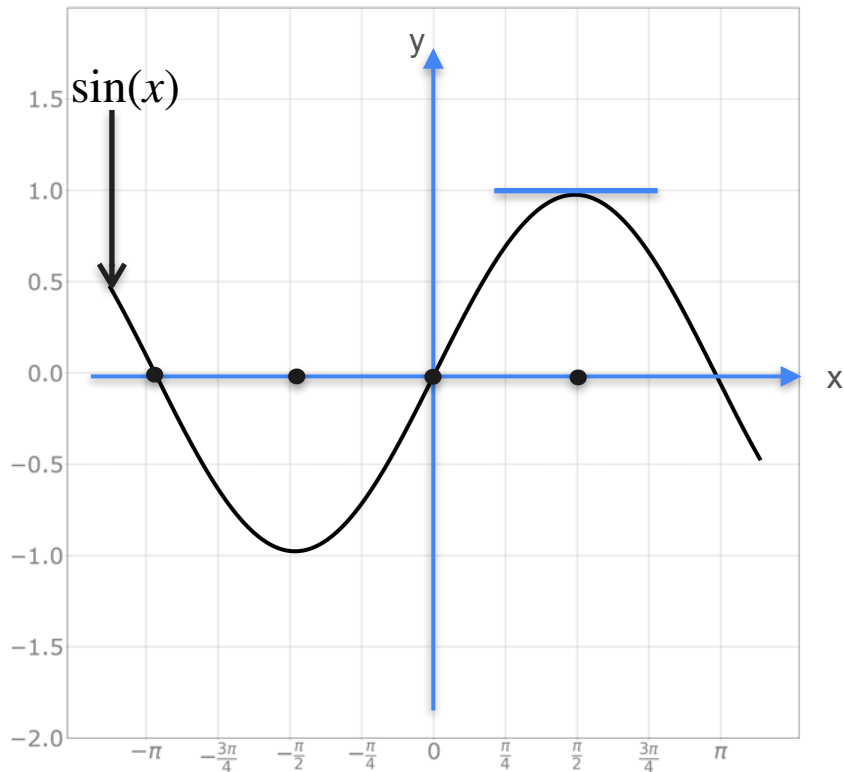
Derivative of Trigonometric Functions



Sine $y = f(x) = \sin(x)$

x	$\pi/2$	$-\pi/2$	0	$-\pi$
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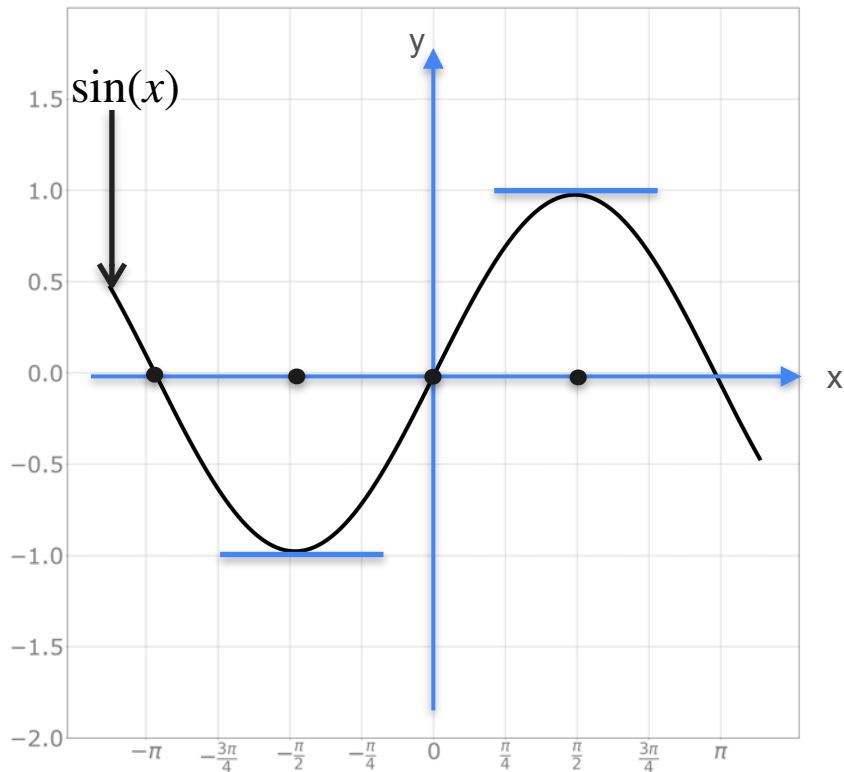
Derivative of Trigonometric Functions



Sine $y = f(x) = \sin(x)$

x	$\pi/2$	$-\pi/2$	0	$-\pi$
Slope	0			

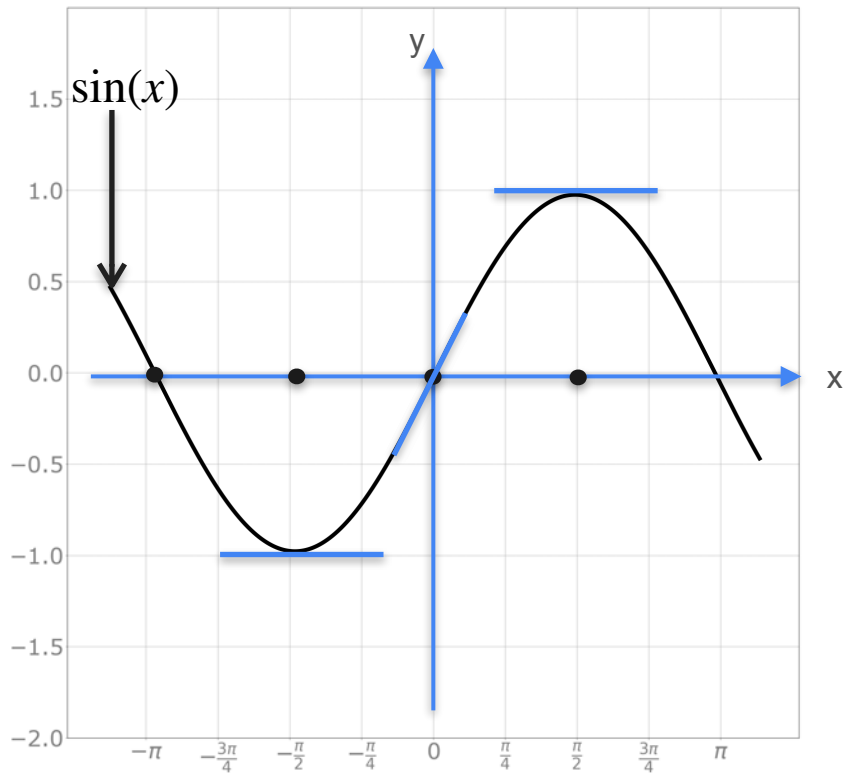
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Sine $y = f(x) = \sin(x)$

x	$\pi/2$	$-\pi/2$	0	$-\pi$
Slope	0	0		

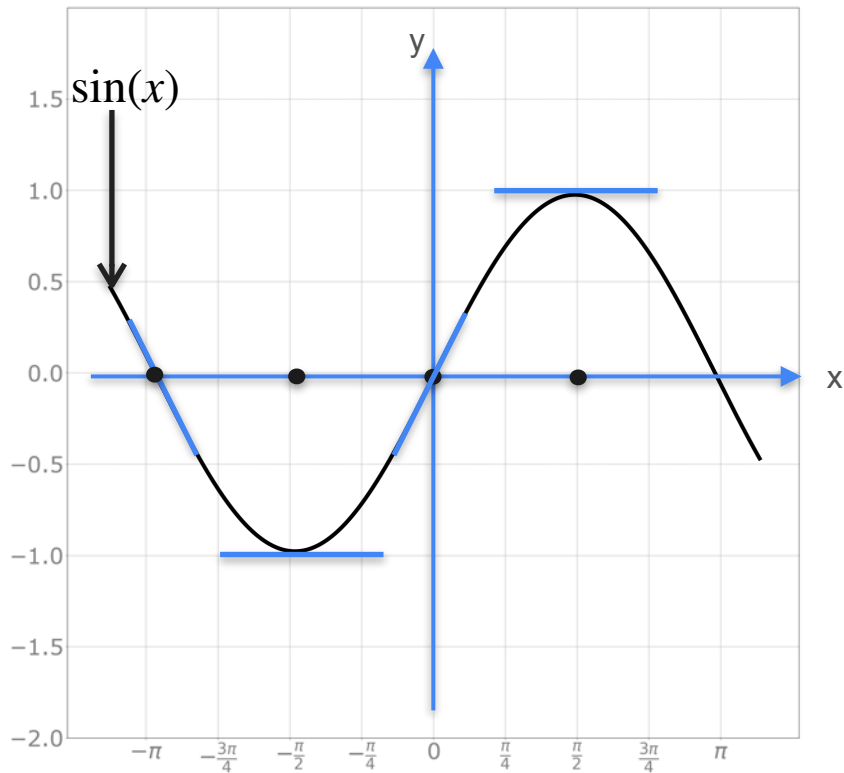
Derivative of Trigonometric Functions



Sine $y = f(x) = \sin(x)$

x	$\pi/2$	$-\pi/2$	0	$-\pi$
Slope	0	0	1	

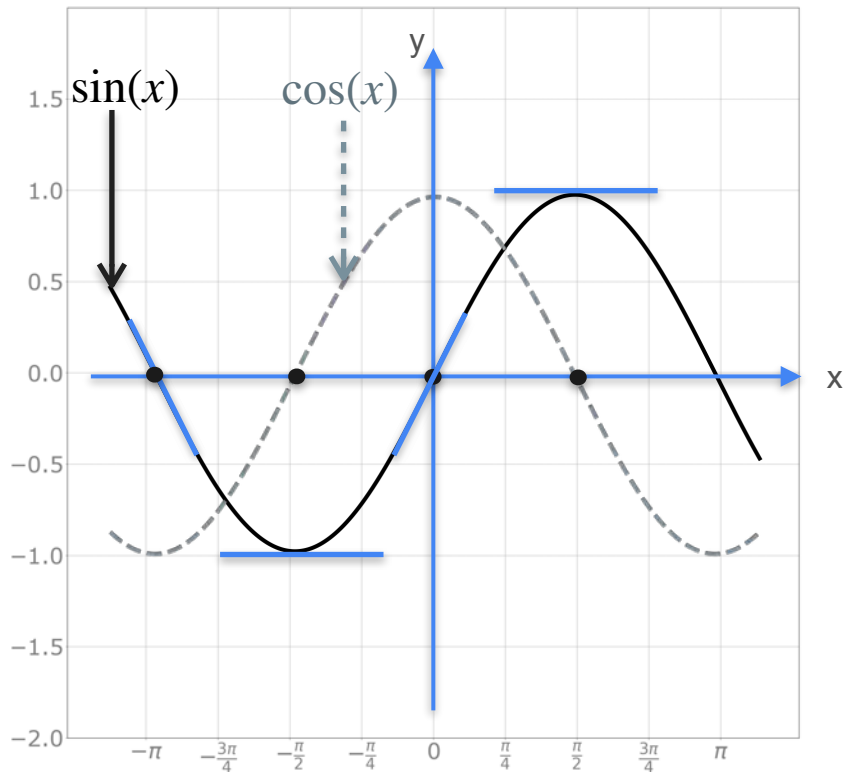
Derivative of Trigonometric Functions



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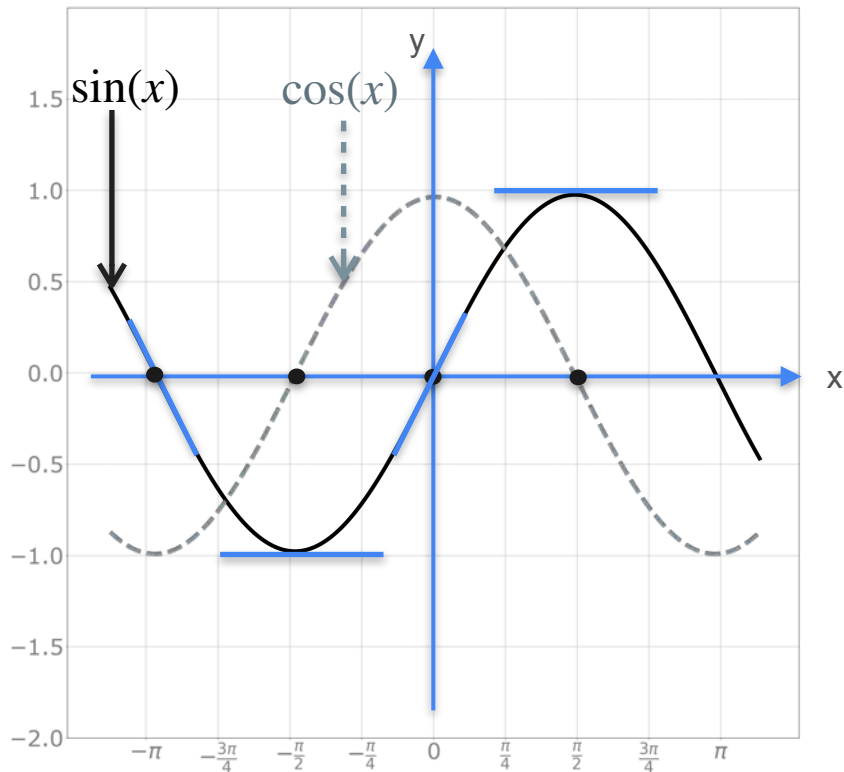
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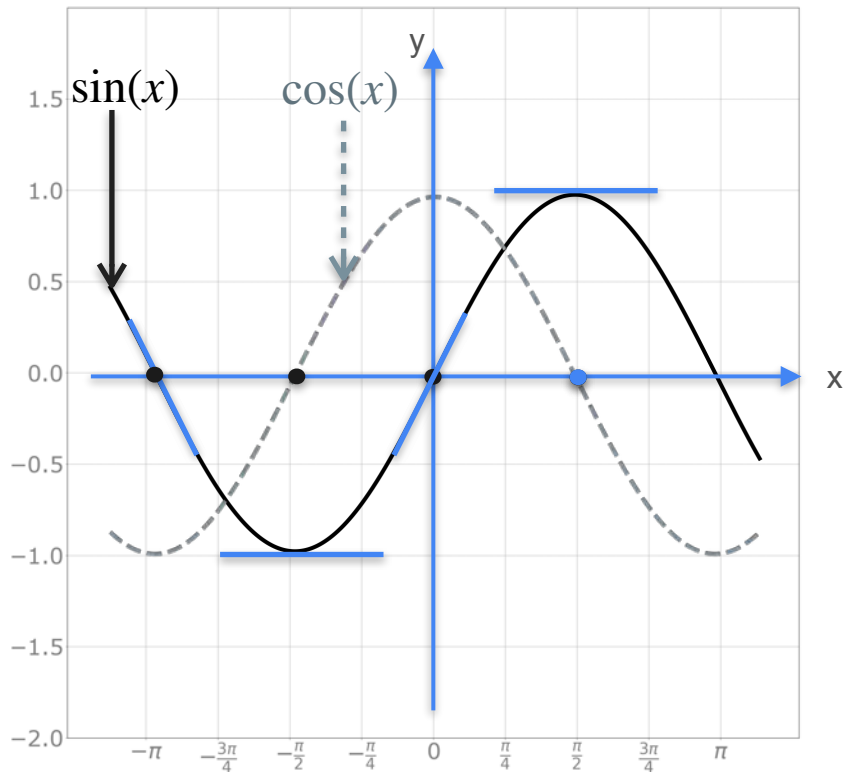
Derivative of Trigonometric Functions



Sine $y = f(x) = \sin(x)$

x	$\pi/2$	$-\pi/2$	0	$-\pi$
Slope	0	0	1	-1
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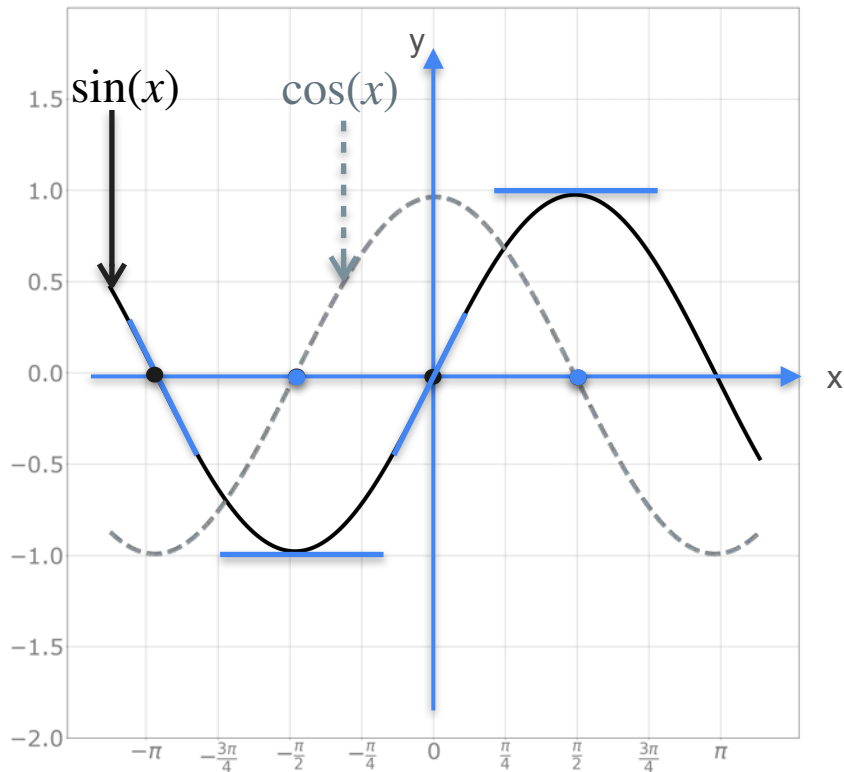
Derivative of Trigonometric Functions



Sine $y = f(x) = \sin(x)$

x	$\pi/2$	$-\pi/2$	0	$-\pi$
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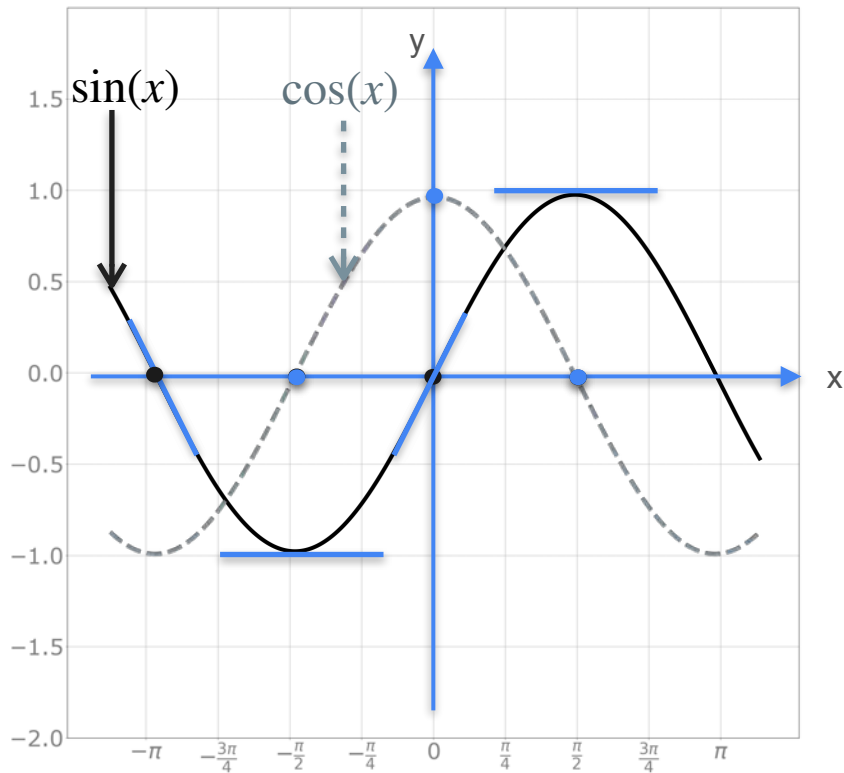
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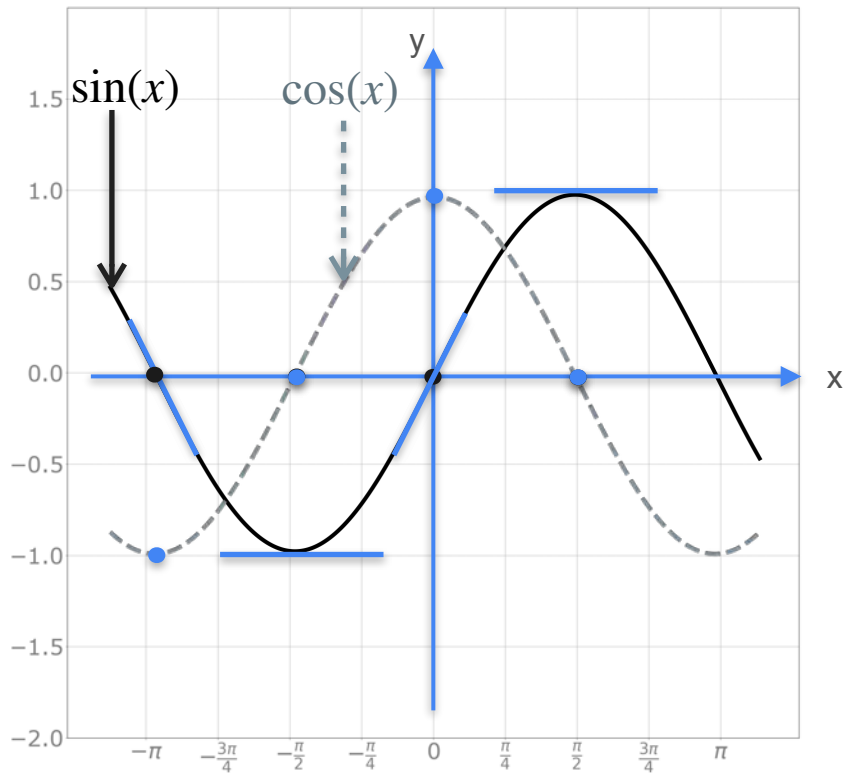
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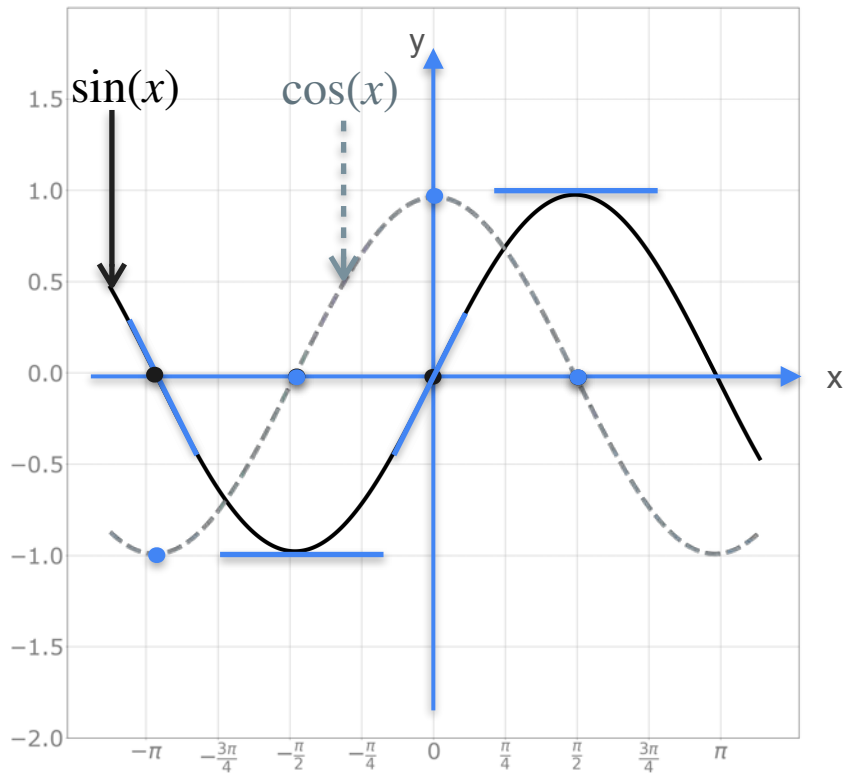
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Derivative of Trigonometric Functions

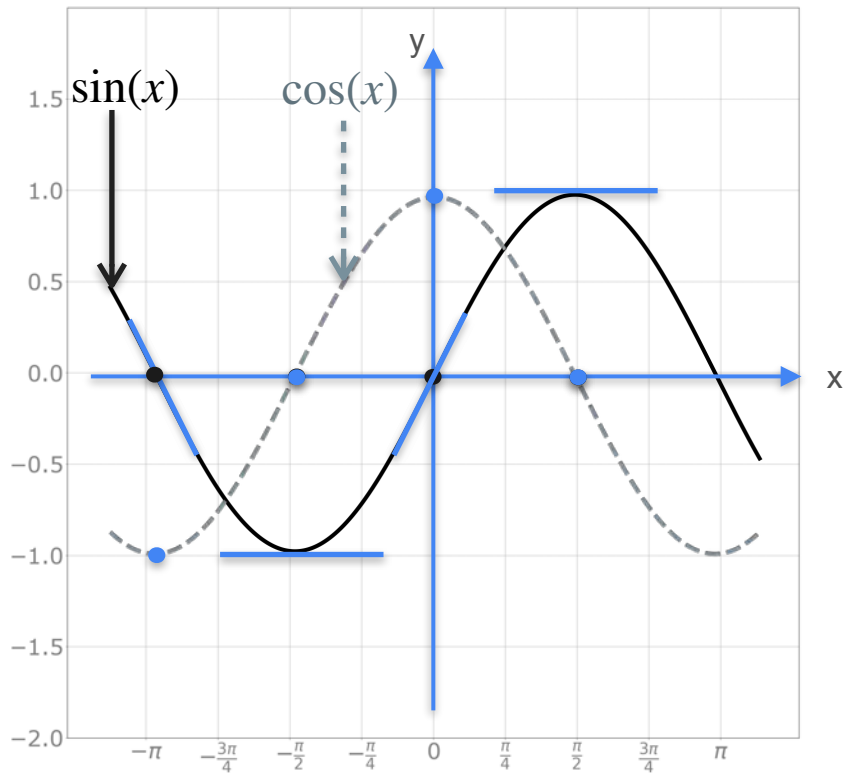


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$$f(x) = \sin(x) \Rightarrow f'(x) = \cos(x)$$

Derivative of Trigonometric Functions



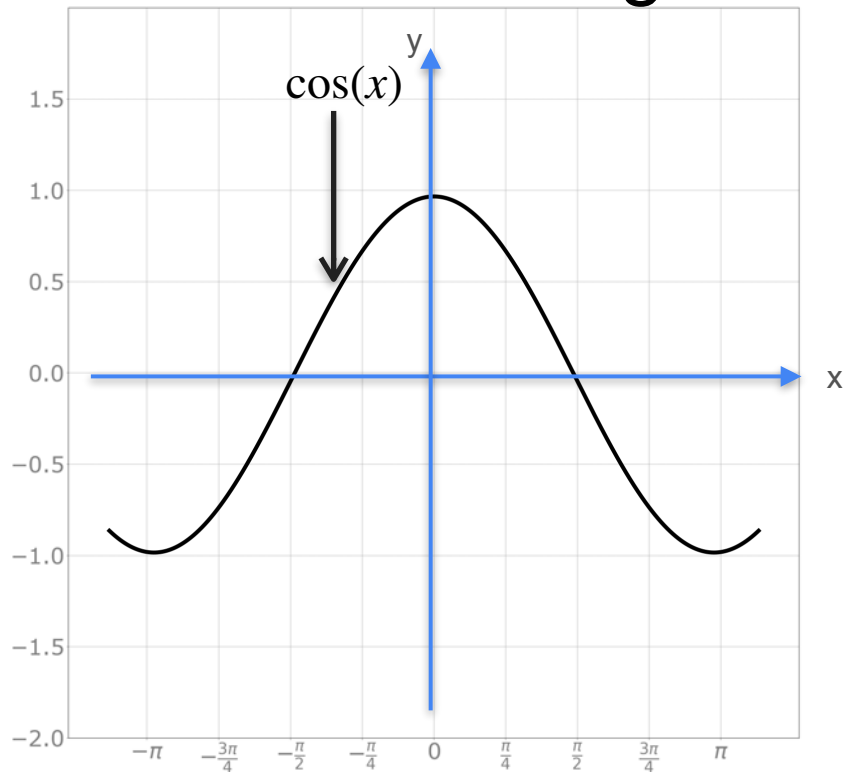
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$$f(x) = \sin(x) \xrightarrow{??} f'(x) = \cos(x)$$

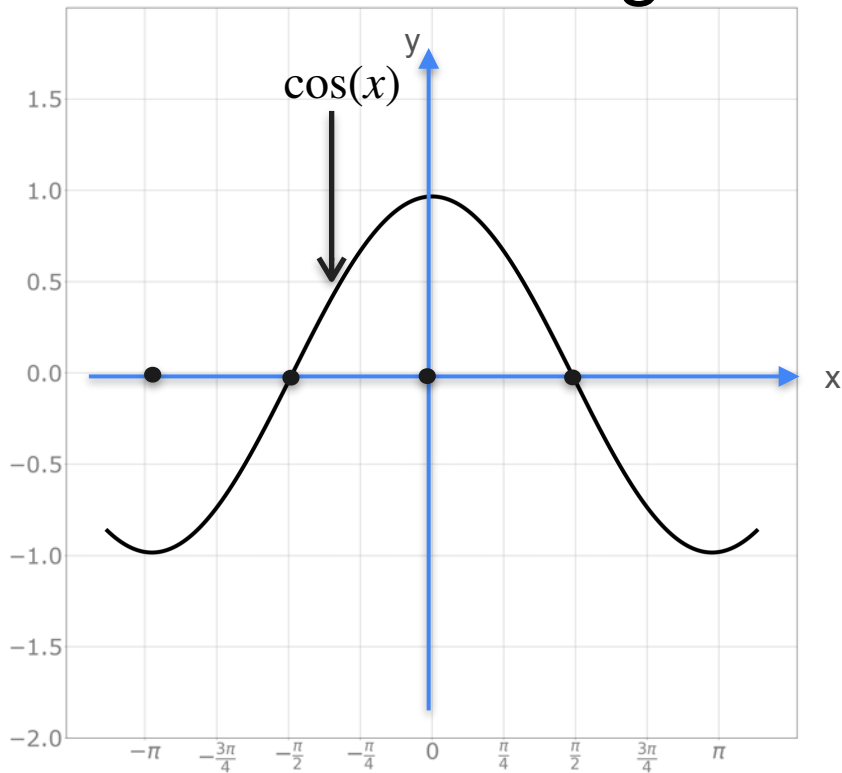
Derivative of Trigonometric Functions

Derivative of Trigonometric Functions



Cosine $y = f(x) = \cos(x)$

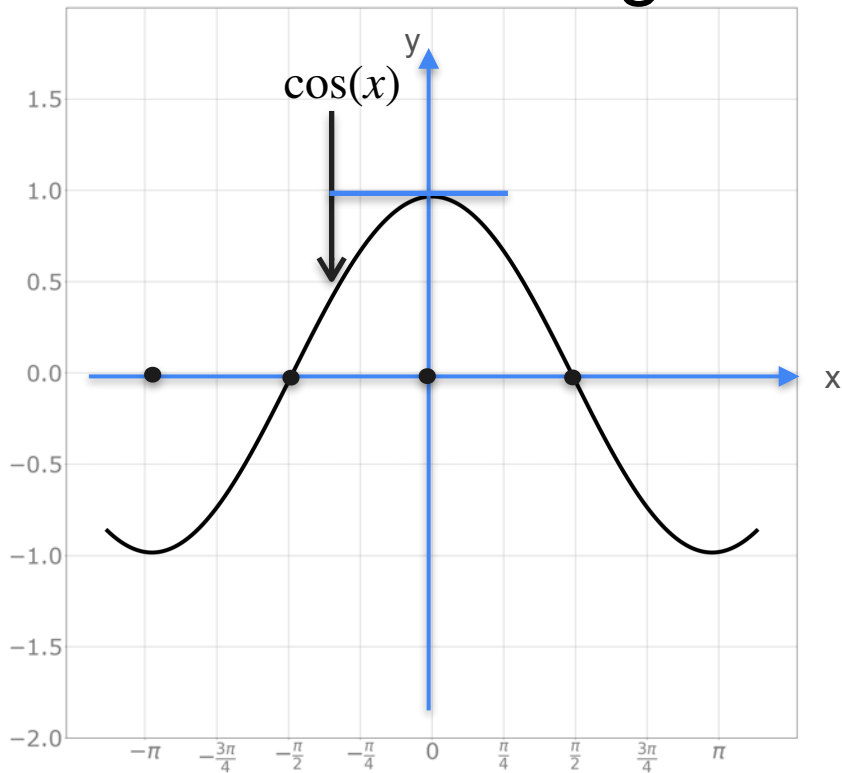
Derivative of Trigonometric Functions



Cosine $y = f(x) = \cos(x)$

x	0	$-\pi$	$\pi/2$	$-\pi/2$
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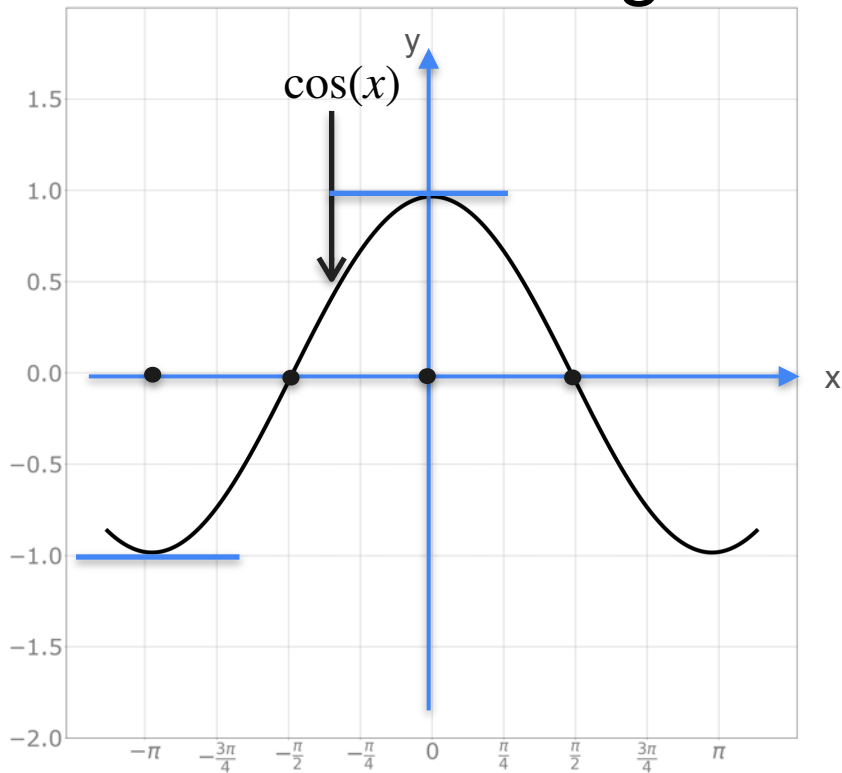
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Cosine $y = f(x) = \cos(x)$

x	0	$-\pi$	$\pi/2$	$-\pi/2$
Slope	0			

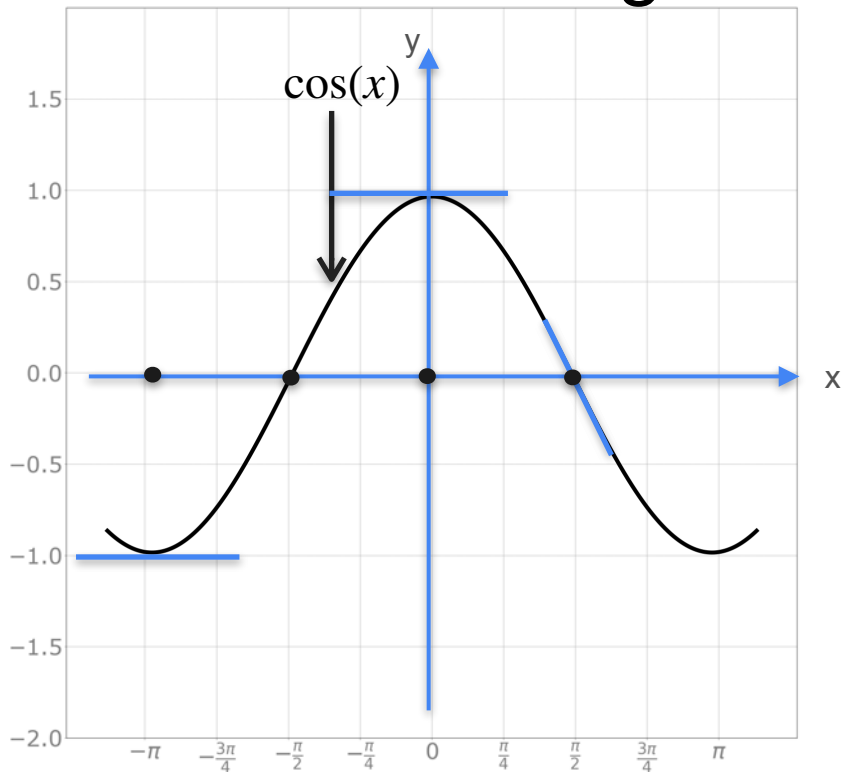
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x	0	$-\pi$	$\pi/2$	$-\pi/2$
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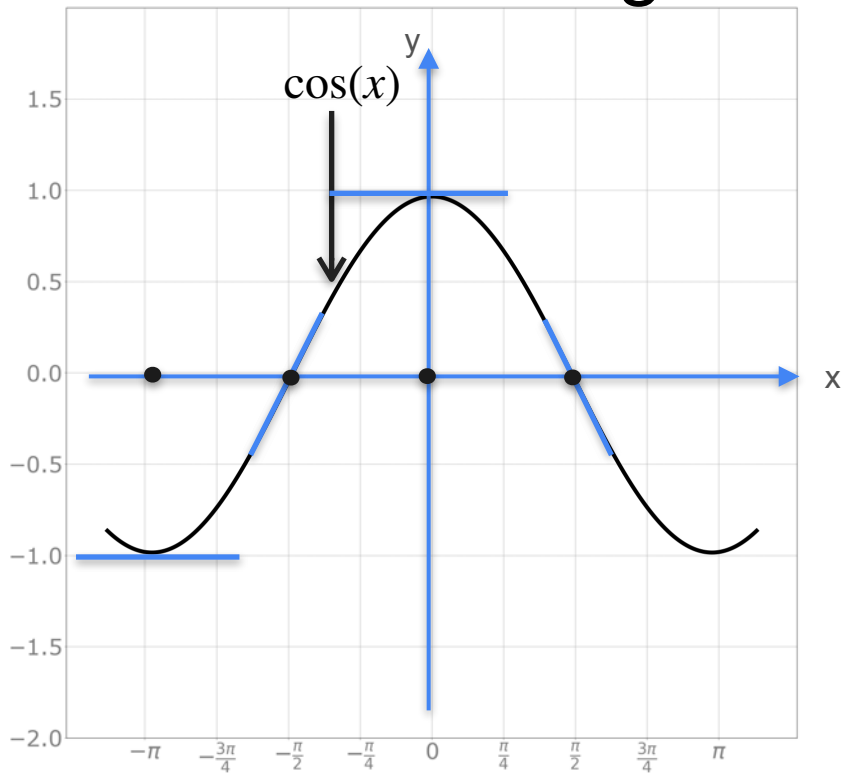
Derivative of Trigonometric Functions



Cosine $y = f(x) = \cos(x)$

x	0	$-\pi$	$\pi/2$	$-\pi/2$
Slope	0	0	-1	

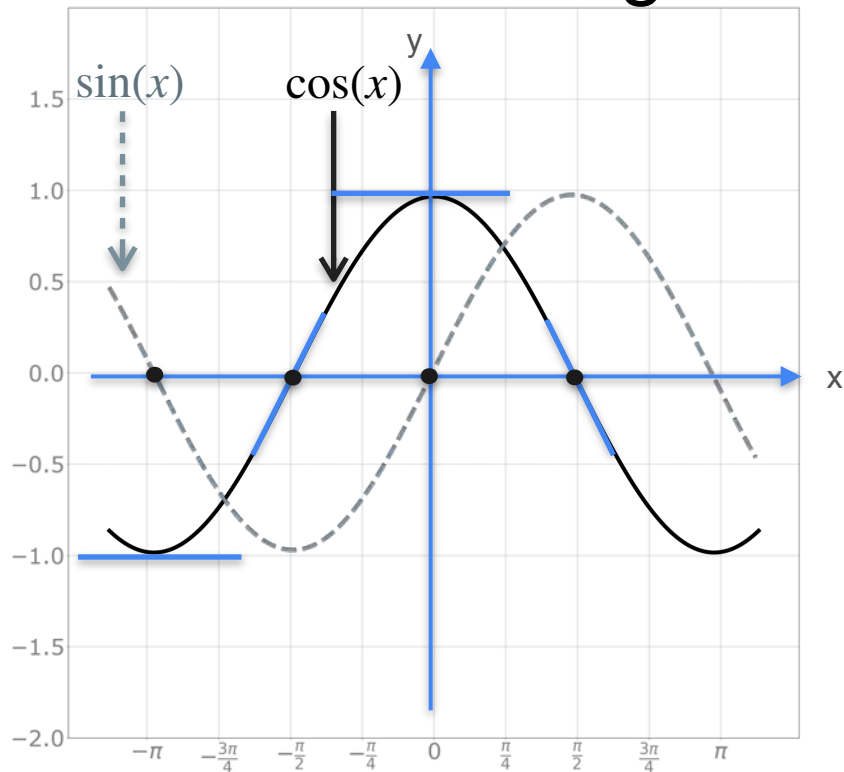
Derivative of Trigonometric Functions



Cosine $y = f(x) = \cos(x)$

x	0	$-\pi$	$\pi/2$	$-\pi/2$
Slope	0	0	-1	1

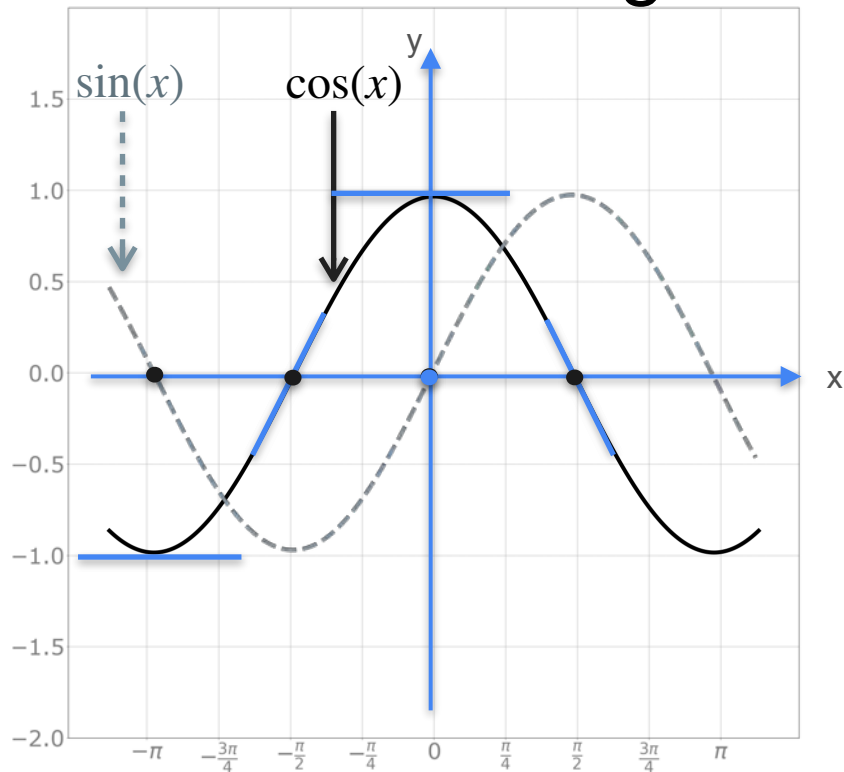
Derivative of Trigonometric Functions



Cosine $y = f(x) = \cos(x)$

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Slope	0	0	-1	1
$\sin(x)$				

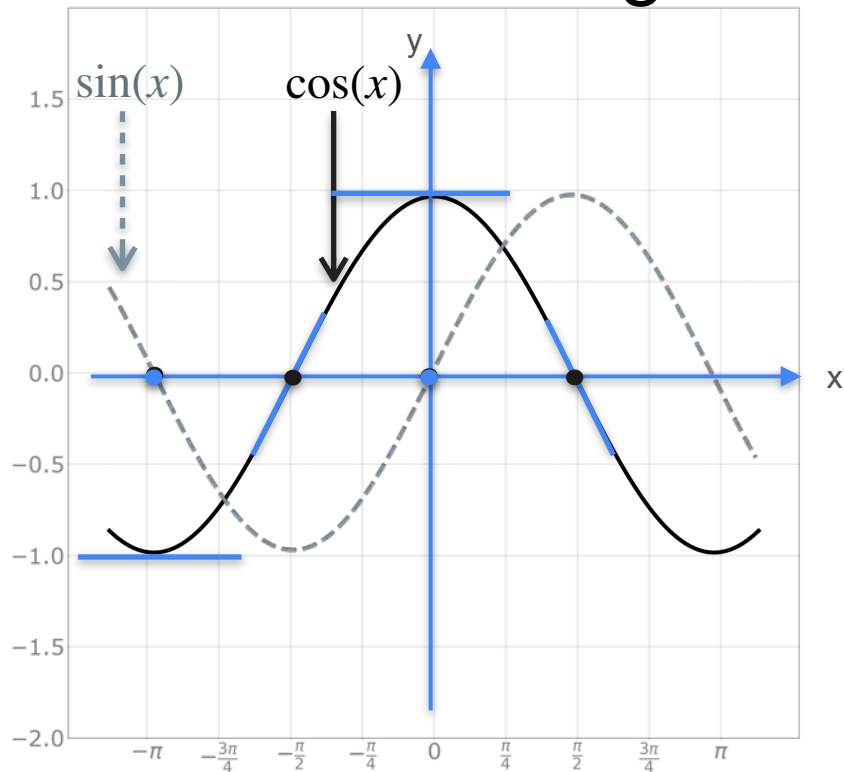
Derivative of Trigonometric Functions



Cosine $y = f(x) = \cos(x)$

x	0	$-\pi$	$\pi/2$	$-\pi/2$
Slope	0	0	-1	1
$\sin(x)$	0			

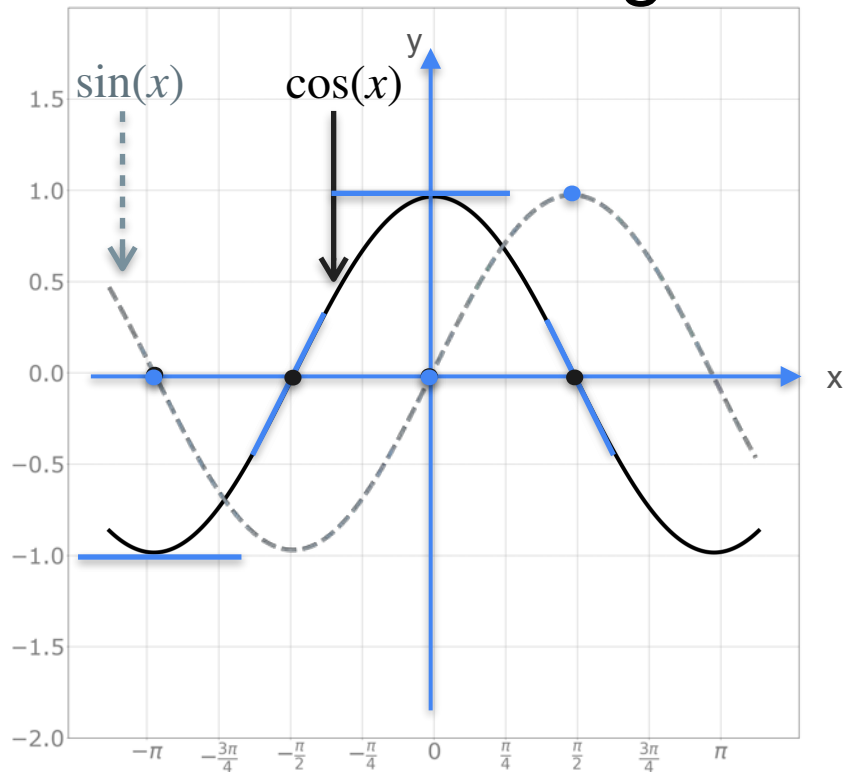
Derivative of Trigonometric Functions



Cosine $y = f(x) = \cos(x)$

x	0	$-\pi$	$\pi/2$	$-\pi/2$
Slope	0	0	-1	1
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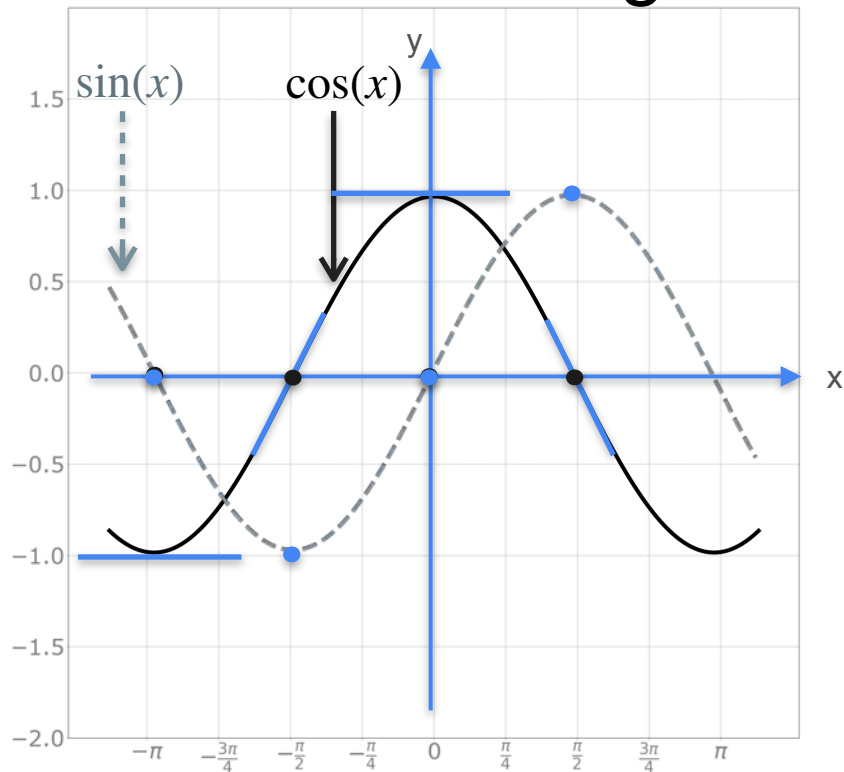
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x	0	$-\pi$	$\pi/2$	$-\pi/2$
Slope	0	0	-1	1
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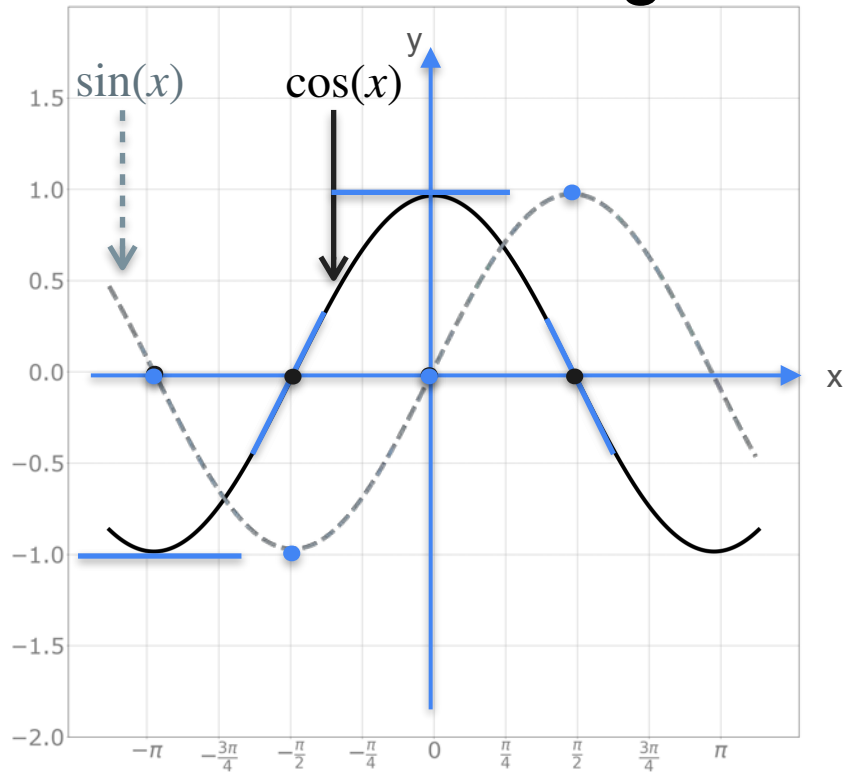
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Derivative of Trigonometric Functions

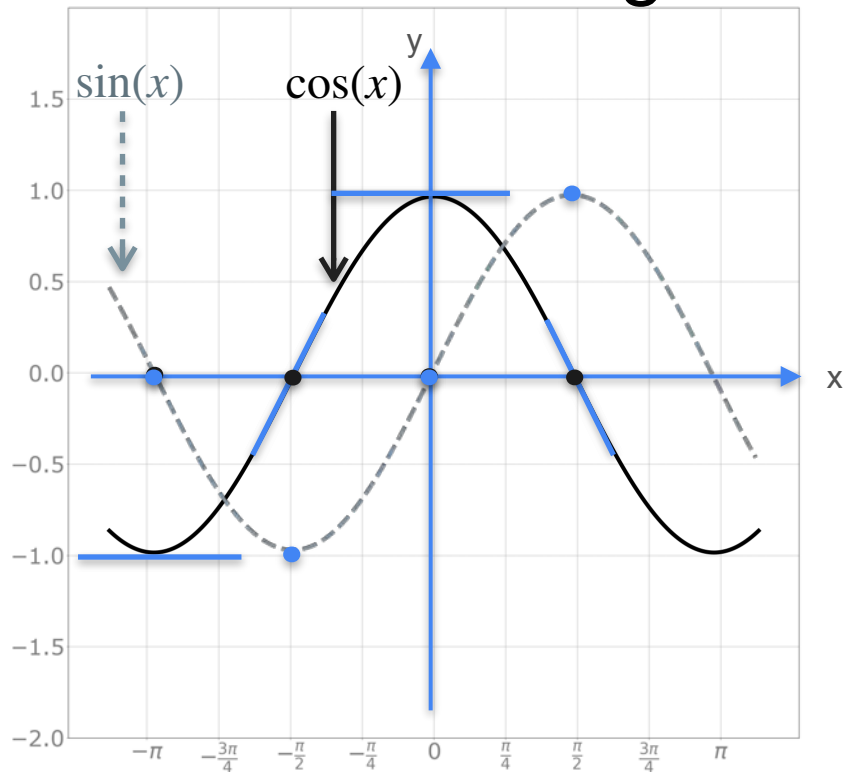


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x	0	$-\pi$	$\pi/2$	$-\pi/2$
Slope	0	0	-1	1
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$$f(x) = \cos(x) \Rightarrow f'(x) = -\sin(x)$$

Derivative of Trigonometric Functions



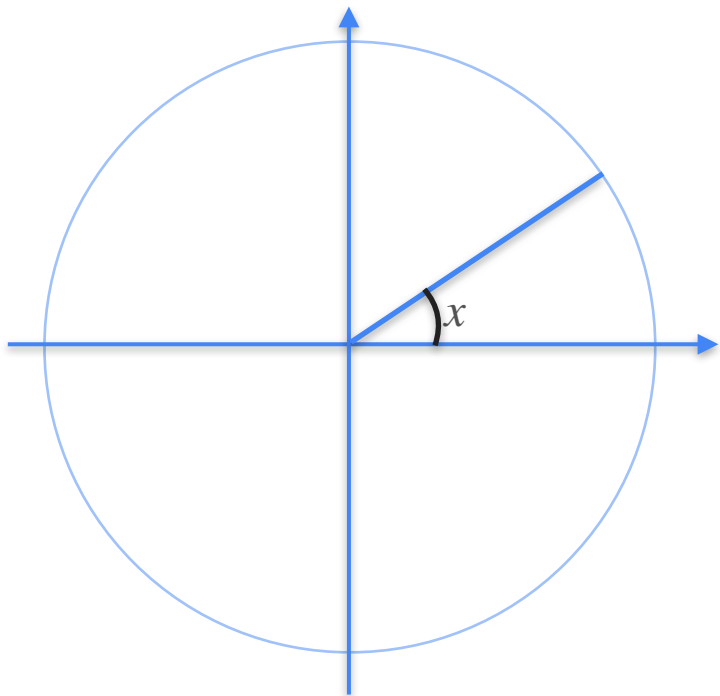
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x	0	$-\pi$	$\pi/2$	$-\pi/2$
Slope	0	0	-1	1
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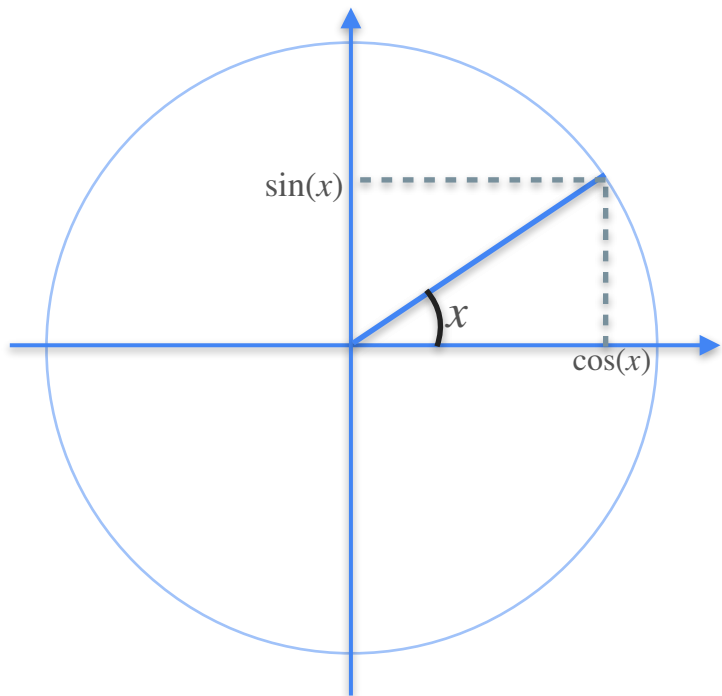
$$f(x) = \cos(x) \xrightarrow{??} f'(x) = -\sin(x)$$

Derivative of Trigonometric Functions

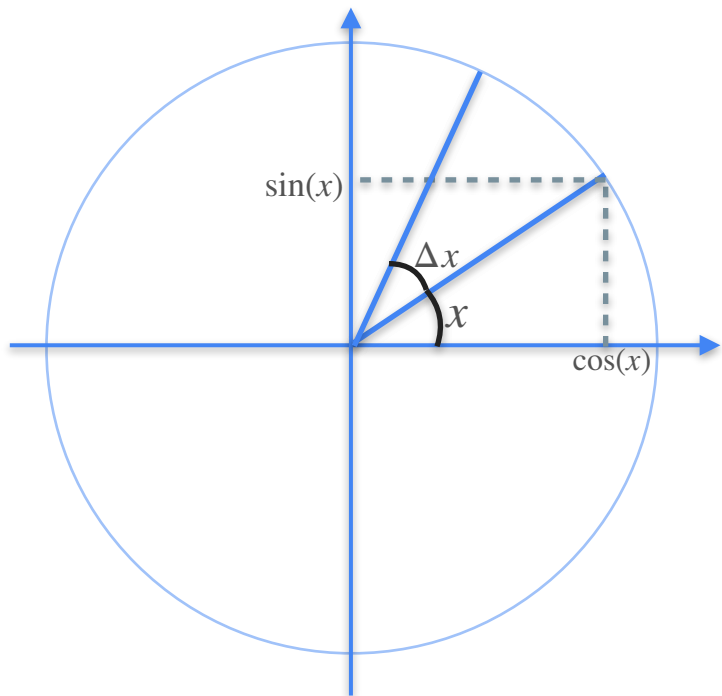
Derivative of Trigonometric Functions



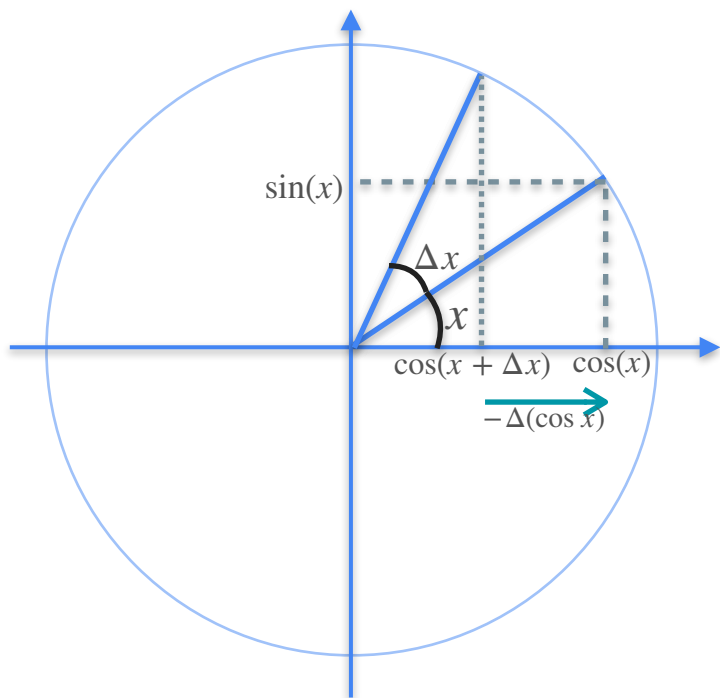
Derivative of Trigonometric Functions



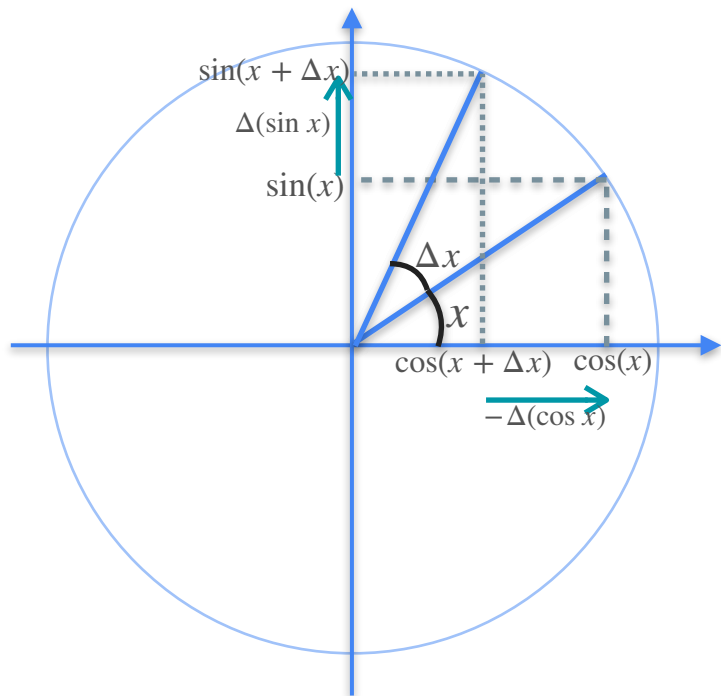
Derivative of Trigonometric Functions



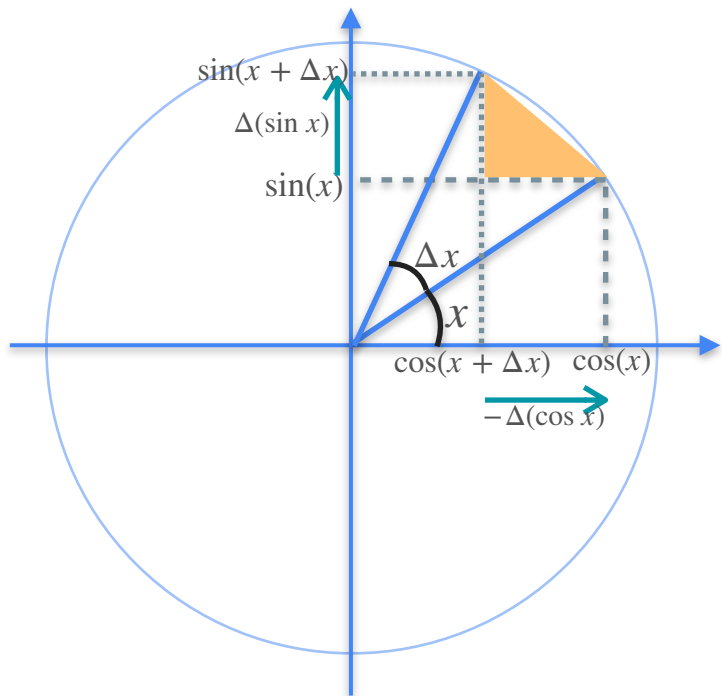
Derivative of Trigonometric Functions



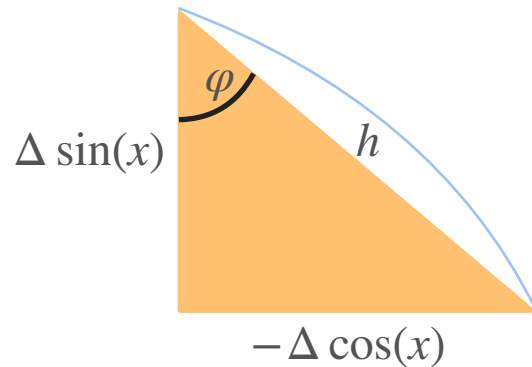
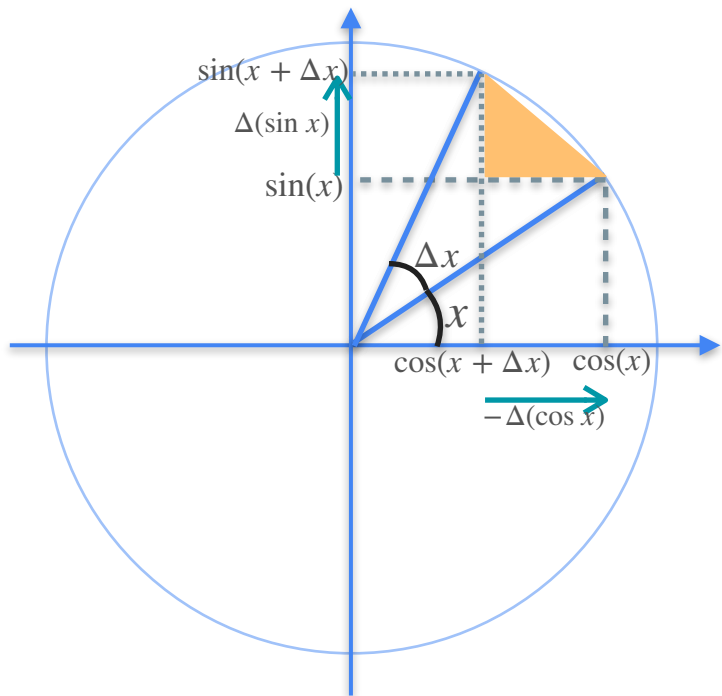
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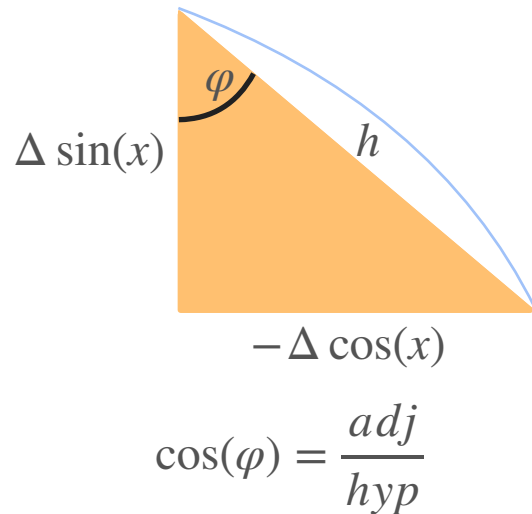
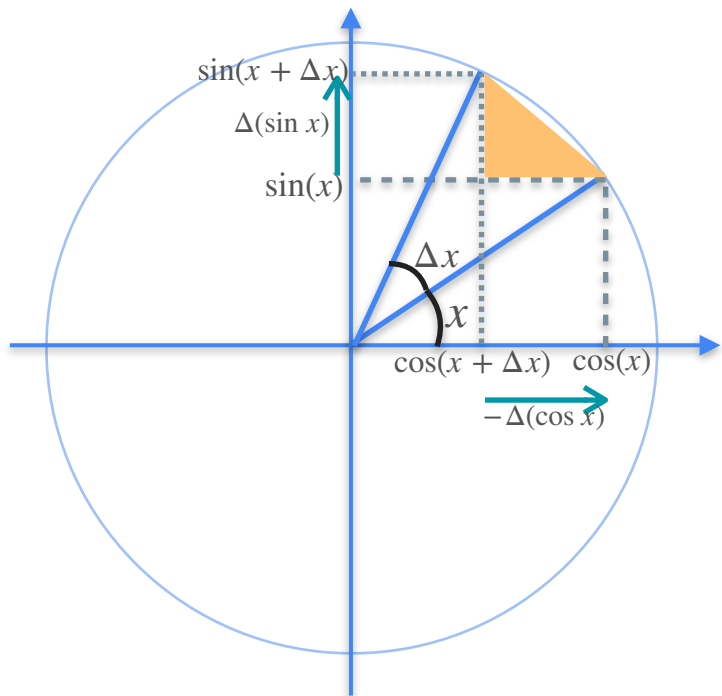
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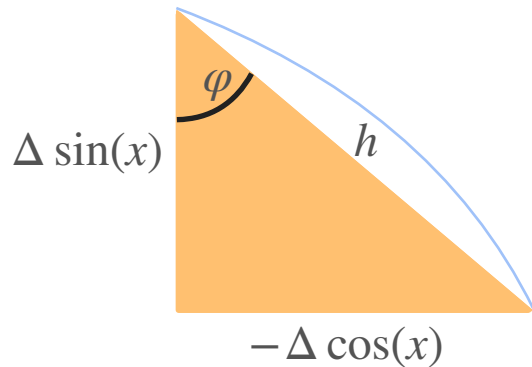
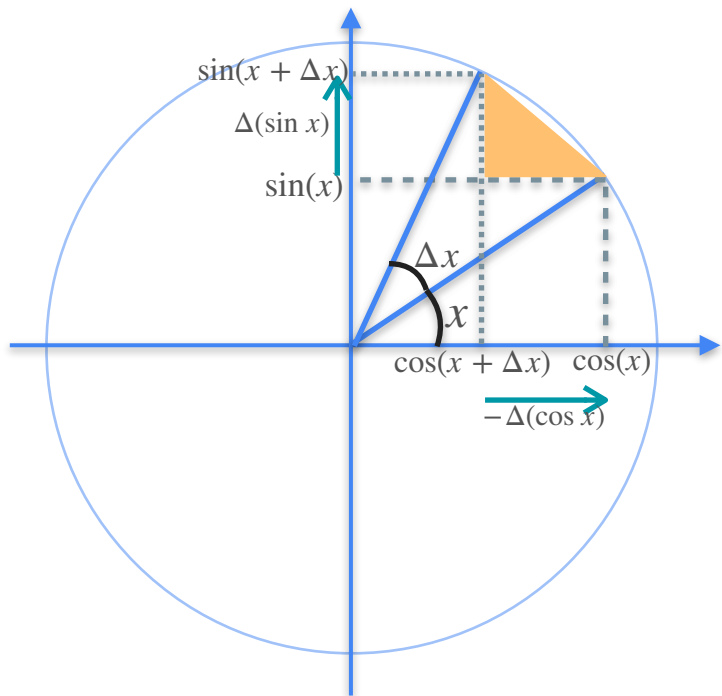
Derivative of Trigonometric Functions



Derivative of Trigonometric Functions

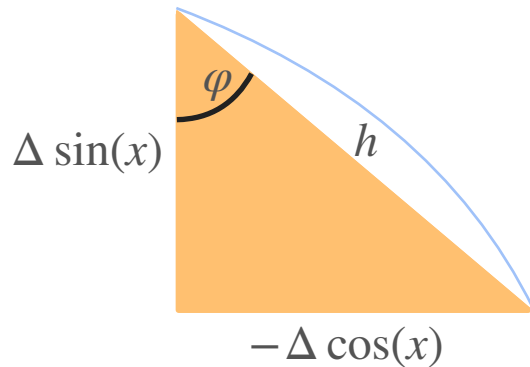
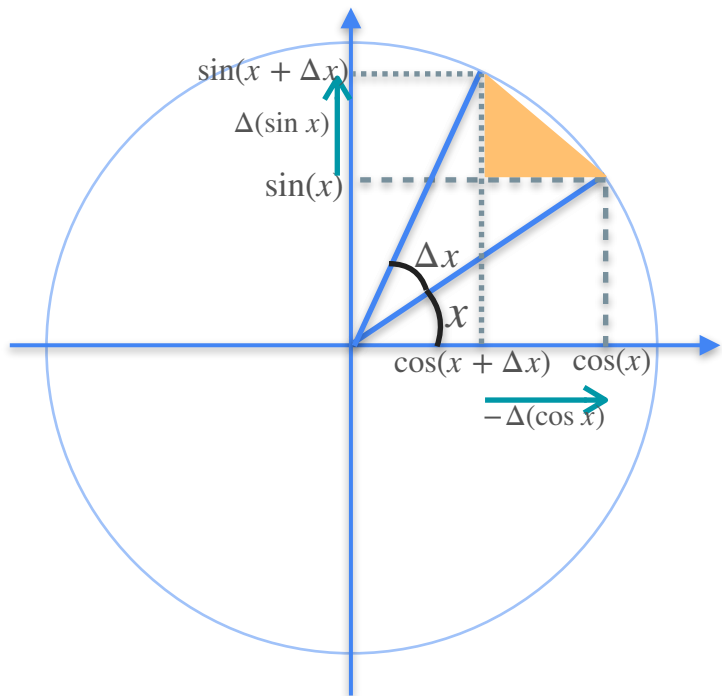


Derivative of Trigonometric Functions



$$\cos(\varphi) = \frac{\text{adj}}{\text{hyp}} = \frac{\Delta(\sin x)}{h}$$

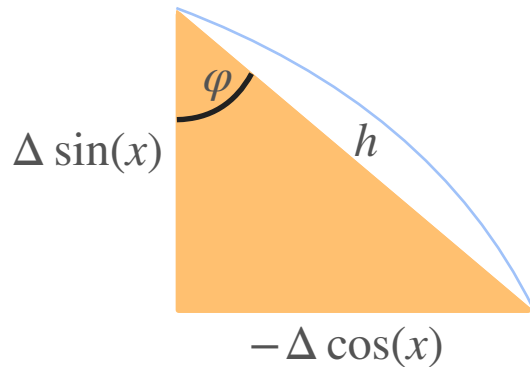
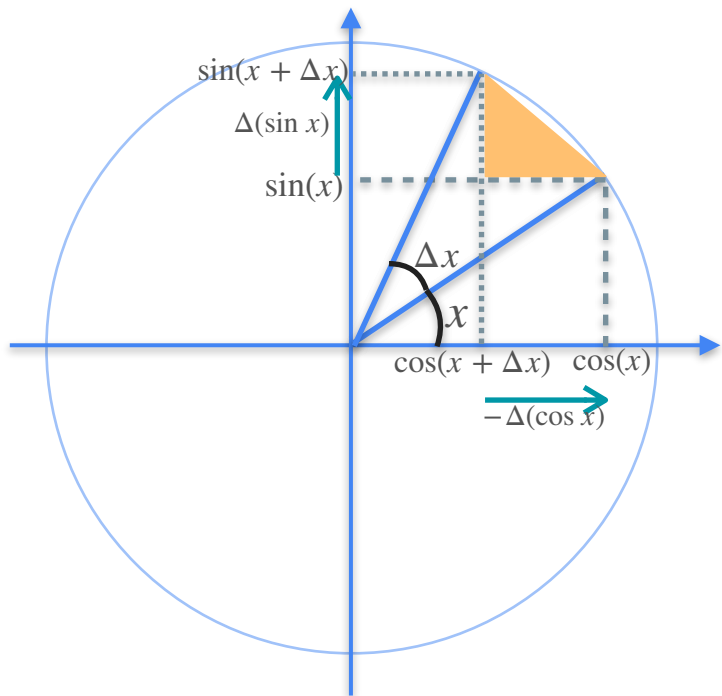
Derivative of Trigonometric Functions



$$\Delta(\sin x) = h \cos(\phi)$$

$$\cos(\phi) = \frac{\text{adj}}{\text{hyp}} = \frac{\Delta(\sin x)}{h}$$

Derivative of Trigonometric Functions

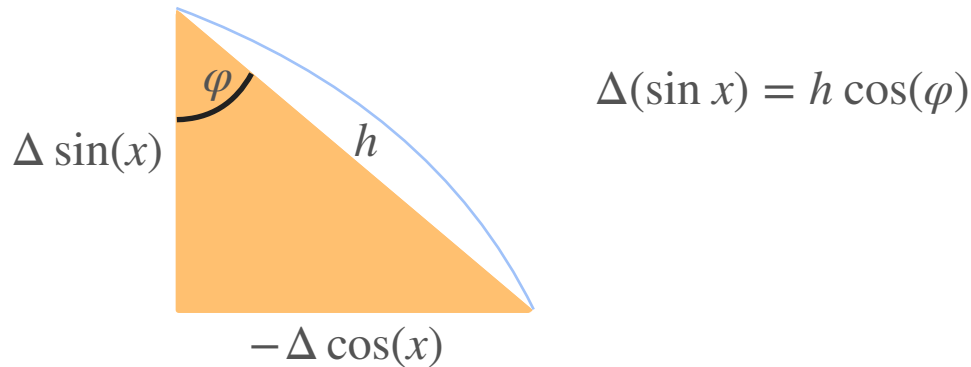
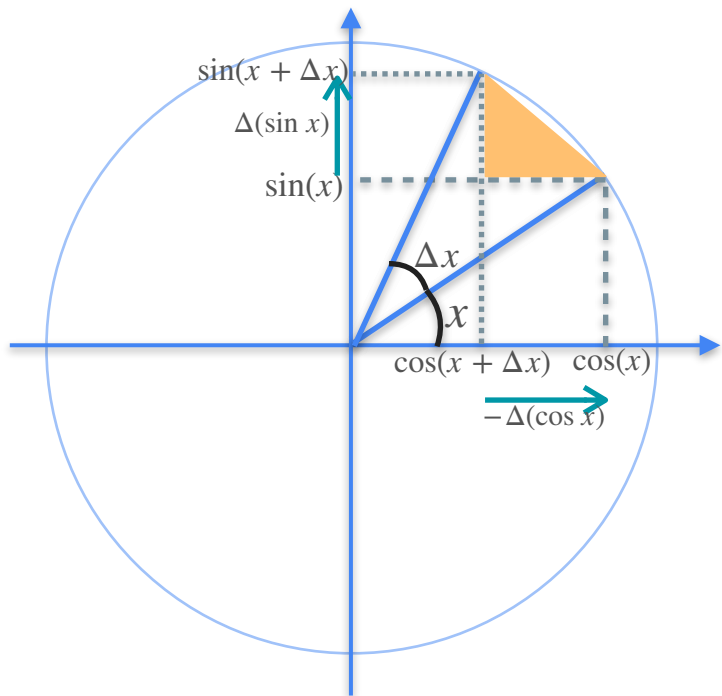


$$\Delta(\sin x) = h \cos(\varphi)$$

$$\cos(\varphi) = \frac{\text{adj}}{\text{hyp}} = \frac{\Delta(\sin x)}{h}$$

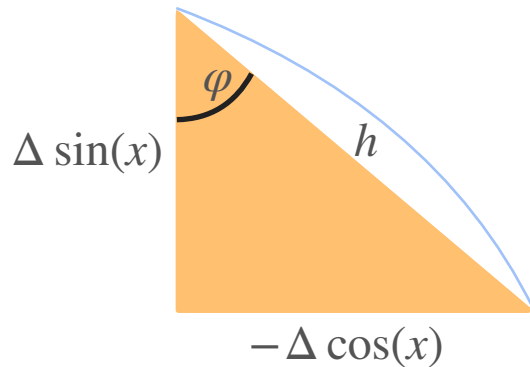
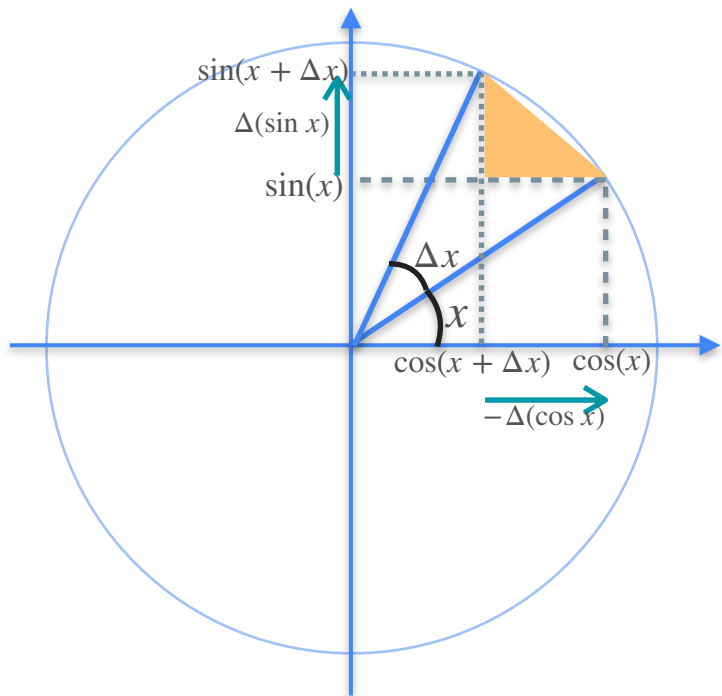
$$\sin(\varphi) = \frac{\text{opp}}{\text{hyp}}$$

Derivative of Trigonometric Functions



$$\cos(\varphi) = \frac{\text{adj}}{\text{hyp}} = \frac{\Delta(\sin x)}{h}$$
$$\sin(\varphi) = \frac{\text{opp}}{\text{hyp}} = \frac{-\Delta(\cos x)}{h}$$

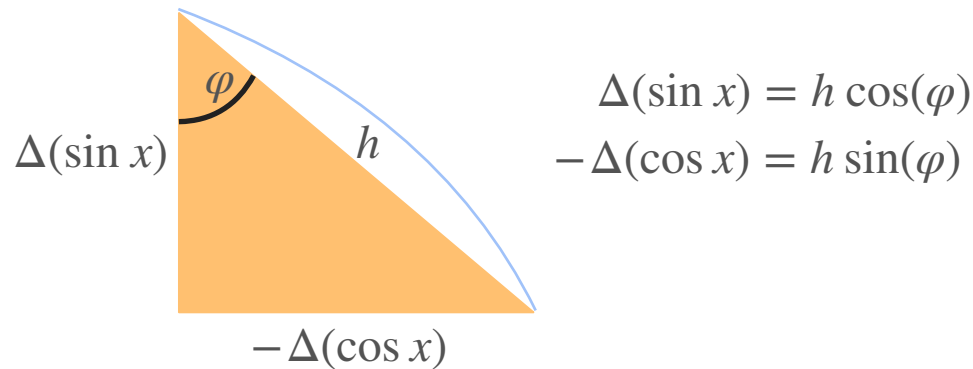
Derivative of Trigonometric Functions



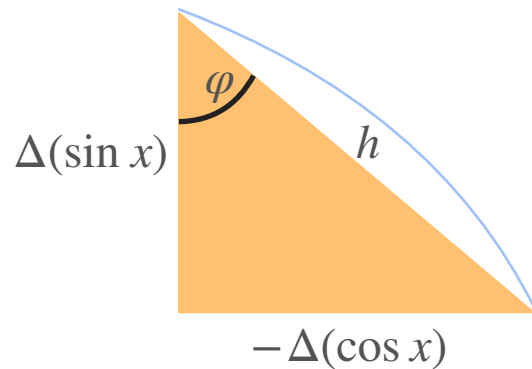
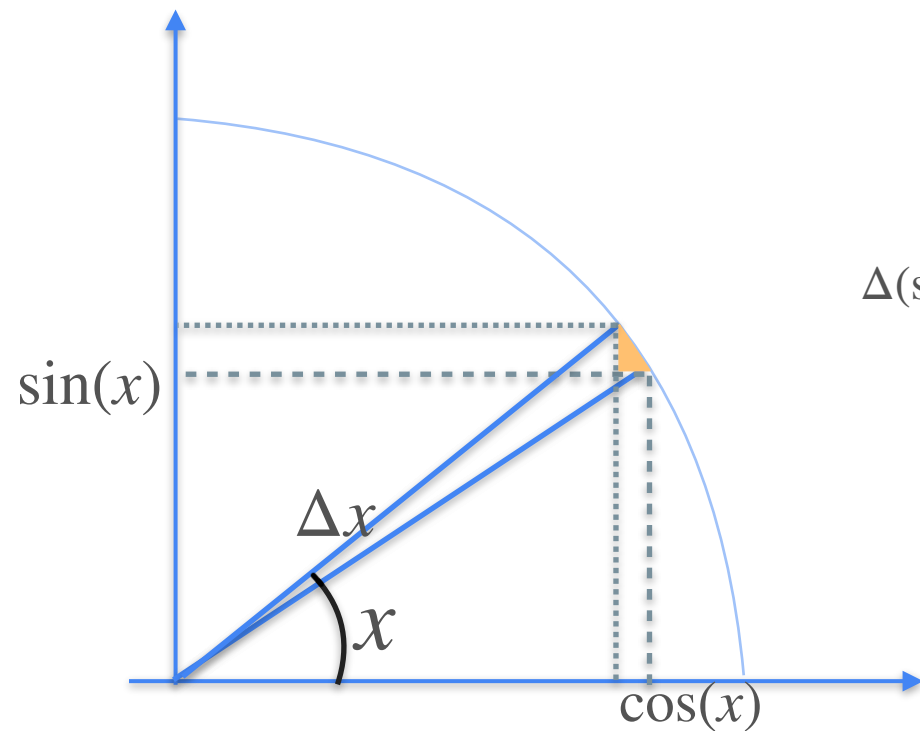
$$\begin{aligned}\Delta(\sin x) &= h \cos(\varphi) \\ -\Delta(\cos x) &= h \sin(\varphi)\end{aligned}$$

$$\begin{aligned}\cos(\varphi) &= \frac{\text{adj}}{\text{hyp}} = \frac{\Delta(\sin x)}{h} \\ \sin(\varphi) &= \frac{\text{opp}}{\text{hyp}} = \frac{-\Delta(\cos x)}{h}\end{aligned}$$

Derivative of Trigonometric Functions

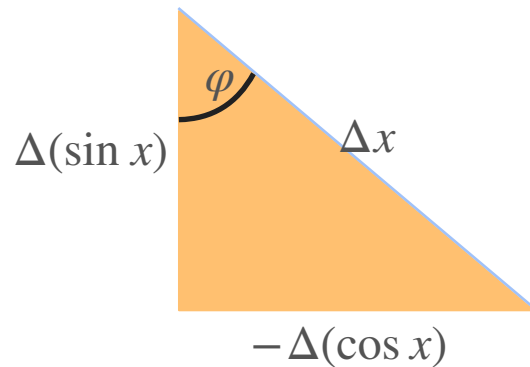
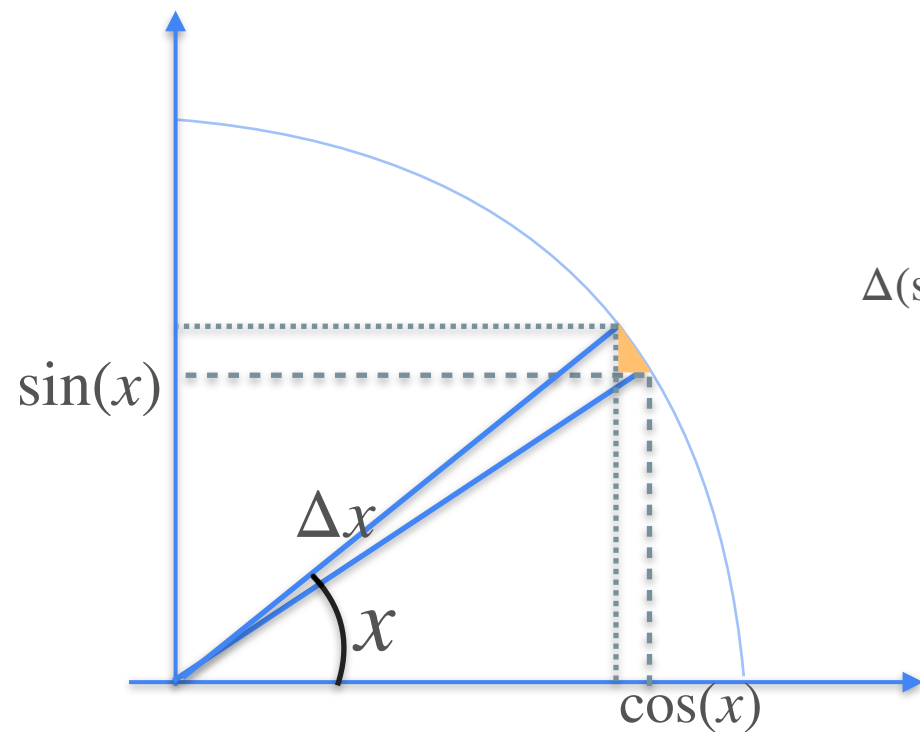


Derivative of Trigonometric Functions



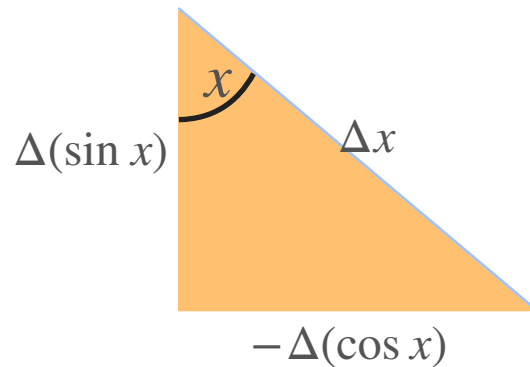
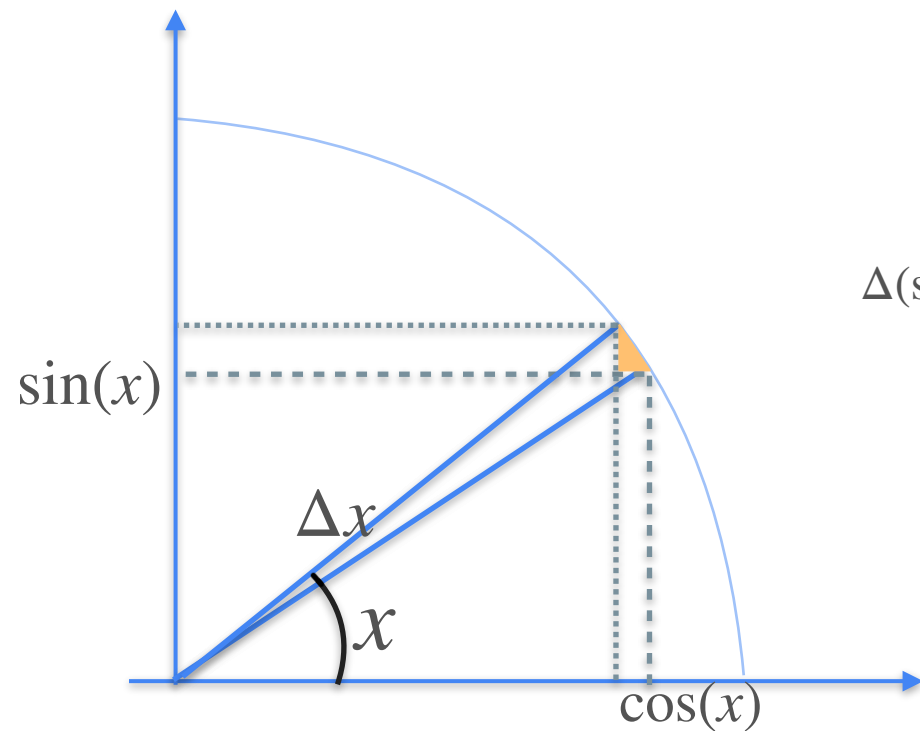
$$\begin{aligned}\Delta(\sin x) &= h \cos(\varphi) \\ -\Delta(\cos x) &= h \sin(\varphi)\end{aligned}$$

Derivative of Trigonometric Functions



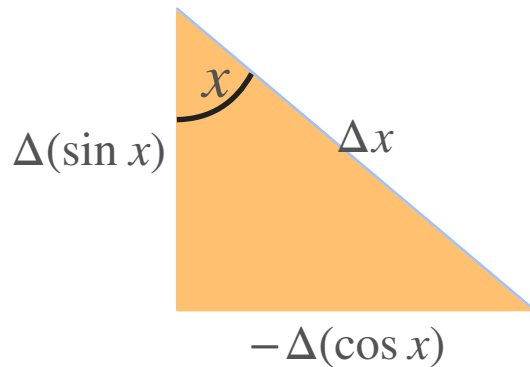
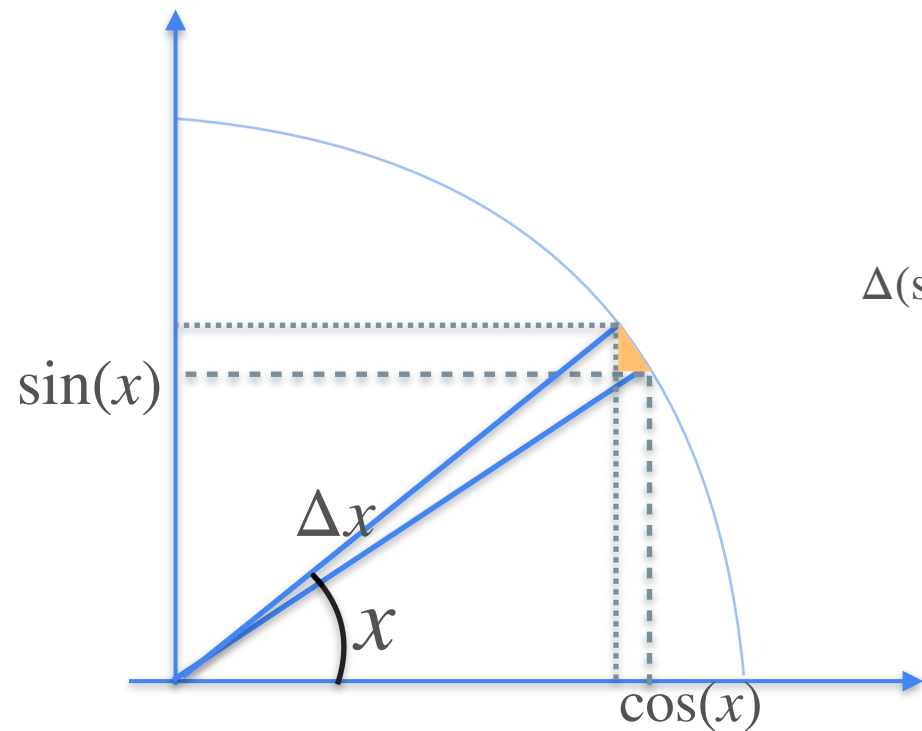
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Derivative of Trigonometric Functions



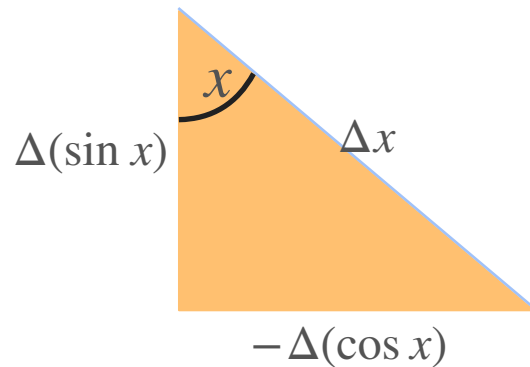
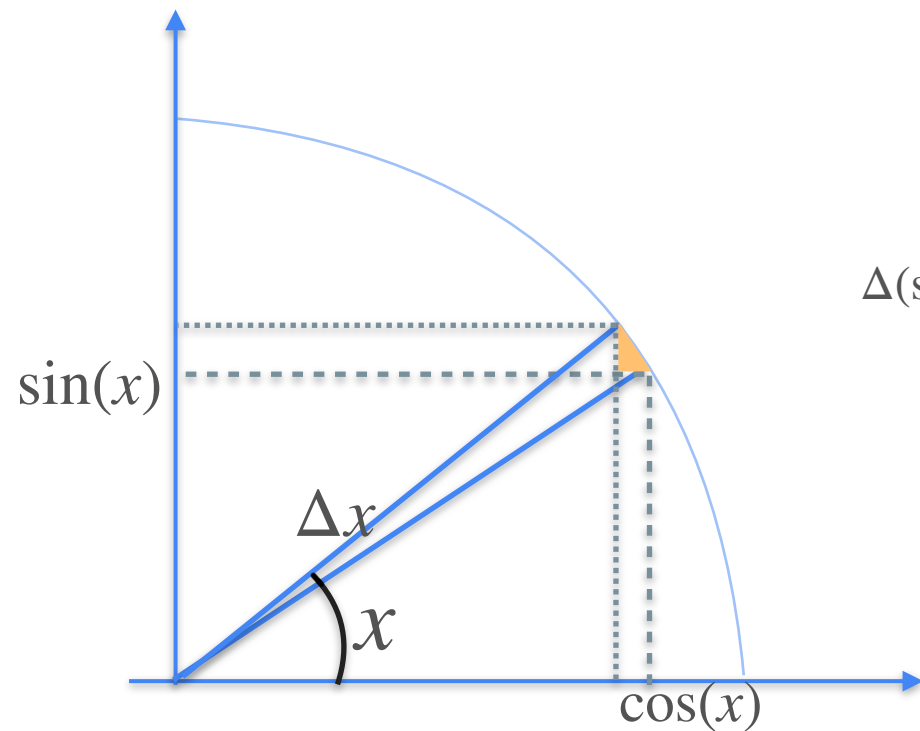
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Derivative of Trigonometric Functions



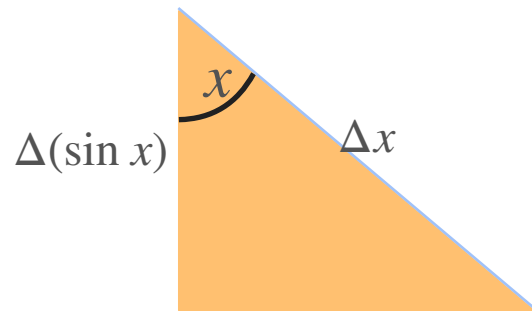
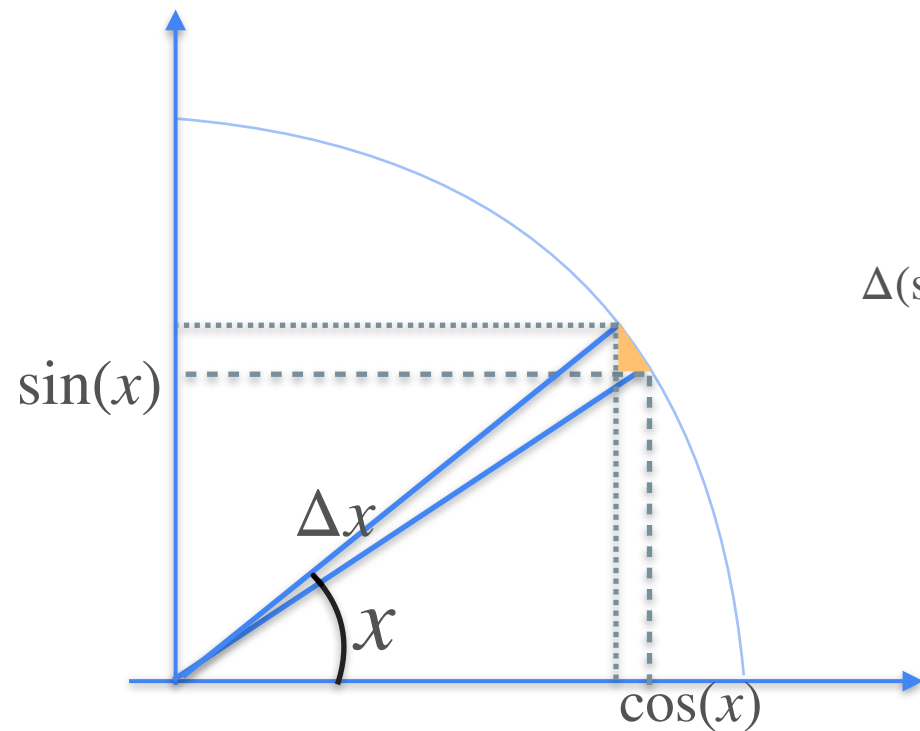
$$\begin{aligned}\Delta(\sin x) &= \Delta x \cos(x) \\ -\Delta(\cos x) &= h \sin(\varphi)\end{aligned}$$

Derivative of Trigonometric Functions



$$\begin{aligned}\Delta(\sin x) &= \Delta x \cos(x) \\ -\Delta(\cos x) &= \Delta x \sin(x)\end{aligned}$$

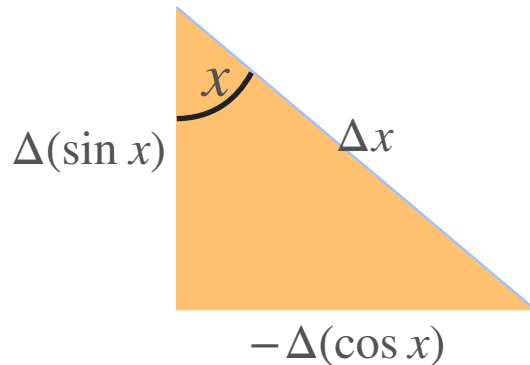
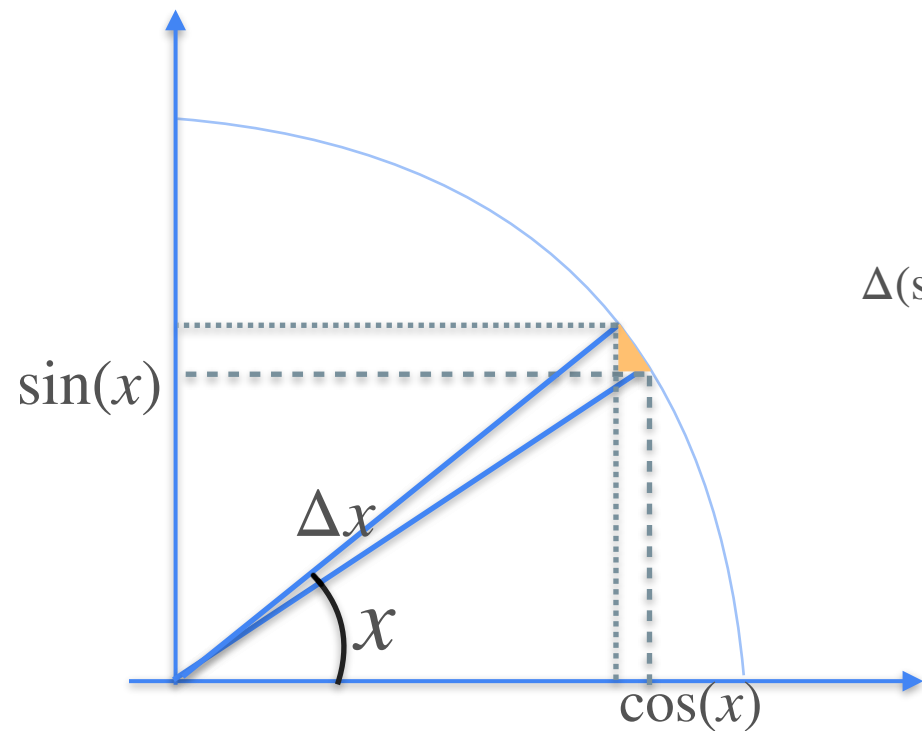
Derivative of Trigonometric Functions



$$\begin{aligned}\Delta(\sin x) &= \Delta x \cos(x) \\ -\Delta(\cos x) &= \Delta x \sin(x)\end{aligned}$$

$$f(x) = \sin(x) \Rightarrow f'(x) = \cos(x)$$

Derivative of Trigonometric Functions



$$\begin{aligned}\Delta(\sin x) &= \Delta x \cos(x) \\ -\Delta(\cos x) &= \Delta x \sin(x)\end{aligned}$$

$$f(x) = \sin(x) \Rightarrow f'(x) = \cos(x)$$

$$g(x) = \cos(x) \Rightarrow g'(x) = -\sin(x)$$



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Derivatives and Optimization

Meaning of the exponential (e)

$$e = 2.71828182\dots$$

$$e = 2.71828182\dots$$

$$\frac{n}{\left(1 + \frac{1}{n}\right)^n}$$

$$e = 2.71828182\dots$$

n	1
$\left(1 + \frac{1}{n}\right)^n$	2

$$\left(1 + \frac{1}{1}\right)^1$$

$$e = 2.71828182\dots$$

n	1	10
$\left(1 + \frac{1}{n}\right)^n$	2	2.594

$$\left(1 + \frac{1}{10}\right)^{10}$$

$$e = 2.71828182\dots$$

n	1	10	100
$\left(1 + \frac{1}{n}\right)^n$	2	2.594	2.705

$$\left(1 + \frac{1}{100}\right)^{100}$$

$$e = 2.71828182\dots$$

n	1	10	100	1000
$\left(1 + \frac{1}{n}\right)^n$	2	2.594	2.705	2.717

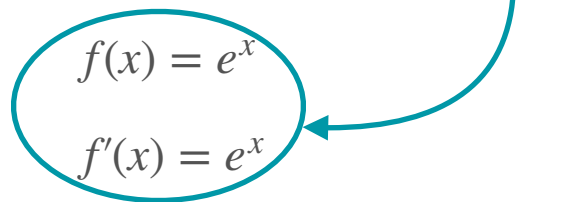
$$\left(1 + \frac{1}{1000}\right)^{1000}$$

$$e = 2.71828182\dots$$

n	1	10	100	1000	∞
$\left(1 + \frac{1}{n}\right)^n$	2	2.594	2.705	2.717	2.718

$$e = 2.71828182\dots$$

n	1	10	100	1000	∞
$\left(1 + \frac{1}{n}\right)^n$	2	2.594	2.705	2.717	e



A teal oval contains the functions $f(x) = e^x$ and $f'(x) = e^x$. A teal arrow points from this oval to the cell containing e in the table above.

$$f(x) = e^x$$
$$f'(x) = e^x$$

$$e = 2.71828182\dots$$



n	1	10	100	1000	∞
$\left(1 + \frac{1}{n}\right)^n$	2	2.594	2.705	2.717	e

$f(x) = e^x$
 $f'(x) = e^x$

Choosing a Bank



Choosing a Bank



Bank 1

Interests

100% every year

Choosing a Bank



Bank 1

Interests

100% every year

(all your money once a year)

Choosing a Bank



Bank 1

Interests

100% every year

(all your money once a year)



Bank 2

Choosing a Bank



Bank 1

Interests

100% every year

(all your money once a year)



Bank 2

Interests

50% every 6 months

Choosing a Bank



Bank 1

Interests

100% every year

(all your money once a year)



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Choosing a Bank



Bank 1

Interests

100% every year

(all your money once a year)



Bank 2

Interests

50% every 6 months

(half of your money twice a year)



Bank 3

Choosing a Bank



Bank 1

Interests

100% every year

(all your money once a year)



Bank 2

Interests

50% every 6 months

(half of your money twice a year)



Bank 3

Interests

33.3% every 4 months

Choosing a Bank



Bank 1

Interests

100% every year

(all your money once a year)



Bank 2

Interests

50% every 6 months

(half of your money twice a year)



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Which bank is better?



Bank 1

Interests

100% every year

(all your money once a year)



Bank 2

Interests

50% every 6 months

(half of your money twice a year)



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Which bank is better?



Bank 1

Interests

100% every year

(all your money once a year)



Bank 2

Interests

50% every 6 months

(half of your money twice a year)



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Which bank is better?



Bank 1

Interests

100% every year

(all your money once a year)



Bank 2

Interests

50% every 6 months

(half of your money twice a year)



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

You have \$1



Bank 1

Interests

100% every year

(all your money once a year)



Bank 2

Interests

50% every 6 months

(half of your money twice a year)



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

You have \$1



Bank 1

Interests

100% every year

(all your money once a year)

Now	US\$1
In 1 year	US\$2



Bank 2

Interests

50% every 6 months

(half of your money twice a year)



Bank 3



Interests

33.3% every 4 months

(A third of your money three times a year)

You have \$1



Bank 1

Interests

100% every year

(all your money once a year)

Now	US\$1
In 1 year	US\$2

↪ +100%



Bank 2

Interests

50% every 6 months

(half of your money twice a year)



Bank 3



Interests

33.3% every 4 months

(A third of your money three times a year)

You have \$1



Bank 1

Interests

100% every year

(all your money once a year)

Now	US\$1
In 1 year	US\$2

↪ +100%



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	US\$1
In 1 year	?



Bank 3



Interests

33.3% every 4 months

(A third of your money three times a year)

You have \$1



Bank 1

Interests

100% every year

(all your money once a year)

Now	US\$1
In 1 year	US\$2

↪ +100%



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	US\$1
In 1 year	?



Bank 3



Interests

33.3% every 4 months

(A third of your money three times a year)

Now	US\$1
In 1 year	?

You have \$1



Bank 1

Interests

100% every year

(all your money once a year)

Now	US\$1
In 1 year	US\$2

↪ +100%



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	US\$1
In 1 year	?

↪ ?



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Now	US\$1
In 1 year	?

You have \$1



Bank 1

Interests

100% every year

(all your money once a year)

Now	US\$1
In 1 year	US\$2

↪ +100%



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	US\$1
In 1 year	?



Bank 3



Interests

33.3% every 4 months

(A third of your money three times a year)

Now	US\$1
In 1 year	?

How Much Do I Have After 1 Year?



Bank 1

Interests

100% every year



Bank 2

Interests

50% every 6 months

How Much Do I Have After 1 Year?



Bank 1

Now



Interests

100% every year



Bank 2

Interests

50% every 6 months

How Much Do I Have After 1 Year?



Bank 1

Interests

100% every year



Bank 2

Interests

50% every 6 months

Now



1

How Much Do I Have After 1 Year?



Bank 1

Interests

100% every year



Bank 2

Interests

50% every 6 months

Now



1

In 1 year



How Much Do I Have After 1 Year?



Bank 1

Interests

100% every year



Bank 2

Interests

50% every 6 months

Now



1

In 1 year



How Much Do I Have After 1 Year?



Bank 1

Interests

100% every year



Bank 2

Interests

50% every 6 months

Now



1

In 1 year



2

How Much Do I Have After 1 Year?



Bank 1

Interests

100% every year



Bank 2

Interests

50% every 6 months

Now



1

In 1 year



2

$$(1 + 1)^1$$

How Much Do I Have After 1 Year?



Bank 1

Interests

100% every year

Now



1

In 1 year



2

$$(1 + 1)^1$$



Bank 2

Interests

50% every 6 months

Now



How Much Do I Have After 1 Year?



Bank 1

Interests

100% every year

Now



1

In 1 year



2

$$(1 + 1)^1$$



Bank 2

Interests

50% every 6 months

Now



1

How Much Do I Have After 1 Year?



Bank 1

Now



1

In 1 year



2

$$(1 + 1)^1$$

Interests

100% every year



Bank 2

Now



1

In 6 months



Interests

50% every 6 months

How Much Do I Have After 1 Year?



Bank 1

Now



1

In 1 year



2

$$(1 + 1)^1$$

Interests

100% every year



Bank 2

Now



1

In 6 months



Interests

50% every 6 months

How Much Do I Have After 1 Year?



Bank 1

Now



1

In 1 year



2

$$(1 + 1)^1$$



Bank 2

Now



1

In 6 months



1.5

Interests

50% every 6 months

How Much Do I Have After 1 Year?



Bank 1

Now



1

In 1 year



2

$$(1 + 1)^1$$

Interests

100% every year



Bank 2

Now



1

In 6 months



1.5

$$1 + \frac{1}{2}$$

Interests

50% every 6 months

How Much Do I Have After 1 Year?



Bank 1

Now



1

In 1 year



2

$$(1 + 1)^1$$

Interests
100% every year



Bank 2

Now



1

In 6 months



1.5

$$1 + \frac{1}{2}$$

In 1 year



Interests
50% every 6 months

How Much Do I Have After 1 Year?



Bank 1

Now



1

In 1 year



2

$$(1 + 1)^1$$

Interests
100% every year



Bank 2

Now



1

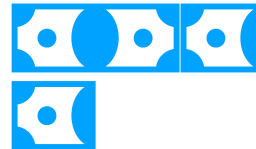
In 6 months



1.5

$$1 + \frac{1}{2}$$

In 1 year



Interests
50% every 6 months

How Much Do I Have After 1 Year?



Bank 1

Now



1

In 1 year



2

$$(1 + 1)^1$$

Interests
100% every year



Bank 2

Now



1

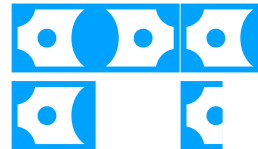
In 6 months



1.5

$$1 + \frac{1}{2}$$

In 1 year



Interests
50% every 6 months

How Much Do I Have After 1 Year?



Bank 1

Now



1

In 1 year



2

$$(1 + 1)^1$$



Bank 2

Now



1

In 6 months



1.5

$$1 + \frac{1}{2}$$

In 1 year



Interests
100% every year

Interests
50% every 6 months

How Much Do I Have After 1 Year?



Bank 1

Now



1

In 1 year



2

$$(1 + 1)^1$$

Interests
100% every year



Bank 2

Now



1

In 6 months



1.5

$$1 + \frac{1}{2}$$

In 1 year



2.25

Interests
50% every 6 months

How Much Do I Have After 1 Year?



Bank 1

Now



1

In 1 year



2

$$(1 + 1)^1$$

Interests
100% every year



Bank 2

Now



1

In 6 months



1.5

$$1 + \frac{1}{2}$$

In 1 year



2.25

$$\left(1 + \frac{1}{2}\right)^2$$

Interests
50% every 6 months

How Much Do I Have After 1 Year?



Bank 1

Now



1

In 1 year



2

$$(1 + 1)^1$$

Interests

100% every year



Bank 2

Now



1

In 6 months



1.5

$$1 + \frac{1}{2}$$

In 1 year



2.25

$$\left(1 + \frac{1}{2}\right)^2$$

Interests

50% every 6 months

Compound interest



Bank 1

Now



1

In 1 year



2

$$(1 + 1)^1$$

Interests

100% every year



Bank 2

Now



1

In 6 months



1.5

$$1 + \frac{1}{2}$$

In 1 year



2.25

$$\left(1 + \frac{1}{2}\right)^2$$

Interests

50% every 6 months

How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now

In 4 months

In 8 months

In 1 year

How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

In 4 months

In 8 months

In 1 year

How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

In 4 months



In 8 months

In 1 year

How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

In 4 months



In 8 months

In 1 year

How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

In 4 months



1.33

In 8 months

In 1 year

How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

In 4 months



1.33

In 8 months



In 1 year

How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

In 4 months



1.33

In 8 months



In 1 year

How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

In 4 months



1.33

In 8 months



1.77

In 1 year

How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

In 4 months



1.33

In 8 months



1.77

In 1 year



How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

In 4 months



1.33

In 8 months



1.77

In 1 year



How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

In 4 months



1.33

In 8 months



1.77

In 1 year



2.37

How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

In 4 months



1.33

In 8 months



1.77

In 1 year



2.37

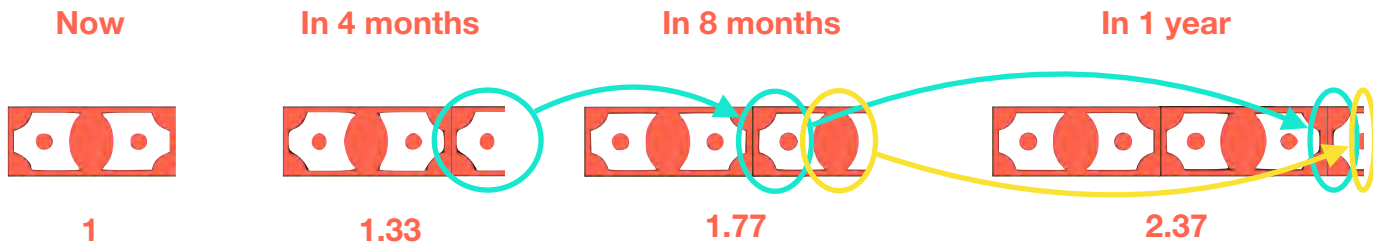
How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months



How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

1

Original
money

In 4 months



1.33

In 8 months



1.77

In 1 year



2.37

How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months

Now



1

1

Original
money

In 4 months



1.33

$$1 + \frac{1}{3}$$

Add 1/3 of
the money

In 8 months



1.77

In 1 year



2.37

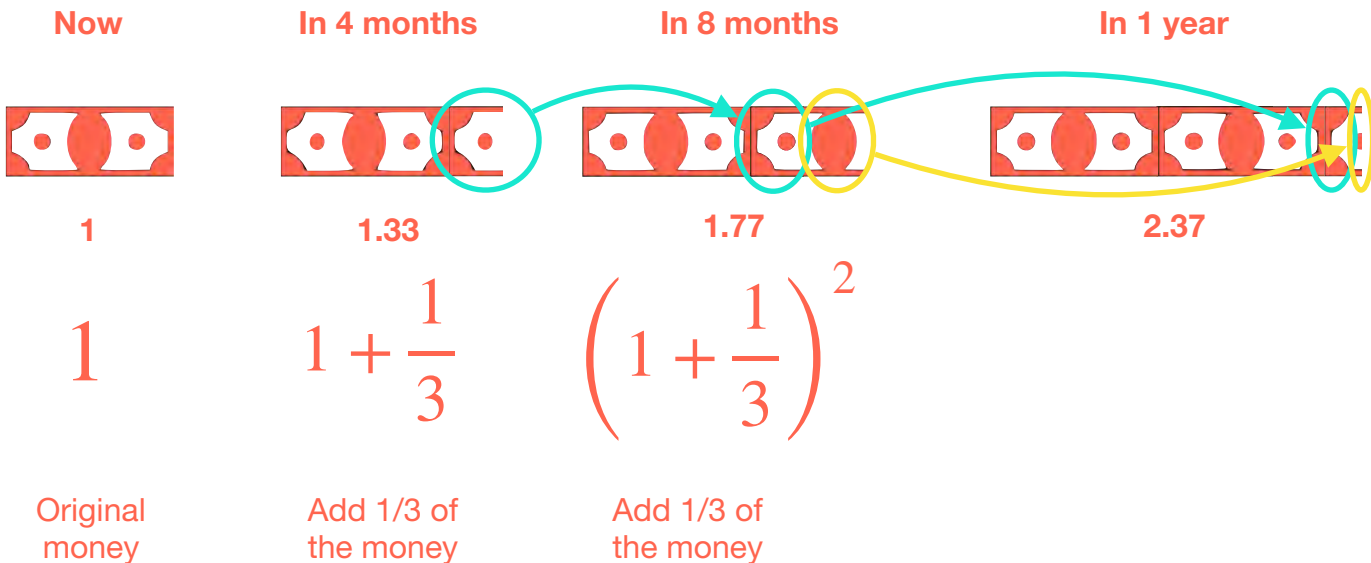
How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months



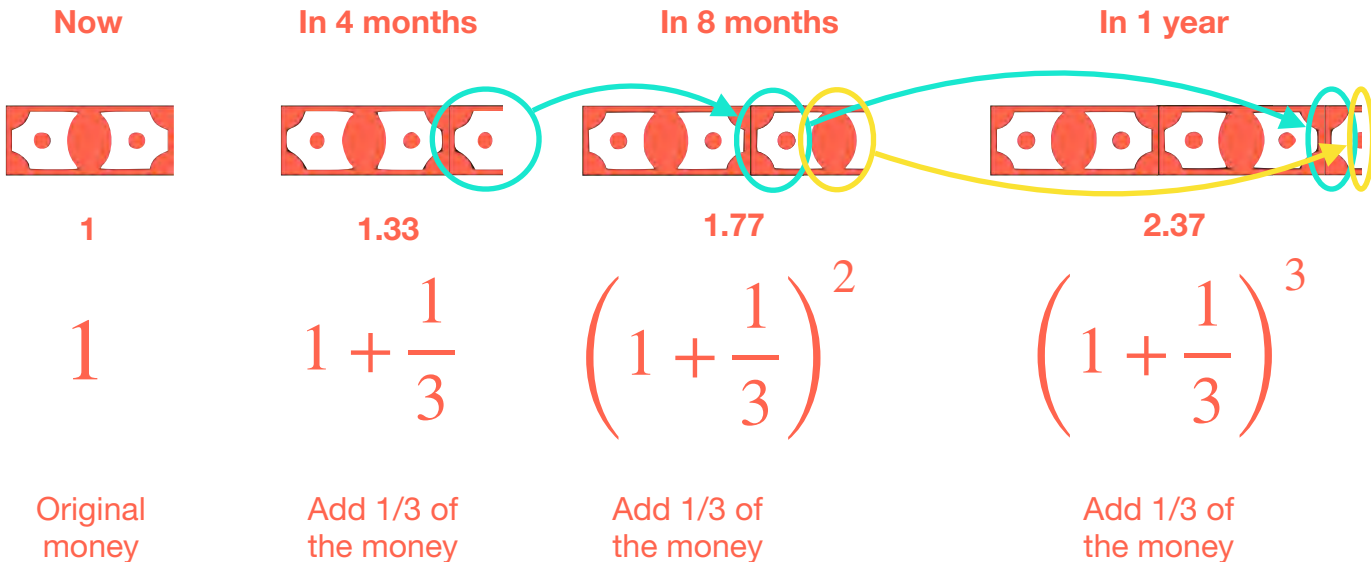
How Much Do I Have After 1 Year?



Bank 3

Interests

33.3% every 4 months



Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)



Bank 2

Interests

50% every 6 months

(half of your money twice a year)



Bank 3



Interests

33.3% every 4 months

(A third of your money three times a year)

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1
In 1 year	\$2



Bank 2

Interests

50% every 6 months

(half of your money twice a year)



Bank 3



Interests

33.3% every 4 months

(A third of your money three times a year)

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1
In 1 year	\$2



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	\$1
In 1 year	\$2.25



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1
In 1 year	\$2



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	\$1
In 1 year	\$2.25



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Now	\$1
In 1 year	\$2.37

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1
In 1 year	\$2

↪ +100%



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	\$1
In 1 year	\$2.25



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Now	\$1
In 1 year	\$2.37

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1		
In 1 year	\$2		
In 2	\$4		
In 3	\$8		
In 4	\$16		



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	\$1
In 1 year	\$2.25



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Now	\$1
In 1 year	\$2.37

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1	
In 1 year	\$2	
In 2	\$4	
In 3	\$8	
In 4	\$16	



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	\$1	
In 1 year	\$2.25	



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Now	\$1	
In 1 year	\$2.37	

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1
In 1 year	\$2
In 2	\$4
In 3	\$8
In 4	\$16

Diagram showing four upward arrows, each labeled +100%, indicating the growth from one period to the next.



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	\$1
In 1 year	\$2.25
In 2	\$5.06
In 3	\$11.39
In 4	\$25.63

Diagram showing four upward arrows, each labeled +125%, indicating the growth from one period to the next.



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Now	\$1
In 1 year	\$2.37

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1
In 1 year	\$2
In 2	\$4
In 3	\$8
In 4	\$16

Diagram showing four upward arrows, each labeled +100%, indicating the growth from one period to the next.



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	\$1
In 1 year	\$2.25
In 2	\$5.06
In 3	\$11.39
In 4	\$25.63

Diagram showing four upward arrows, each labeled +125%, indicating the growth from one period to the next.



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Now	\$1
In 1 year	\$2.37

Diagram showing a single upward arrow labeled +137%, indicating the total growth over one year.

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1	
In 1 year	\$2	+100%
In 2	\$4	+100%
In 3	\$8	+100%
In 4	\$16	+100%



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	\$1	
In 1 year	\$2.25	+125%
In 2	\$5.06	+125%
In 3	\$11.39	+125%
In 4	\$25.63	+125%



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Now	\$1	
In 1 year	\$2.37	+137%
In 2	\$5.62	+137%
In 3	\$13.32	+137%
In 4	\$31.57	+137%

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1
...	↑ --



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	\$1
...	↑ --



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Now	\$1
...	↑ --

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1
...	↑ --

$$(1 + 1)^1 = 2$$



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	\$1
...	↑ --



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Now	\$1
...	↑ --

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1
...	...

$$(1 + 1)^1 = 2$$



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	\$1
...	...

$$\left(1 + \frac{1}{2}\right)^2 = 2.25$$



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Now	\$1
...	...

Which Bank Is Better?



Bank 1

Interests

100% every year

(all your money once a year)

Now	\$1
.....	

$$(1 + 1)^1 = 2$$



Bank 2

Interests

50% every 6 months

(half of your money twice a year)

Now	\$1
.....	

$$\left(1 + \frac{1}{2}\right)^2 = 2.25$$



Bank 3

Interests

33.3% every 4 months

(A third of your money three times a year)

Now	\$1
.....	

$$\left(1 + \frac{1}{3}\right)^3 = 2.37$$

How Much Do I Have After 1 Year?



Bank 12

Interests

1/12 every month

Now

In 1 month

In 2 months

...

In 12 months

How Much Do I Have After 1 Year?



How Much Do I Have After 1 Year?

 Bank 12 Interests 1/12 every month	Now	In 1 month	In 2 months	...	In 12 months
	1	$1 + \frac{1}{12}$			
	Original money	Add 1/12 of the money			

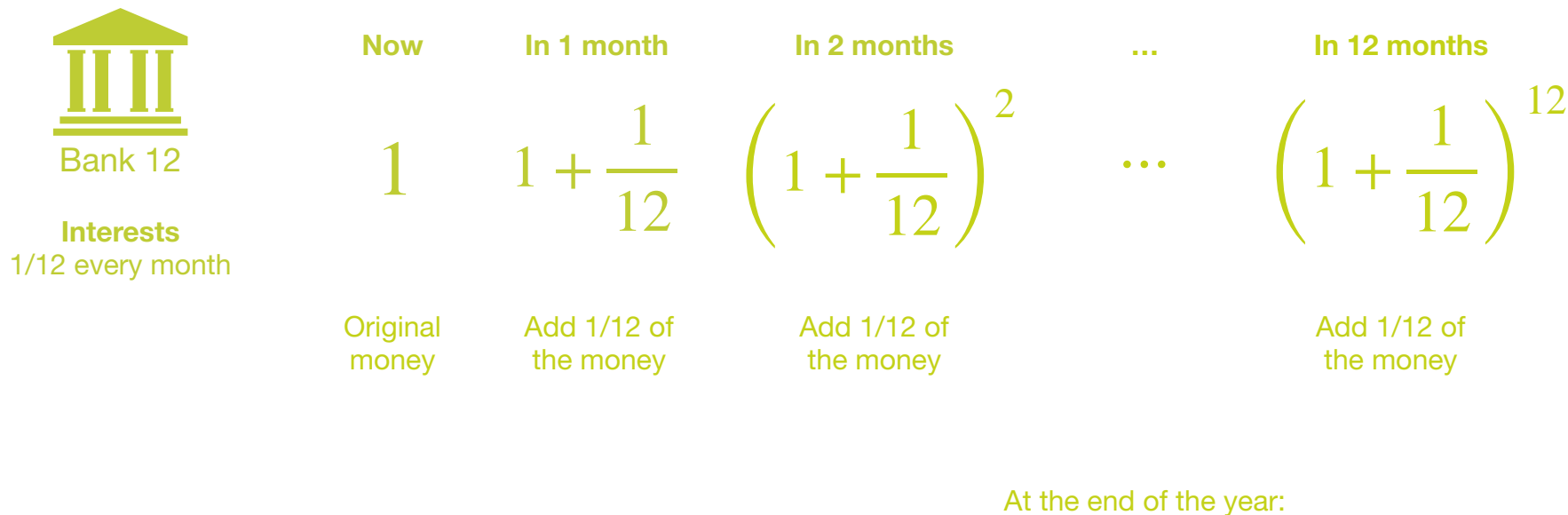
How Much Do I Have After 1 Year?

 Bank 12 Interests 1/12 every month	Now	In 1 month	In 2 months	...	In 12 months
	1	$1 + \frac{1}{12}$	$\left(1 + \frac{1}{12}\right)^2$		$\left(1 + \frac{1}{12}\right)^{12}$
	Original money	Add 1/12 of the money	Add 1/12 of the money		Add 1/12 of the money

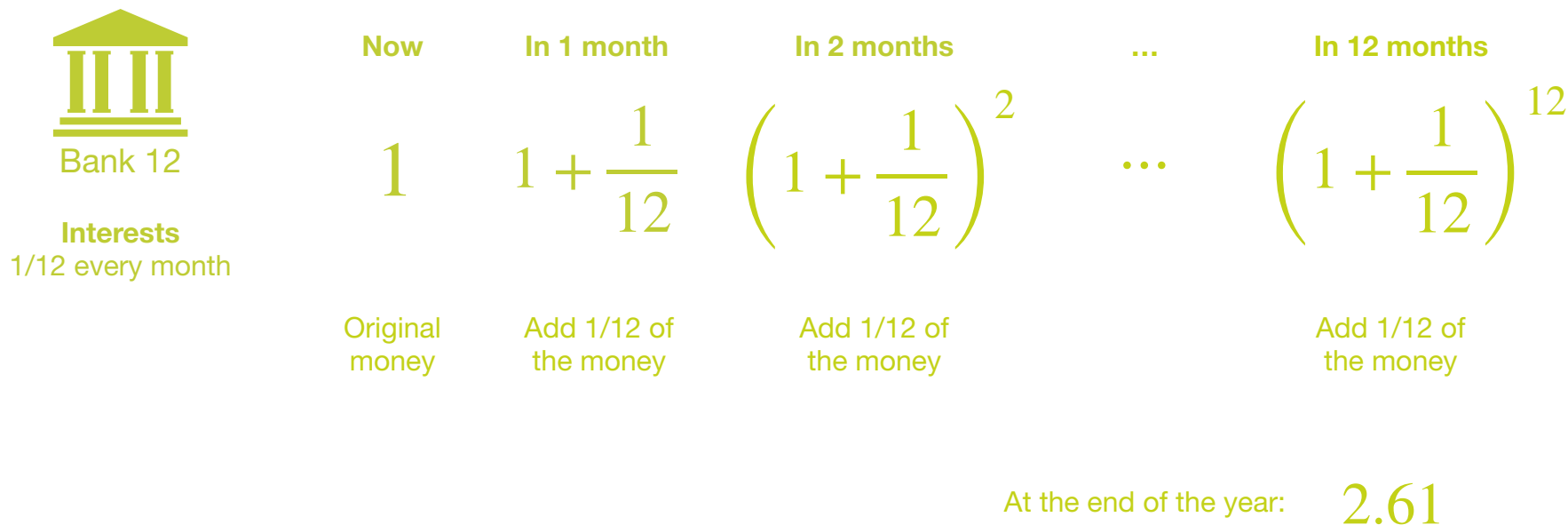
How Much Do I Have After 1 Year?

 Bank 12 Interests 1/12 every month	Now	In 1 month	In 2 months	...	In 12 months
	1	$1 + \frac{1}{12}$	$\left(1 + \frac{1}{12}\right)^2$	...	$\left(1 + \frac{1}{12}\right)^{12}$
	Original money	Add 1/12 of the money	Add 1/12 of the money		

How Much Do I Have After 1 Year?



How Much Do I Have After 1 Year?



In General



Bank n

Interests

$1/n$ every interval

In General



Bank n

Interests

$1/n$ every interval

1	2	3	.	.	.	.	.	.	k	.	.	.	.	.	.	n-2	n-1	n
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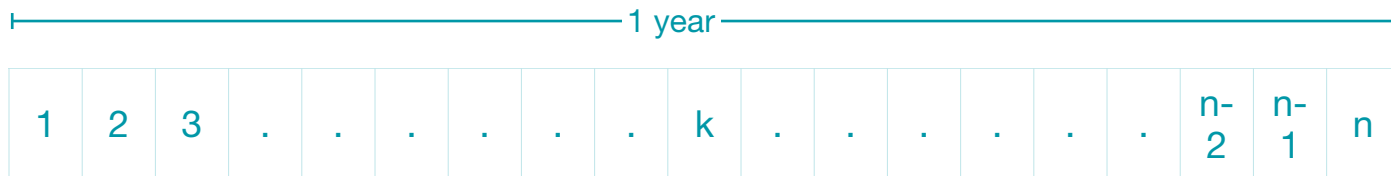
In General



Bank n

Interests

$1/n$ every interval



In General



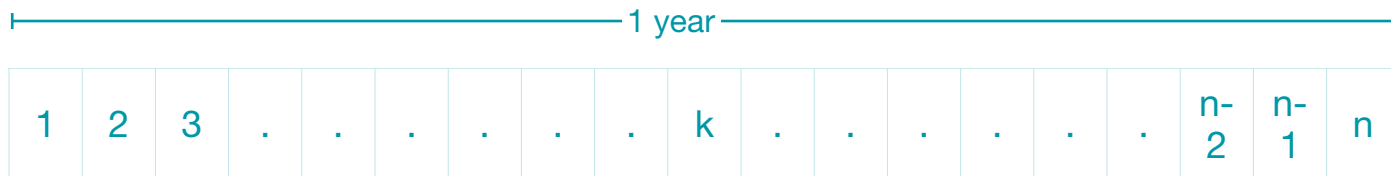
Bank n

Interests

$1/n$ every interval

Now

1



In General



Bank n

1 year

1	2	3	.	.	.	.	.	.	k	.	.	.	.	.	.	.	n-2	n-1	n
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----	-----	---

Interests

1/n every interval

Now

In 1 interval

$$1 \quad 1 + \frac{1}{n}$$

In General



Bank n

1 year

1	2	3	.	.	.	.	.	.	k	.	.	.	.	.	.	.	n-2	n-1	n
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----	-----	---

Interests

1/n every interval

Now

In 1 interval

In 2 intervals

$$1$$

$$1 + \frac{1}{n}$$

$$\left(1 + \frac{1}{n}\right)^2$$

In General



Bank n

1 year

1	2	3	.	.	.	.	.	.	k	.	.	.	.	.	.	.	n-2	n-1	n
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----	-----	---

Interests

1/n every interval

Now

In 1 interval

In 2 intervals

1

$$1 + \frac{1}{n}$$

$$\left(1 + \frac{1}{n}\right)^2$$

↙

$$\left(1 + \frac{1}{n}\right) + \frac{1}{n} \left(1 + \frac{1}{n}\right)$$

In General



Bank n

1 year

1	2	3	.	.	.	.	.	.	k	.	.	.	.	.	.	.	n-2	n-1	n
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----	-----	---

Interests

1/n every interval

Now

In 1 interval

In 2 intervals

1

$$1 + \frac{1}{n}$$

$$\left(1 + \frac{1}{n}\right)^2$$

↙

$$\left(1 + \frac{1}{n}\right) + \frac{1}{n} \left(1 + \frac{1}{n}\right)$$

Money

In General



Bank n

1 year

1	2	3	.	.	.	.	.	.	k	.	.	.	.	.	.	.	n-2	n-1	n
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----	-----	---

Interests

1/n every interval

Now

In 1 interval

In 2 intervals

1

$1 + \frac{1}{n}$

$\left(1 + \frac{1}{n}\right)^2$

$\left(1 + \frac{1}{n}\right) + \frac{1}{n} \left(1 + \frac{1}{n}\right)$

Money

Interest

In General



Bank n

1 year

1	2	3	.	.	.	.	.	.	k	.	.	.	.	.	.	.	n-2	n-1	n
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----	-----	---

Interests

1/n every interval

Now

In 1 interval

In 2 intervals

$$1 \quad 1 + \frac{1}{n} \quad \left(1 + \frac{1}{n}\right)^2$$

$$\left(1 + \frac{1}{n}\right) + \frac{1}{n} \left(1 + \frac{1}{n}\right)$$

Money

Interest

In General



Bank n

1 year

1	2	3	.	.	.	.	.	.	k	.	.	.	.	.	.	.	n-2	n-1	n
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----	-----	---

Interests

1/n every interval

Now

1

In 1 interval

$$1 + \frac{1}{n}$$

In 2 intervals

$$\left(1 + \frac{1}{n}\right)^2$$

In 3 intervals

$$\left(1 + \frac{1}{n}\right)^3$$

$$\left(1 + \frac{1}{n}\right) + \frac{1}{n} \left(1 + \frac{1}{n}\right)$$

Money

Interest

In General



Bank n

1 year

1	2	3	.	.	.	.	.	.	k	.	.	.	.	.	.	.	n-2	n-1	n
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----	-----	---

Interests

1/n every interval

Now

In 1 interval

In 2 intervals

In 3 intervals

...

1

$1 + \frac{1}{n}$

$\left(1 + \frac{1}{n}\right)^2$

$\left(1 + \frac{1}{n}\right)^3$

$\left(1 + \frac{1}{n}\right) + \frac{1}{n} \left(1 + \frac{1}{n}\right)$

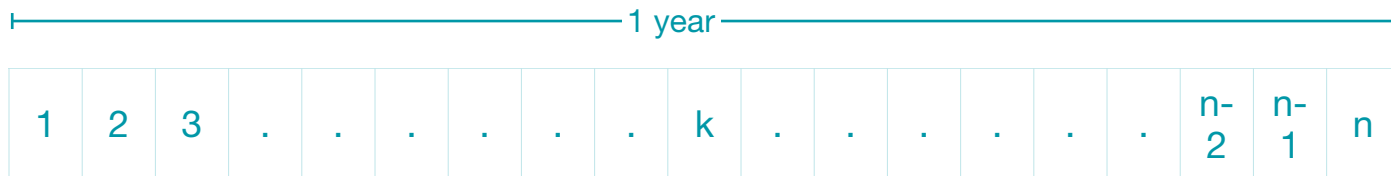
Money

Interest

In General



Bank n



Interests

1/n every interval

Now

1

In 1 interval

$$1 + \frac{1}{n}$$

In 2 intervals

$$\left(1 + \frac{1}{n}\right)^2$$

In 3 intervals

$$\left(1 + \frac{1}{n}\right)^3$$

...

In k intervals

$$\left(1 + \frac{1}{n}\right)^k$$

$$\left(1 + \frac{1}{n}\right) + \frac{1}{n} \left(1 + \frac{1}{n}\right)$$

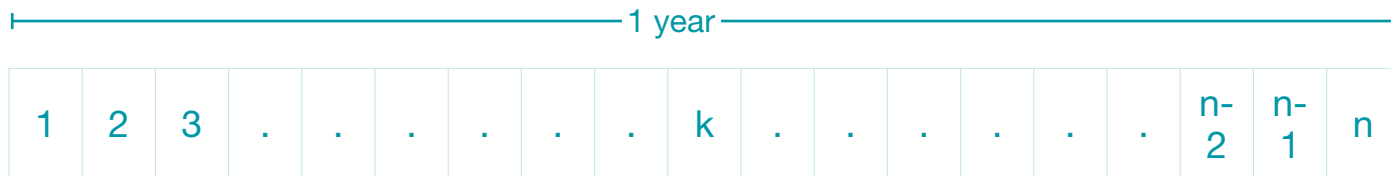
Money

Interest

In General



Bank n



Interests

$1/n$ every interval

Now

1

In 1 interval

$$1 + \frac{1}{n}$$

In 2 intervals

$$\left(1 + \frac{1}{n}\right)^2$$

In 3 intervals

$$\left(1 + \frac{1}{n}\right)^3$$

...

In k intervals

$$\left(1 + \frac{1}{n}\right)^k$$

...

$$\left(1 + \frac{1}{n}\right) + \frac{1}{n} \left(1 + \frac{1}{n}\right)$$

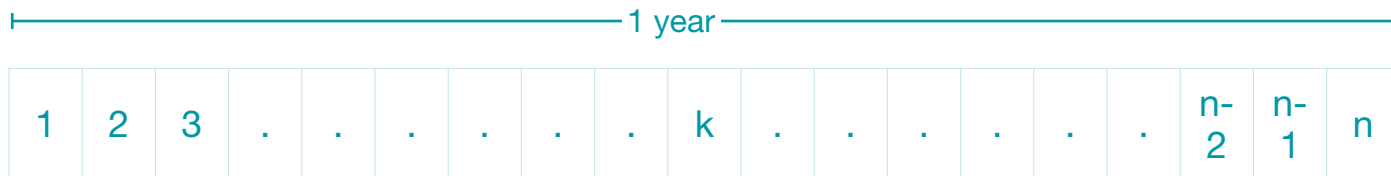
Money

Interest

In General



Bank n



Interests

1/n every interval

Now	In 1 interval	In 2 intervals	In 3 intervals	...	In k intervals	...	In 1 year (n intervals)
1	$1 + \frac{1}{n}$	$\left(1 + \frac{1}{n}\right)^2$	$\left(1 + \frac{1}{n}\right)^3$		$\left(1 + \frac{1}{n}\right)^k$		$\left(1 + \frac{1}{n}\right)^n$
	$\left(1 + \frac{1}{n}\right)$ Money $+ \frac{1}{n} \left(1 + \frac{1}{n}\right)$ Interest						

In General



Bank n

Interests

$1/n$ every interval

In General



Bank n

Interests

$1/n$ every interval

In 1 year

$$\left(1 + \frac{1}{n}\right)^n$$

In General



Bank n

Interests

1/n every interval

In 1 year

$$\left(1 + \frac{1}{n}\right)^n$$

↑
1/n of your money

In General



Bank n

Interests

$1/n$ every interval

In 1 year

$$\left(1 + \frac{1}{n} \right)^n$$

n ← n times

↑
 $1/n$ of your money

A Lot of Banks

A Lot of Banks

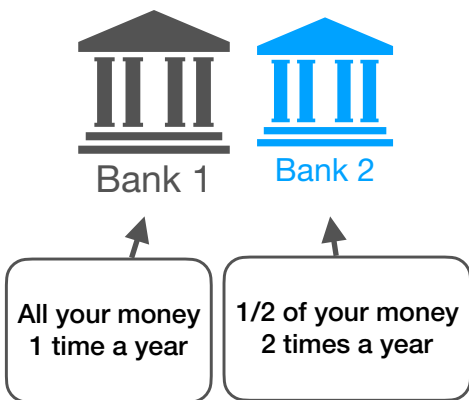


Bank 1

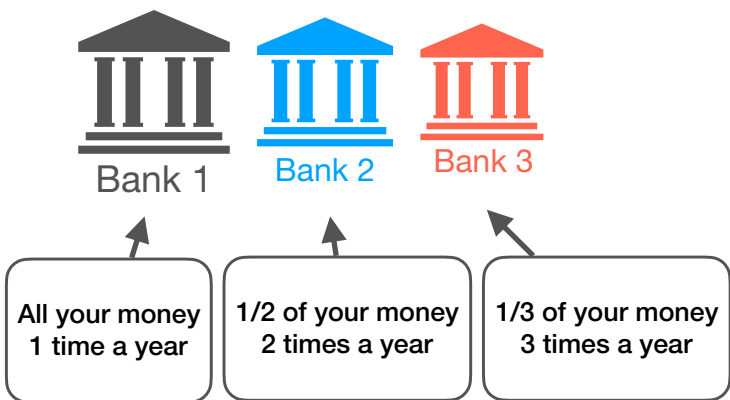


All your money
1 time a year

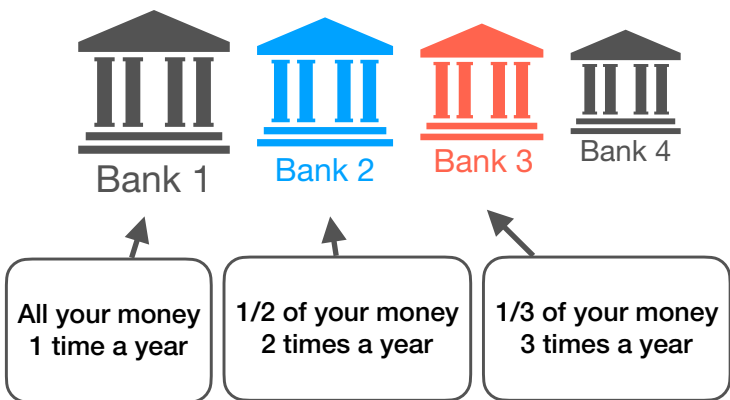
A Lot of Banks



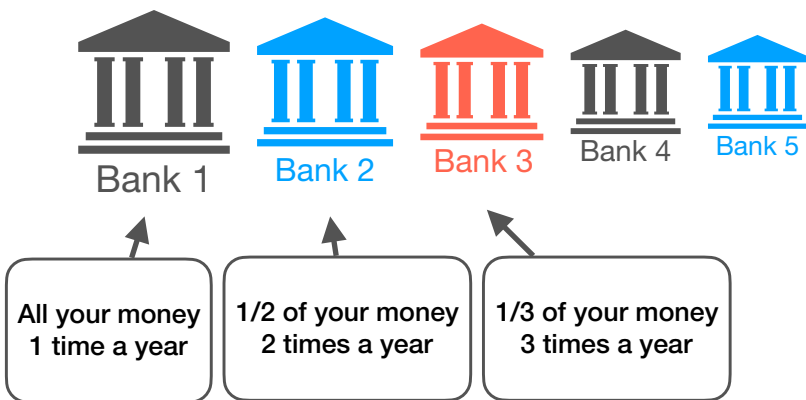
A Lot of Banks



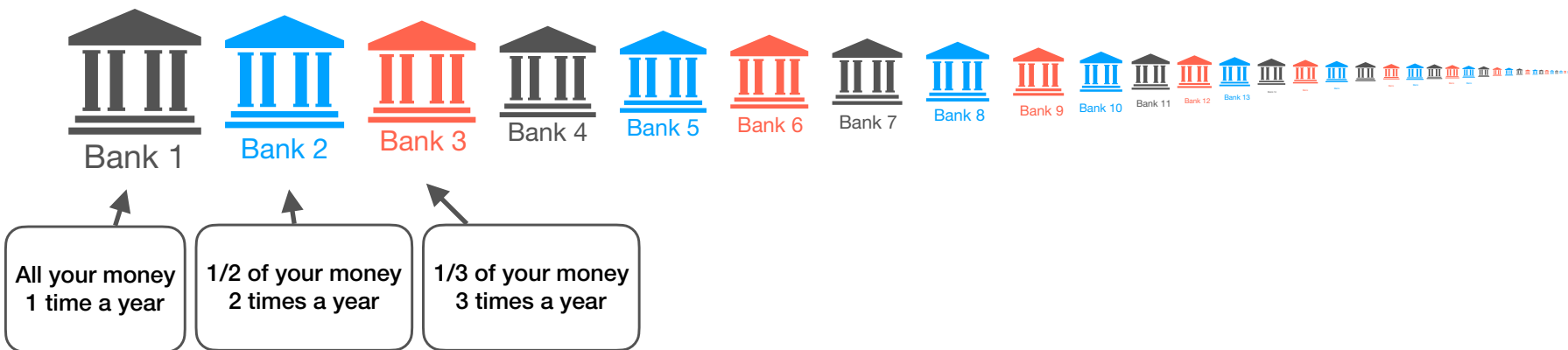
A Lot of Banks



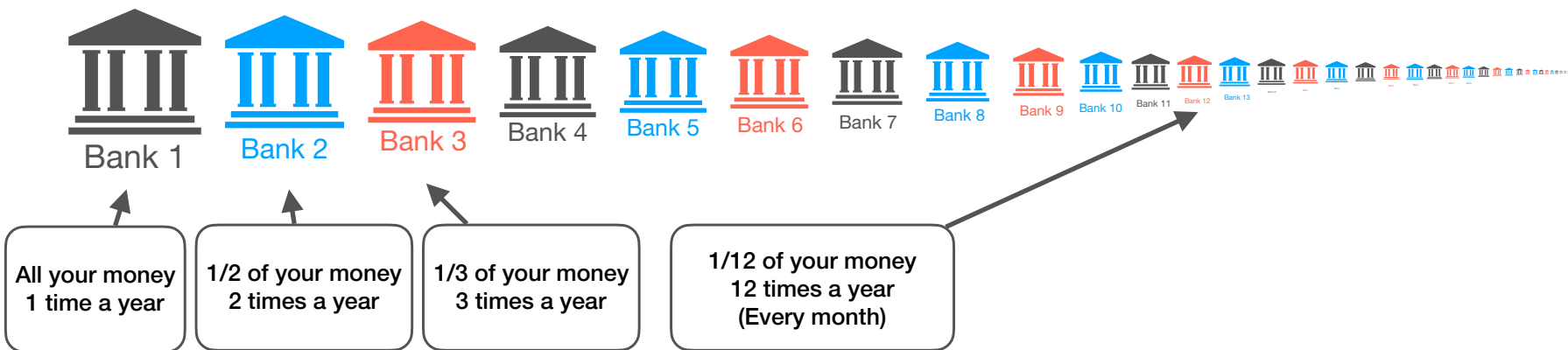
A Lot of Banks



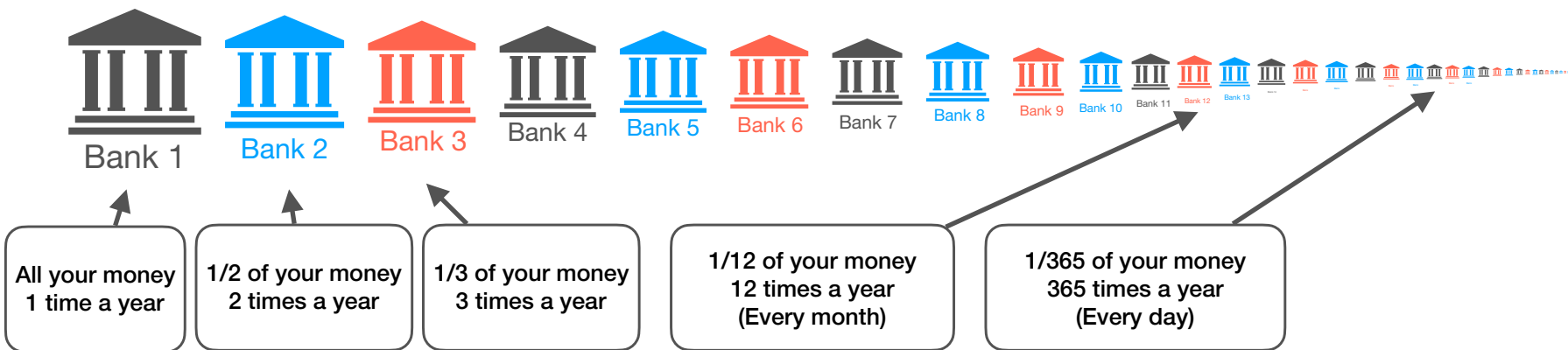
A Lot of Banks



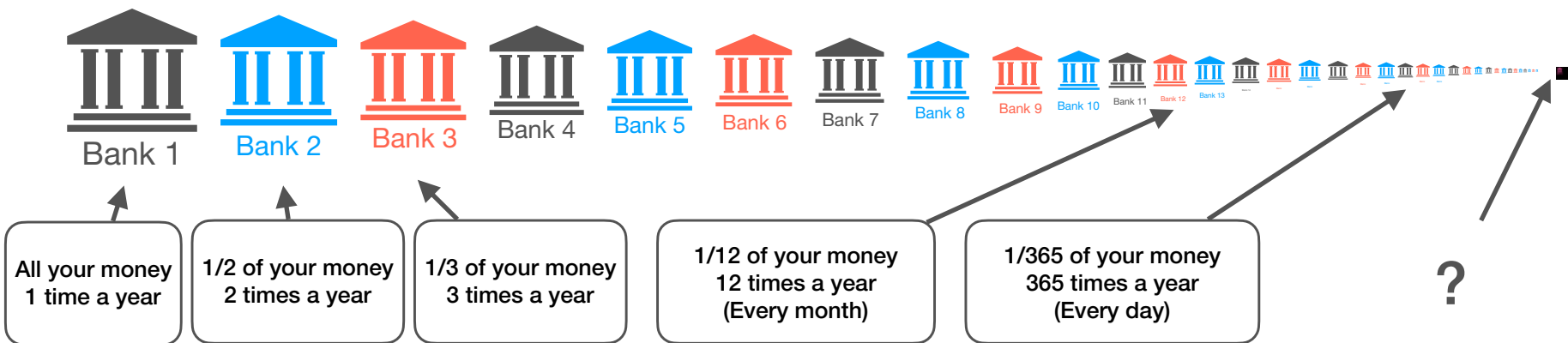
A Lot of Banks



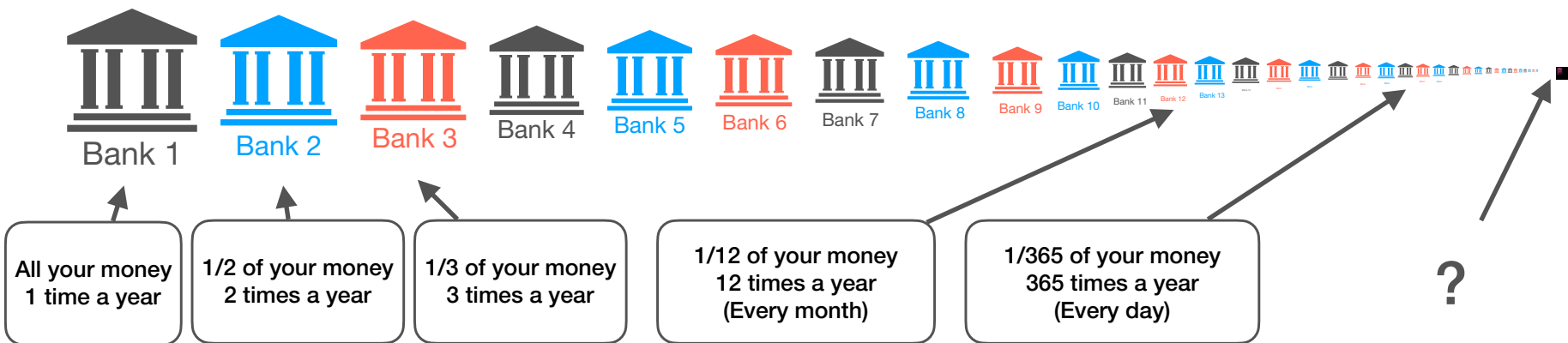
A Lot of Banks



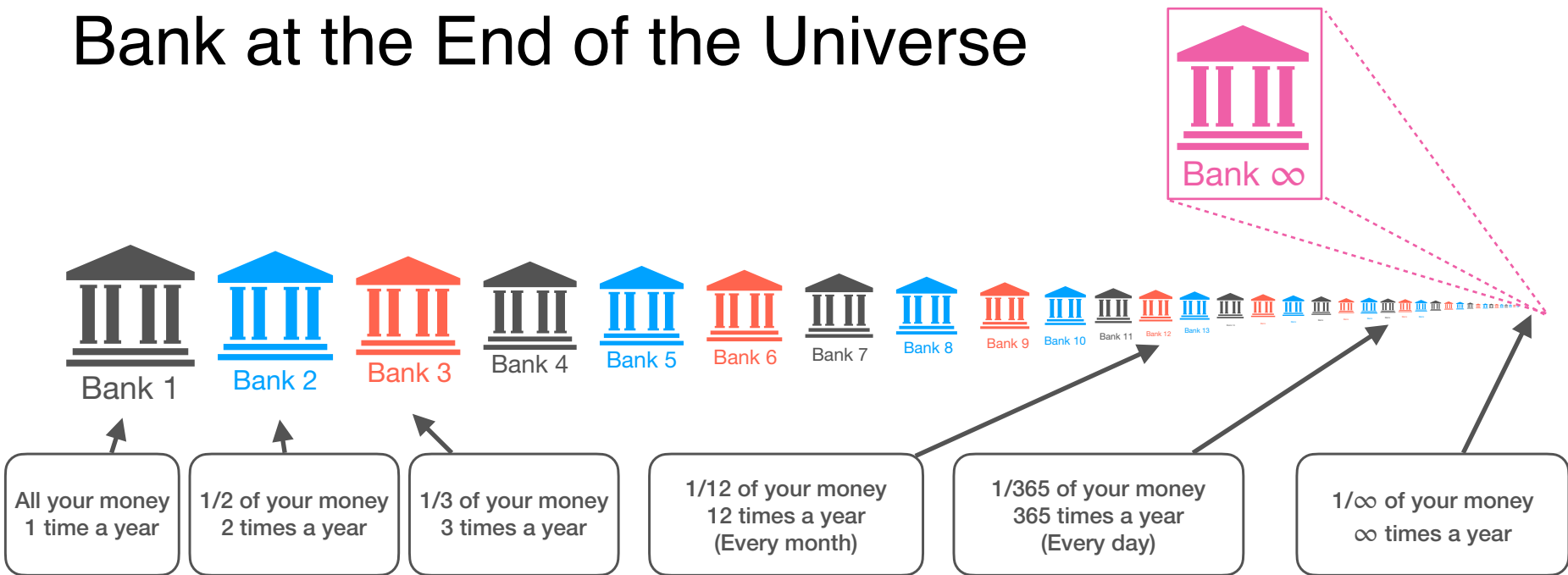
A Lot of Banks



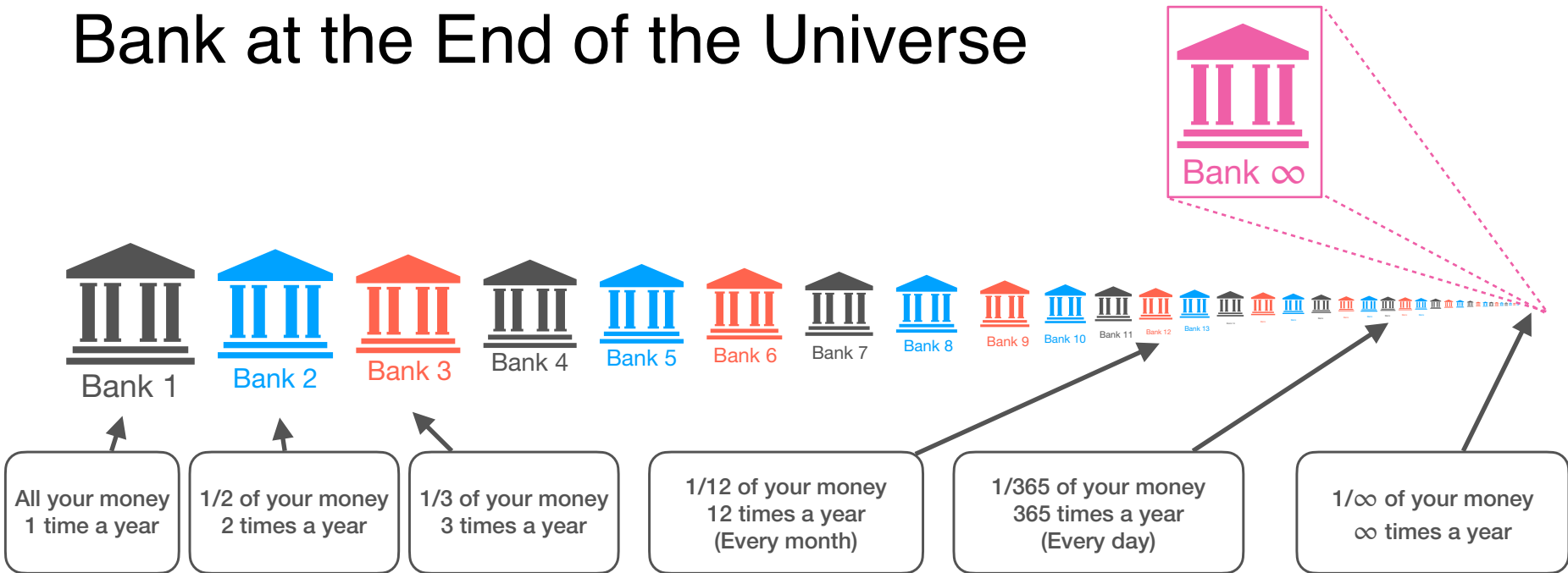
Bank at the end of the universe



Bank at the End of the Universe



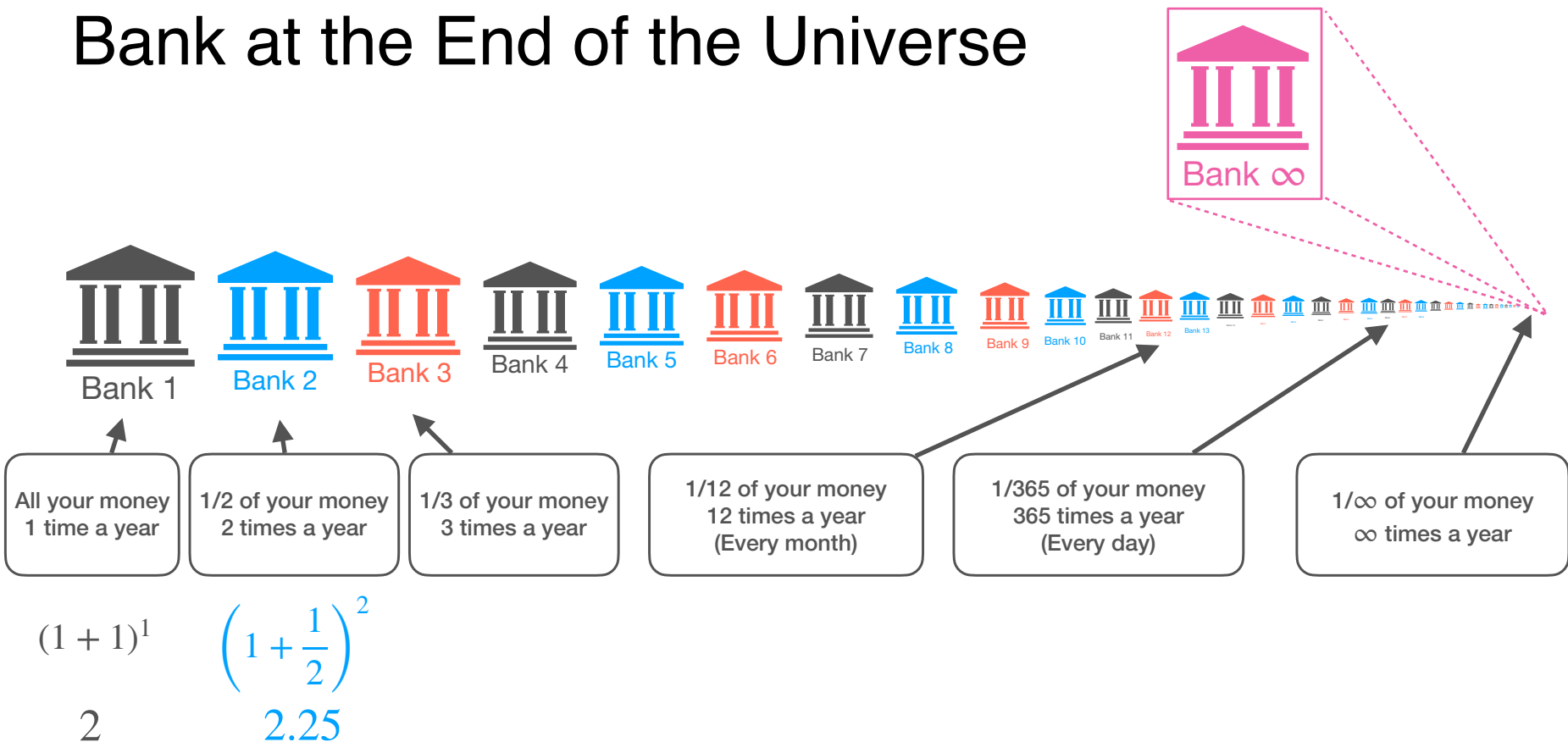
Bank at the End of the Universe



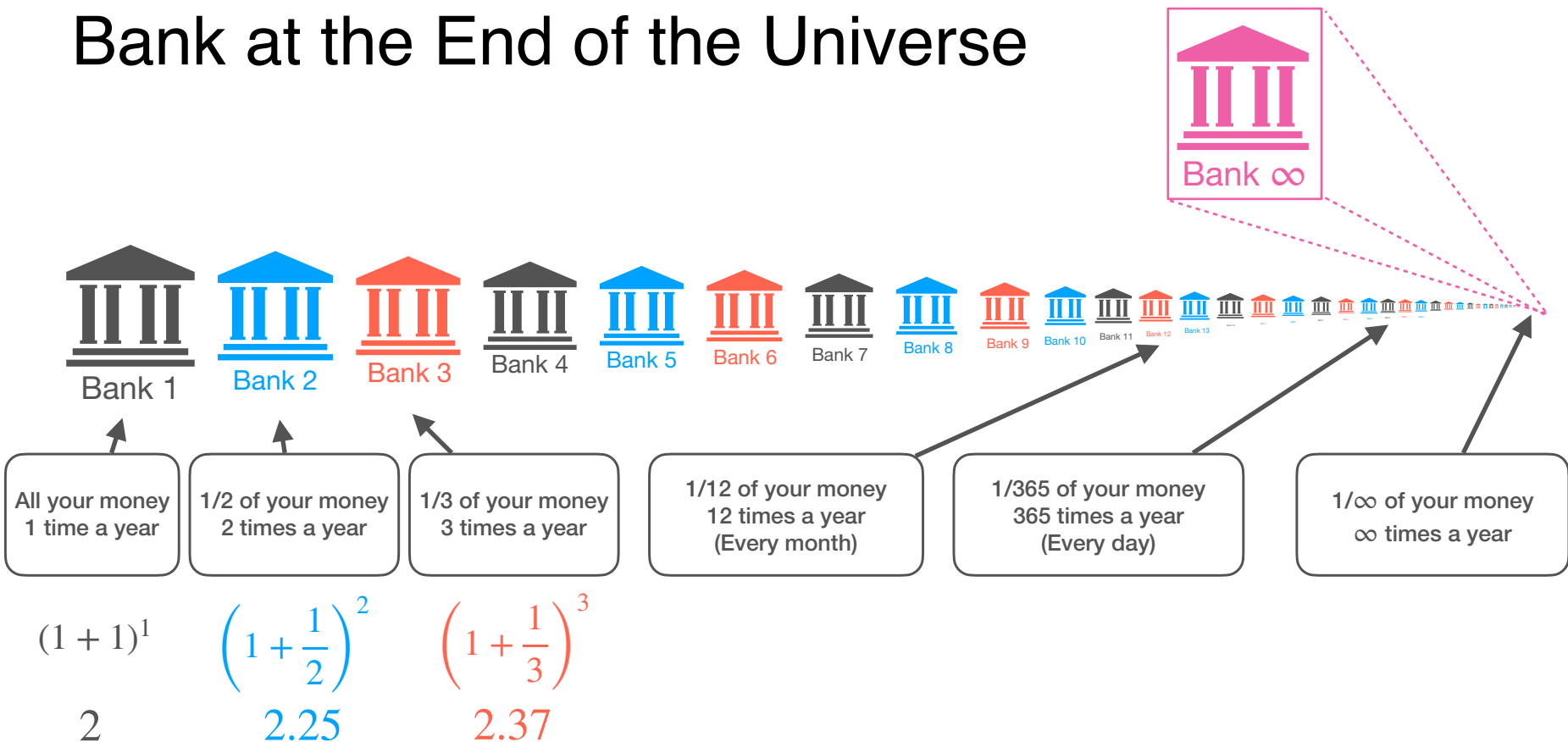
$$(1 + 1)^1$$

$$2$$

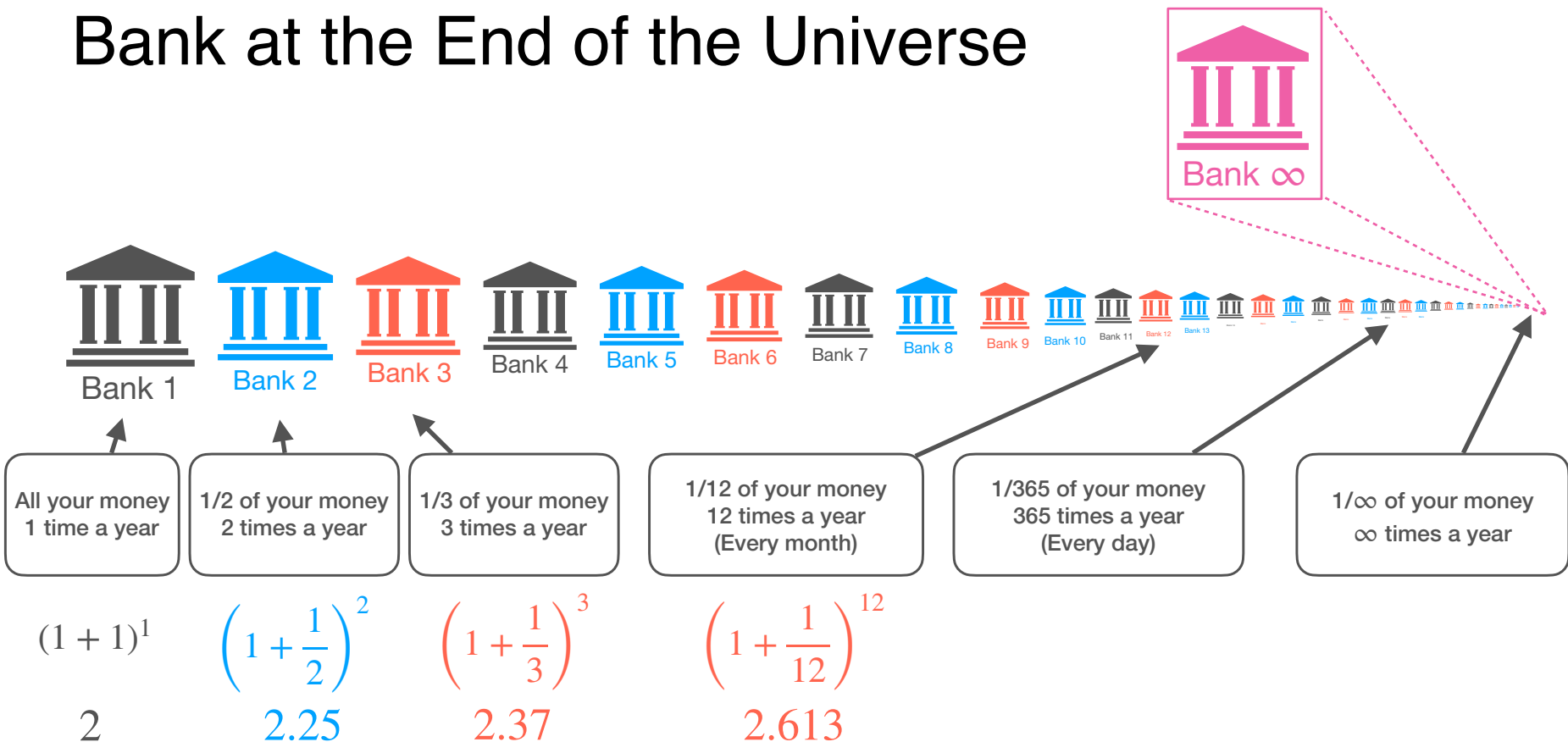
Bank at the End of the Universe



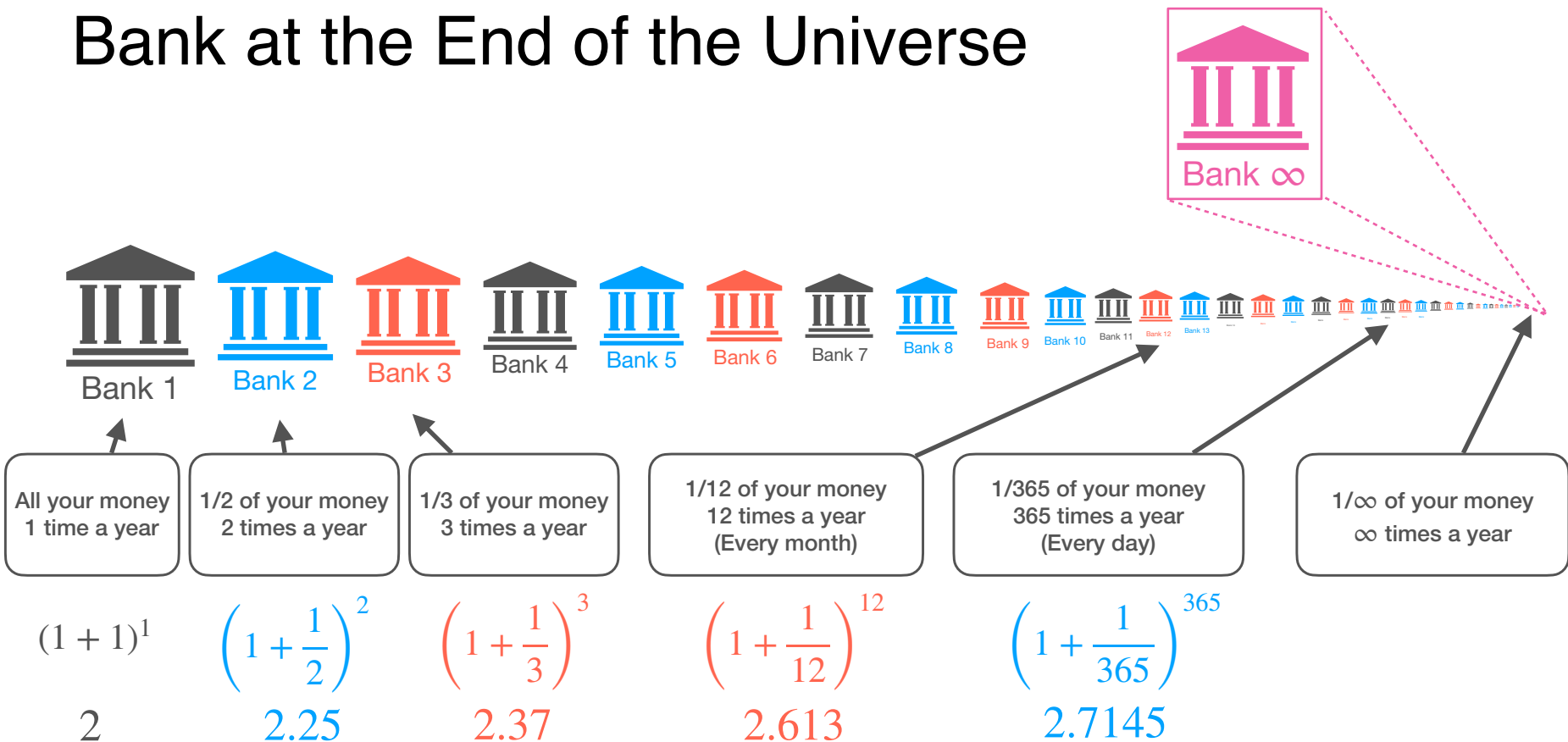
Bank at the End of the Universe



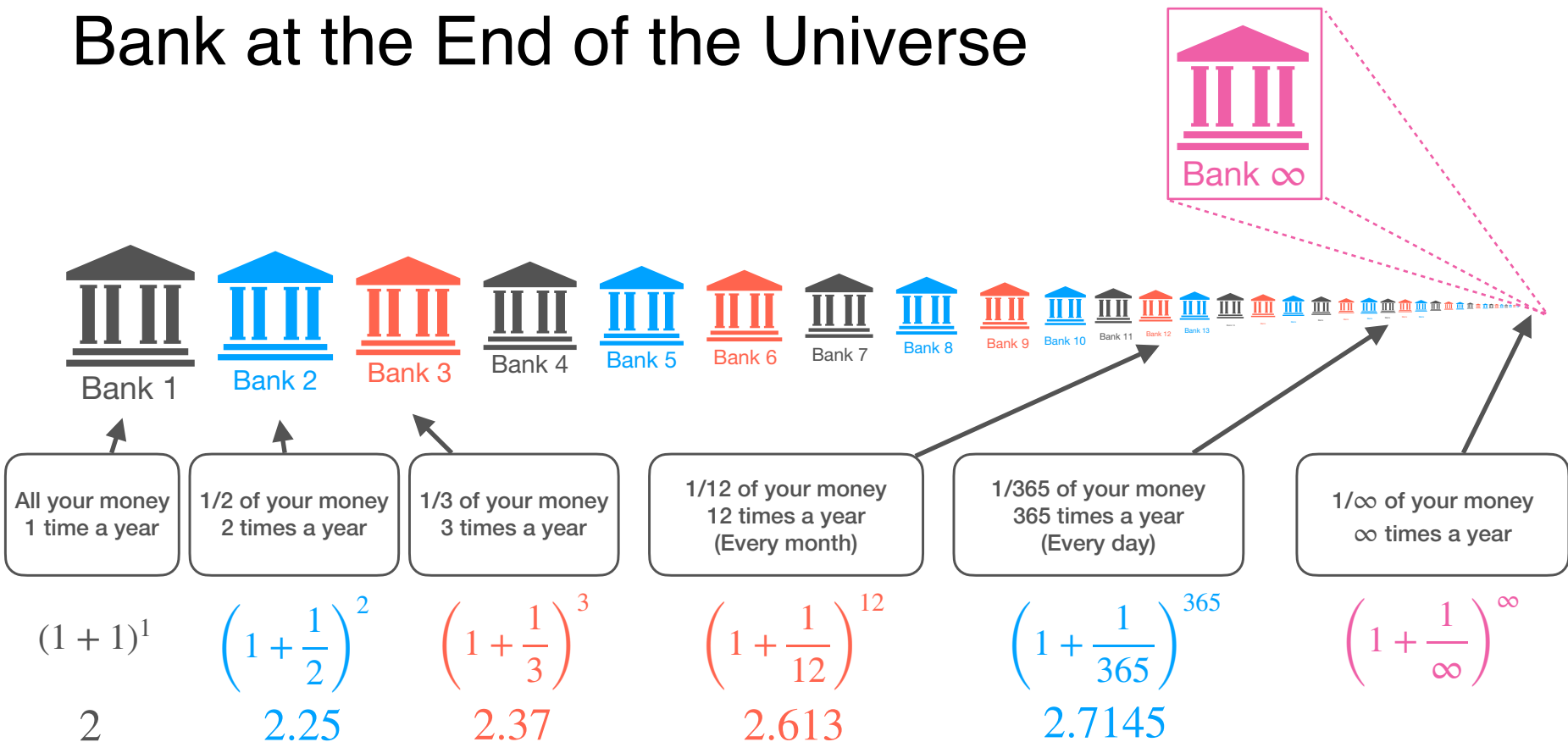
Bank at the End of the Universe



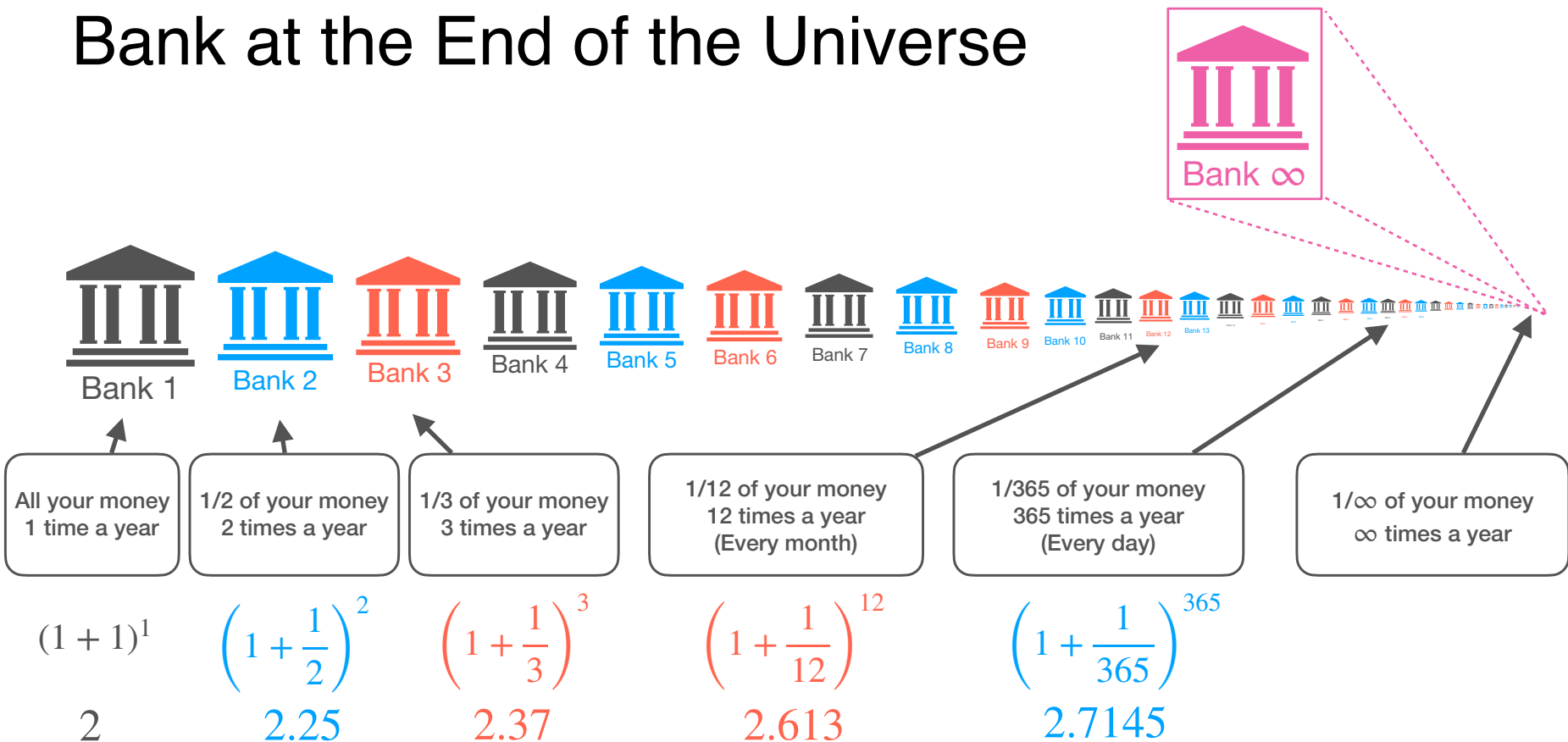
Bank at the End of the Universe



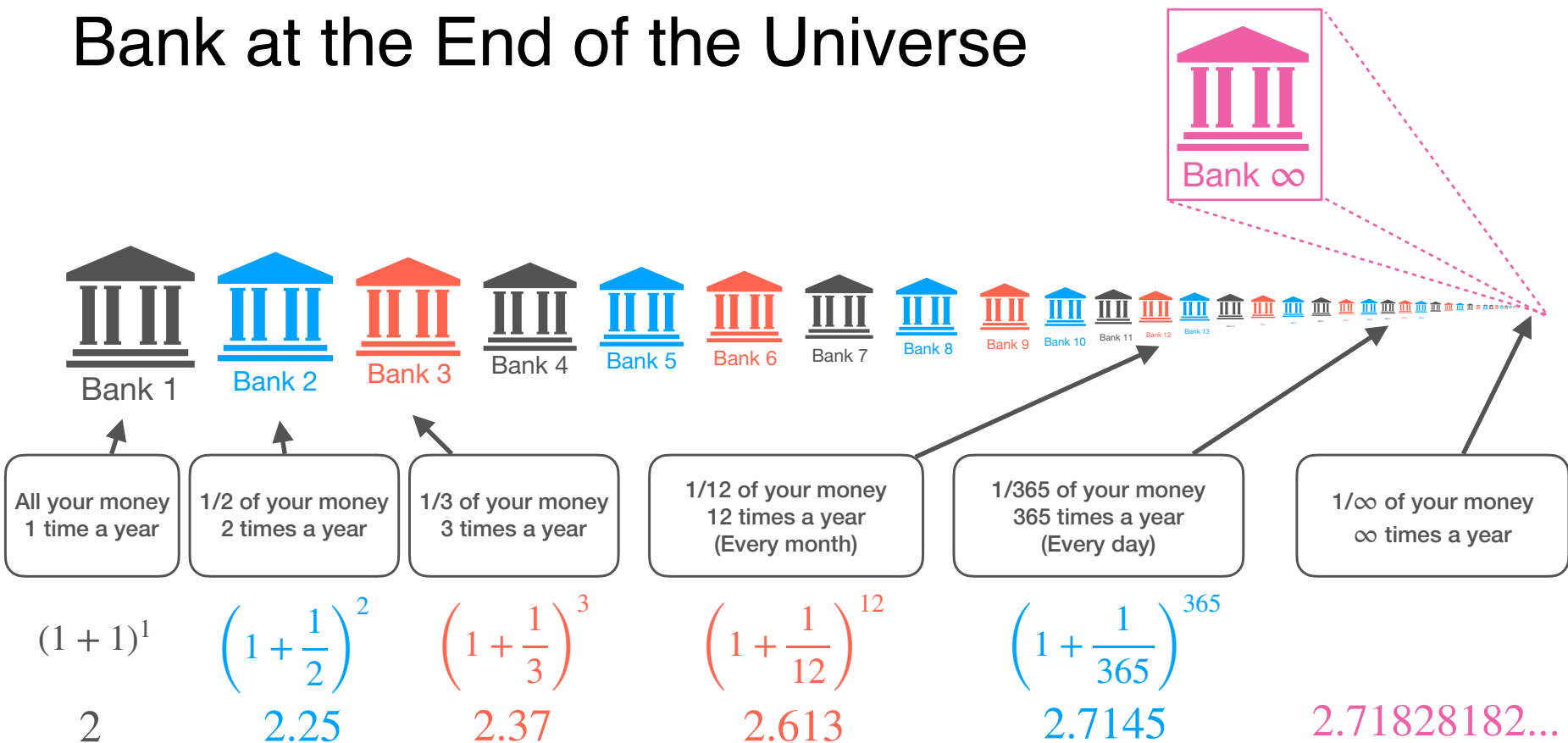
Bank at the End of the Universe



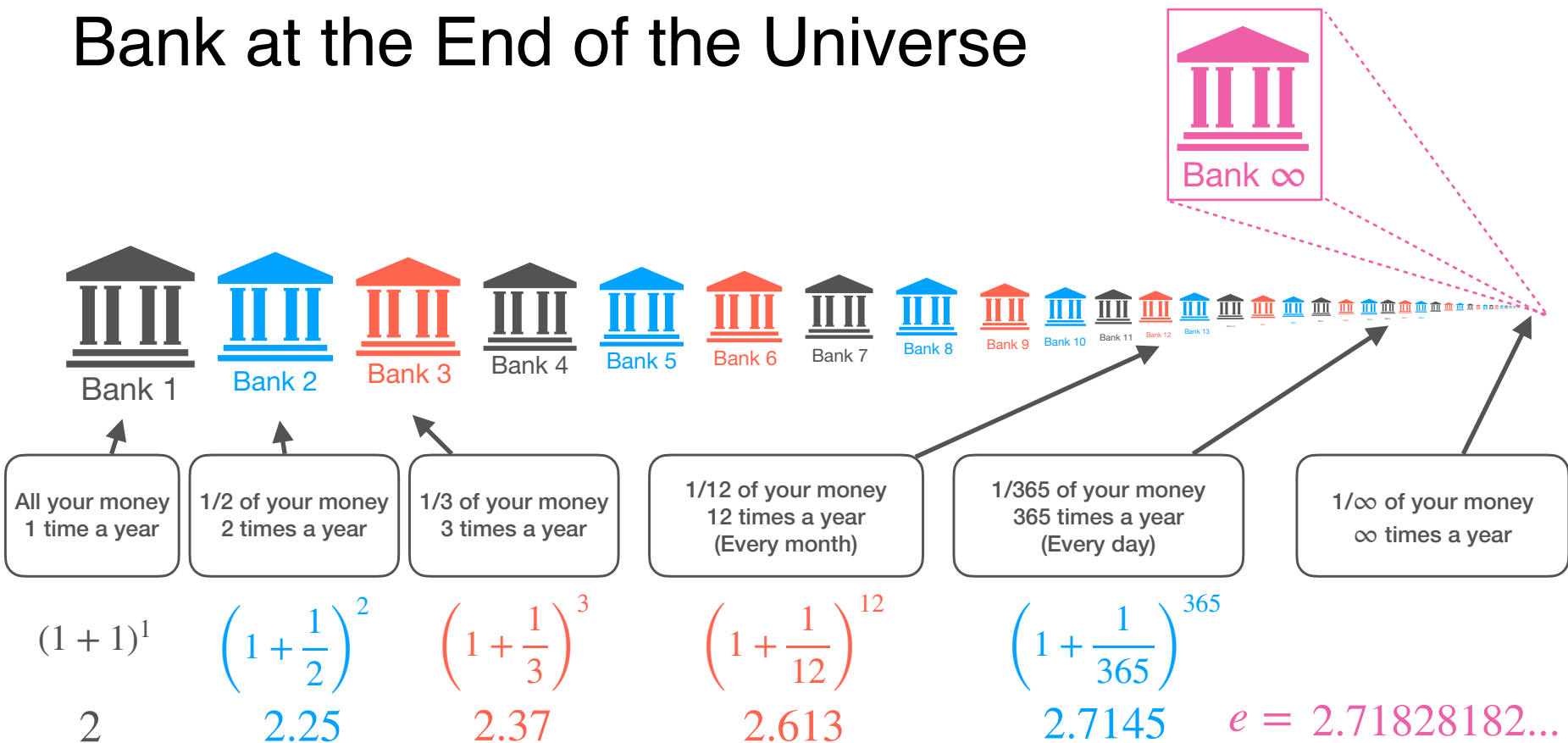
Bank at the End of the Universe



Bank at the End of the Universe



Bank at the End of the Universe



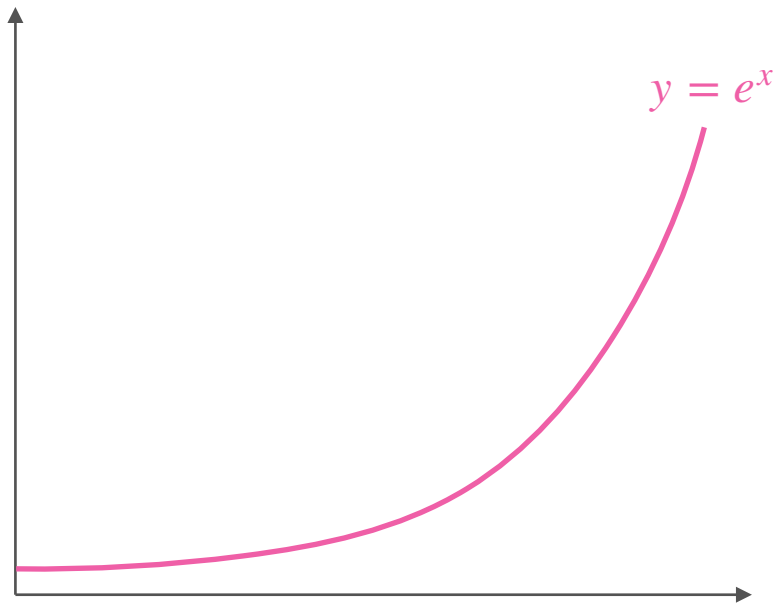


DeepLearning.AI

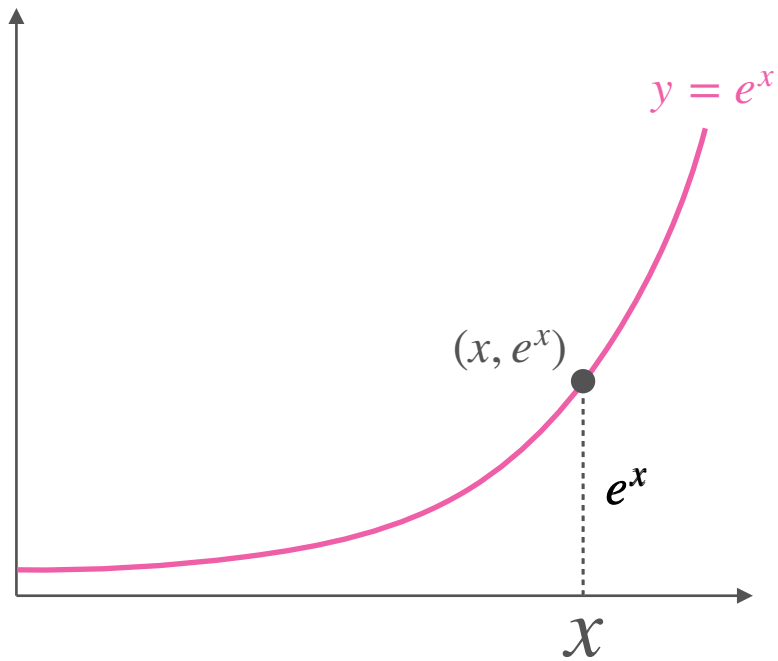
Derivatives and Optimization

The derivative of e^x

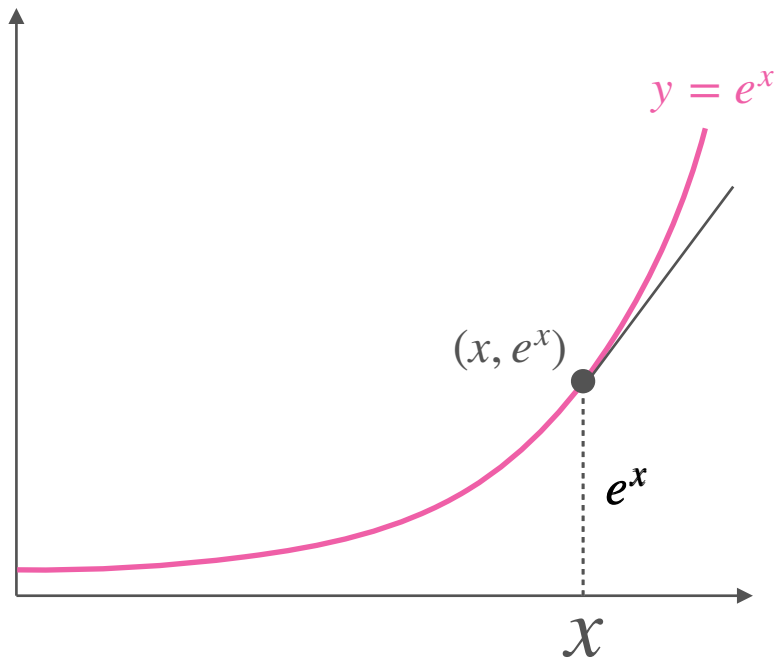
The Derivative of e^x Is Also e^x



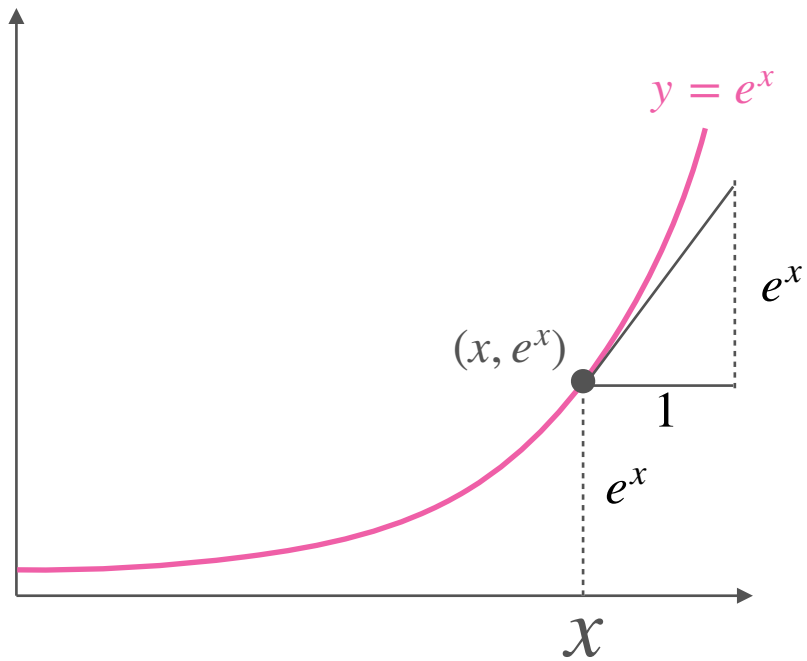
The Derivative of e^x Is Also e^x



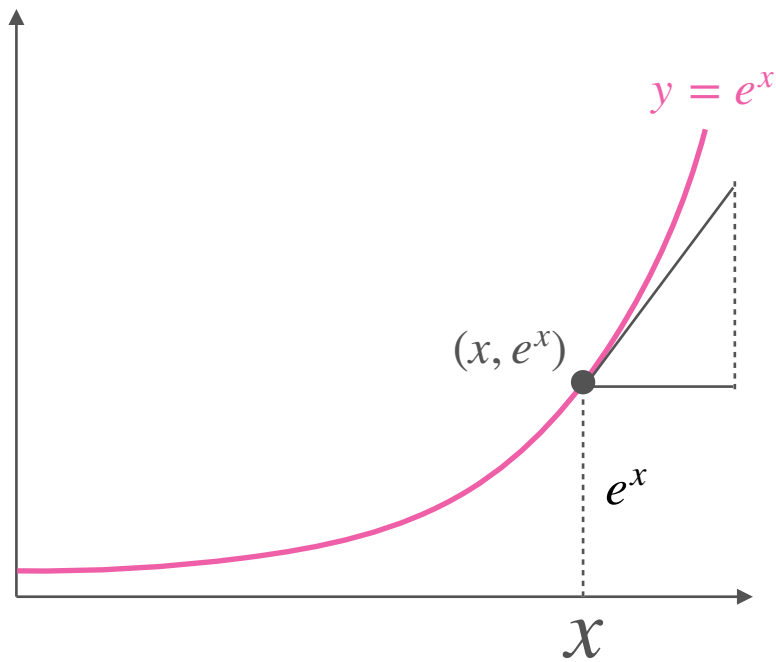
The Derivative of e^x Is Also e^x



The Derivative of e^x Is Also e^x

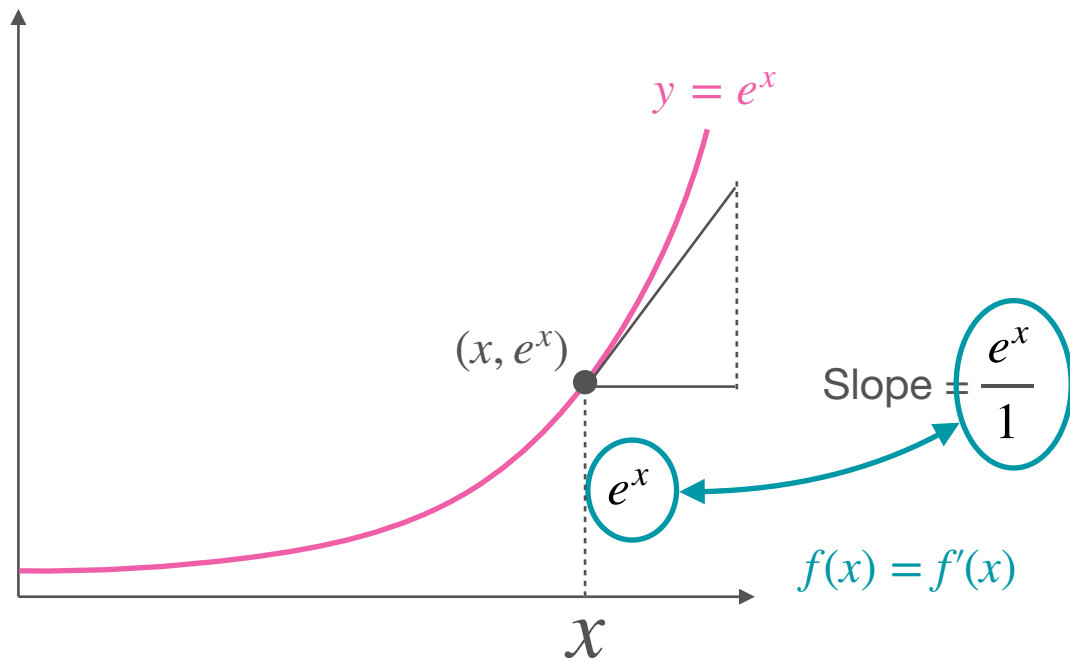


The Derivative of e^x Is Also e^x



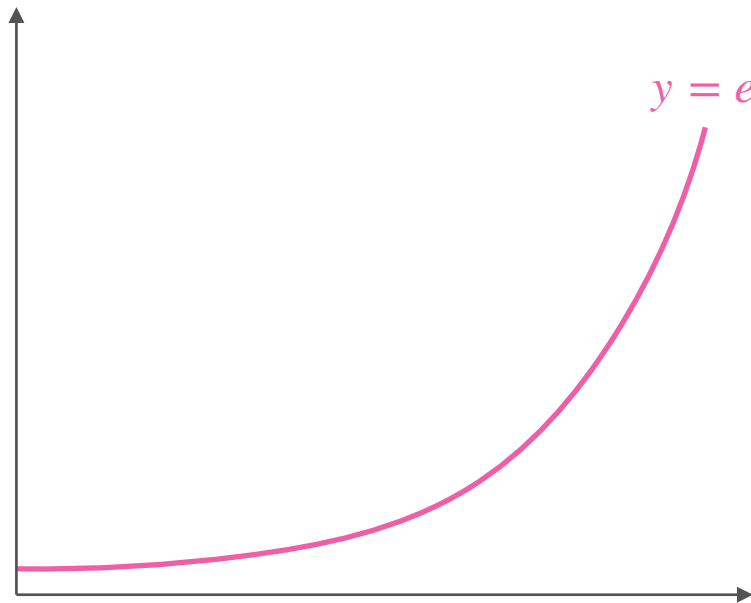
$$\text{Slope} = \frac{e^x}{1}$$

The Derivative of e^x Is Also e^x



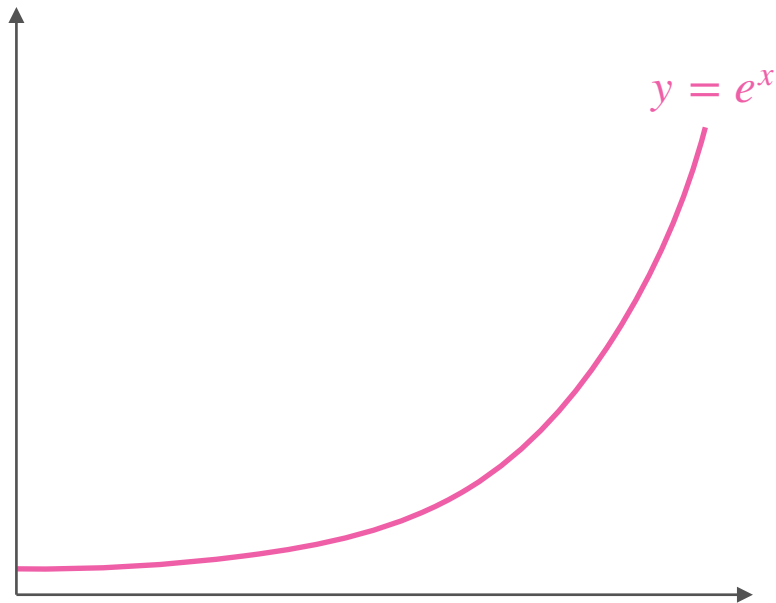
Derivative of e^x

Derivative of e^x



Exponential: $y = f(x) = e^x$

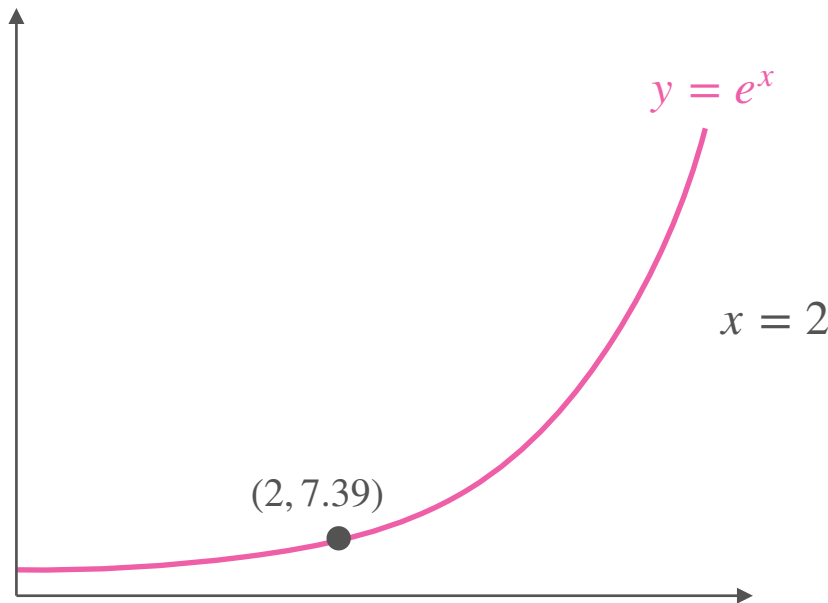
Derivative of e^x



Exponential: $y = f(x) = e^x$

Slope: $\frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$

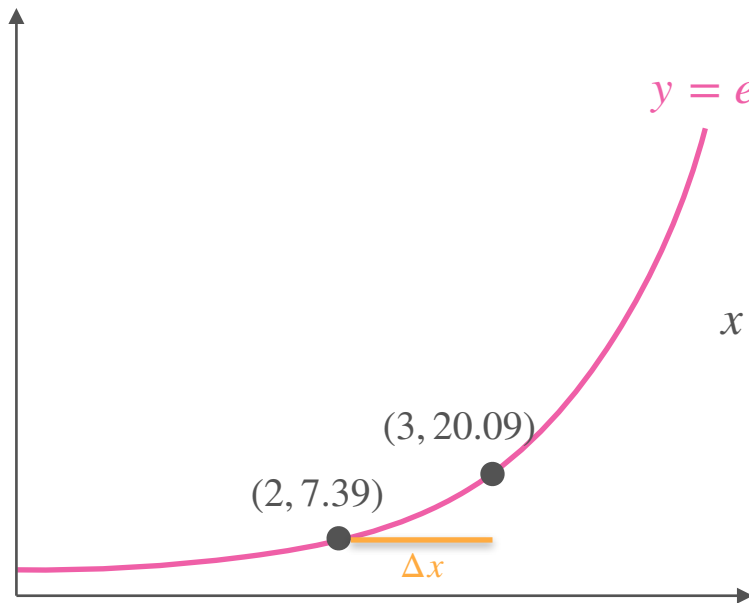
Derivative of e^x



Exponential: $y = f(x) = e^x$

Slope: $\frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$

Derivative of e^x



$$y = e^x$$

Exponential: $y = f(x) = e^x$

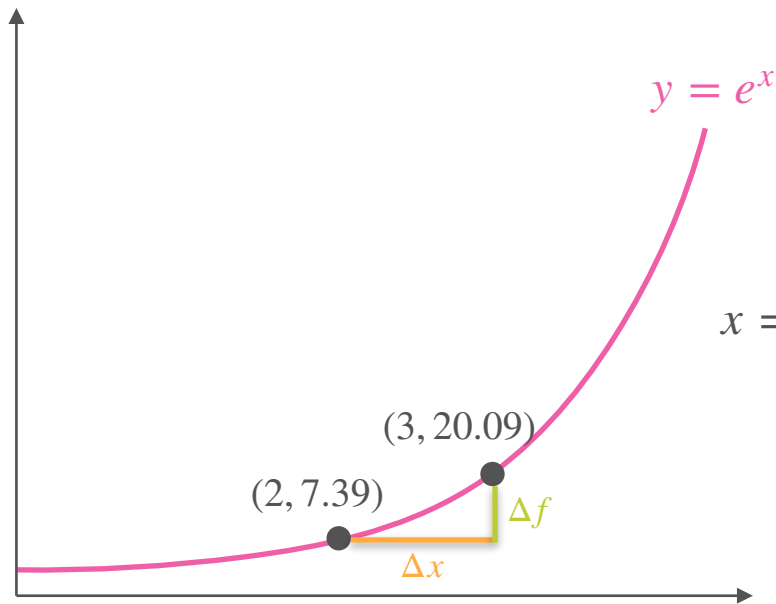
$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

$$x = 2$$

$$\Delta x$$

$$1.0$$

Derivative of e^x



Exponential: $y = f(x) = e^x$

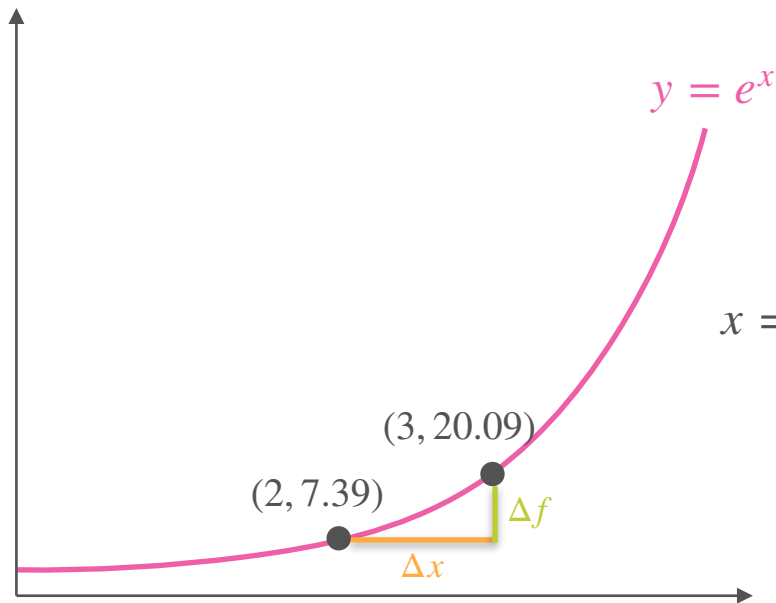
Slope: $\frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$

$x = 2$

Δx	1.0
Δf	12.70

$$e^{2+1} - e^2 = 20.09 - 7.39$$

Derivative of e^x



Exponential: $y = f(x) = e^x$

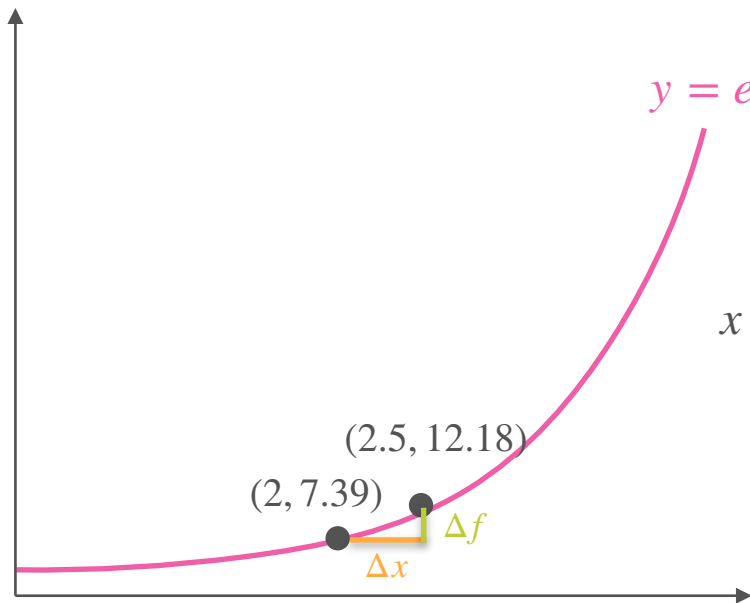
Slope: $\frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$

$x = 2$

Δx	1.0
Δf	12.70
Slope	12.70

$$\frac{e^{2+1} - e^2 = 20.09 - 7.39}{1}$$

Derivative of e^x



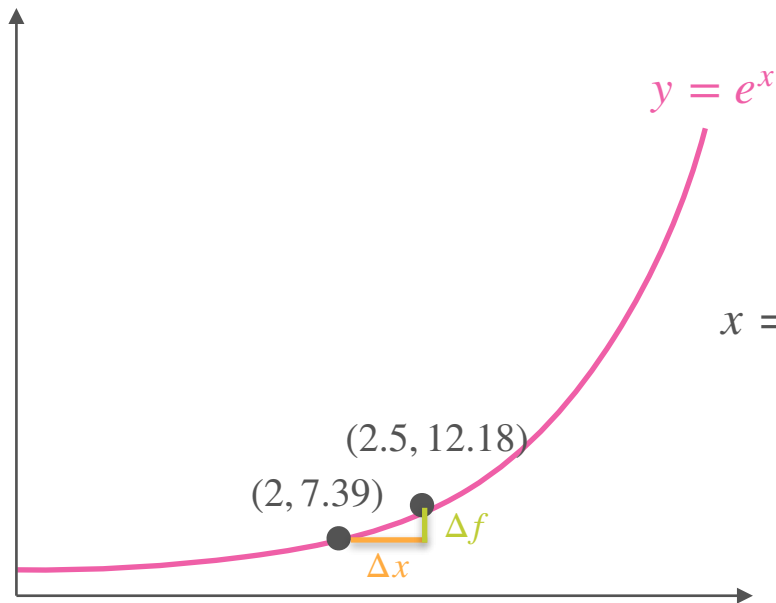
Exponential: $y = f(x) = e^x$

Slope: $\frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$

$x = 2$

Δx	1.0	1/2
Δf	12.70	
Slope	12.70	

Derivative of e^x



Exponential: $y = f(x) = e^x$

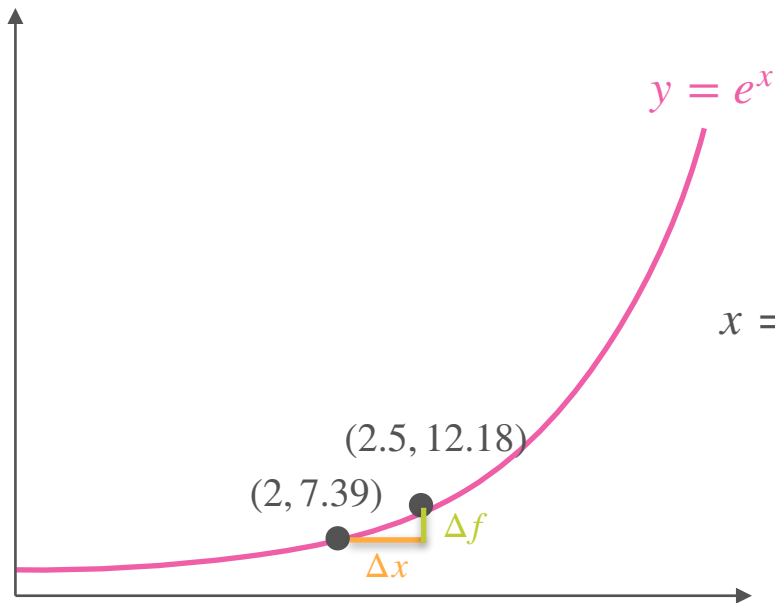
Slope: $\frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$

$x = 2$

Δx	1.0	1/2
Δf	12.70	4.79
Slope	12.70	

$$e^{2+0.5} - e^2 = 12.18 - 7.39$$

Derivative of e^x



Exponential: $y = f(x) = e^x$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

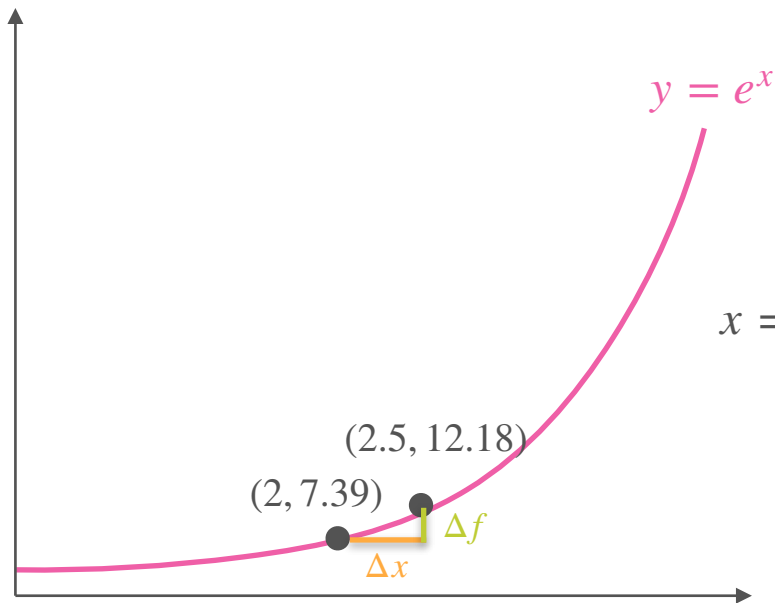
$x = 2$

Δx	1.0	1/2
Δf	12.70	4.79
Slope	12.70	9.59

$$e^{2+0.5} - e^2 = 12.18 - 7.39$$

$$\frac{4.79}{0.5}$$

Derivative of e^x



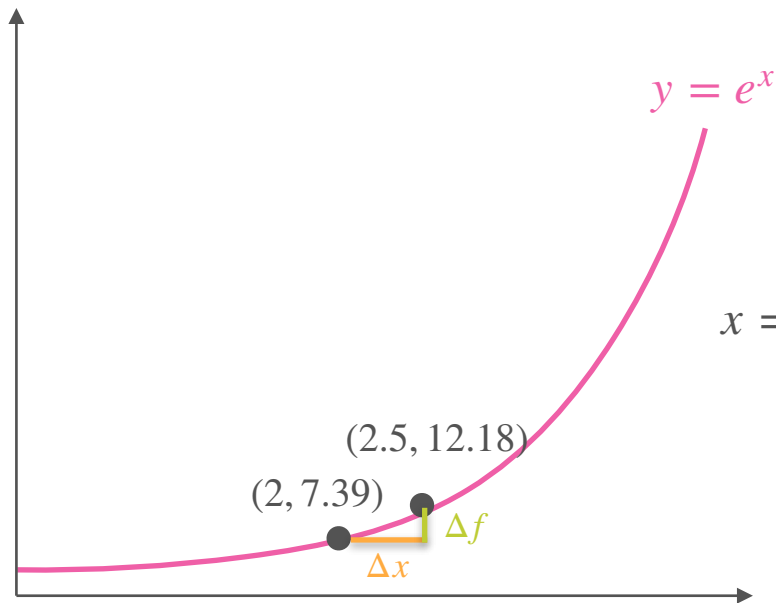
Exponential: $y = f(x) = e^x$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

$x = 2$

Δx	1.0	1/2	1/4
Δf	12.70	4.79	
Slope	12.70	9.59	

Derivative of e^x



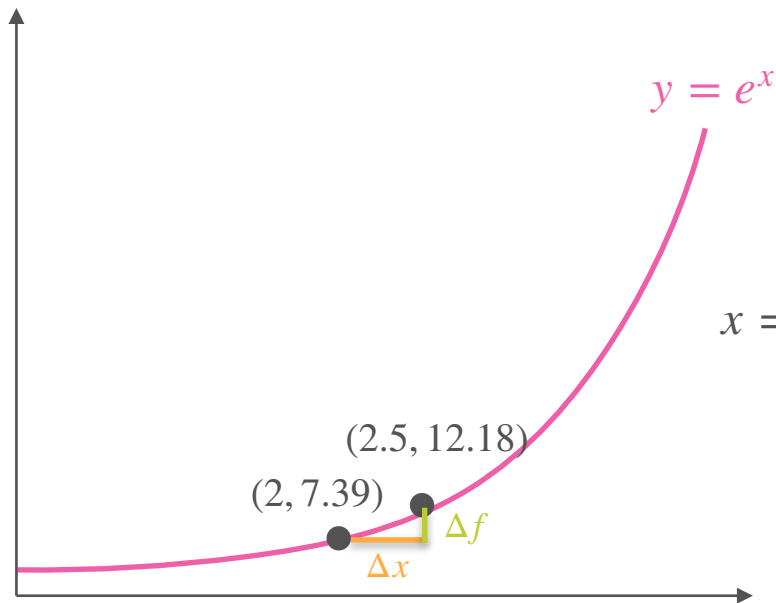
Exponential: $y = f(x) = e^x$

Slope: $\frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$

$x = 2$

Δx	1.0	1/2	1/4
Δf	12.70	4.79	2.10
Slope	12.70	9.59	

Derivative of e^x

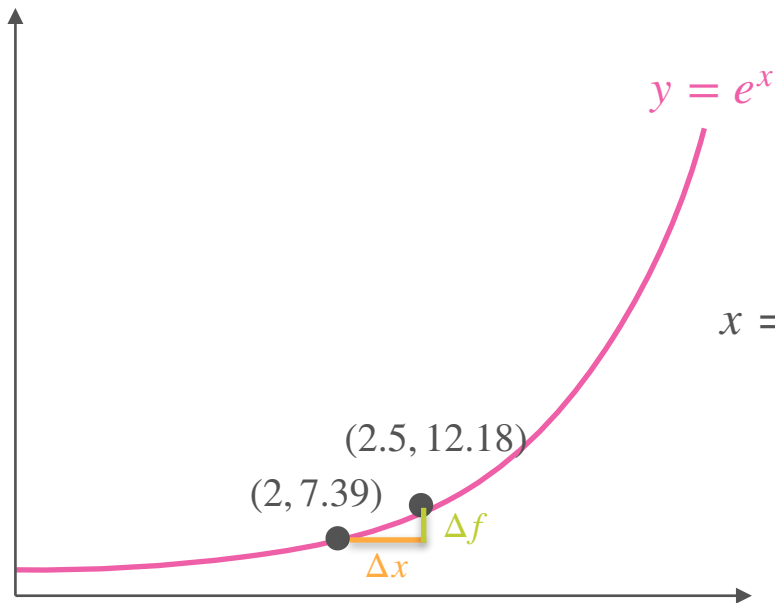


Exponential: $y = f(x) = e^x$

Slope: $\frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$

Δx	1.0	1/2	1/4
Δf	12.70	4.79	2.10
Slope	12.70	9.59	8.39

Derivative of e^x



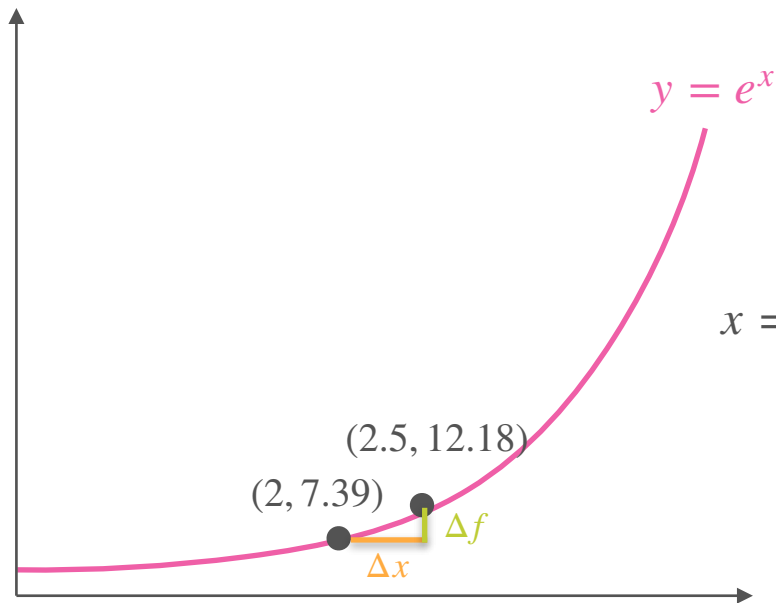
Exponential: $y = f(x) = e^x$

Slope: $\frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$

$x = 2$

Δx	1.0	1/2	1/4	1/8
Δf	12.70	4.79	2.10	
Slope	12.70	9.59	8.39	

Derivative of e^x

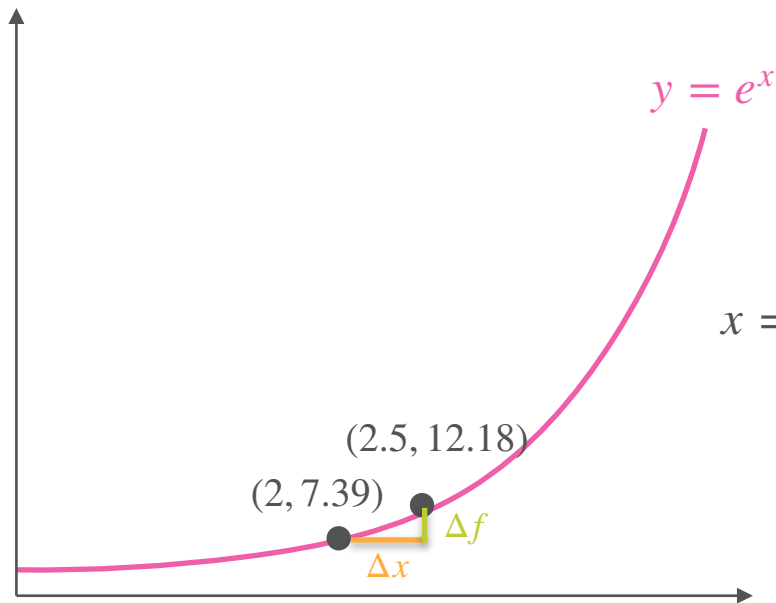


Exponential: $y = f(x) = e^x$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8
Δf	12.70	4.79	2.10	0.98
Slope	12.70	9.59	8.39	

Derivative of e^x

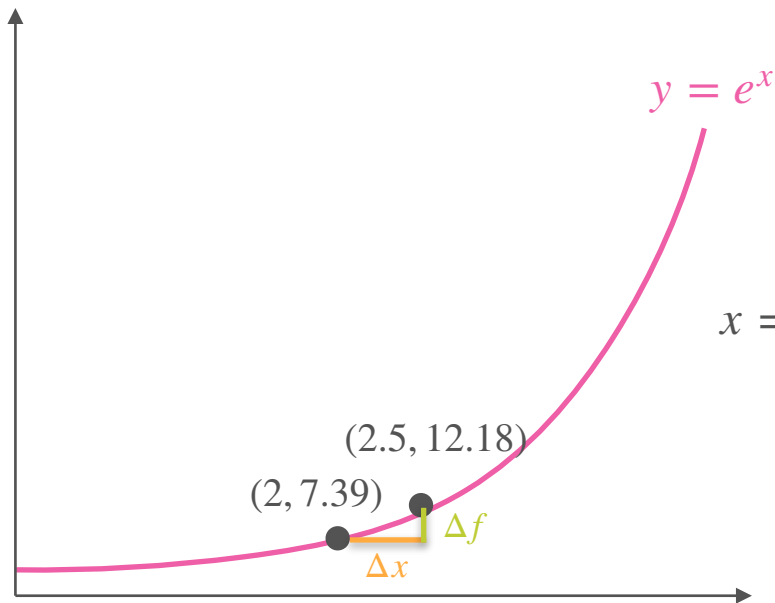


Exponential: $y = f(x) = e^x$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8
Δf	12.70	4.79	2.10	0.98
Slope	12.70	9.59	8.39	7.87

Derivative of e^x



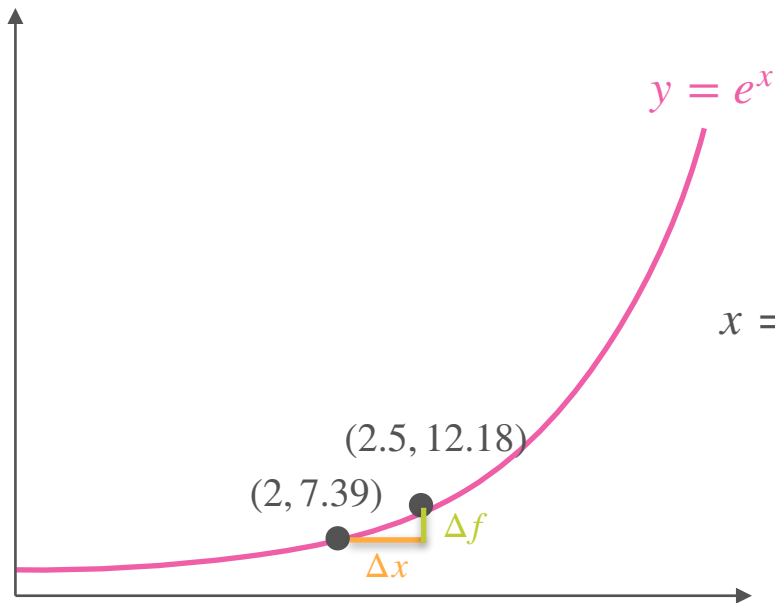
Exponential: $y = f(x) = e^x$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

$x = 2$

Δx	1.0	1/2	1/4	1/8	1/16
Δf	12.70	4.79	2.10	0.98	
Slope	12.70	9.59	8.39	7.87	

Derivative of e^x



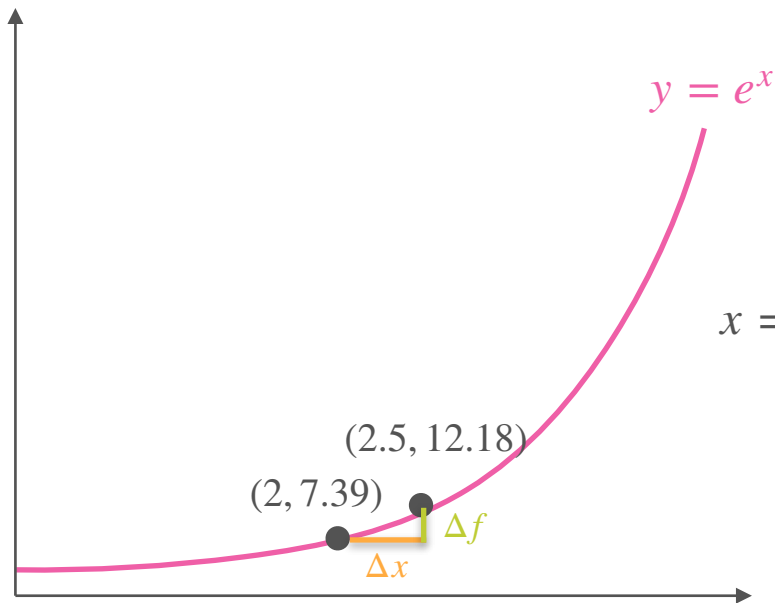
Exponential: $y = f(x) = e^x$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

$x = 2$

Δx	1.0	1/2	1/4	1/8	1/16
Δf	12.70	4.79	2.10	0.98	0.48
Slope	12.70	9.59	8.39	7.87	

Derivative of e^x

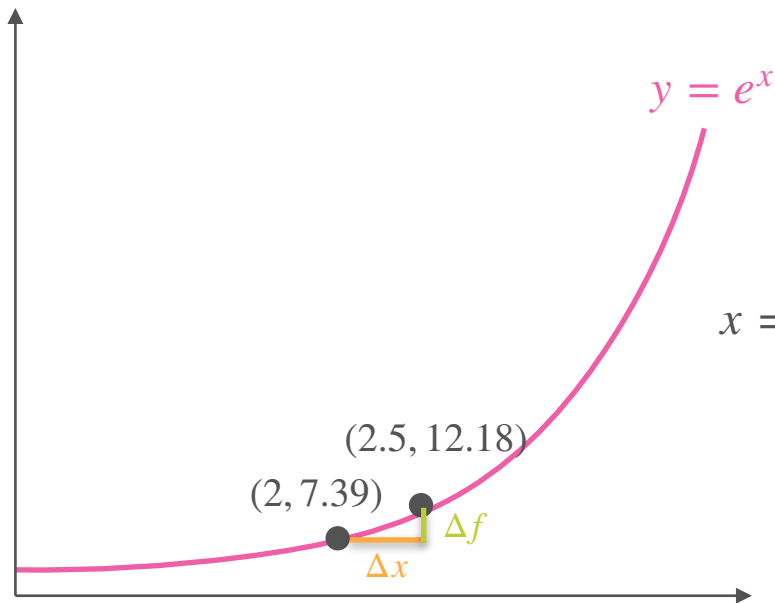


Exponential: $y = f(x) = e^x$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8	1/16
Δf	12.70	4.79	2.10	0.98	0.48
Slope	12.70	9.59	8.39	7.87	7.62

Derivative of e^x



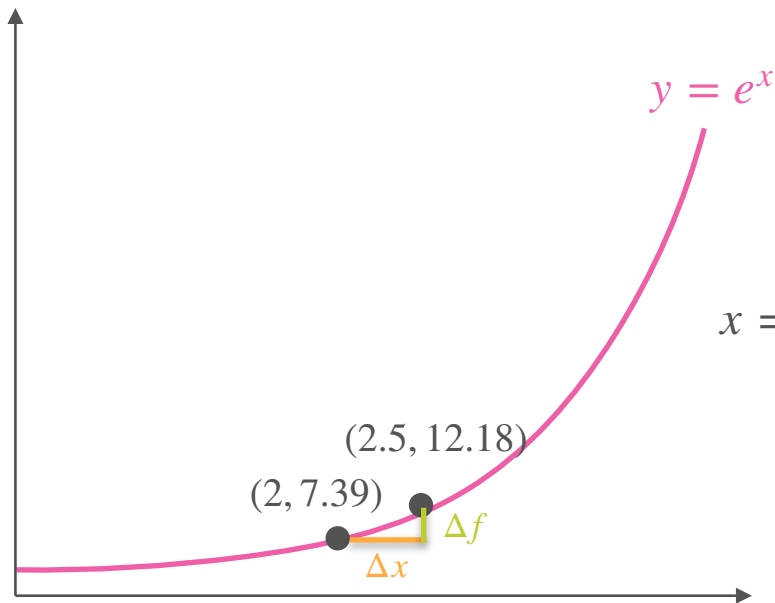
Exponential: $y = f(x) = e^x$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

$x = 2$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	12.70	4.79	2.10	0.98	0.48	
Slope	12.70	9.59	8.39	7.87	7.62	

Derivative of e^x

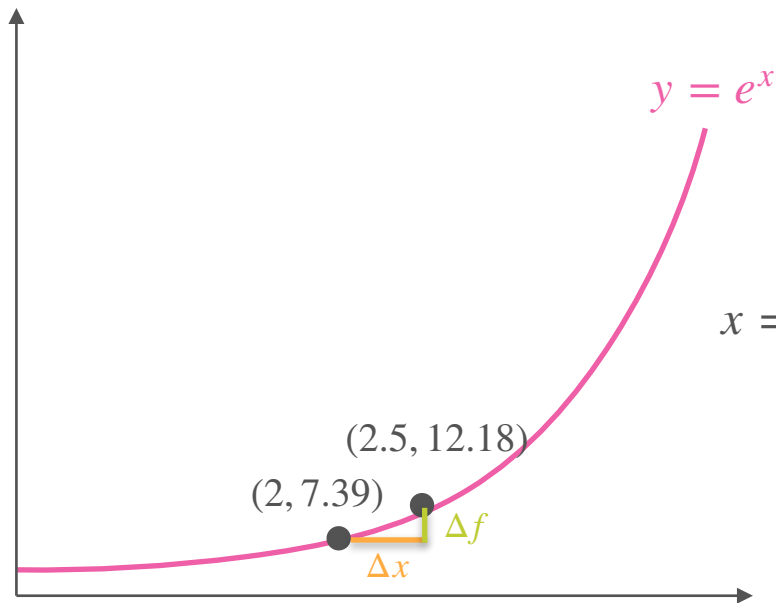


Exponential: $y = f(x) = e^x$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	12.70	4.79	2.10	0.98	0.48	0.007
Slope	12.70	9.59	8.39	7.87	7.62	

Derivative of e^x

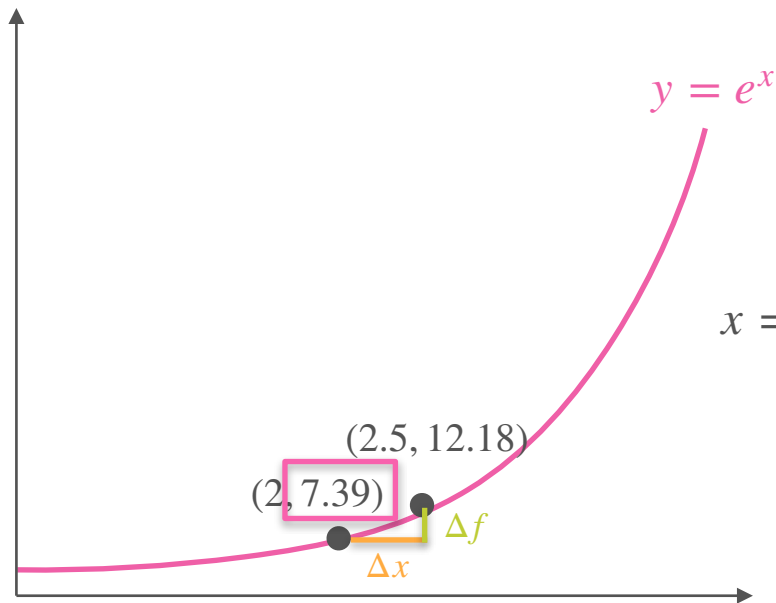


Exponential: $y = f(x) = e^x$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	12.70	4.79	2.10	0.98	0.48	0.007
Slope	12.70	9.59	8.39	7.87	7.62	7.39

Derivative of e^x

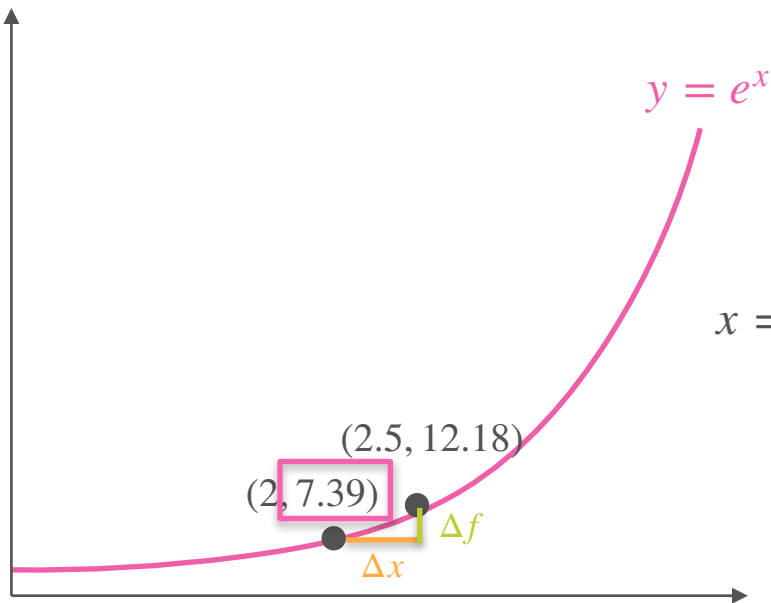


Exponential: $y = f(x) = e^x$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	12.70	4.79	2.10	0.98	0.48	0.007
Slope	12.70	9.59	8.39	7.87	7.62	7.39

Derivative of e^x



Exponential: $y = f(x) = e^x$

$$\text{Slope: } \frac{\Delta f}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

Δx	1.0	1/2	1/4	1/8	1/16	1/1000
Δf	12.70	4.79	2.10	0.98	0.48	0.007
Slope	12.70	9.59	8.39	7.87	7.62	7.39

$$e^2$$

(Reading Item) Derivative of a^x

$$a^x = e^{x \log(a)}$$

$$\frac{d}{dx} a^x = \log(a) e^{x \log(a)} = \log(a) a^x$$



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Derivatives and Optimization

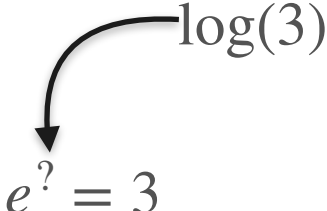
The derivative of $\log(x)$

Logarithm

Logarithm

$$e^? = 3$$

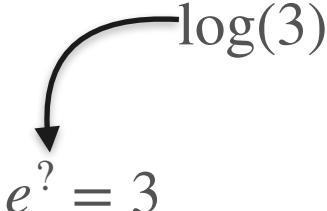
Logarithm



A diagram illustrating the relationship between a logarithm and an exponential equation. A curved arrow points from the expression $\log(3)$ to the question mark in the equation $e^? = 3$.

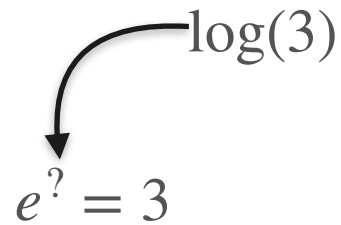
$$\log(3)$$
$$e^? = 3$$

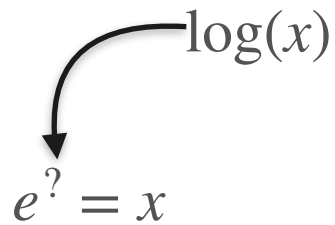
Logarithm


$$e^? = 3$$

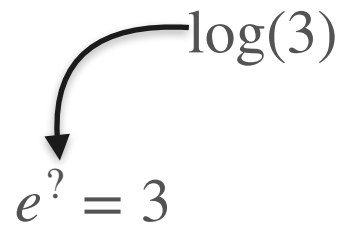
$$e^? = x$$

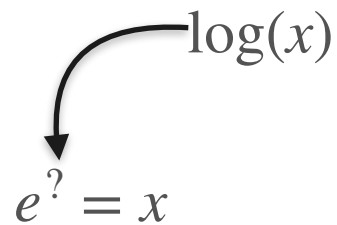
Logarithm


$$e^? = 3$$


$$e^? = x$$

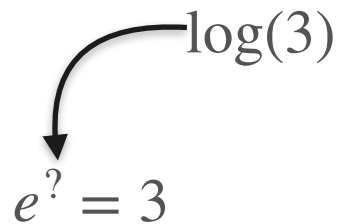
Logarithm


$$e^? = 3$$

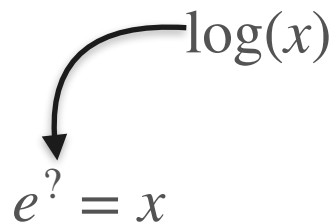

$$e^? = x$$

$$f(x) = e^x$$

Logarithm

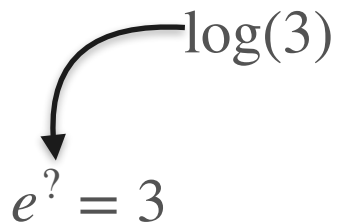

$$e^? = 3$$

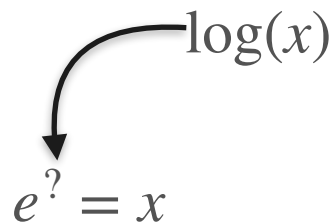
$$f(x) = e^x$$


$$e^? = x$$

$$f^{-1}(y) = \log(y)$$

Logarithm


$$e^? = 3$$

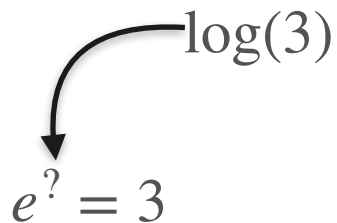

$$e^? = x$$

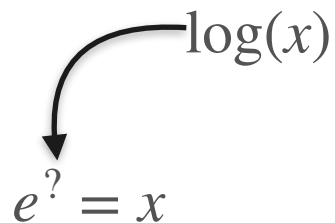
$$f(x) = e^x$$

$$f^{-1}(y) = \log(y)$$

$$e^{\log(x)} = x$$

Logarithm


$$e^? = 3$$


$$e^? = x$$

$$f(x) = e^x$$

$$f^{-1}(y) = \log(y)$$

$$e^{\log(x)} = x$$

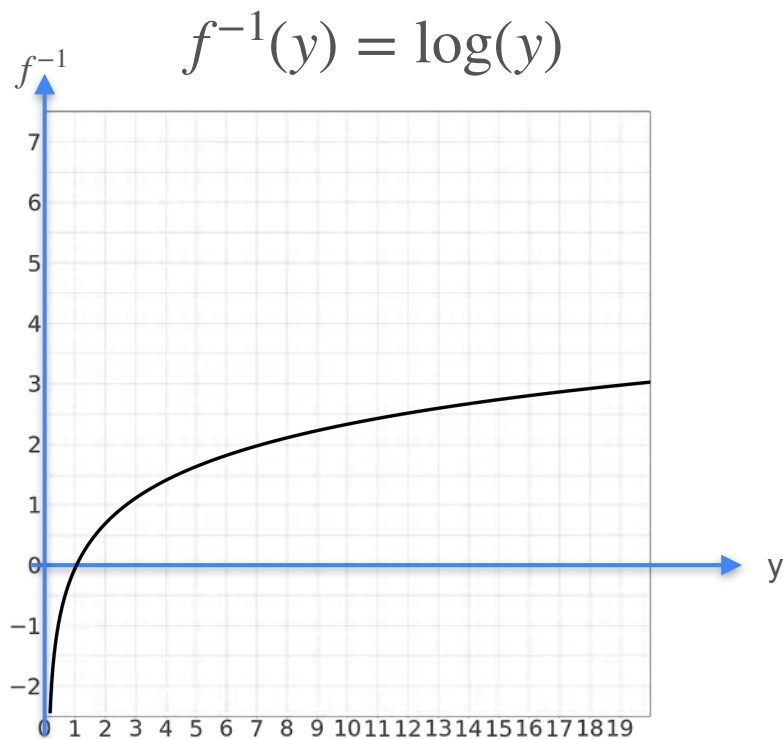
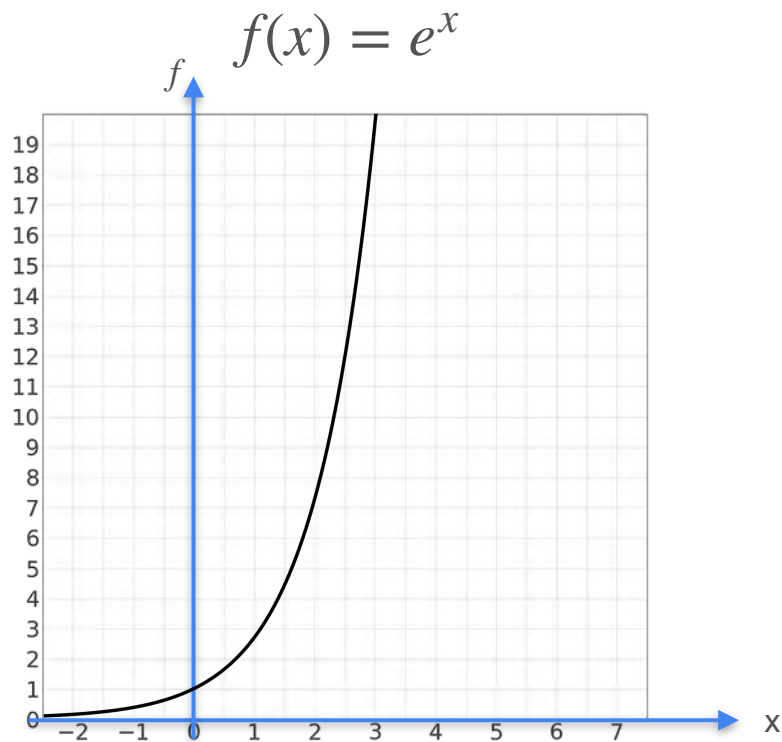
$$\log(e^y) = y$$

Logarithm

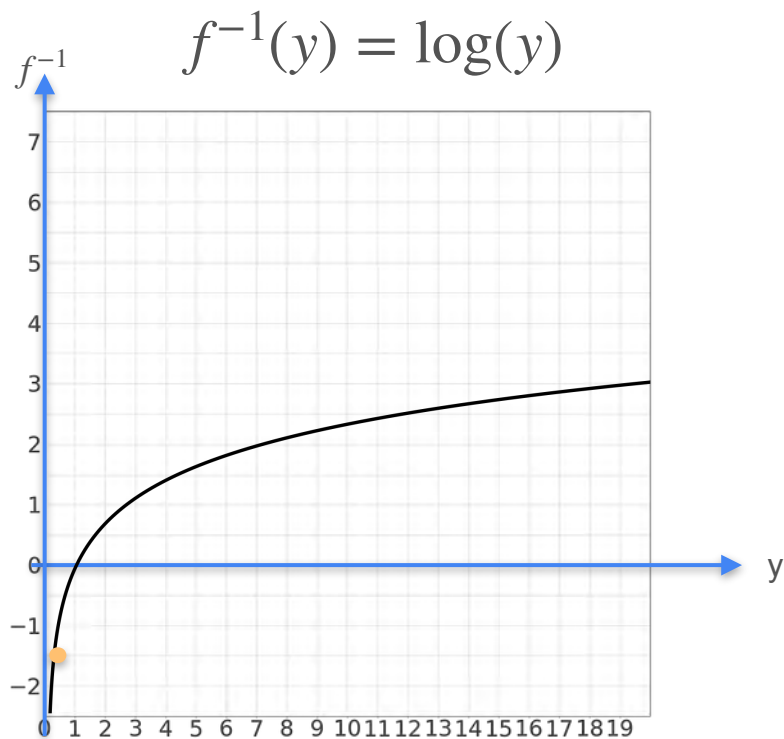
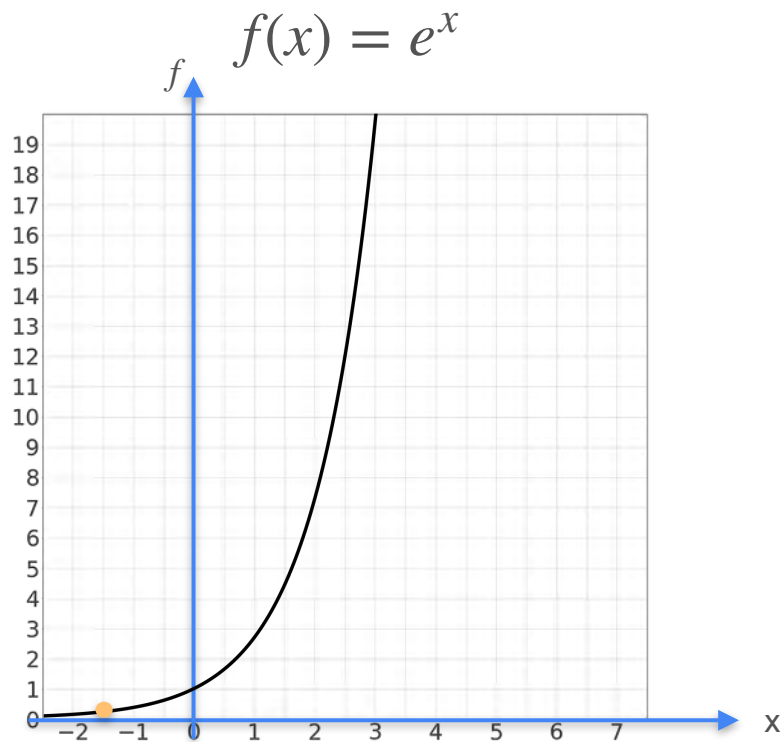
$$f(x) = e^x$$

$$f^{-1}(y) = \log(y)$$

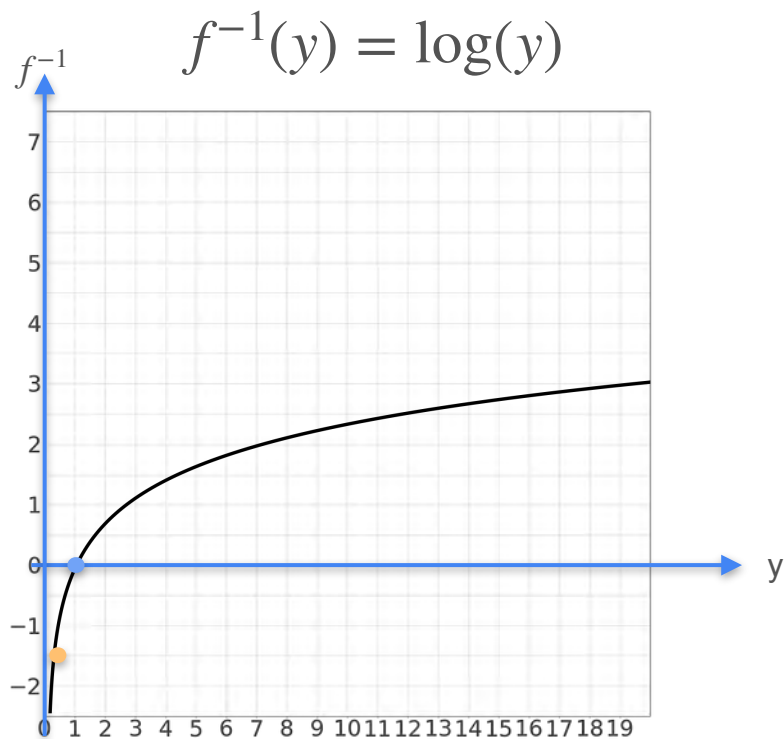
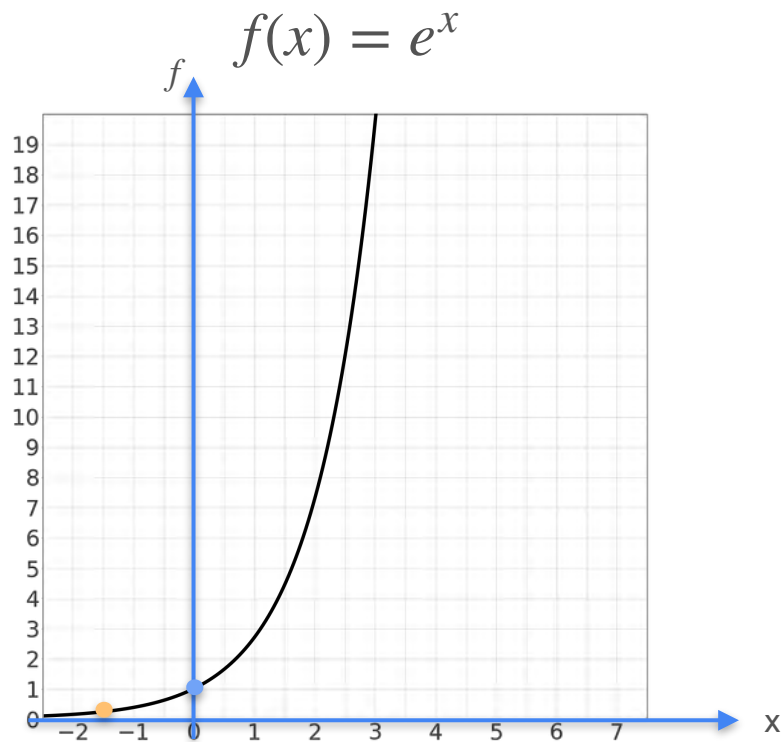
Logarithm



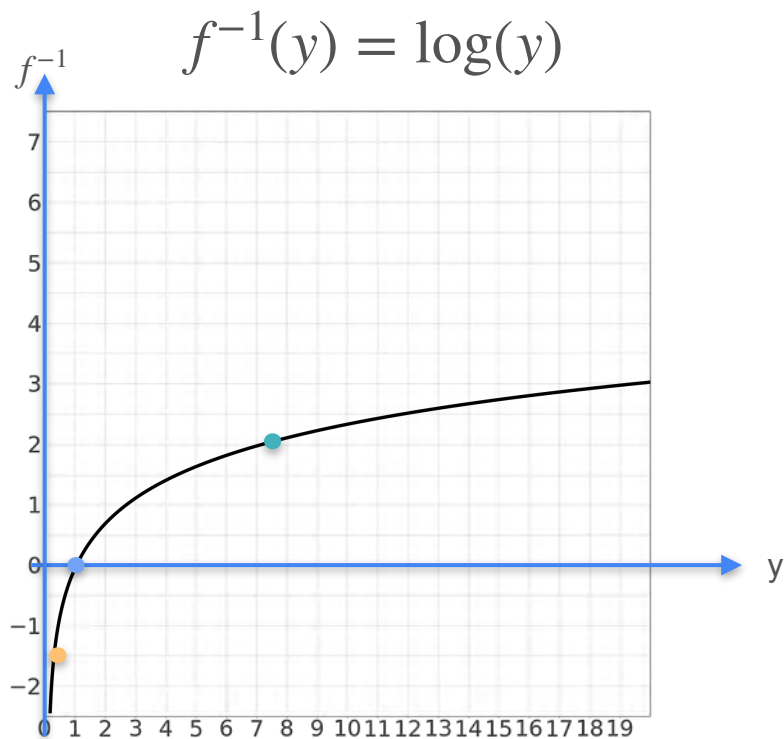
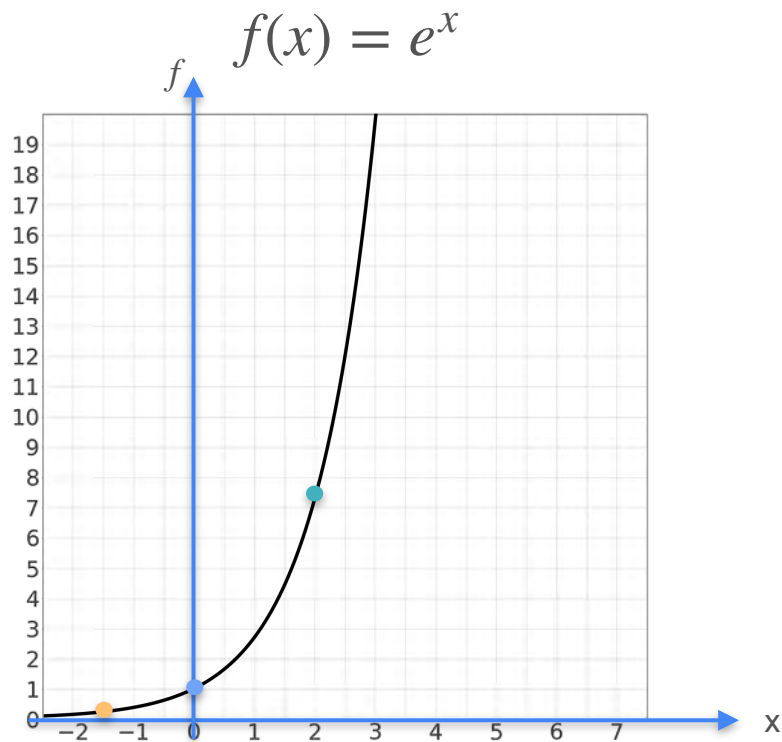
Logarithm



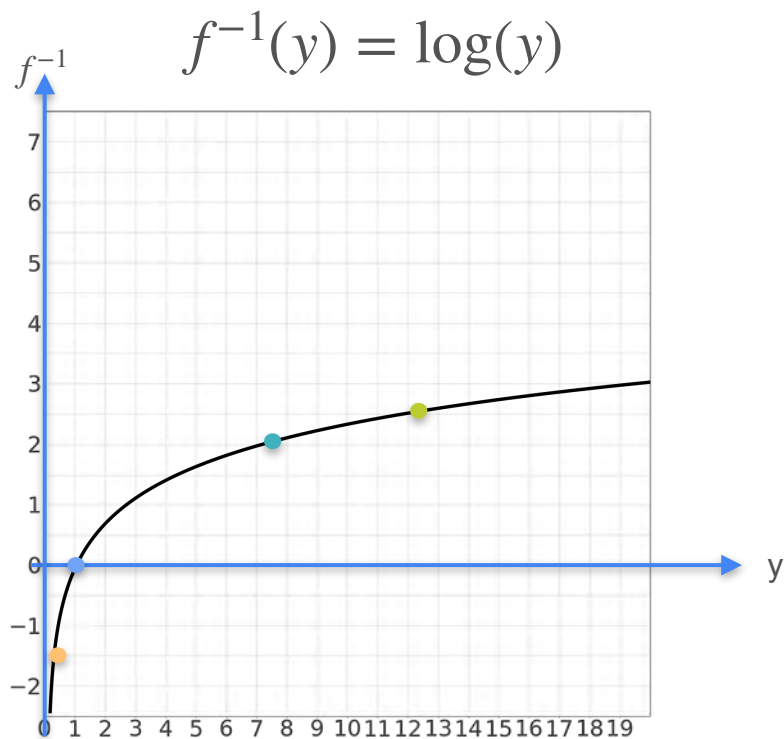
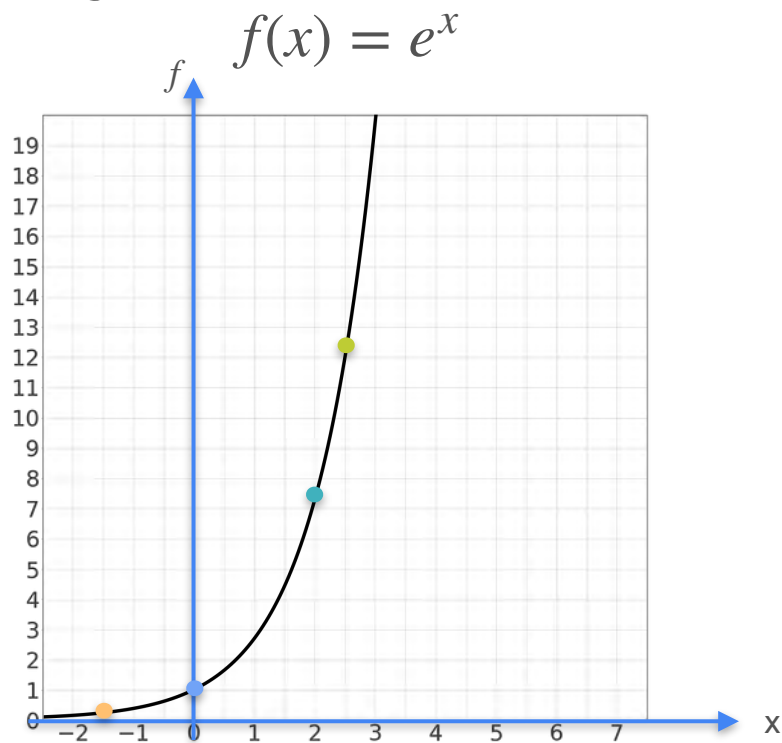
Logarithm



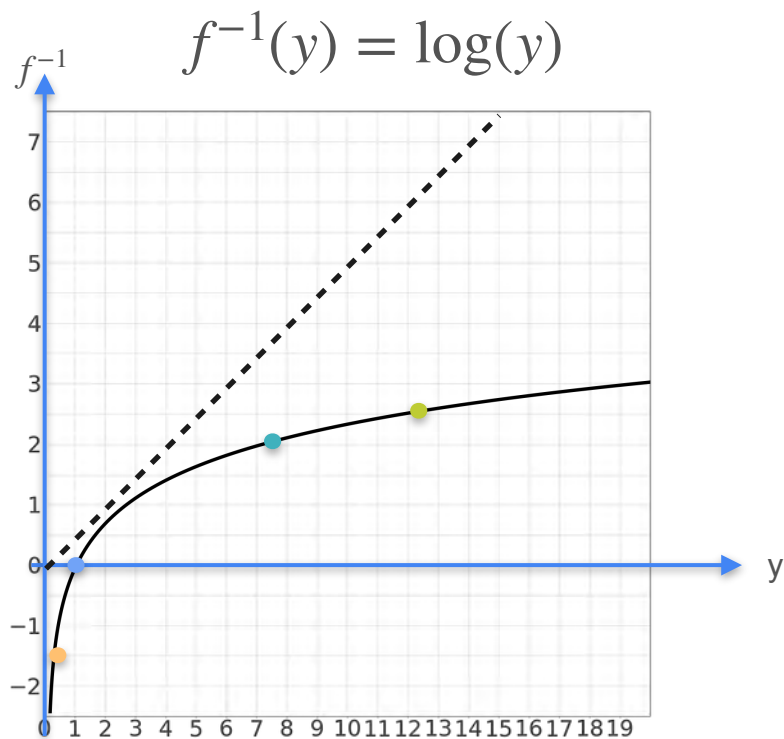
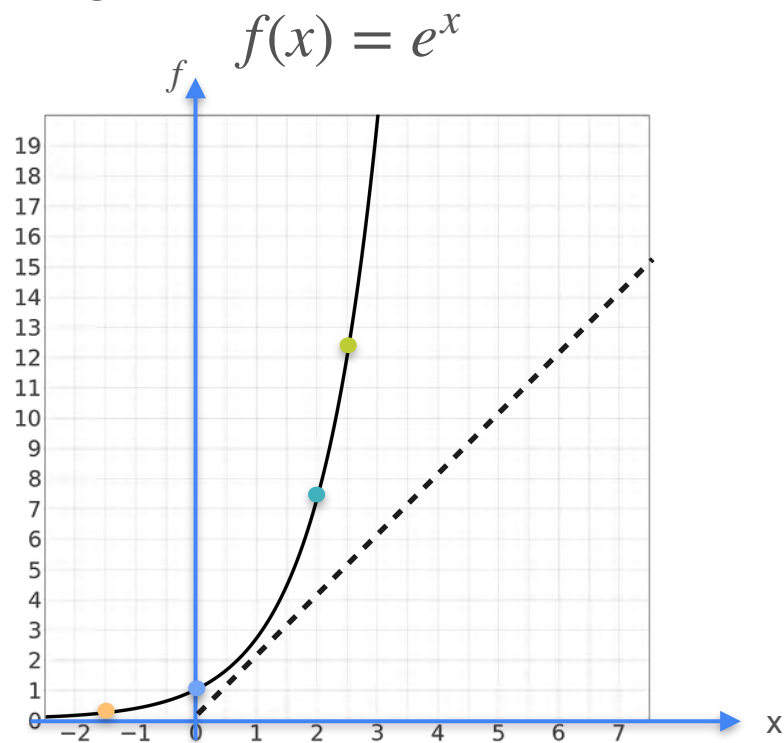
Logarithm



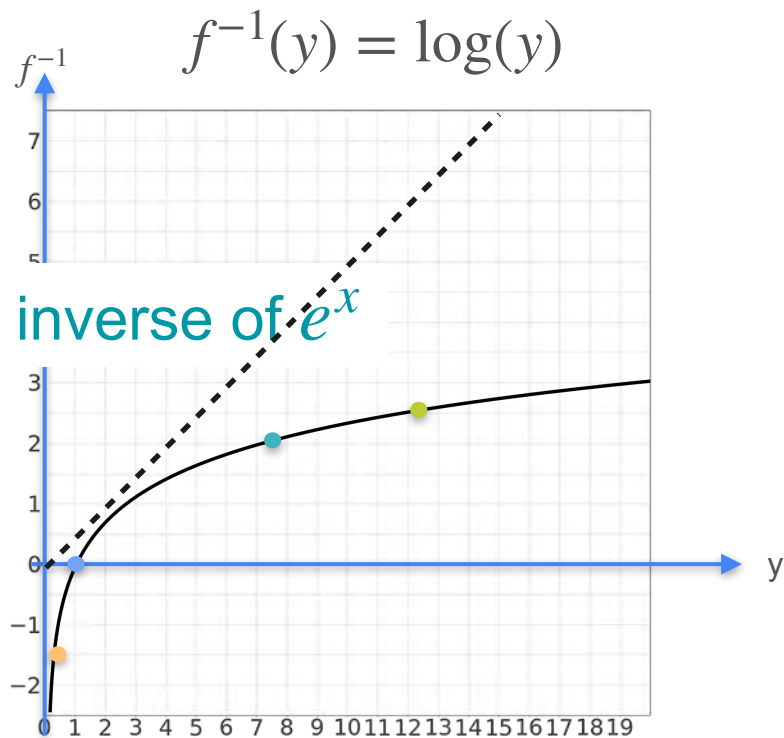
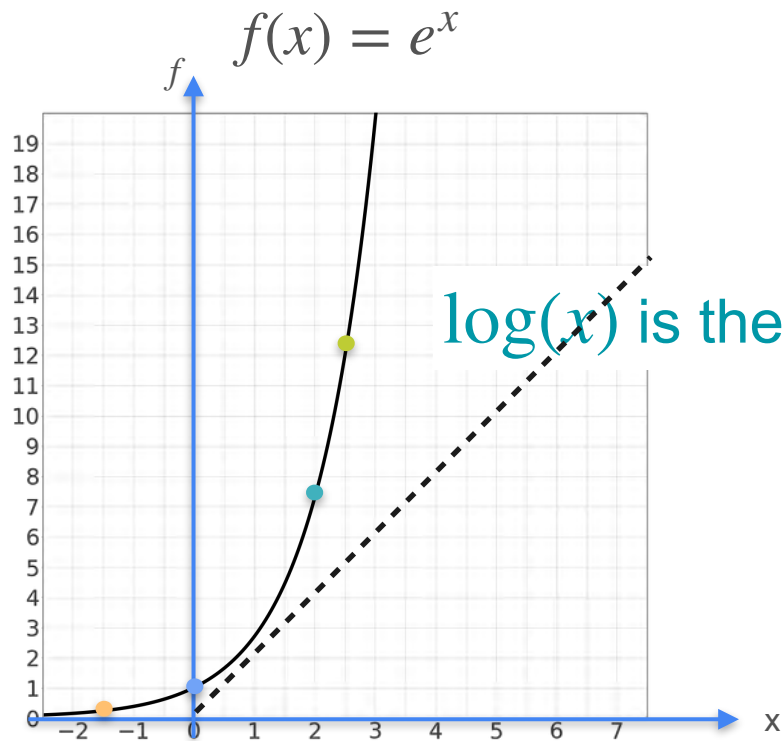
Logarithm



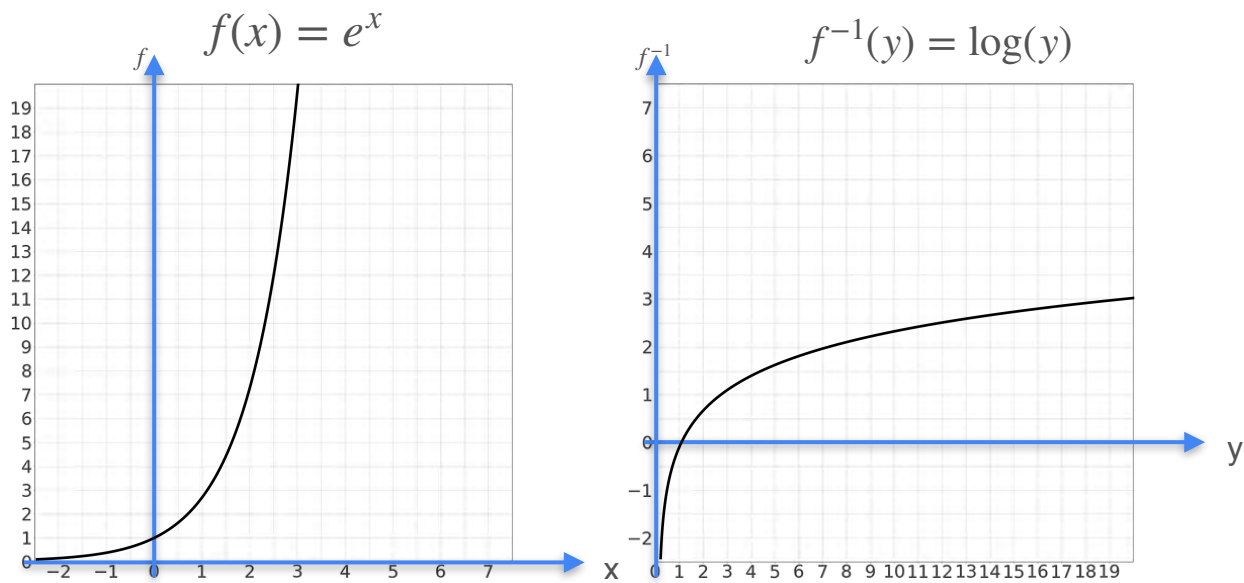
Logarithm



Logarithm

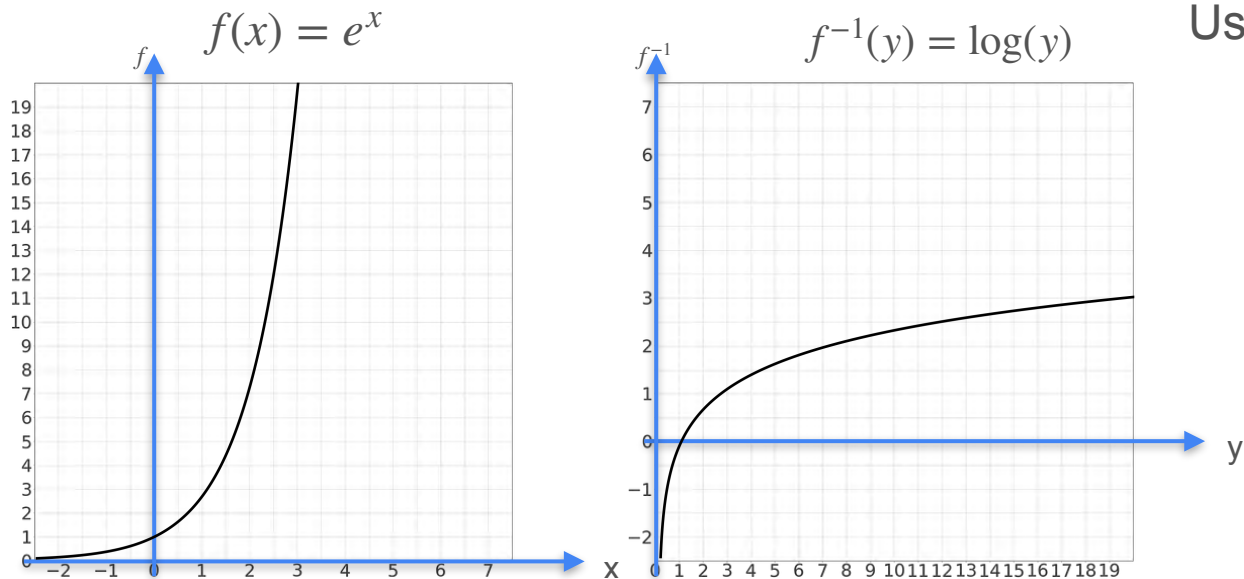


Logarithm



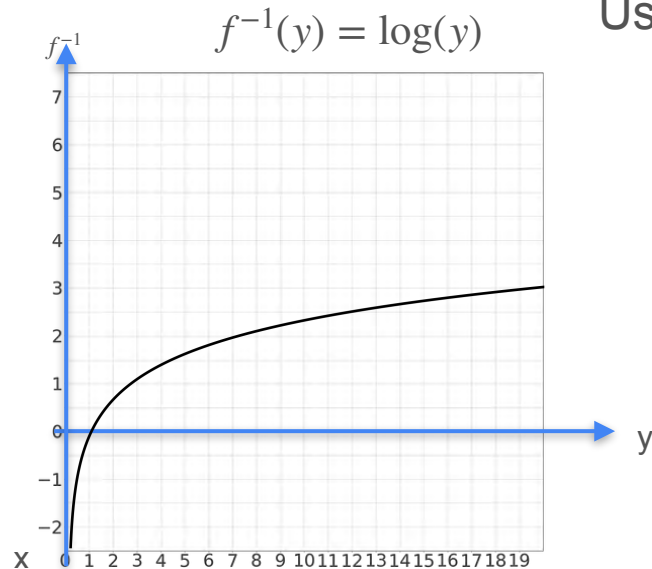
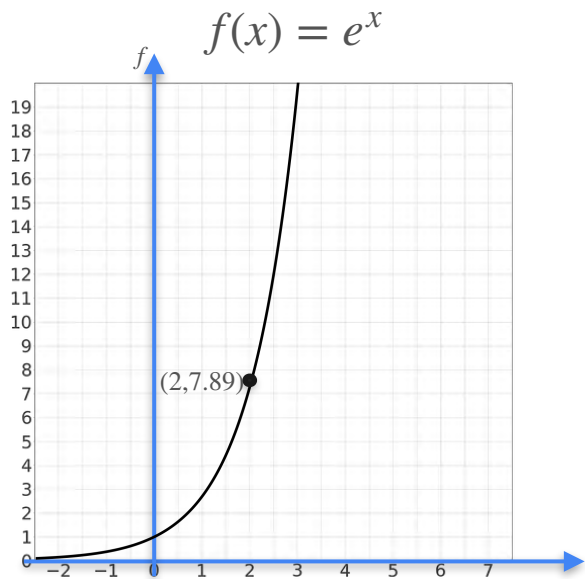
Logarithm

Using the result for inverses



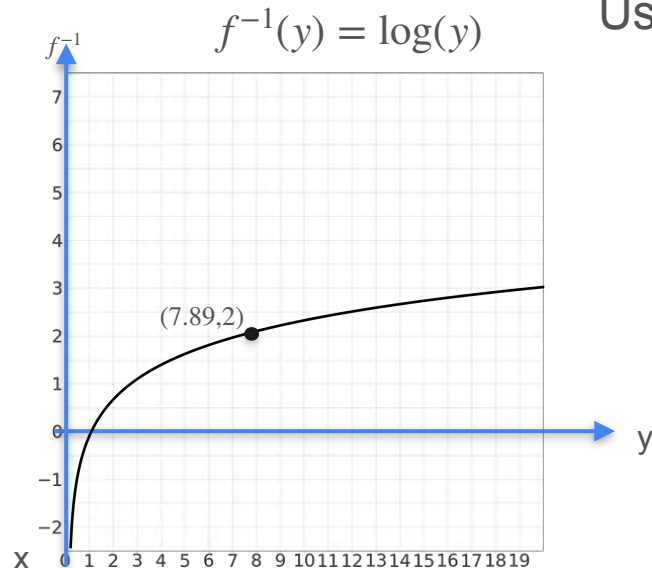
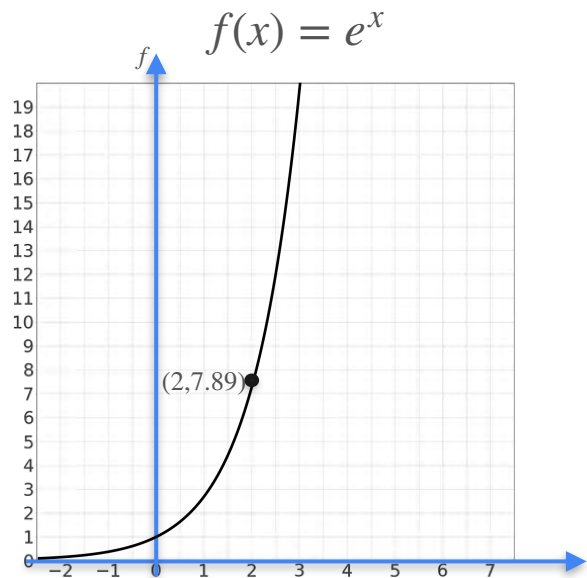
Logarithm

Using the result for inverses



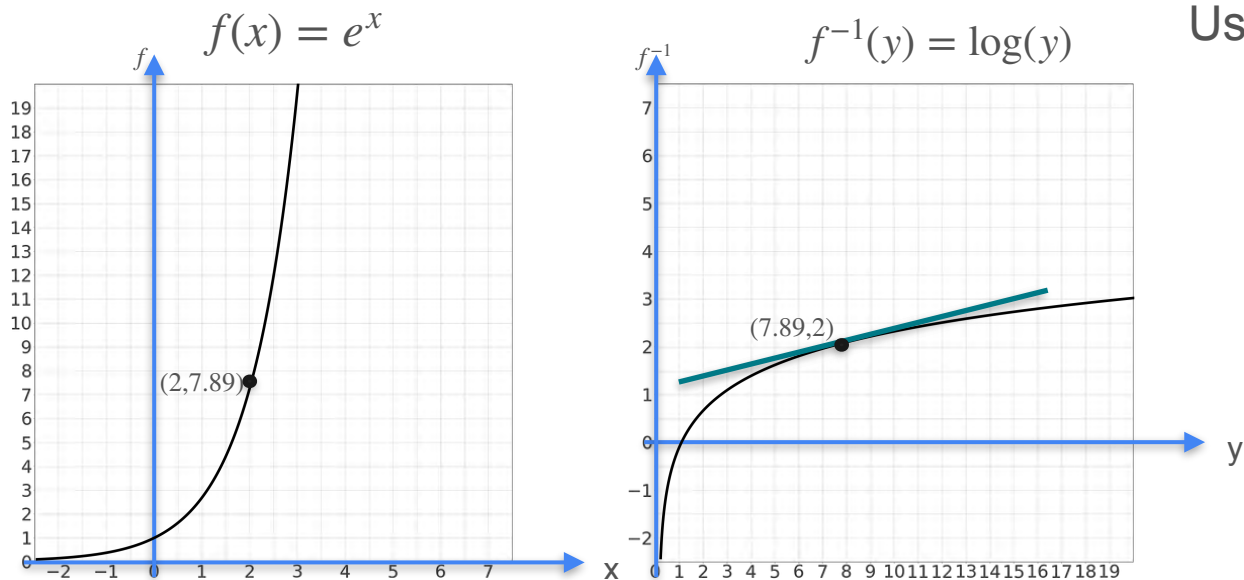
Logarithm

Using the result for inverses



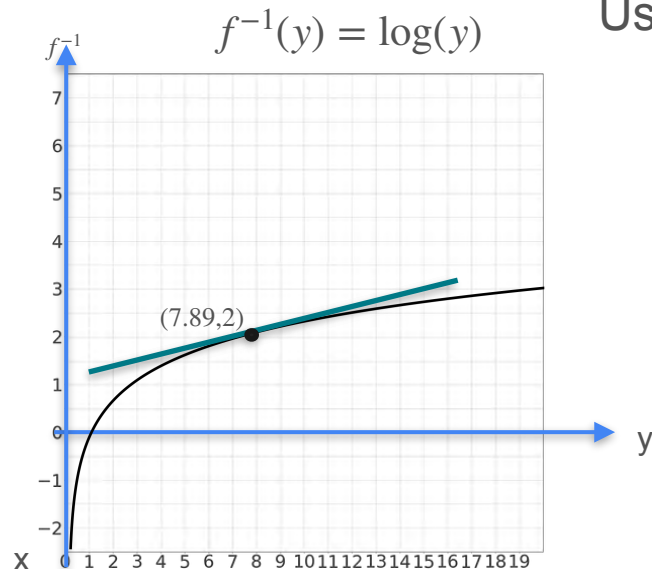
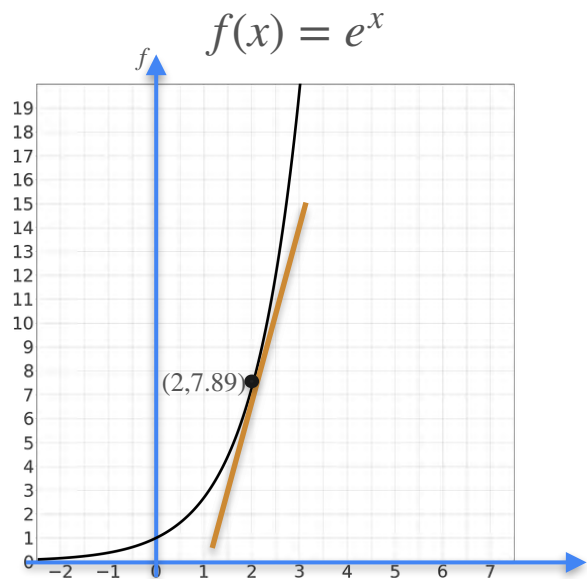
Logarithm

Using the result for inverses



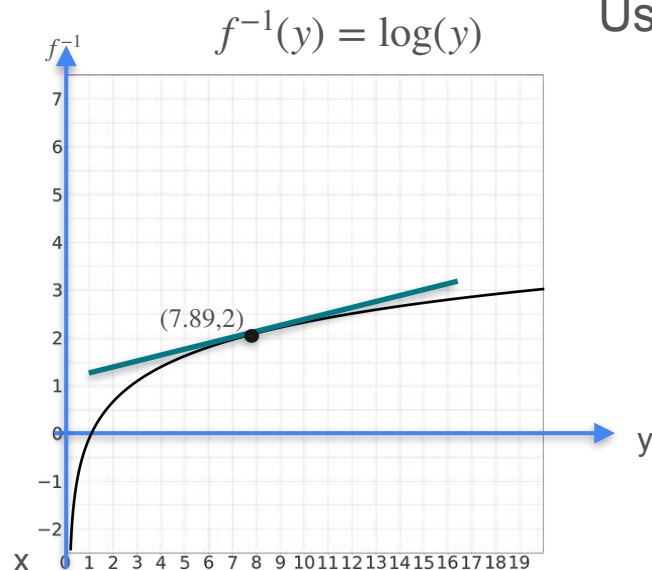
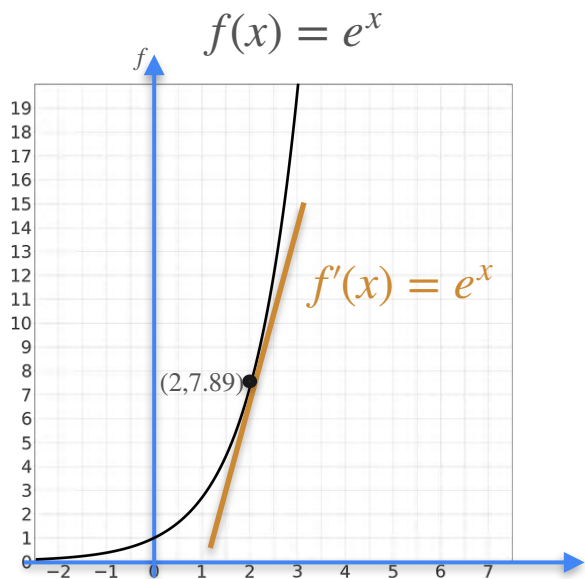
Logarithm

Using the result for inverses



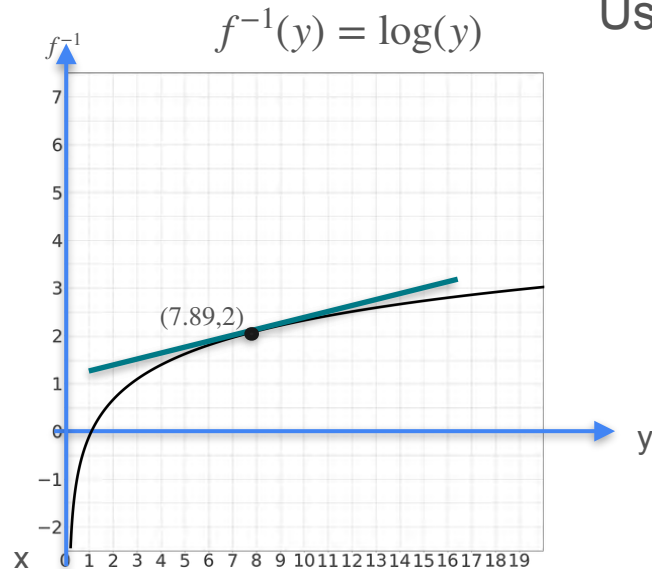
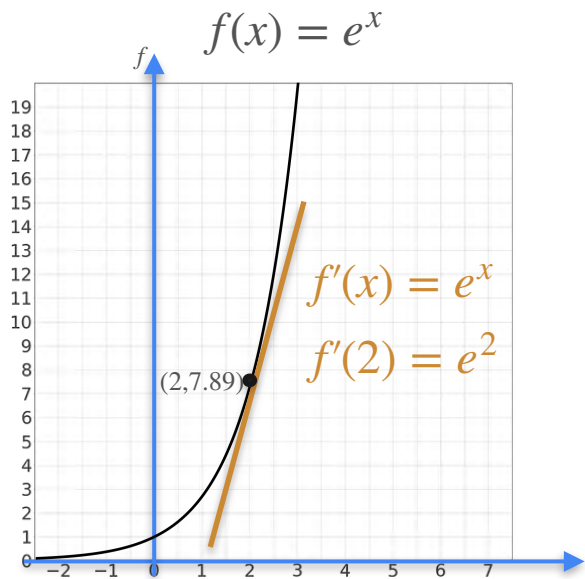
Logarithm

Using the result for inverses



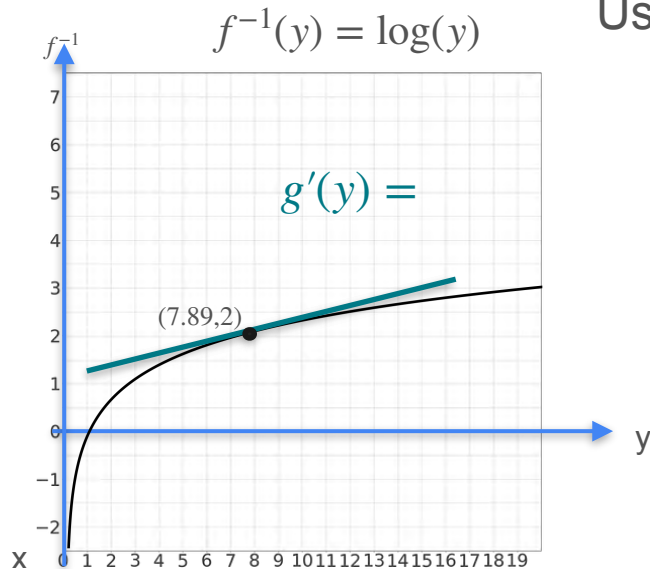
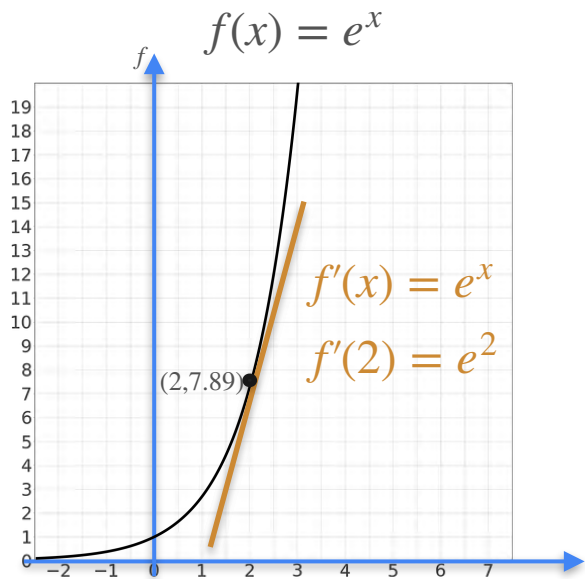
Logarithm

Using the result for inverses



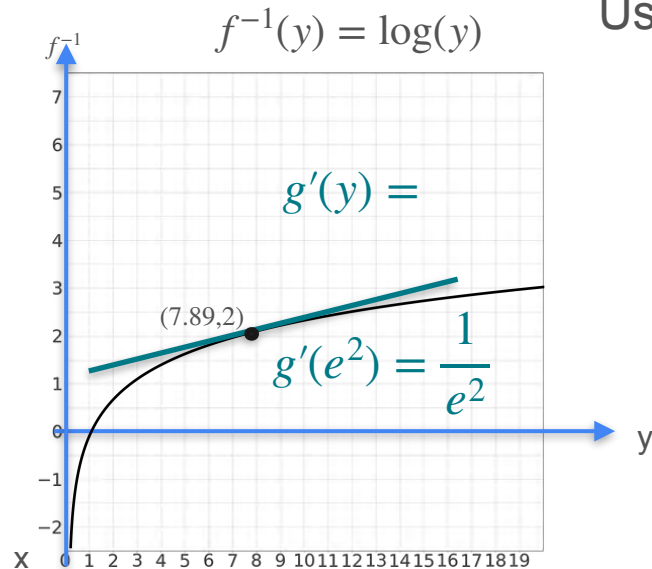
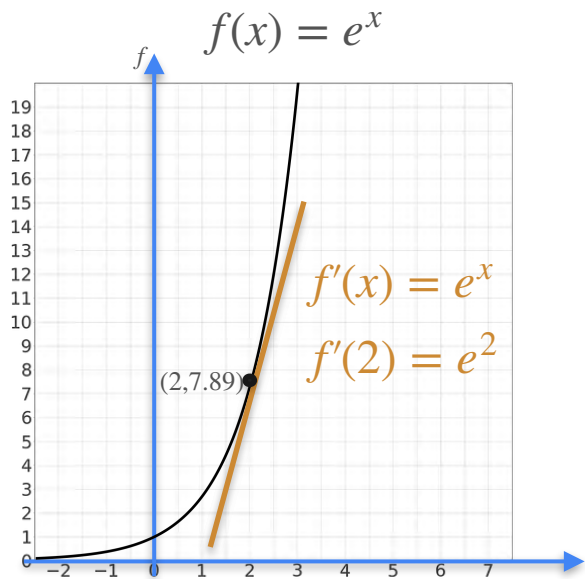
Logarithm

Using the result for inverses



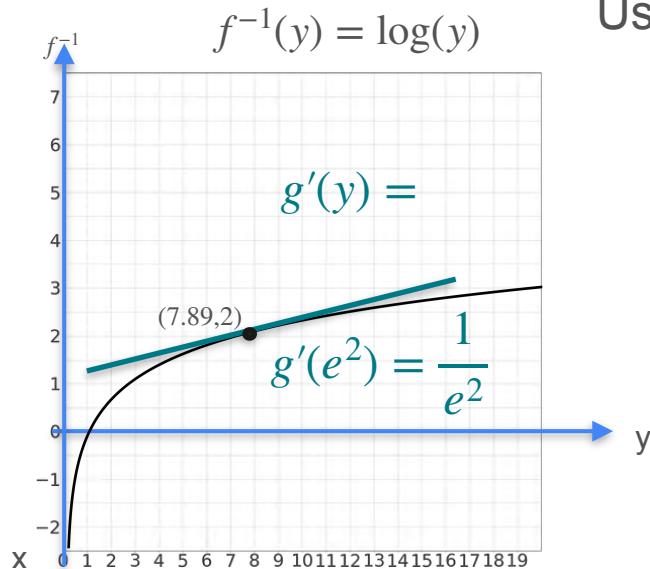
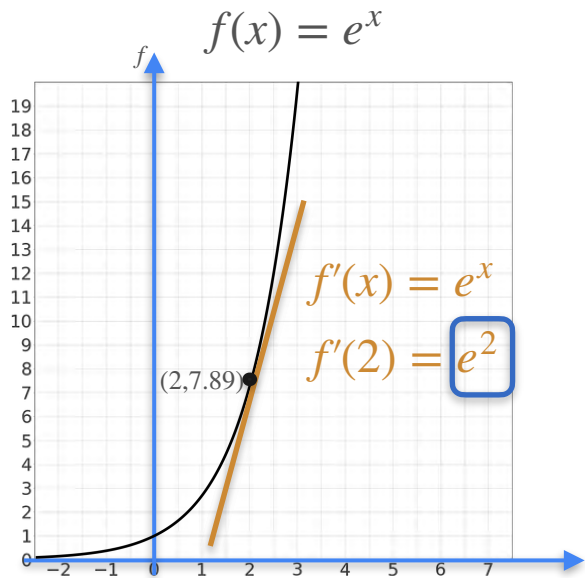
Logarithm

Using the result for inverses



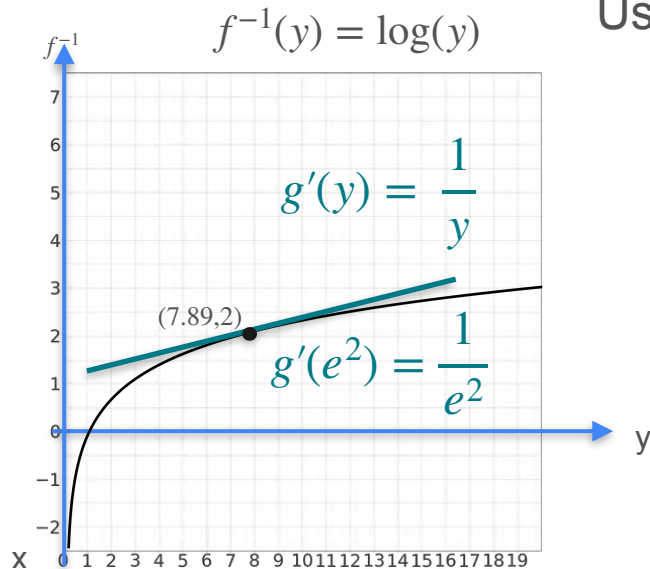
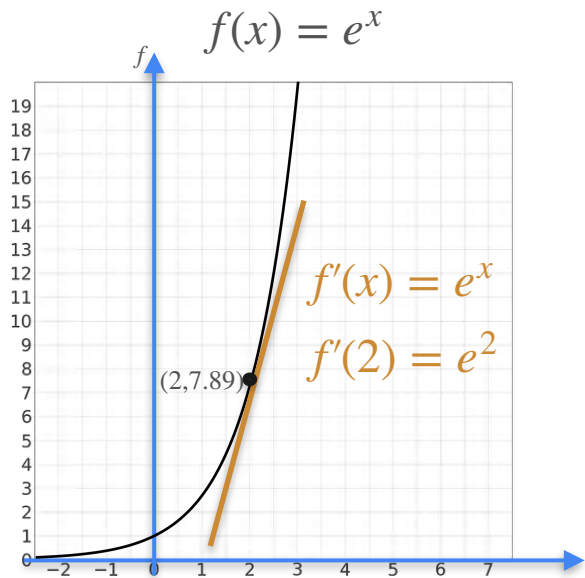
Logarithm

Using the result for inverses

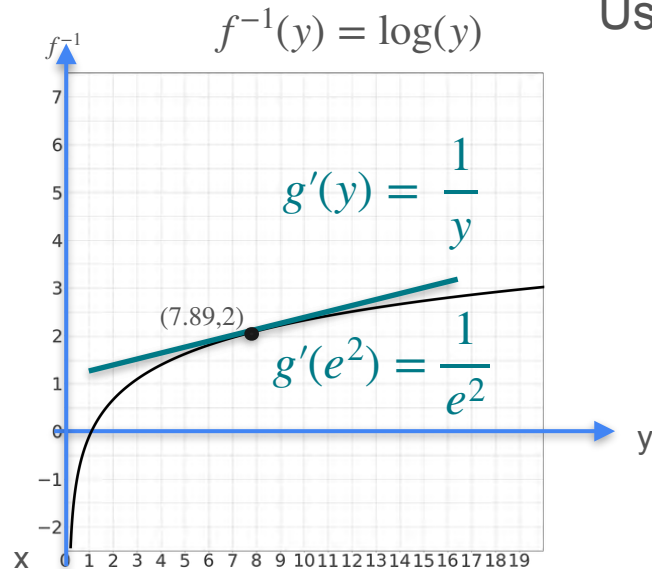
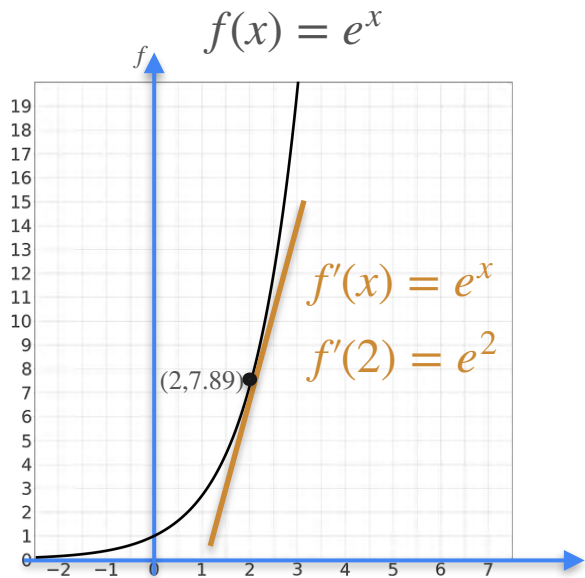


Logarithm

Using the result for inverses



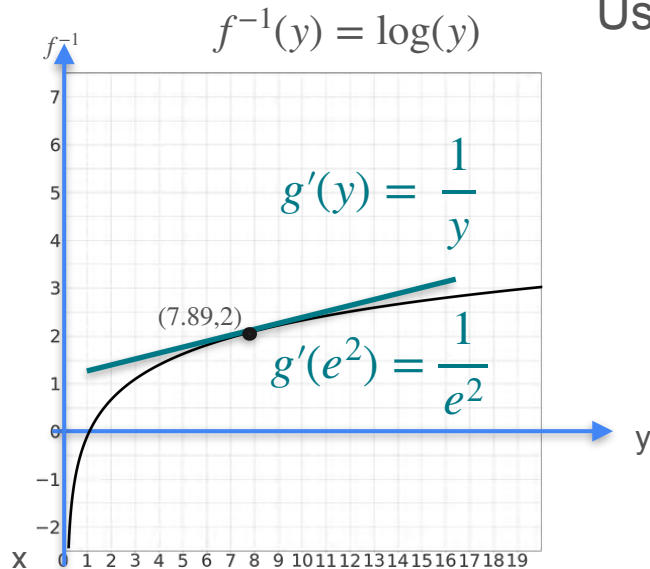
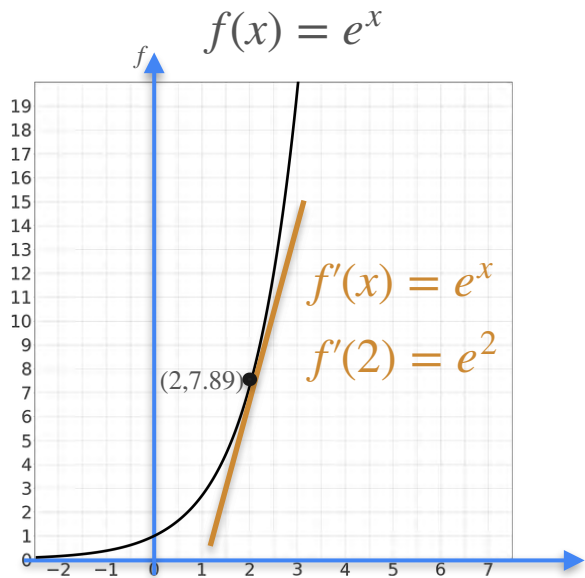
Logarithm



Using the result for inverses

$$\frac{d}{dy} f^{-1}(y) = \frac{1}{f'(x)}$$

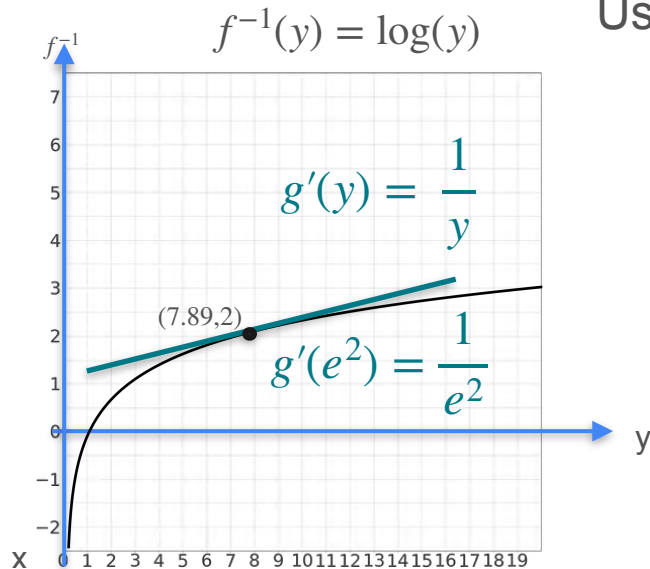
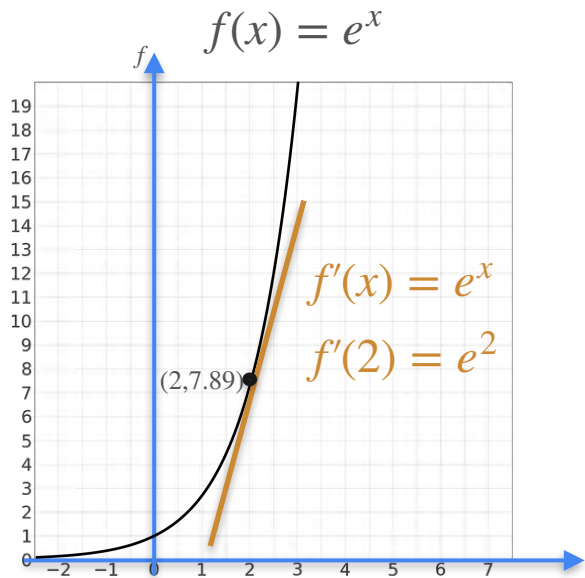
Logarithm



Using the result for inverses

$$\frac{d}{dy} f^{-1}(y) = \frac{1}{f'(f^{-1}(y))}$$

Logarithm

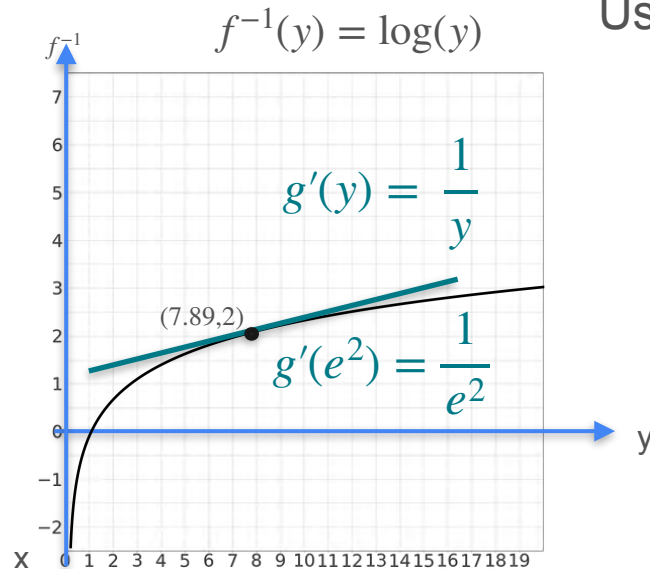
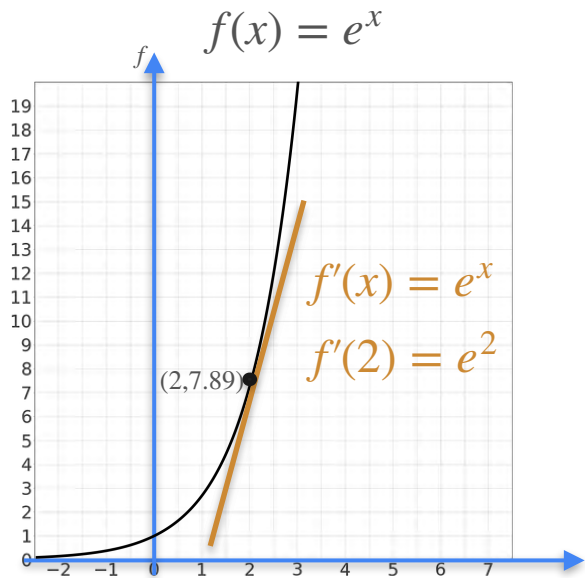


Using the result for inverses

$$\frac{d}{dy} f^{-1}(y) = \frac{1}{f'(f^{-1}(y))}$$

$\log(y)$

Logarithm

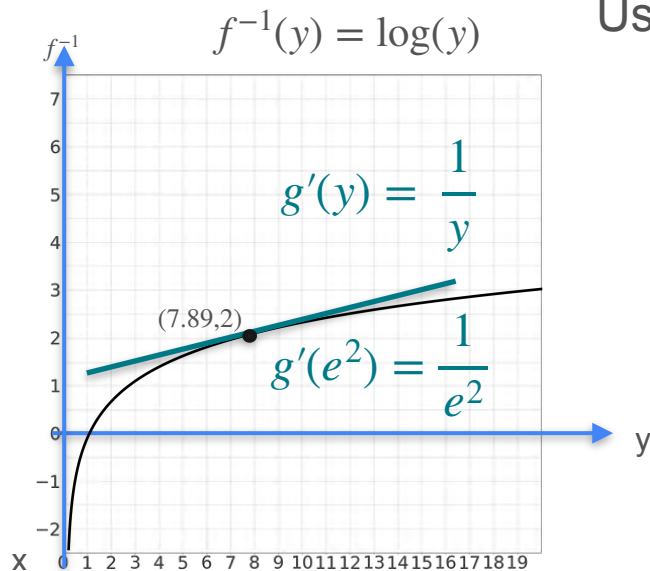
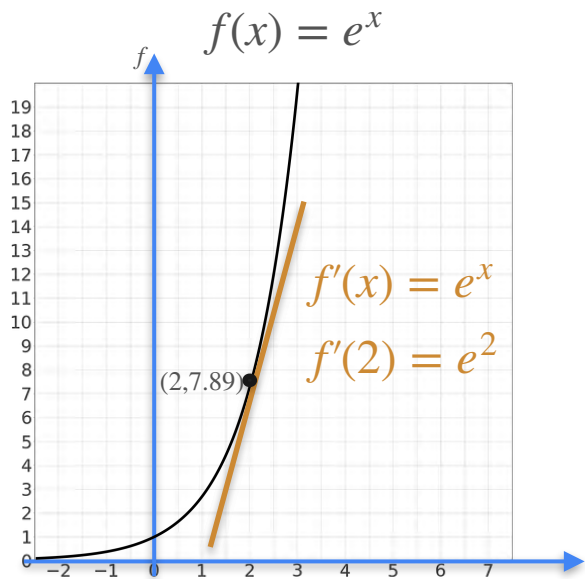


Using the result for inverses

$$\frac{d}{dy} f^{-1}(y) = \frac{1}{f'(f^{-1}(y))}$$

$$\frac{d}{dy} \log(y)$$

Logarithm

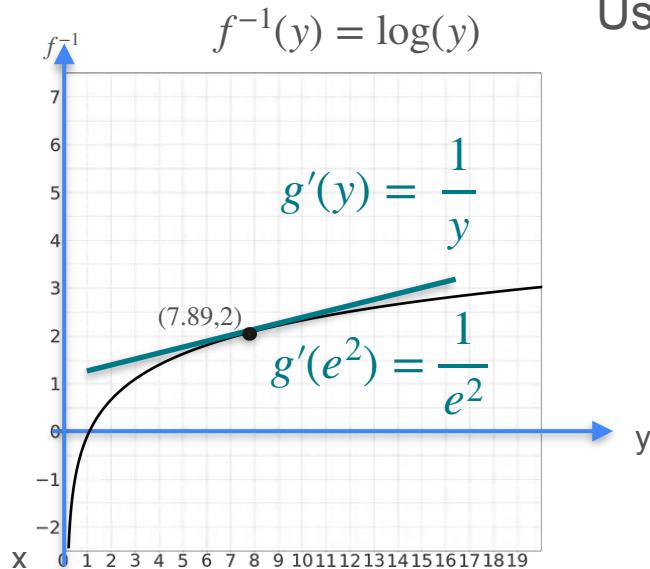
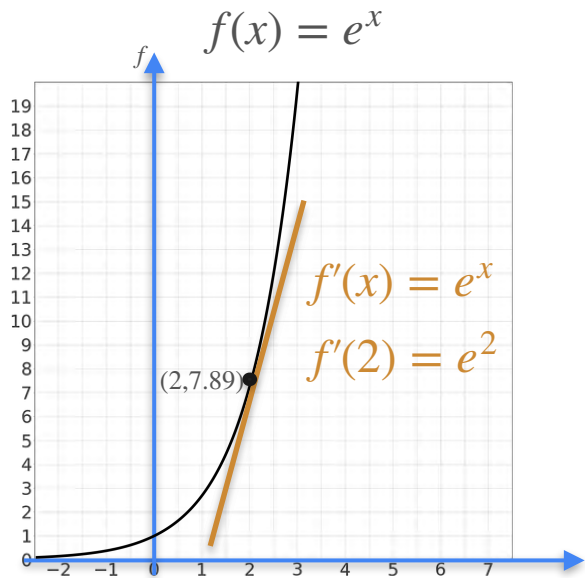


Using the result for inverses

$$\frac{d}{dy} f^{-1}(y) = \frac{1}{f'(f^{-1}(y))}$$

$$\frac{d}{dy} \log(y) = \frac{1}{e^2}$$

Logarithm

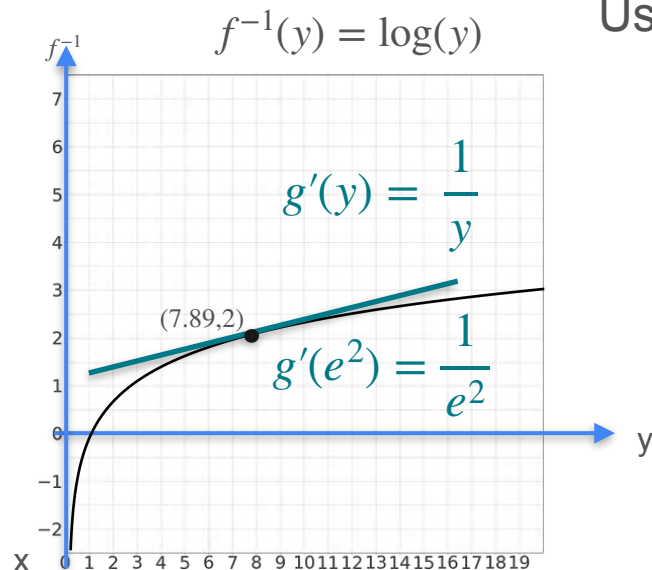
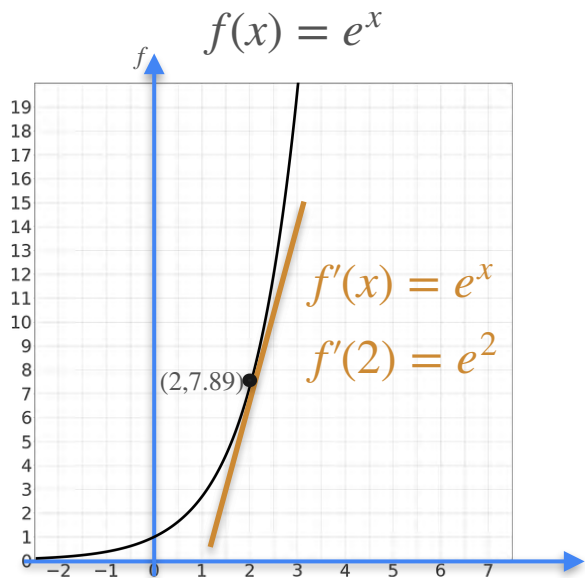


Using the result for inverses

$$\frac{d}{dy} f^{-1}(y) = \frac{1}{f'(f^{-1}(y))}$$

$$\frac{d}{dy} \log(y) = \frac{1}{e^{\log(y)}}$$

Logarithm

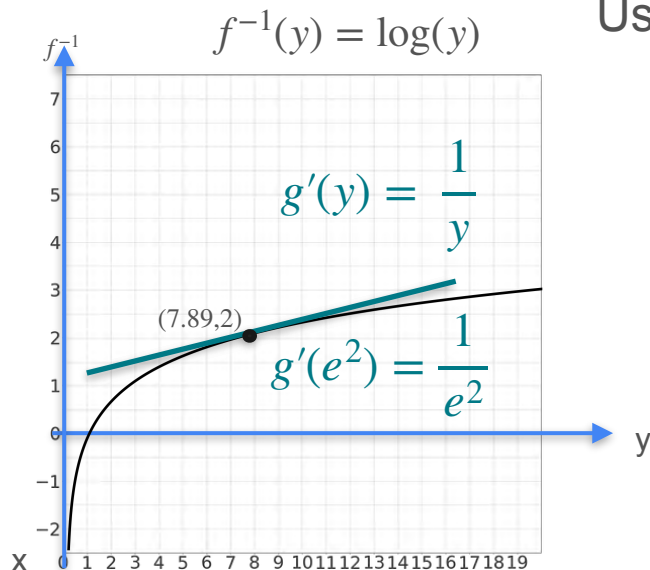
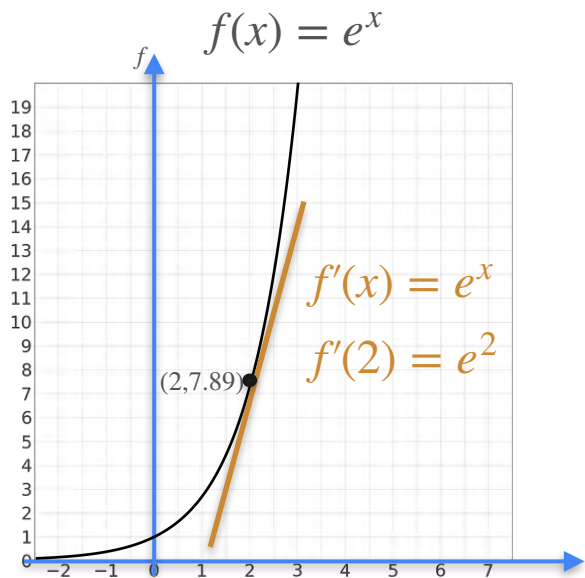


Using the result for inverses

$$\frac{d}{dy} f^{-1}(y) = \frac{1}{f'(f^{-1}(y))}$$

$$\begin{aligned} \frac{d}{dy} \log(y) &= \frac{1}{e^{\log(y)}} \\ &= \frac{1}{y} \end{aligned}$$

Logarithm



Using the result for inverses

$$\frac{d}{dy} f^{-1}(y) = \frac{1}{f'(f^{-1}(y))}$$

$$\begin{aligned} \frac{d}{dy} \log(y) &= \frac{1}{e^{\log(y)}} \\ &= \frac{1}{y} \end{aligned}$$

$$\frac{d}{dy} \log(y) = \frac{1}{y}$$



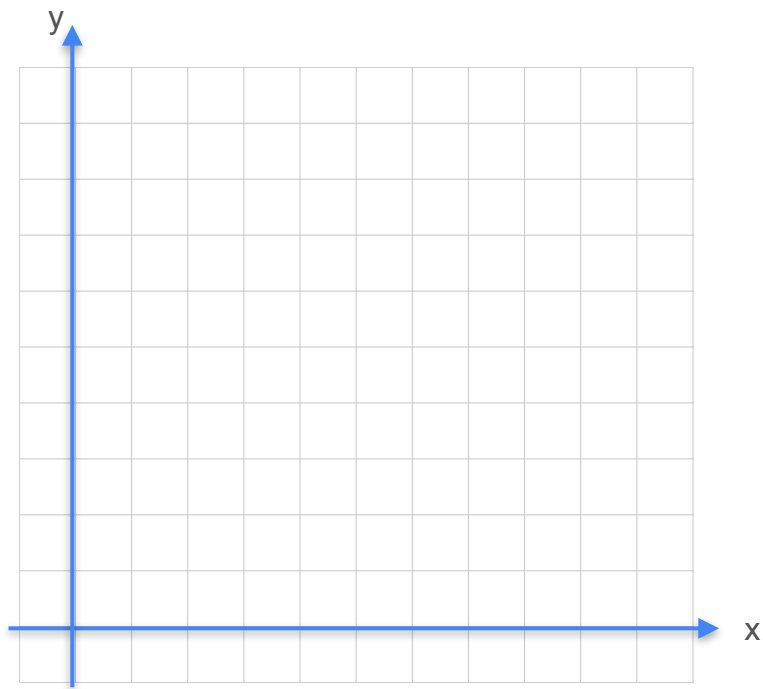
DeepLearning.AI

Derivatives and Optimization

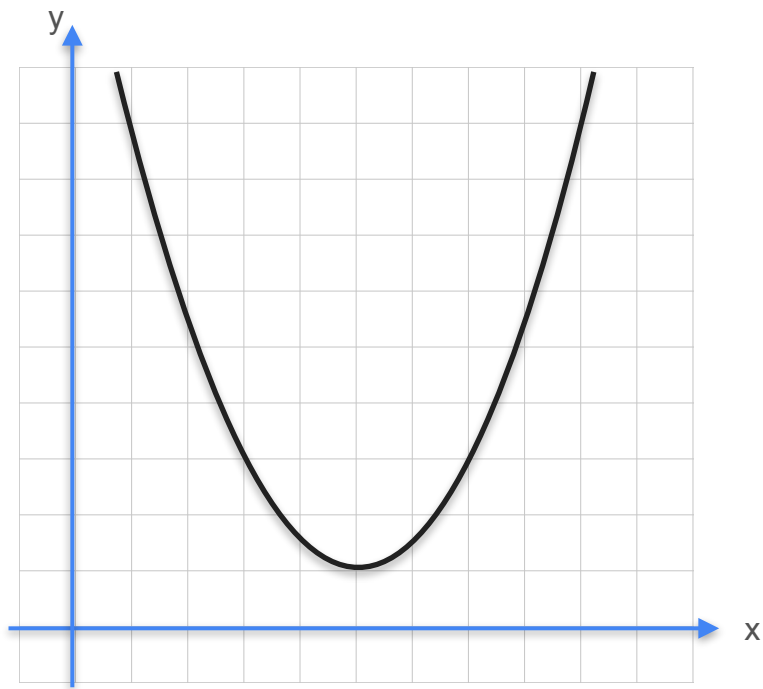
Existence of the derivative

Differentiable Function

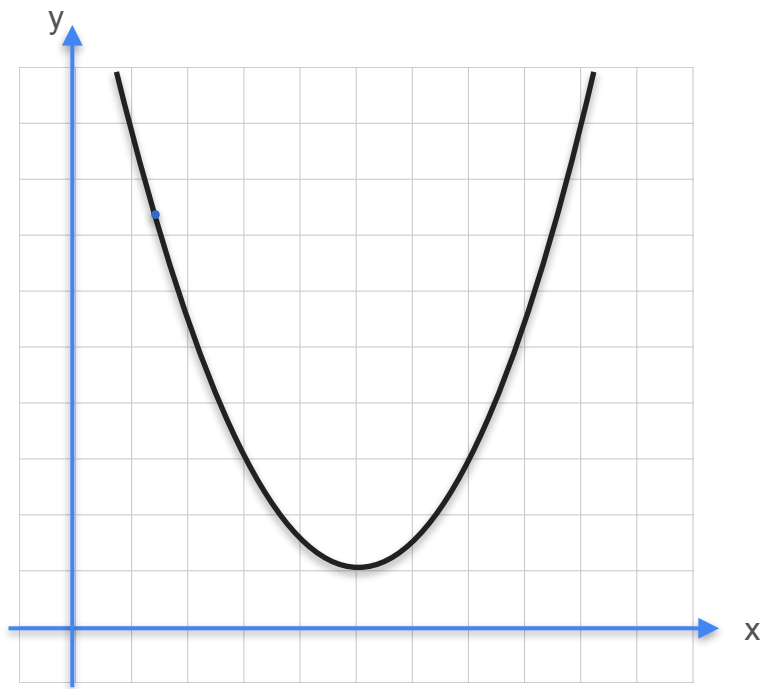
Differentiable Function



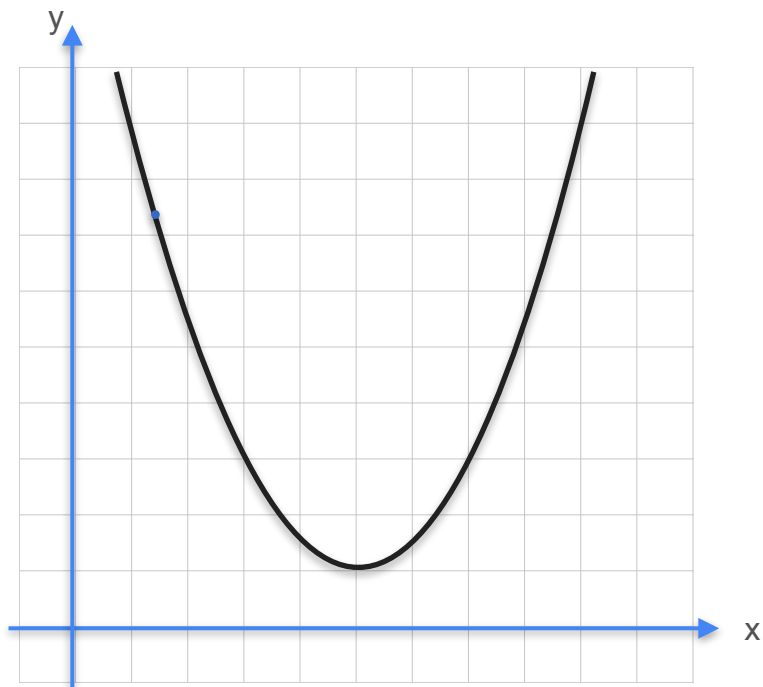
Differentiable Function



Differentiable Function



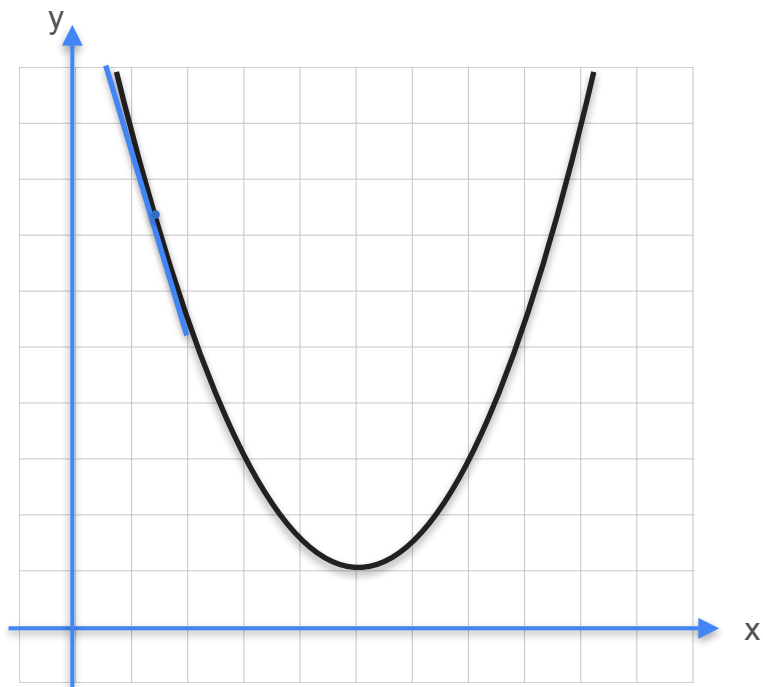
Differentiable Function



For a function to be differentiable at a point:

The derivative has to exist for that point

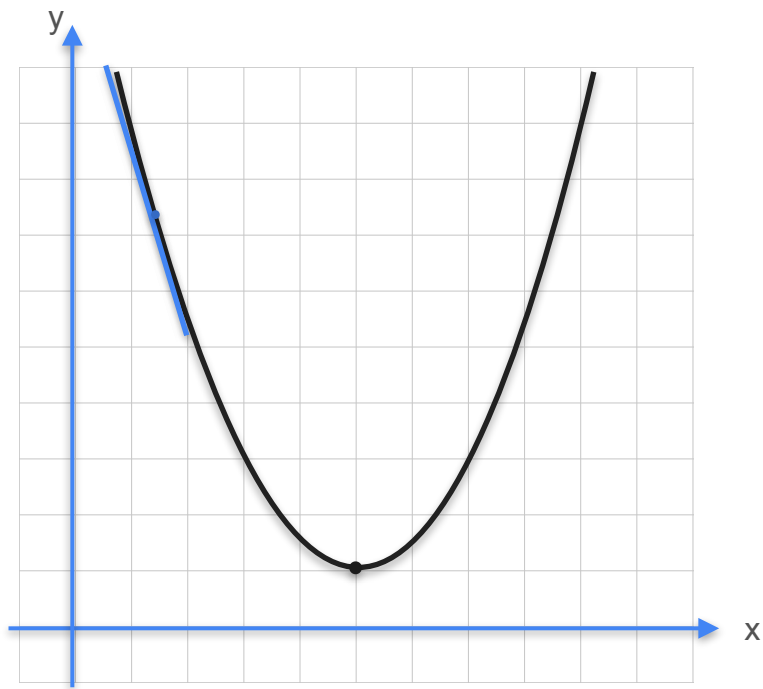
Differentiable Function



For a function to be differentiable at a point:

The derivative has to exist for that point

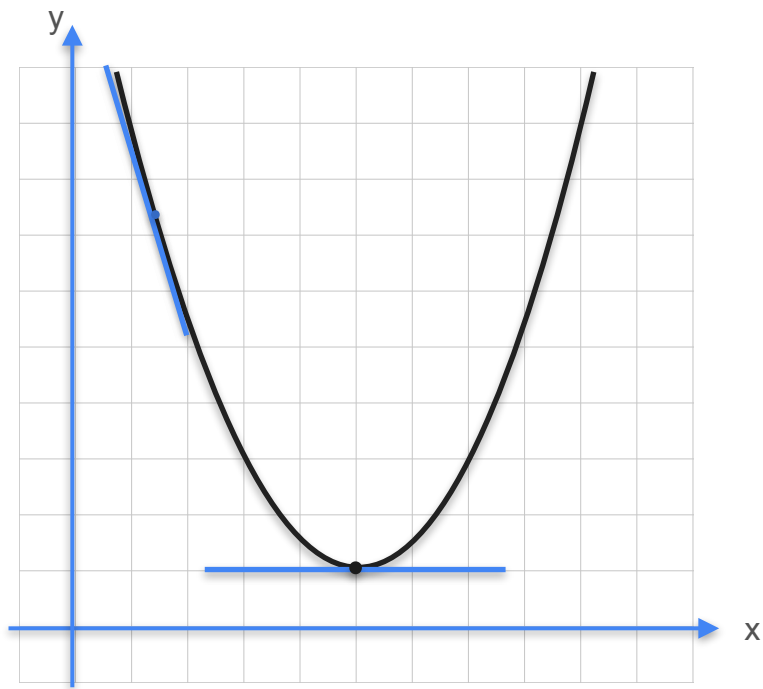
Differentiable Function



For a function to be differentiable at a point:

The derivative has to exist for that point

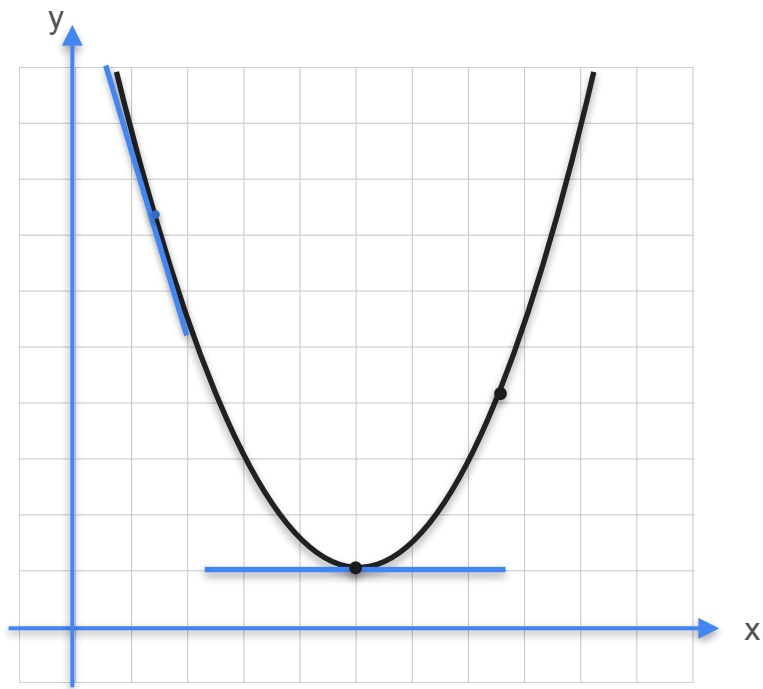
Differentiable Function



For a function to be differentiable at a point:

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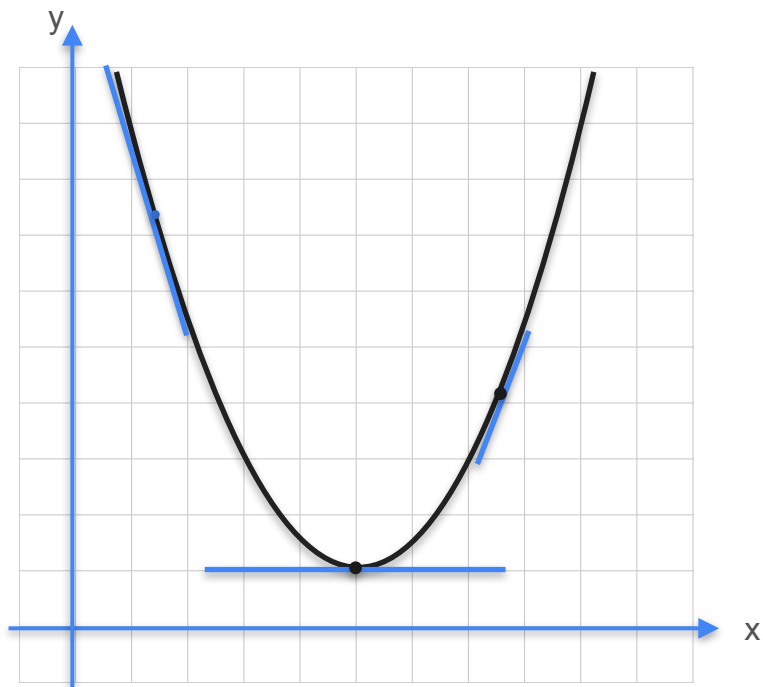
Differentiable Function



For a function to be differentiable at a point:

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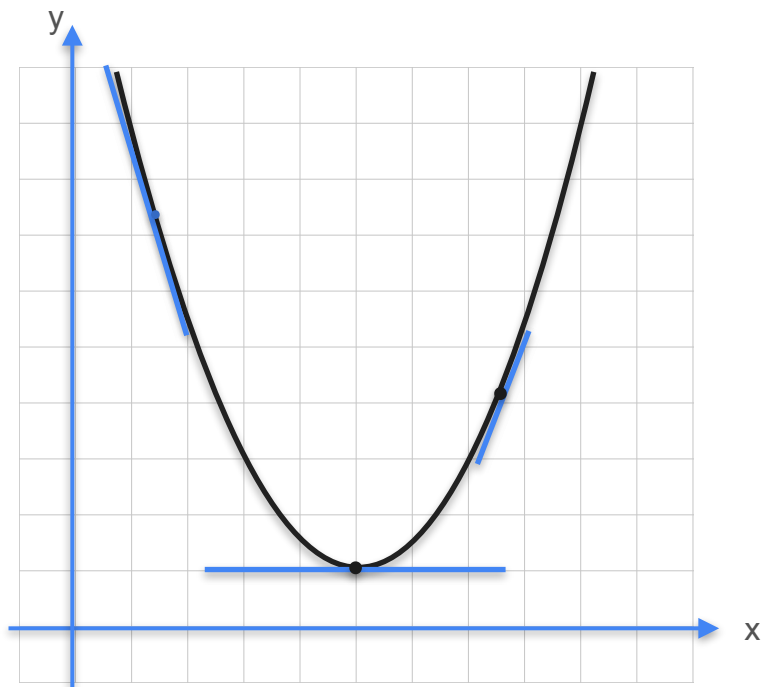
Differentiable Function



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Differentiable Function



For a function to be differentiable at a point:

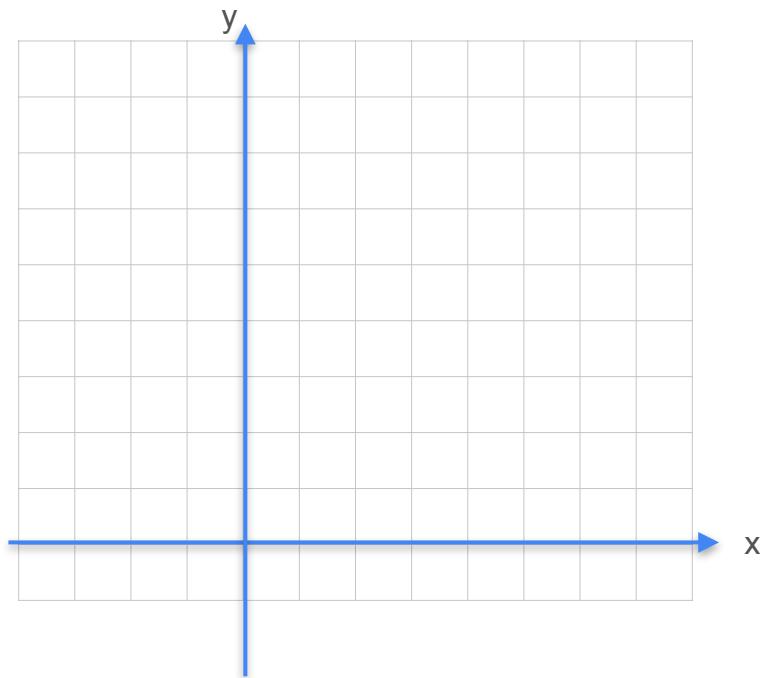
The derivative has to exist for that point

For a function to be differentiable at an interval:

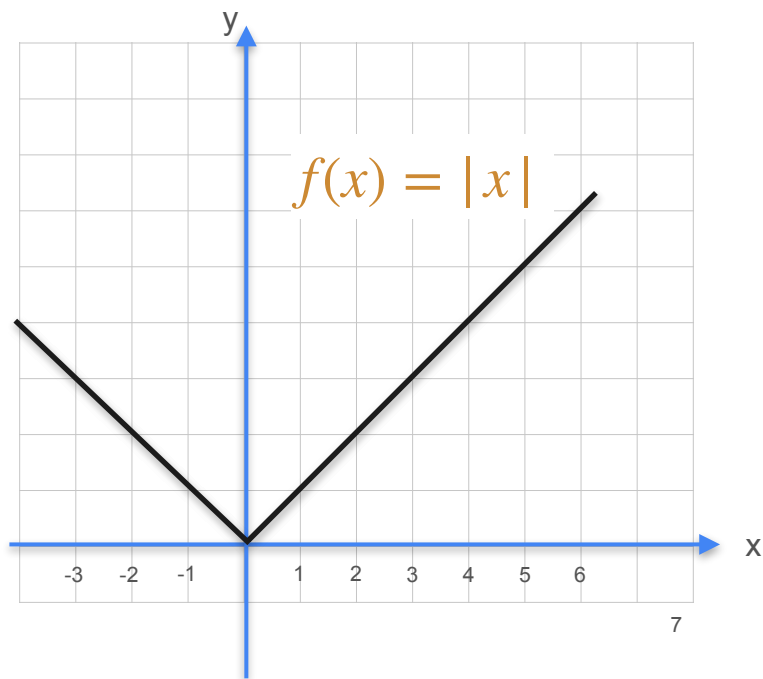
The derivative has to exist for *every* point in the interval

Non Differentiable Functions

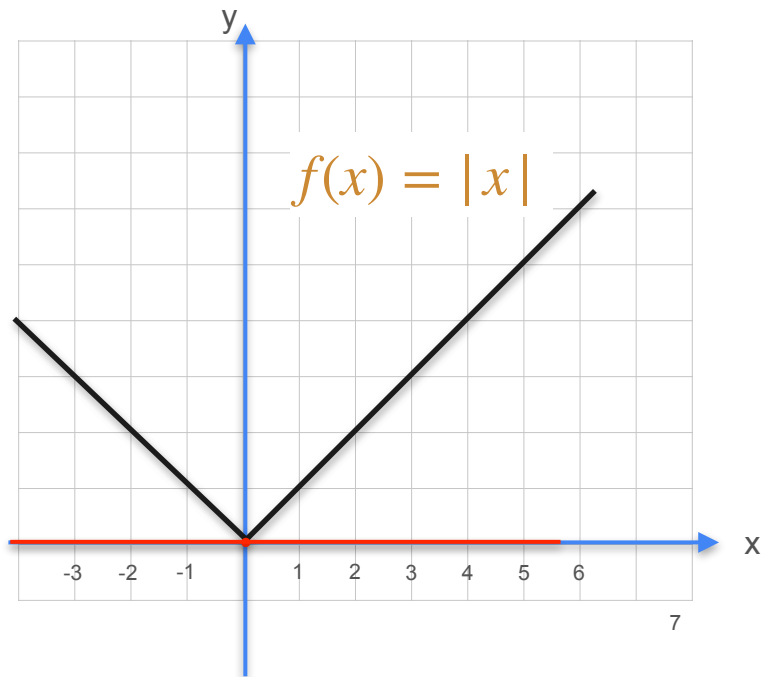
Non Differentiable Functions



Non Differentiable Functions



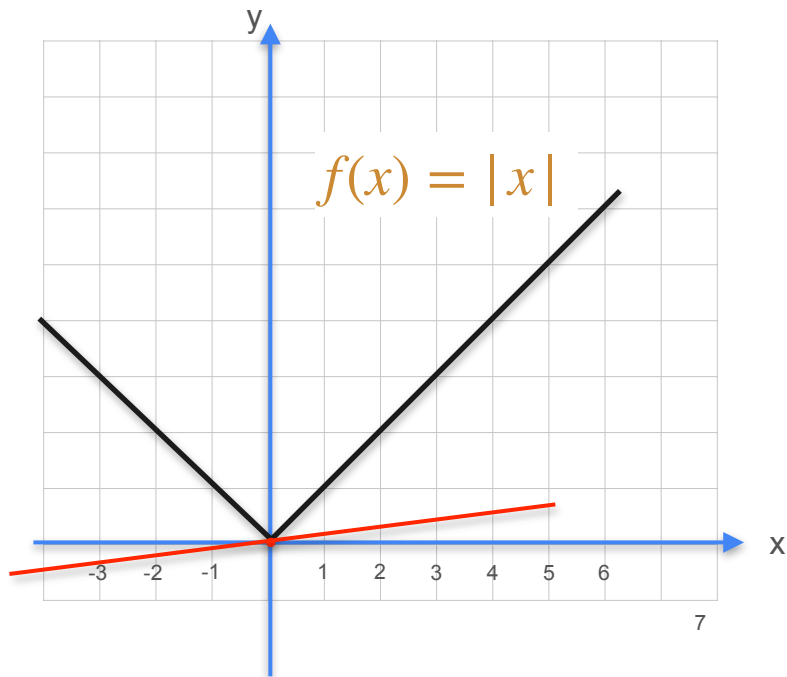
Non Differentiable Functions



The absolute value function.

$$f(x) = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

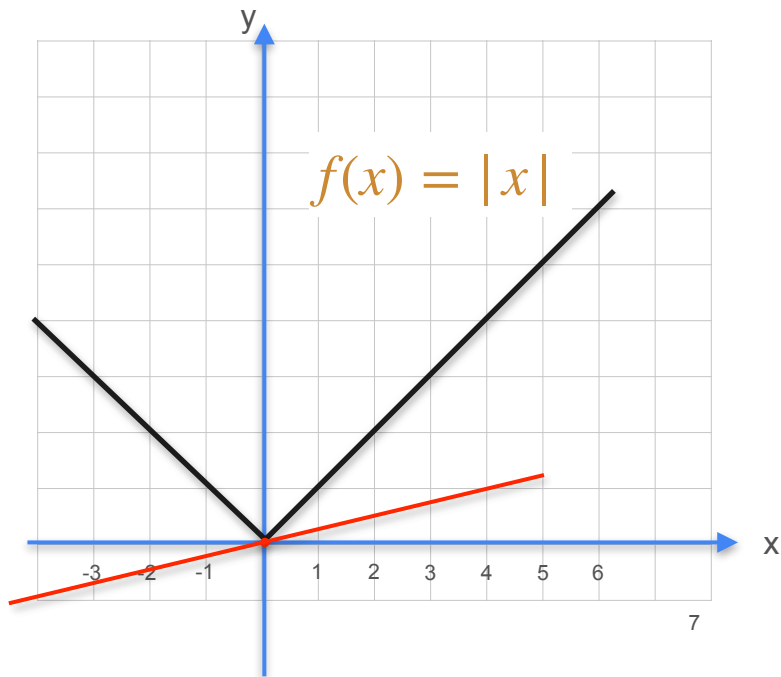
Non Differentiable Functions



The absolute value function.

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Non Differentiable Functions



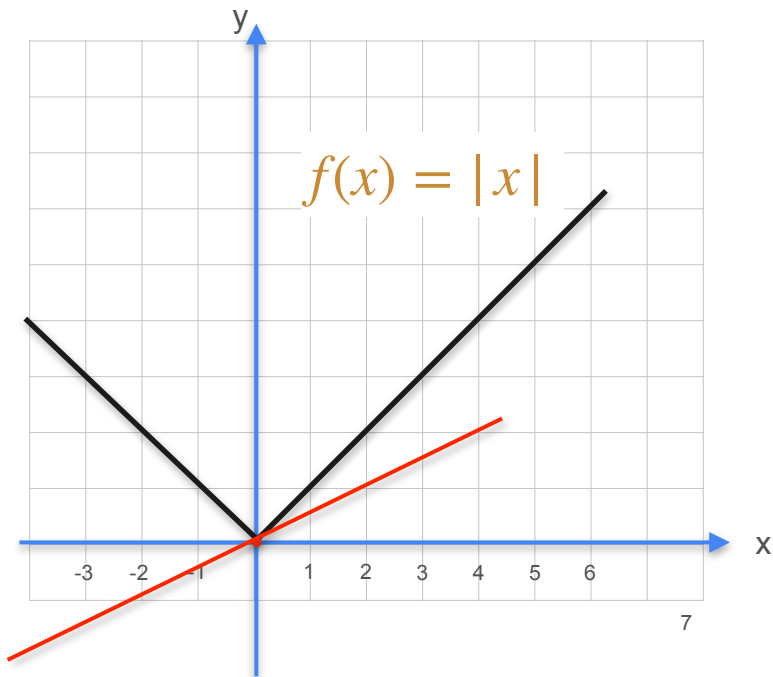
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Non Differentiable Functions

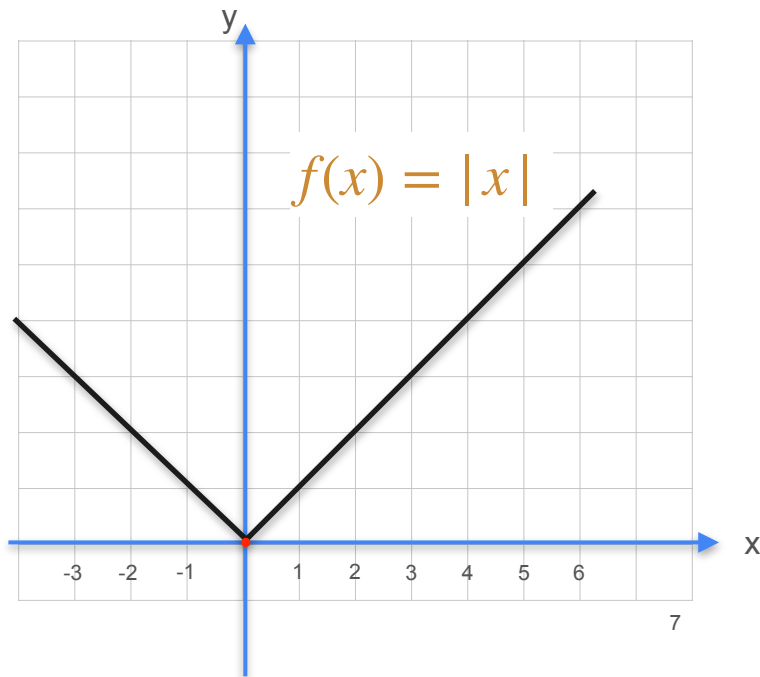
The absolute value function.

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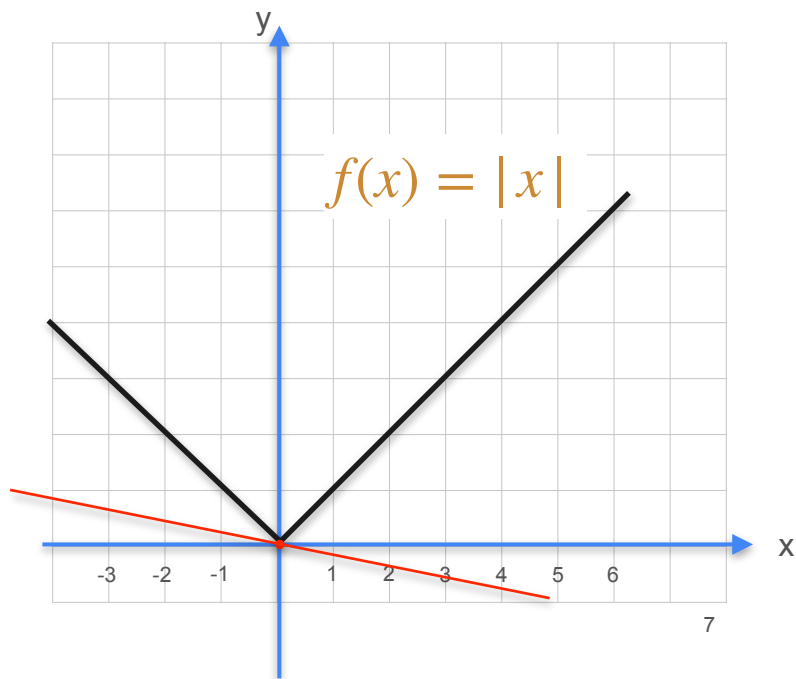


Non Differentiable Functions

The absolute value function.

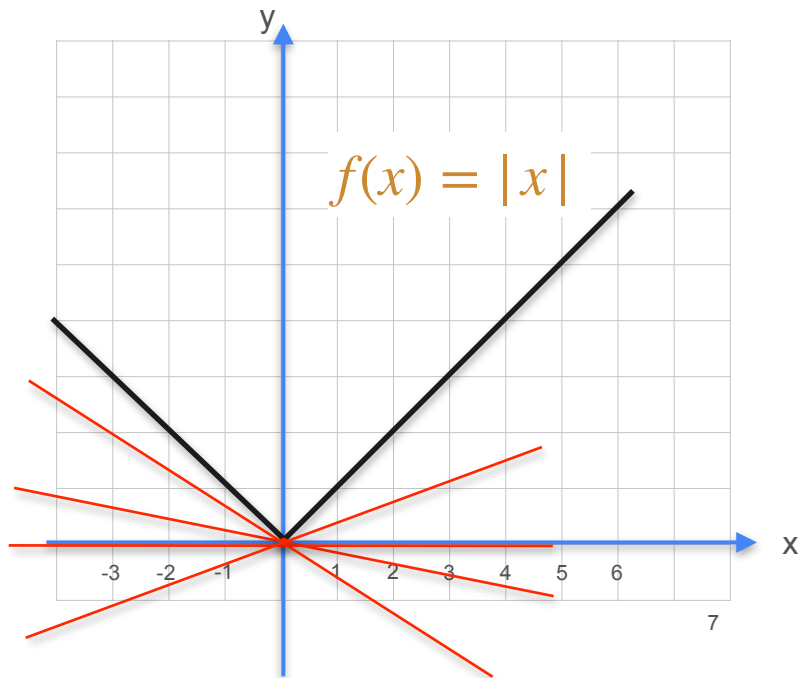


Non Differentiable Functions



The absolute value function.

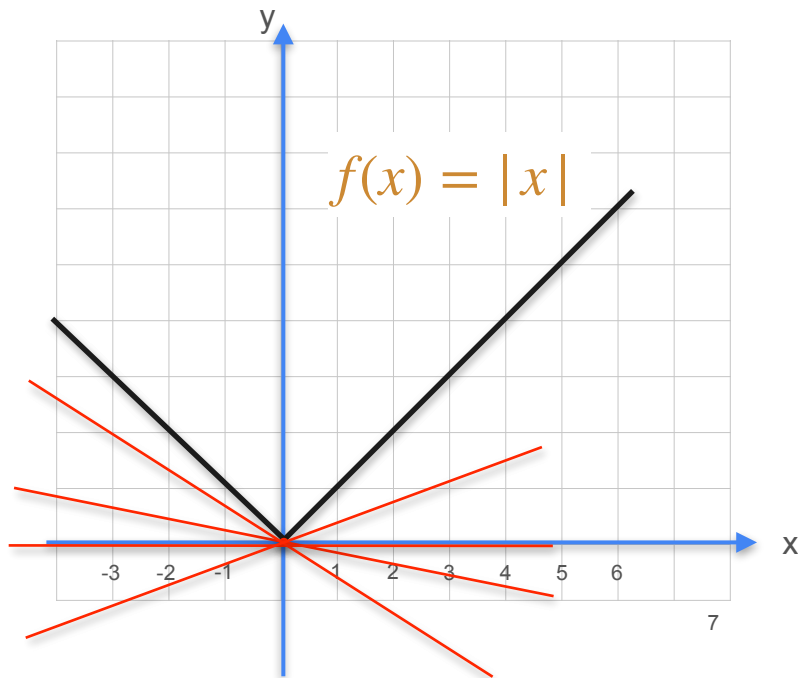
Non Differentiable Functions



The absolute value function.

At $x = 0$, the derivative does not exist

Non Differentiable Functions



The absolute value function.

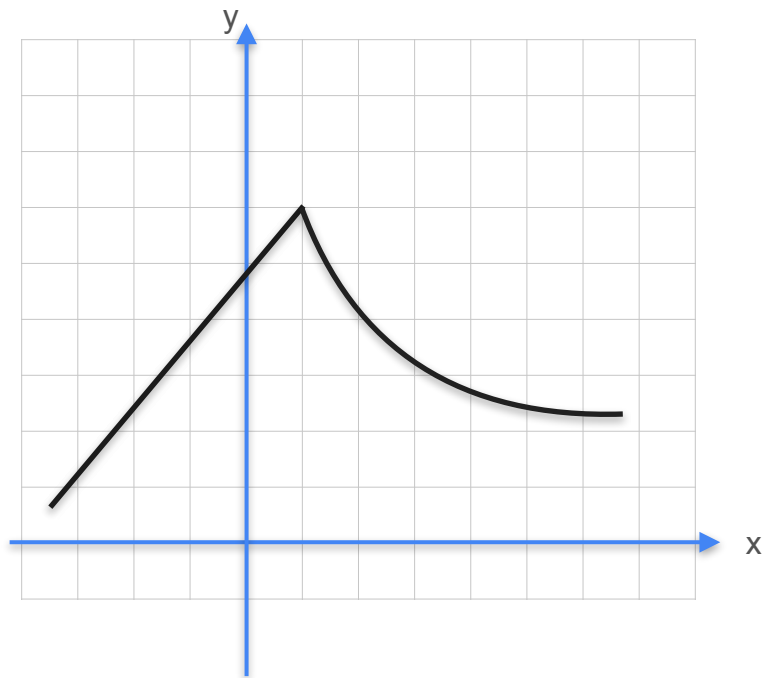
$$f(x) = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

At $x = 0$, the derivative does not exist

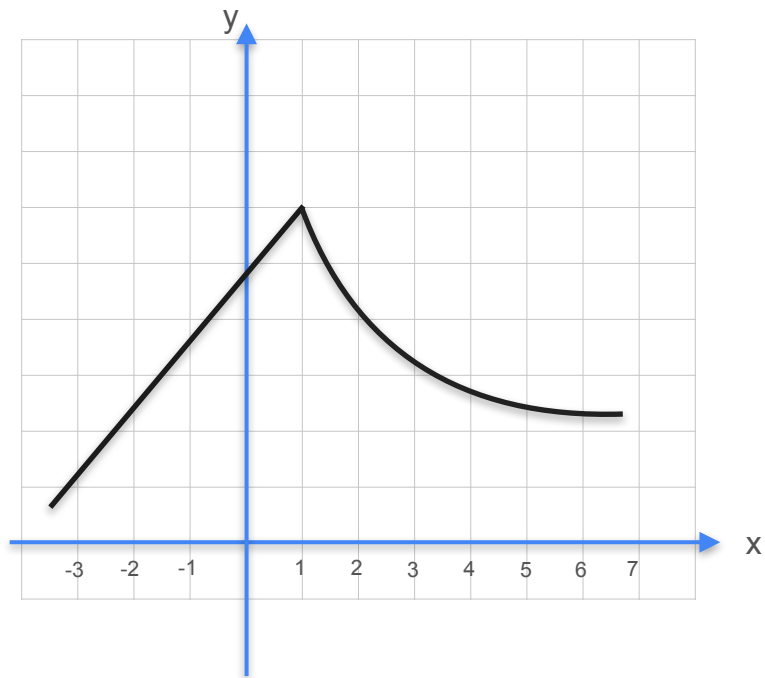
Generally, when a function has a corner or a cusp, the function is not differentiable at that point.

Non Differentiable Functions - Quiz 1

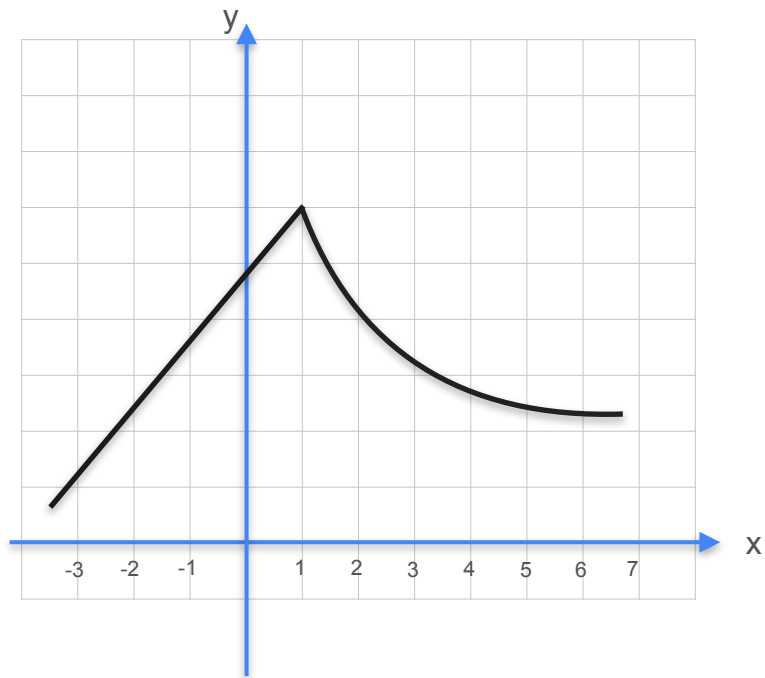
Non Differentiable Functions - Quiz 1



Non Differentiable Functions - Quiz 1

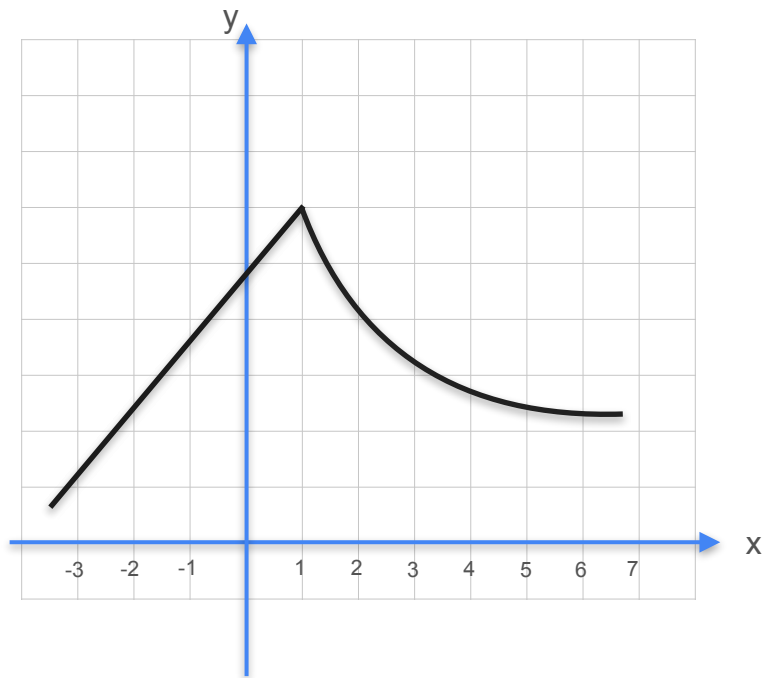


Non Differentiable Functions - Quiz 1



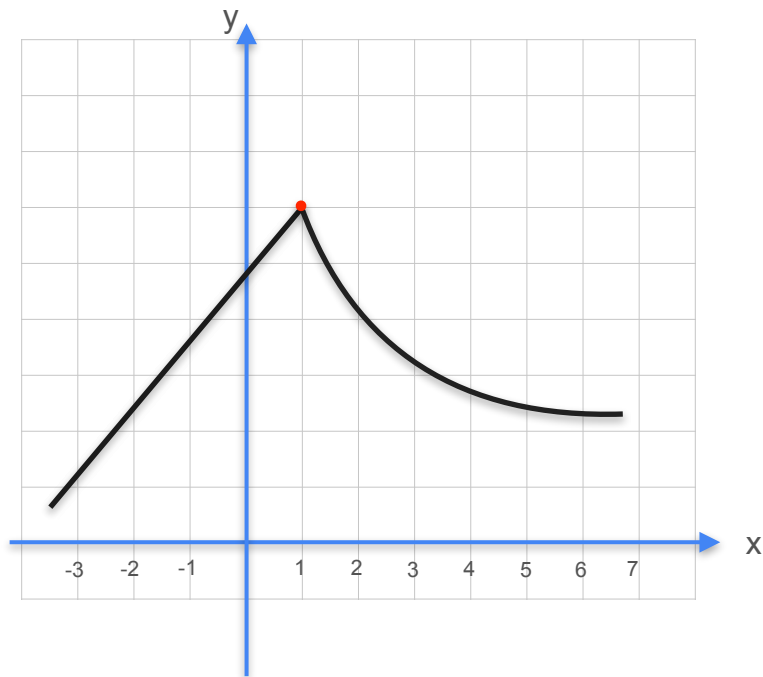
At which point in this function does the derivative not exist?

Non Differentiable Functions - Quiz 1



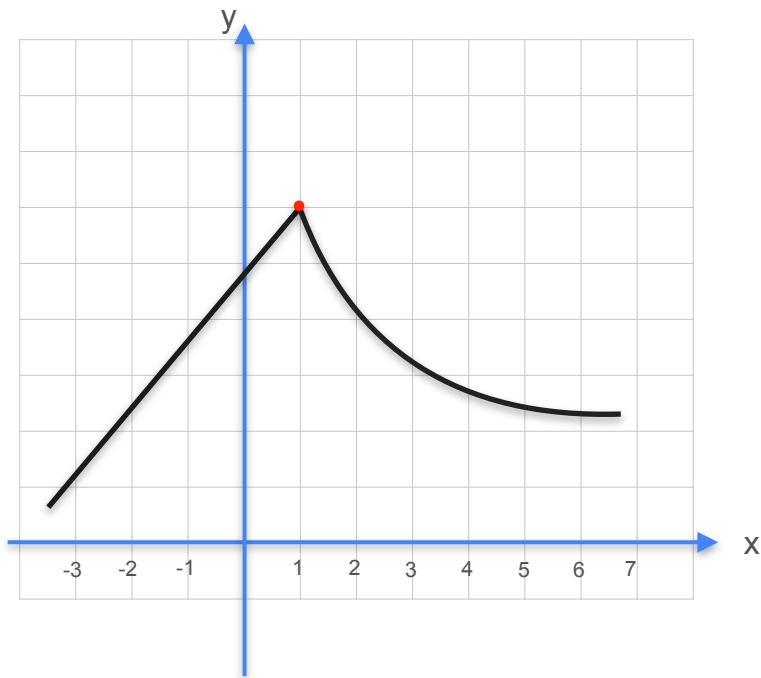
At which point in this function does the derivative not exist?

Non Differentiable Functions - Quiz 1



At which point in this function does the derivative not exist?

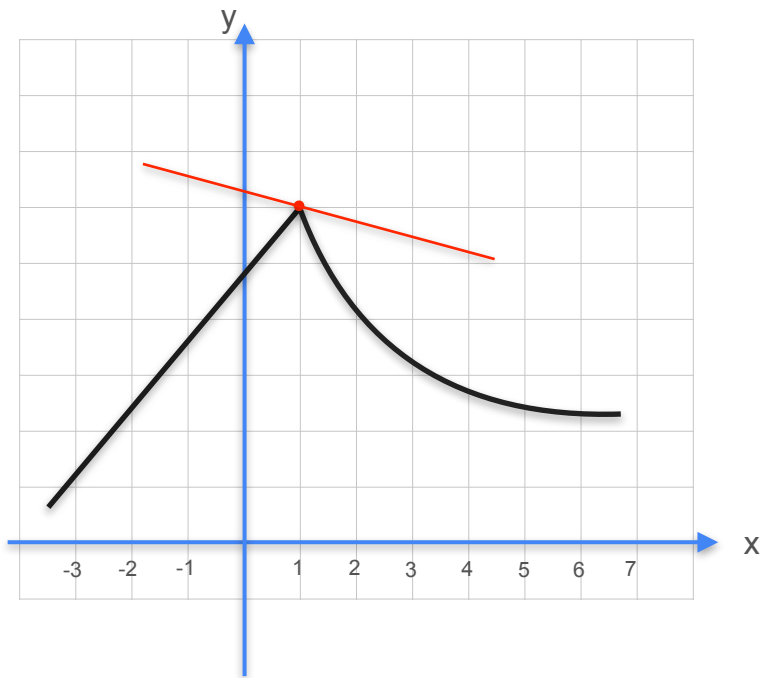
Non Differentiable Functions - Quiz 1



At which point in this function does the derivative not exist?

The entire function is non-differentiable because a derivative does not exist for all points in the domain.

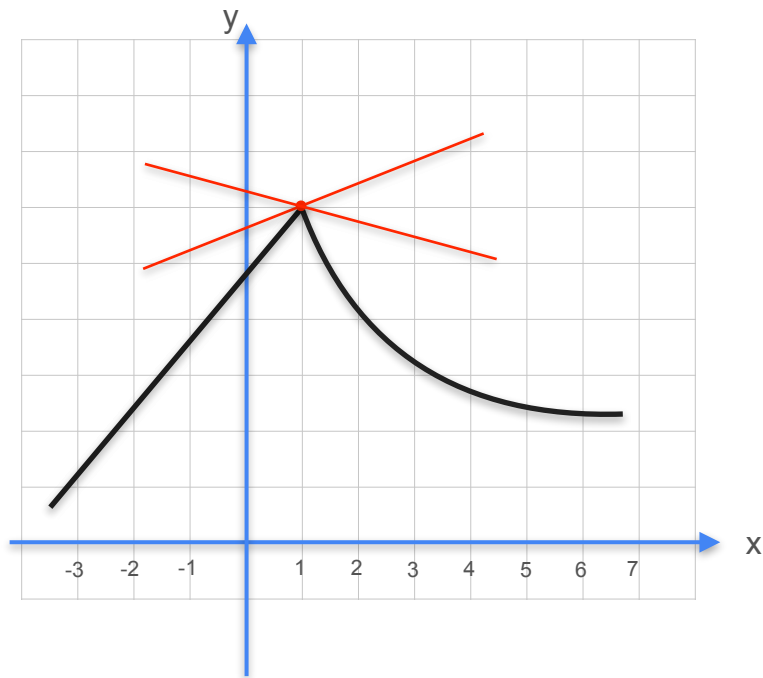
Non Differentiable Functions - Quiz 1



At which point in this function does the derivative not exist?

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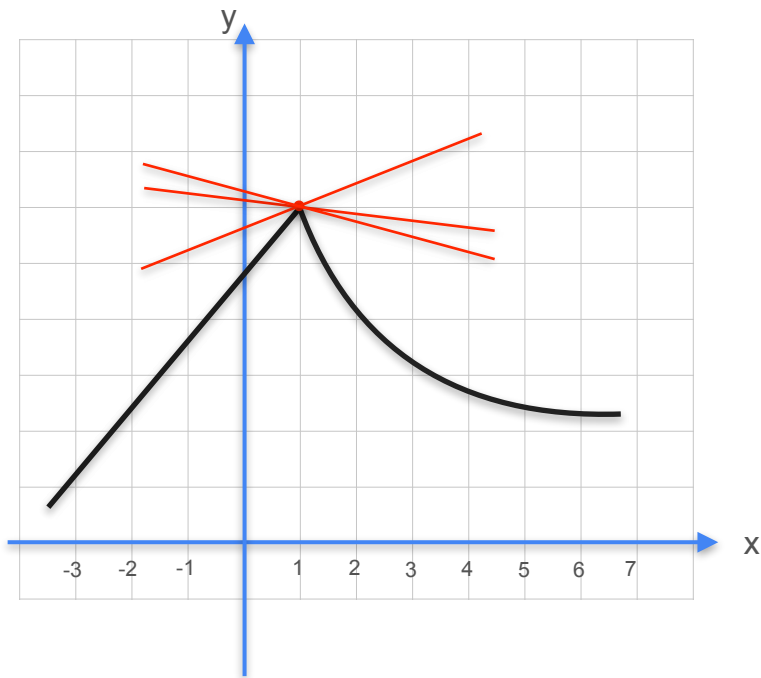
Non Differentiable Functions - Quiz 1



At which point in this function does the derivative not exist?

The entire function is non-differentiable because a derivative does not exist for all points in the domain.

Non Differentiable Functions - Quiz 1

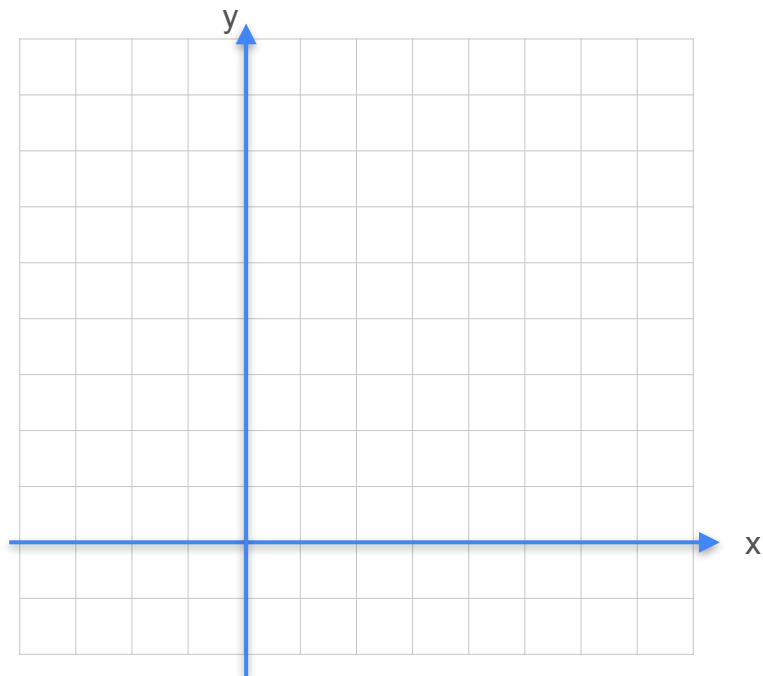


At which point in this function does the derivative not exist?

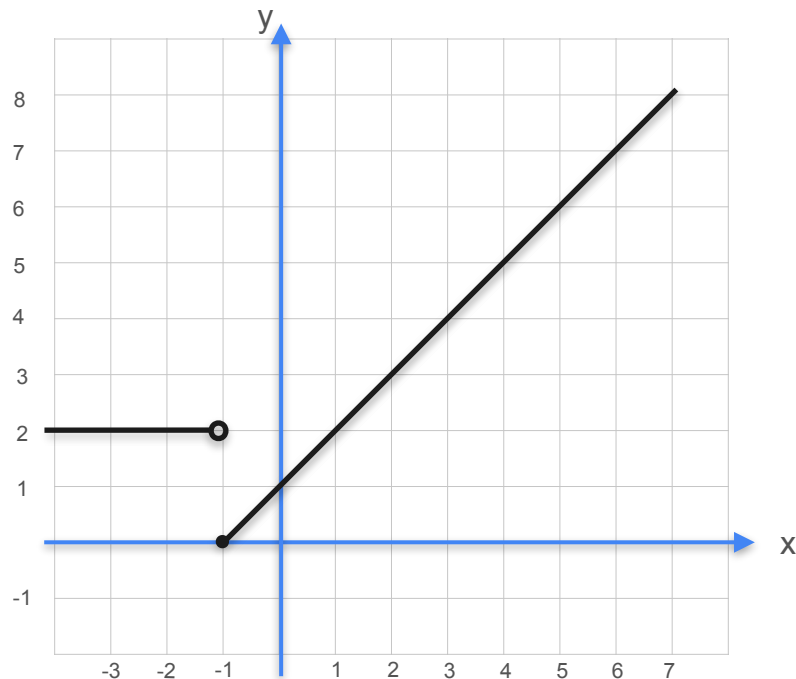
The entire function is non-differentiable because a derivative does not exist for all points in the domain.

Non Differentiable Functions - Quiz 2

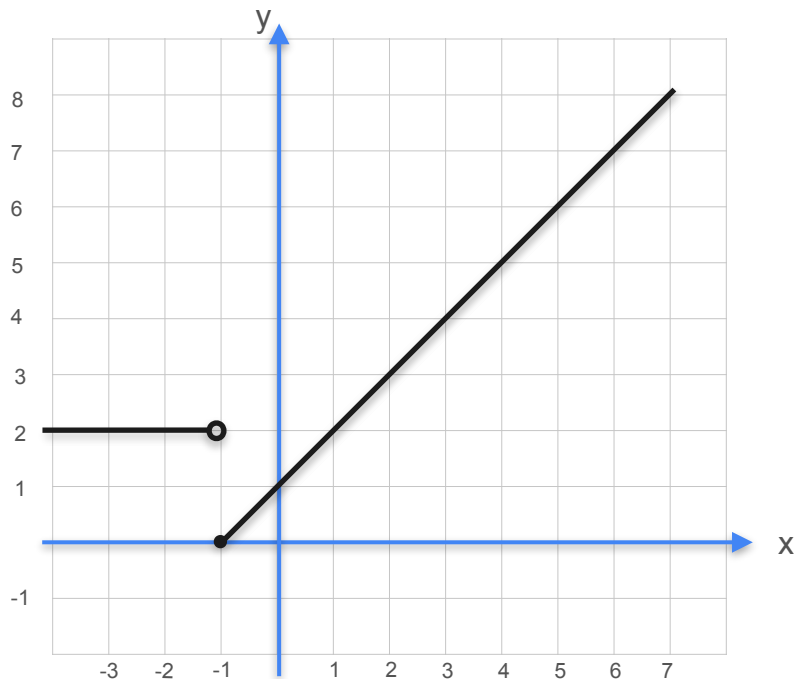
Non Differentiable Functions - Quiz 2



Non Differentiable Functions - Quiz 2

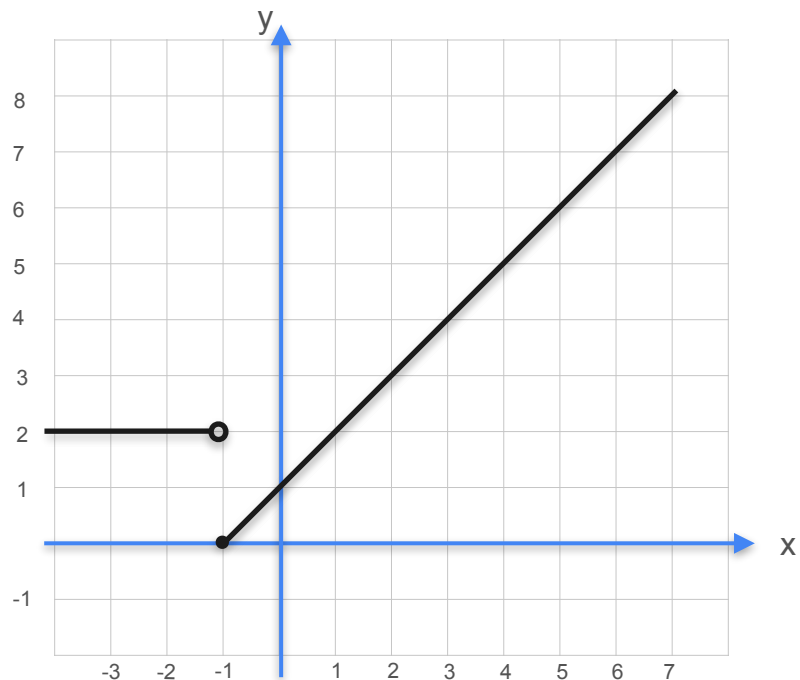


Non Differentiable Functions - Quiz 2

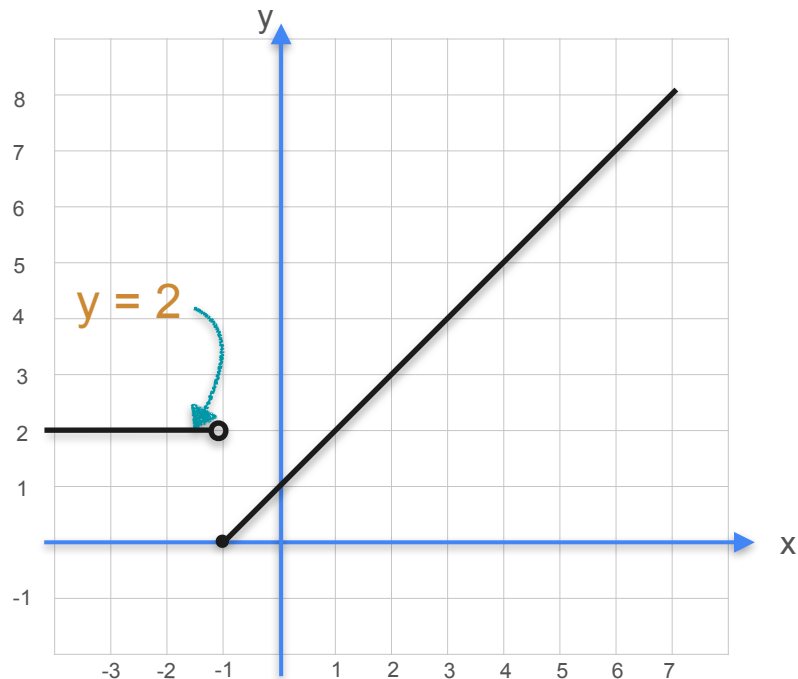


At which point in this function does the derivative not exist?

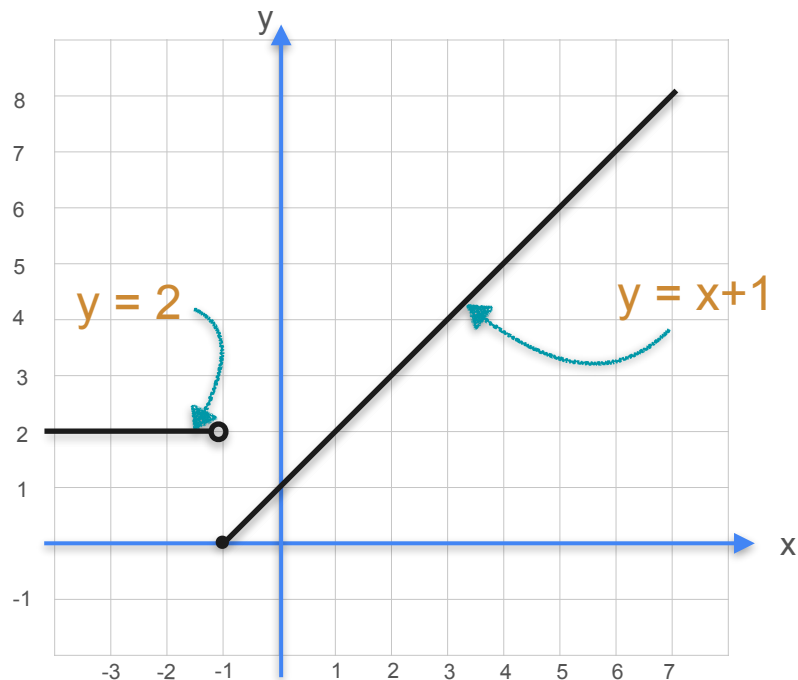
Non Differentiable Functions



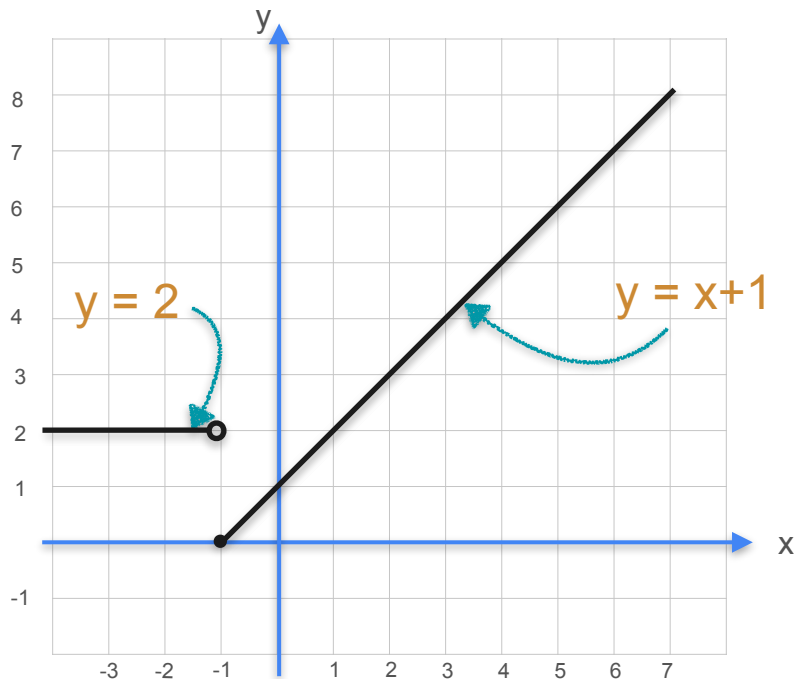
Non Differentiable Functions



Non Differentiable Functions



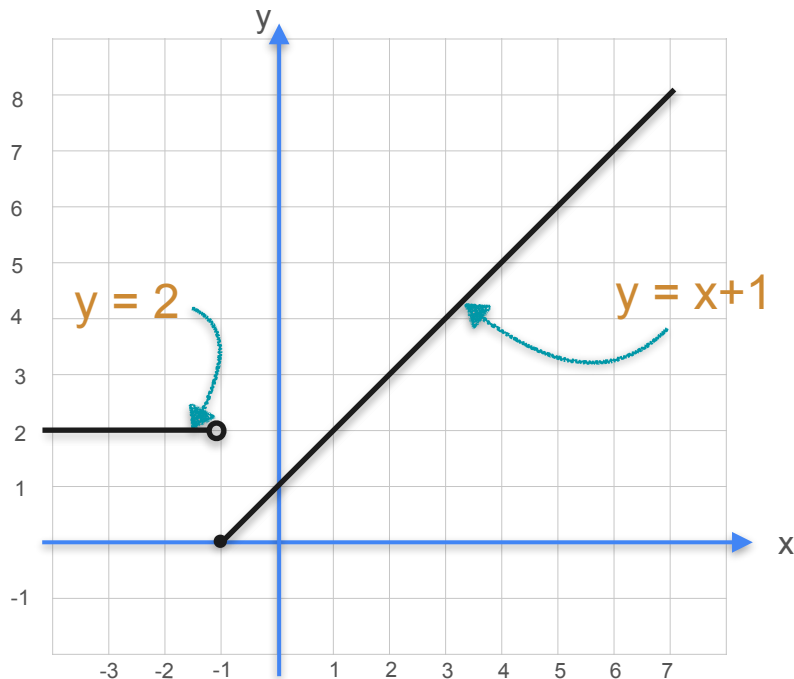
Non Differentiable Functions



This is a piece-wise function.

$$f(x) = \begin{cases} 2, & \text{if } x < -1 \\ x + 1, & \text{if } x \geq -1 \end{cases}$$

Non Differentiable Functions

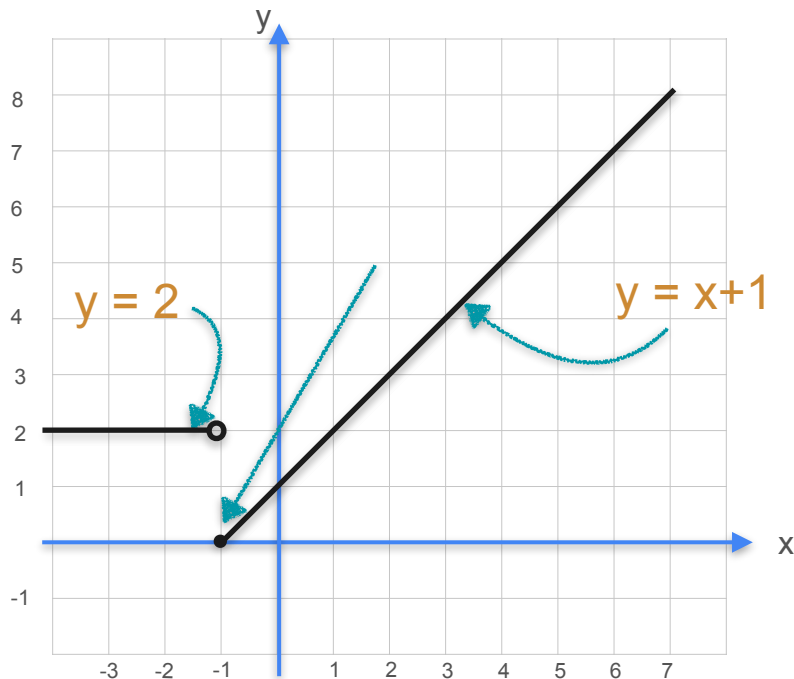


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Jump Discontinuity

Non Differentiable Functions

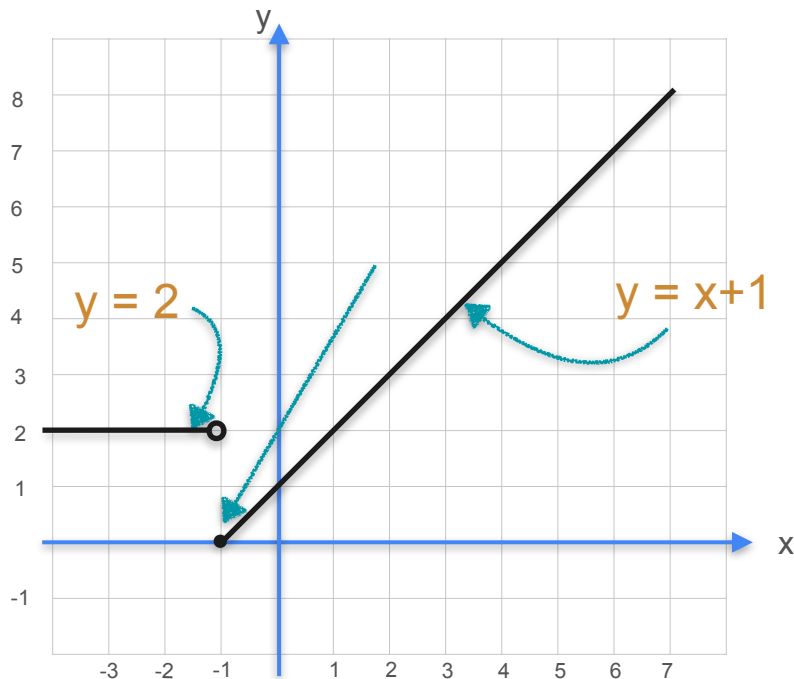


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Non Differentiable Functions



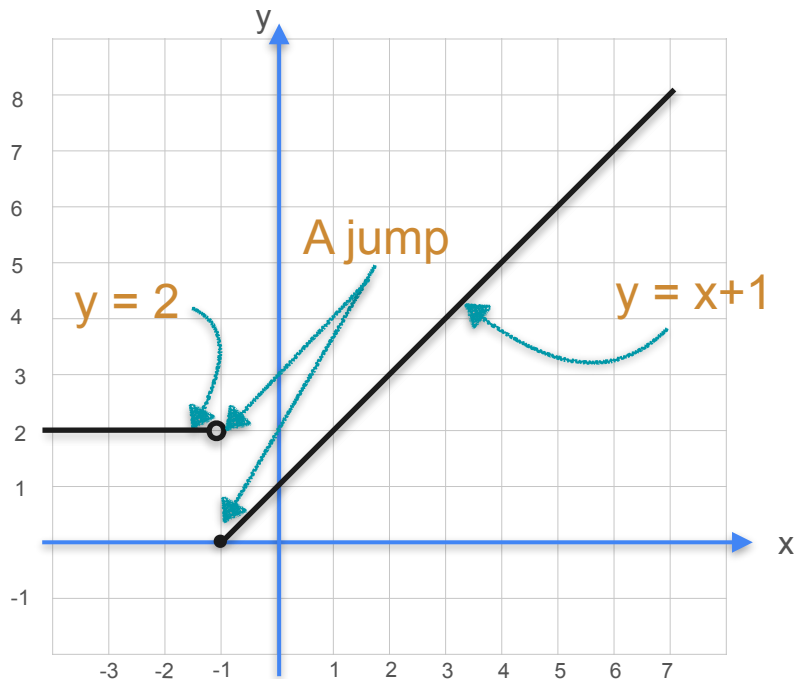
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Jump Discontinuity

The graph of the function does not appear to be continuous

Non Differentiable Functions



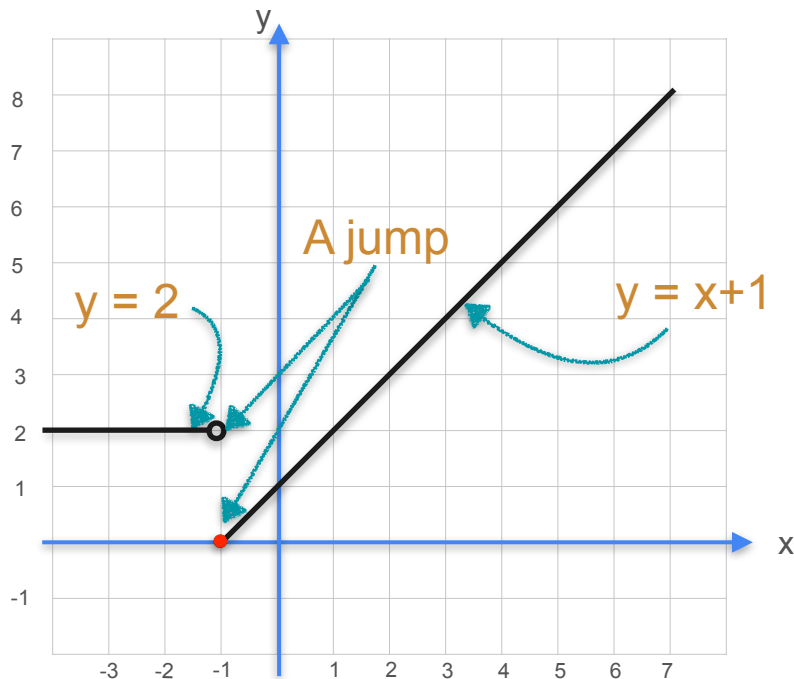
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Non Differentiable Functions



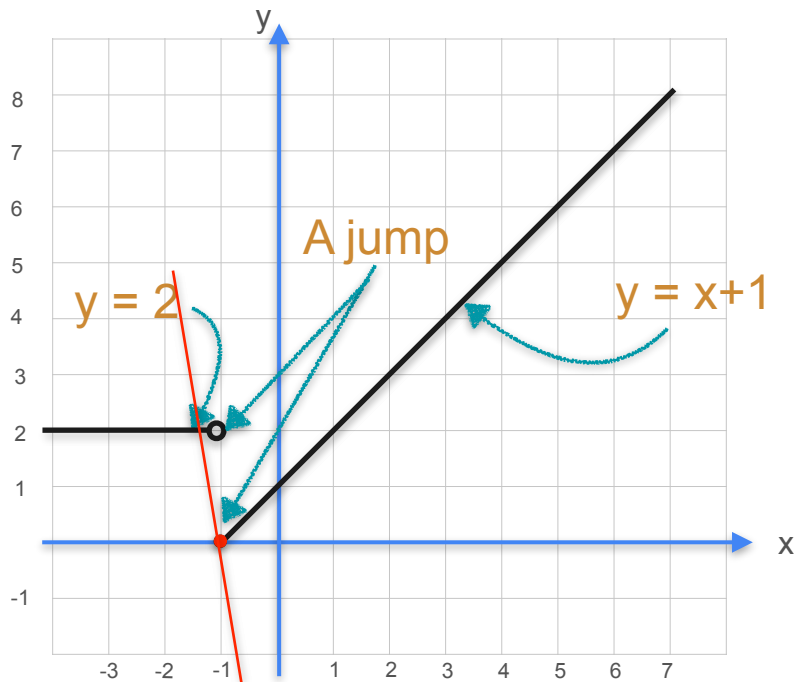
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Non Differentiable Functions



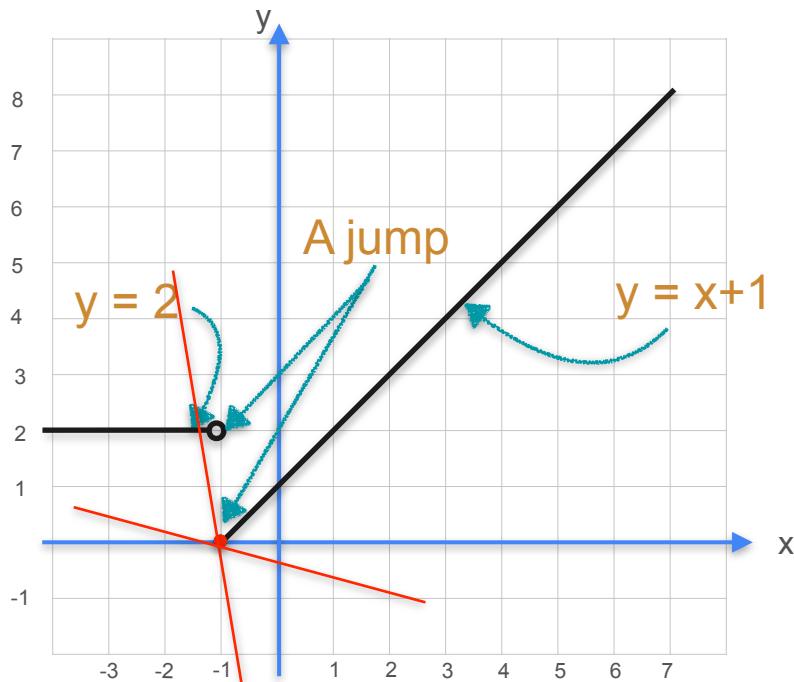
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Non Differentiable Functions



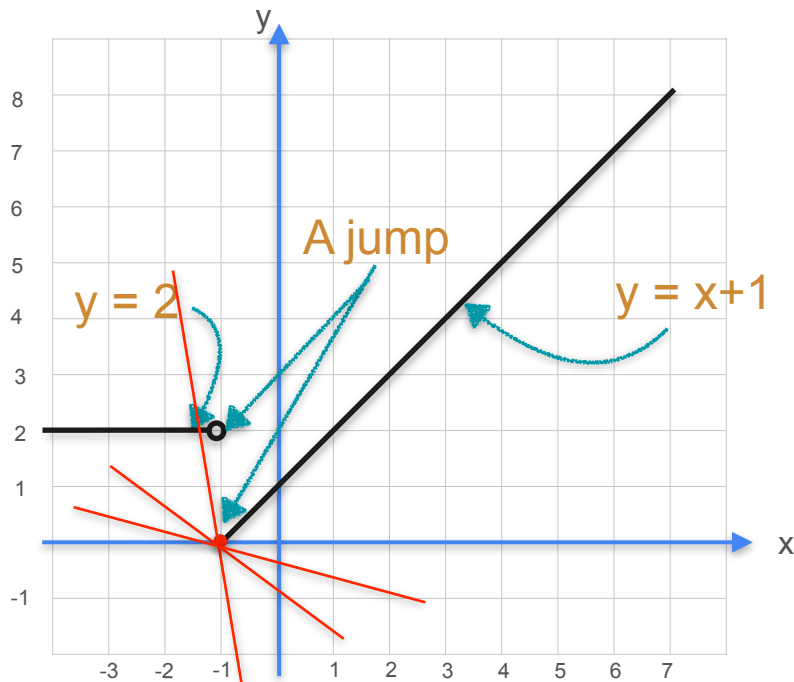
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Non Differentiable Functions



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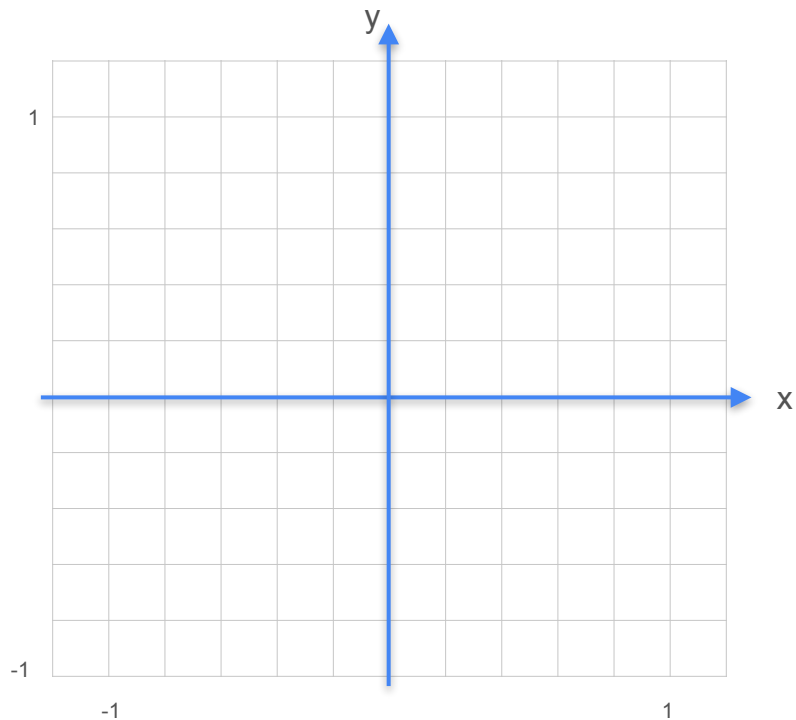
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Jump Discontinuity

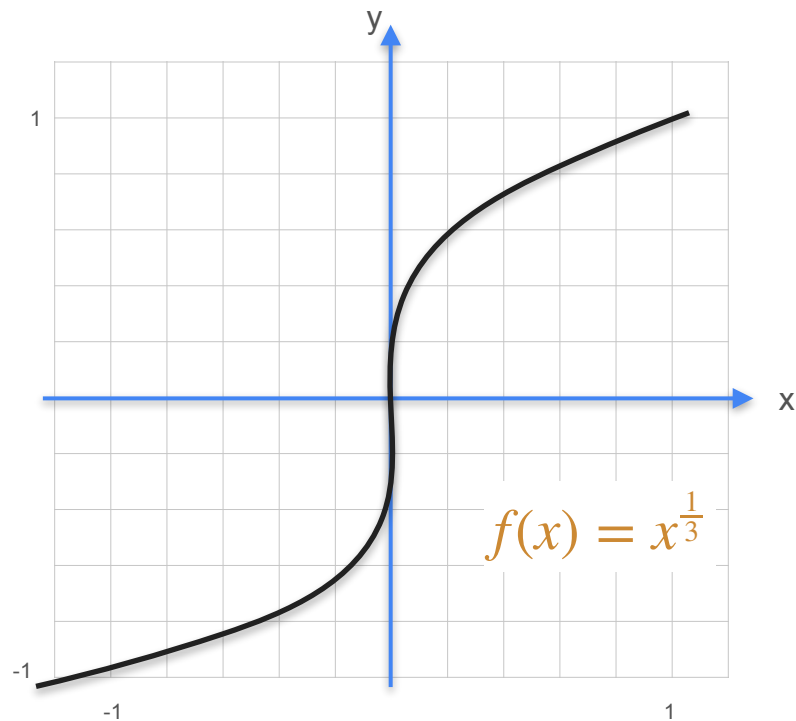
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Non Differentiable Functions

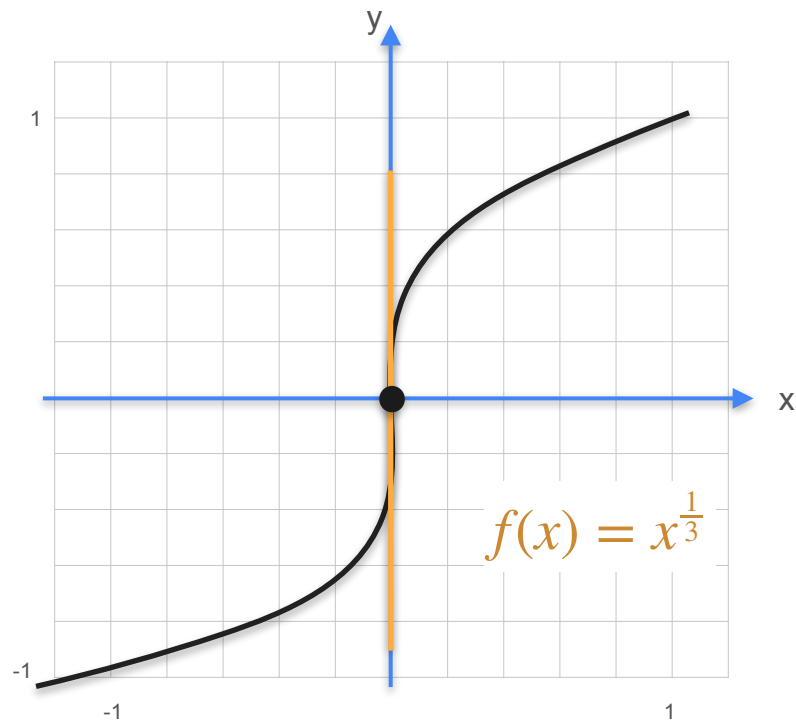
Non Differentiable Functions



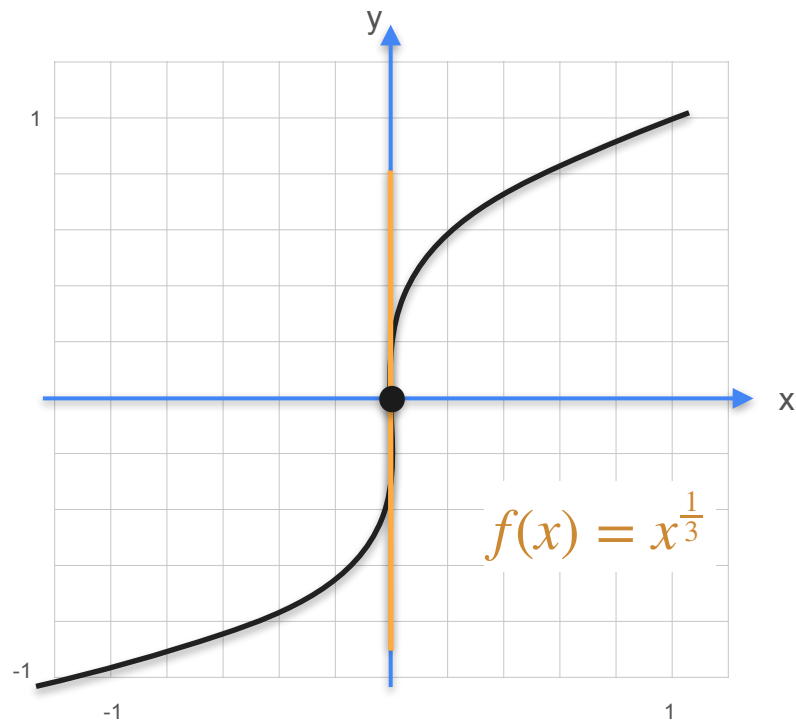
Non Differentiable Functions



Non Differentiable Functions

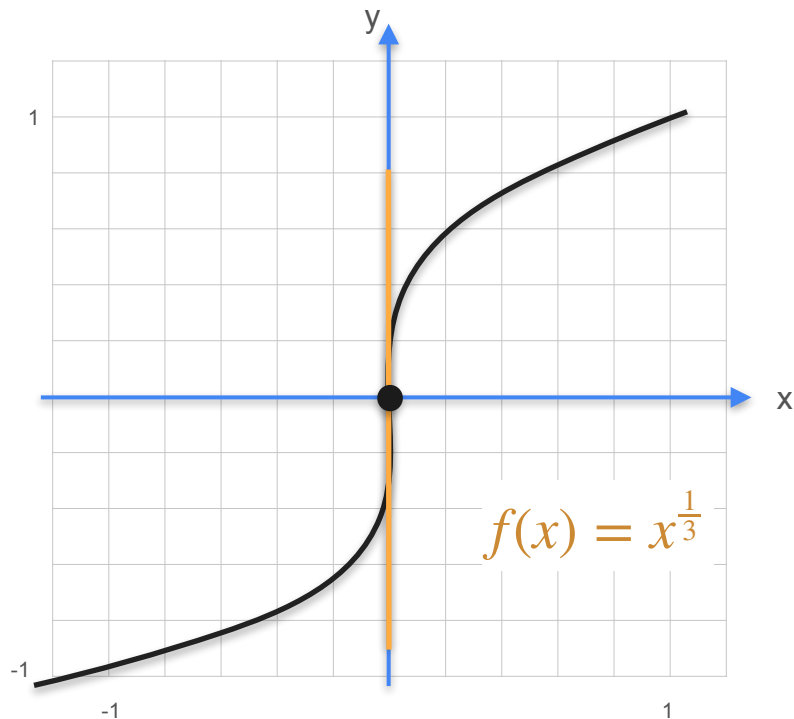


Non Differentiable Functions



Vertical tangents

Non Differentiable Functions

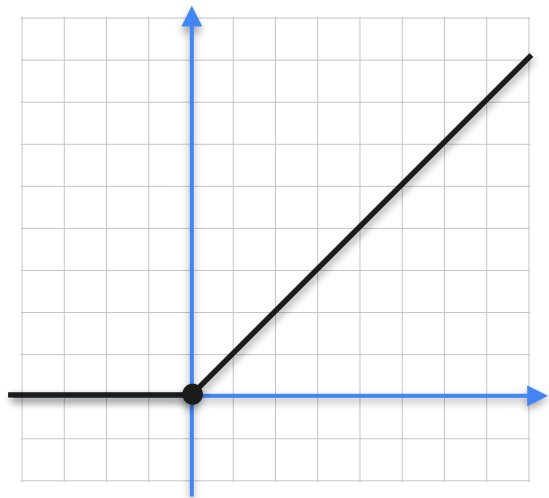


Vertical tangents

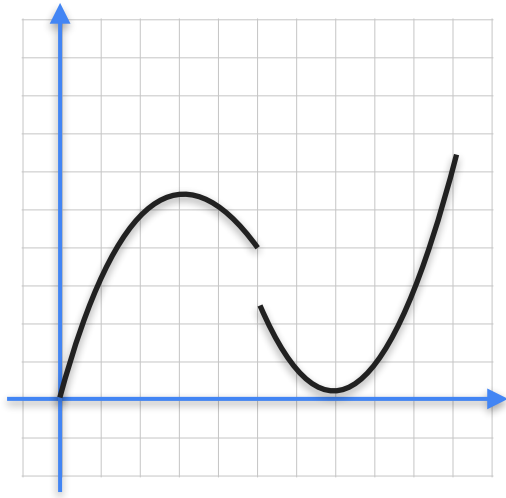
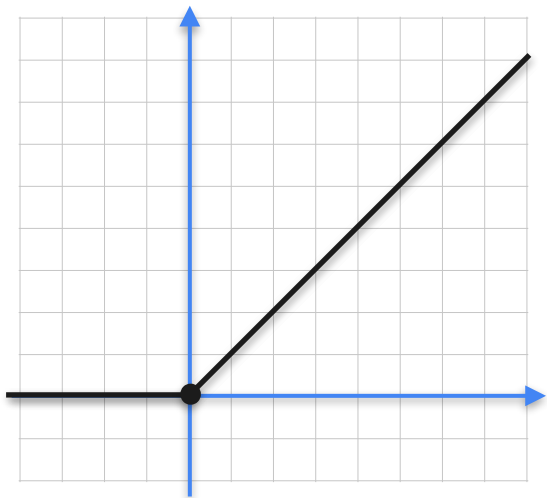
At $x = 0$, this graph has a tangent line that runs straight up parallel to the y-axis

Recap: Non-Differentiable Functions

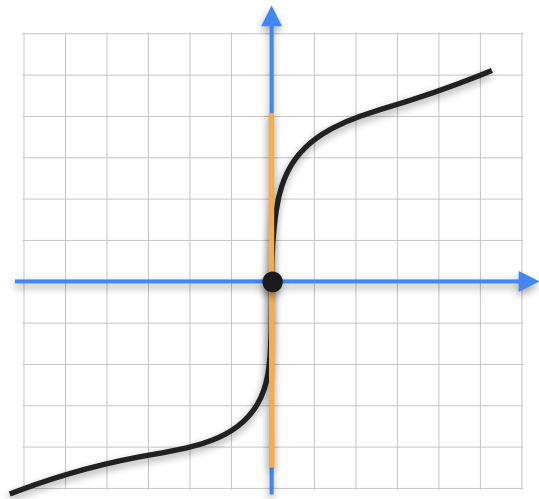
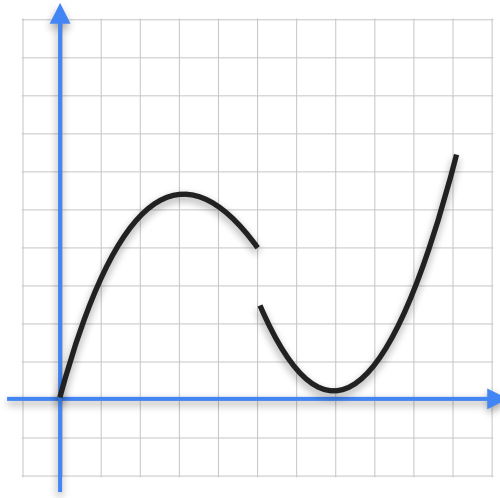
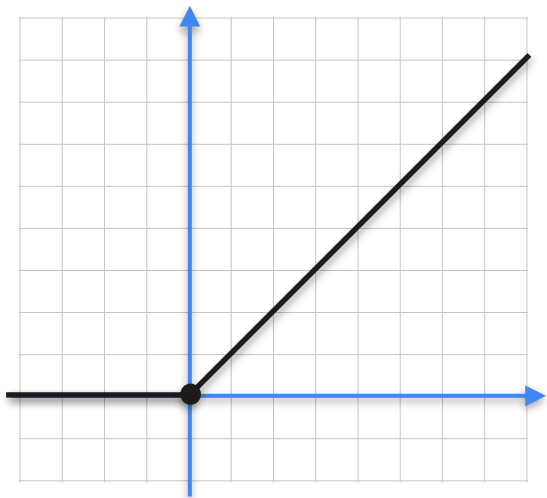
Recap: Non-Differentiable Functions



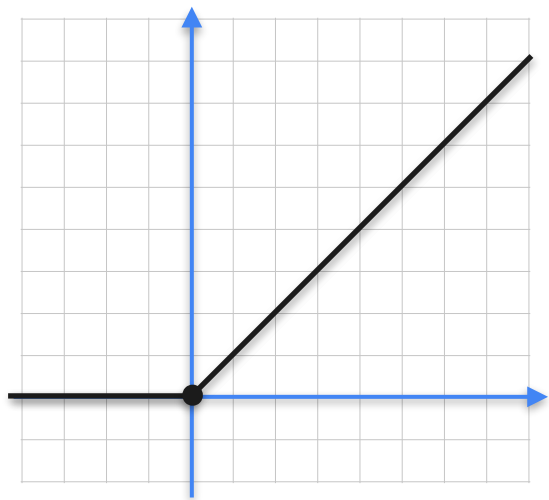
Recap: Non-Differentiable Functions



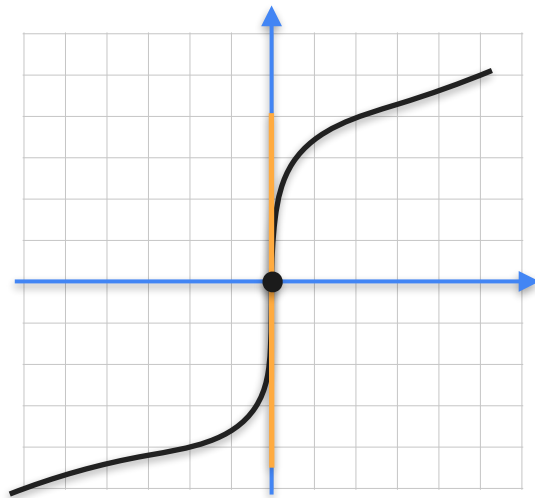
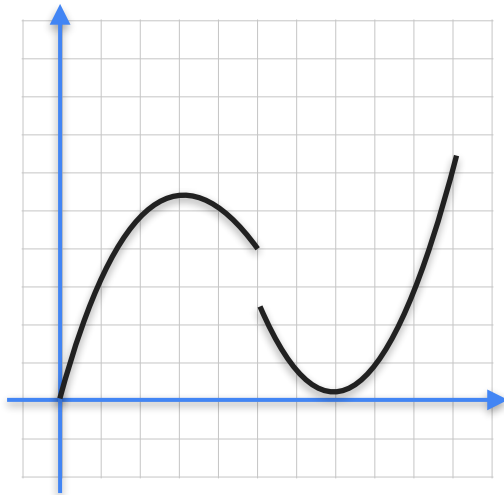
Recap: Non-Differentiable Functions



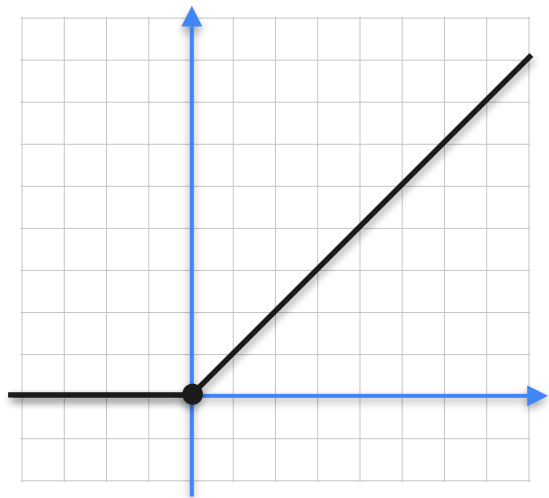
Recap: Non-Differentiable Functions



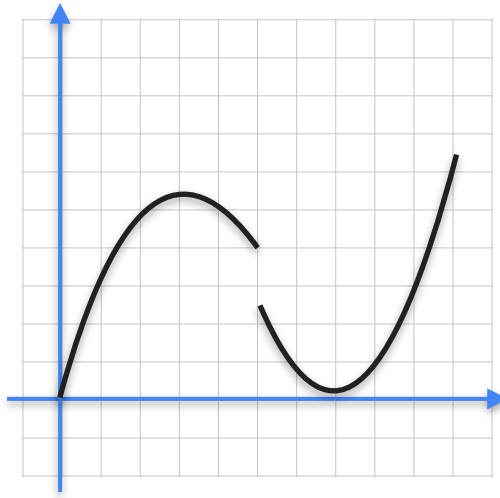
Corners/Cusps



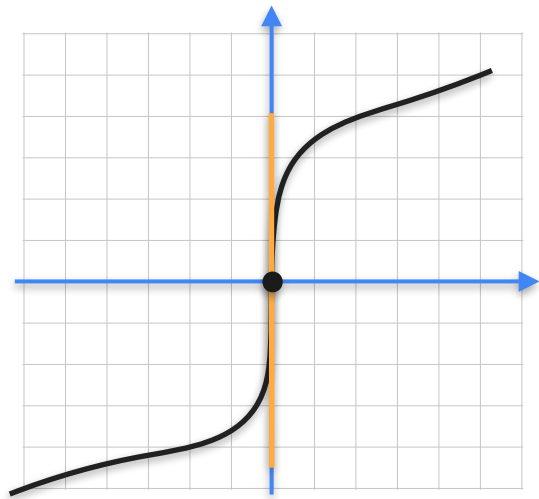
Recap: Non-Differentiable Functions



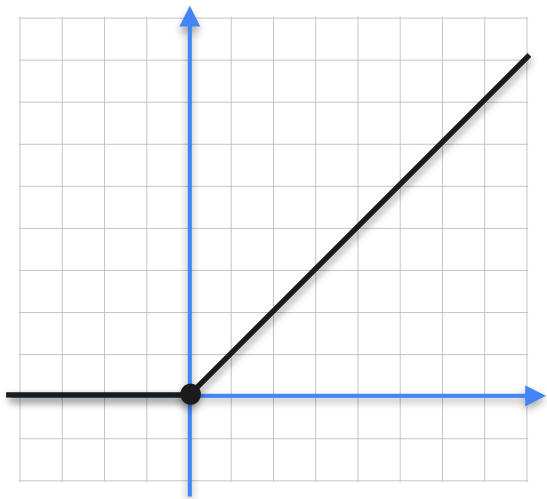
Corners/Cusps



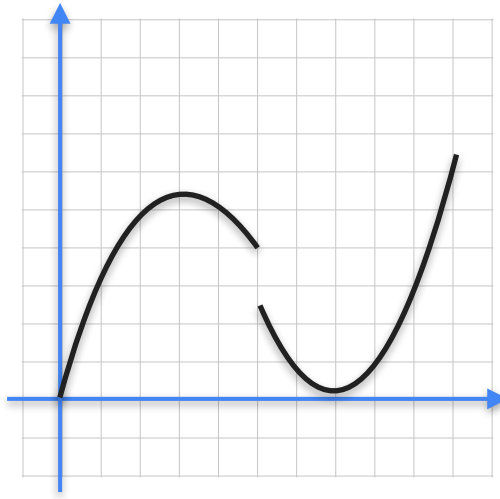
Jump Discontinuity



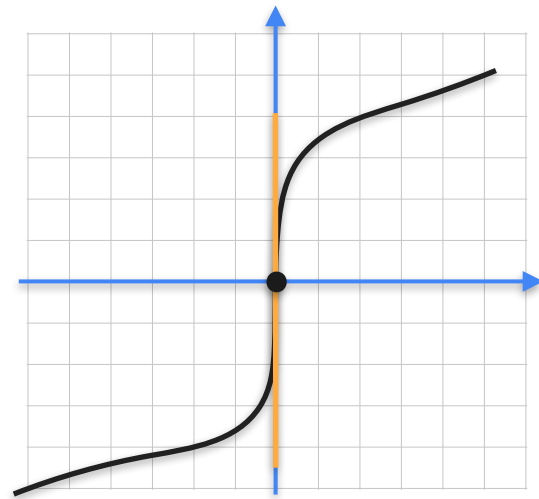
Recap: Non-Differentiable Functions



Corners/Cusps



Jump Discontinuity



Vertical tangents



DeepLearning.AI

Derivatives and Optimization

Properties of the derivative: Multiplication by scalars

The Sum Rule

$$f = 4g$$

The Sum Rule

$$f' = 4g$$

The Sum Rule

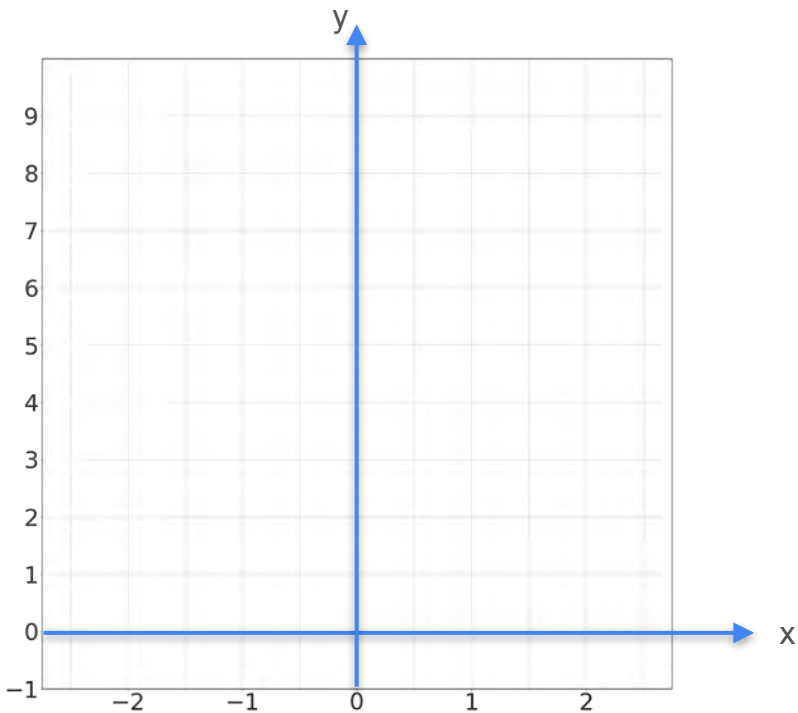
$$f' = 4g'$$

The Sum Rule

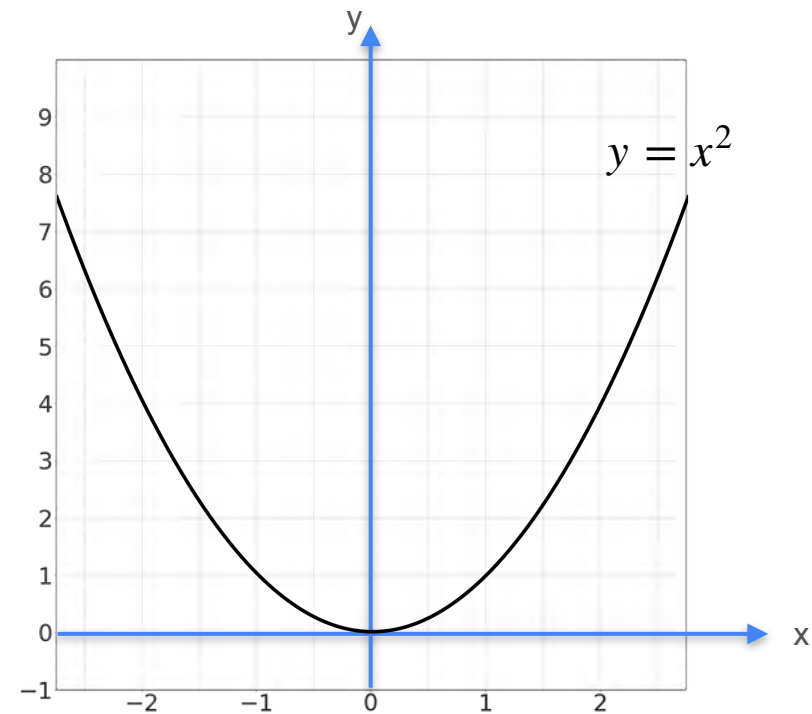
$$f' = c g'$$

Multiplication by a Scalar

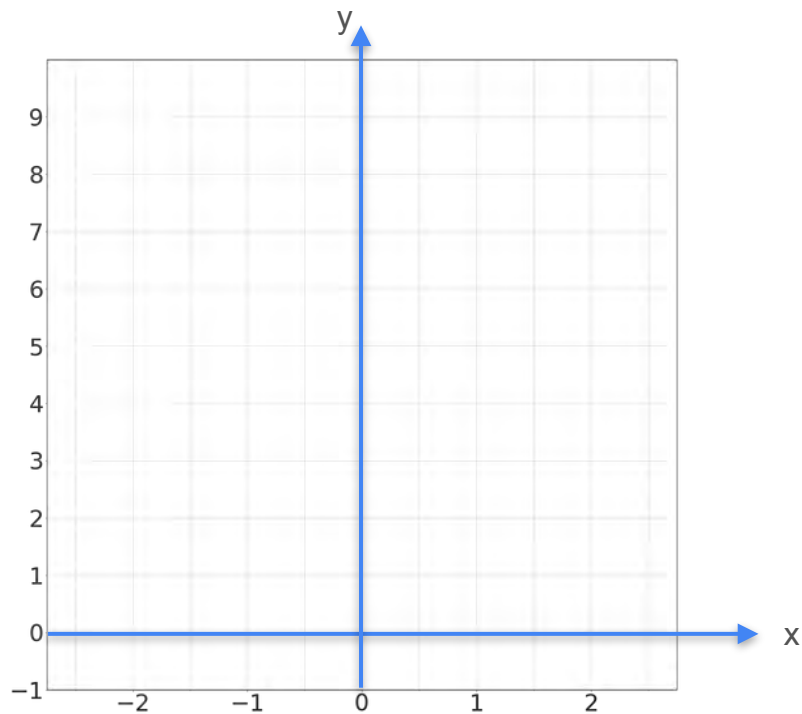
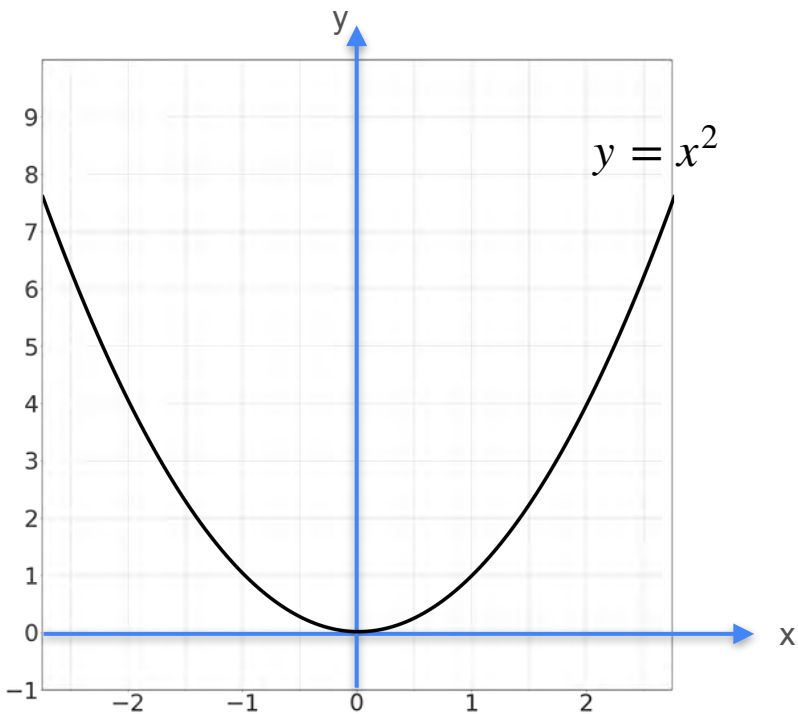
Multiplication by a Scalar



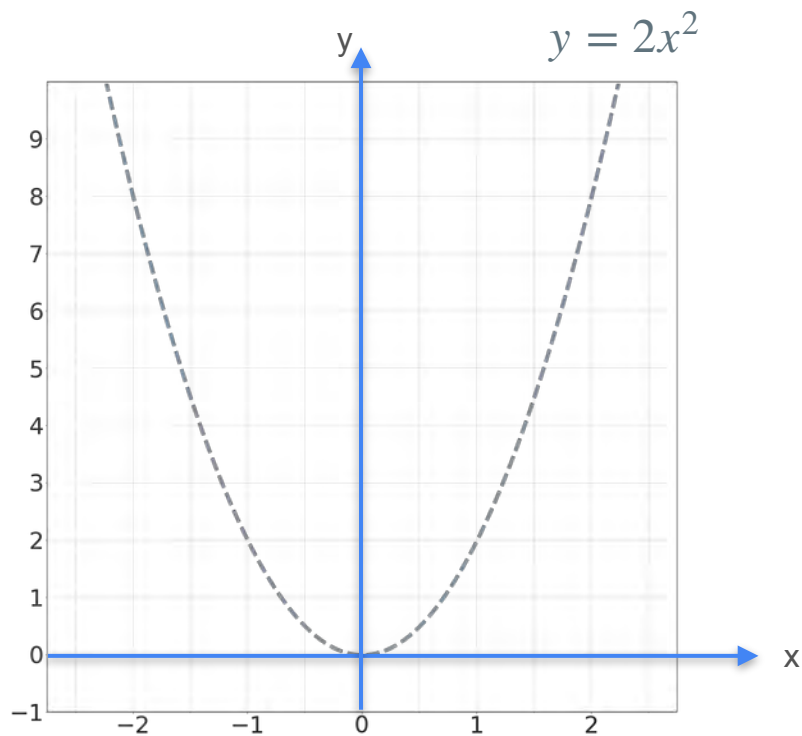
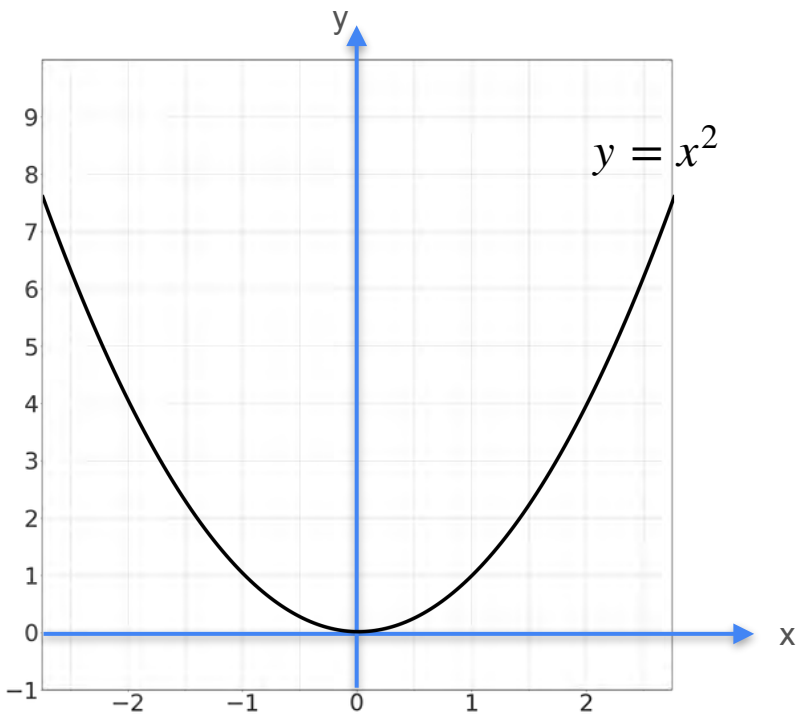
Multiplication by a Scalar



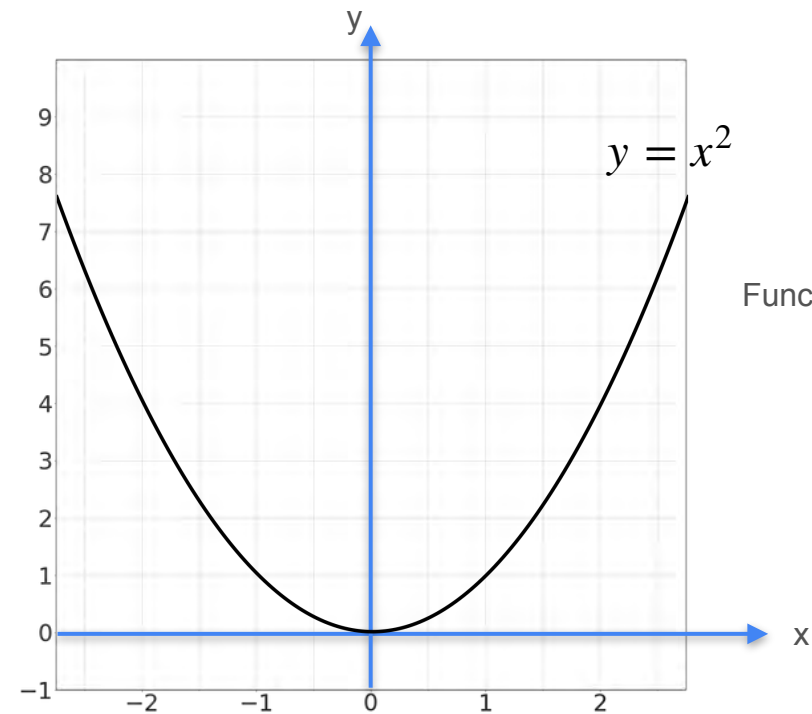
Multiplication by a Scalar



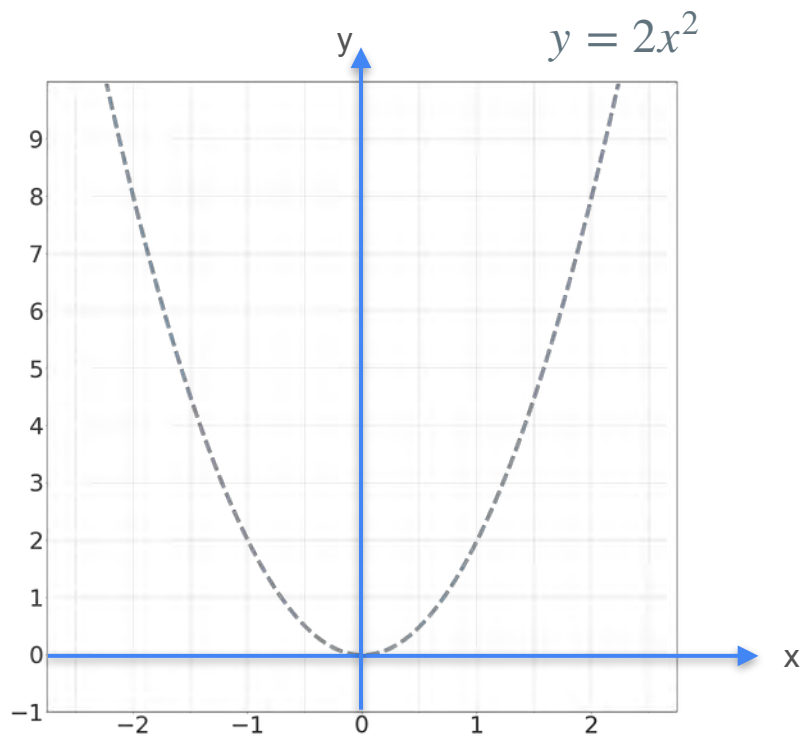
Multiplication by a Scalar



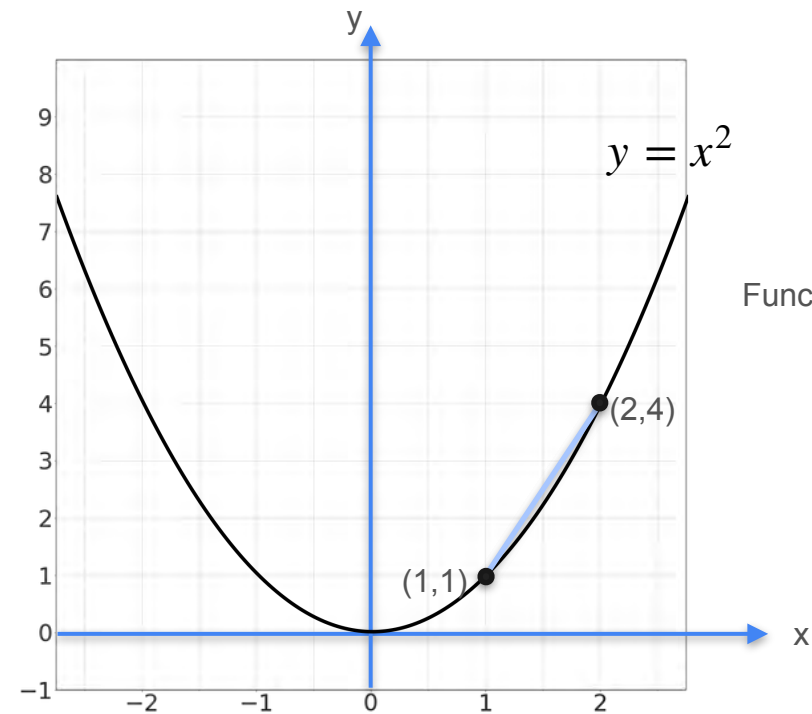
Multiplication by a Scalar



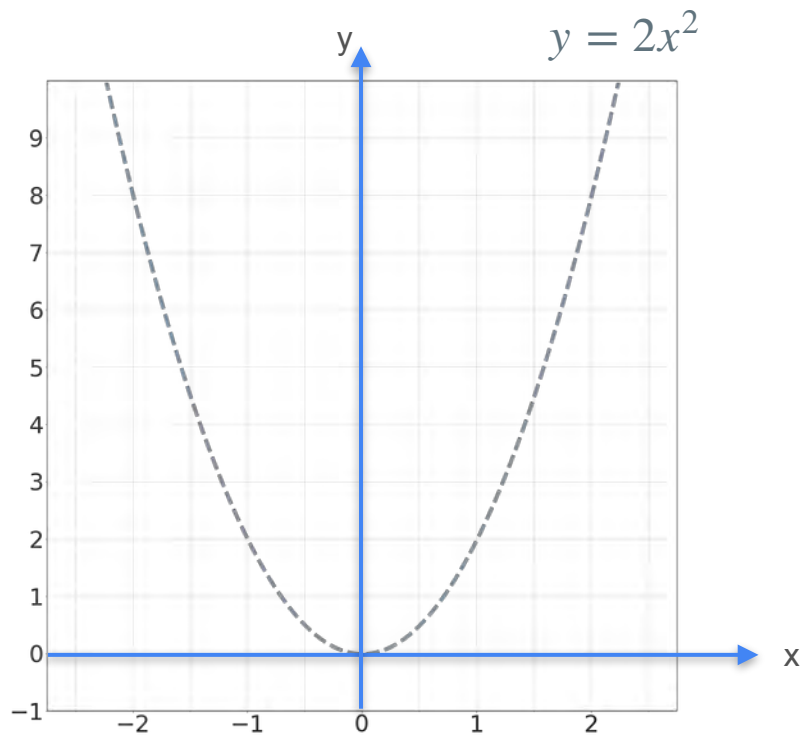
Function multiplies by 2



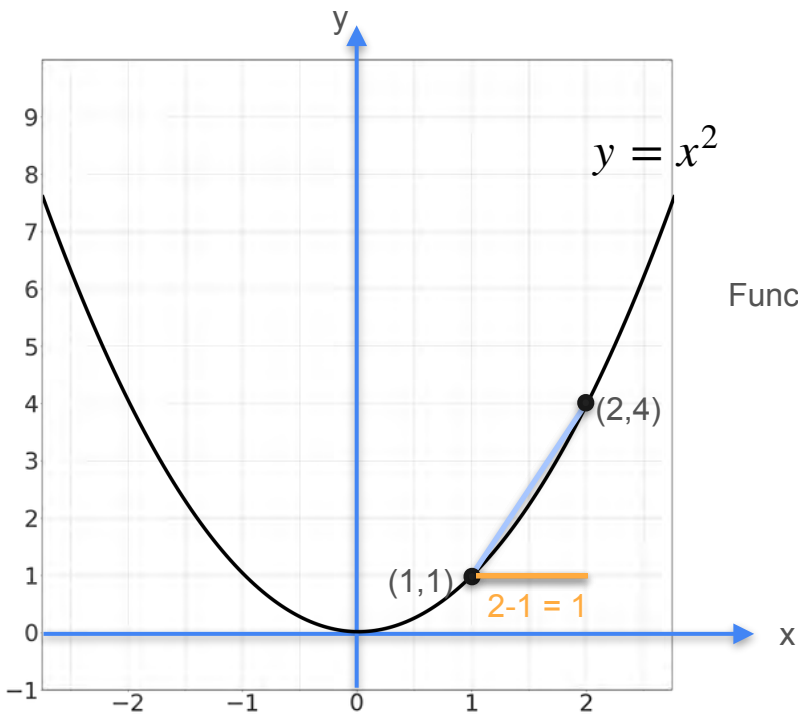
Multiplication by a Scalar



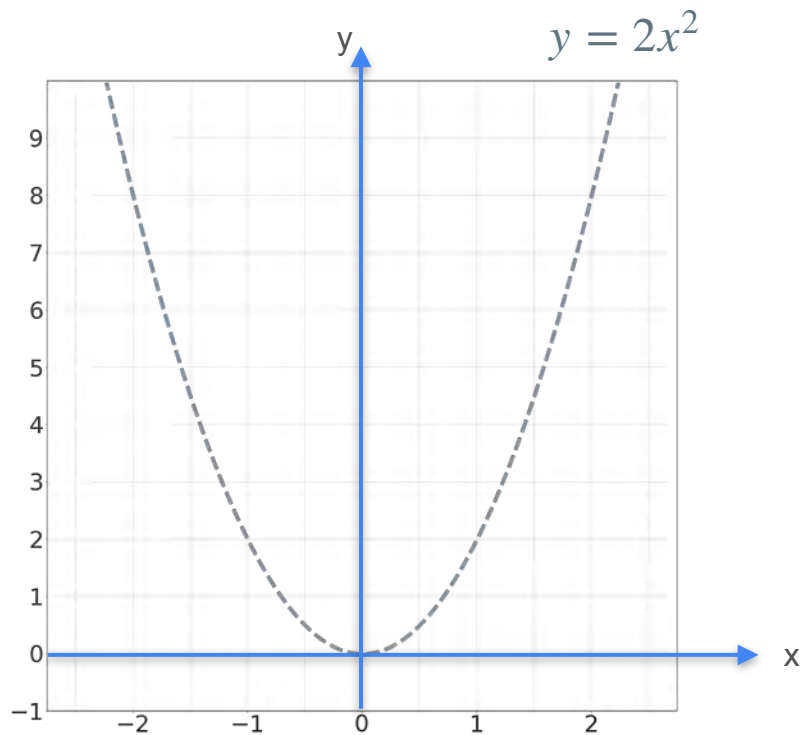
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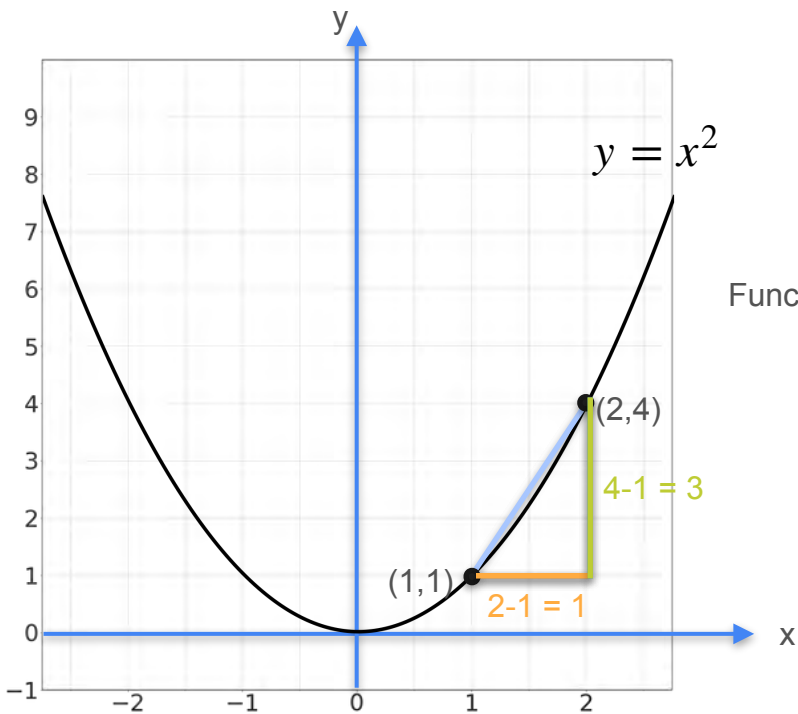
Multiplication by a Scalar



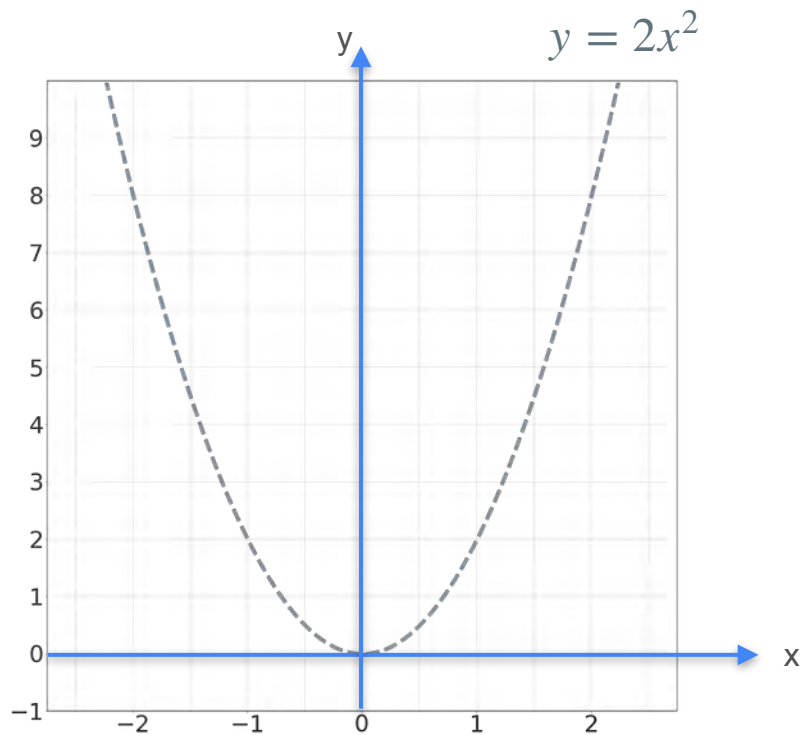
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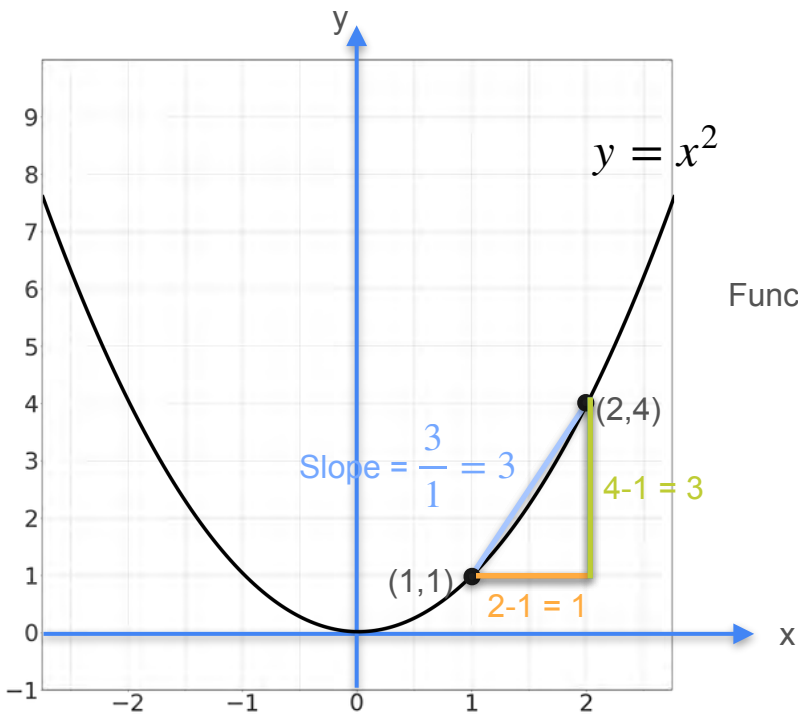
Multiplication by a Scalar



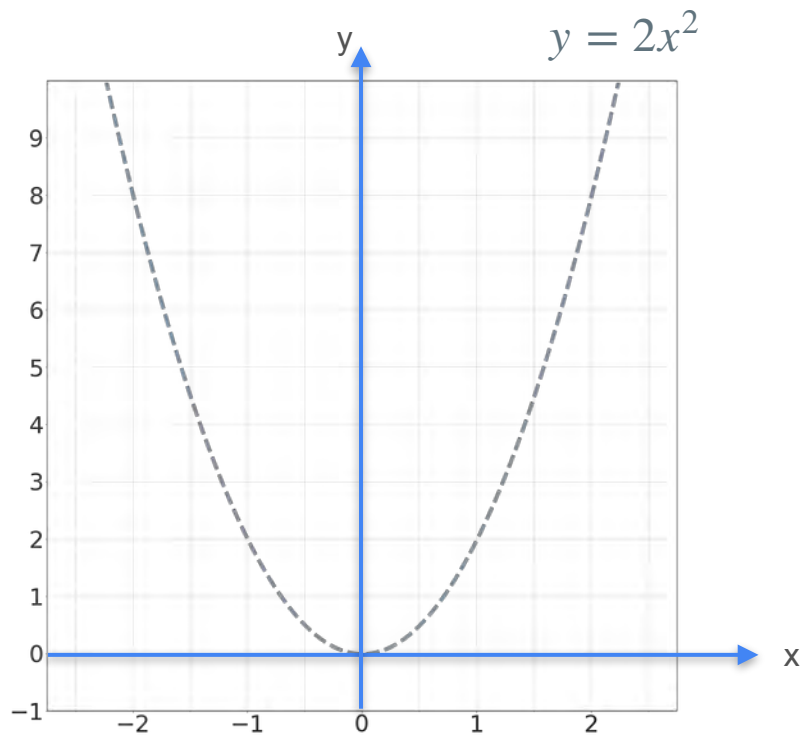
Function multiplies by 2



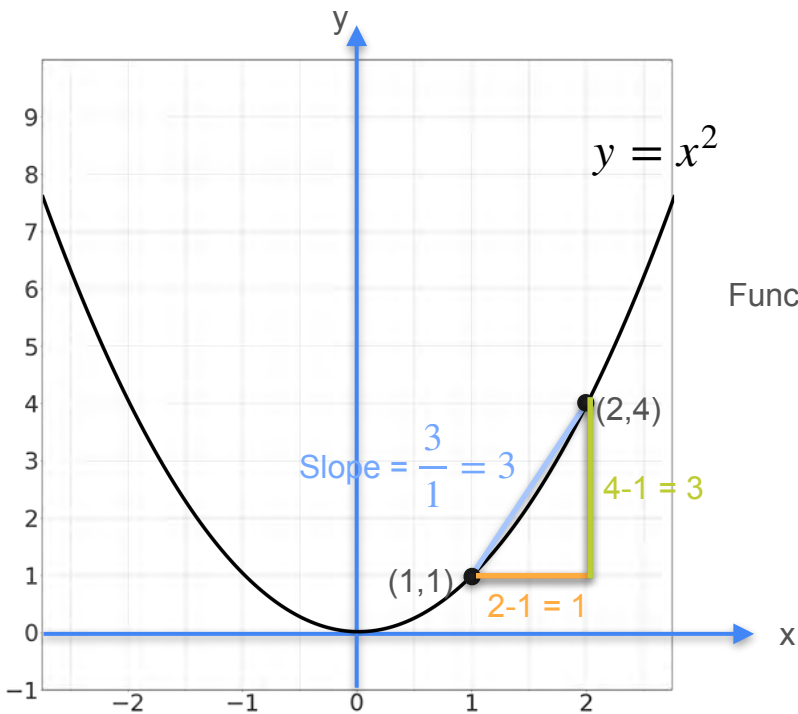
Multiplication by a Scalar



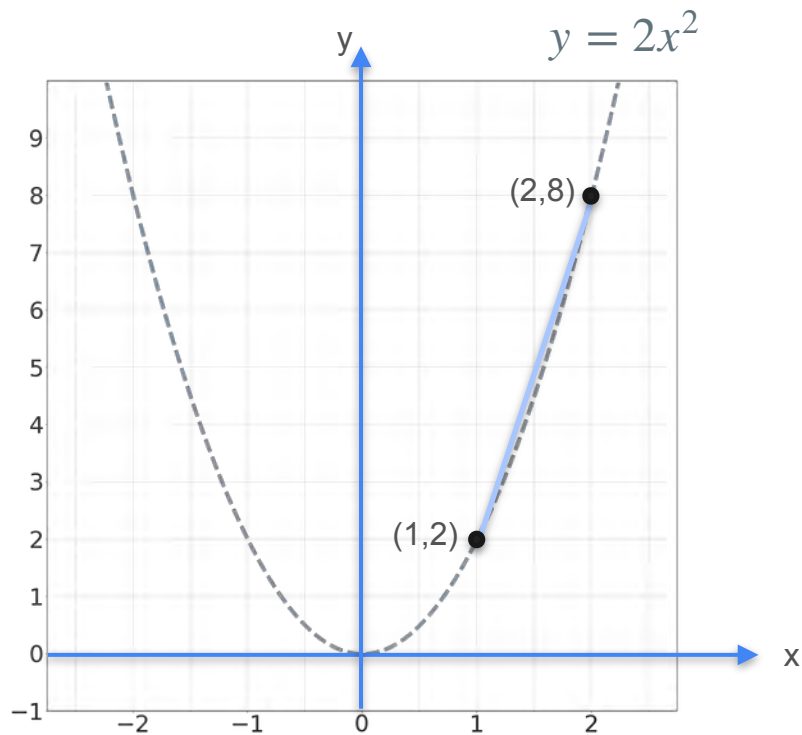
Function multiplies by 2



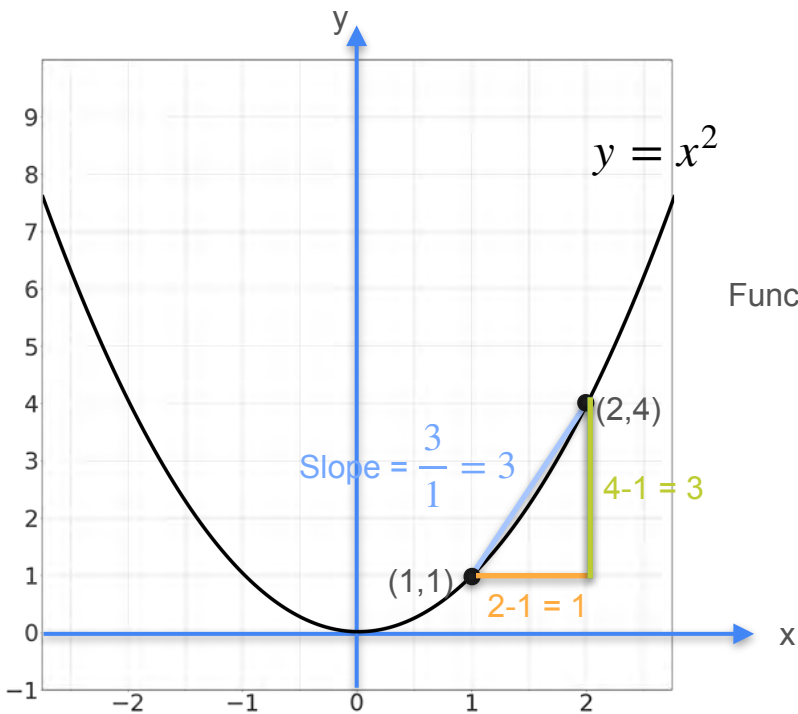
Multiplication by a Scalar



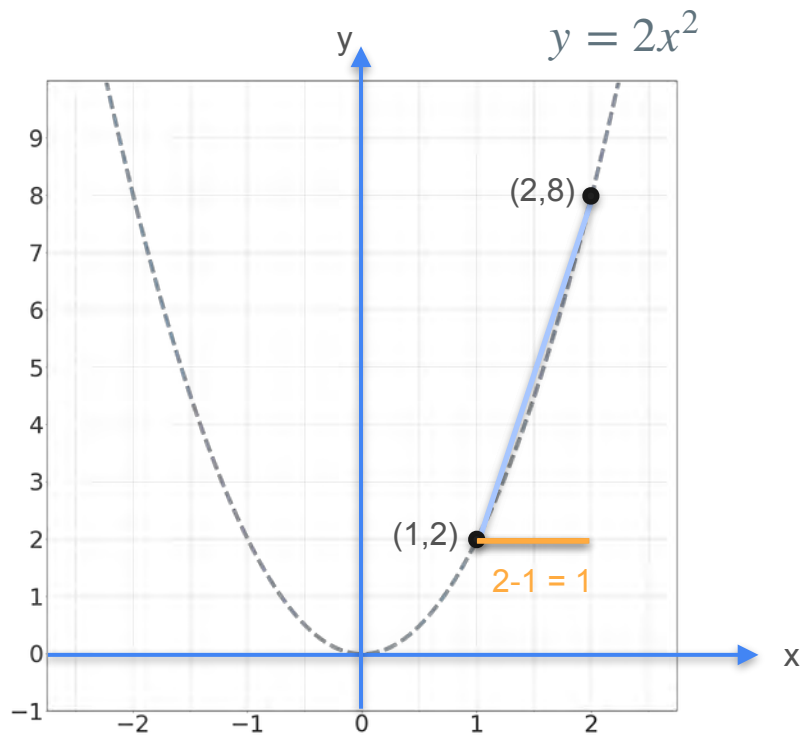
Function multiplies by 2



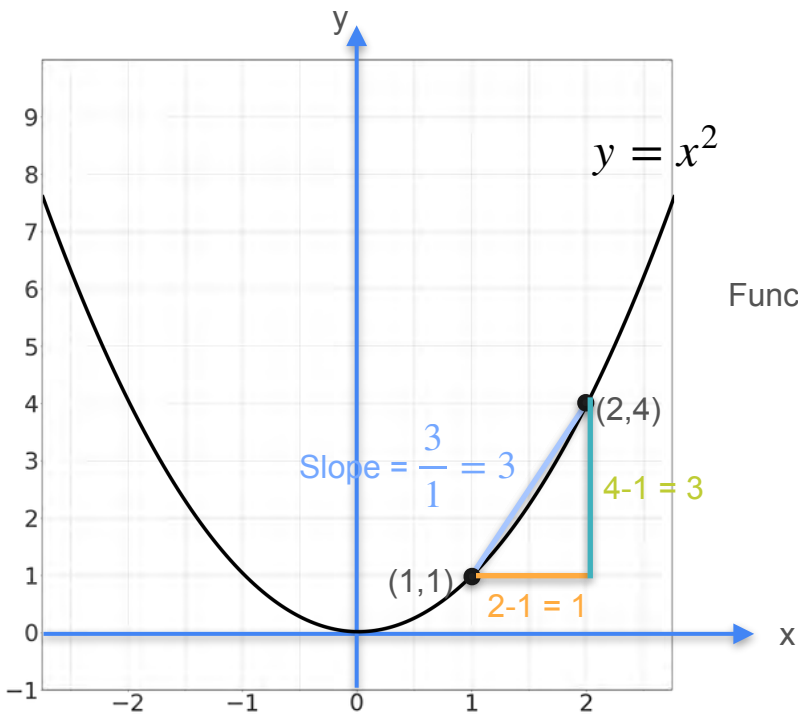
Multiplication by a Scalar



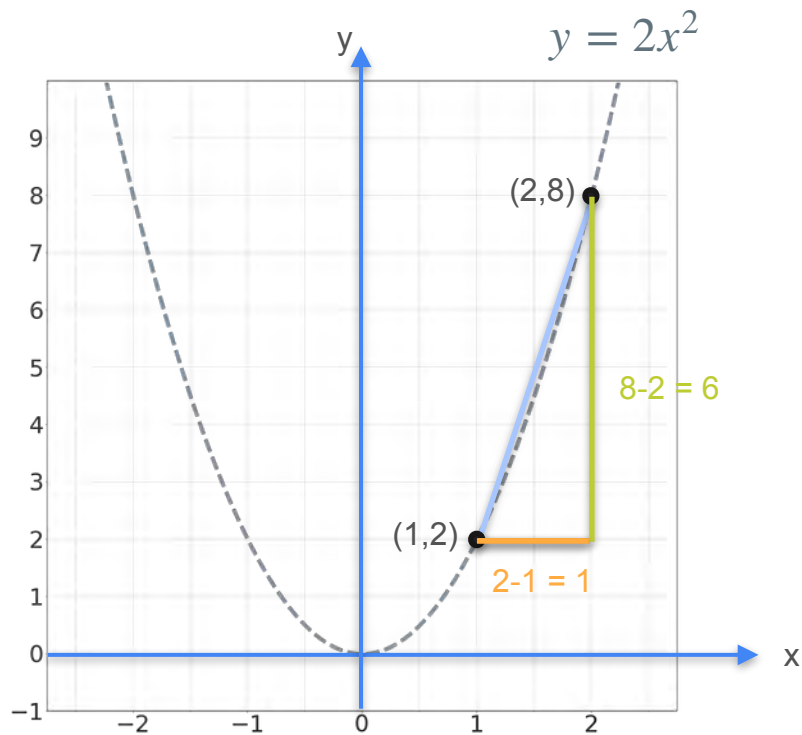
Function multiplies by 2



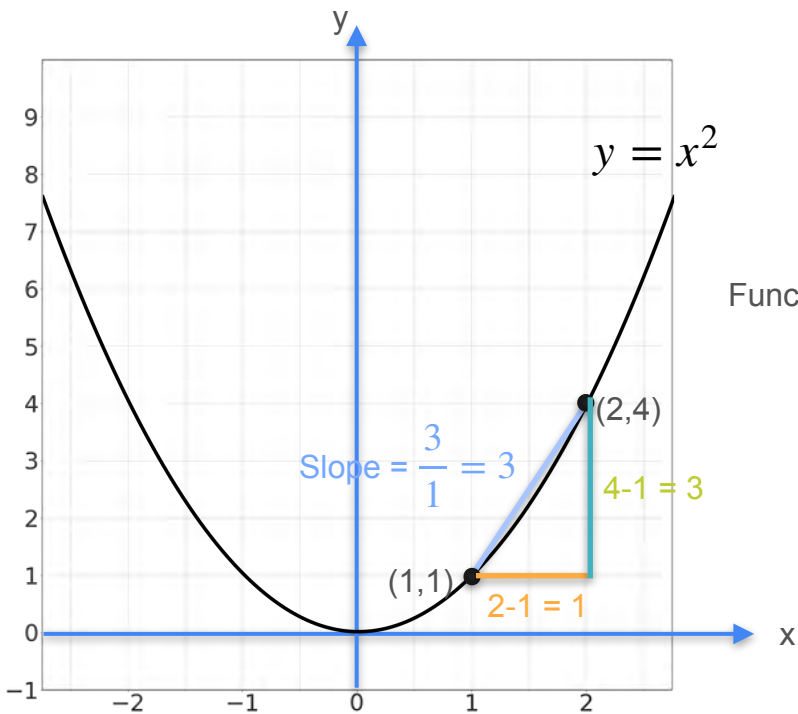
Multiplication by a Scalar



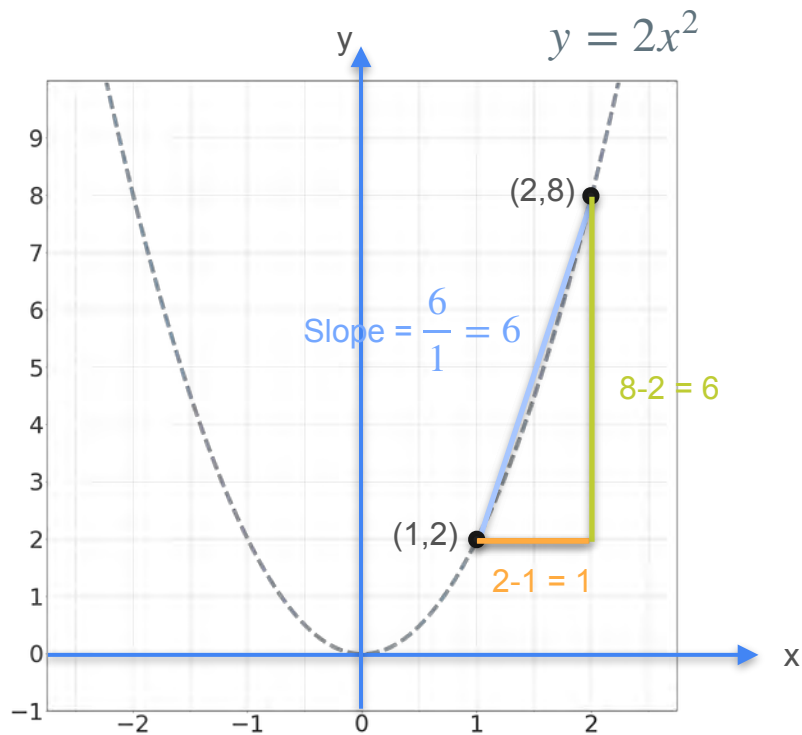
Function multiplies by 2



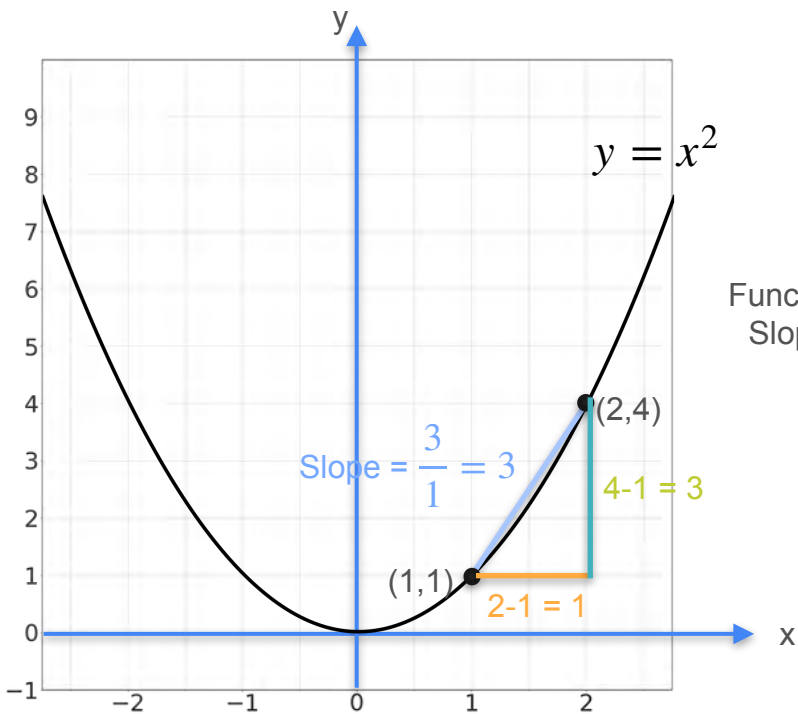
Multiplication by a Scalar



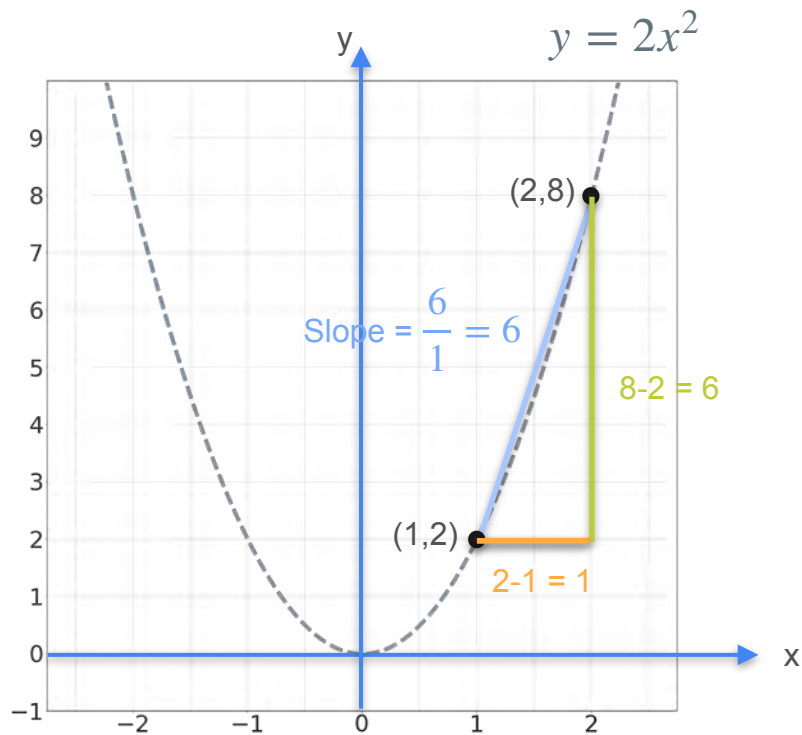
Function multiplies by 2



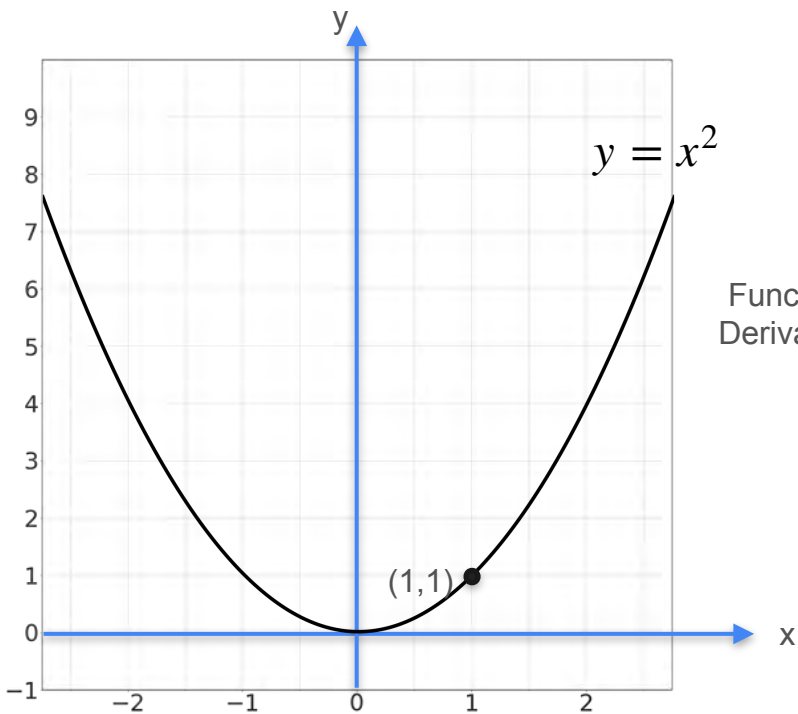
Multiplication by a Scalar



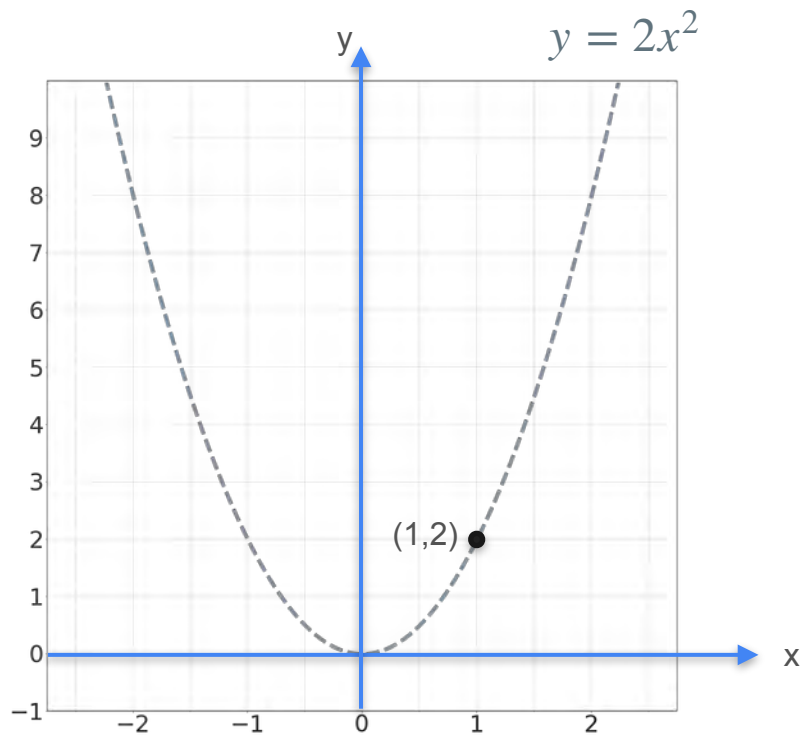
Function multiplies by 2
Slope multiplies by 2



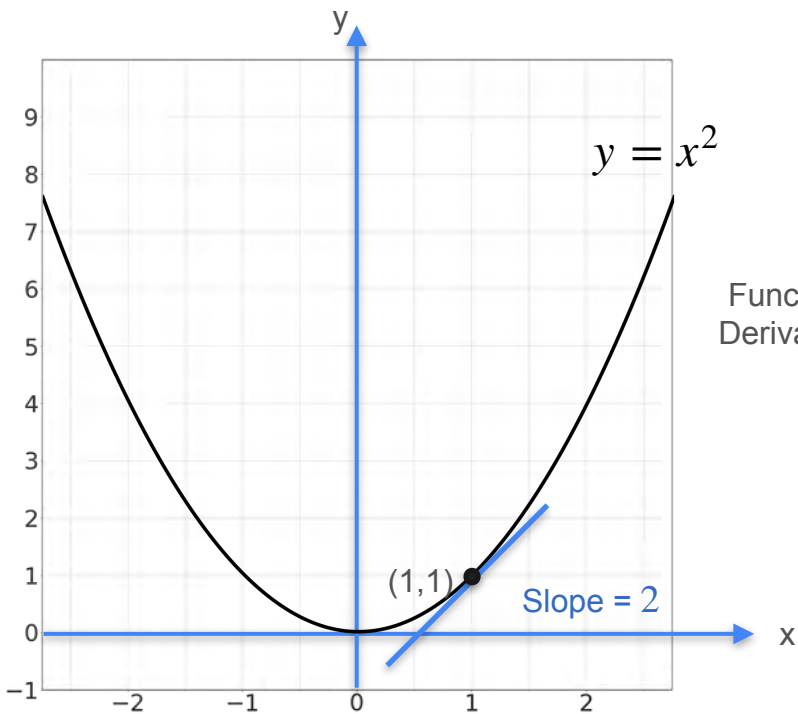
Multiplication by a Scalar



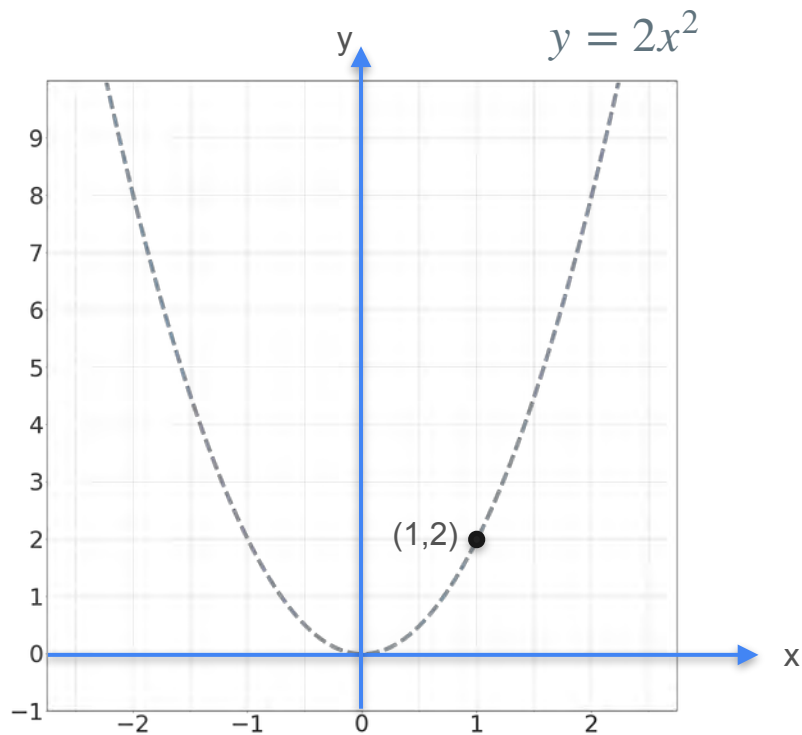
Function multiplies by 2
Derivative multiplies by 2



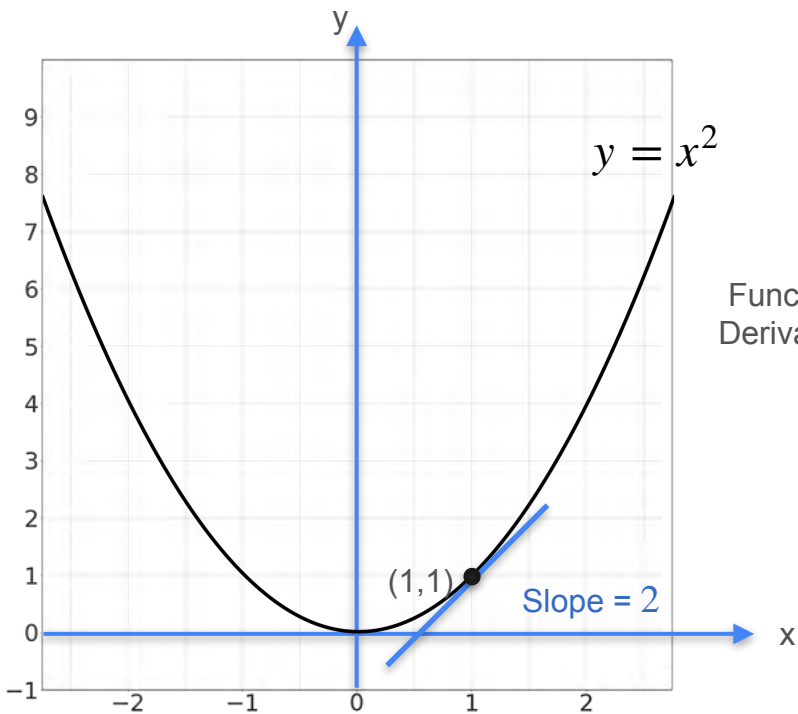
Multiplication by a Scalar



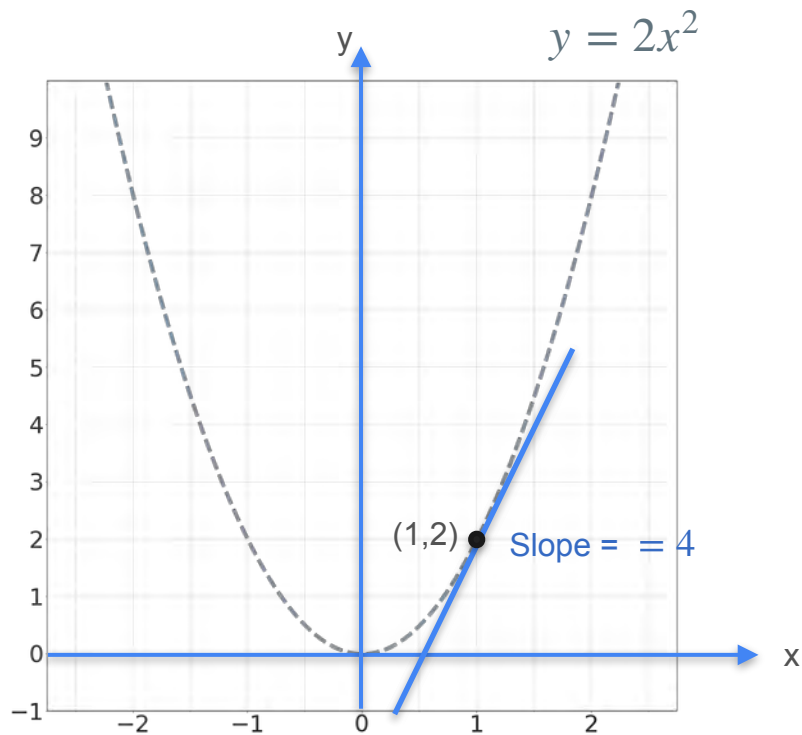
Function multiplies by 2
Derivative multiplies by 2



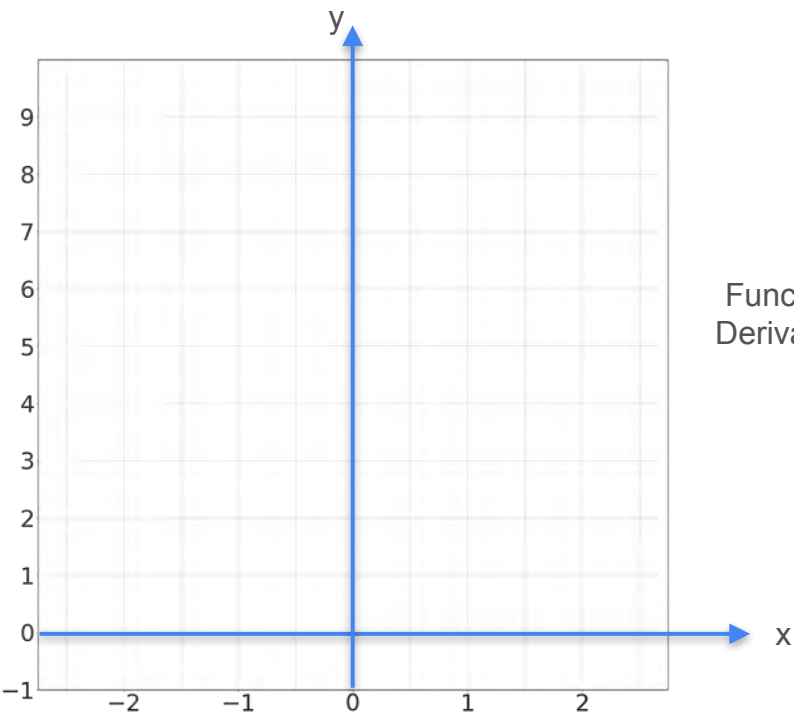
Multiplication by a Scalar



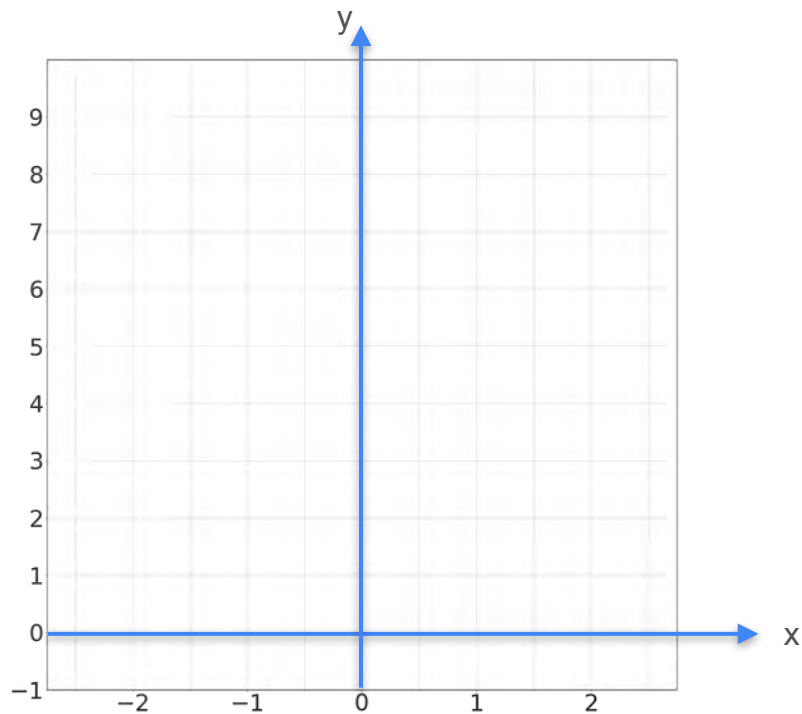
Function multiplies by 2
Derivative multiplies by 2



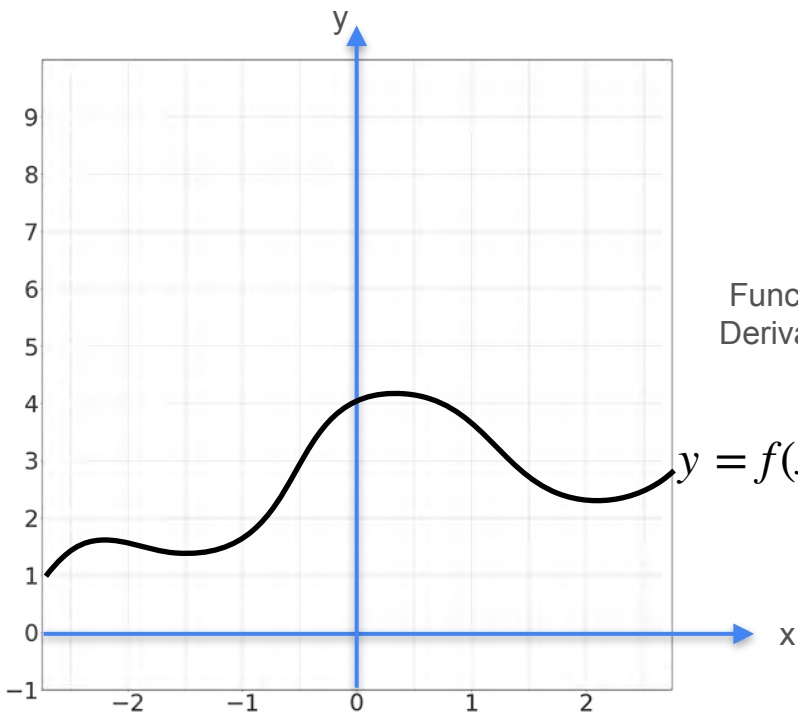
Multiplication by a Scalar



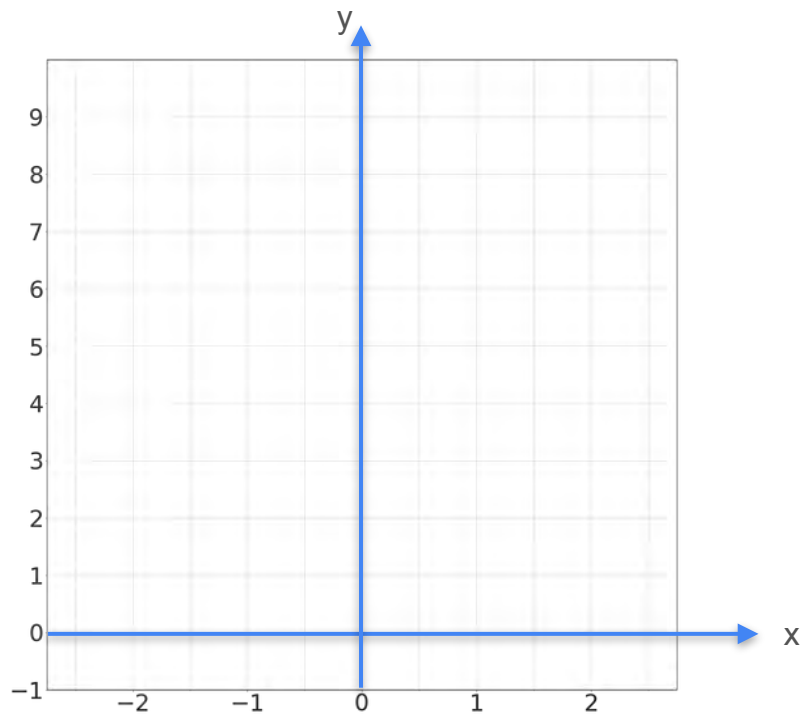
Function multiplies by c
Derivative multiplies by c



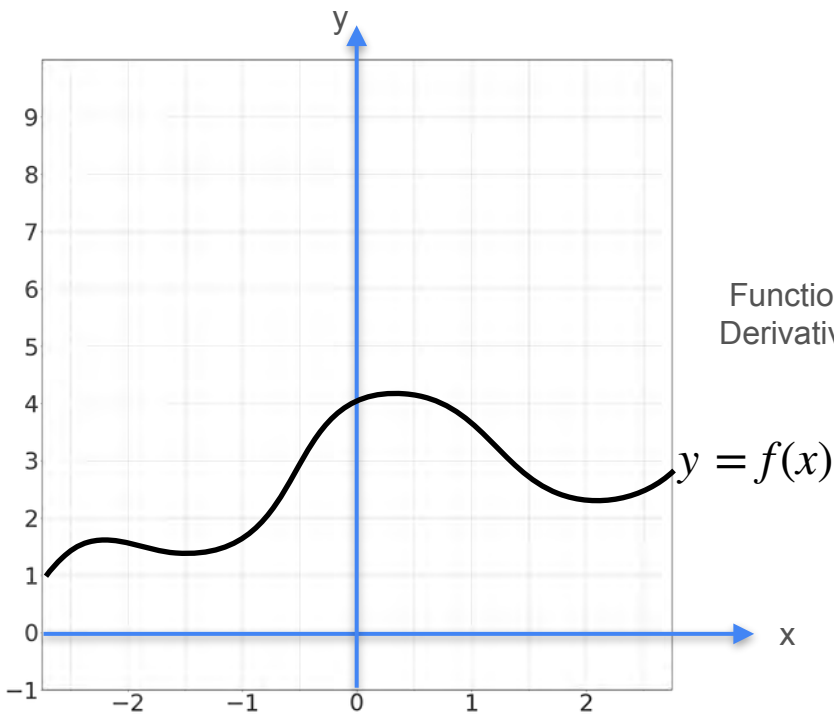
Multiplication by a Scalar



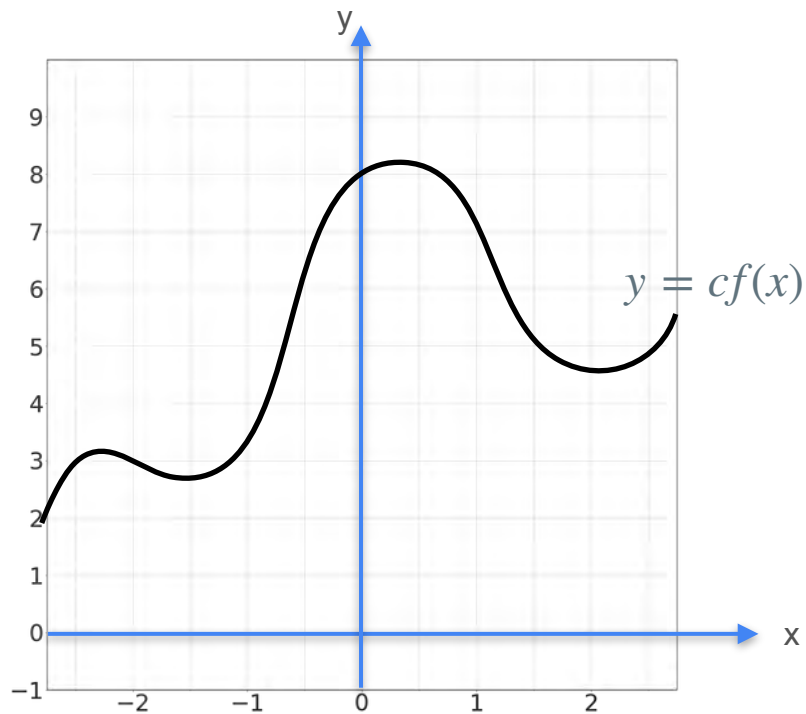
Function multiplies by c
Derivative multiplies by c



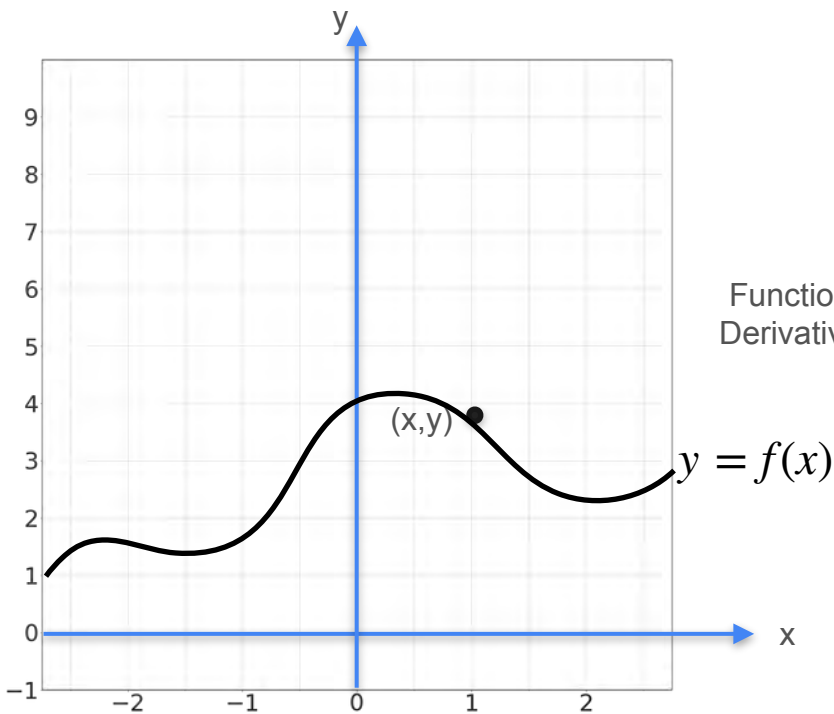
Multiplication by a Scalar



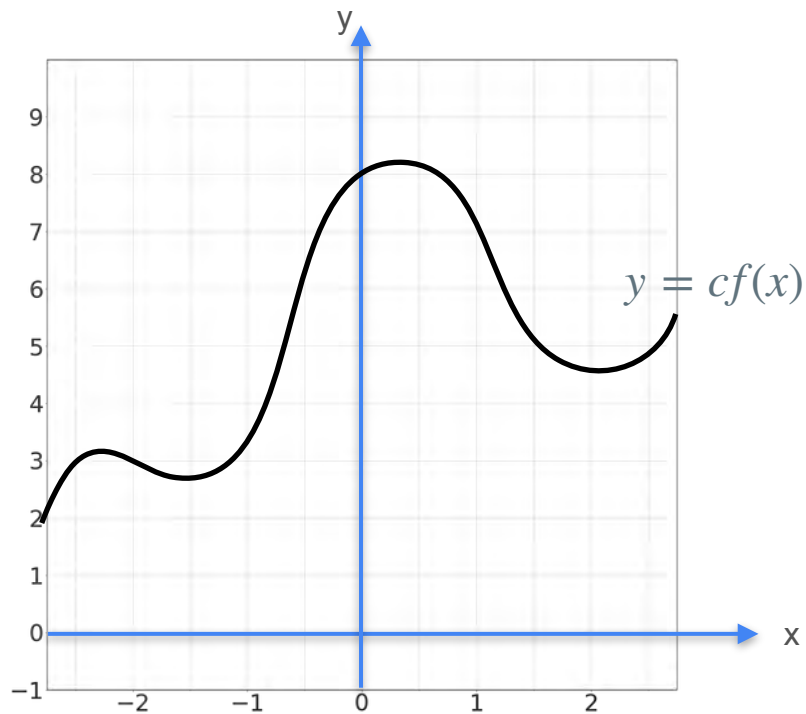
Function multiplies by c
Derivative multiplies by c



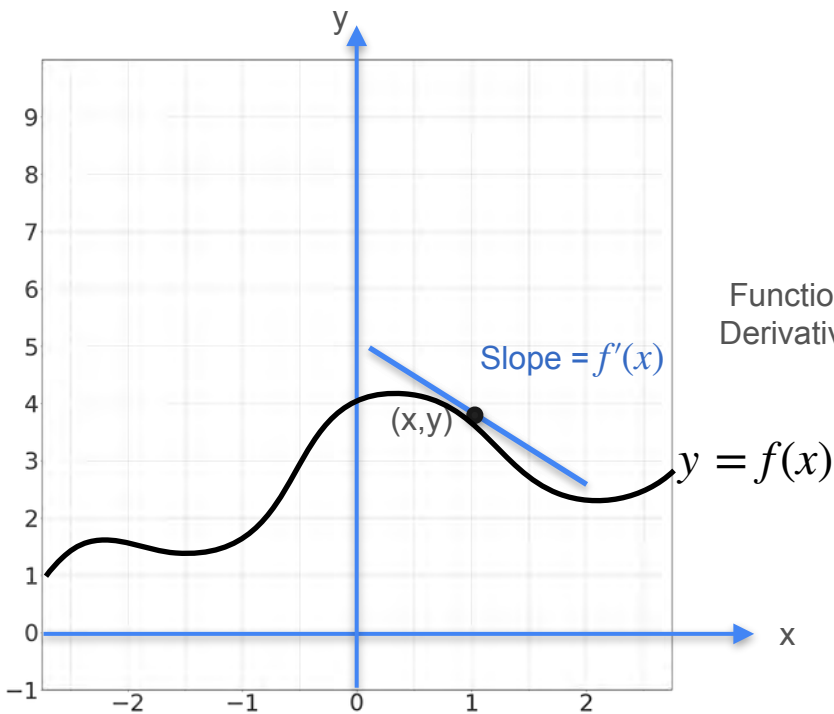
Multiplication by a Scalar



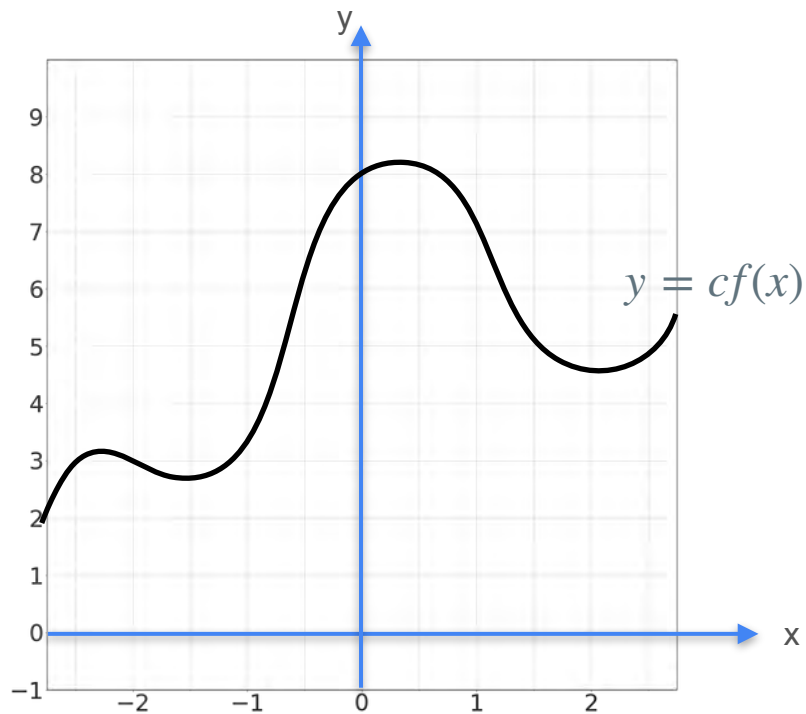
Function multiplies by c
Derivative multiplies by c



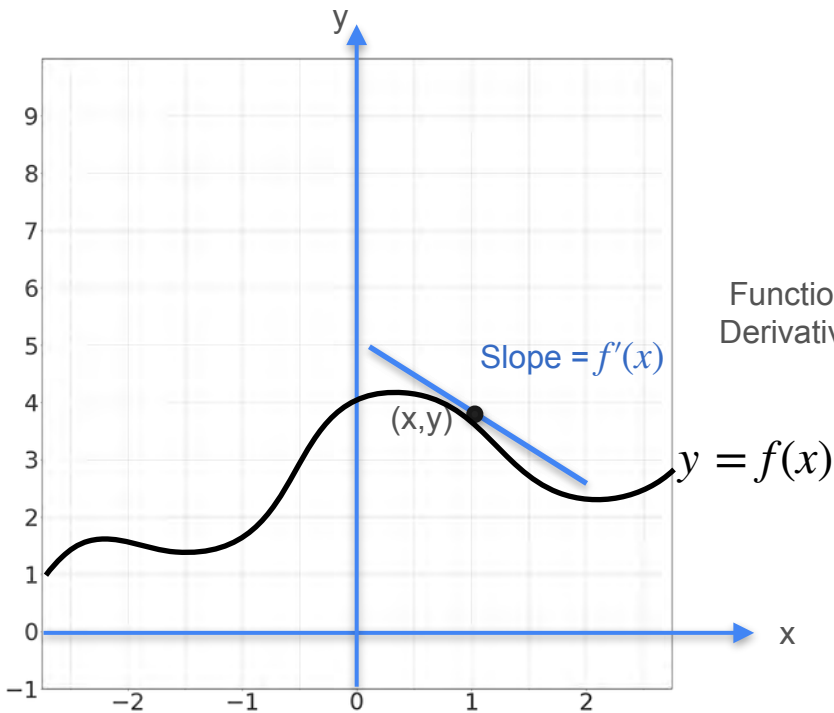
Multiplication by a Scalar



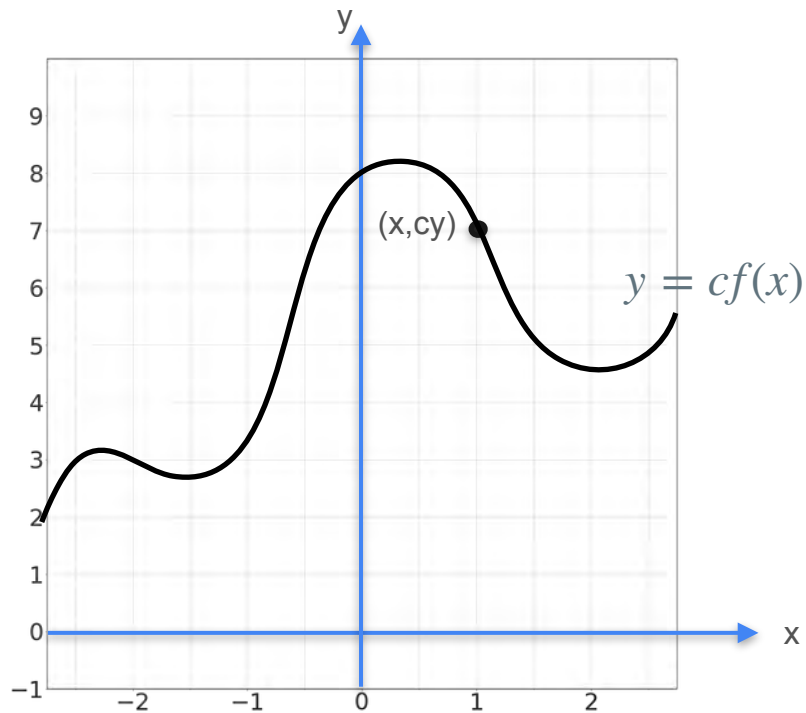
Function multiplies by c
Derivative multiplies by c



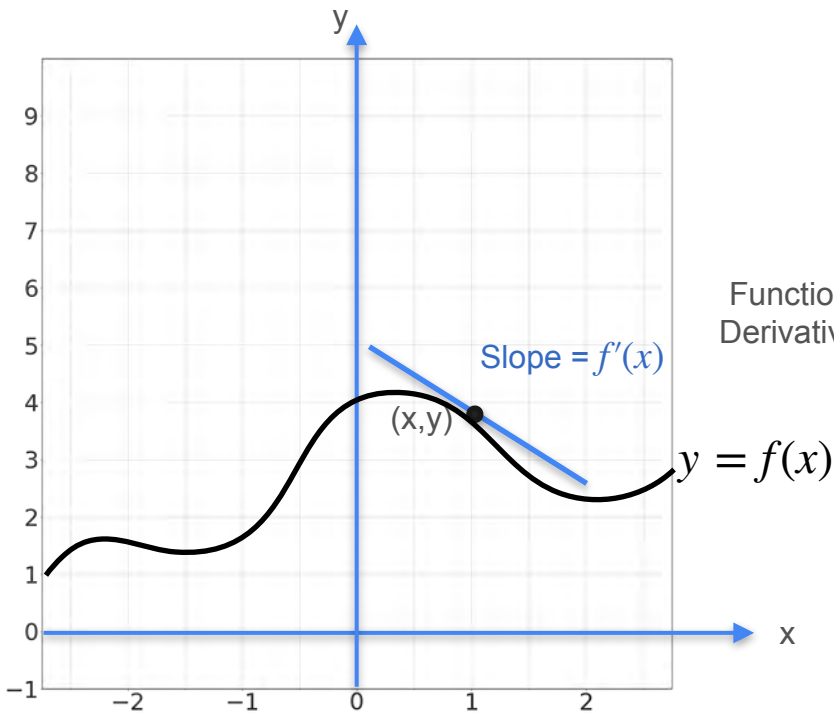
Multiplication by a Scalar



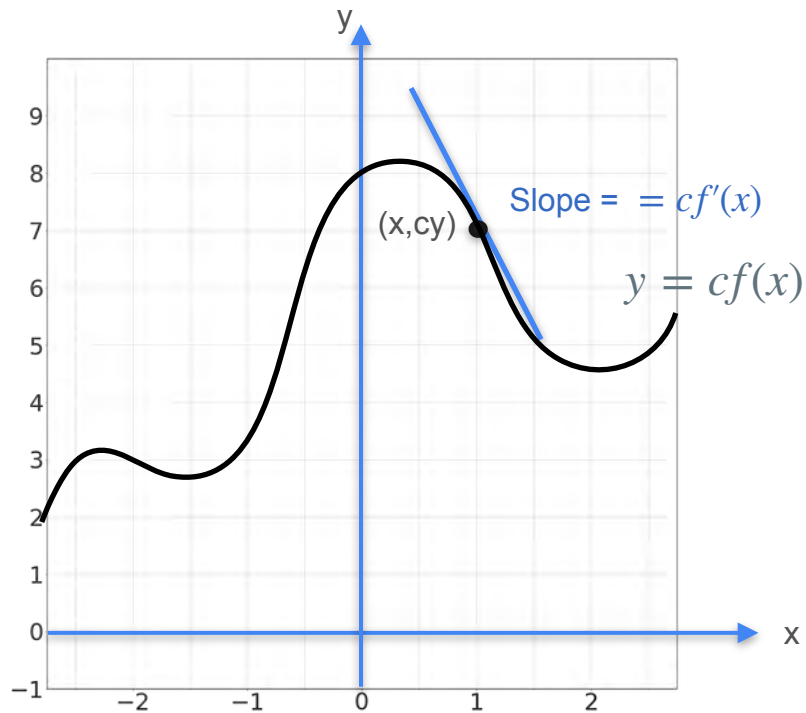
Function multiplies by c
Derivative multiplies by c



Multiplication by a Scalar



Function multiplies by c
Derivative multiplies by c





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Derivatives and Optimization

**Properties of the derivative:
The sum rule**

The Sum Rule

$$f = g + h$$

The Sum Rule

$$f' = g + h$$

The Sum Rule

$$f' = g' + h$$

The Sum Rule

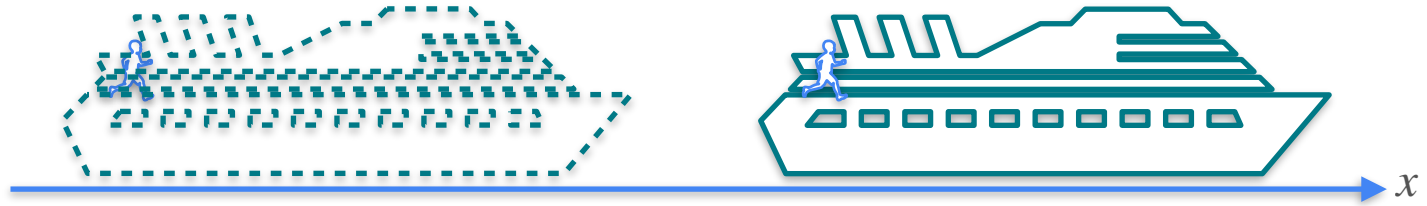
$$f' = g' + h'$$

Sum Rule

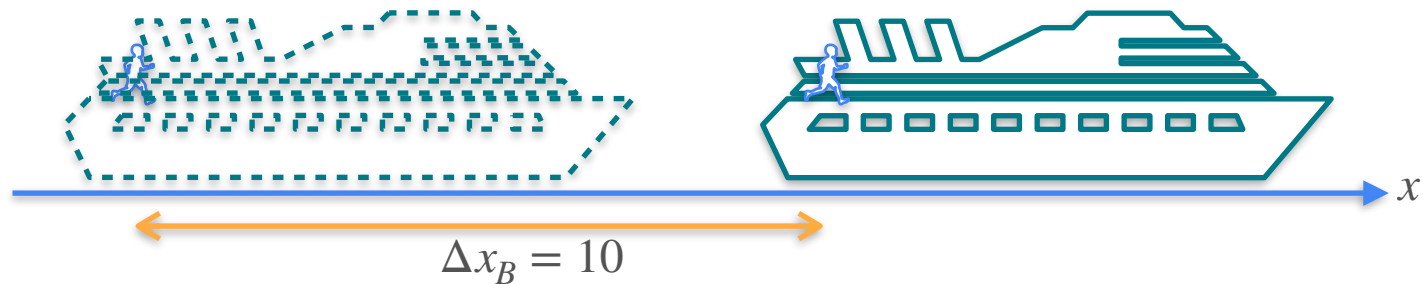
Sum Rule



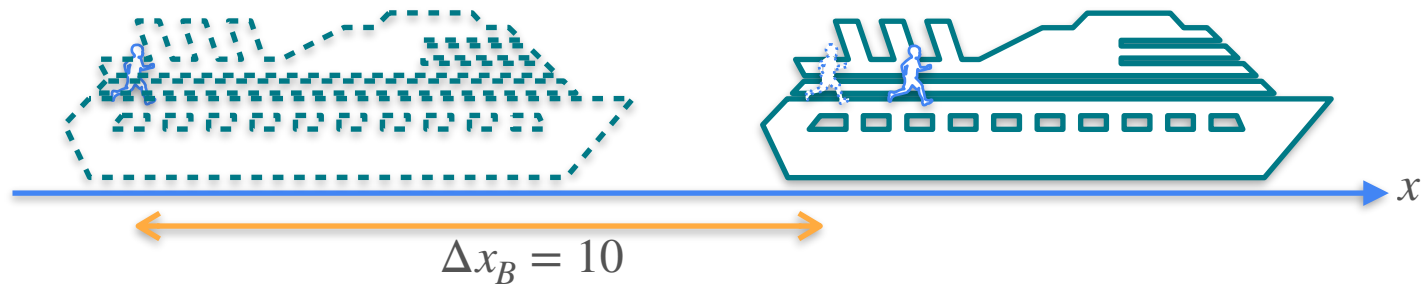
Sum Rule



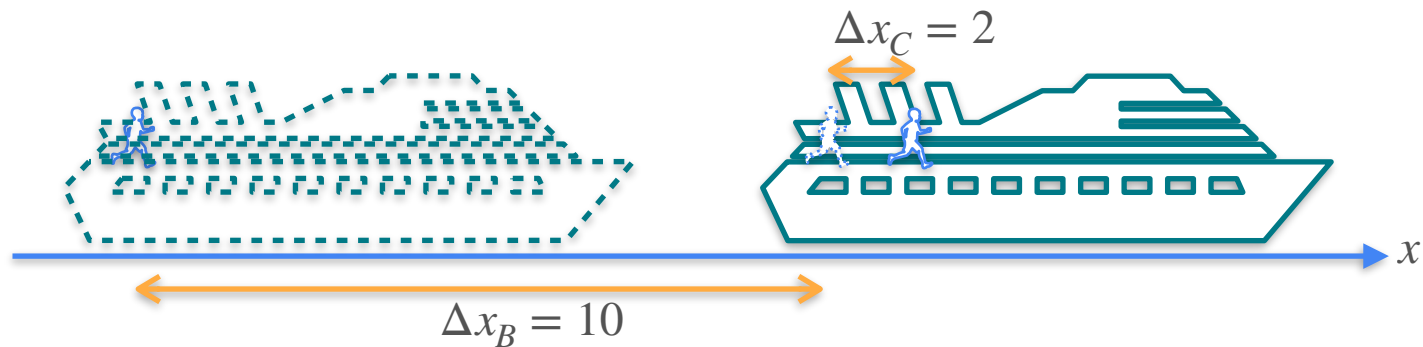
Sum Rule



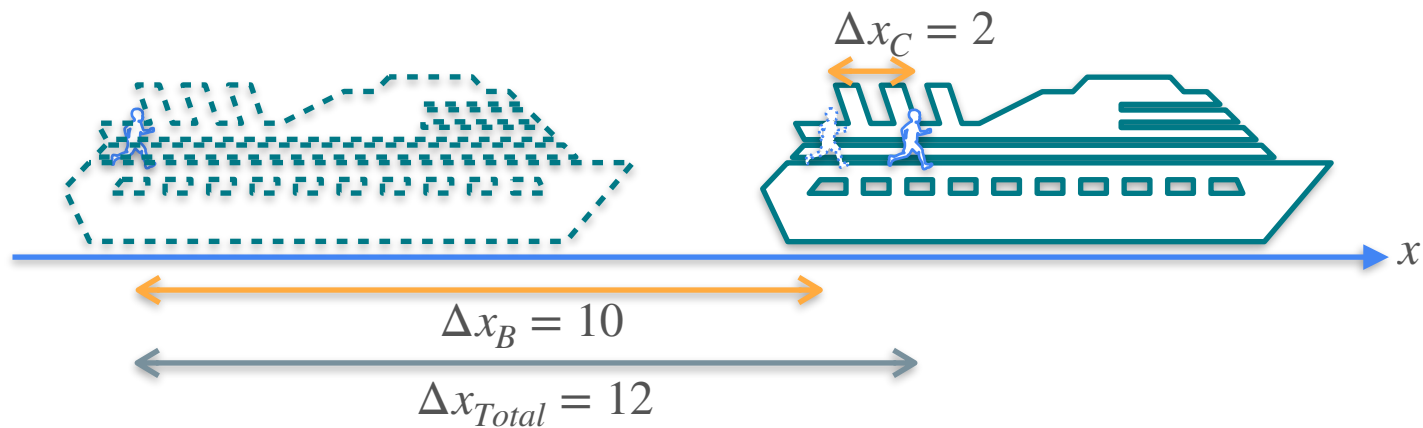
Sum Rule



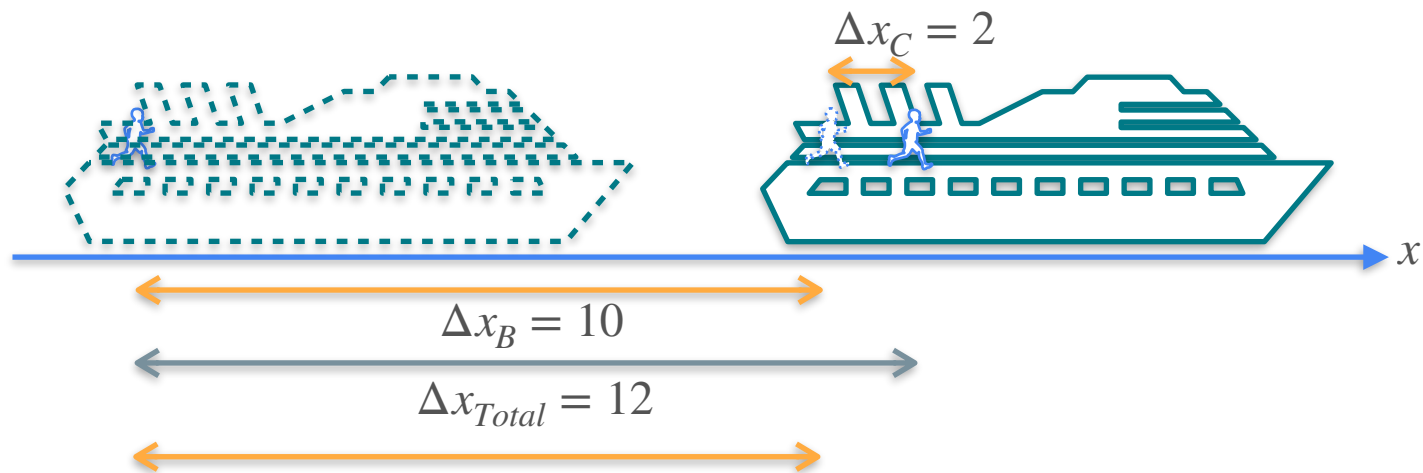
Sum Rule



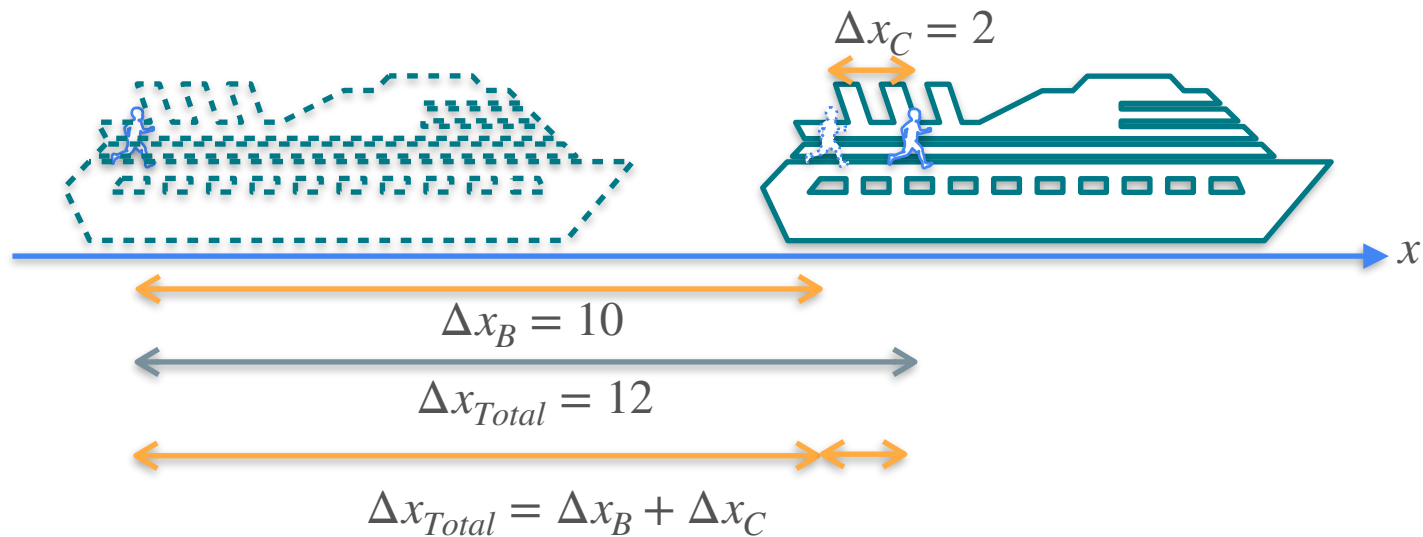
Sum Rule



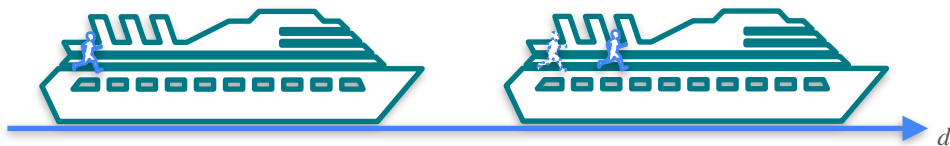
Sum Rule



Sum Rule



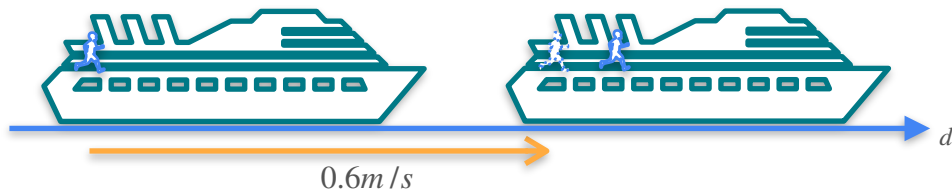
Quiz:



- Speed of the boat: 0.6 m/s
- Speed of the child inside the boat: 0.05 m/s
- Both travel in the same direction.

What is the speed of the child with respect to the earth?

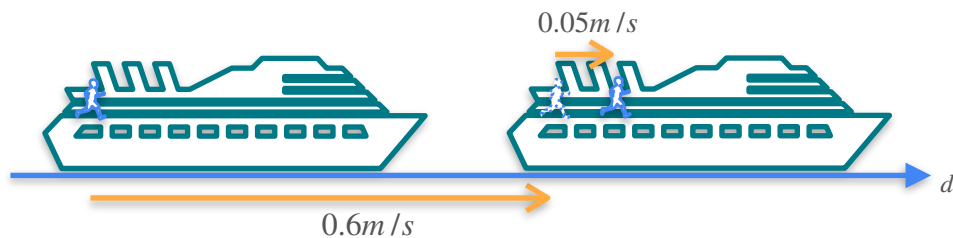
Quiz:



- Speed of the boat: 0.6 m/s
- Speed of the child inside the boat: 0.05 m/s
- Both travel in the same direction.

What is the speed of the child with respect to the earth?

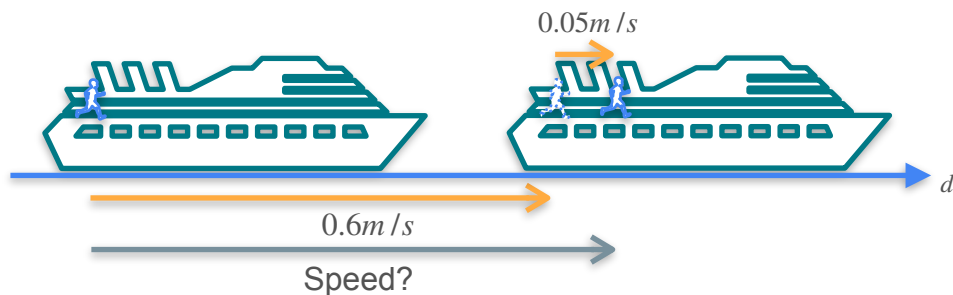
Quiz:



- Speed of the boat: 0.6 m/s
- Speed of the child inside the boat: 0.05 m/s
- Both travel in the same direction.

What is the speed of the child with respect to the earth?

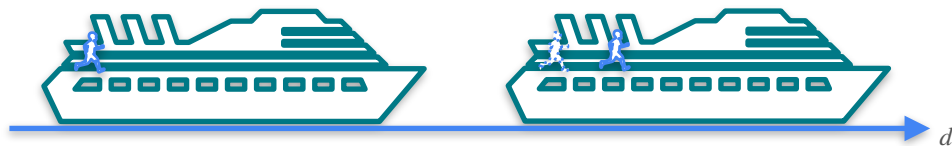
Quiz:



- Speed of the boat: 0.6 m/s
- Speed of the child inside the boat: 0.05 m/s
- Both travel in the same direction.

What is the speed of the child with respect to the earth?

Solution:

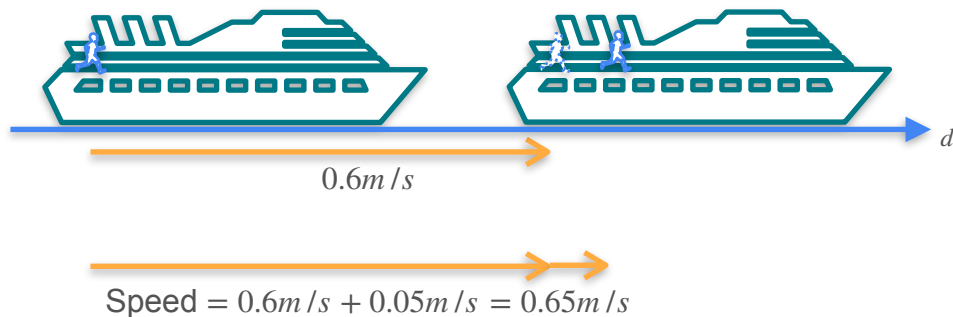



$$\text{Speed} = 0.6 \text{ m/s} + 0.05 \text{ m/s} = 0.65 \text{ m/s}$$

- Speed of the boat: 0.6 m/s
- Speed of the child inside the boat: 0.05 m/s
- Both travel in the same direction.

What is the speed of the child with respect to the earth?

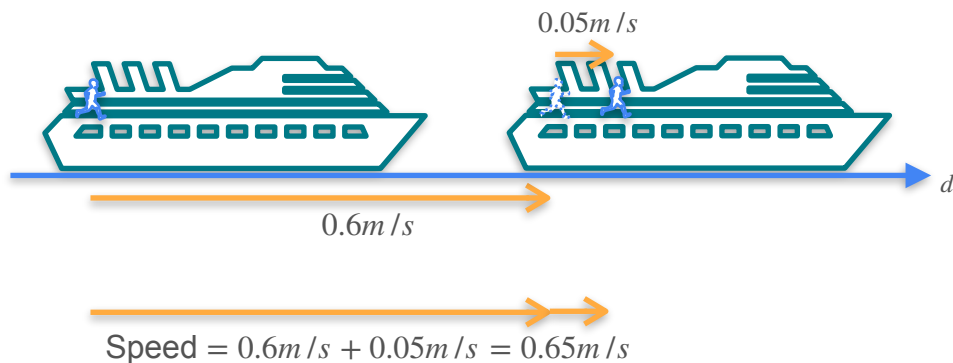
Solution:



- Speed of the boat: 0.6 m/s
- Speed of the child inside the boat: 0.05 m/s
- Both travel in the same direction.

What is the speed of the child with respect to the earth?

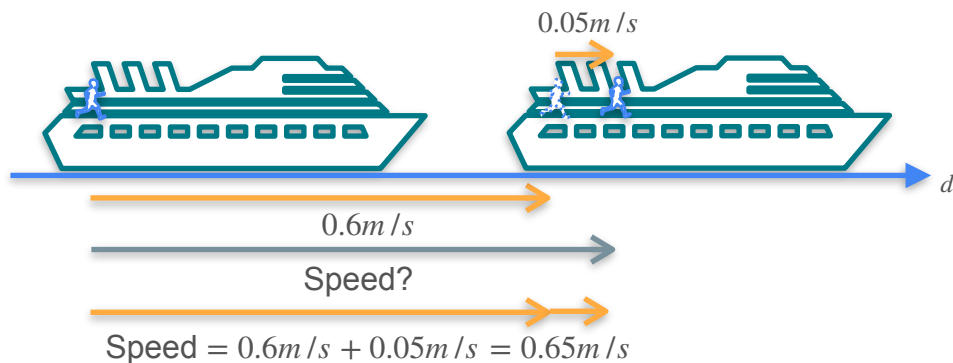
Solution:



- Speed of the boat: 0.6 m/s
- Speed of the child inside the boat: 0.05 m/s
- Both travel in the same direction.

What is the speed of the child with respect to the earth?

Solution:

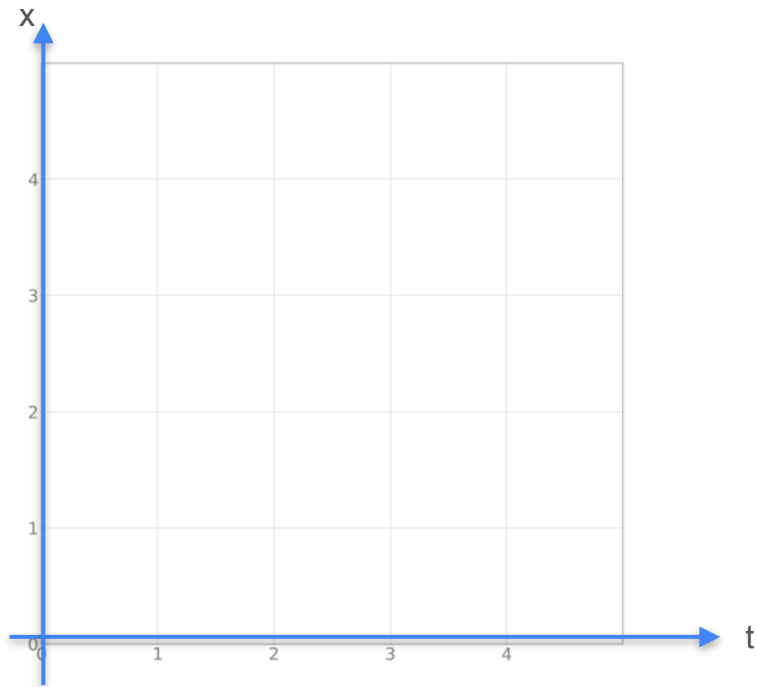


- Speed of the boat: 0.6 m/s
- Speed of the child inside the boat: 0.05 m/s
- Both travel in the same direction.

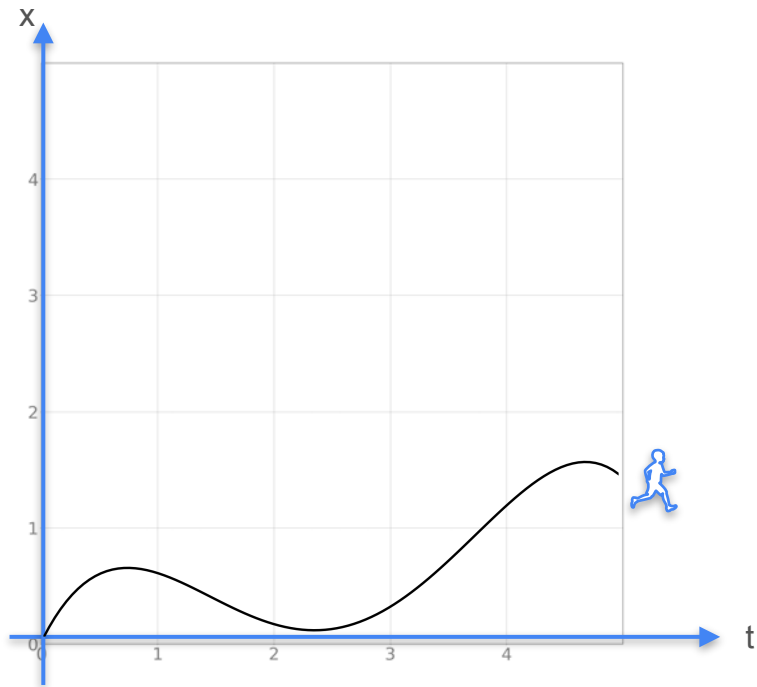
What is the speed of the child with respect to the earth?

Sum Rule

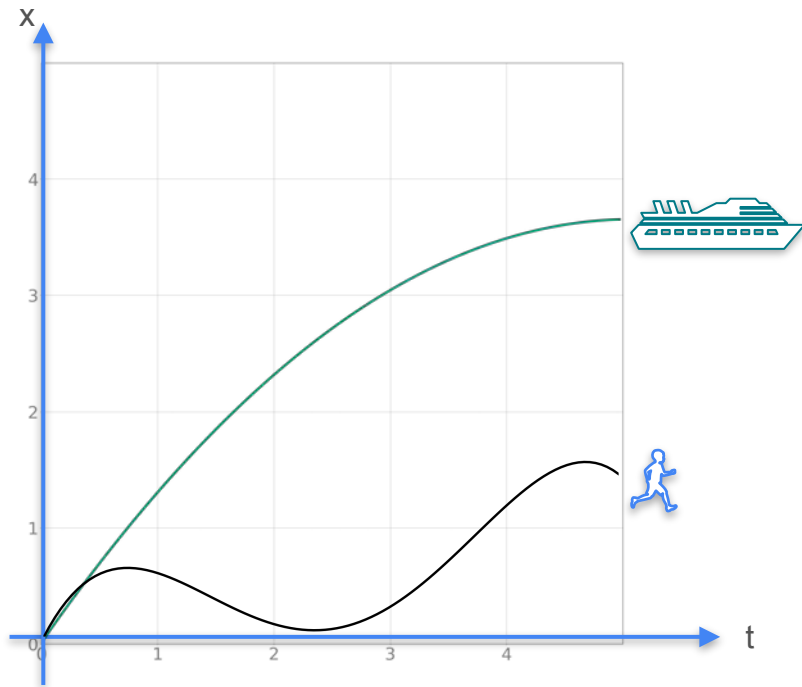
Sum Rule



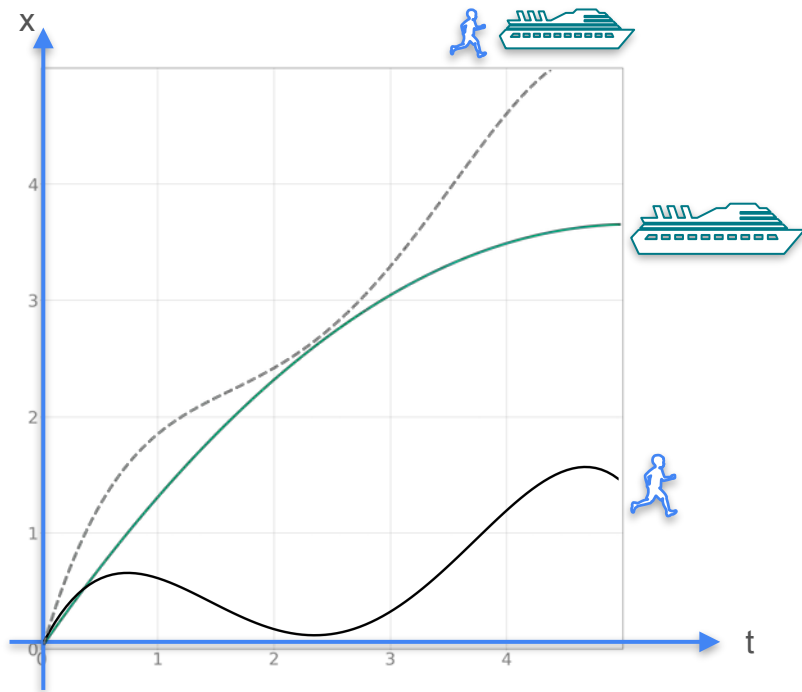
Sum Rule



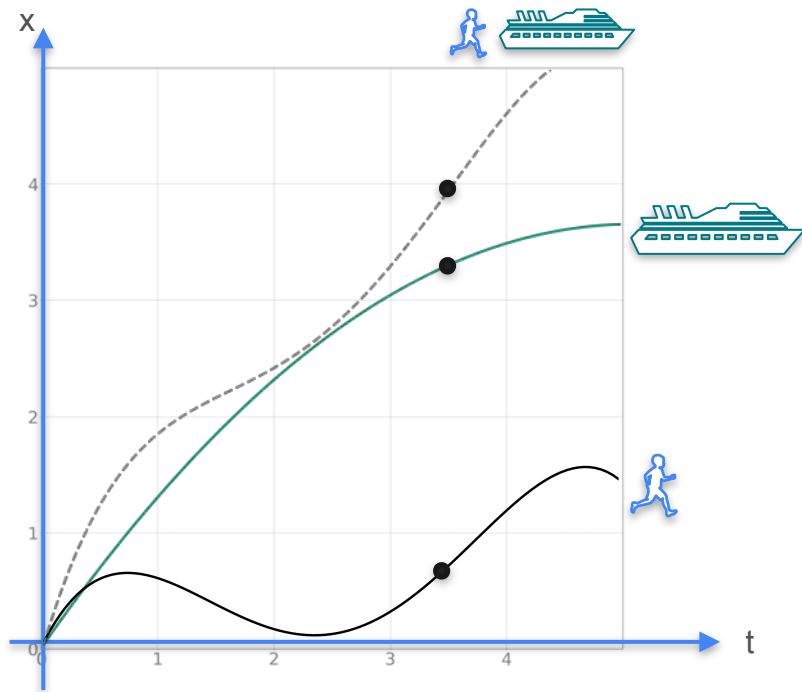
Sum Rule



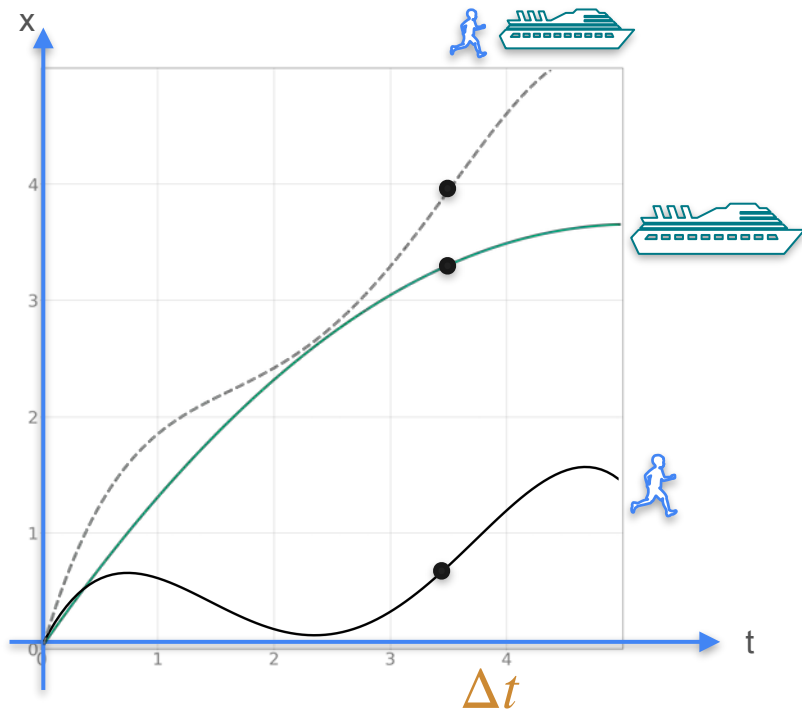
Sum Rule



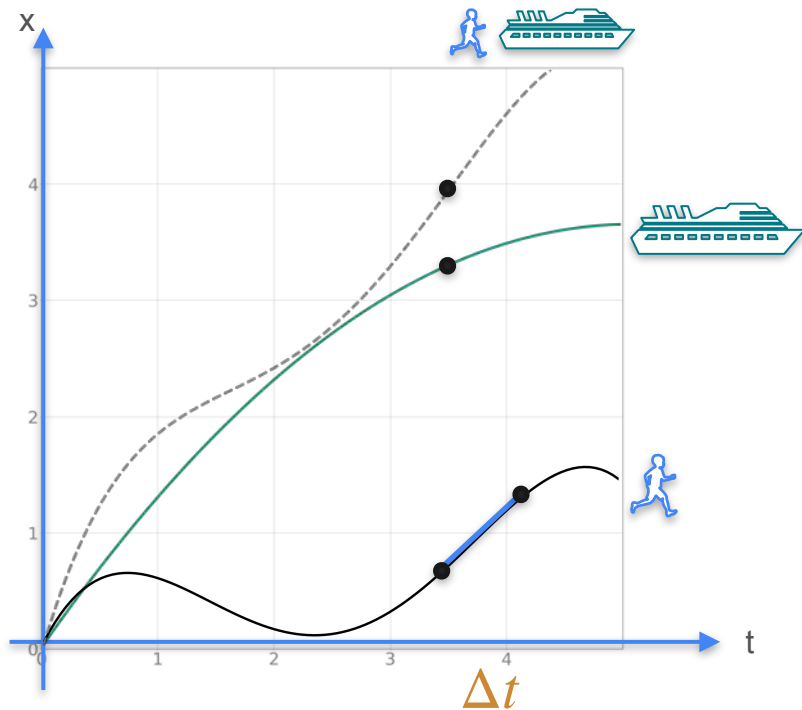
Sum Rule



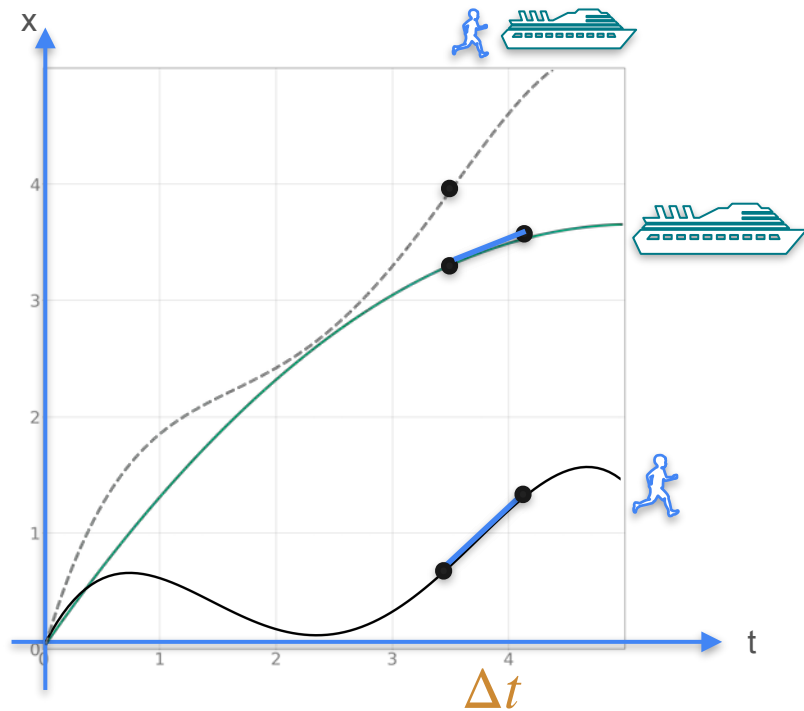
Sum Rule



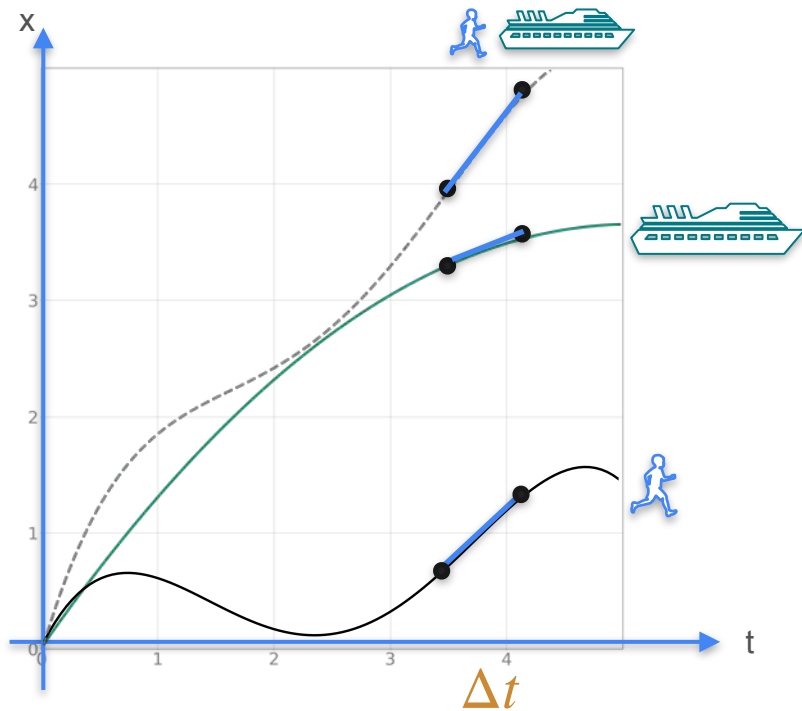
Sum Rule



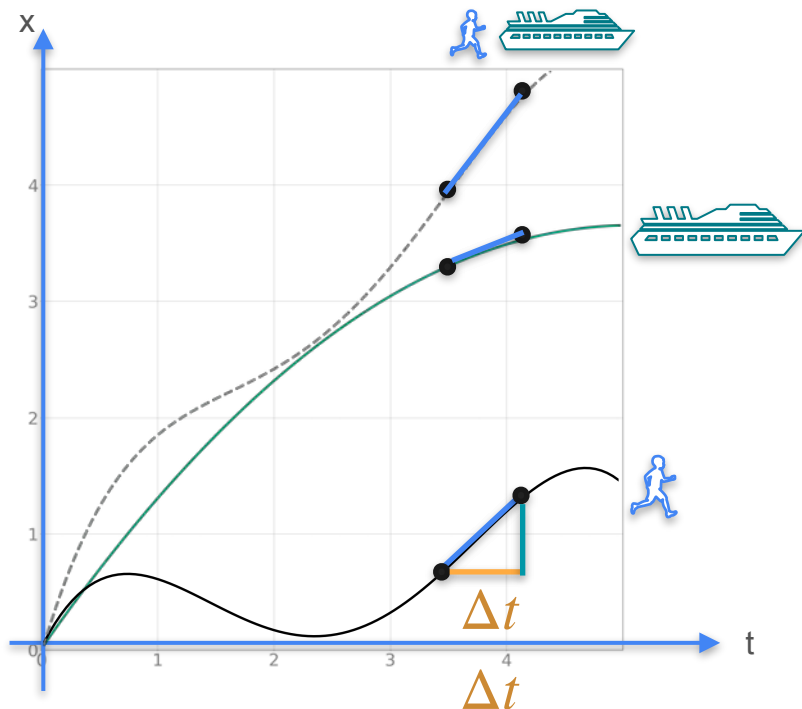
Sum Rule



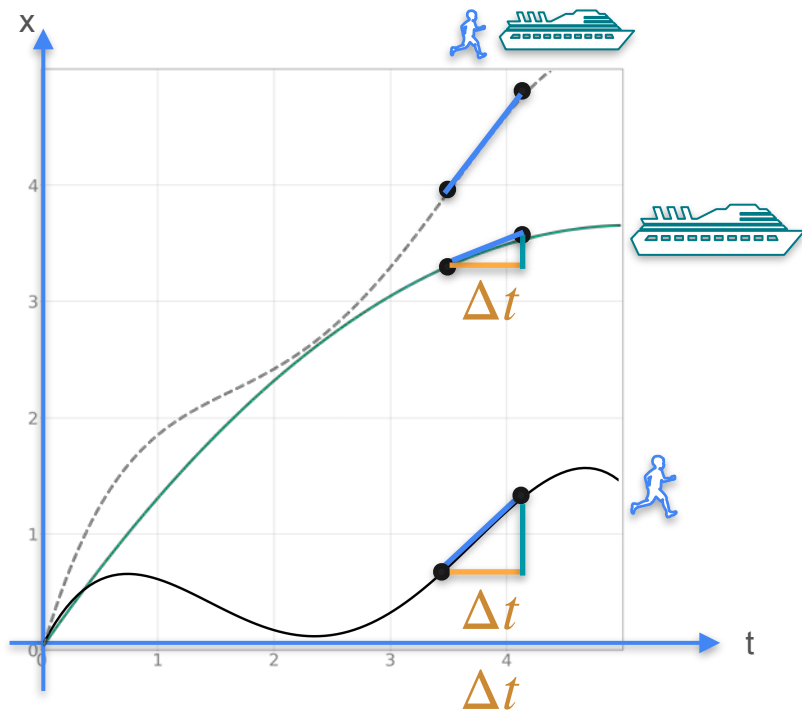
Sum Rule



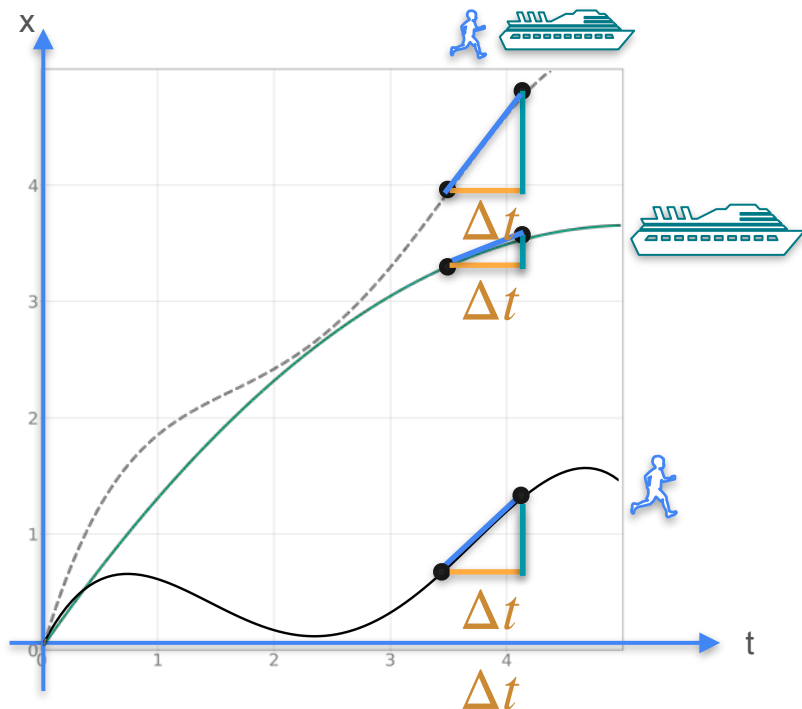
Sum Rule



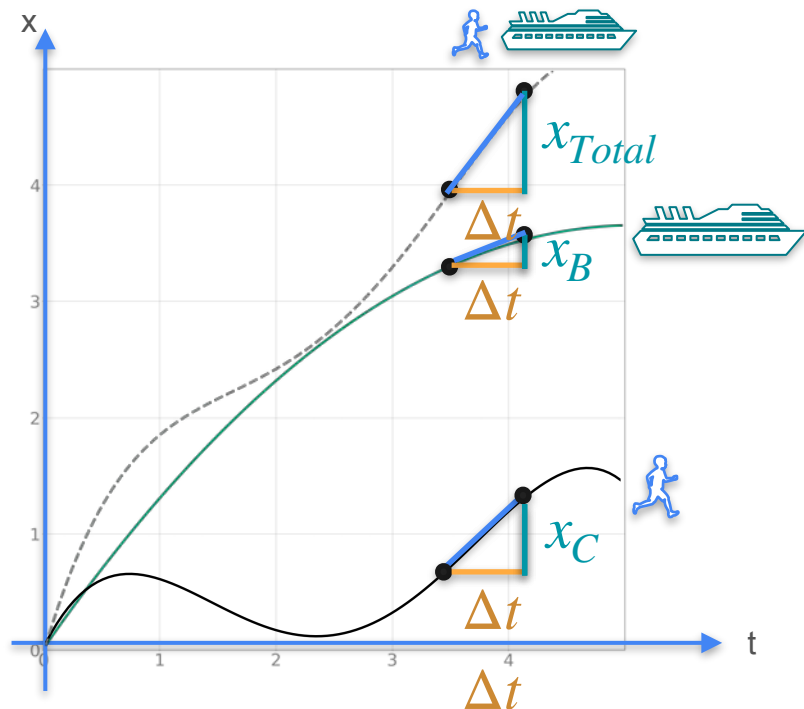
Sum Rule



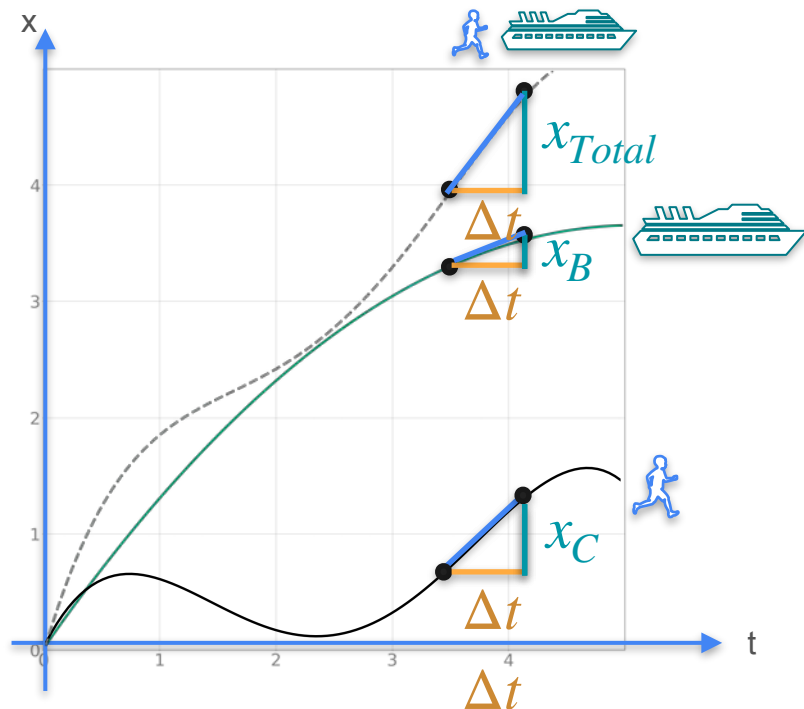
Sum Rule



Sum Rule

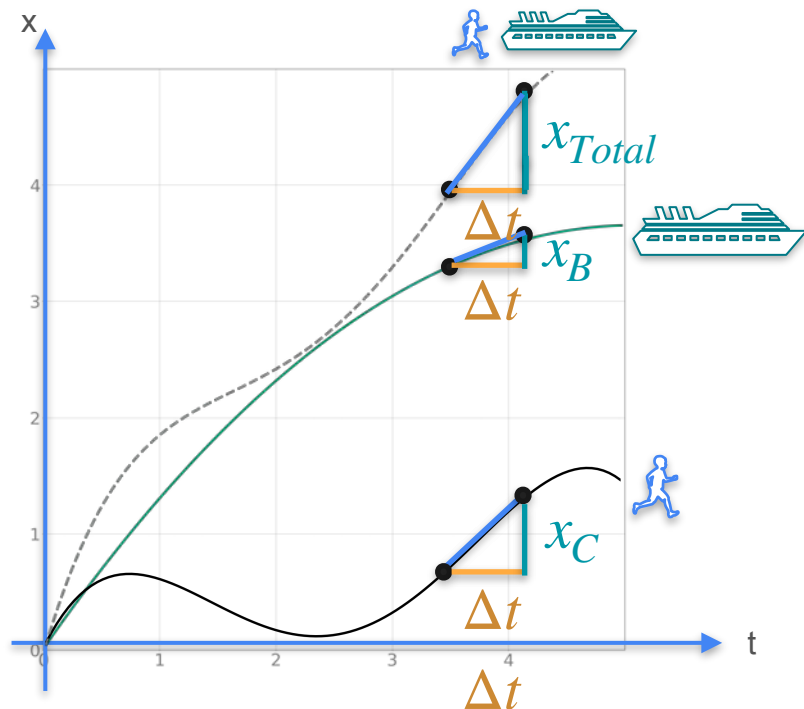


Sum Rule



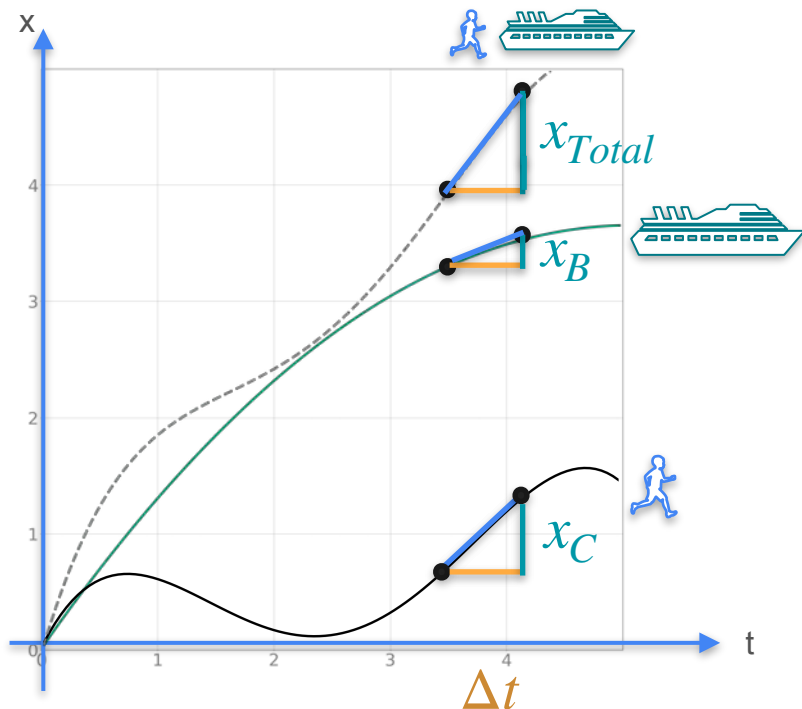
$$x_{Total} = x_B + x_C$$

Sum Rule



$$x_{Total} = x_B + x_C$$

Sum Rule

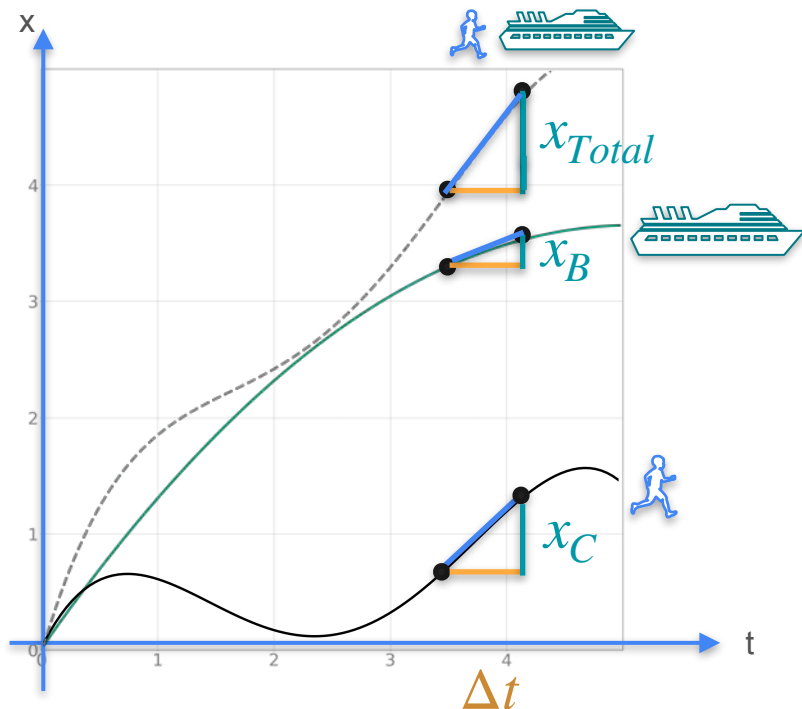


$$x_{Total} = x_B + x_C$$



$$\frac{x_{Total}}{\Delta t} = \frac{x_B}{\Delta t} + \frac{x_C}{\Delta t}$$

Sum Rule



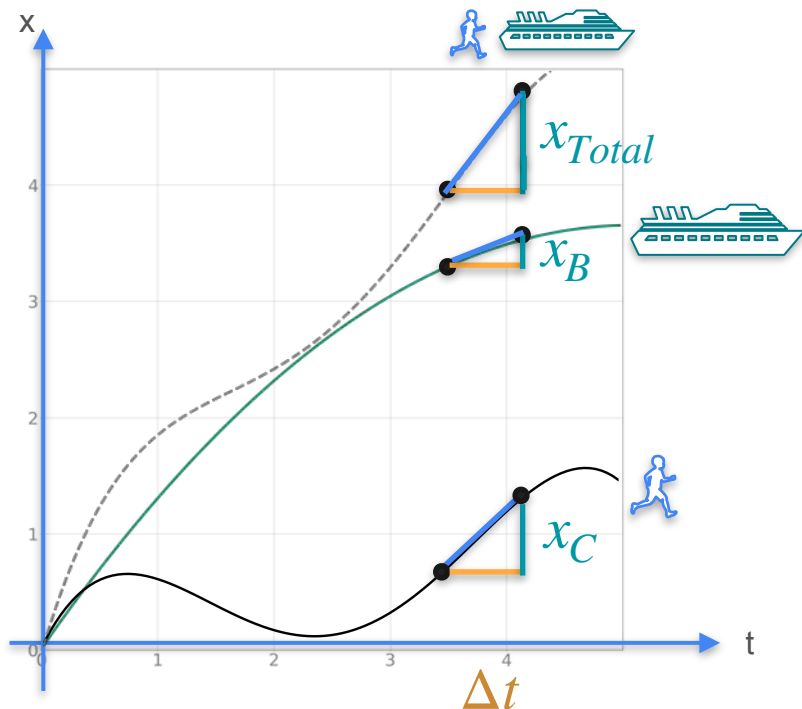
$$x_{Total} = x_B + x_C$$



$$\frac{x_{Total}}{\Delta t} = \frac{x_B}{\Delta t} + \frac{x_C}{\Delta t}$$



Sum Rule



$$x_{Total} = x_B + x_C$$

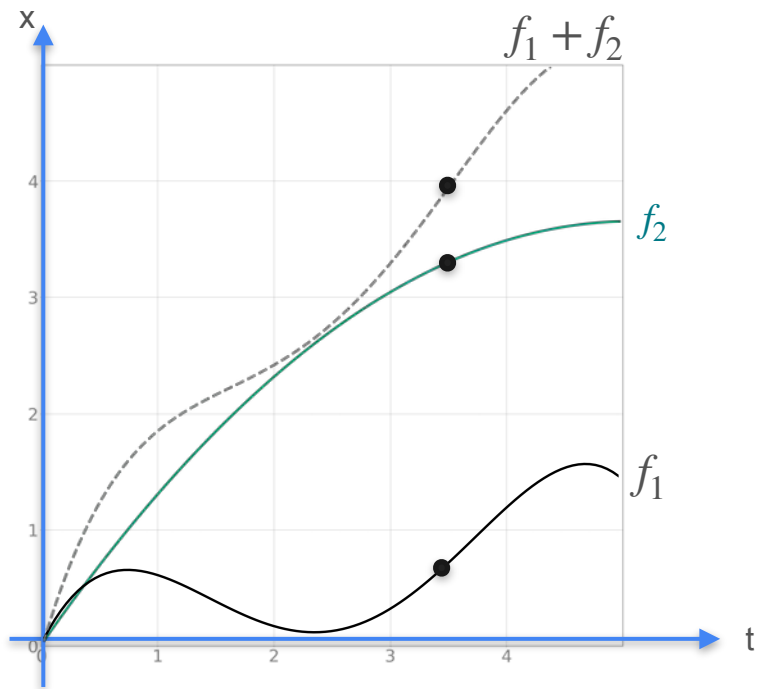


$$\frac{x_{Total}}{\Delta t} = \frac{x_B}{\Delta t} + \frac{x_C}{\Delta t}$$

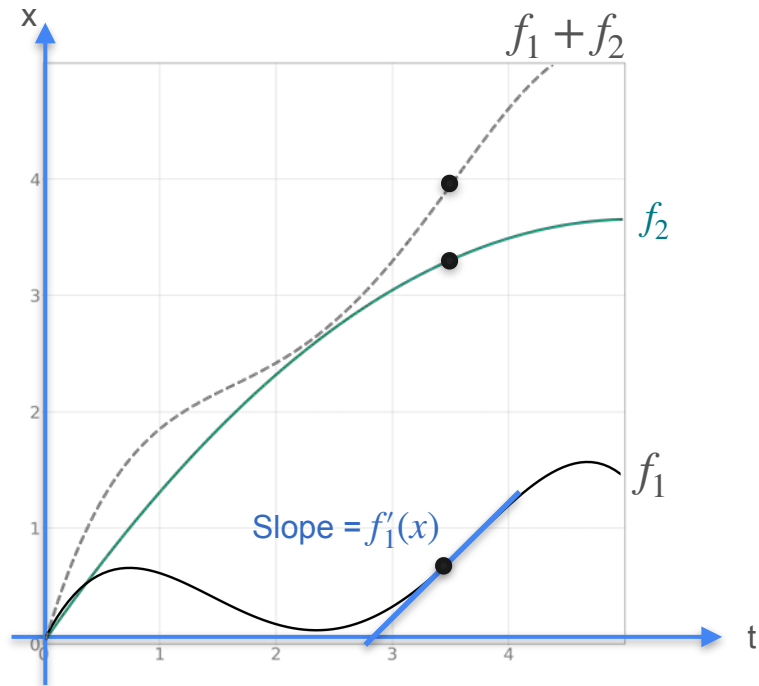


$$v_{Total} = v_B + v_C$$

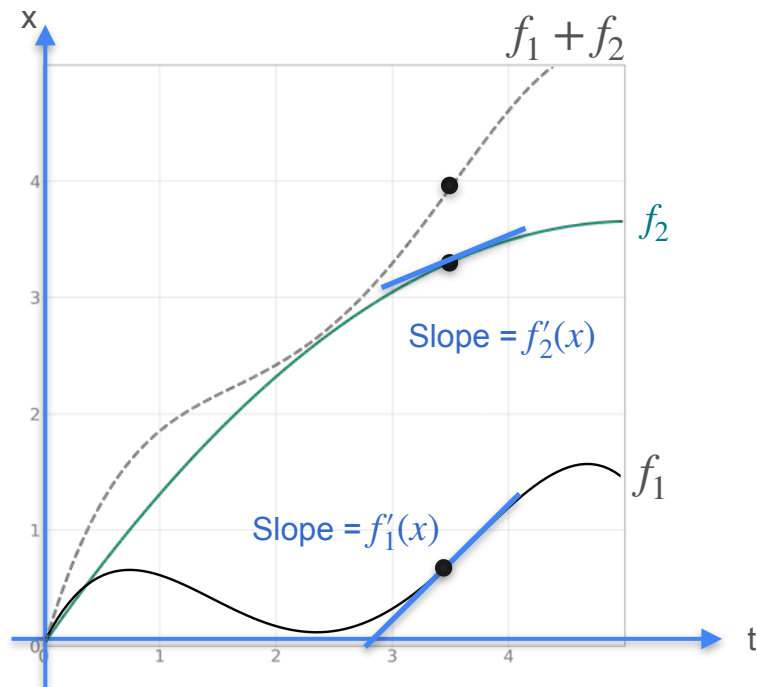
Sum Rule



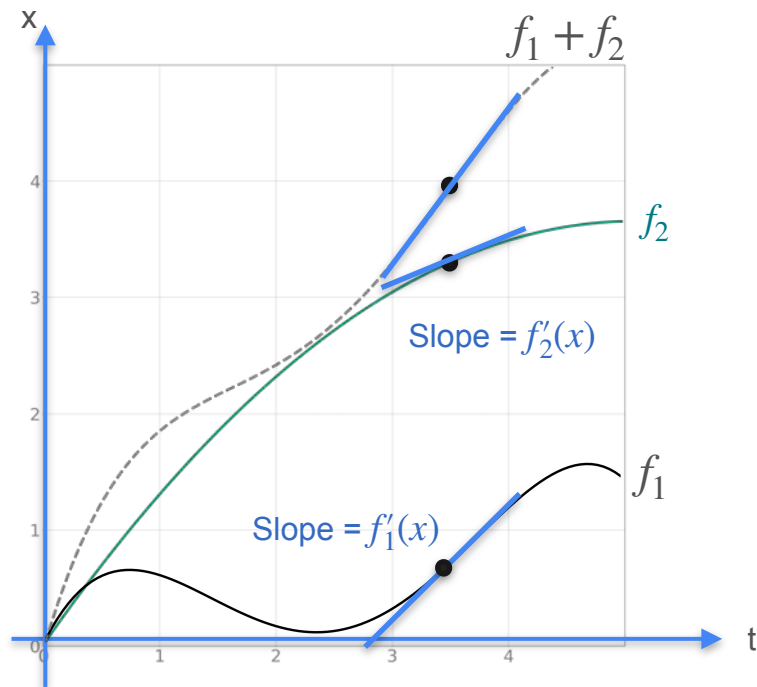
Sum Rule



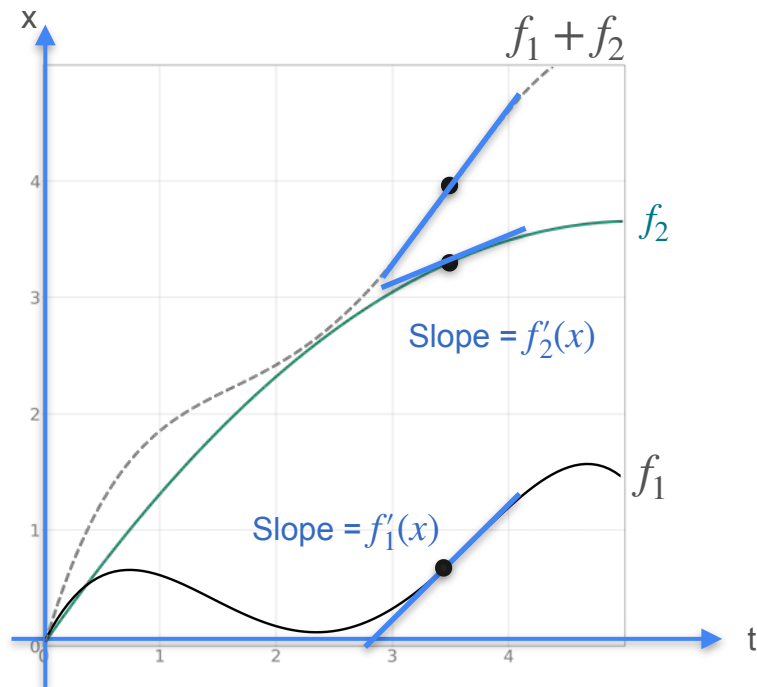
Sum Rule



Sum Rule

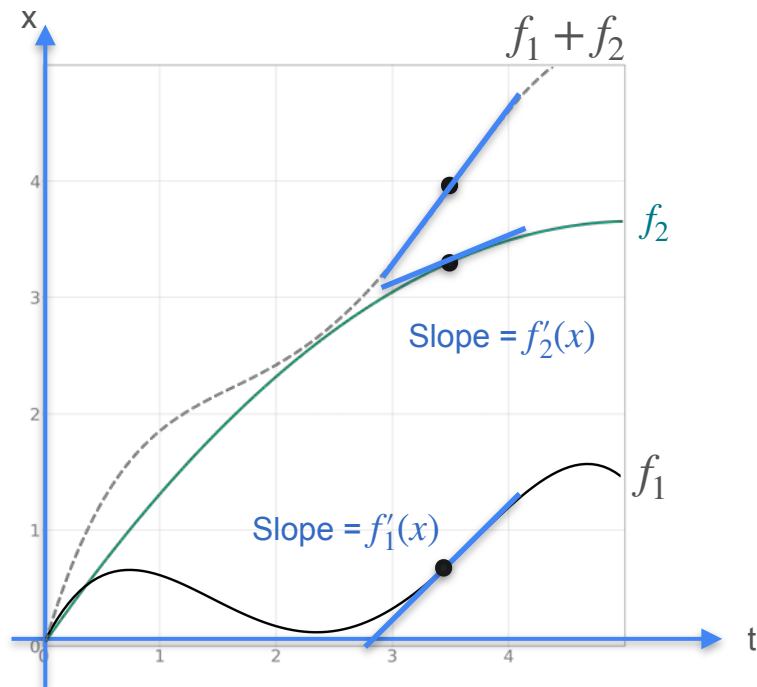


Sum Rule



$$f = f_1 + f_2$$

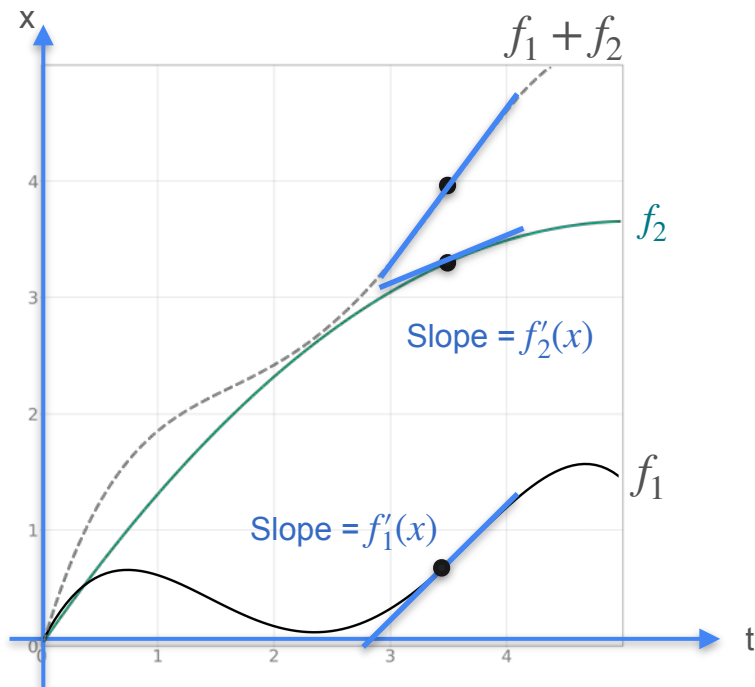
Sum Rule



$$f = f_1 + f_2$$



Sum Rule

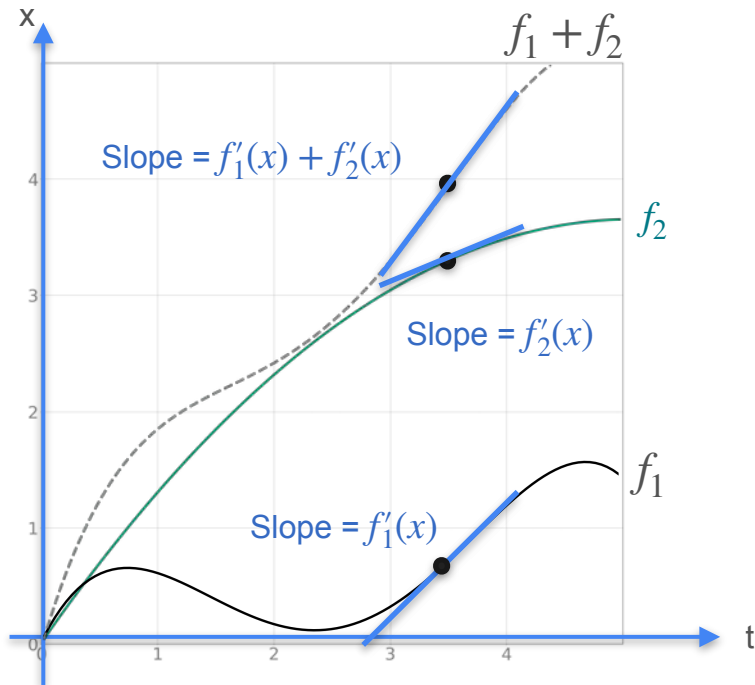


$$f = f_1 + f_2$$



$$f' = f'_1 + f'_2$$

Sum Rule



$$f = f_1 + f_2$$



$$f' = f'_1 + f'_2$$



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Derivatives and Optimization

**Properties of the derivative:
The product rule**

The Product Rule

$$f = gh$$

The Product Rule

$$f = gh$$

The Product Rule

$$f = gh$$

The Product Rule

$$f = gh$$

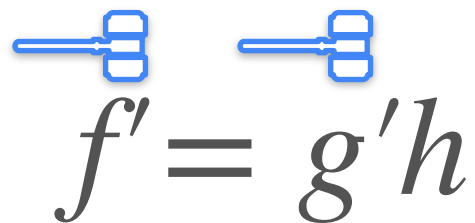
The Product Rule

$$f = gh$$


$$f'$$

The Product Rule

$$f = gh$$



The diagram shows the derivative equation $f' = g'h$ with blue annotations. Above the f' term, there is a blue bracket with a horizontal line extending to the left, indicating the derivative of f with respect to a variable. Similarly, above the $g'h$ term, there is a blue bracket with a horizontal line extending to the left, indicating the derivative of $g'h$ with respect to the same variable.

$$f' = g'h$$

The Product Rule

$$f = gh$$


$$f' = g'h + gh'$$

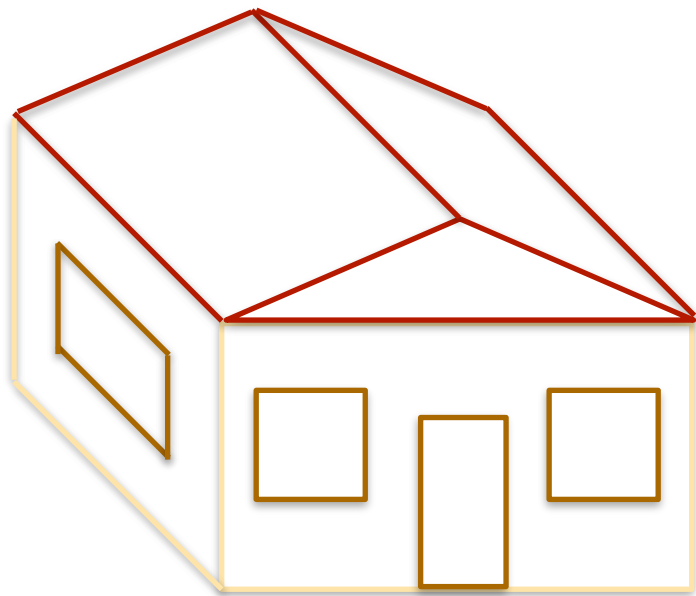
The Product Rule

$$f = gh$$

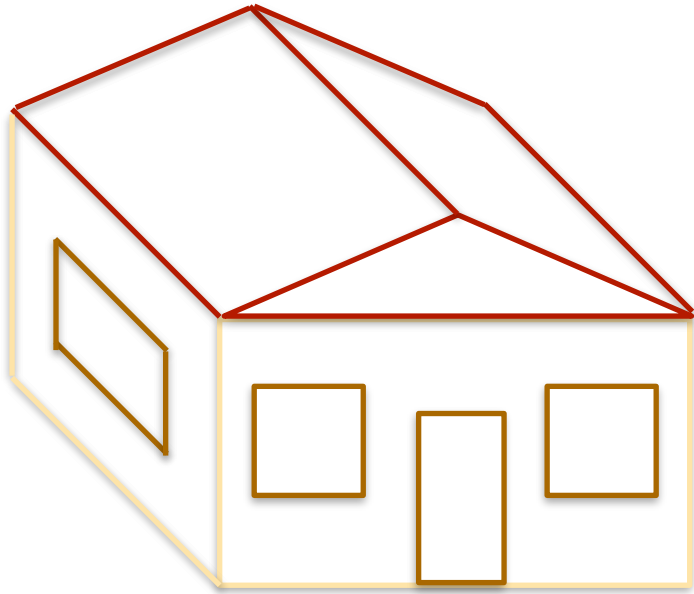
$$f' = g'h + gh'$$

Product Rule

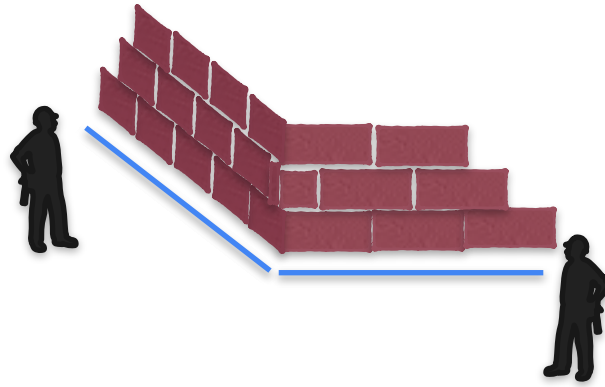
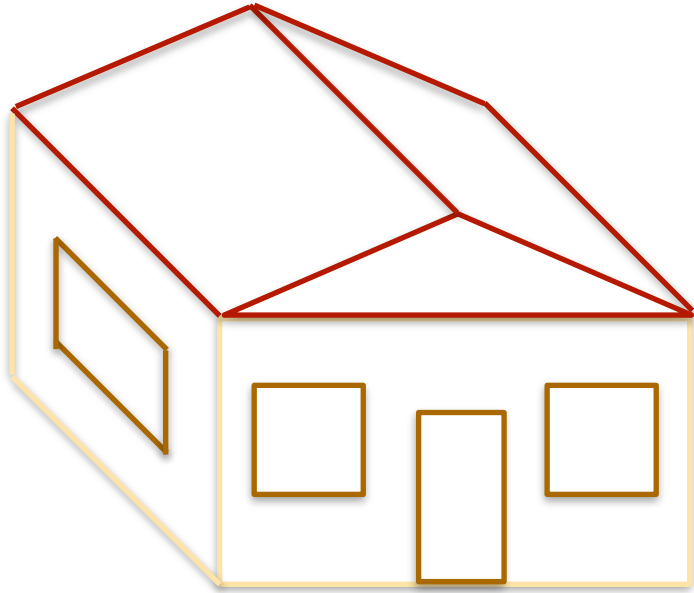
Product Rule



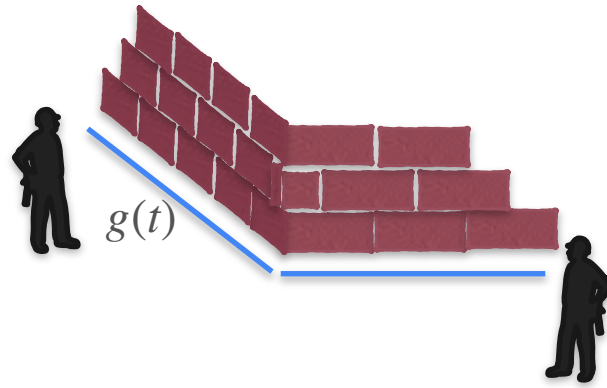
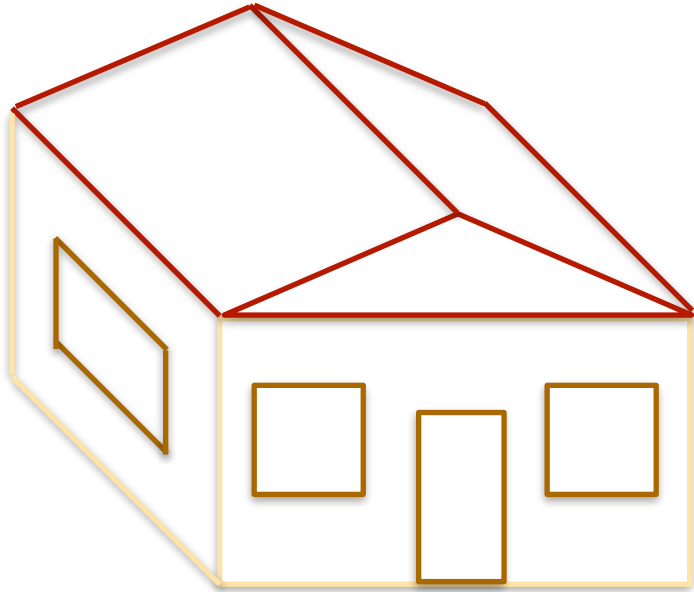
Product Rule



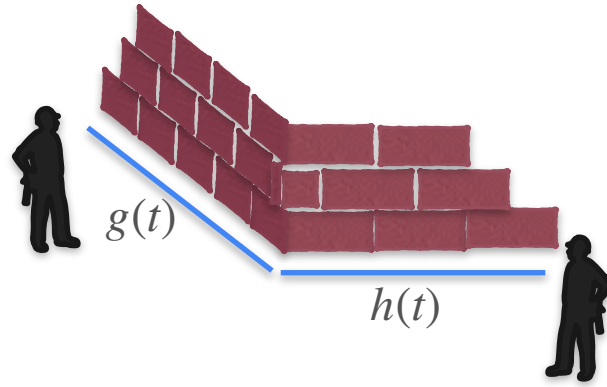
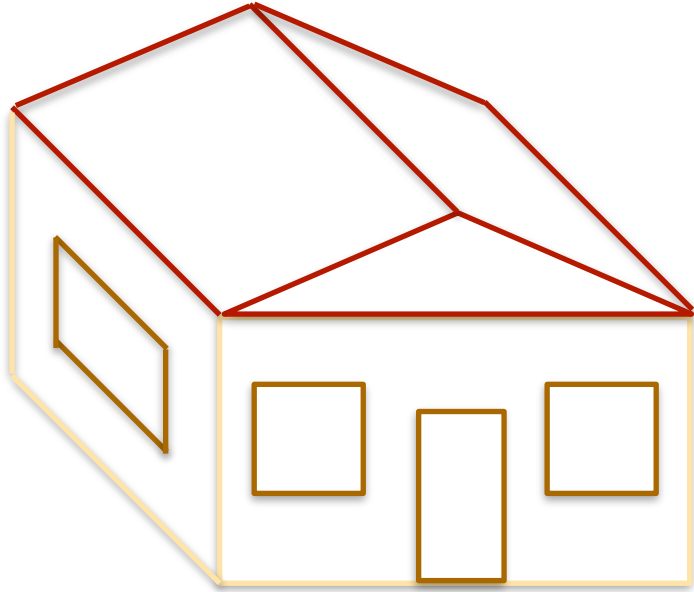
Product Rule



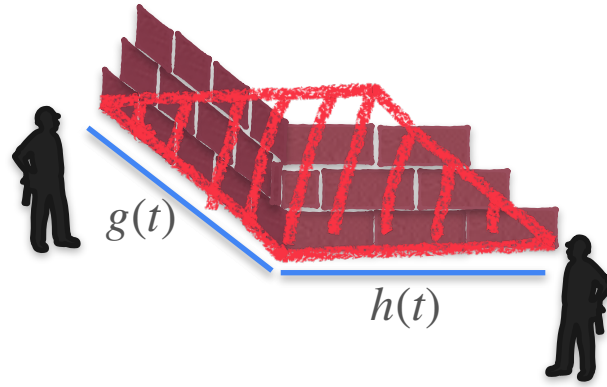
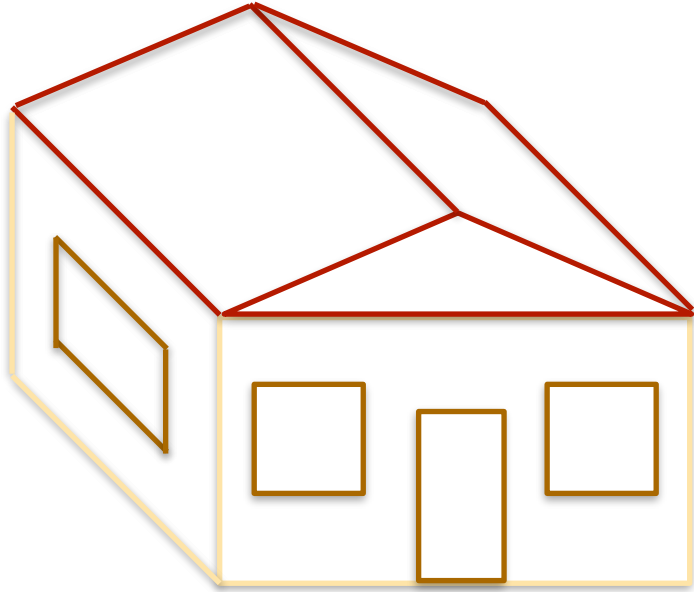
Product Rule



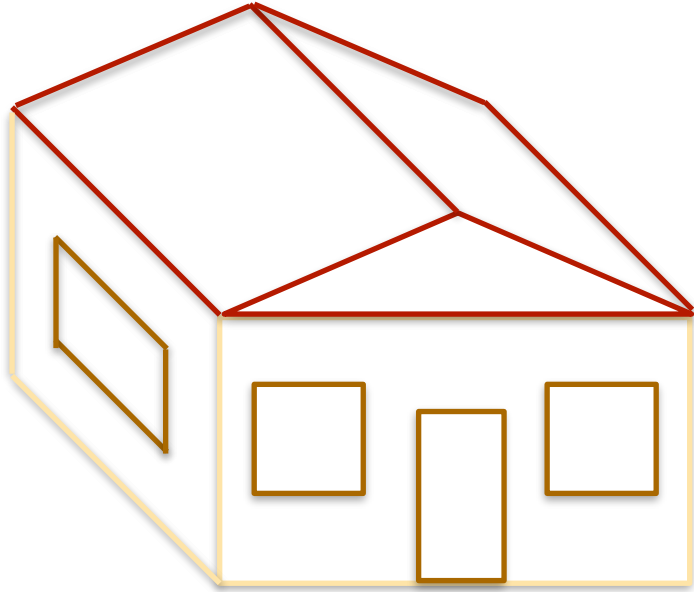
Product Rule



Product Rule

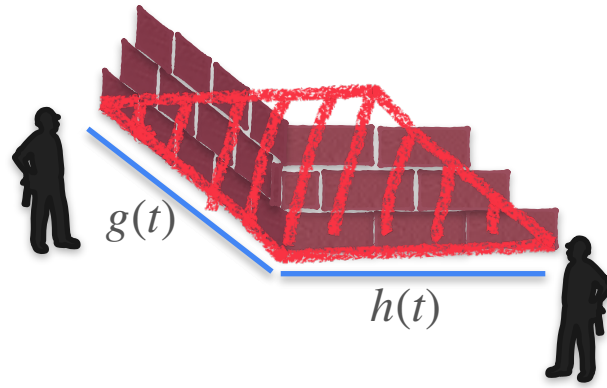


Product Rule



$$f(t) = g(t)h(t)$$

~~xxxx~~



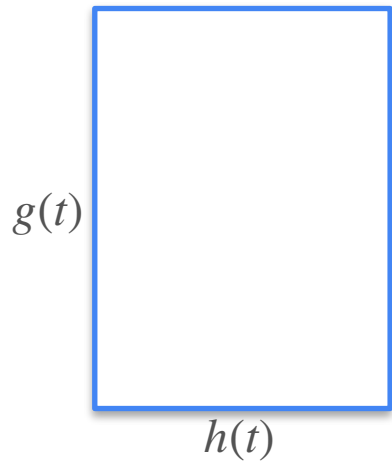
Product Rule

Product Rule

$$y = f(t) = g(t)h(t)$$

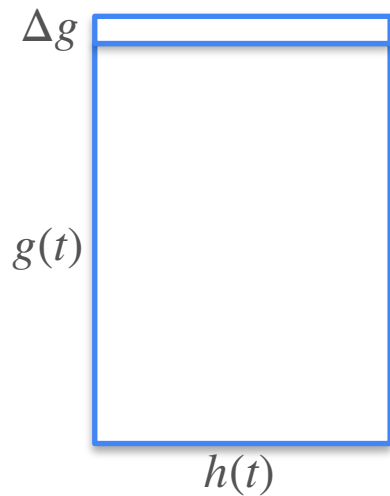
Product Rule

$$y = f(t) = g(t)h(t)$$



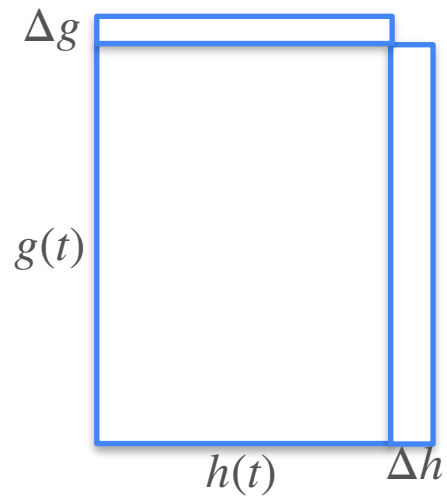
Product Rule

$$y = f(t) = g(t)h(t)$$



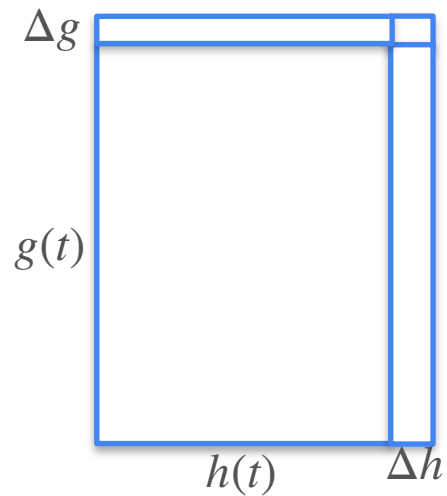
Product Rule

$$y = f(t) = g(t)h(t)$$



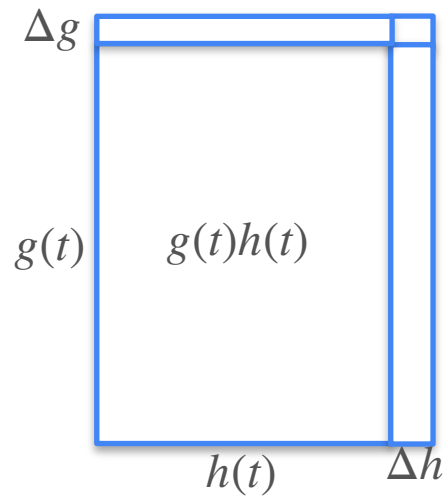
Product Rule

$$y = f(t) = g(t)h(t)$$



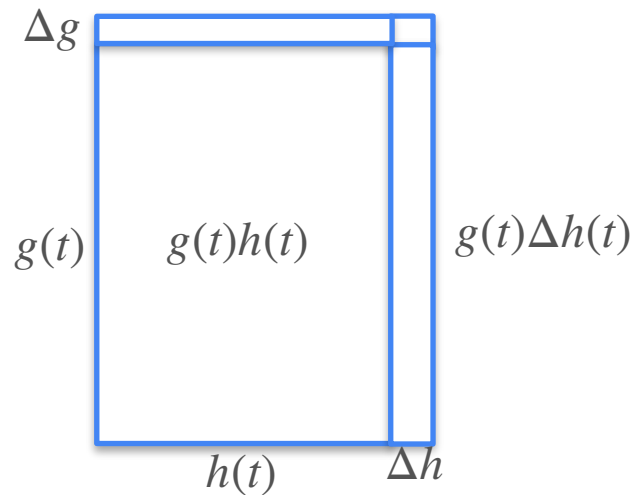
Product Rule

$$y = f(t) = g(t)h(t)$$



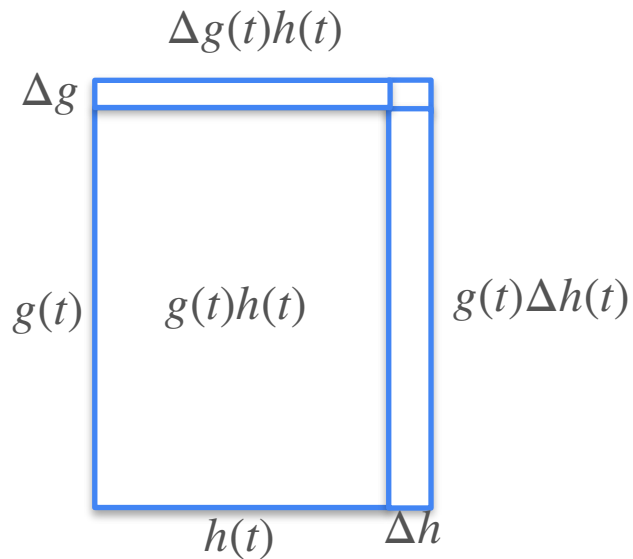
Product Rule

$$y = f(t) = g(t)h(t)$$



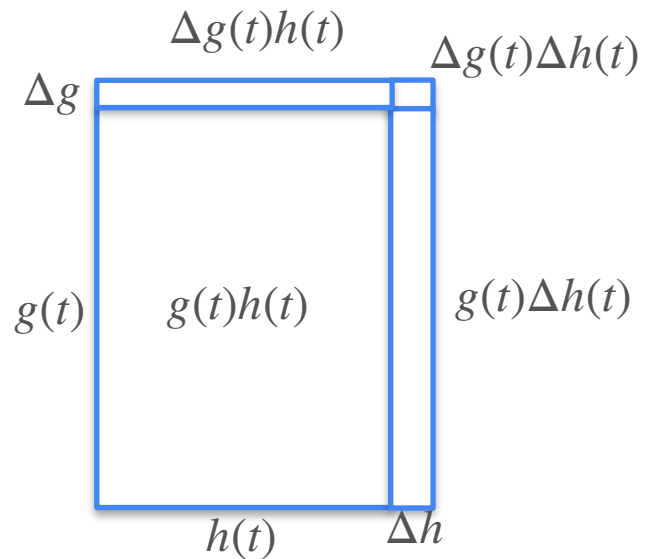
Product Rule

$$y = f(t) = g(t)h(t)$$

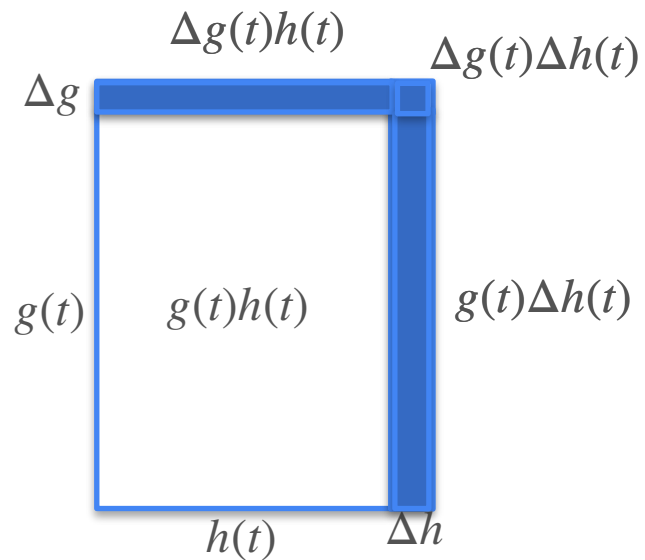


Product Rule

$$y = f(t) = g(t)h(t)$$



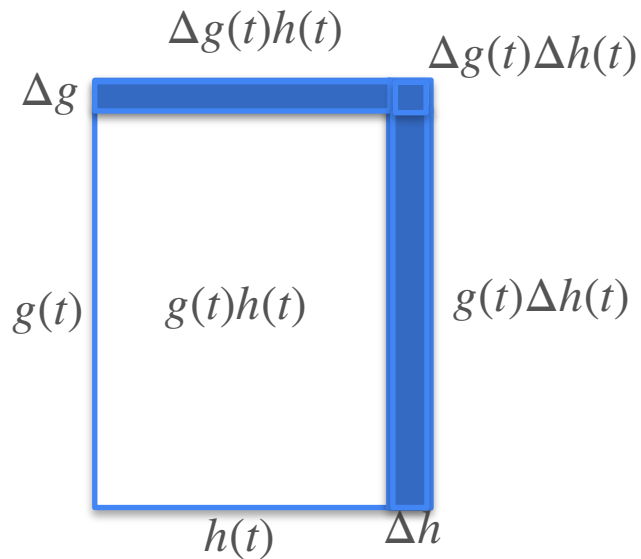
Product Rule



$$y = f(t) = g(t)h(t)$$

$$\frac{\Delta f(t)}{\Delta t}$$

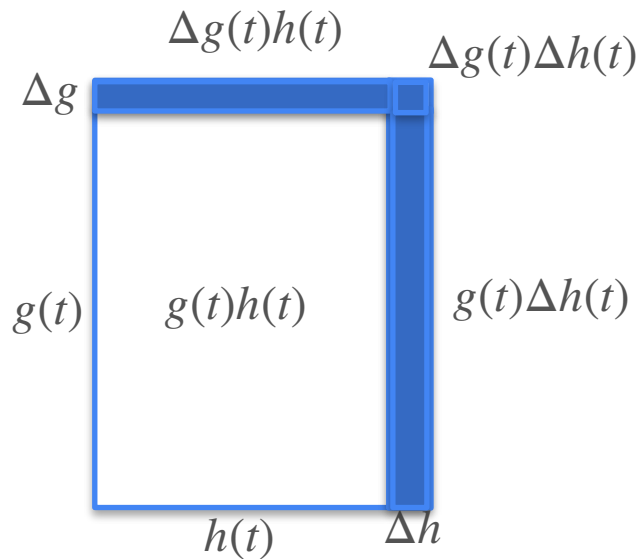
Product Rule



$$y = f(t) = g(t)h(t)$$

$$\frac{\Delta f(t)}{\Delta t} = \frac{\Delta g(t)h(t) + g(t)\Delta h(t) + \Delta g(t)\Delta h(t)}{\Delta t}$$

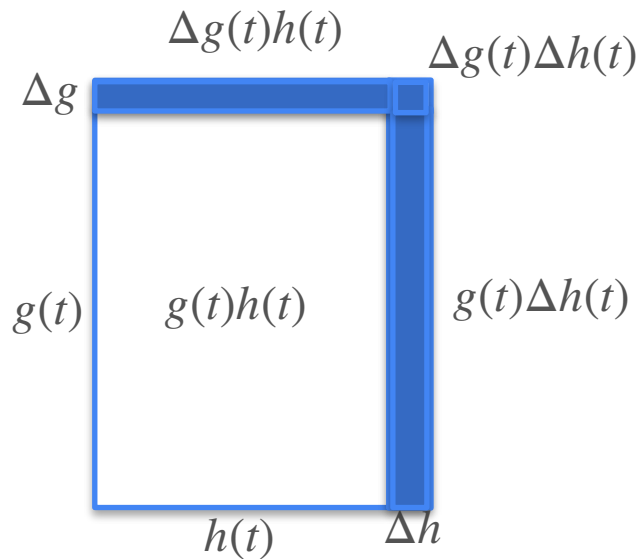
Product Rule



$$y = f(t) = g(t)h(t)$$

$$\begin{aligned}\frac{\Delta f(t)}{\Delta t} &= \frac{\Delta g(t)h(t) + g(t)\Delta h(t) + \Delta g(t)\Delta h(t)}{\Delta t} \\ &= \frac{\Delta g(t)}{\Delta t}h(t) + g(t)\frac{\Delta h(t)}{\Delta t} + \frac{\Delta g(t)\Delta h(t)}{\Delta t}\end{aligned}$$

Product Rule



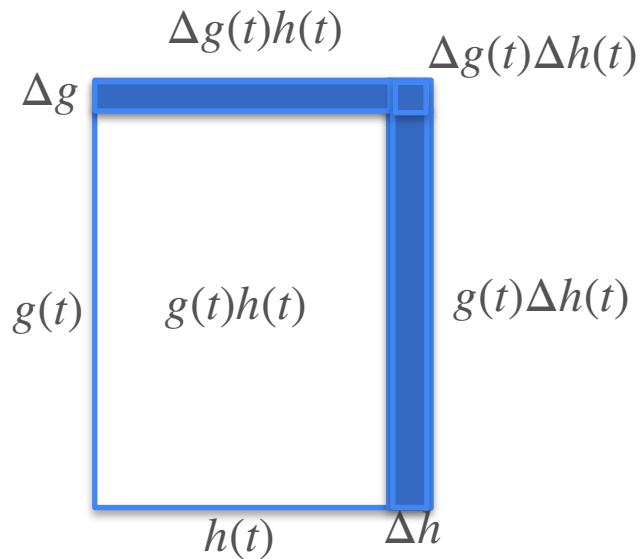
$$y = f(t) = g(t)h(t)$$

$$\frac{\Delta f(t)}{\Delta t} = \frac{\Delta g(t)h(t) + g(t)\Delta h(t) + \Delta g(t)\Delta h(t)}{\Delta t}$$

$$= \frac{\Delta g(t)}{\Delta t}h(t) + g(t)\frac{\Delta h(t)}{\Delta t} + \frac{\Delta g(t)\Delta h(t)}{\Delta t}$$

as $\Delta t \rightarrow 0$

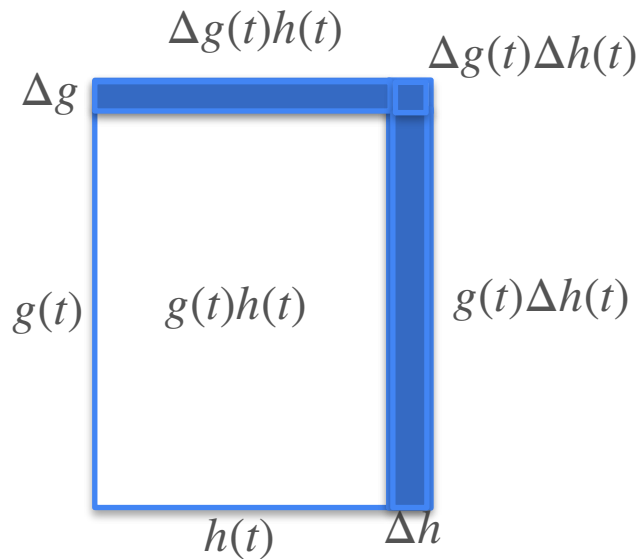
Product Rule



$$y = f(t) = g(t)h(t)$$

$$\begin{aligned} \frac{f'(t)}{\Delta t} &= \frac{\Delta g(t)h(t) + g(t)\Delta h(t) + \Delta g(t)\Delta h(t)}{\Delta t} \\ &= \frac{\Delta g(t)}{\Delta t}h(t) + g(t)\frac{\Delta h(t)}{\Delta t} + \frac{\Delta g(t)\Delta h(t)}{\Delta t} \\ &\quad \text{as } \Delta t \rightarrow 0 \end{aligned}$$

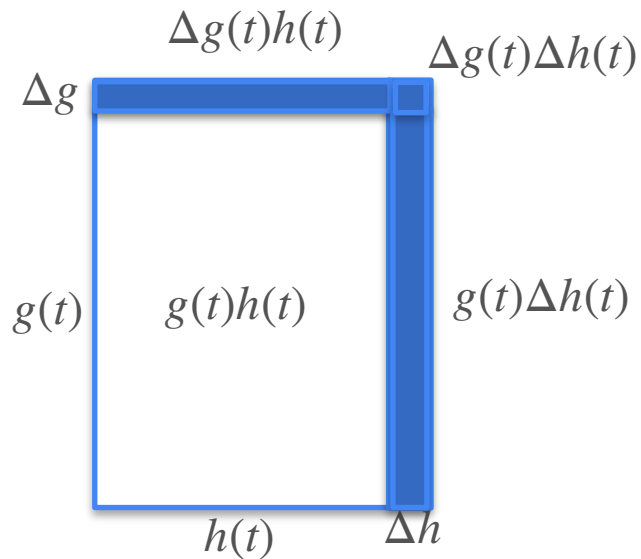
Product Rule



$$y = f(t) = g(t)h(t)$$

$$\begin{aligned} \frac{f'(t)}{\Delta t} &= \frac{\Delta g(t)h(t) + g(t)\Delta h(t) + \Delta g(t)\Delta h(t)}{\Delta t} \\ &= \frac{g'(t)h(t)}{\Delta t} h(t) + g(t) \frac{\Delta h(t)}{\Delta t} + \frac{\Delta g(t)\Delta h(t)}{\Delta t} \\ &\quad \text{as } \Delta t \rightarrow 0 \end{aligned}$$

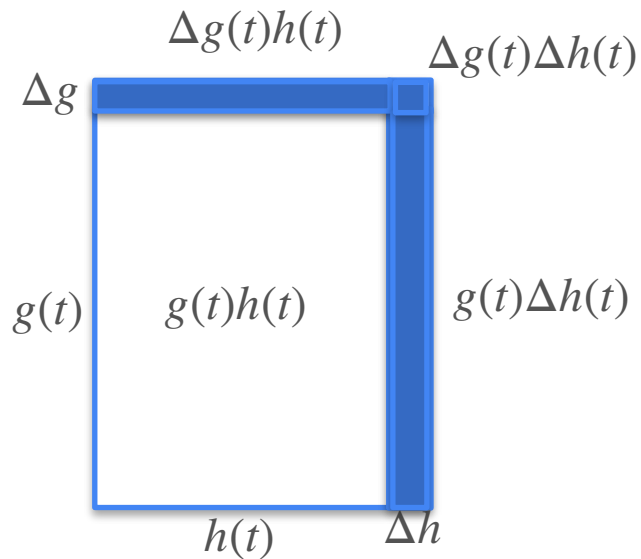
Product Rule



$$y = f(t) = g(t)h(t)$$

$$\begin{aligned} \frac{f'(t)}{\Delta t} &= \frac{\Delta g(t)h(t) + g(t)\Delta h(t) + \Delta g(t)\Delta h(t)}{\Delta t} \\ &= \frac{g'(t)h(t)}{\Delta t} h(t) + g(t) \frac{\Delta h(t)}{\Delta t} + \frac{\Delta g(t)\Delta h(t)}{\Delta t} \\ &\quad \text{as } \Delta t \rightarrow 0 \end{aligned}$$

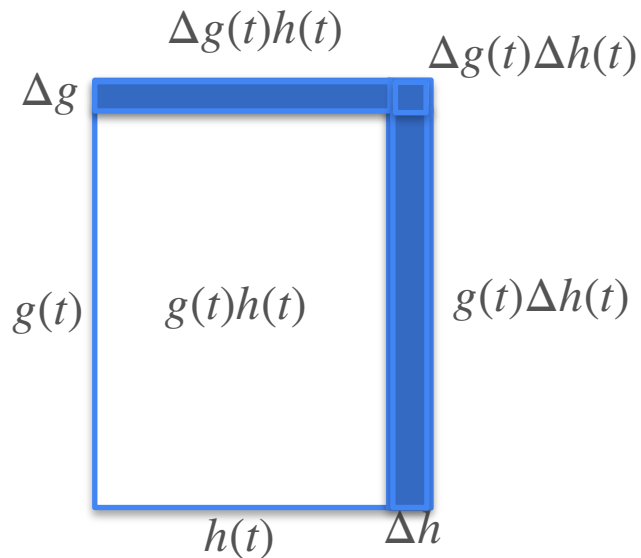
Product Rule



$$y = f(t) = g(t)h(t)$$

$$\begin{aligned} \frac{\Delta f(t)}{\Delta t} &= \frac{\Delta g(t)h(t) + g(t)\Delta h(t) + \Delta g(t)\Delta h(t)}{\Delta t} \\ &= \frac{g'(t)h(t)}{\Delta t} h(t) + g(t) \frac{\Delta h(t)}{\Delta t} + \frac{0}{\Delta t} \\ &\text{as } \Delta t \rightarrow 0 \end{aligned}$$

Product Rule



$$y = f(t) = g(t)h(t)$$

$$\begin{aligned}\frac{\Delta f(t)}{\Delta t} &= \frac{\Delta g(t)h(t) + g(t)\Delta h(t) + \Delta g(t)\Delta h(t)}{\Delta t} \\ &= \frac{g'(t)h(t)}{\Delta t} h(t) + g(t) \frac{\Delta h(t)}{\Delta t} + \frac{0}{\Delta t}\end{aligned}$$

as $\Delta t \rightarrow 0$

$$f'(t) = g'(t)h(t) + g(t)h'(t)$$



DeepLearning.AI

Derivatives and Optimization

**Properties of the derivative:
The chain rule**

The Chain Rule

The Chain Rule

$$h(t)$$

The Chain Rule

$$g(h(t))$$

The Chain Rule

$$\frac{d}{dt} g(h(t))$$

The Chain Rule

$$\frac{d}{dt} g(h(t))$$
$$\frac{dh}{dt}$$

The Chain Rule

$$\frac{d}{dt} g(h(t))$$
$$\frac{dg}{dh} \frac{dh}{dt}$$

The Chain Rule

$$\begin{aligned}\frac{d}{dt} g(h(t)) \\ = \frac{dg}{dh} \cdot \frac{dh}{dt}\end{aligned}$$

The Chain Rule

$$g(h(t))$$

$$\frac{dg}{dh} \frac{dh}{dt}$$

The Chain Rule

$$f(g(h(t)))$$

$$\frac{dg}{dh} \frac{dh}{dt}$$

The Chain Rule

$$\frac{d}{dt} f(g(h(t)))$$
$$\frac{dg}{dh} \frac{dh}{dt}$$

The Chain Rule

$$\frac{d}{dt} f(g(h(t)))$$
$$\frac{df}{dg} \frac{dg}{dh} \frac{dh}{dt}$$

The Chain Rule

$$\begin{aligned} \frac{d}{dt} f(g(h(t))) \\ = \frac{df}{dg} \cdot \frac{dg}{dh} \cdot \frac{dh}{dt} \end{aligned}$$

The Chain Rule

$$\frac{d}{dt} g(h(t)) = \frac{dg}{dh} \cdot \frac{dh}{dt}$$

The Chain Rule

$$\frac{d}{dt} g(h(t)) = \frac{dg}{dh} \cdot \frac{dh}{dt}$$
$$=$$

The Chain Rule

$$\begin{aligned}\frac{d}{dt} g(h(t)) &= \frac{dg}{dh} \cdot \frac{dh}{dt} \\ &= h'(t)\end{aligned}$$

The Chain Rule

$$\begin{aligned}\frac{d}{dt} g(h(t)) &= \frac{dg}{dh} \cdot \frac{dh}{dt} \\ &= g'(h(t)) \cdot h'(t)\end{aligned}$$

The Chain Rule

$$\frac{d}{dt} f(g(h(t))) = \frac{df}{dg} \cdot \frac{dg}{dh} \cdot \frac{dh}{dt}$$

The Chain Rule

$$\frac{d}{dt} f(g(h(t))) = \frac{df}{dg} \cdot \frac{dg}{dh} \cdot \frac{dh}{dt}$$
$$=$$

The Chain Rule

$$\begin{aligned} \frac{d}{dt} f(g(h(t))) &= \frac{df}{dg} \cdot \frac{dg}{dh} \cdot \frac{dh}{dt} \\ &= h'(t) \end{aligned}$$

The Chain Rule

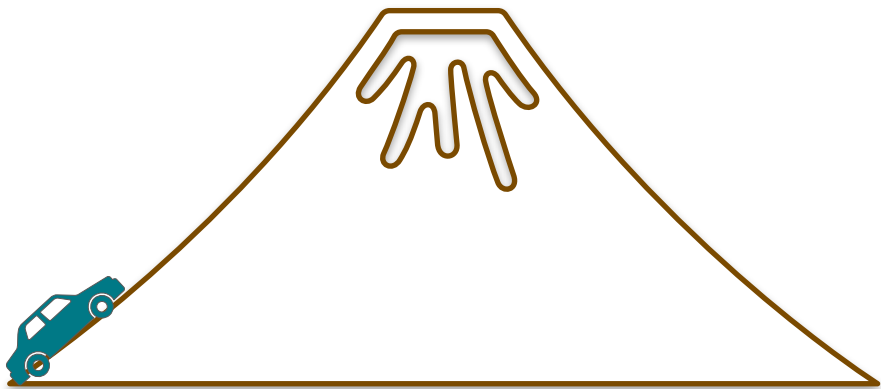
$$\begin{aligned}\frac{d}{dt} f(g(h(t))) &= \frac{df}{dg} \cdot \frac{dg}{dh} \cdot \frac{dh}{dt} \\ &= g'(h(t)) \cdot h'(t)\end{aligned}$$

The Chain Rule

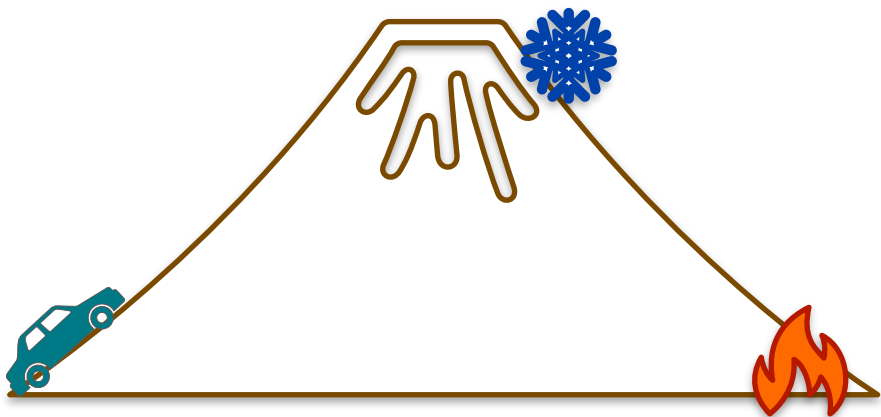
$$\begin{aligned}\frac{d}{dt} f(g(h(t))) &= \frac{df}{dg} \cdot \frac{dg}{dh} \cdot \frac{dh}{dt} \\ &= f'(g(h(t))) \cdot g'(h(t)) \cdot h'(t)\end{aligned}$$

Idea of Chain Rule

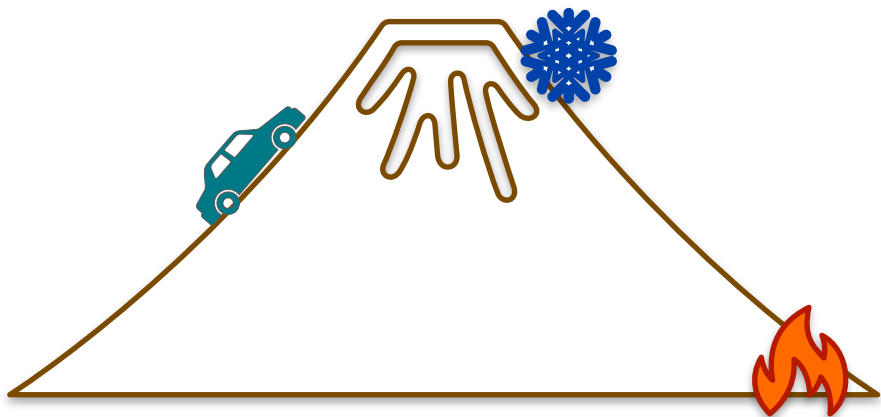
Idea of Chain Rule



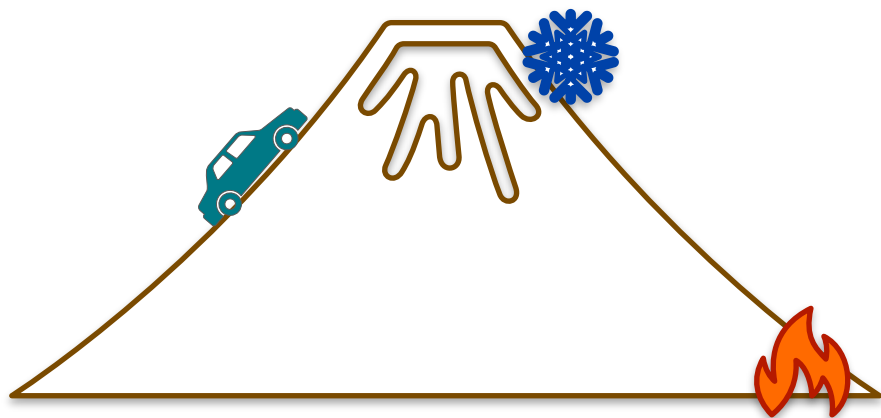
Idea of Chain Rule



Idea of Chain Rule

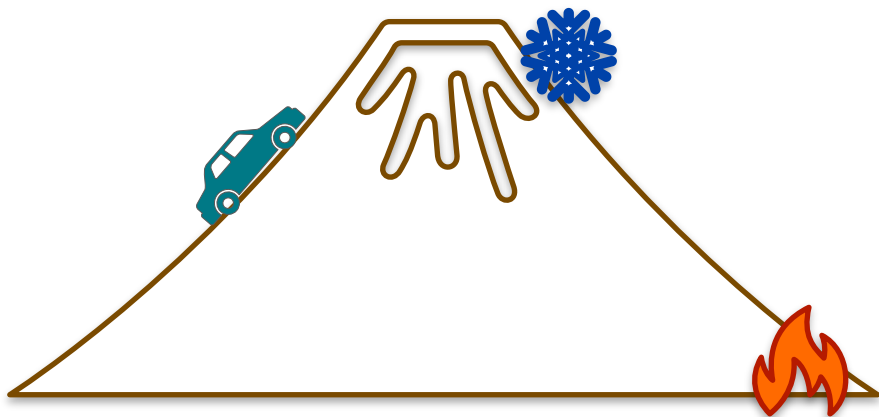


Idea of Chain Rule



Temperature changes w.r.t. height $\frac{dT}{dh}$

Idea of Chain Rule



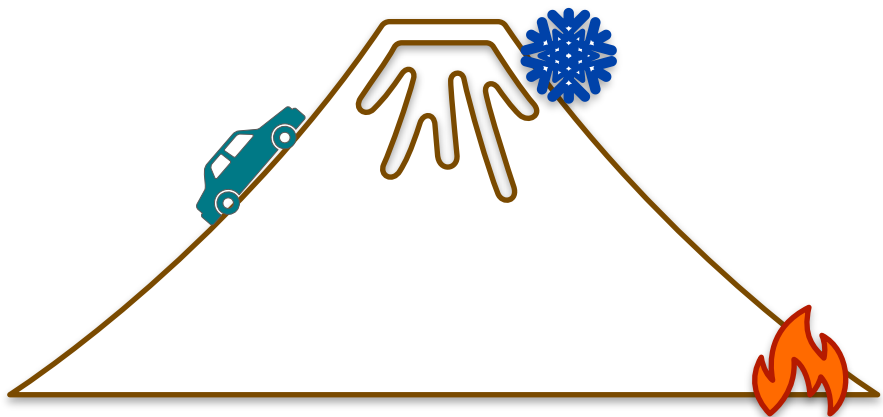
Temperature changes w.r.t. height

$$\frac{dT}{dh}$$

height changes w.r.t. time

$$\frac{dh}{dt}$$

Idea of Chain Rule



Temperature changes w.r.t. height

$$\frac{dT}{dh}$$

height changes w.r.t. time

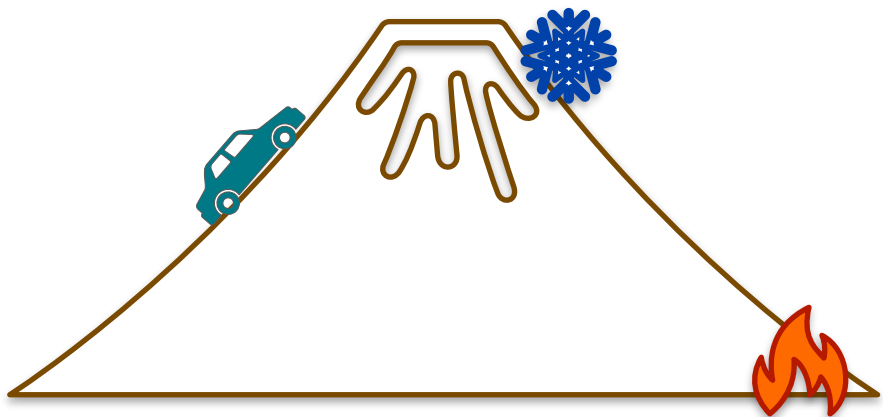
$$\frac{dh}{dt}$$



Temperature changes w.r.t. time

$$\frac{dT}{dt}$$

Idea of Chain Rule



Temperature changes w.r.t. height

$$\frac{dT}{dh}$$

height changes w.r.t. time

$$\frac{dh}{dt}$$



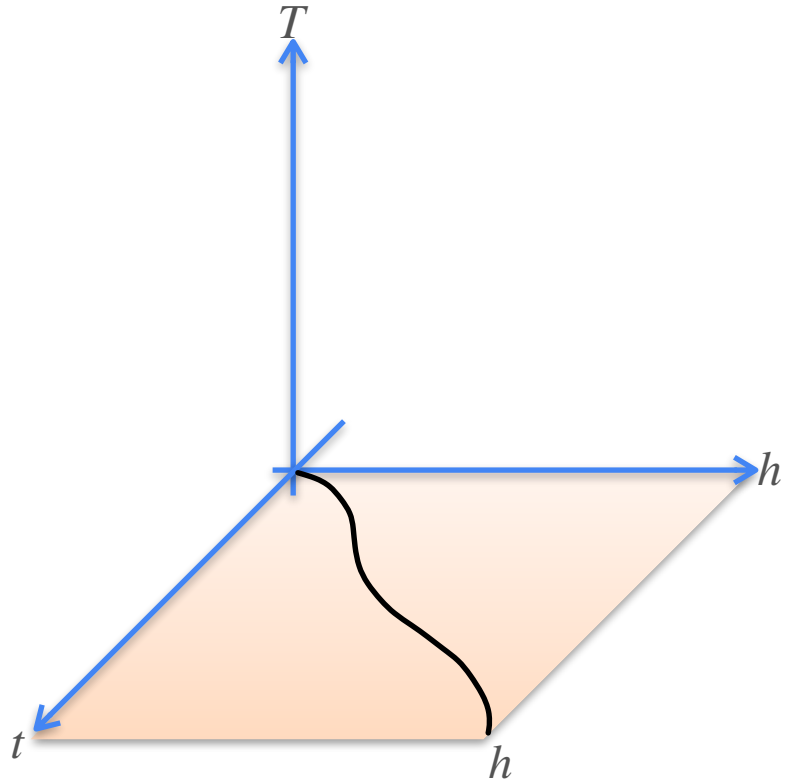
Temperature changes w.r.t. time

$$\frac{dT}{dt}$$

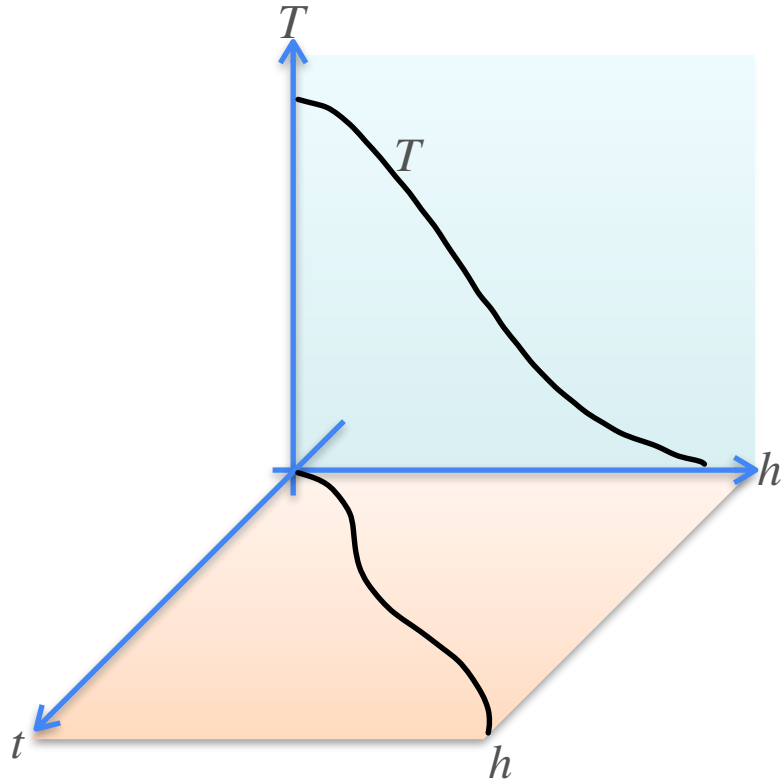
$$\frac{dT}{dt} = \frac{dT}{dh} \frac{dh}{dt}$$

Chain Rule

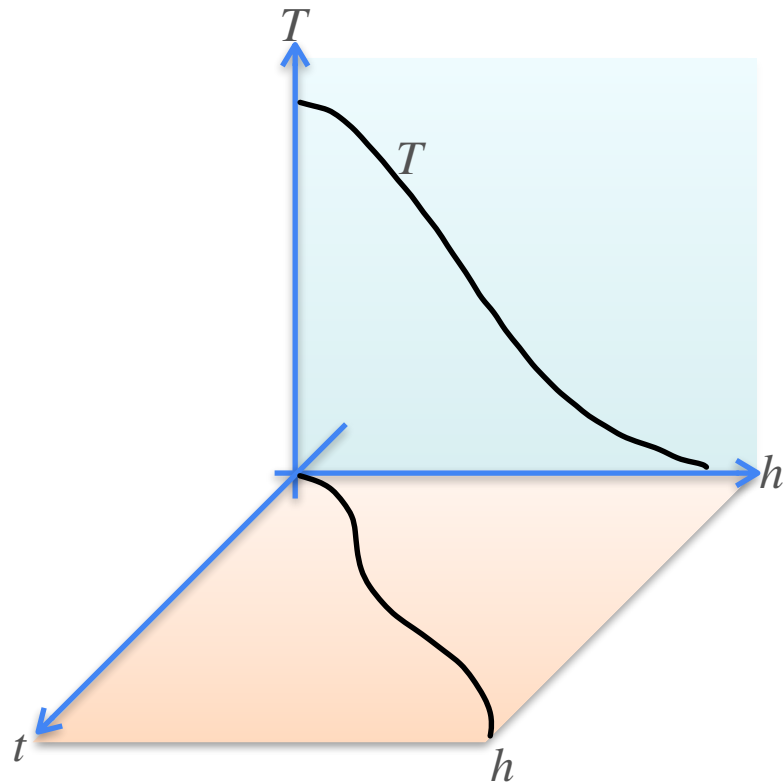
Chain Rule



Chain Rule

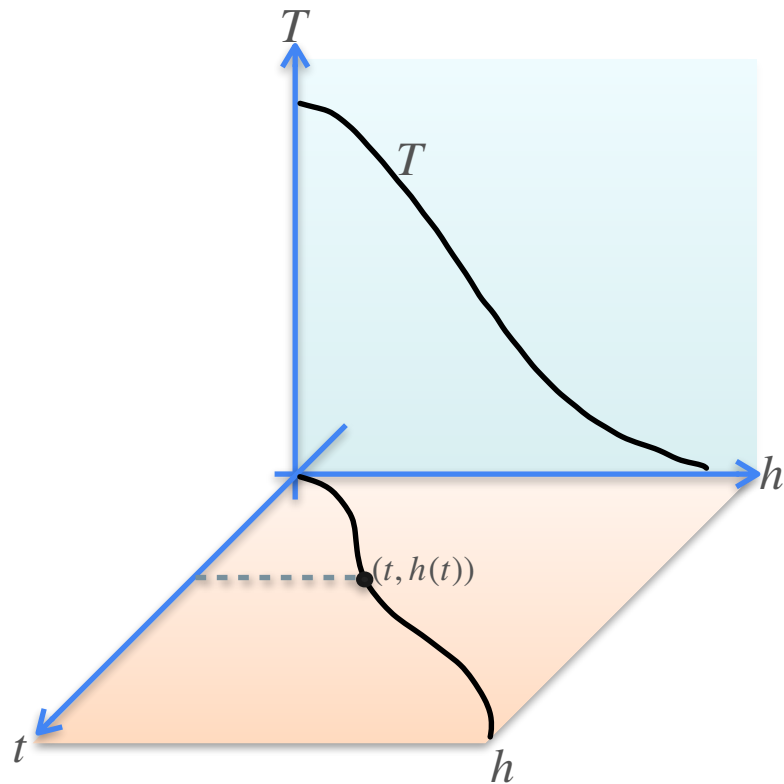


Chain Rule



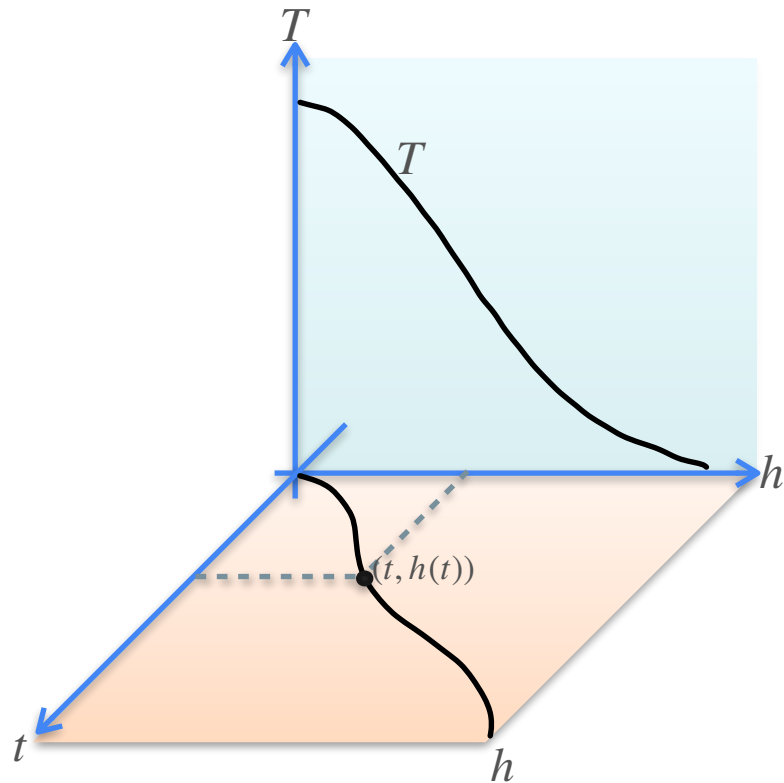
$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

Chain Rule



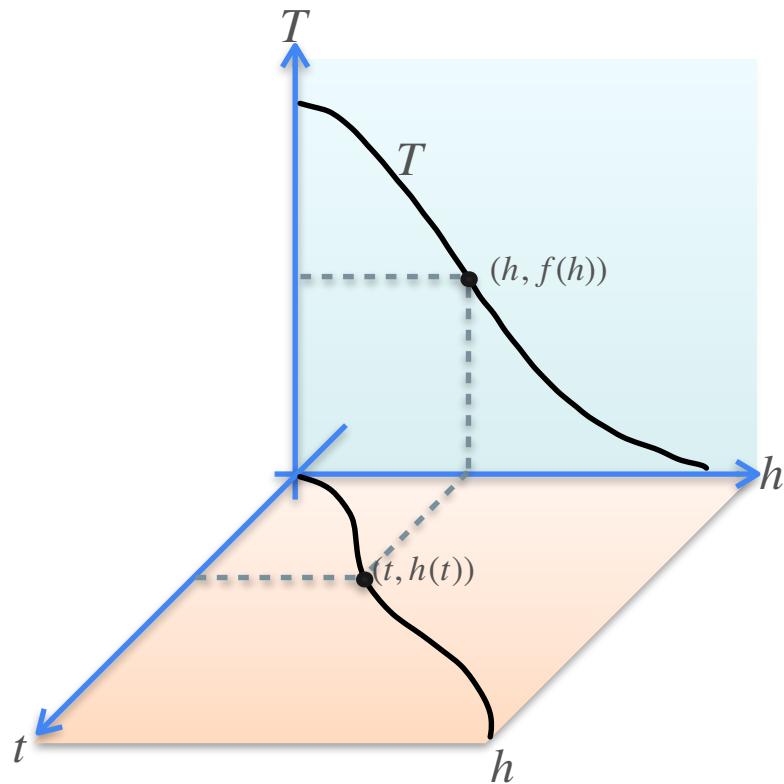
$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

Chain Rule



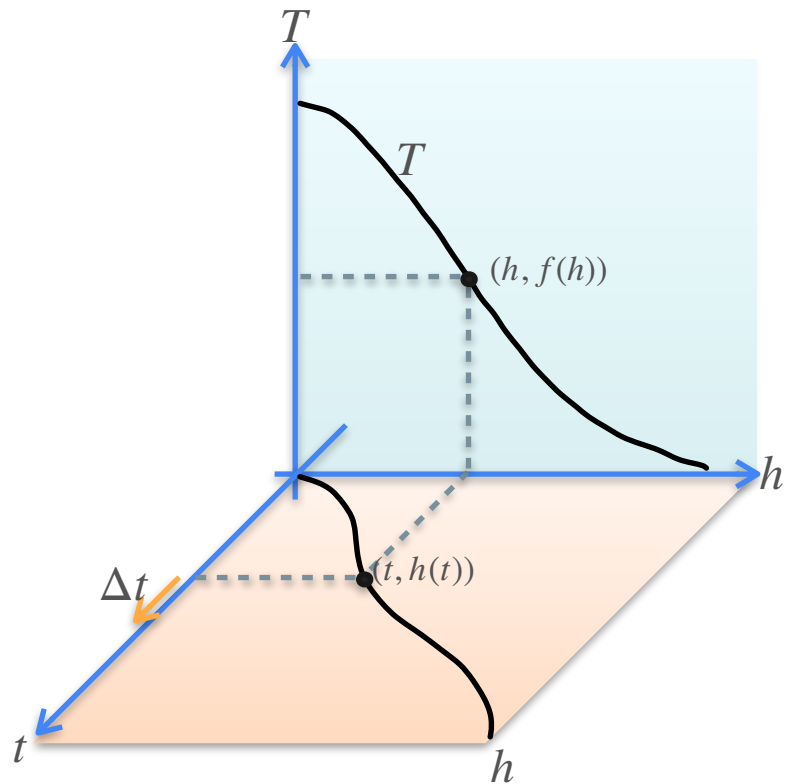
$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

Chain Rule



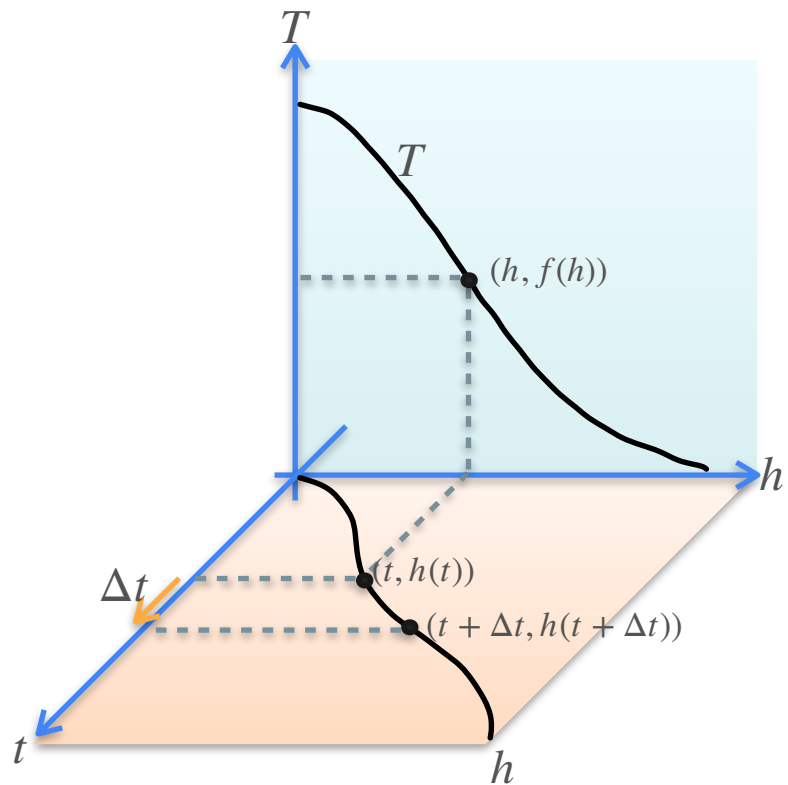
$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

Chain Rule



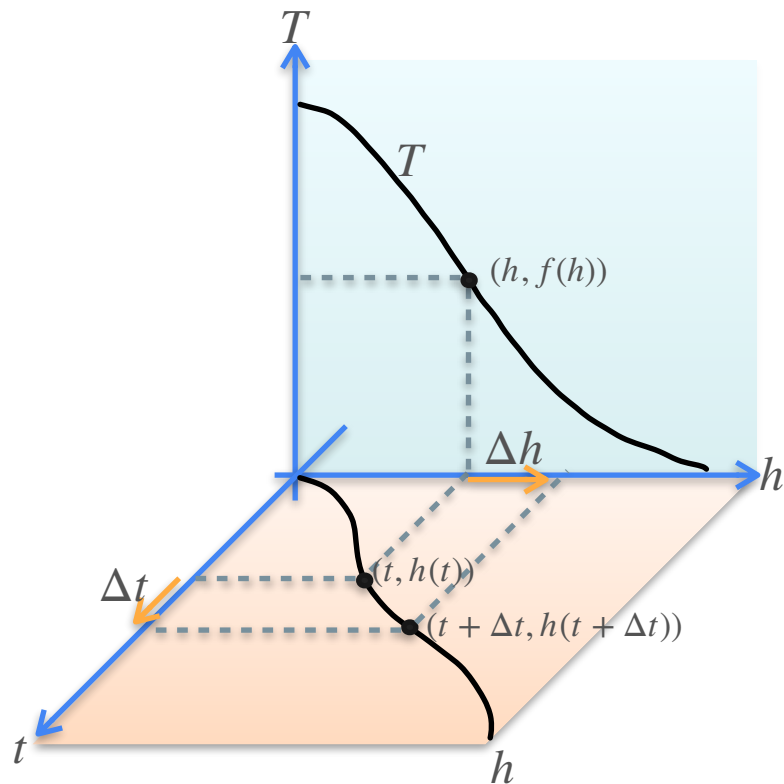
$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

Chain Rule



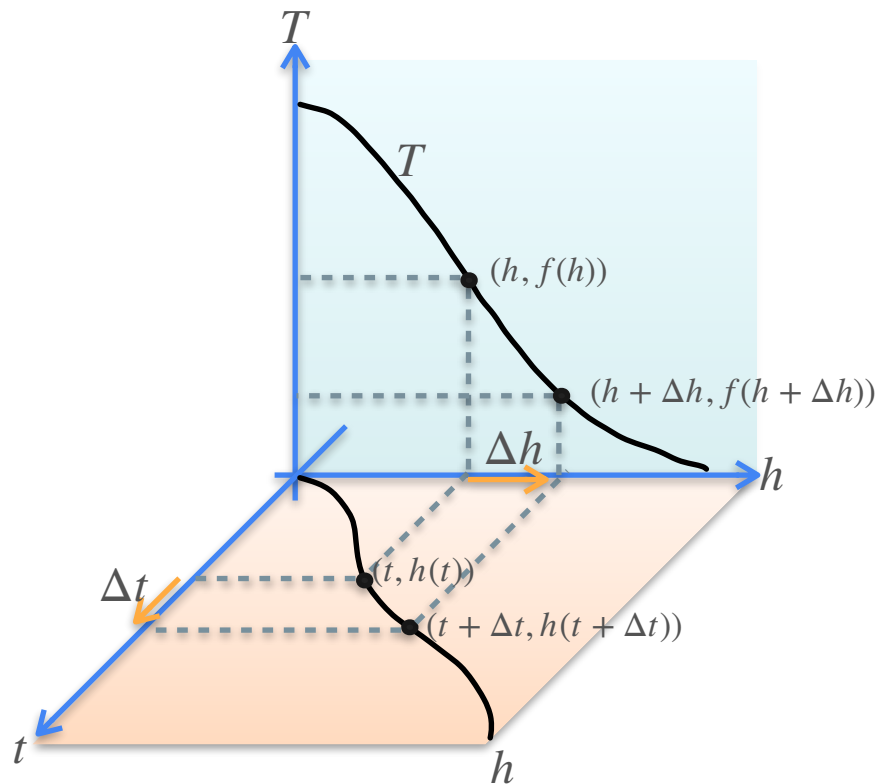
$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

Chain Rule



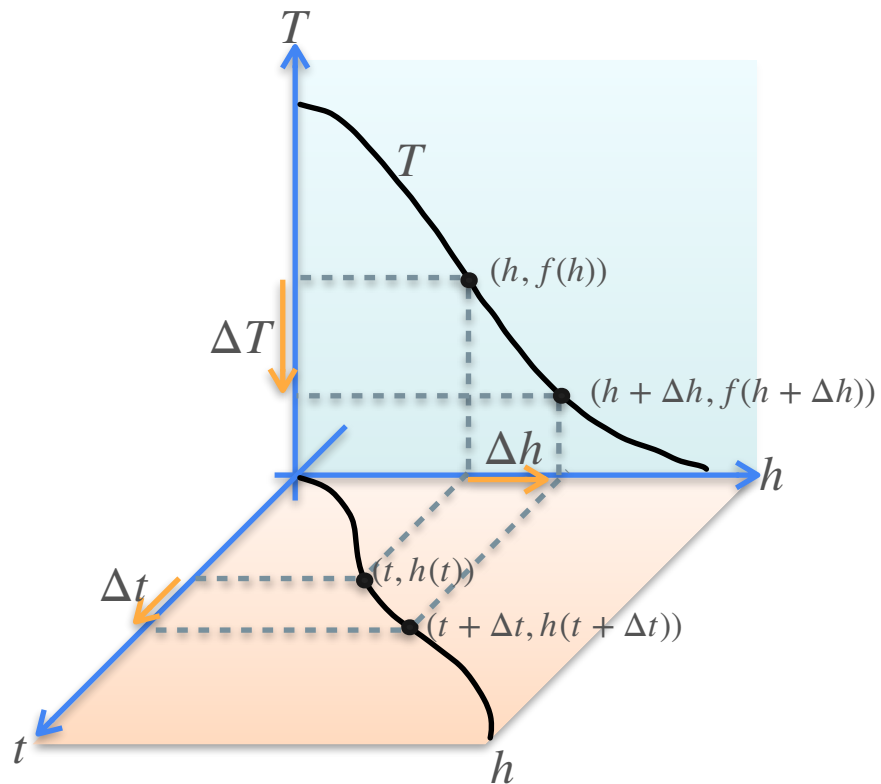
$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

Chain Rule



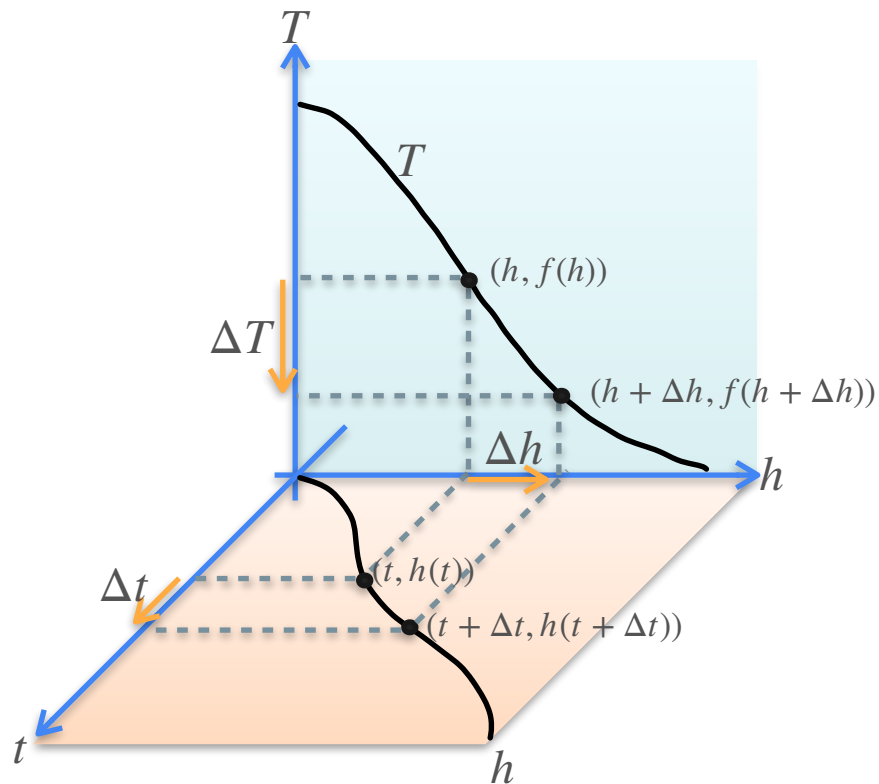
$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

Chain Rule



$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

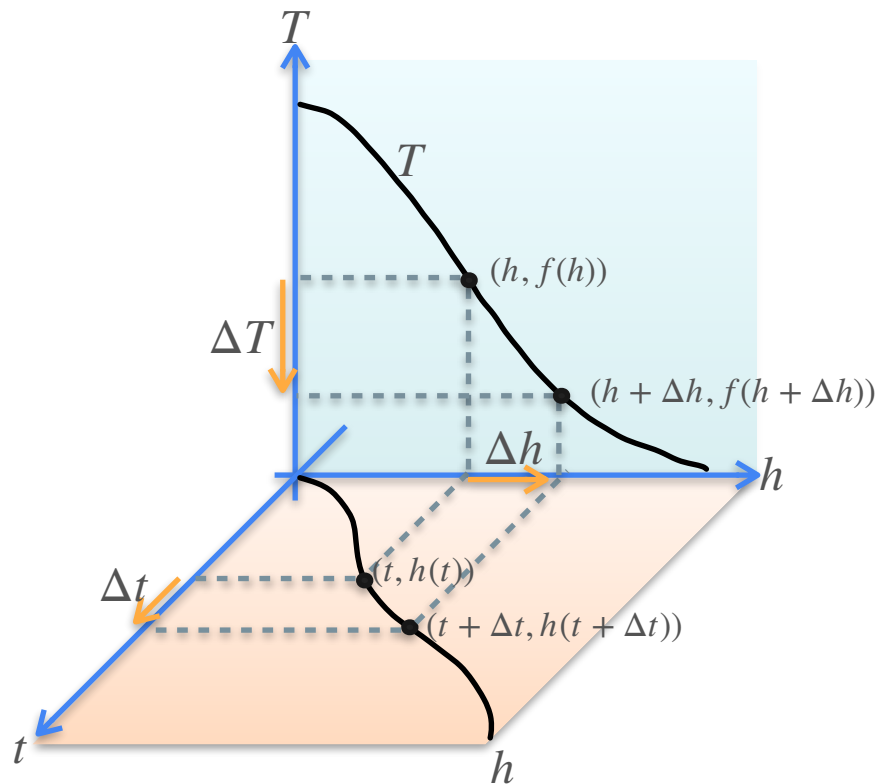
Chain Rule



$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

$$\frac{\Delta T}{\Delta t} =$$

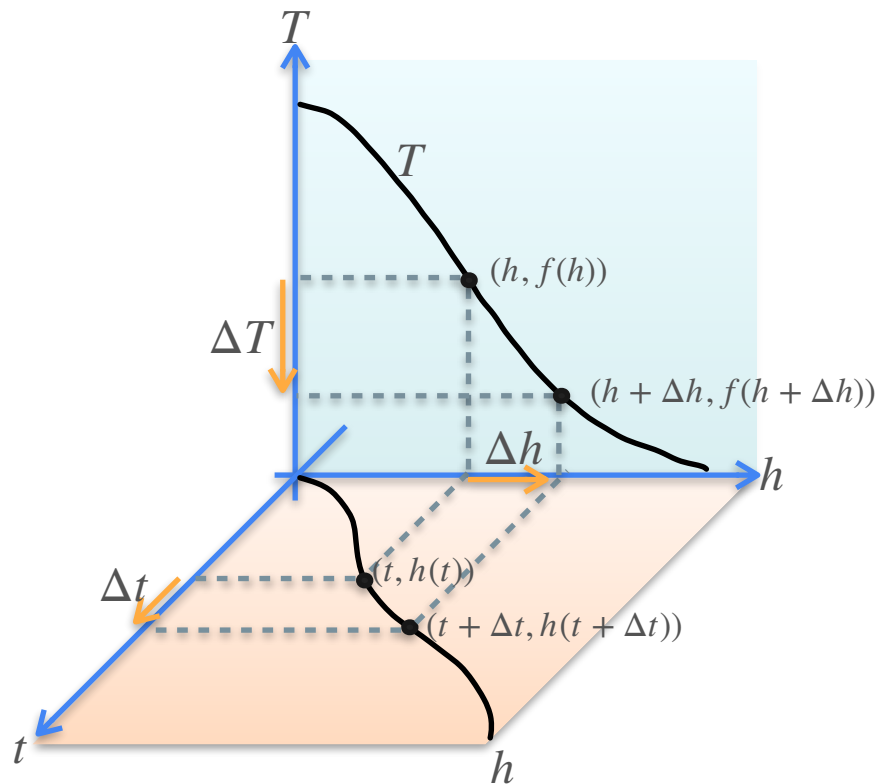
Chain Rule



$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

$$\frac{\Delta T}{\Delta t} = \frac{\Delta h}{\Delta t}$$

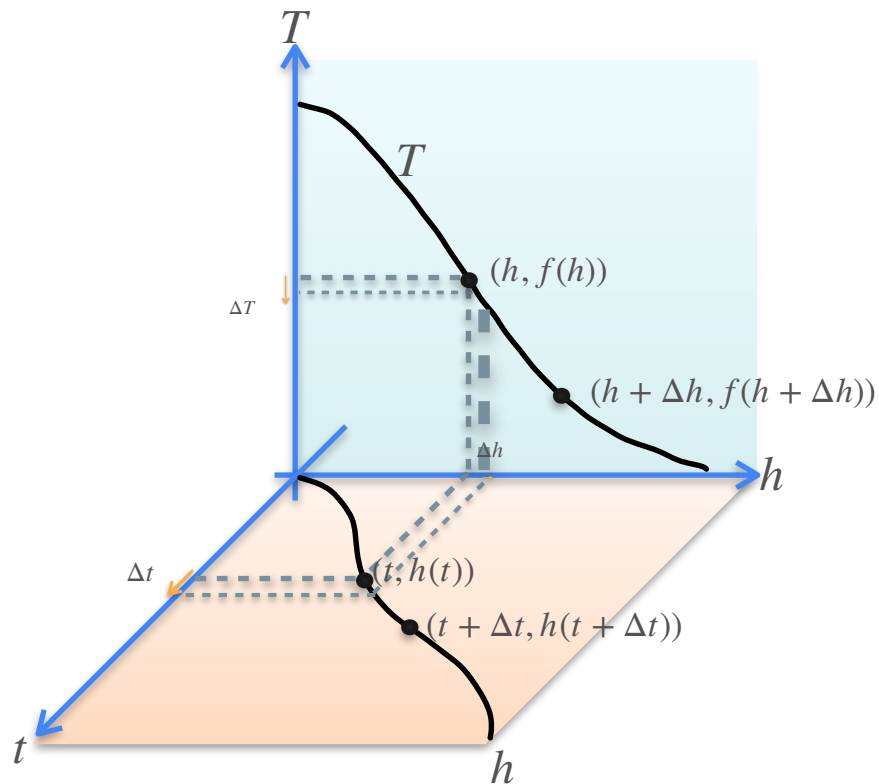
Chain Rule



$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

$$\frac{\Delta T}{\Delta t} = \frac{\Delta T}{\Delta h} \frac{\Delta h}{\Delta t}$$

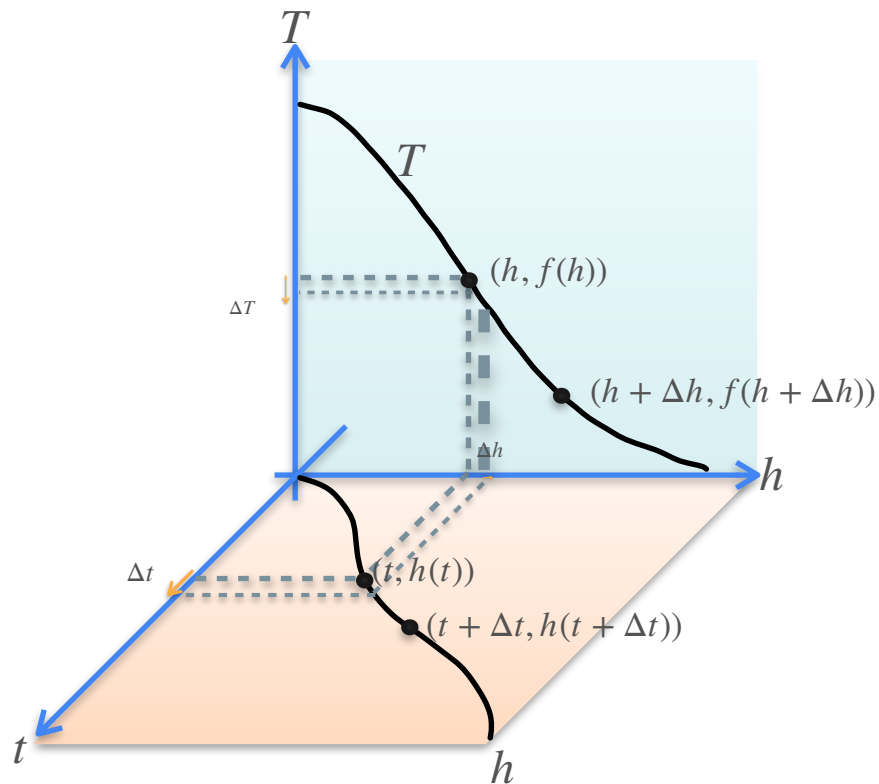
Chain Rule



$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

$$\frac{\Delta T}{\Delta t} = \frac{\Delta T}{\Delta h} \frac{\Delta h}{\Delta t}$$

Chain Rule

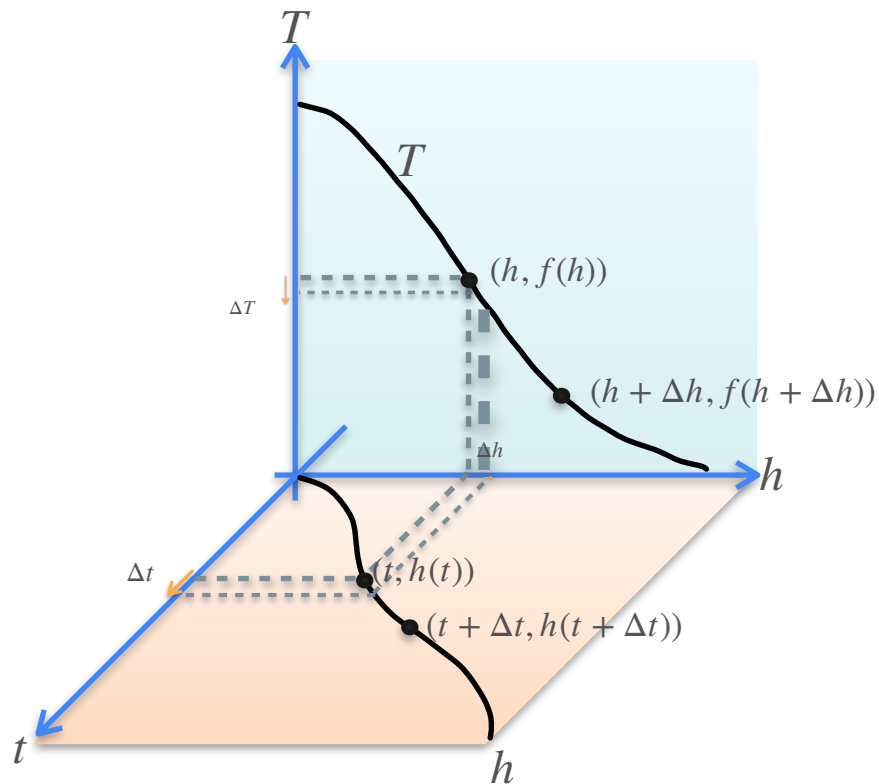


$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

$$\frac{\Delta T}{\Delta t} = \frac{\Delta T}{\Delta h} \frac{\Delta h}{\Delta t}$$

$$\frac{dT}{dt}$$

Chain Rule

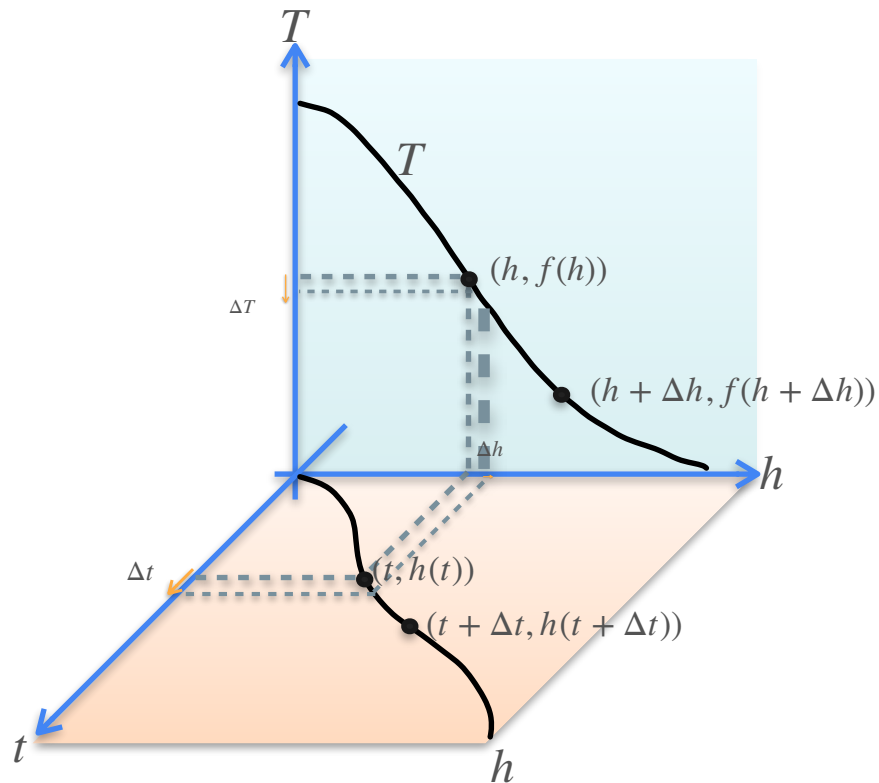


$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

$$\frac{\Delta T}{\Delta t} = \frac{\Delta T}{\Delta h} \frac{\Delta h}{\Delta t}$$

$$\frac{dT}{dt} = \frac{dT}{dh} \frac{dh}{dt}$$

Chain Rule



$$\Delta t \rightarrow \Delta h \rightarrow \Delta T$$

$$\frac{\Delta T}{\Delta t} = \frac{\Delta T}{\Delta h} \frac{\Delta h}{\Delta t}$$

$$\frac{dT}{dt} = \frac{dT}{dh} \frac{dh}{dt}$$



DeepLearning.AI

Derivatives and Optimization

Introduction to optimization

Motivation for Optimization

Motivation for Optimization



Motivation for Optimization



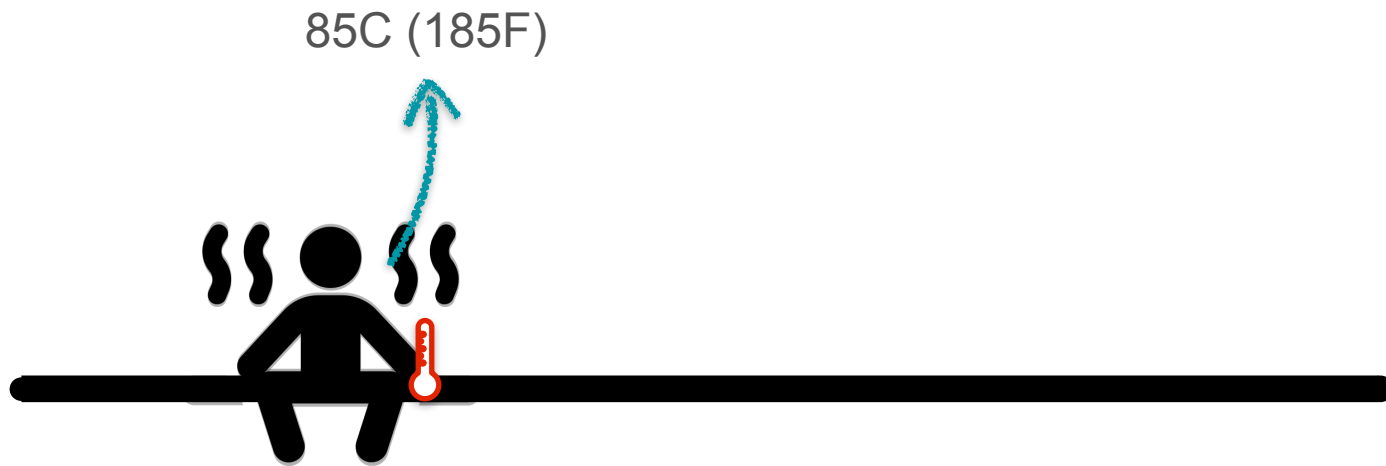
Motivation for Optimization



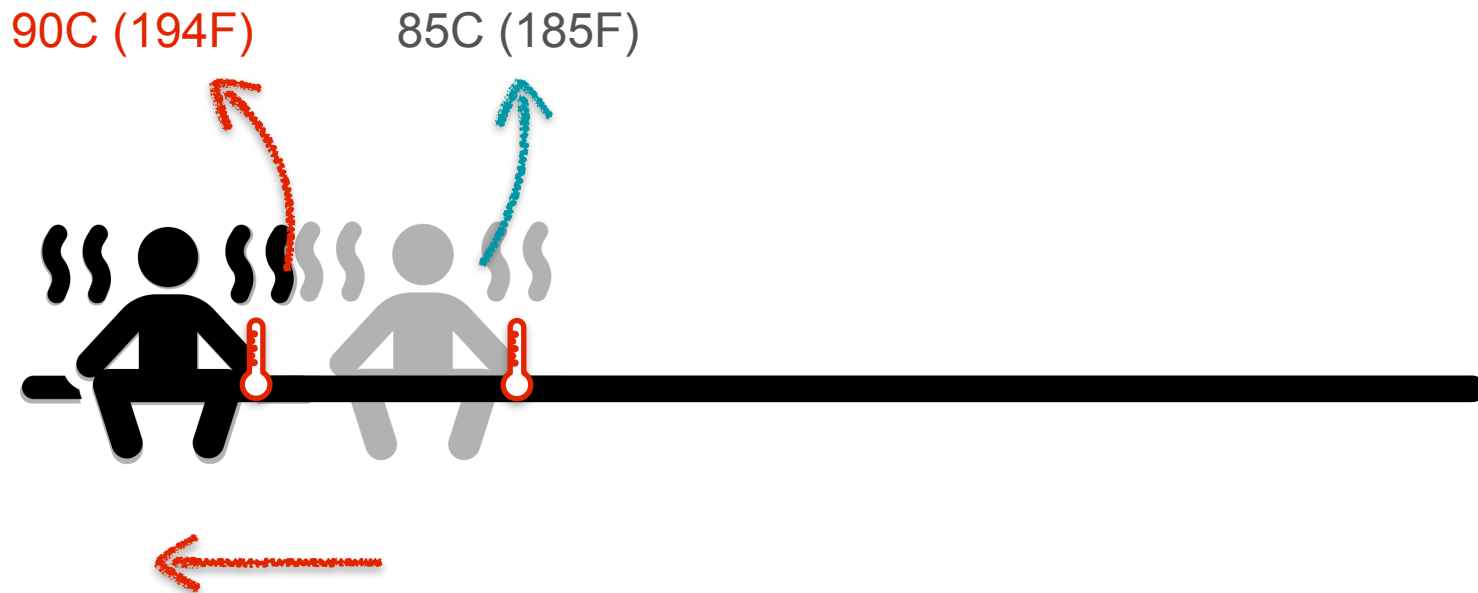
Motivation for Optimization



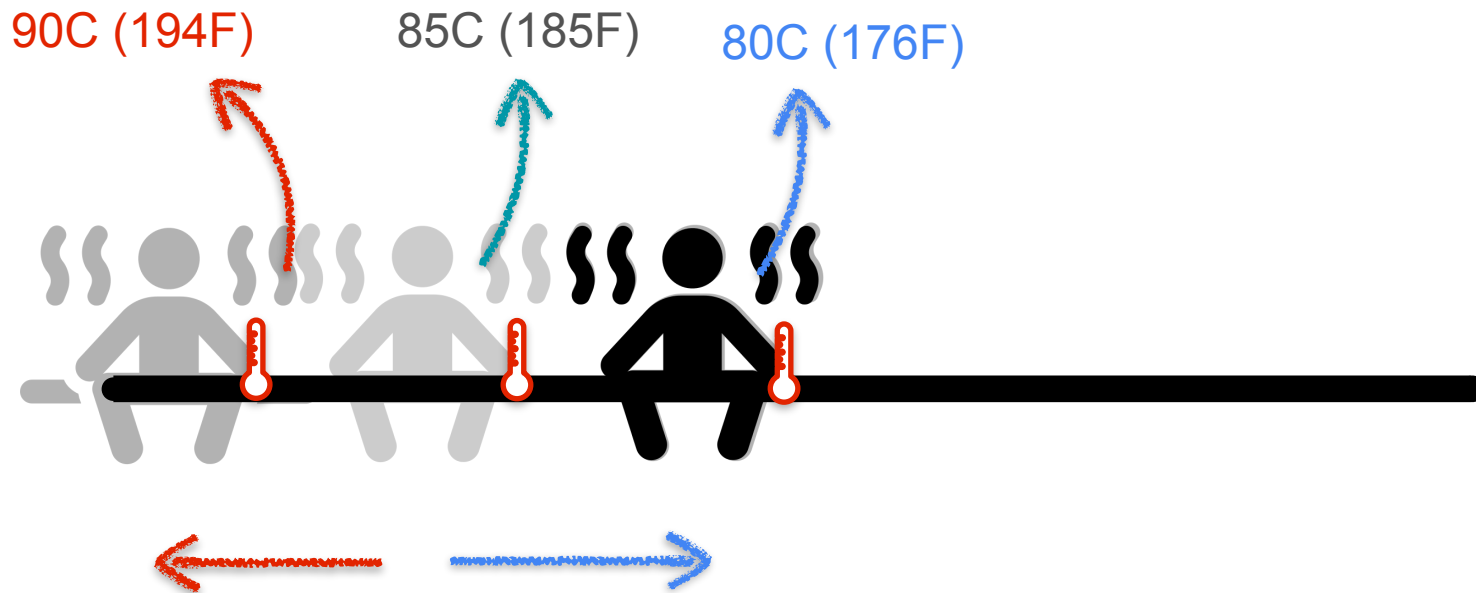
Motivation for Optimization



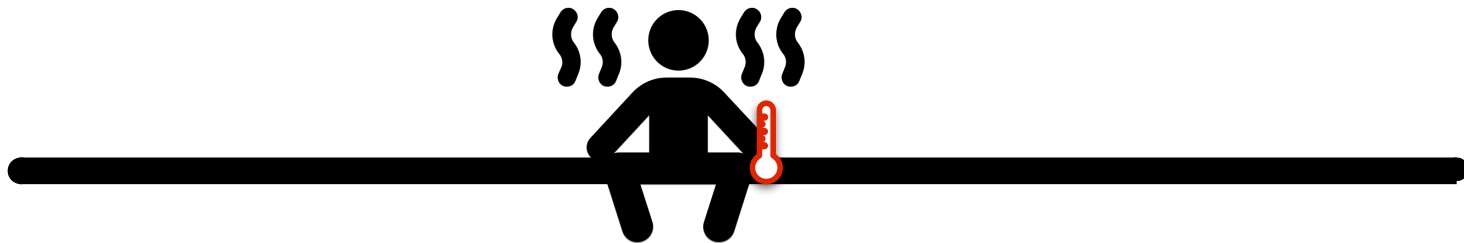
Motivation for Optimization



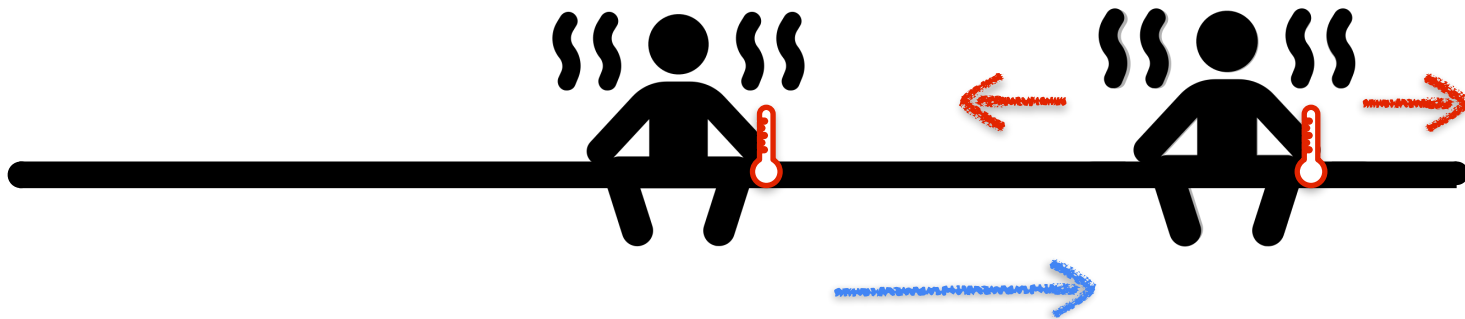
Motivation for Optimization



Motivation for Optimization



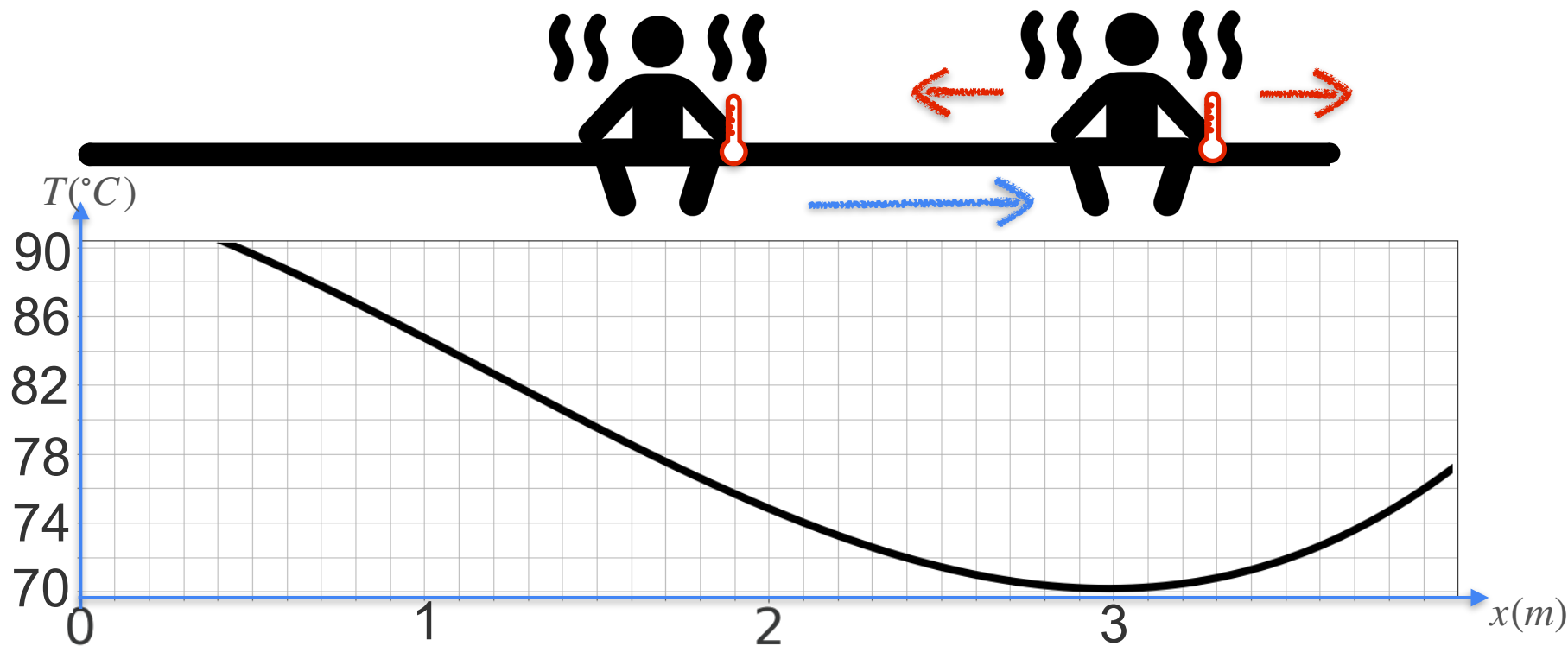
Motivation for Optimization



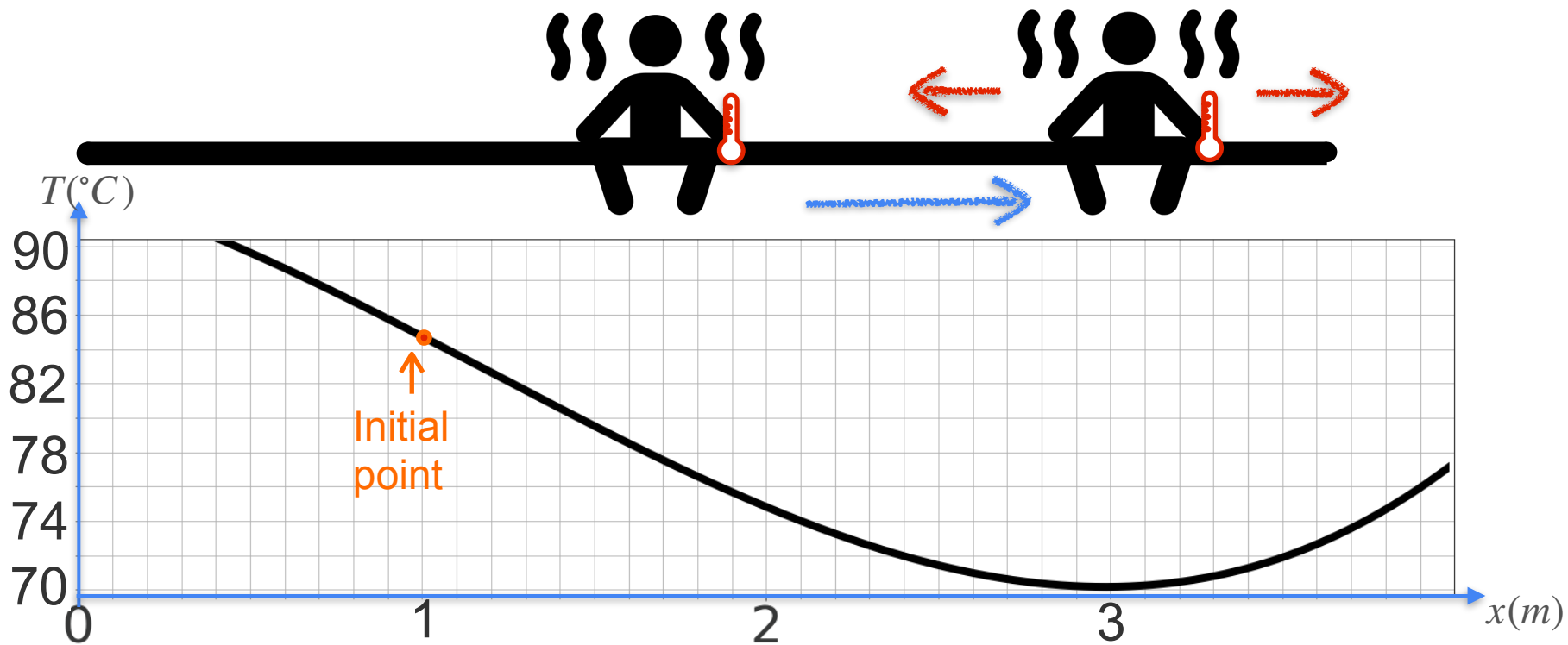
Motivation for Optimization



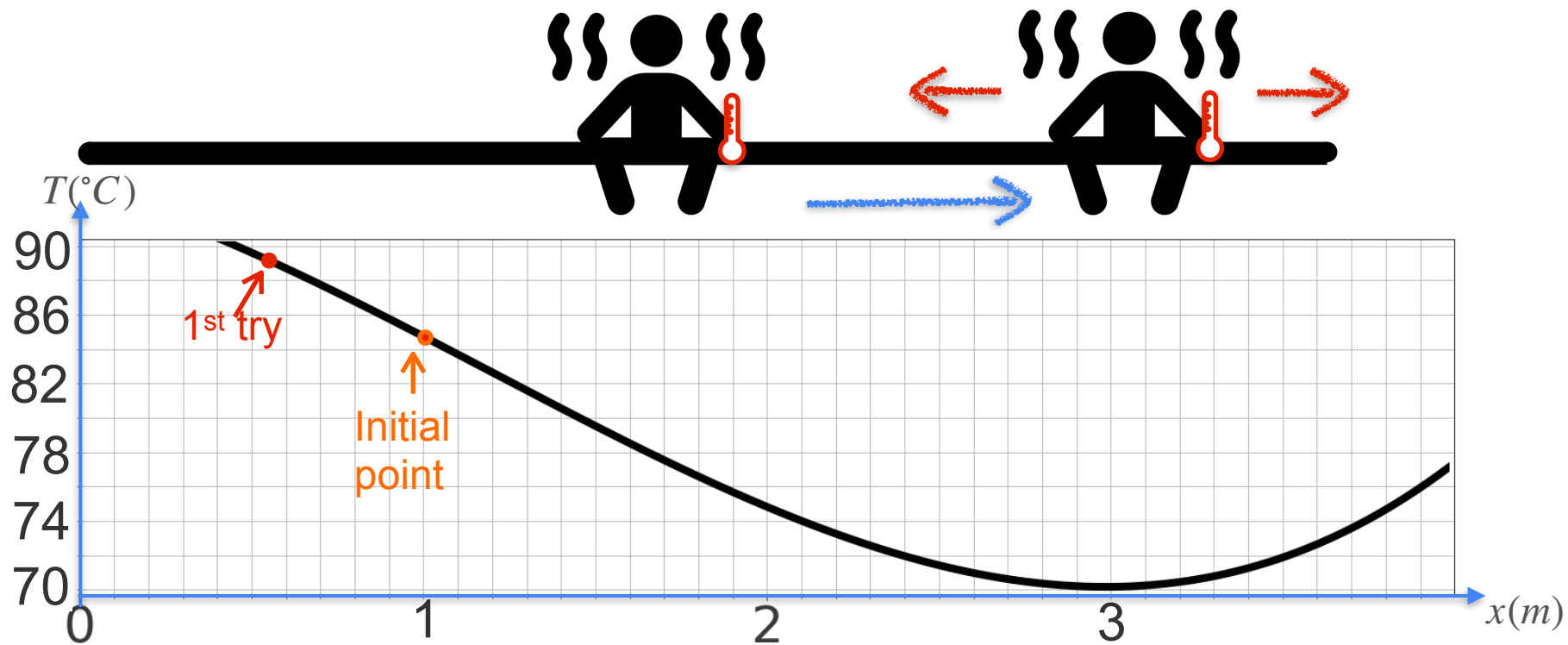
Motivation for Optimization



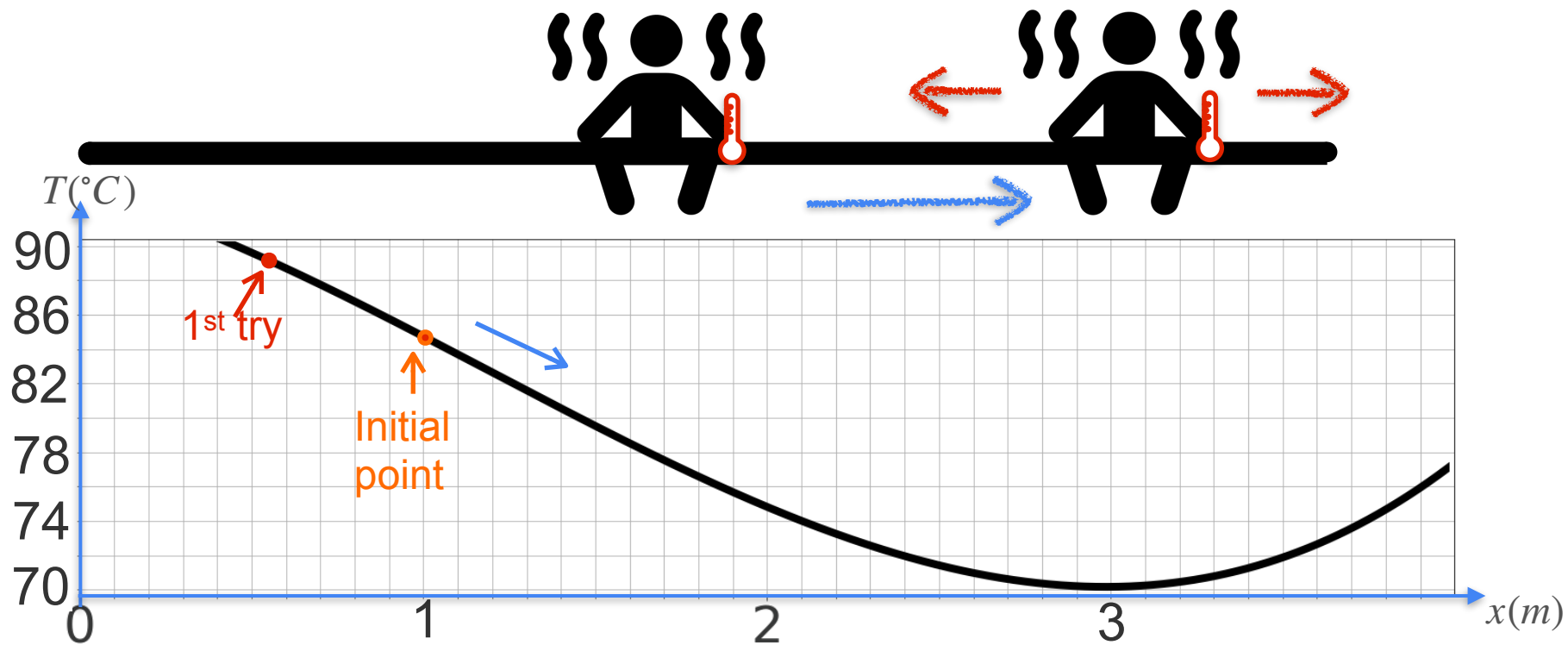
Motivation for Optimization



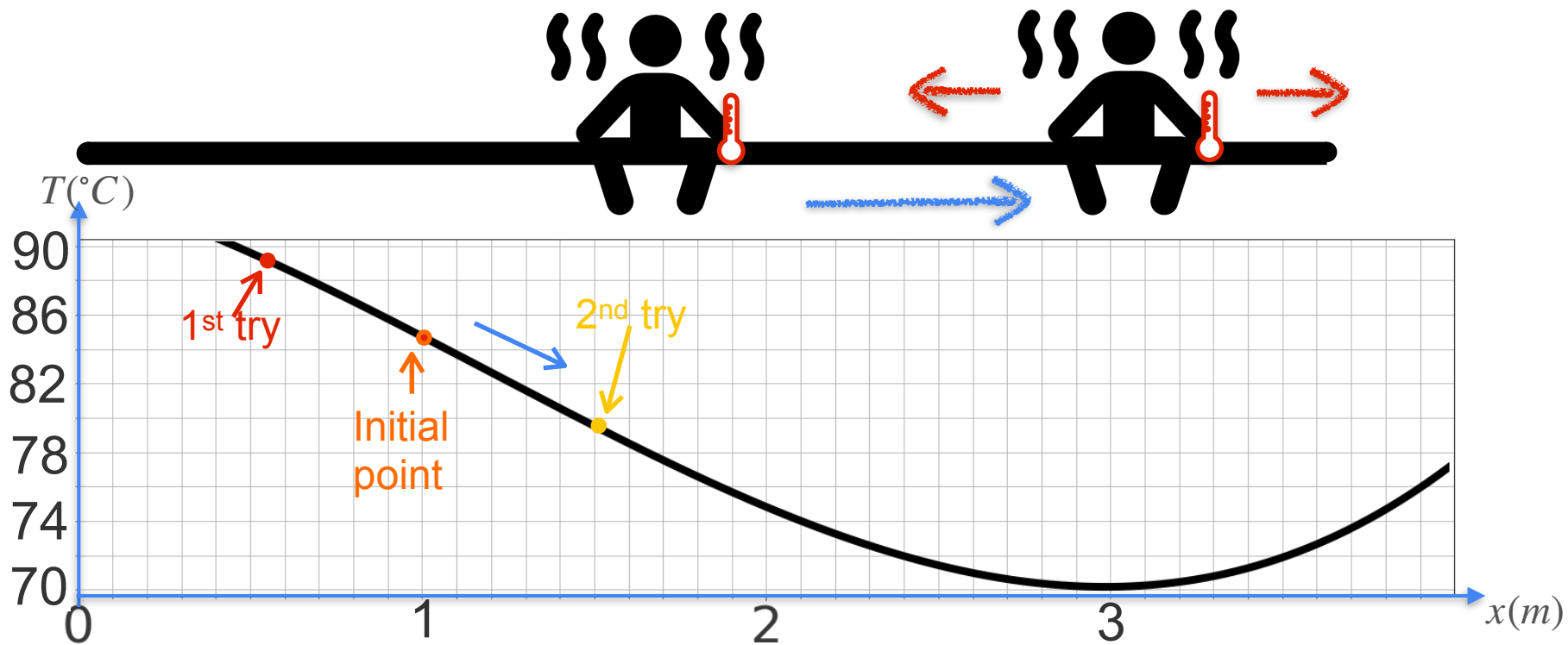
Motivation for Optimization



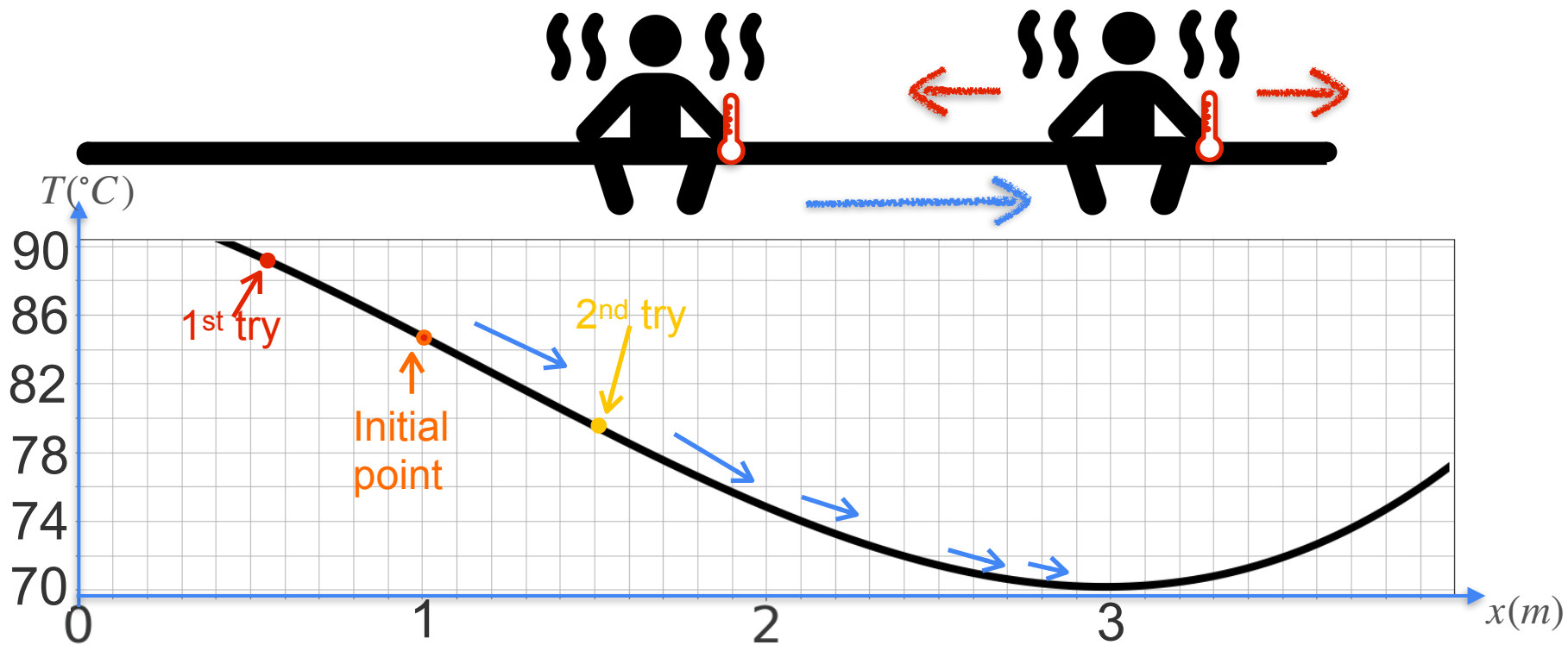
Motivation for Optimization



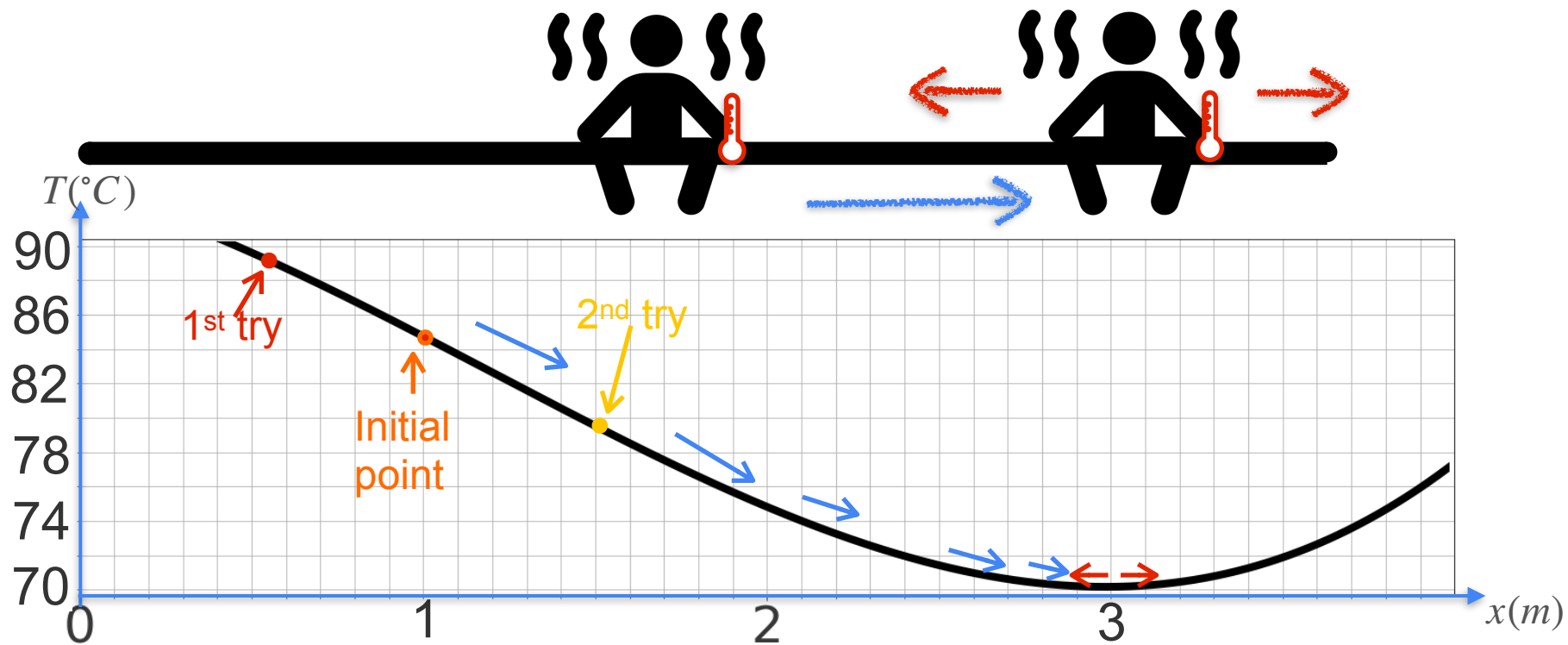
Motivation for Optimization



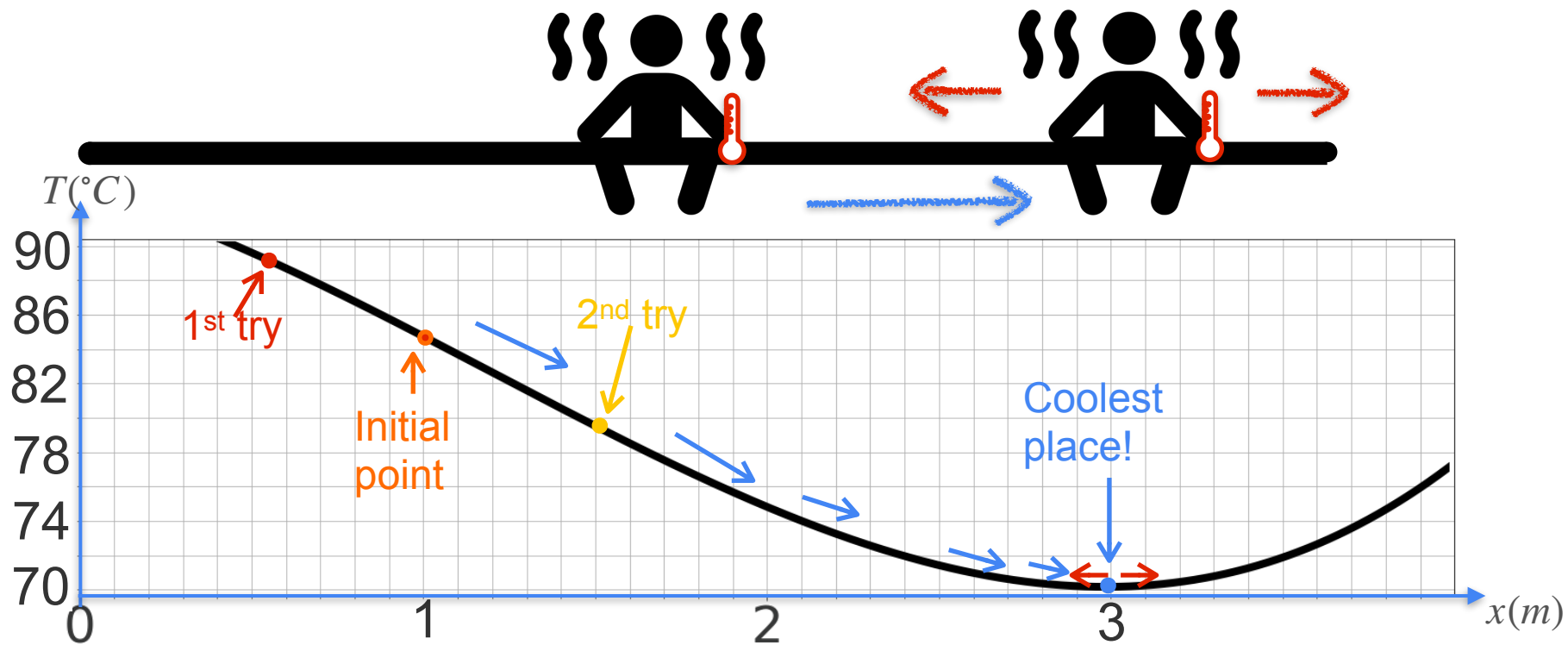
Motivation for Optimization



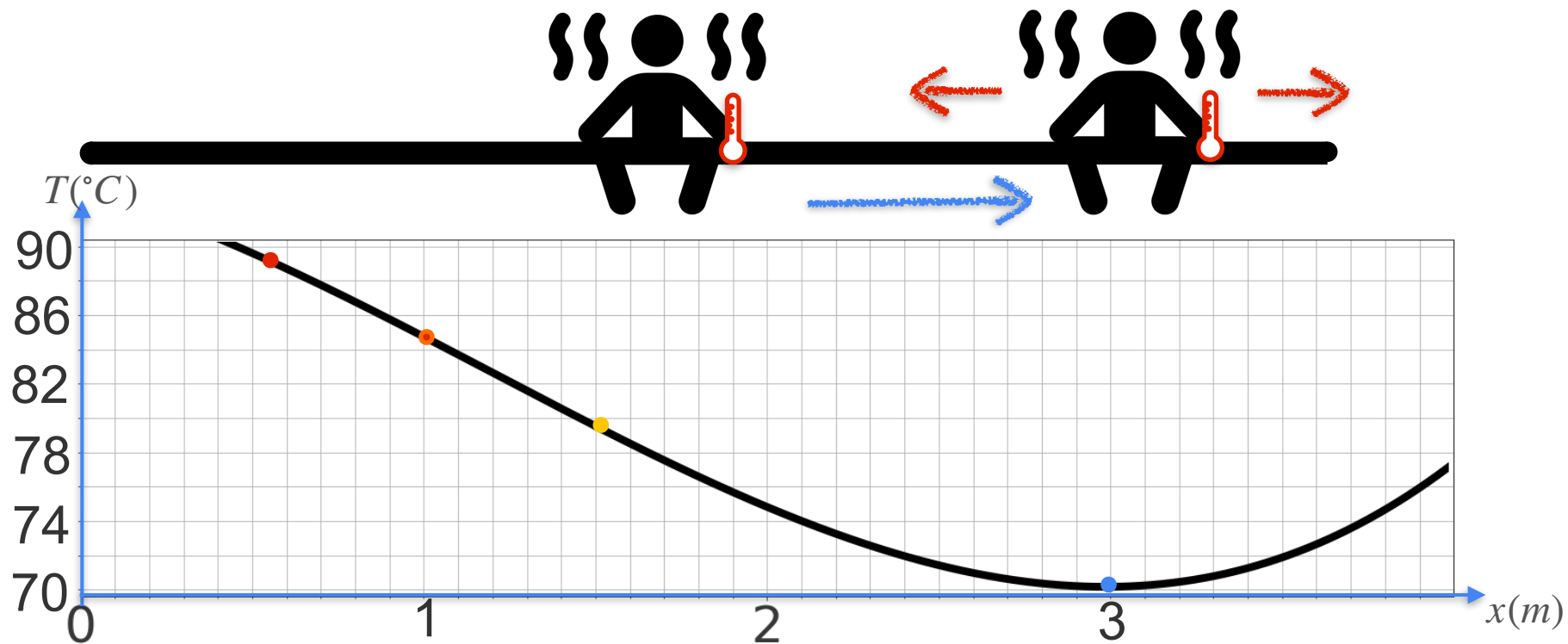
Motivation for Optimization



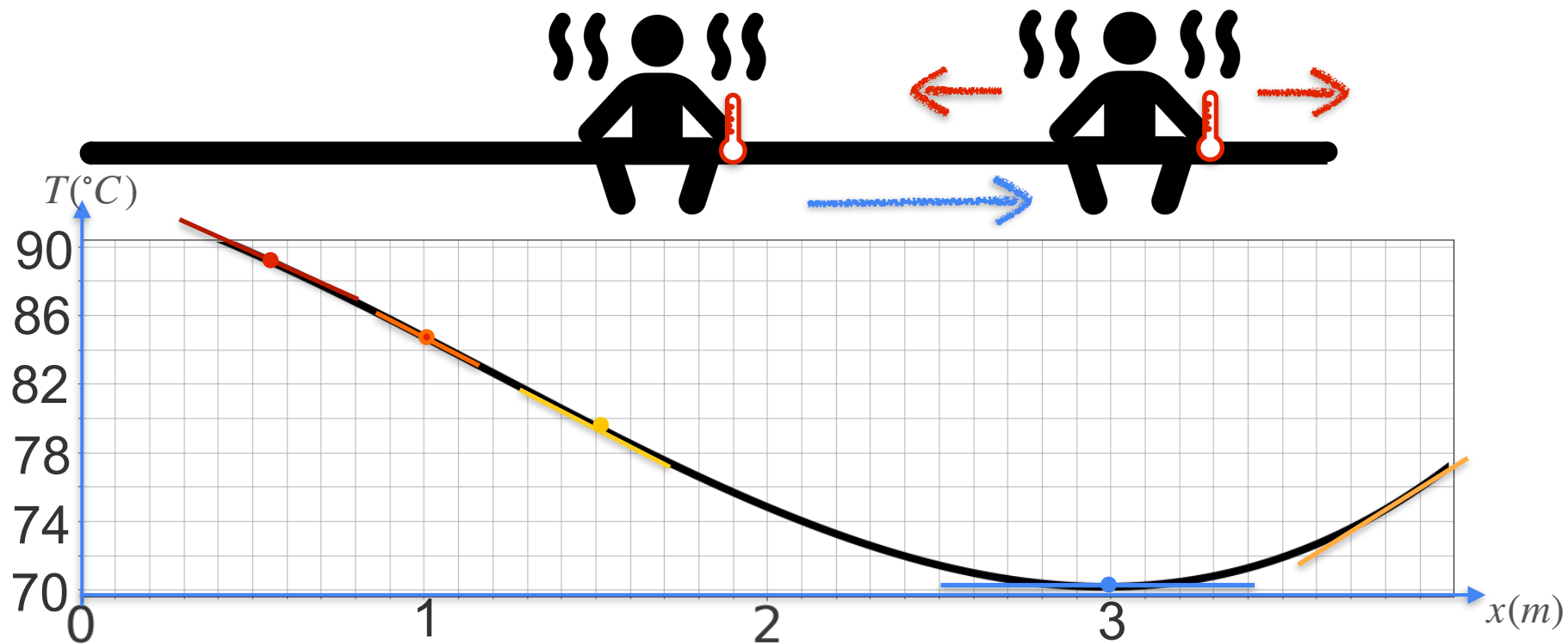
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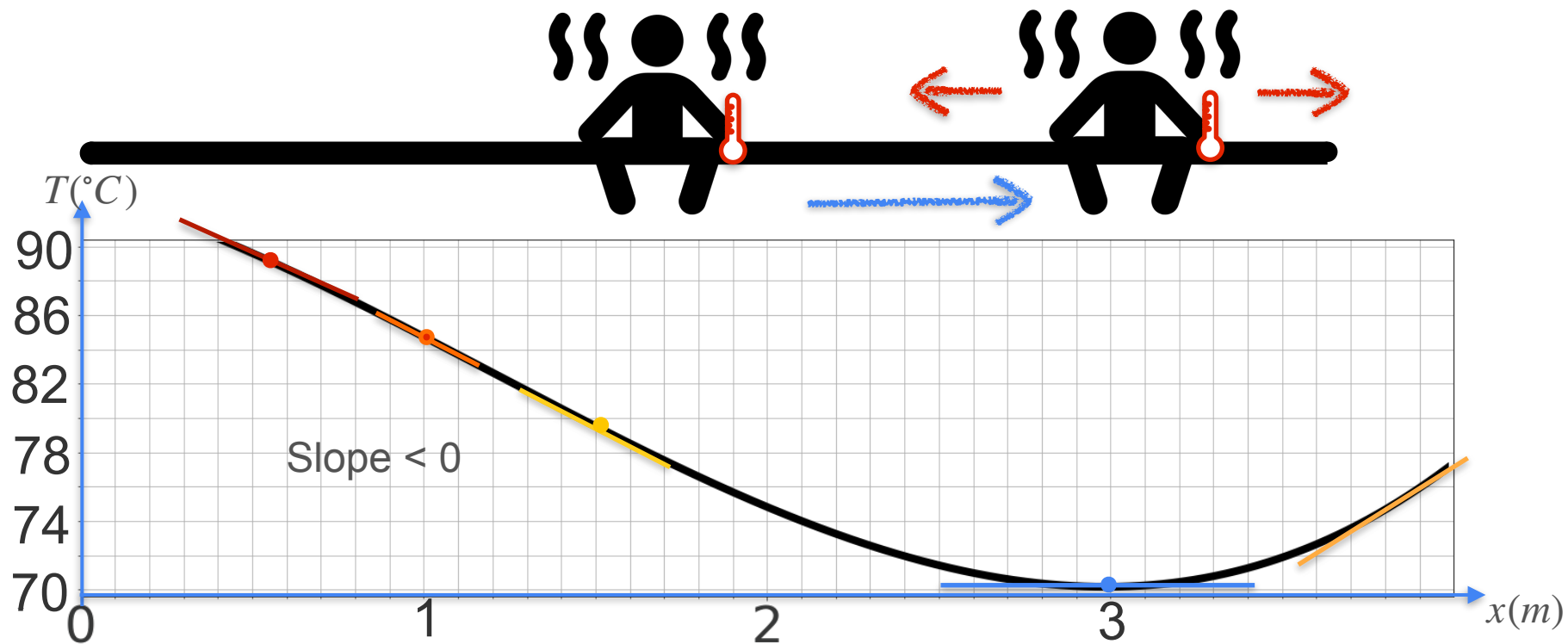
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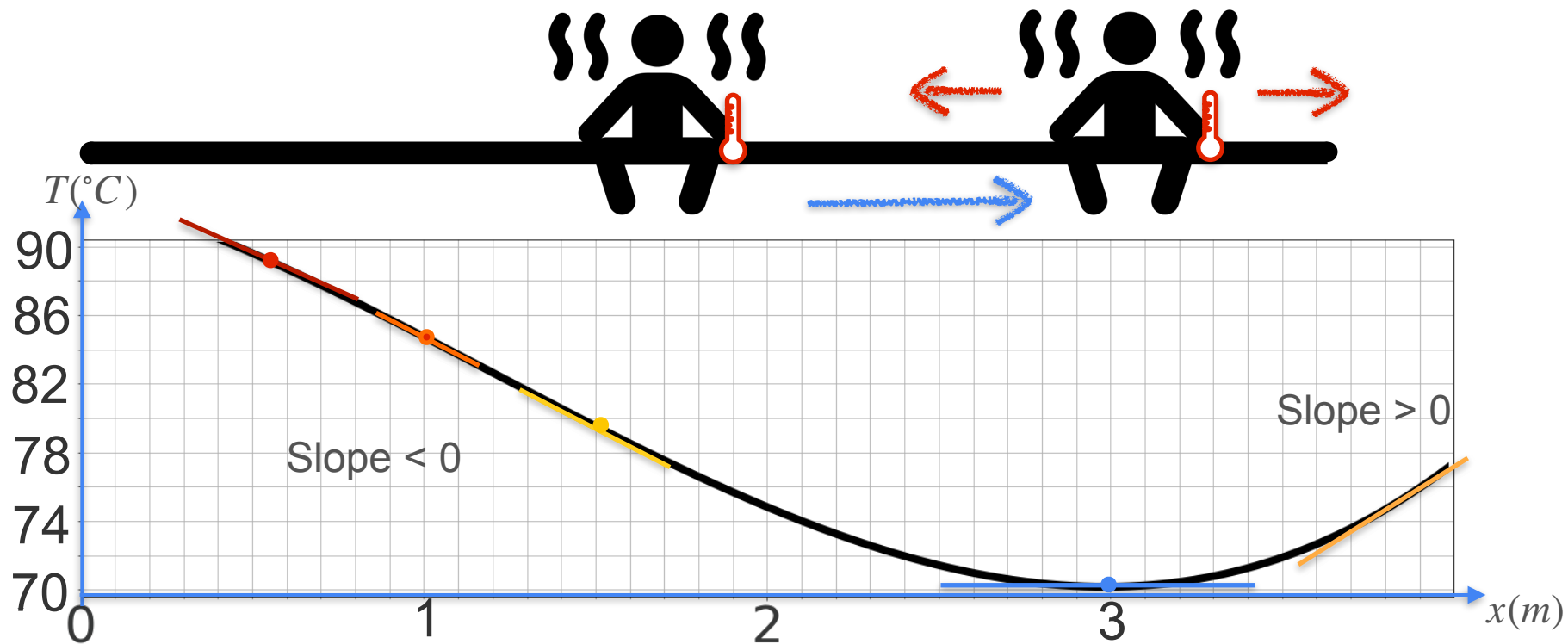
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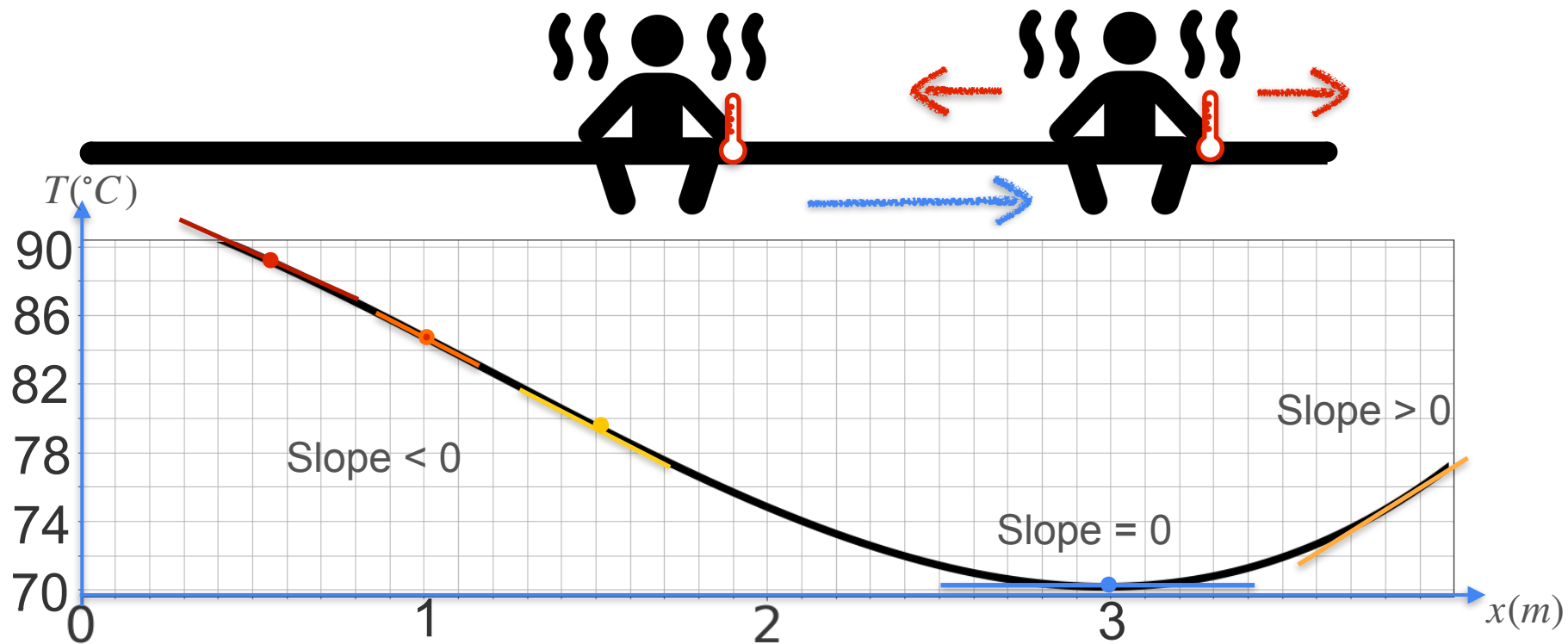
Motivation for Optimization



Motivation for Optimization

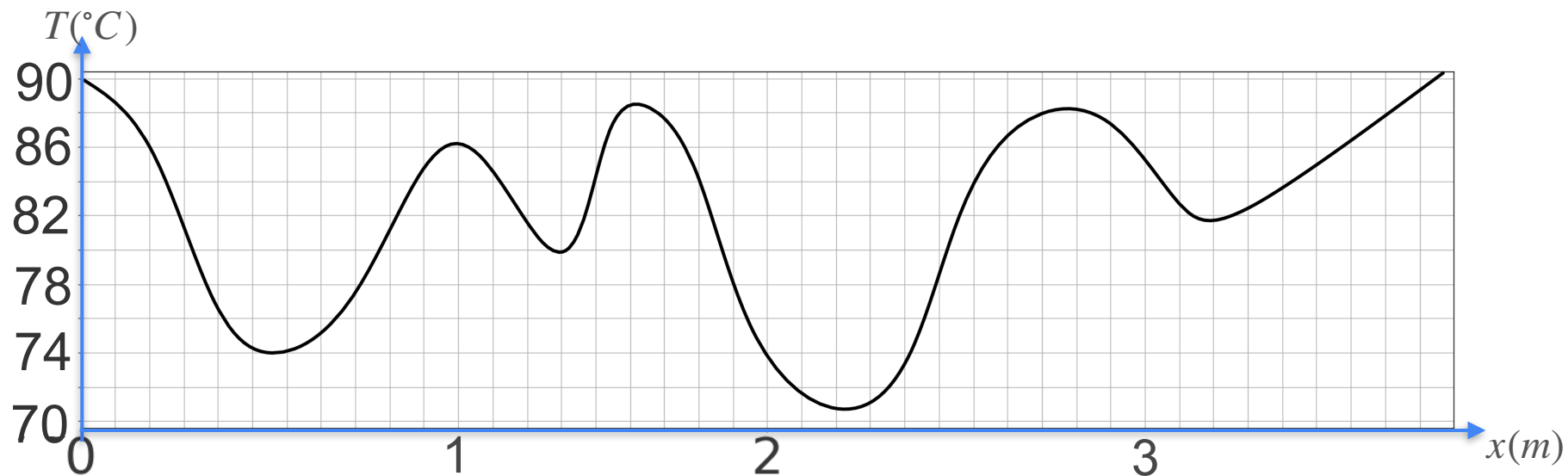


Motivation for Optimization

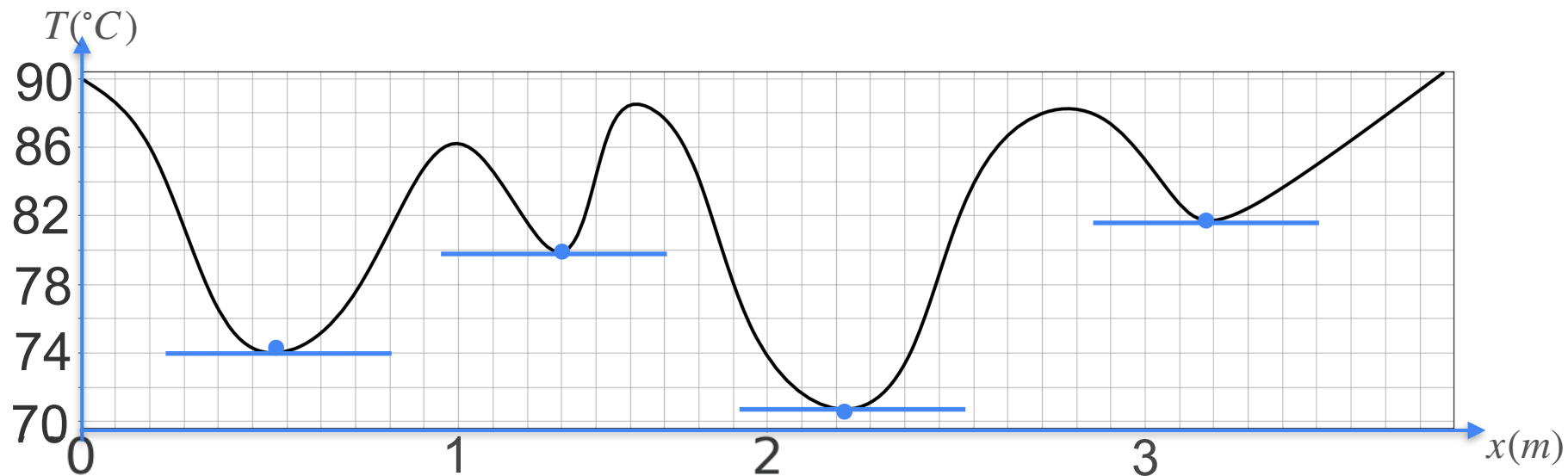


Multiple Minima

Multiple Minima

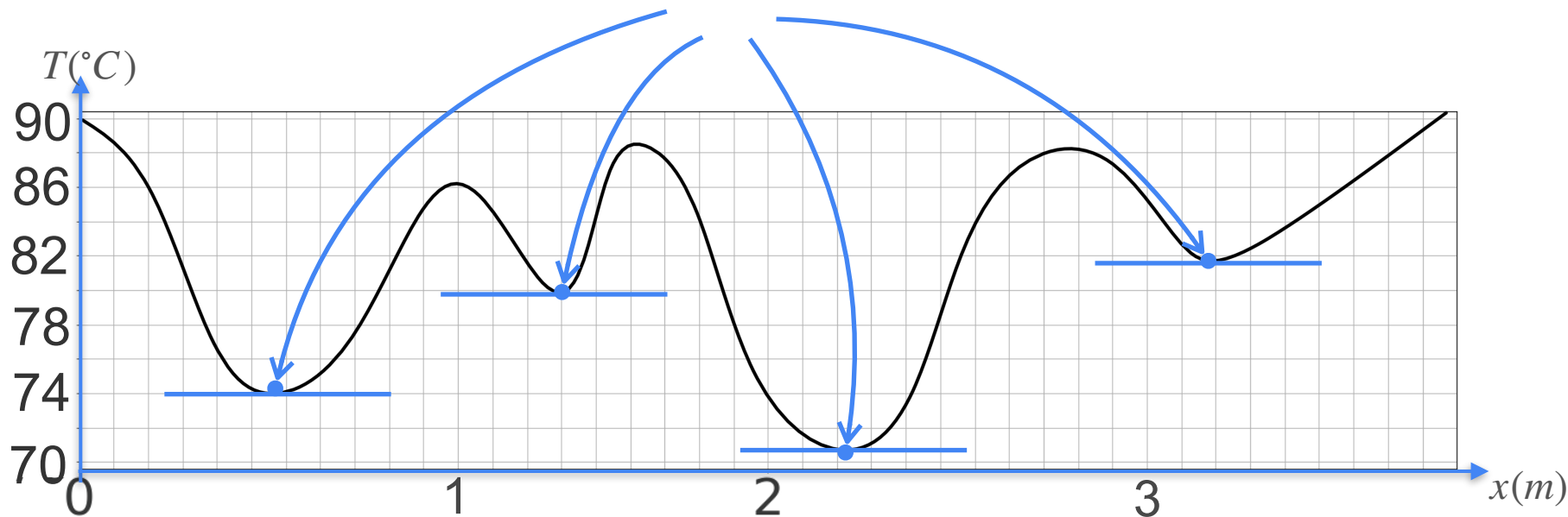


Multiple Minima



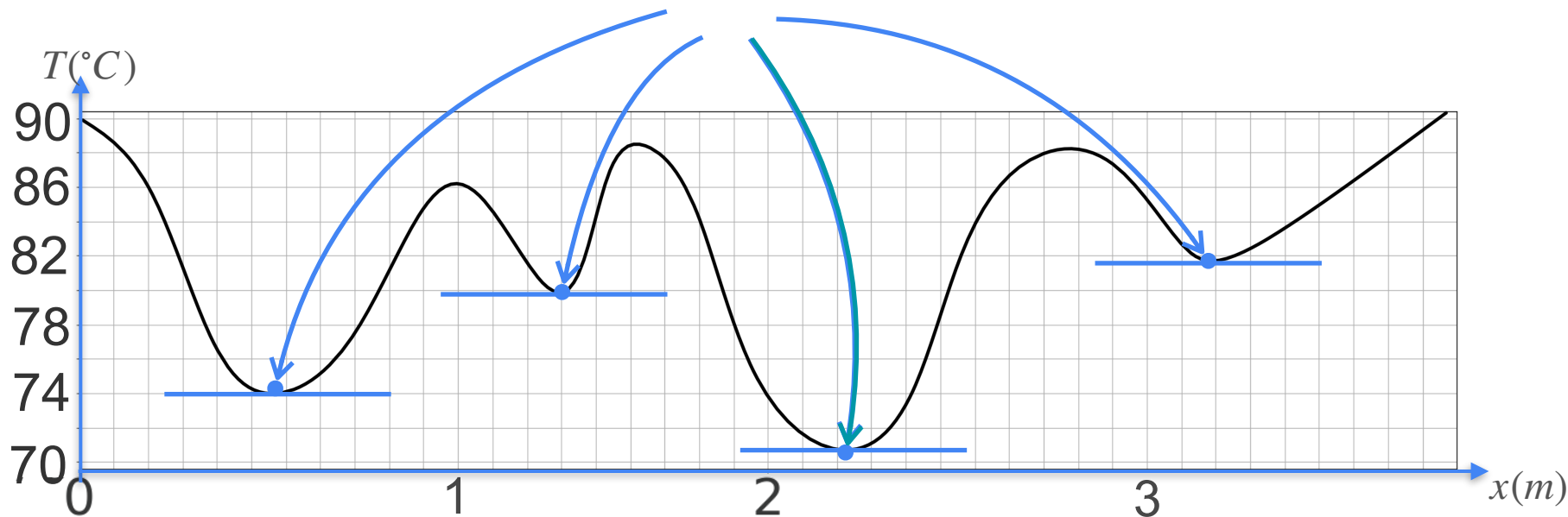
Multiple Minima

Candidates for the minimum are at the points of zero slope



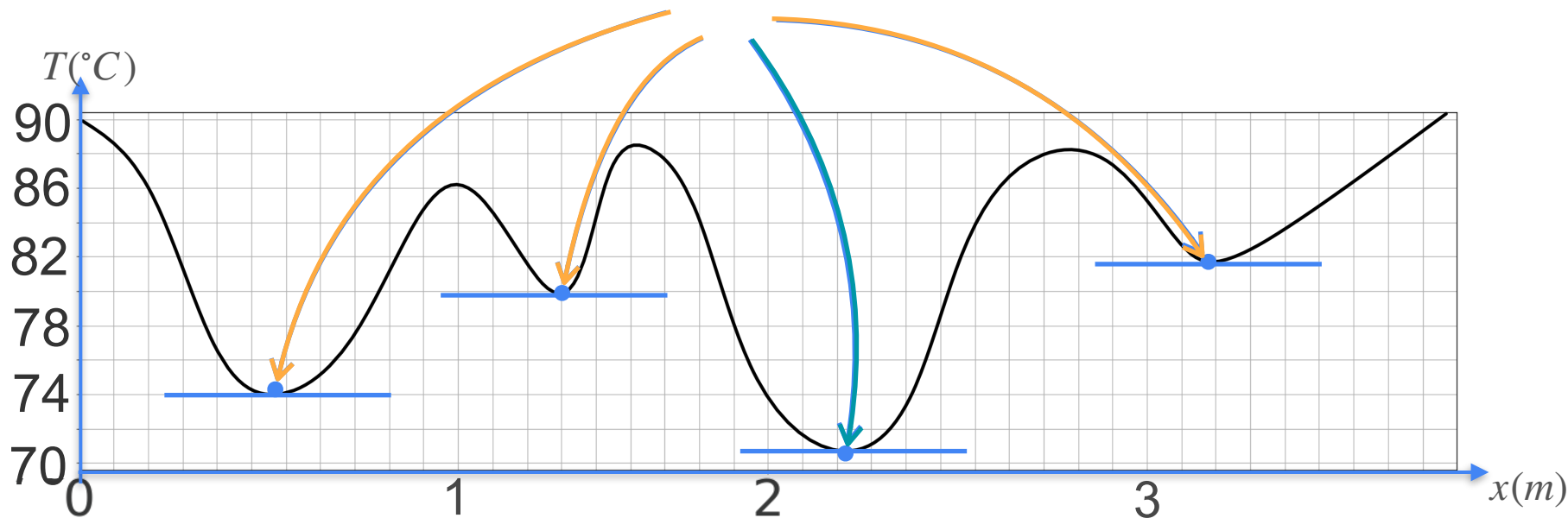
Multiple Minima

Candidates for the minimum are at the points of zero slope



Multiple Minima

Candidates for the minimum are at the points of zero slope





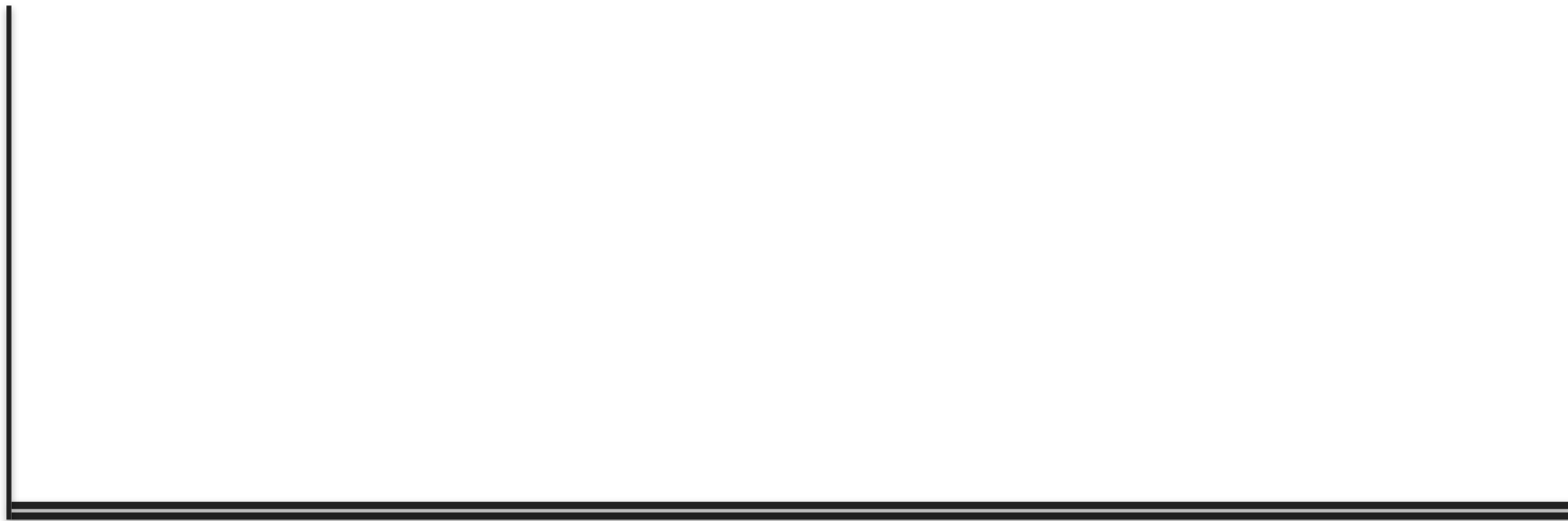
DeepLearning.AI

Derivatives and Optimization

**Optimization of squared loss:
The one powerline problem**

Cost Function Motivation

Cost Function Motivation



Cost Function Motivation



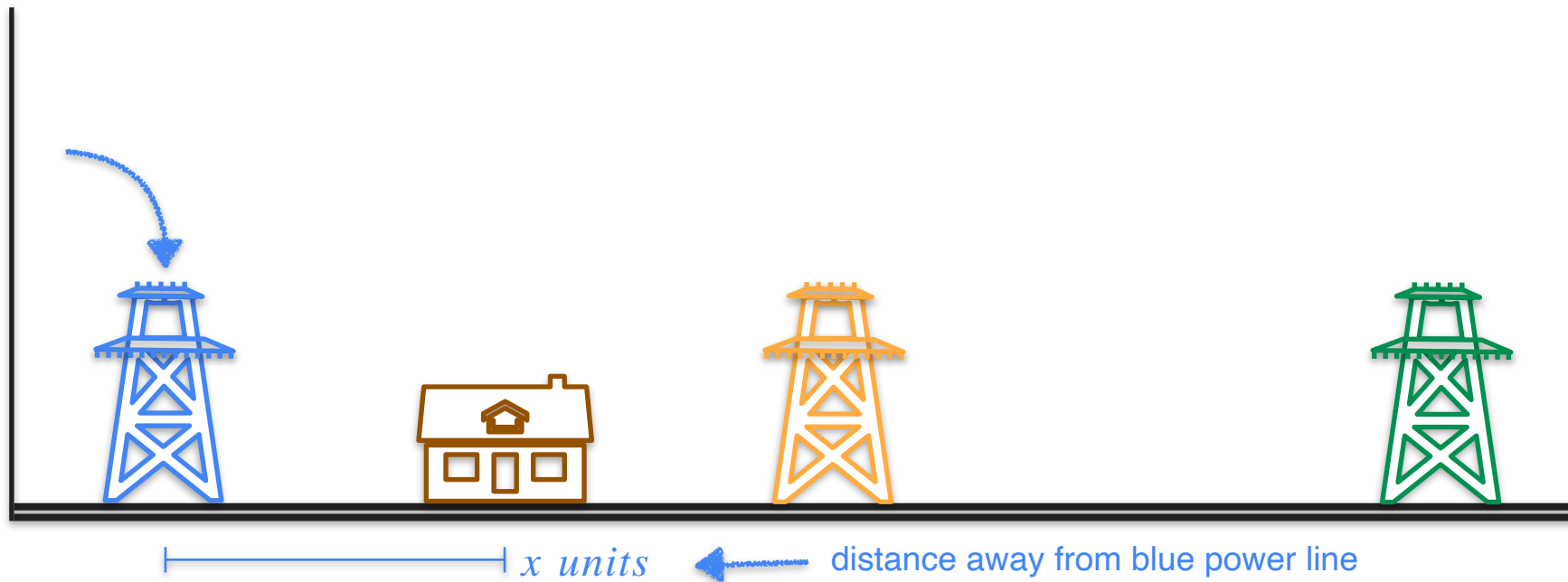
Cost Function Motivation



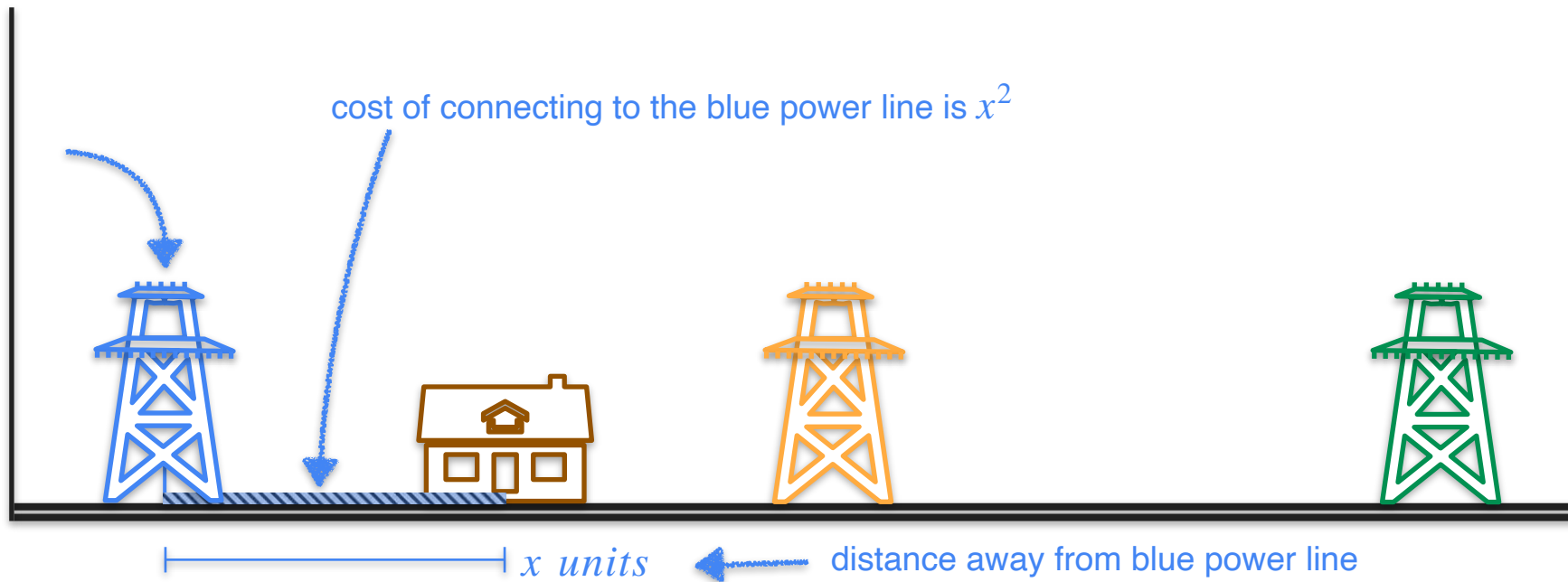
Cost Function Motivation



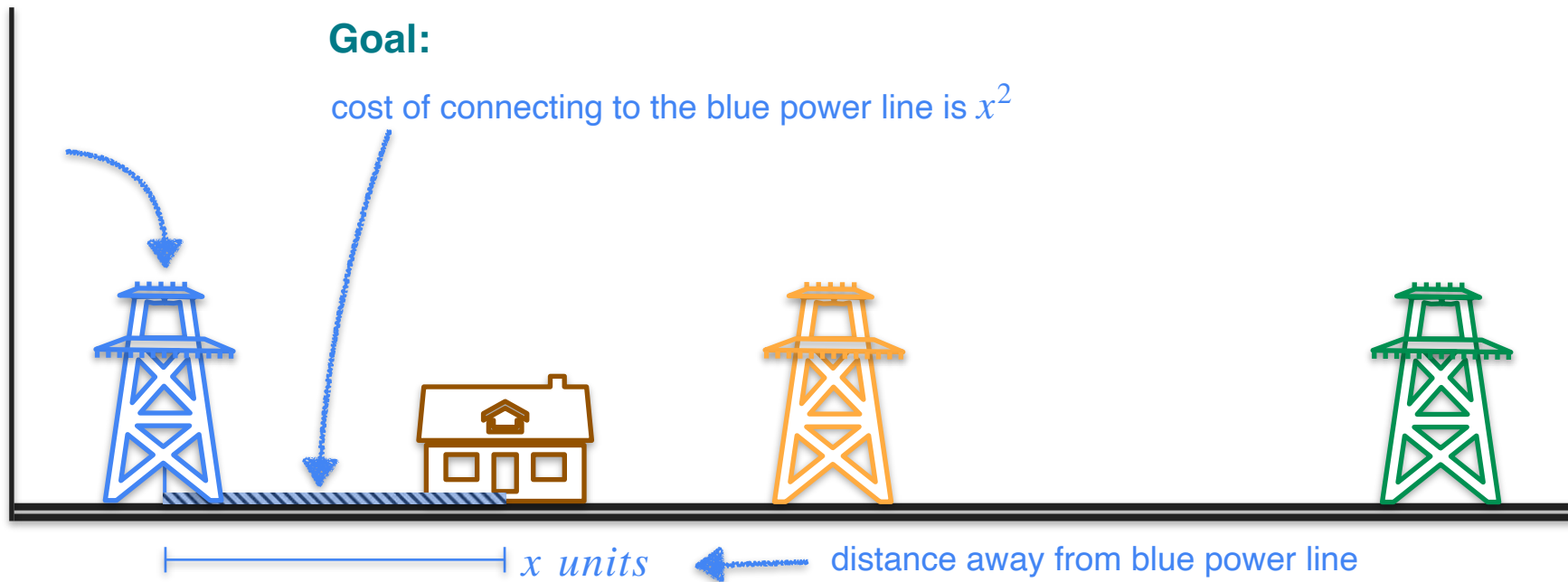
Cost Function Motivation



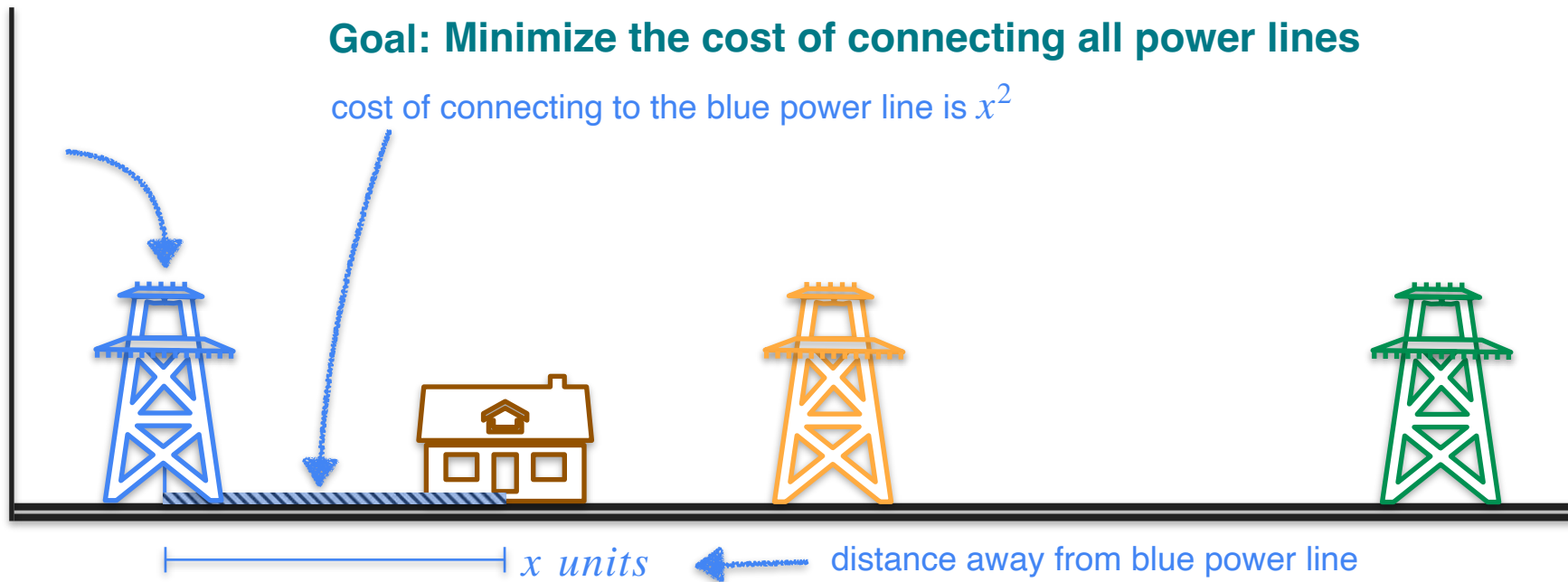
Cost Function Motivation



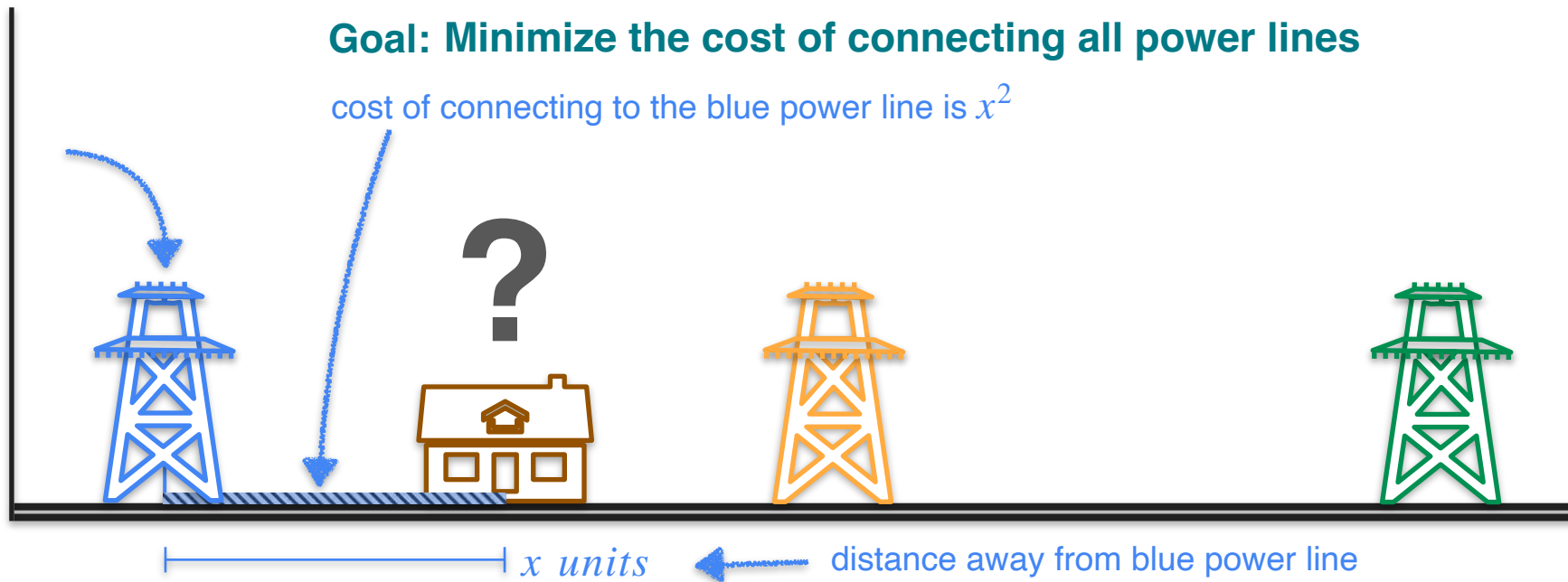
Cost Function Motivation



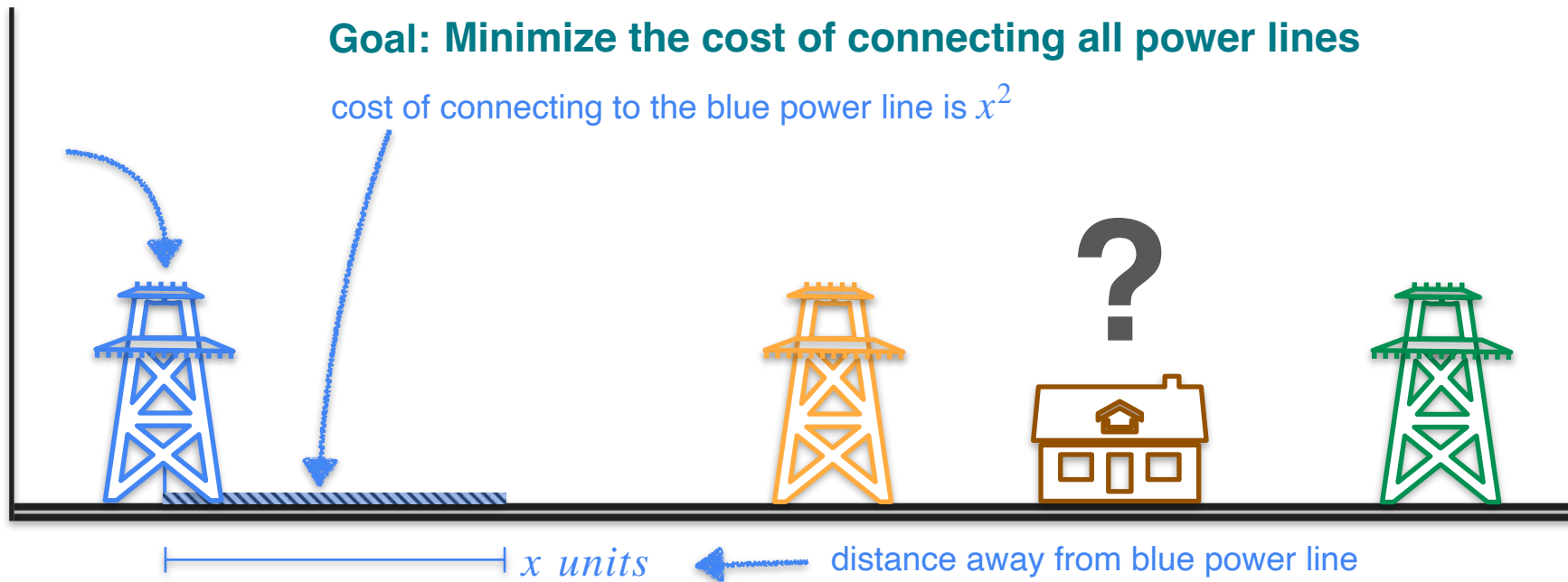
Cost Function Motivation



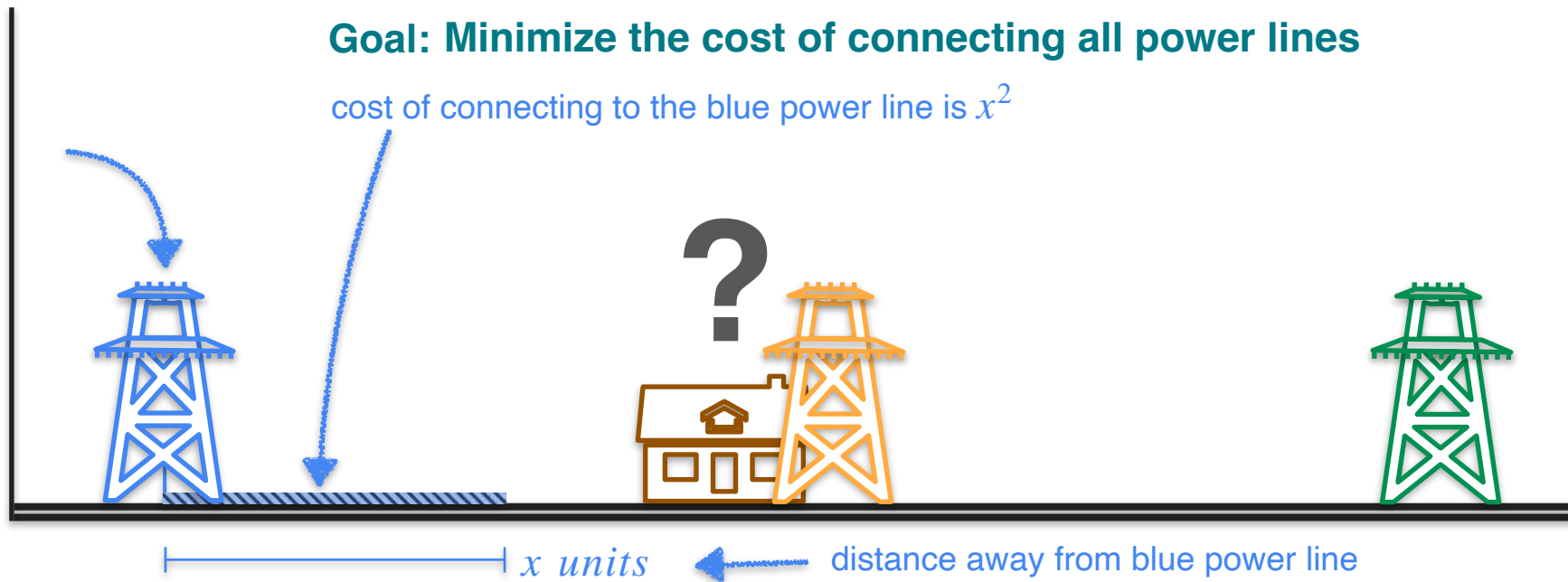
Cost Function Motivation



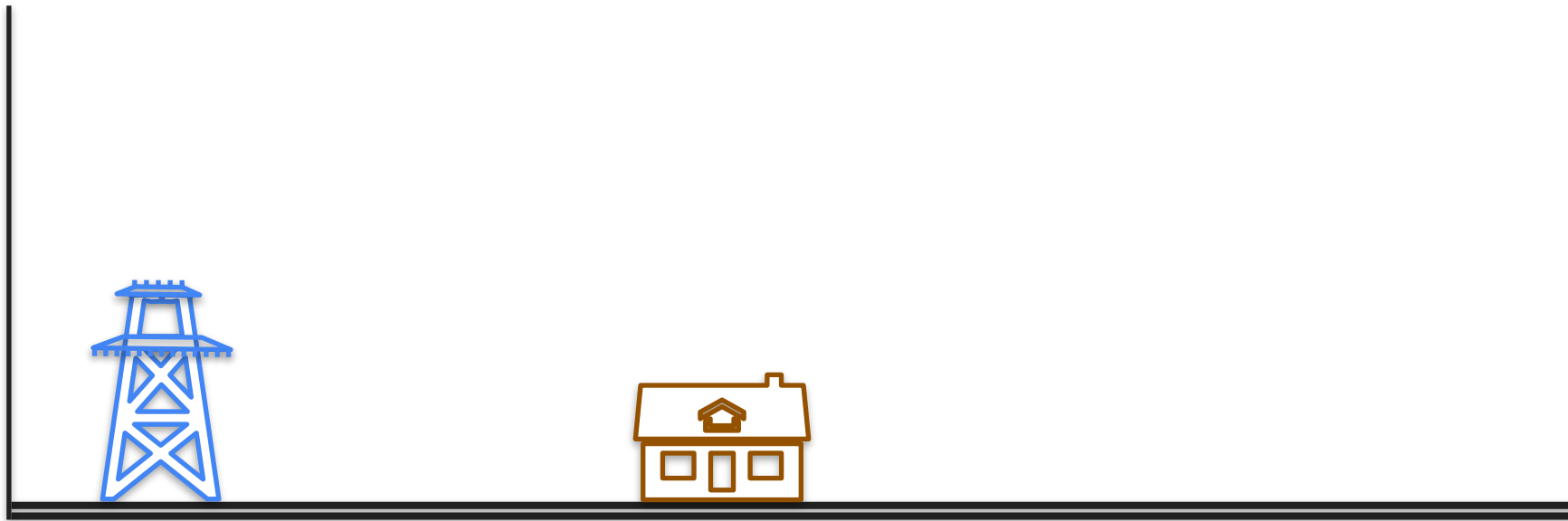
Cost Function Motivation



Cost Function Motivation



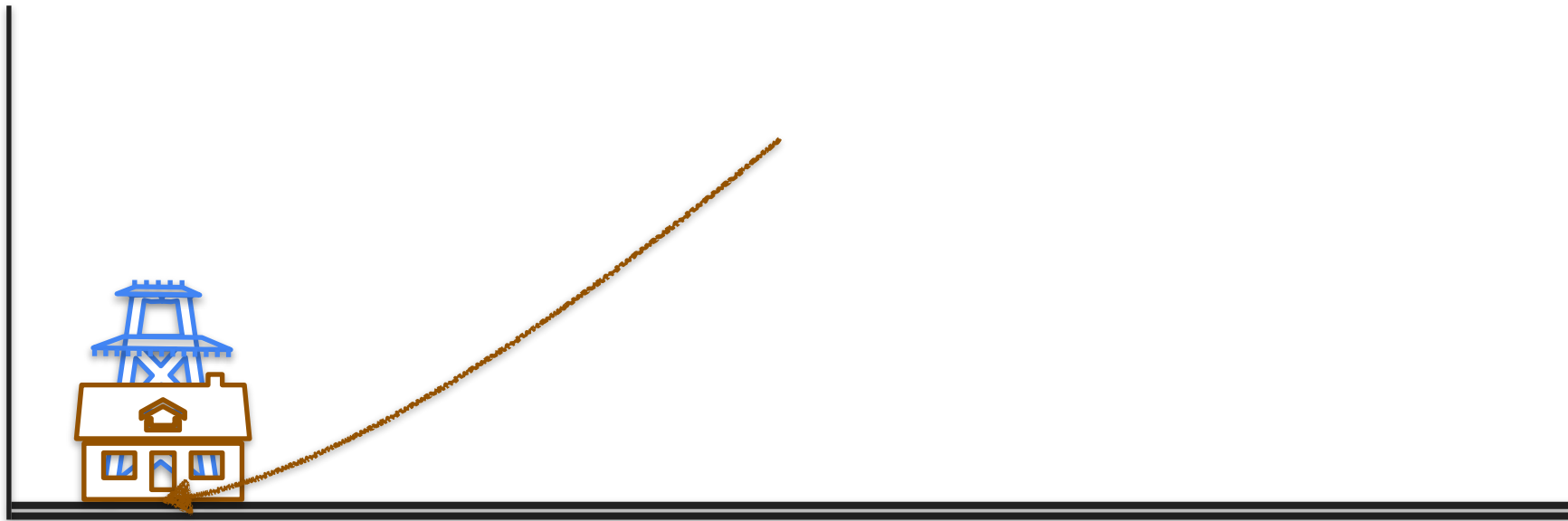
One Power Line Problem



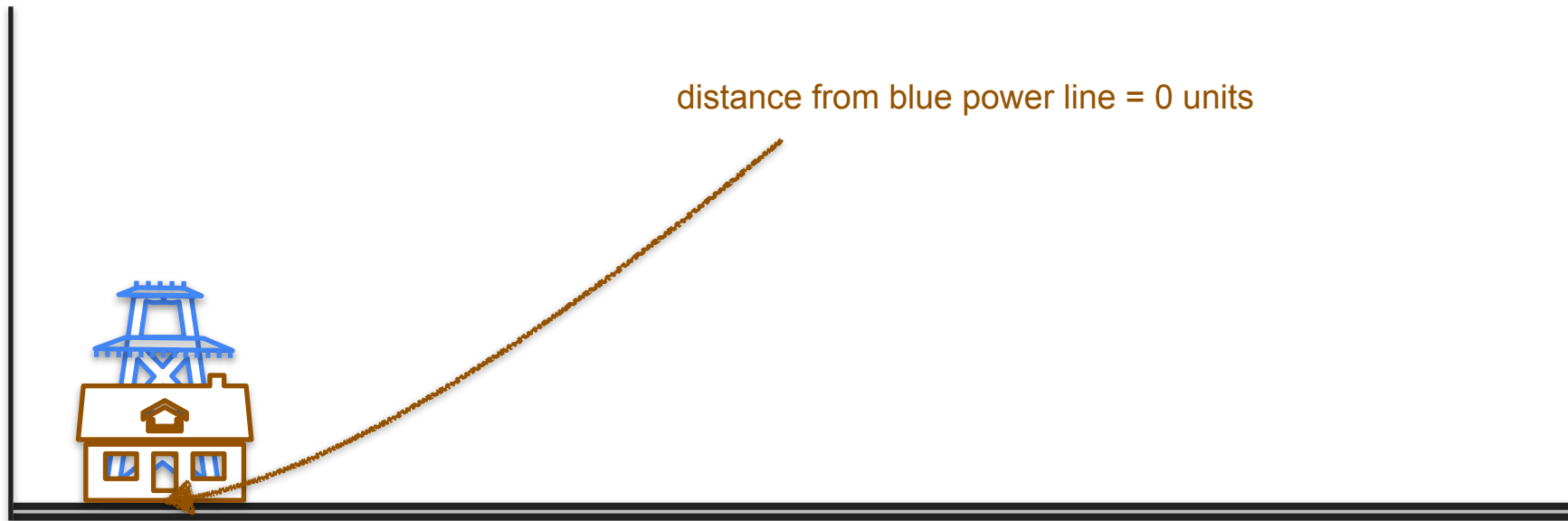
One Power Line Problem



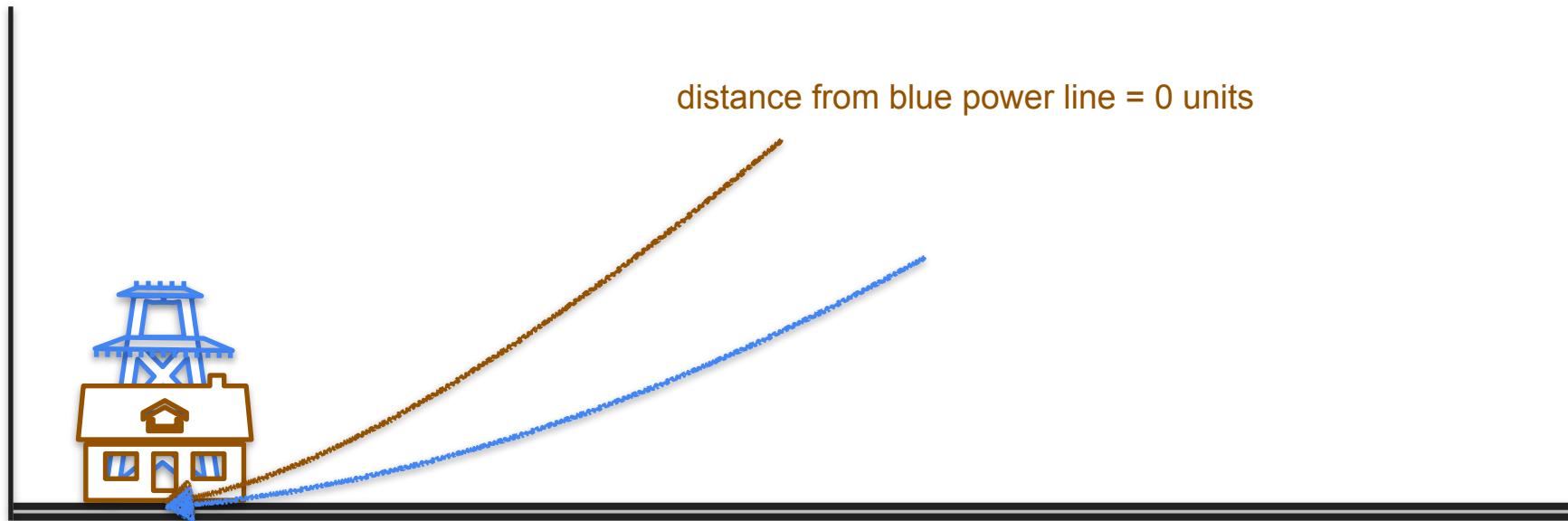
One Power Line Problem



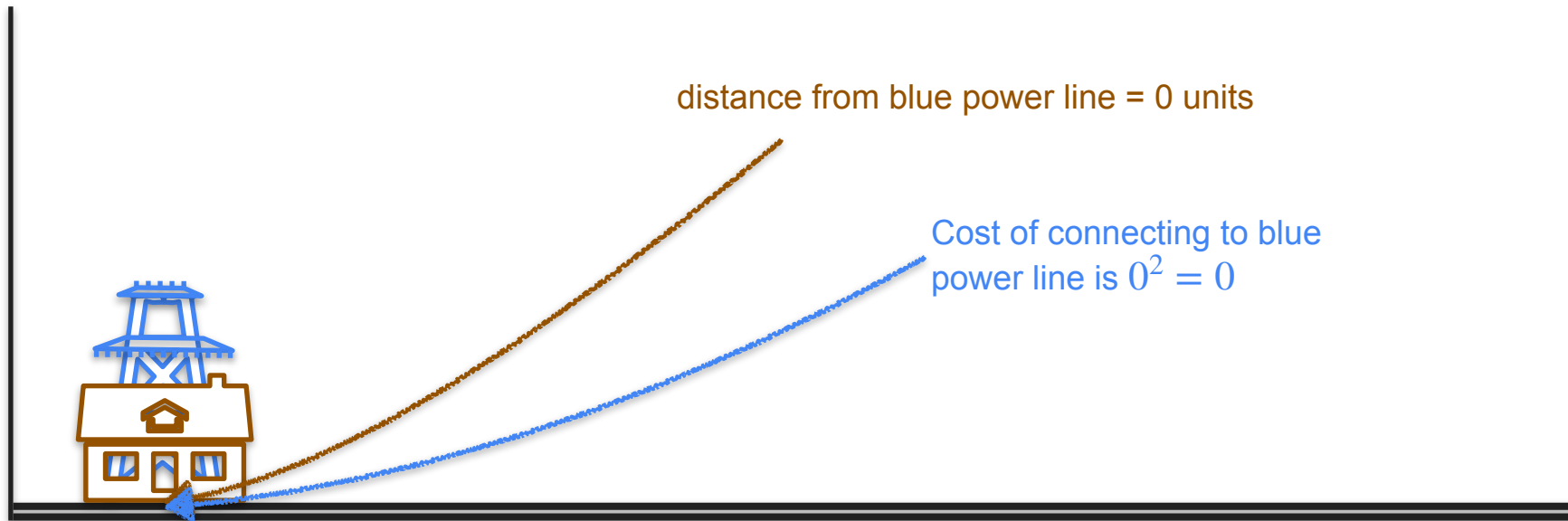
One Power Line Problem



One Power Line Problem



One Power Line Problem



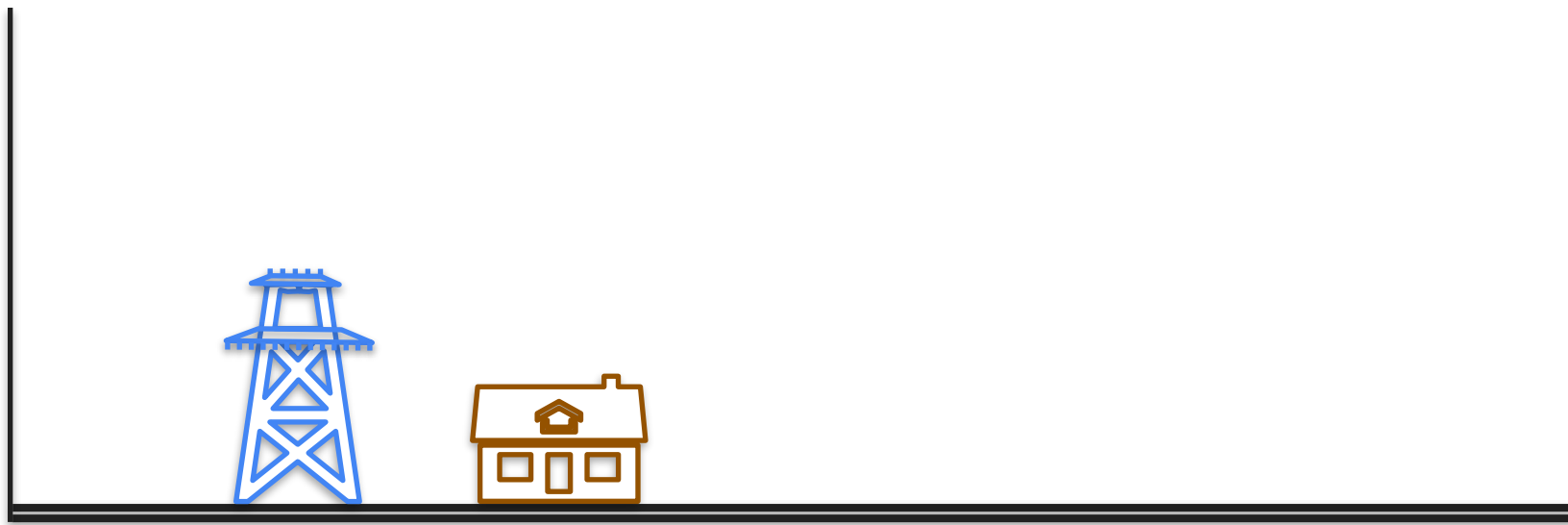


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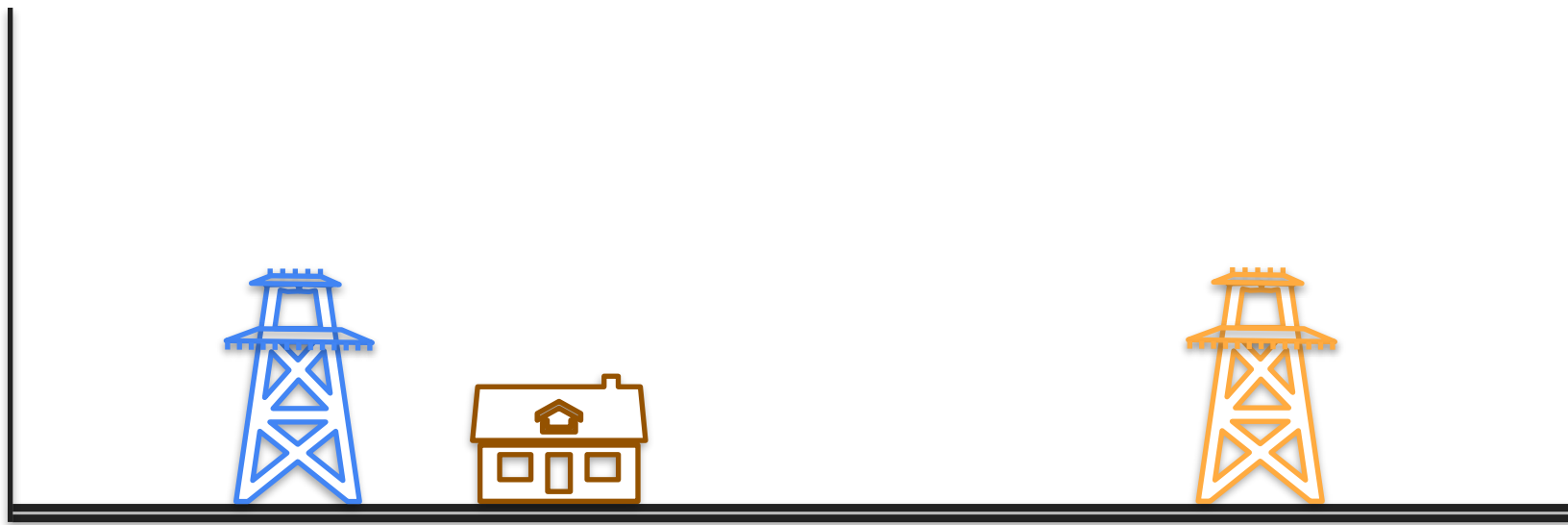
Derivatives and Optimization

**Optimization of squared loss:
The two powerline problem**

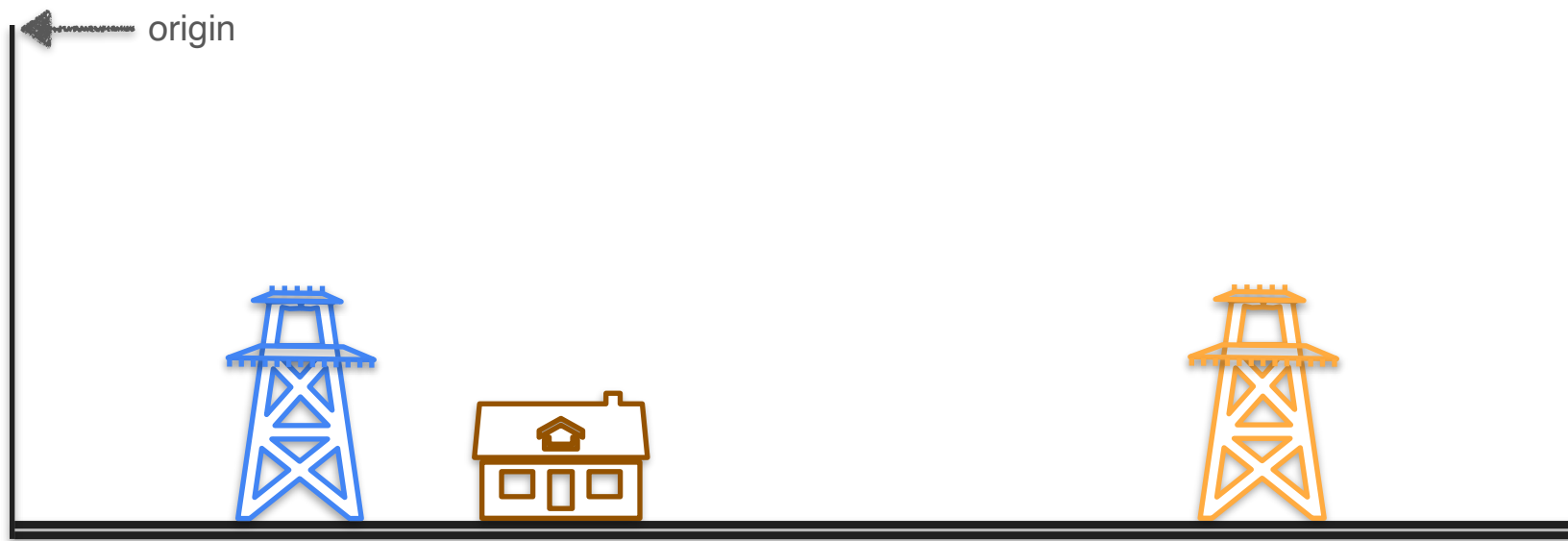
Two Power Line Problem



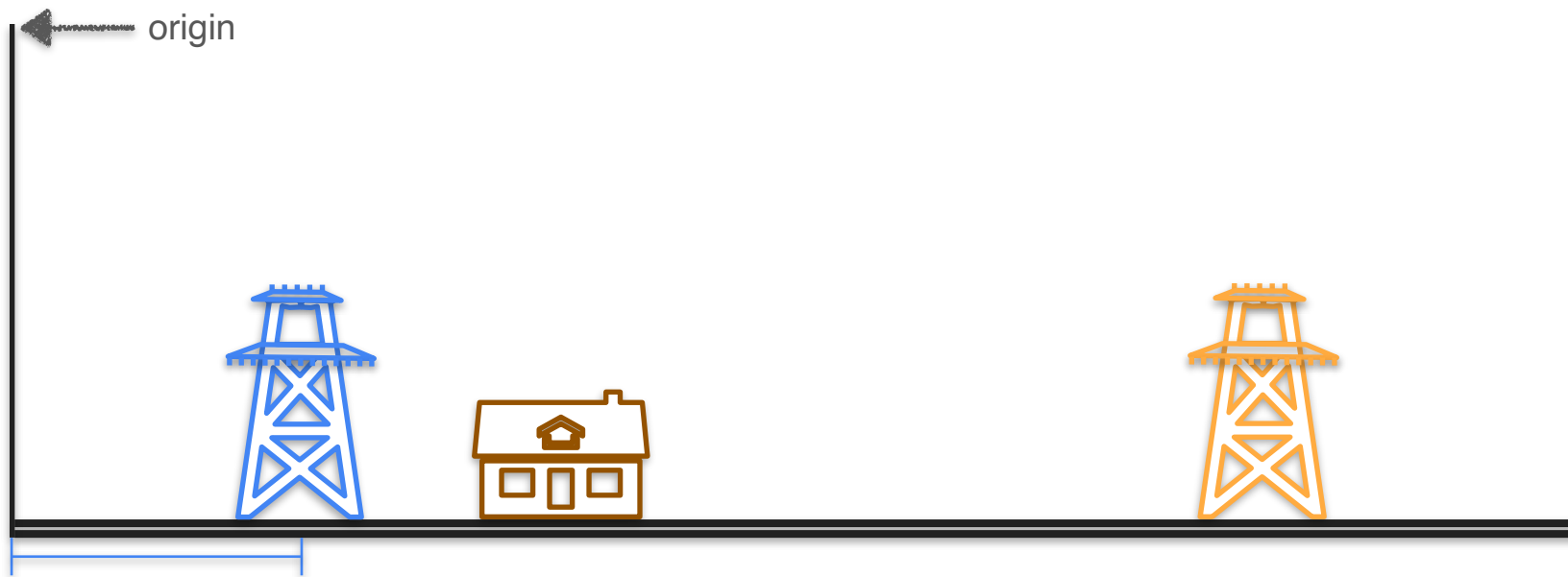
Two Power Line Problem



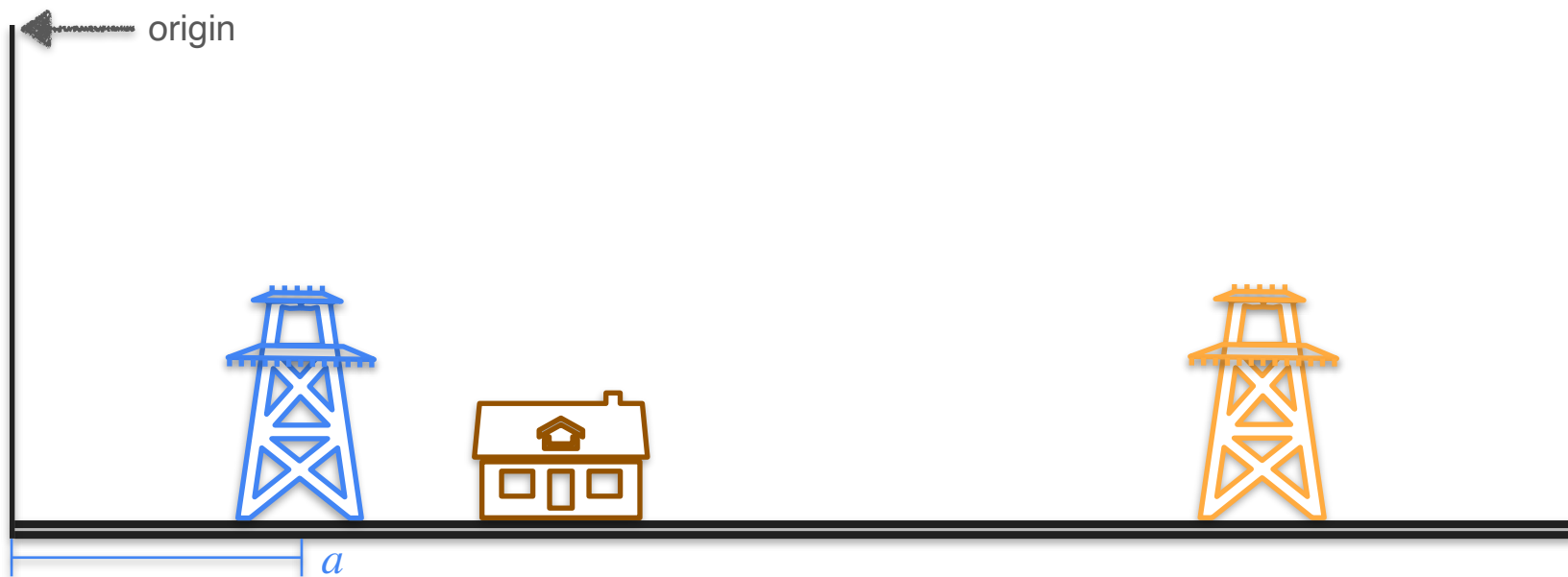
Two Power Line Problem



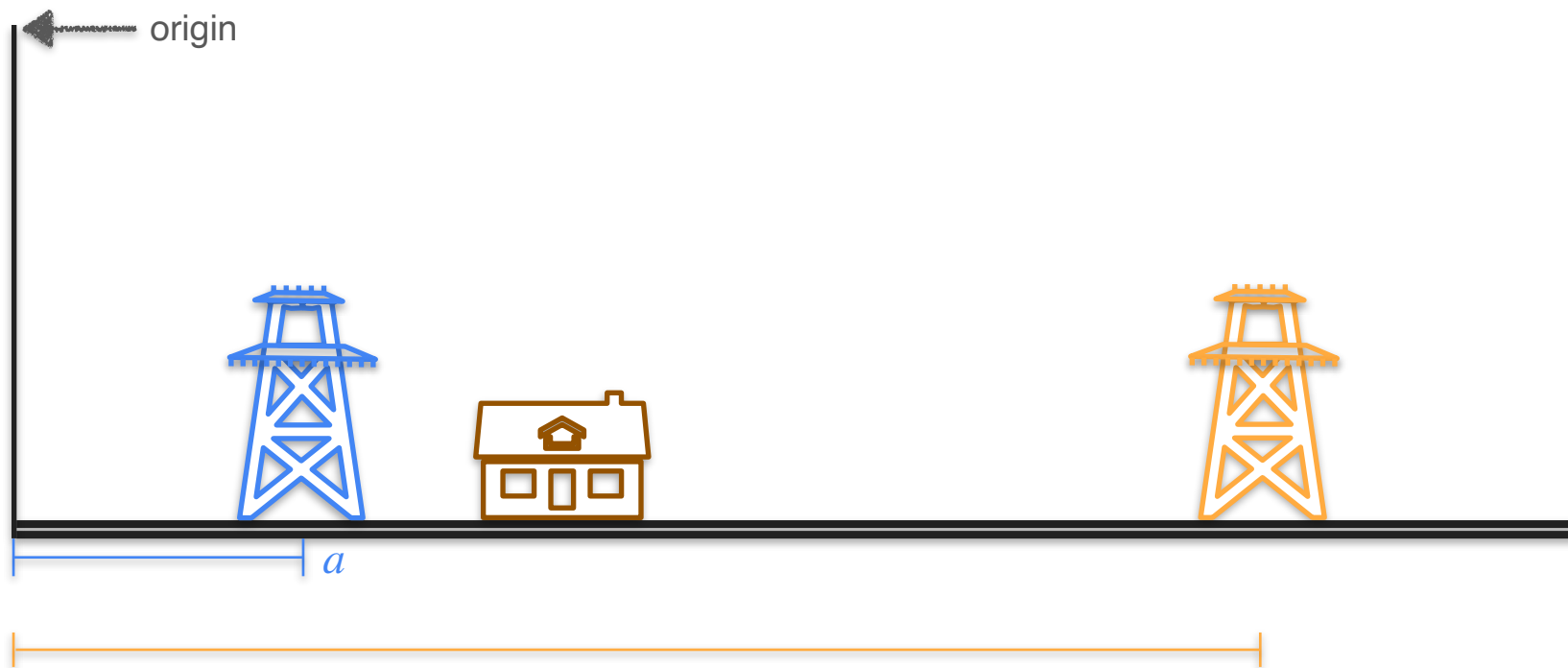
Two Power Line Problem



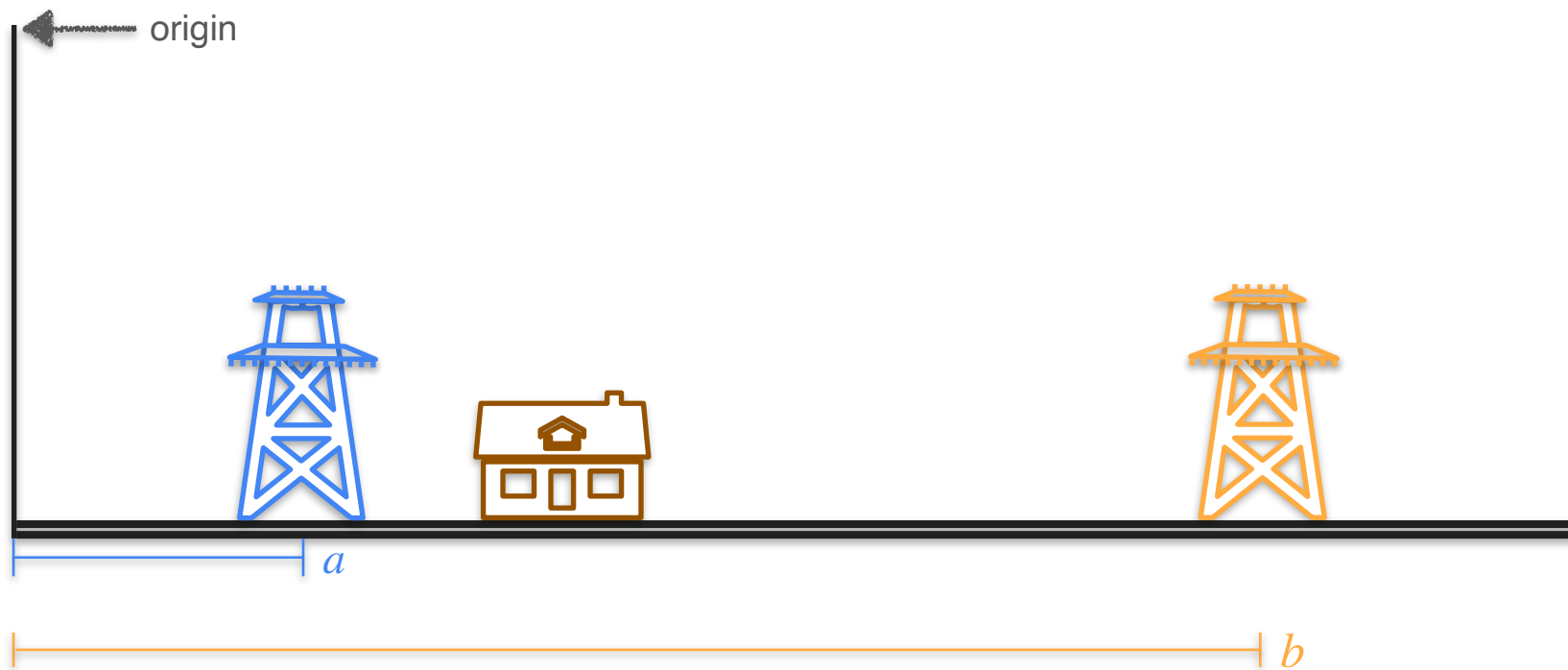
Two Power Line Problem



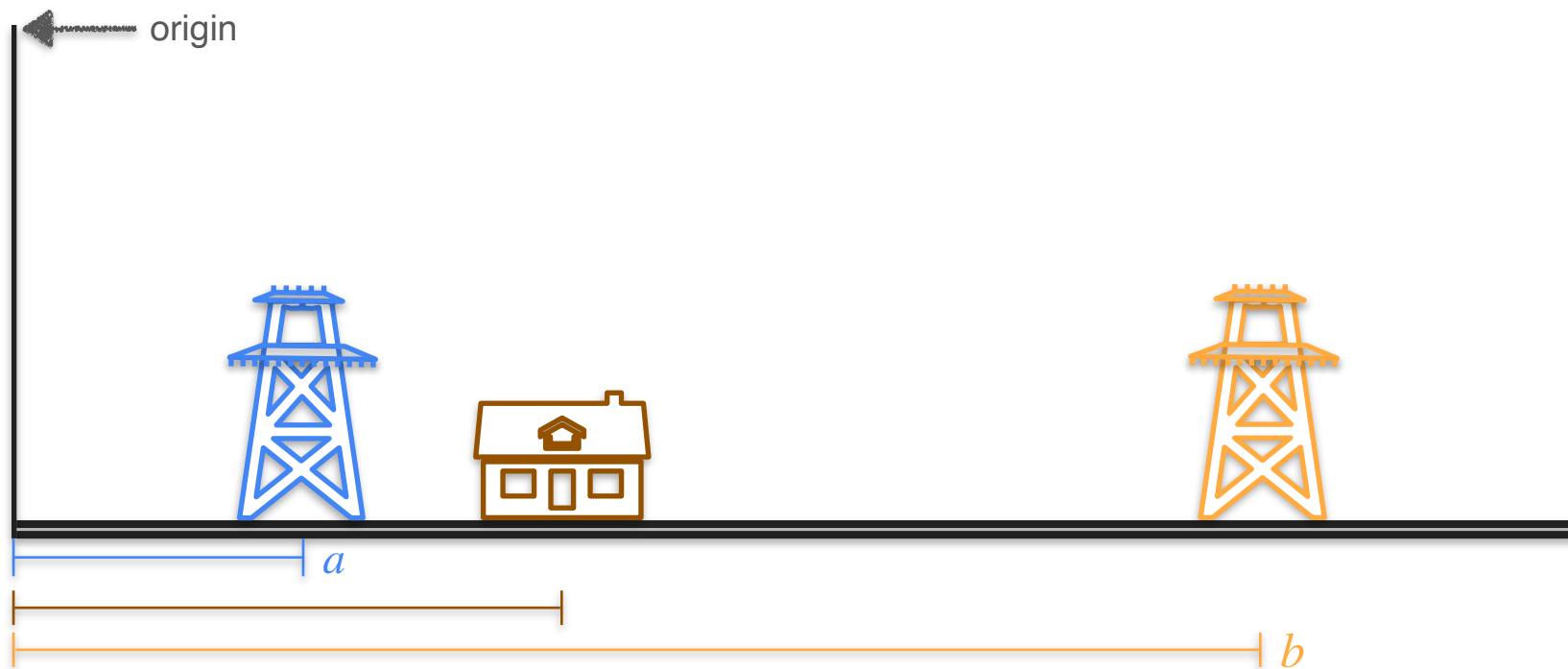
Two Power Line Problem



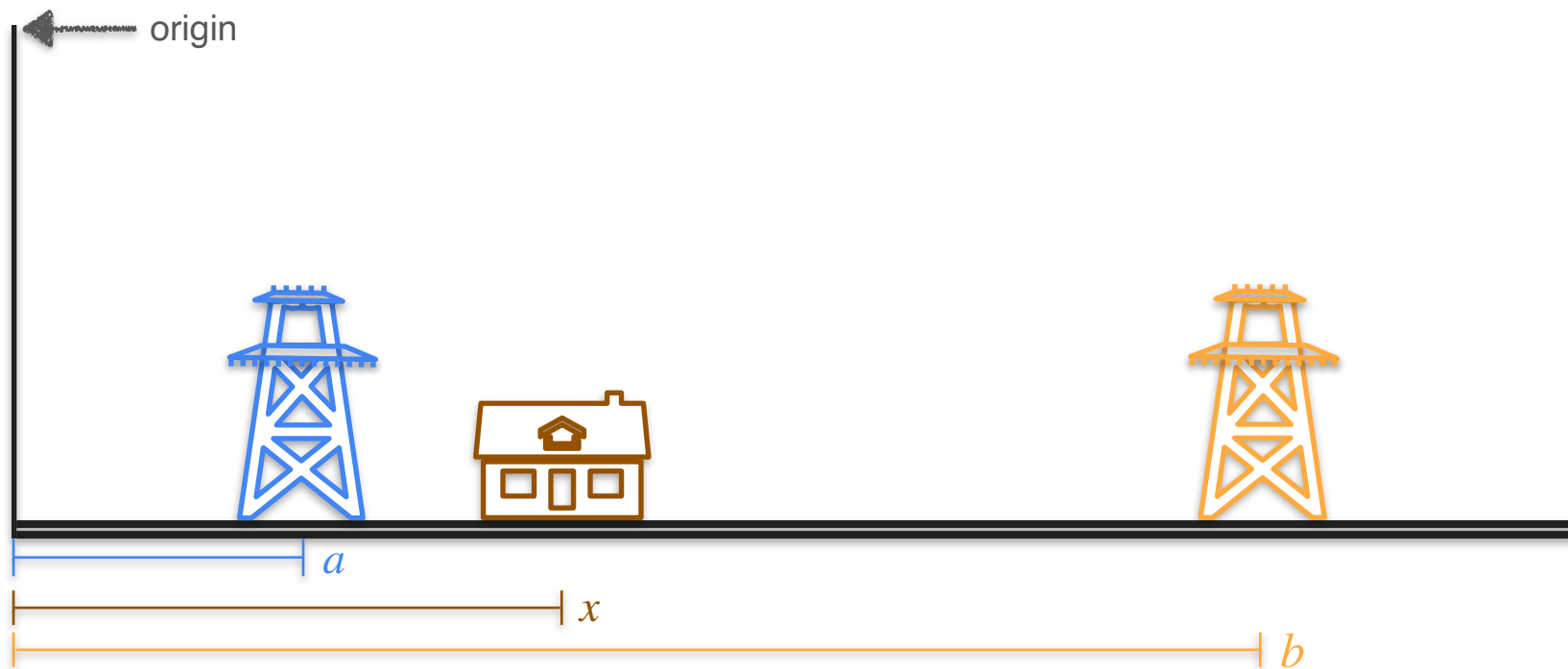
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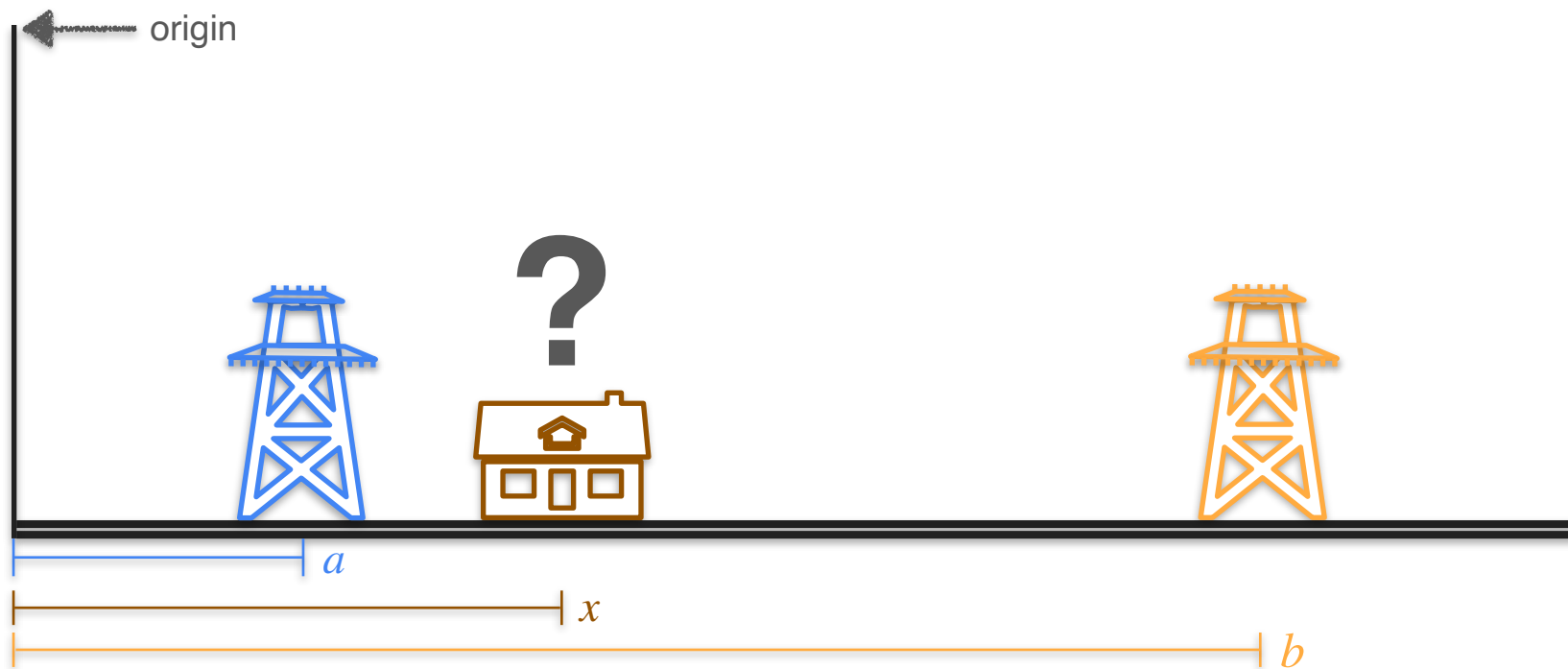
Two Power Line Problem



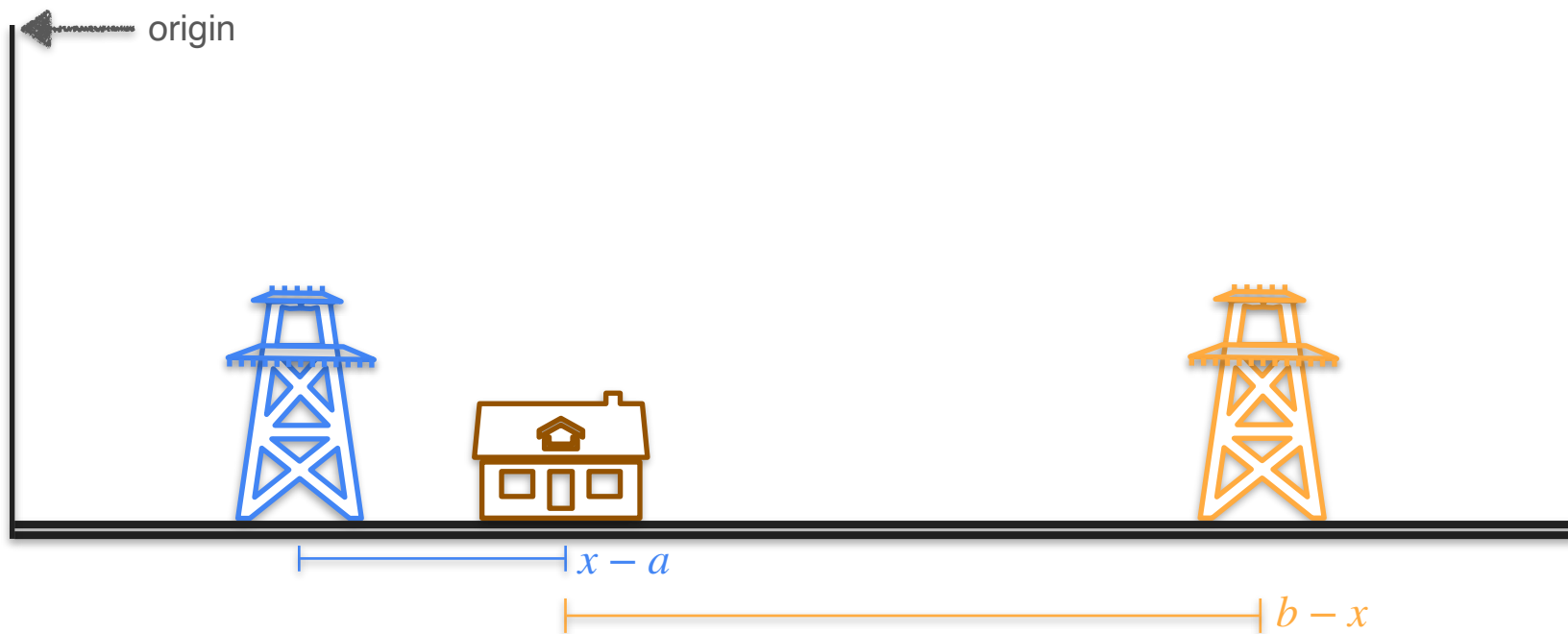
Two Power Line Problem



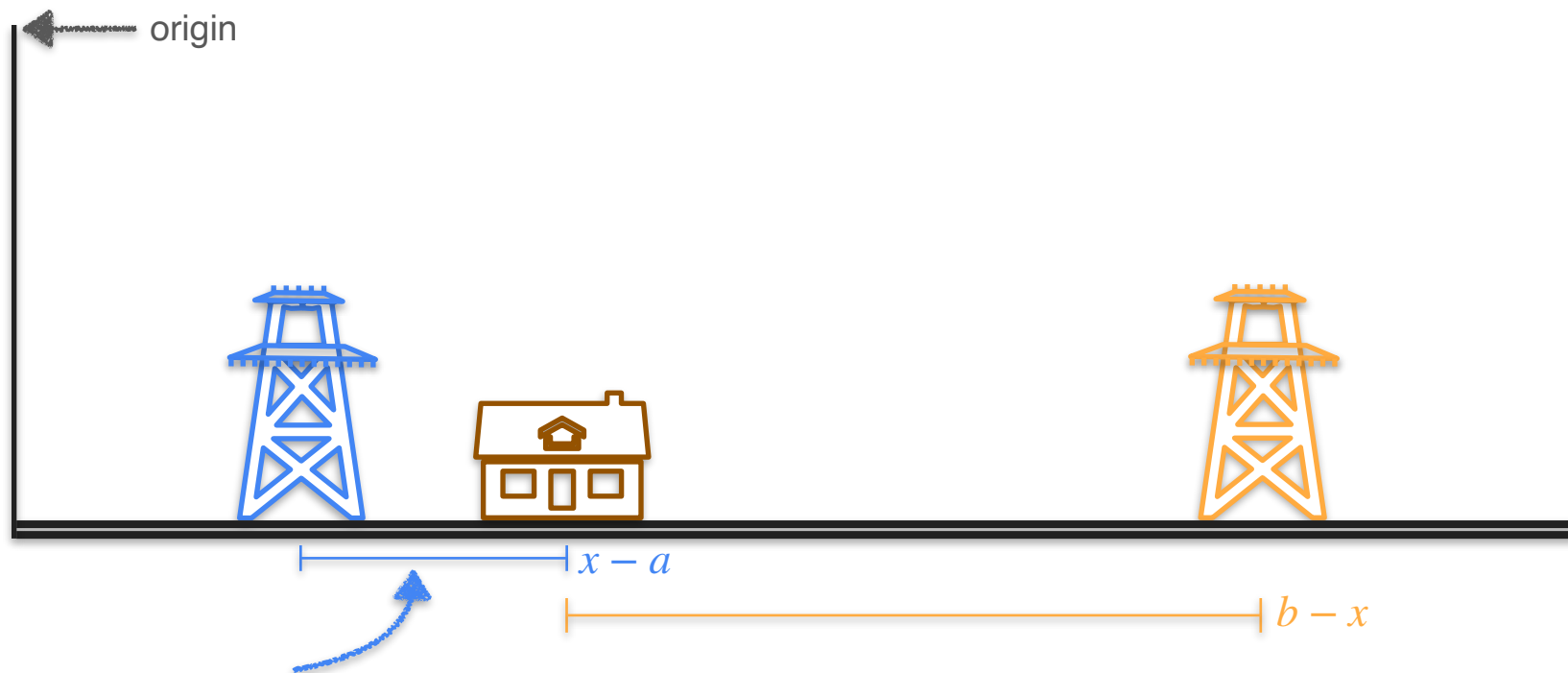
Two Power Line Problem



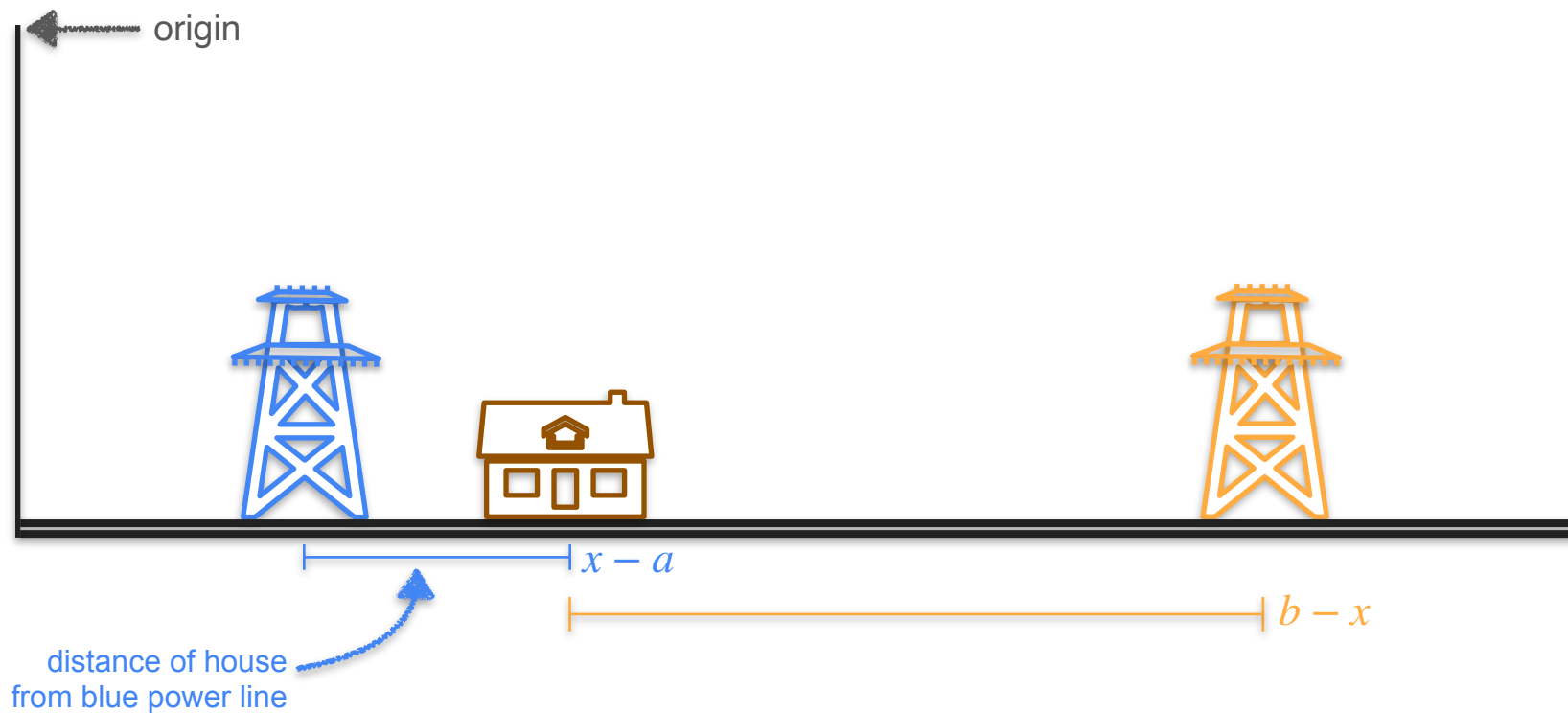
Two Power Line Problem



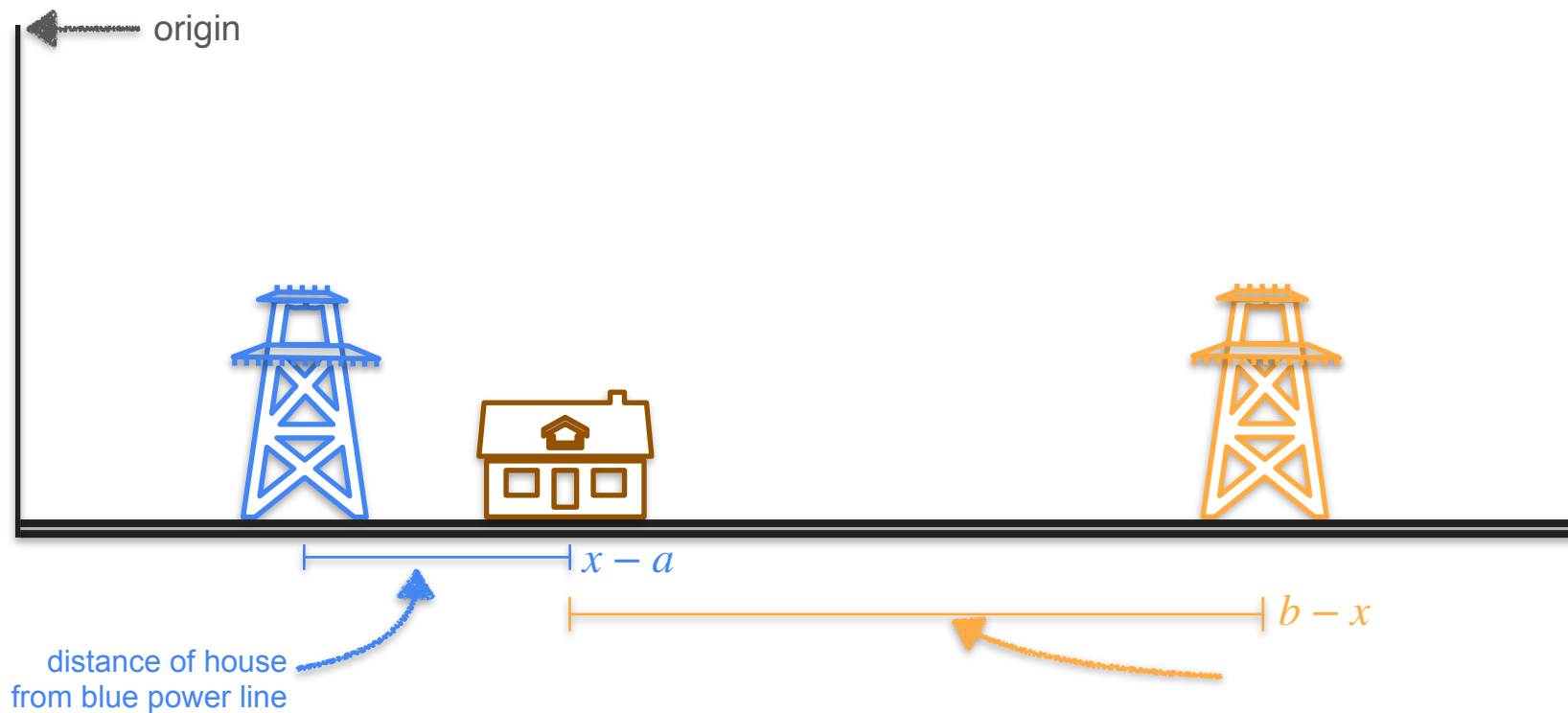
Two Power Line Problem



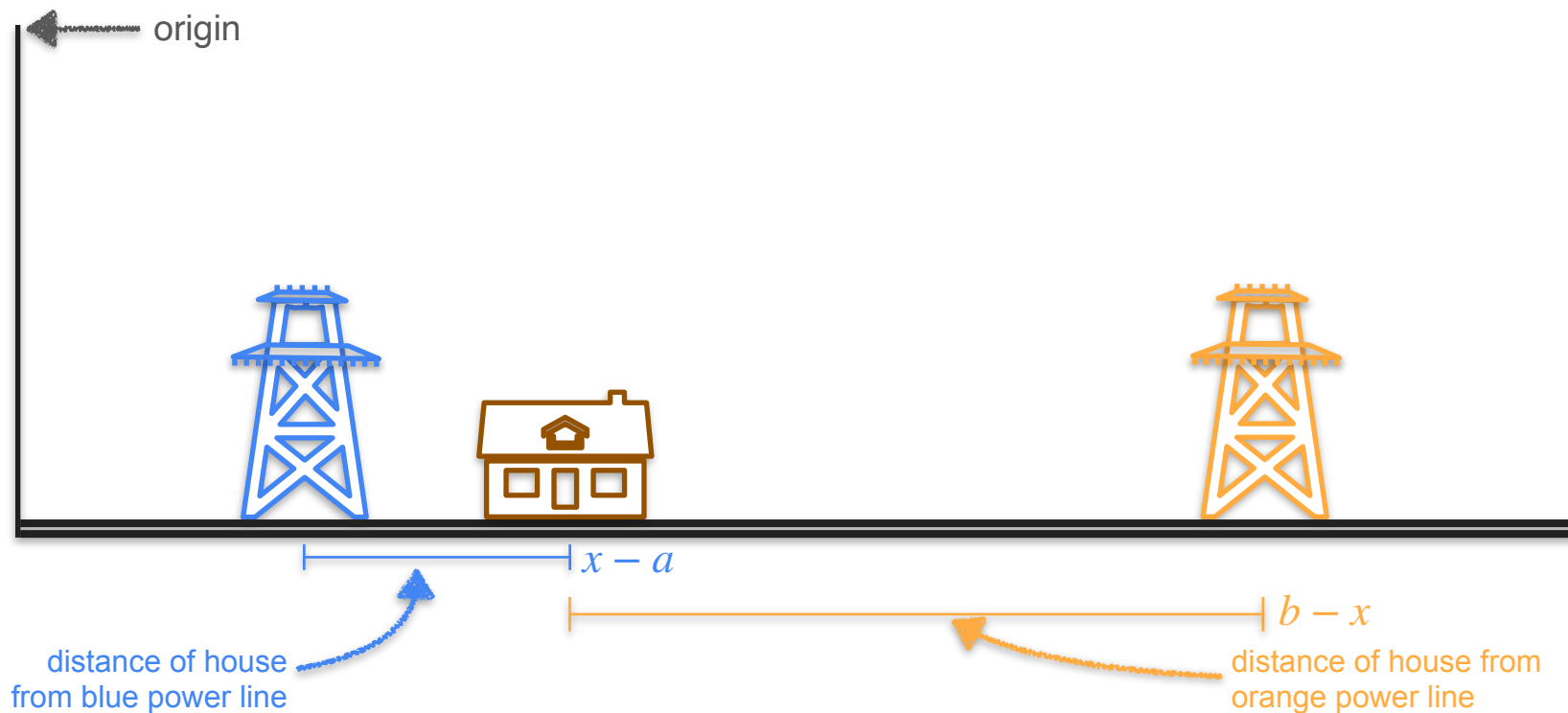
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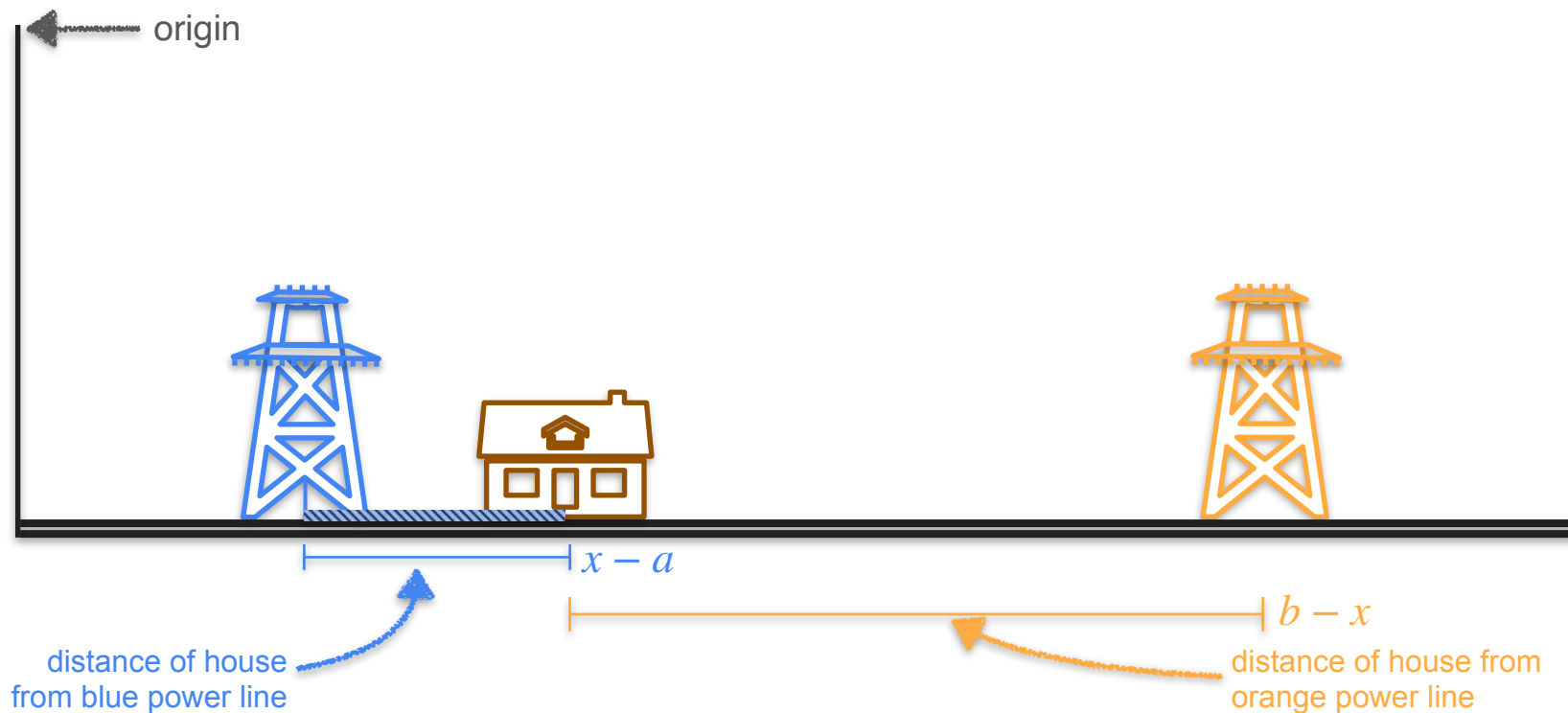
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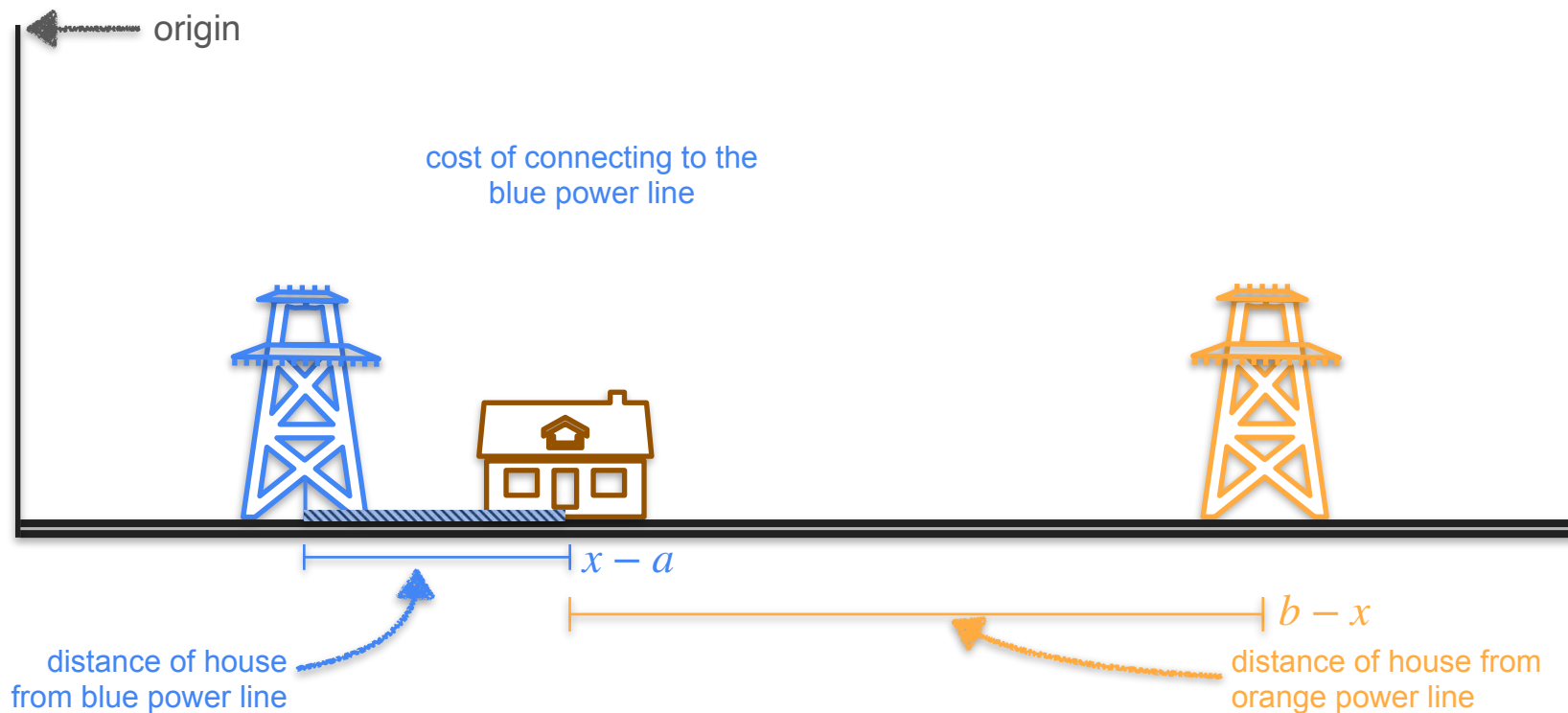
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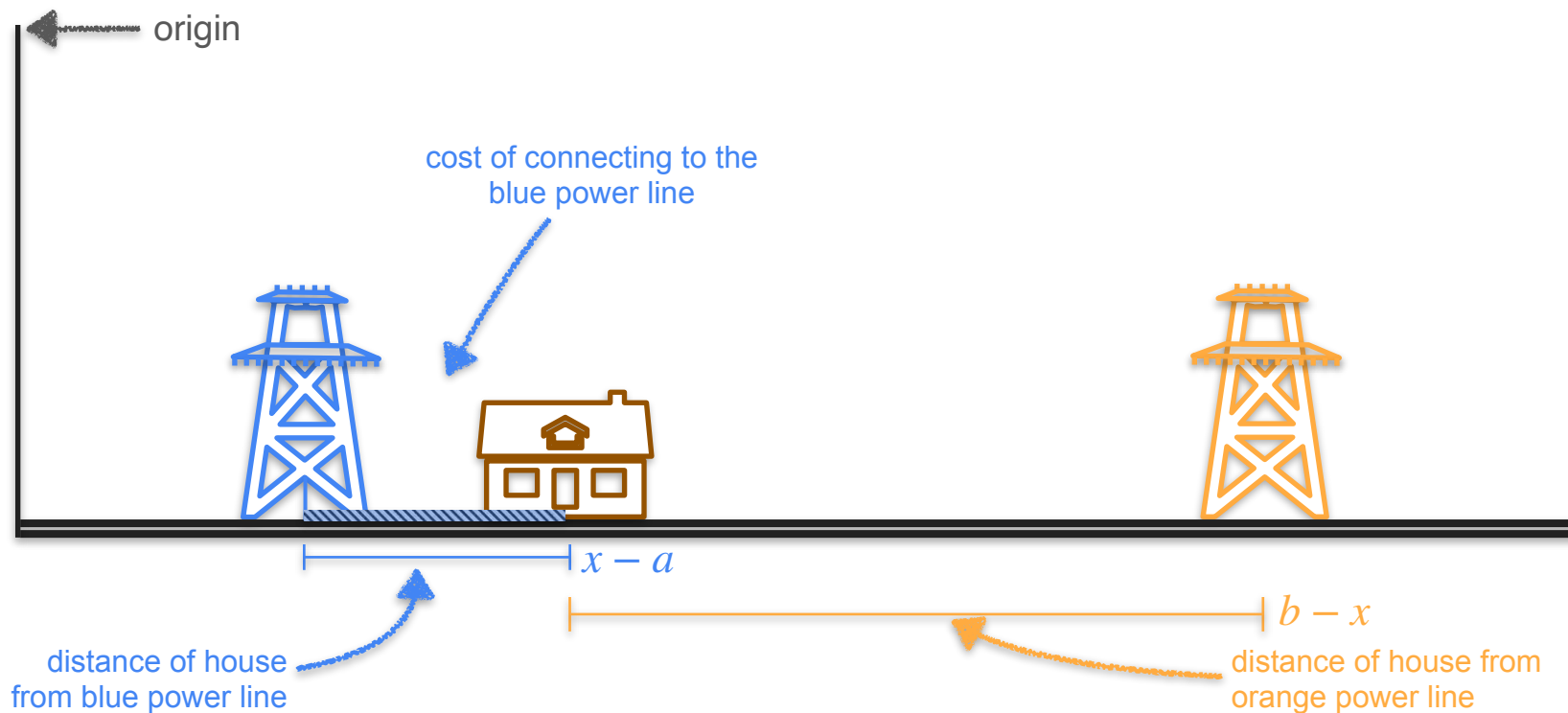
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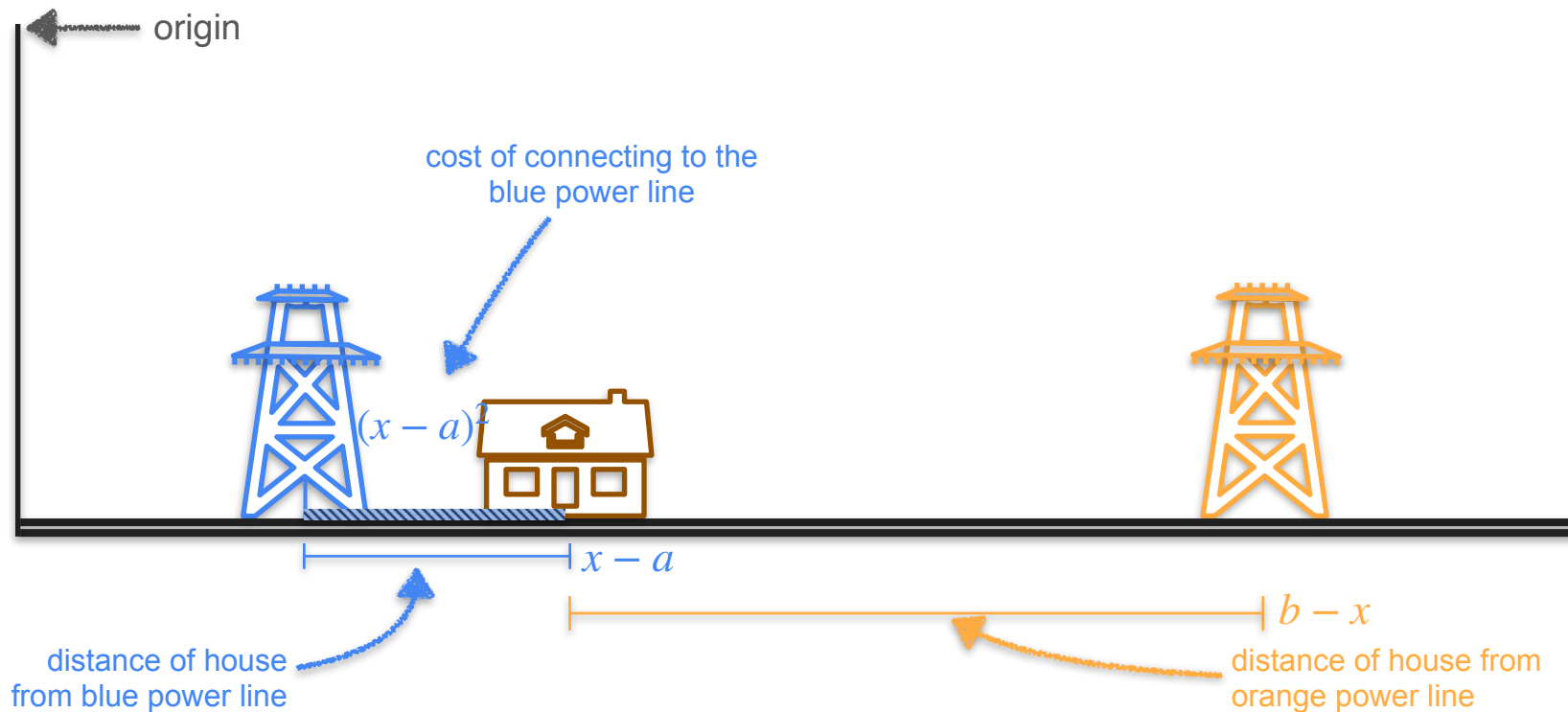
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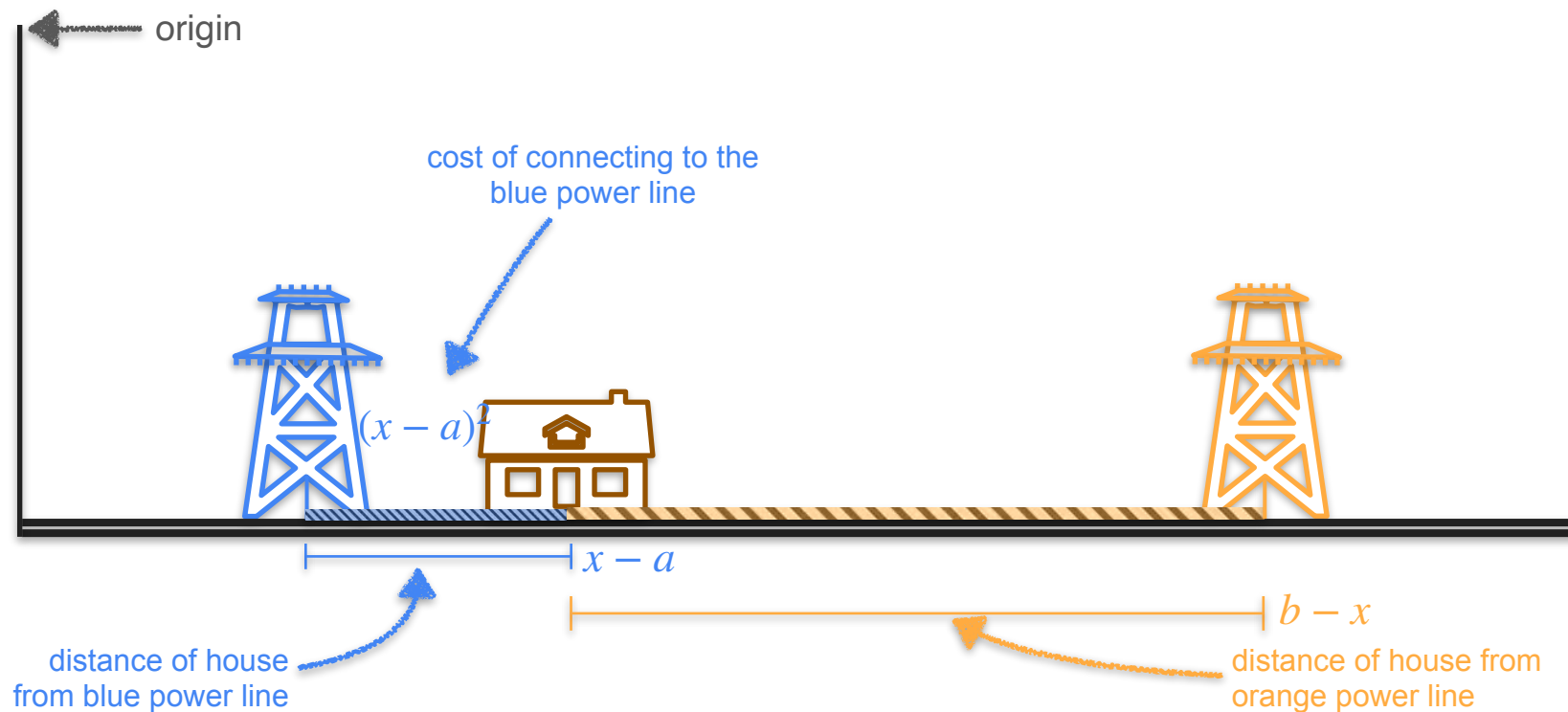
Two Power Line Problem



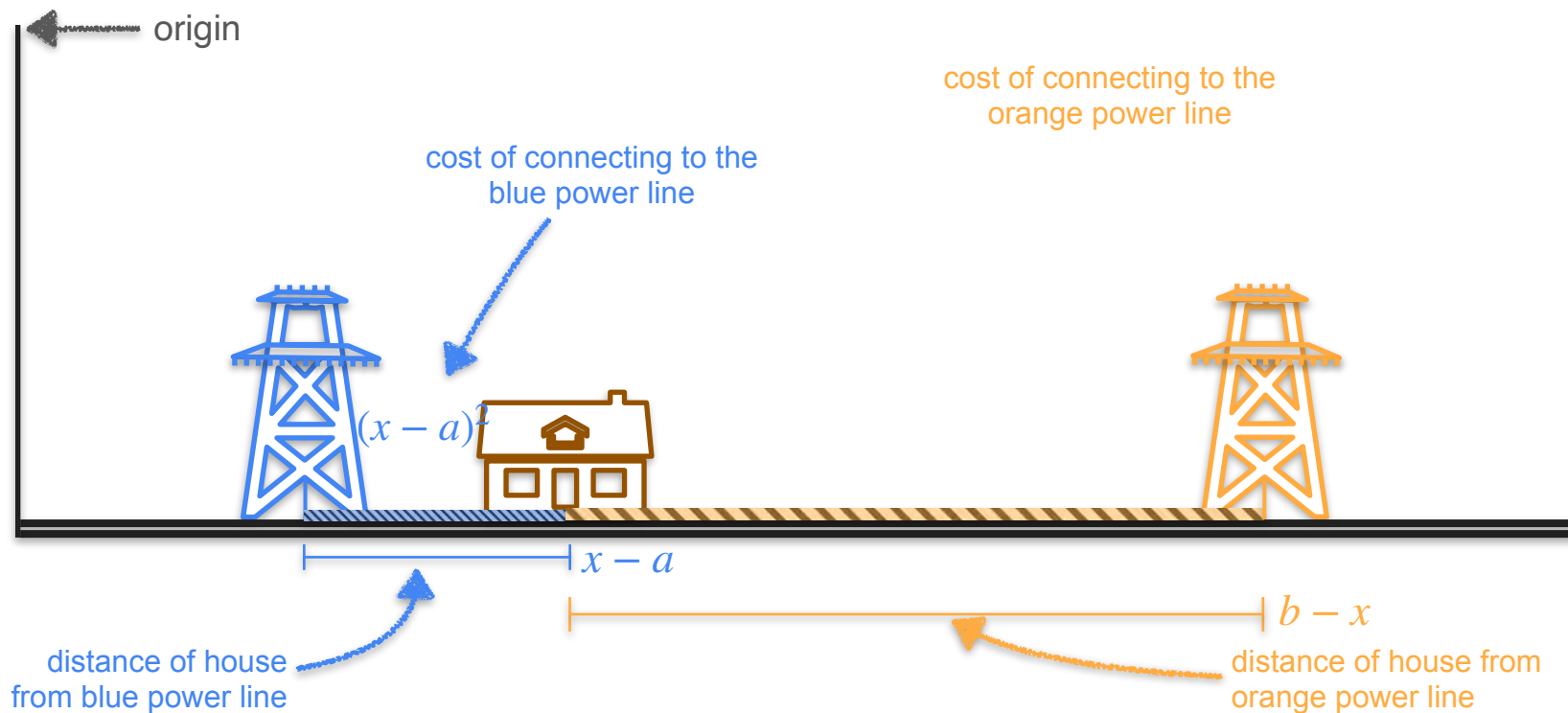
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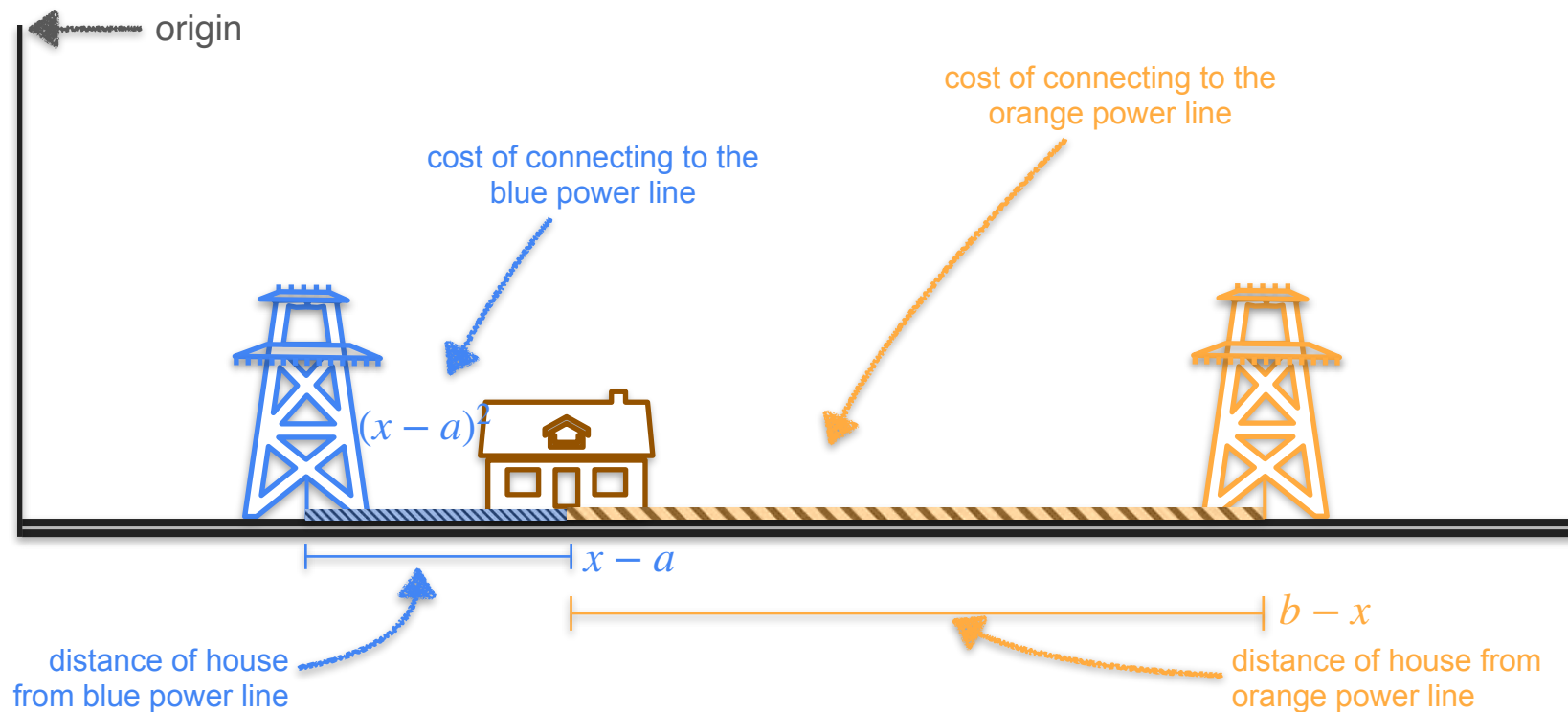
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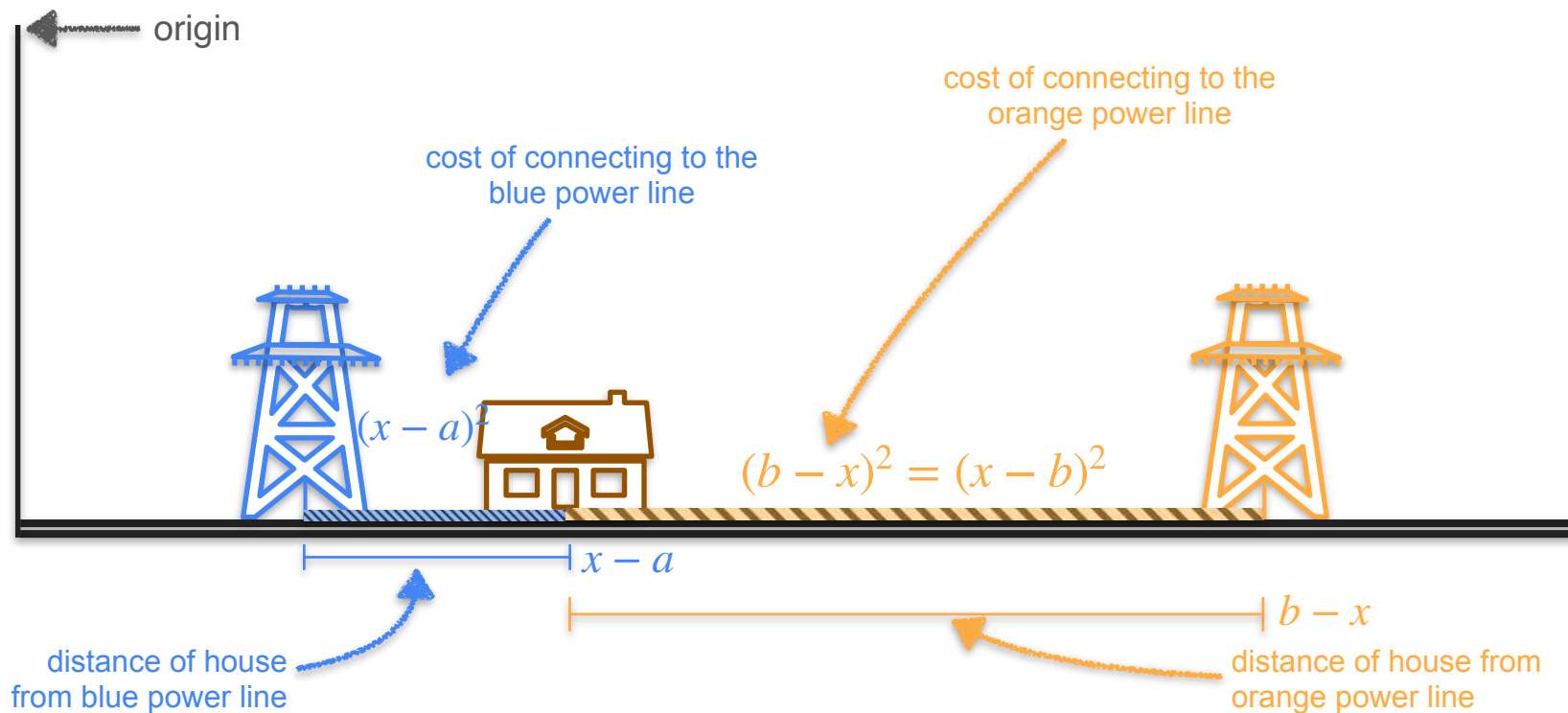
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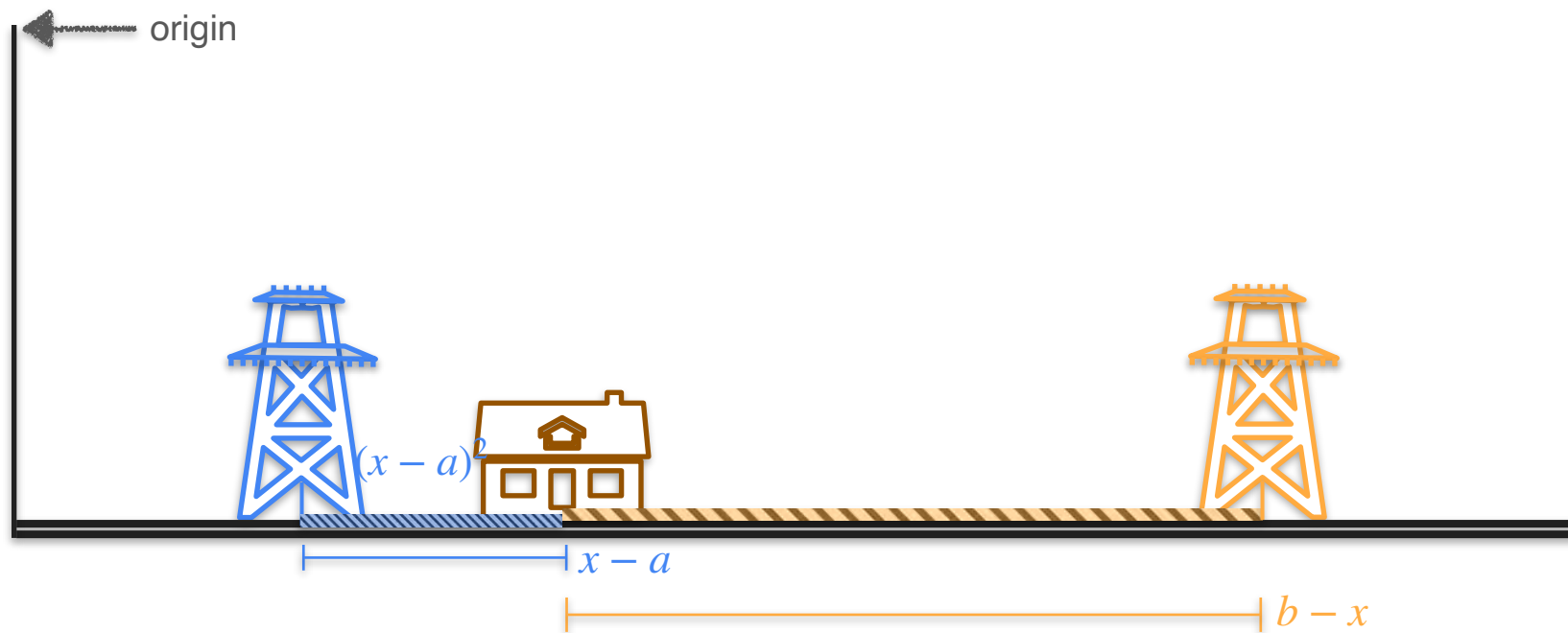
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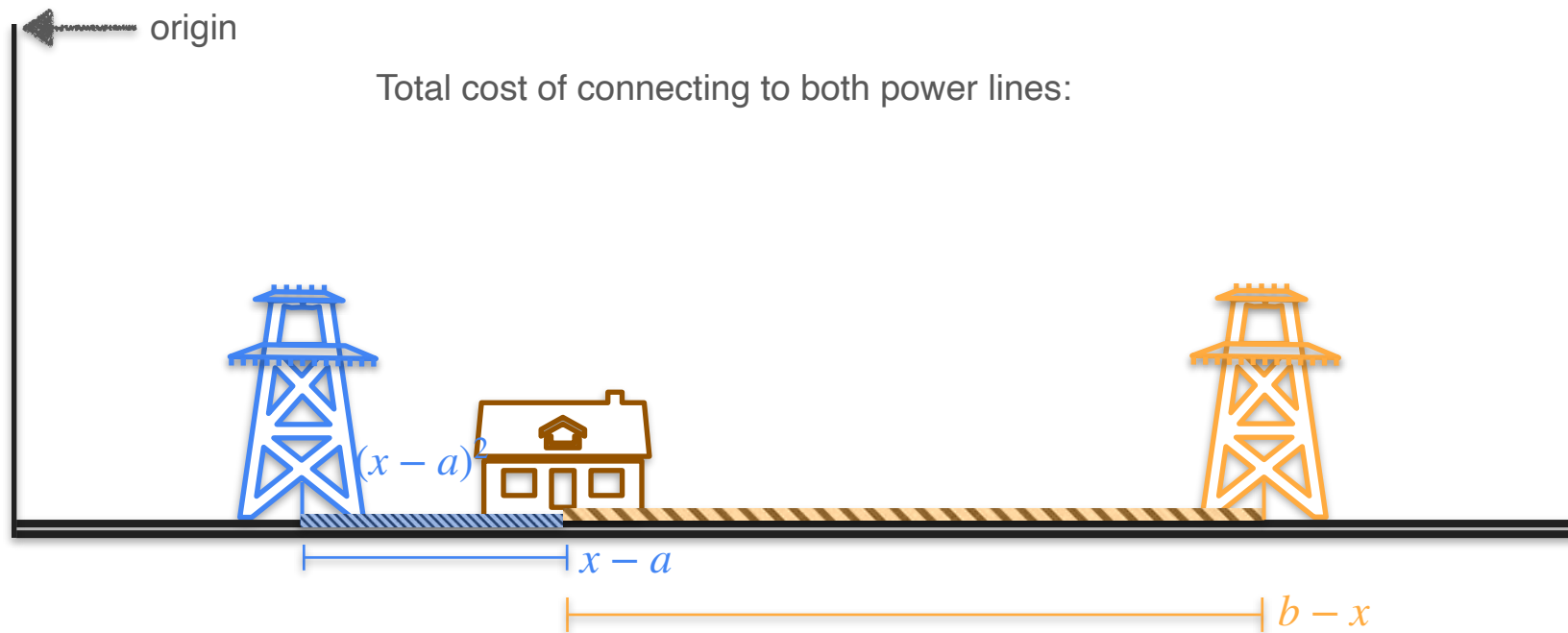
Two Power Line Problem



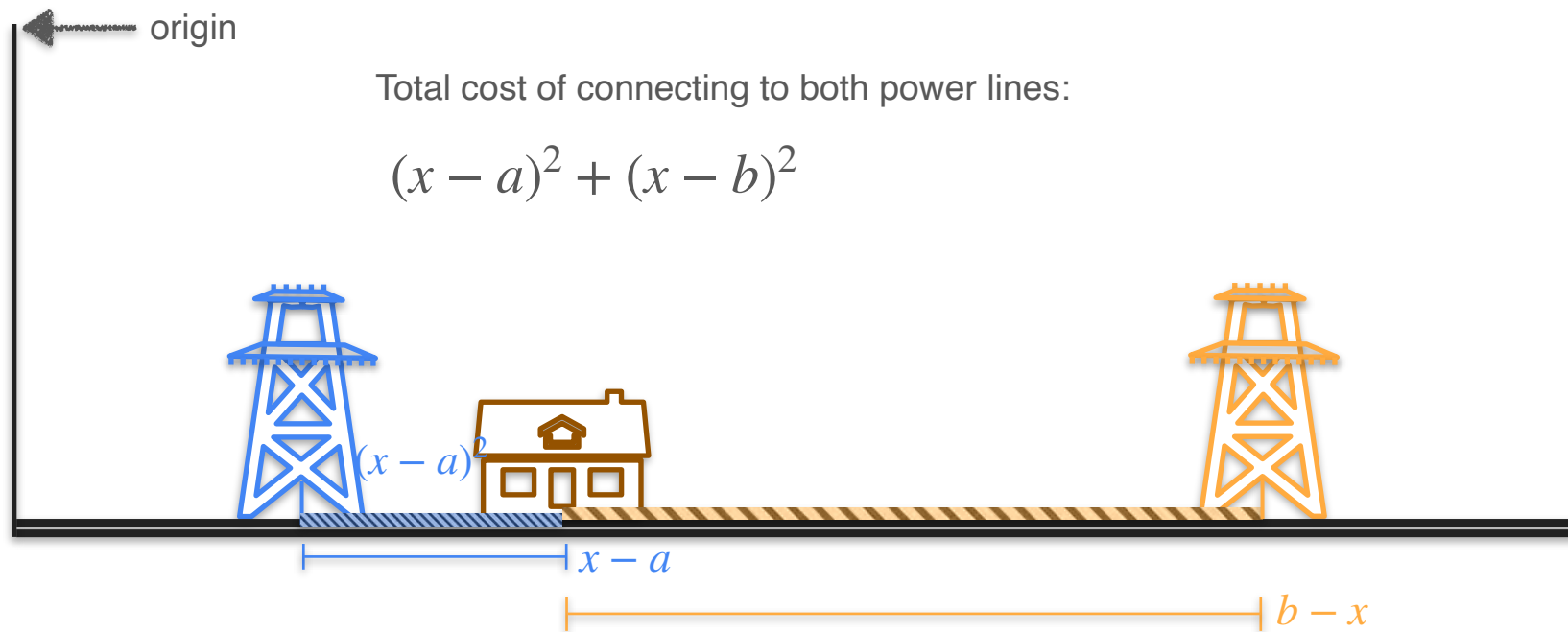
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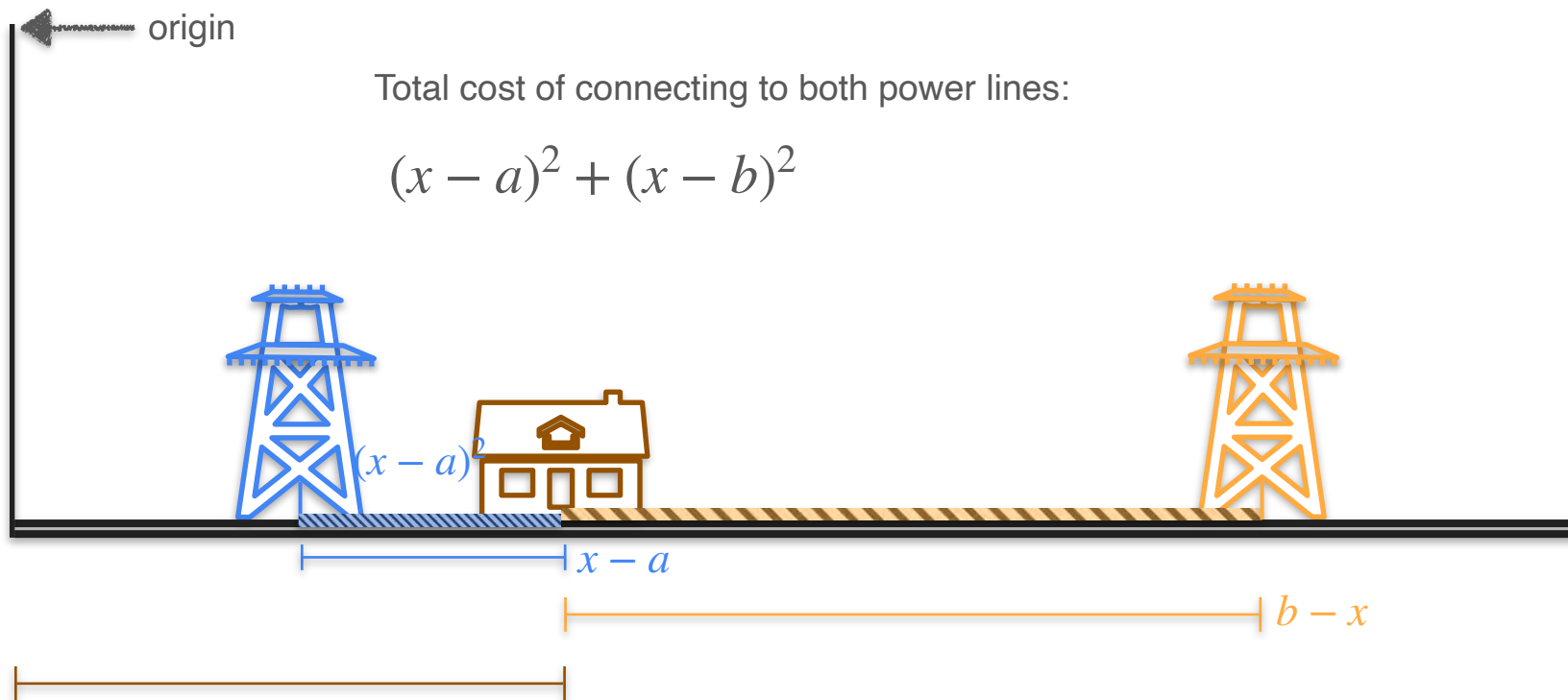
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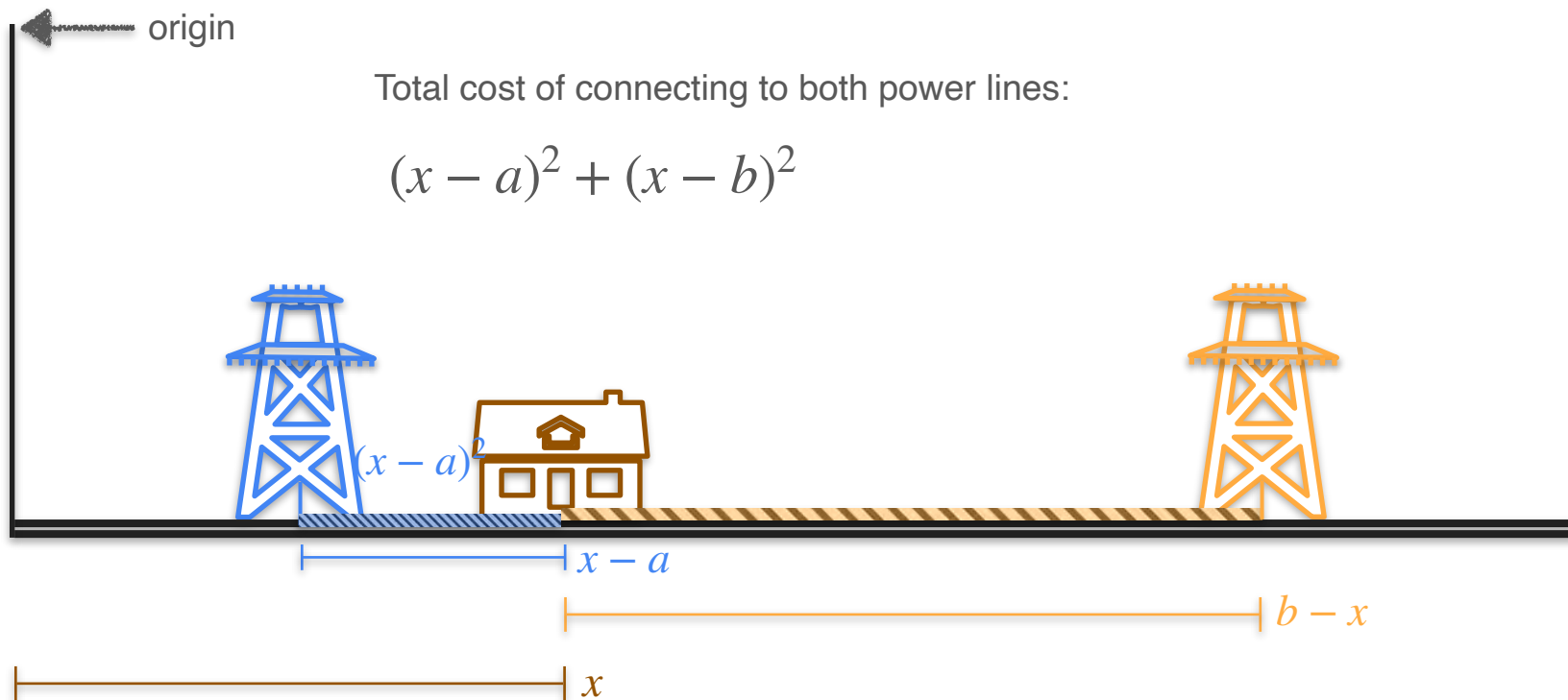
Two Power Line Problem



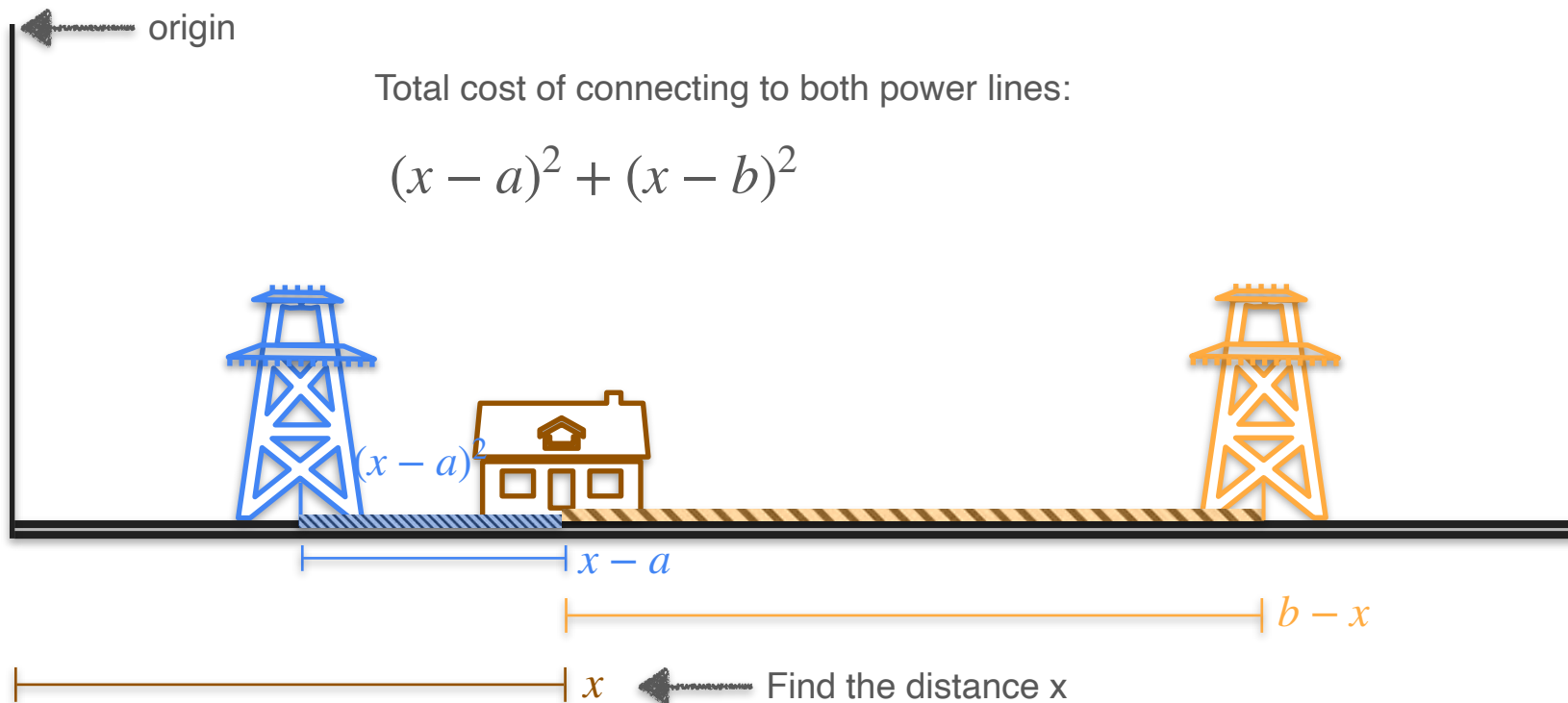
Two Power Line Problem



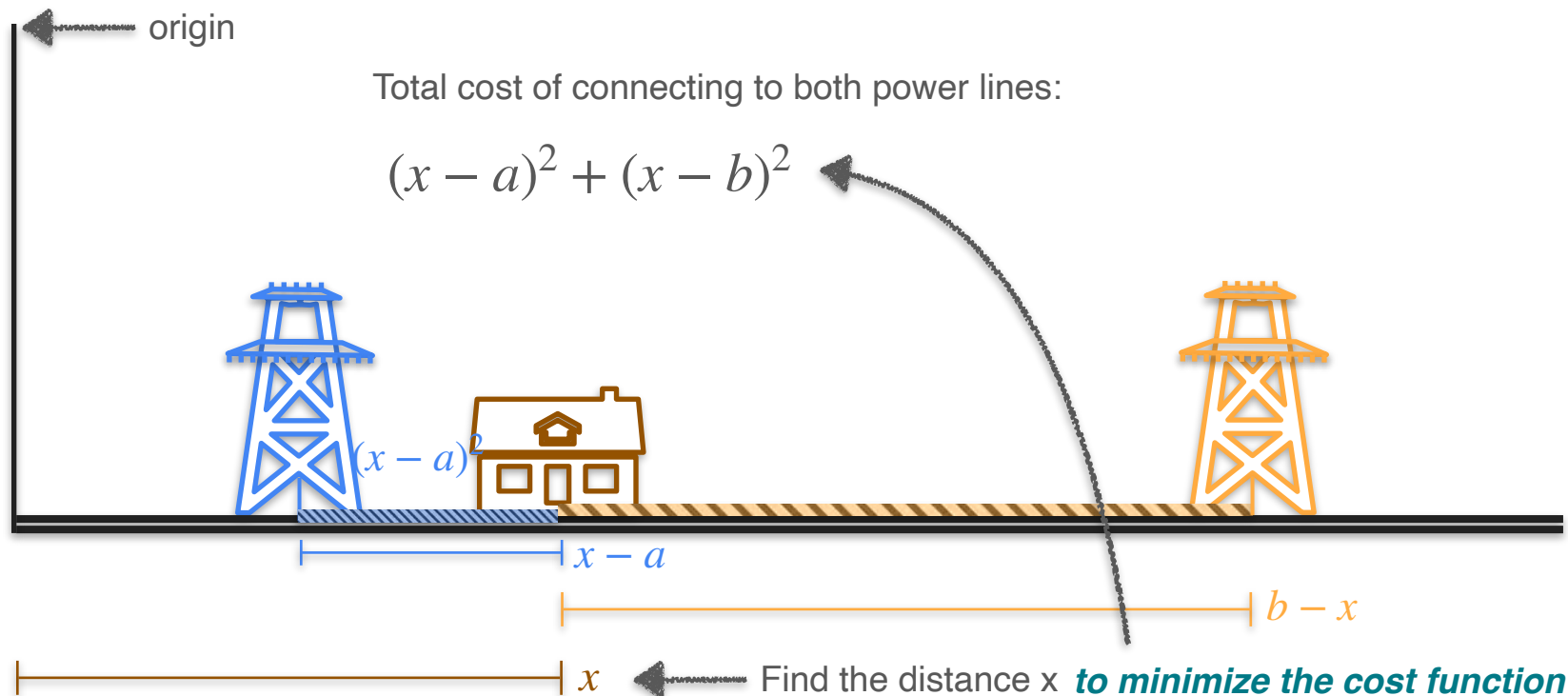
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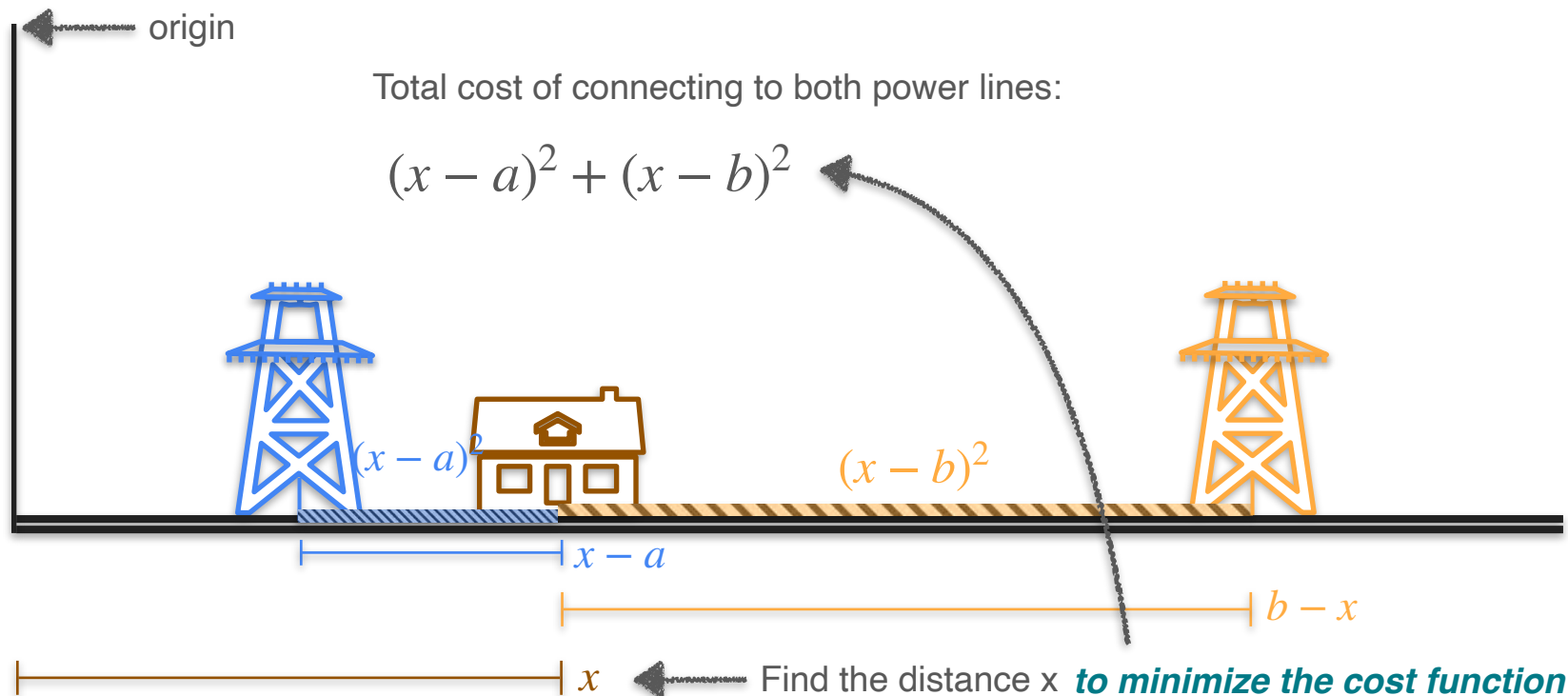
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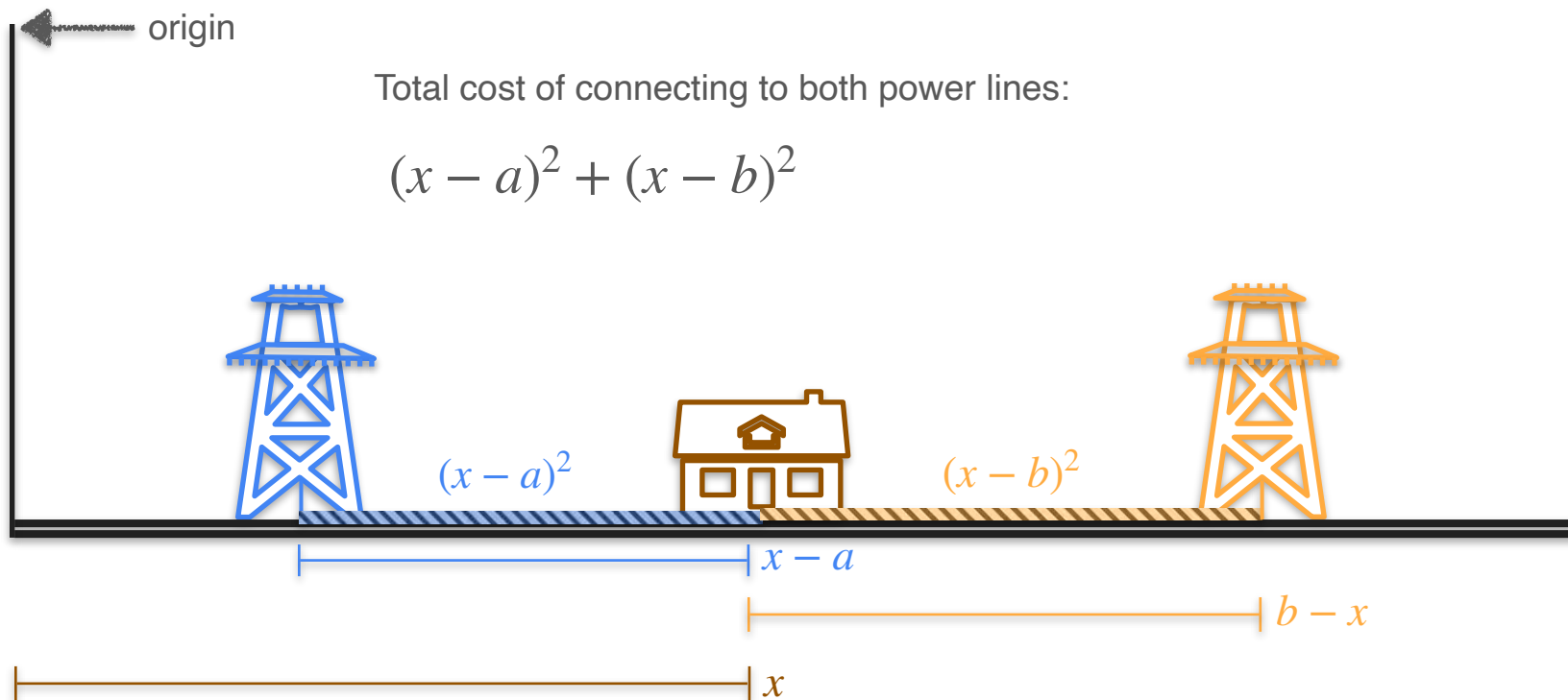
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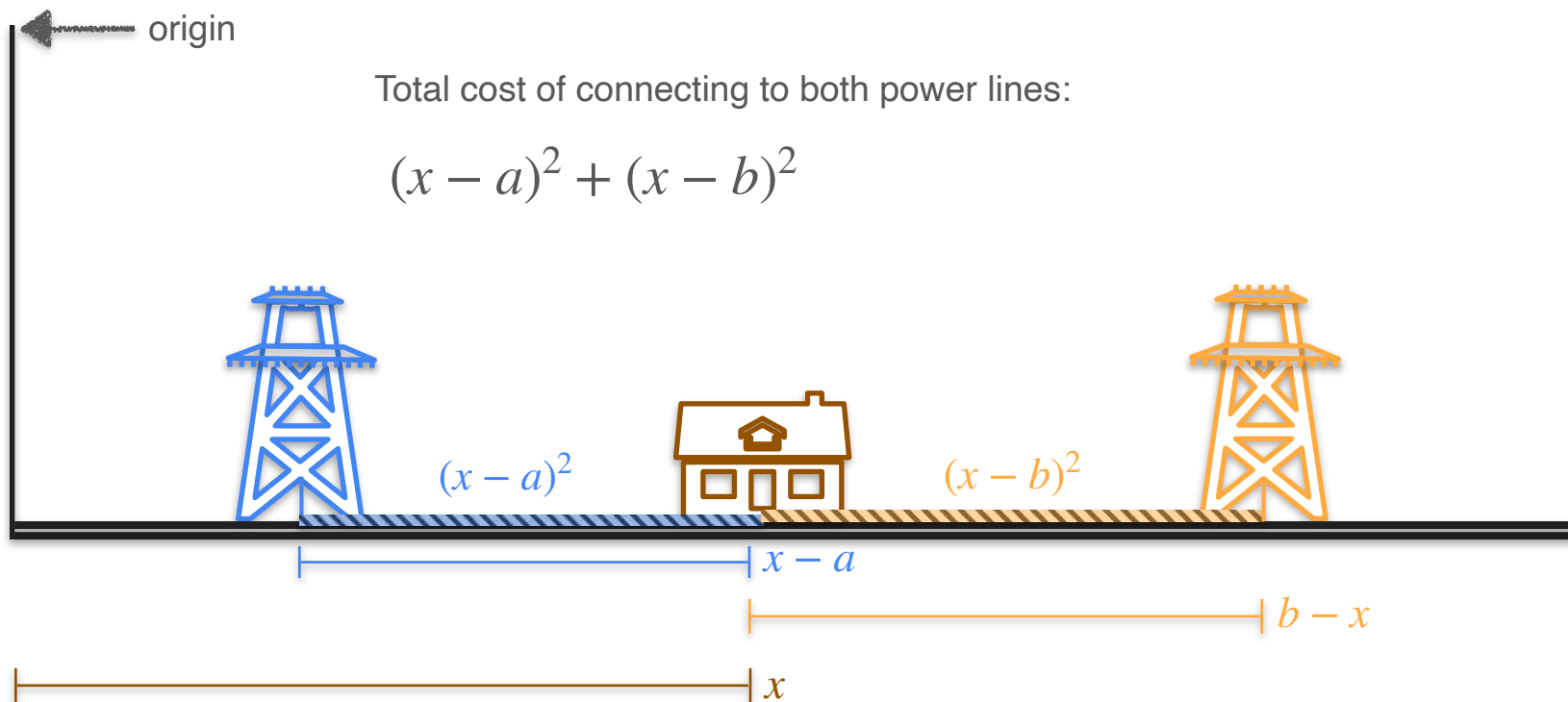
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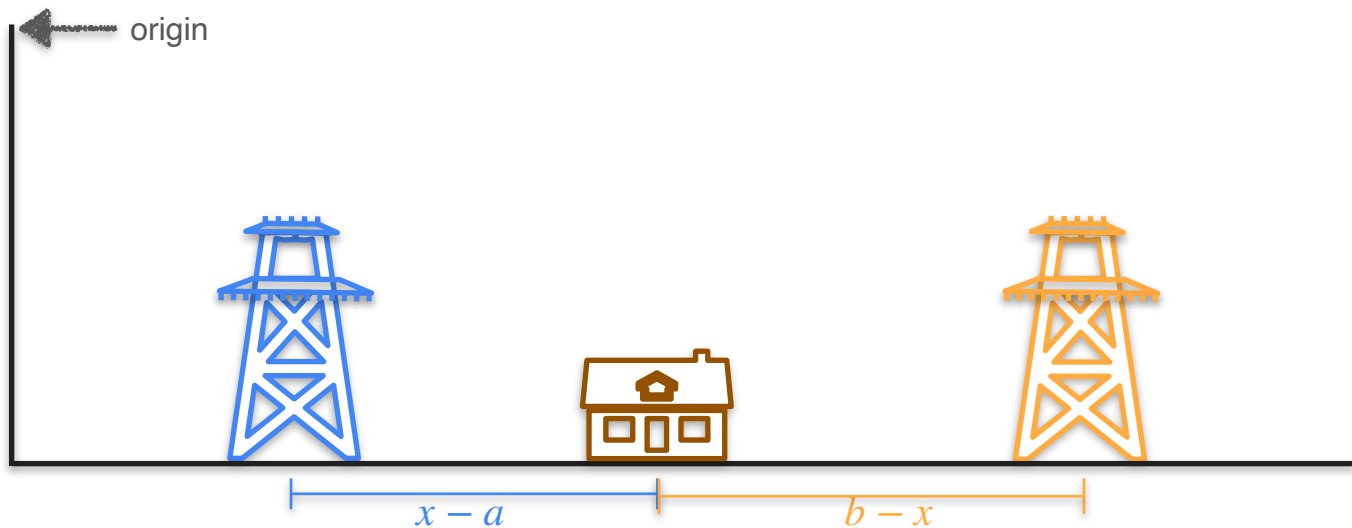
Two Power Line Problem



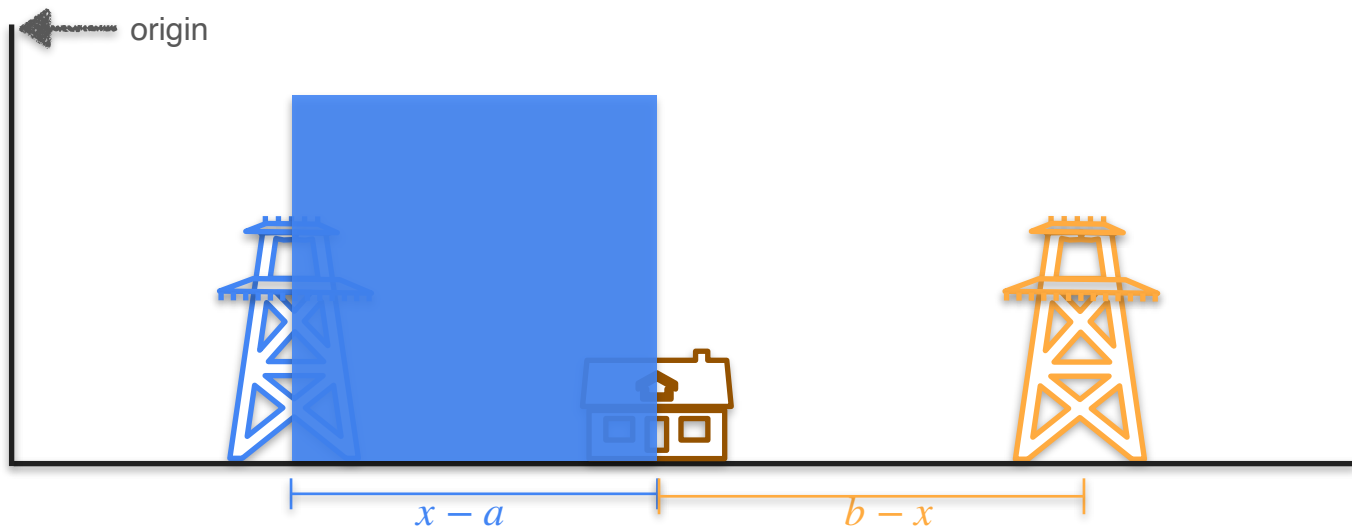
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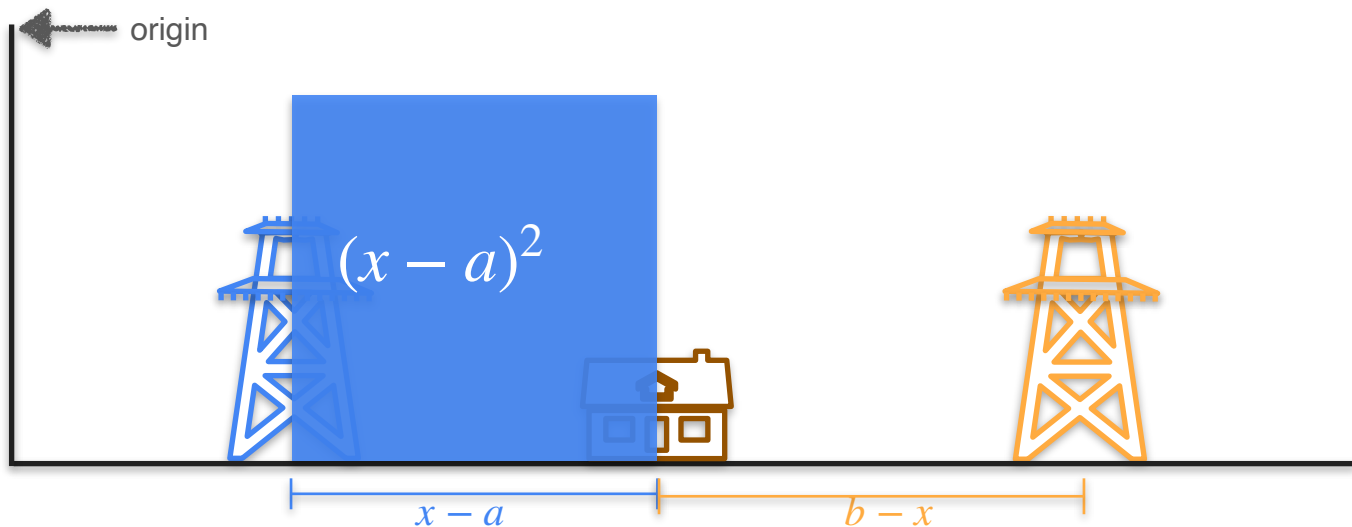
Two Power Line Problem - Square Analogy



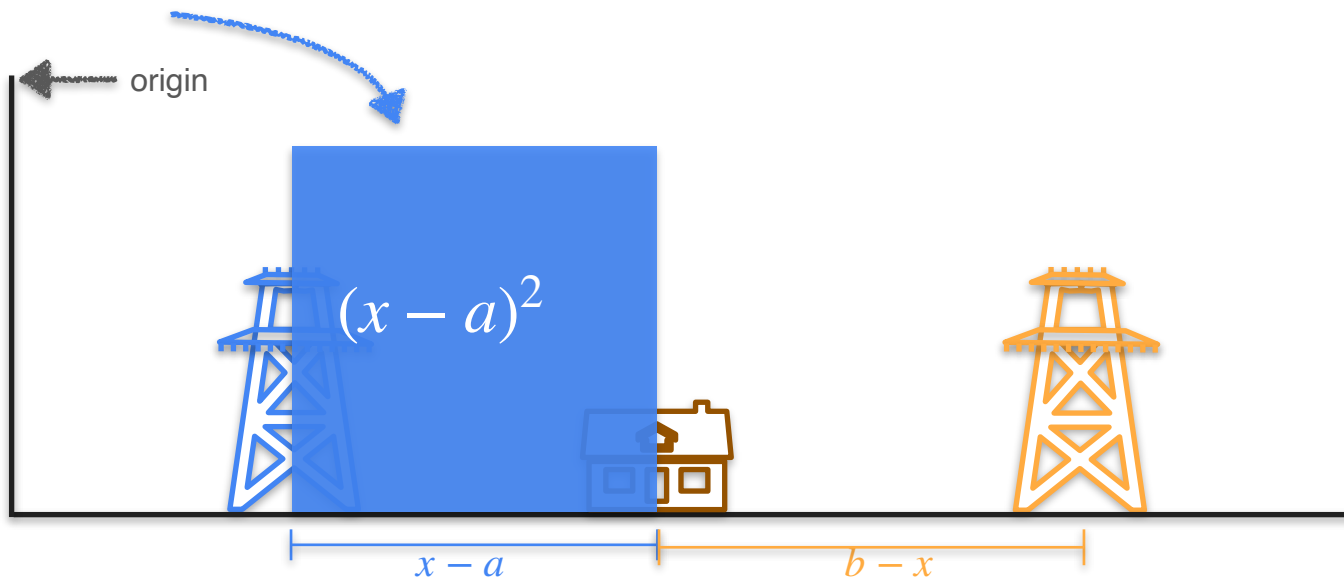
Two Power Line Problem - Square Analogy



Two Power Line Problem - Square Analogy

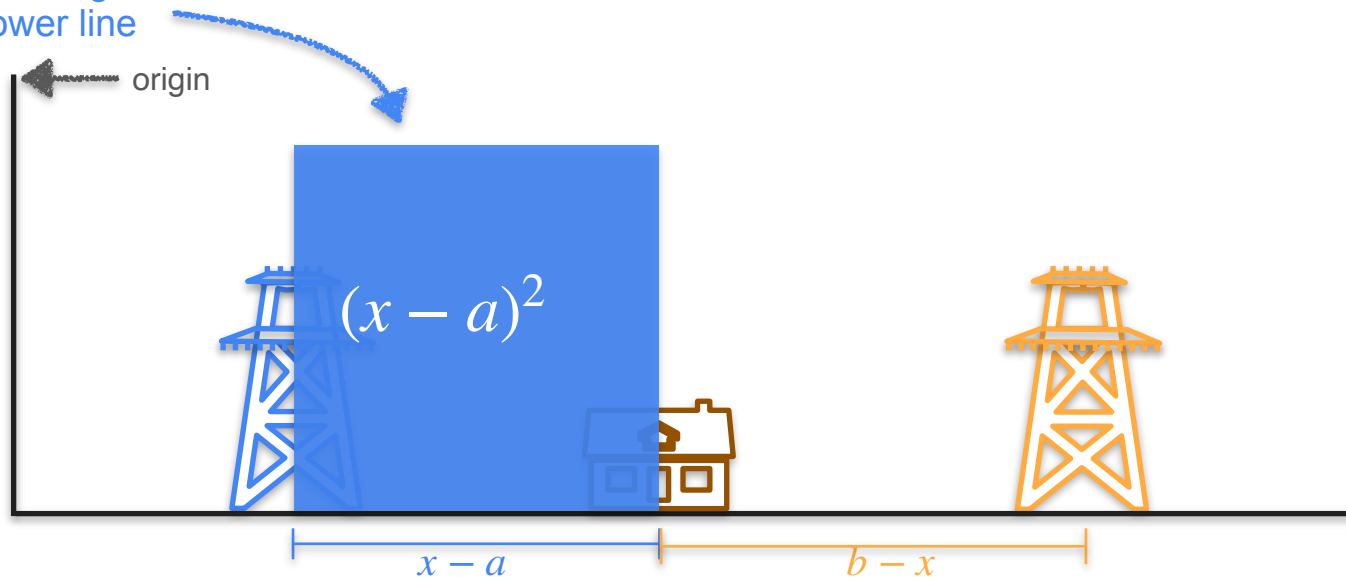


Two Power Line Problem - Square Analogy



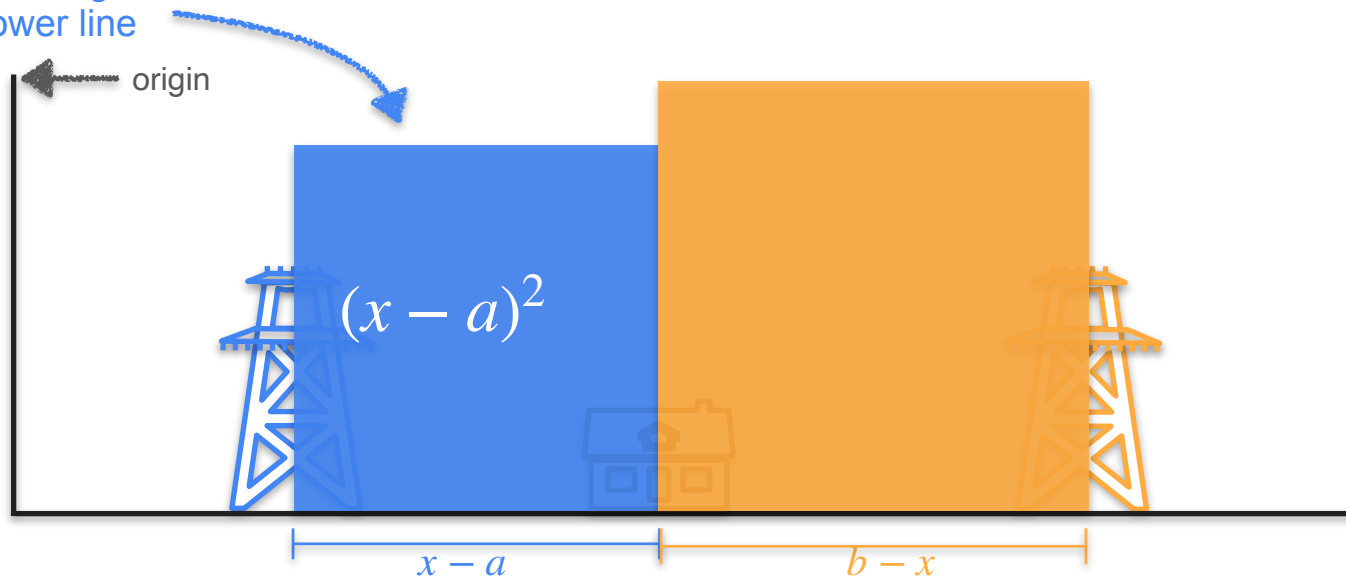
Two Power Line Problem - Square Analogy

cost of connecting to
the blue power line



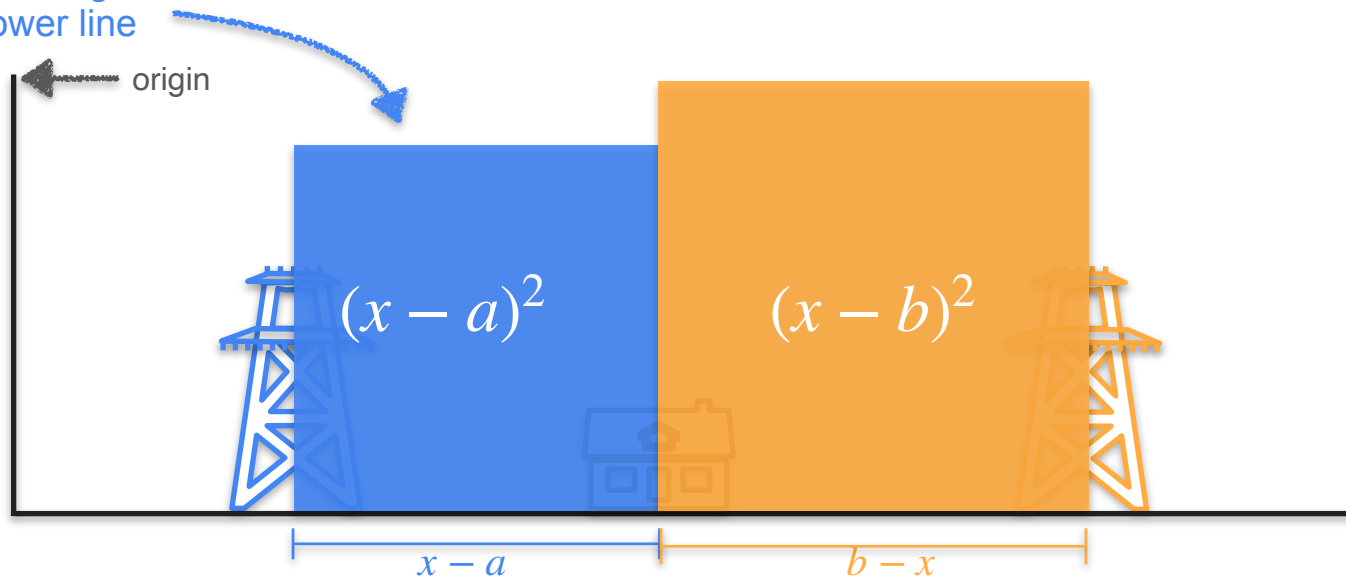
Two Power Line Problem - Square Analogy

cost of connecting to
the blue power line



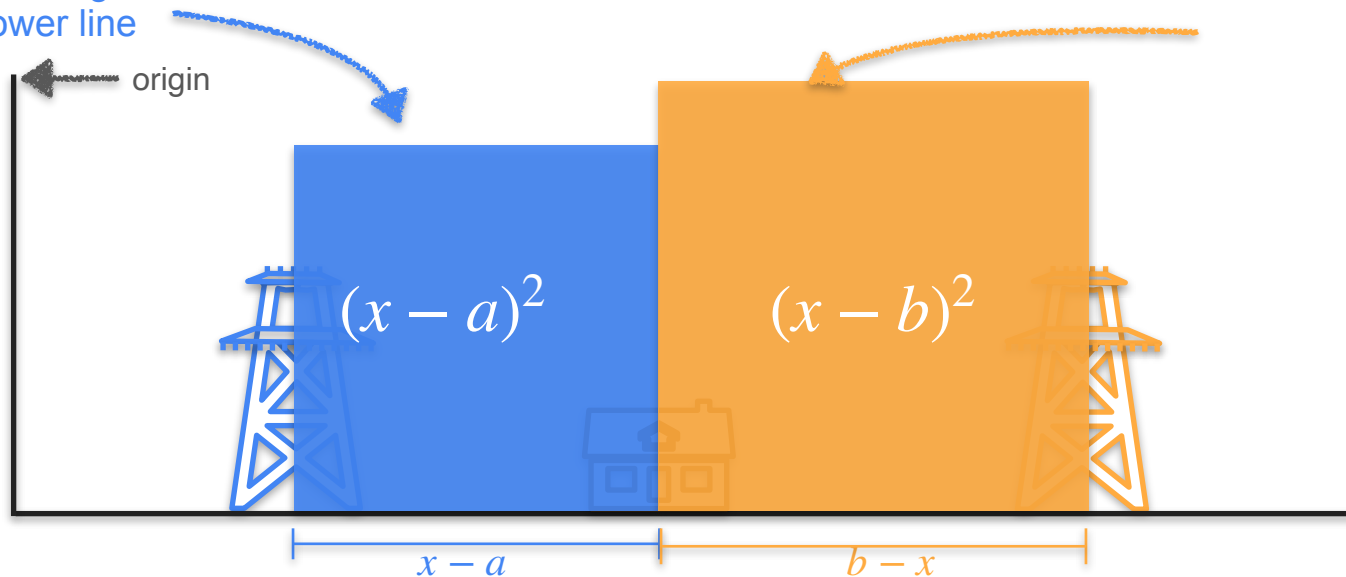
Two Power Line Problem - Square Analogy

cost of connecting to
the blue power line

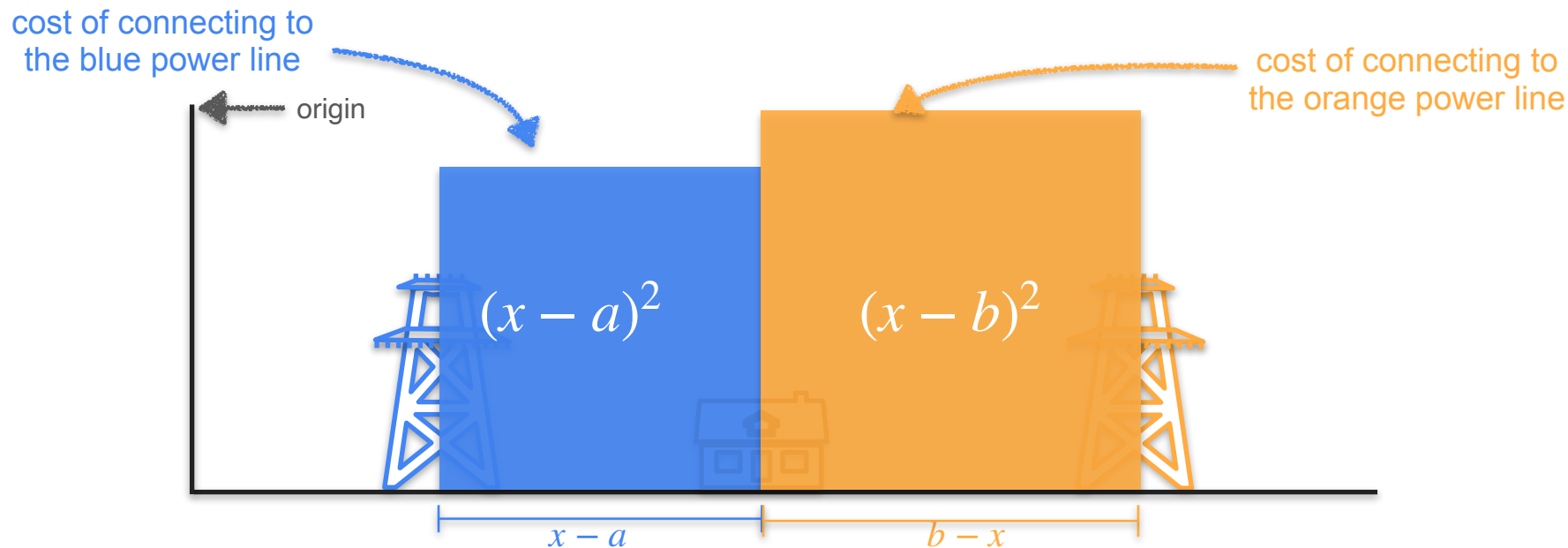


Two Power Line Problem - Square Analogy

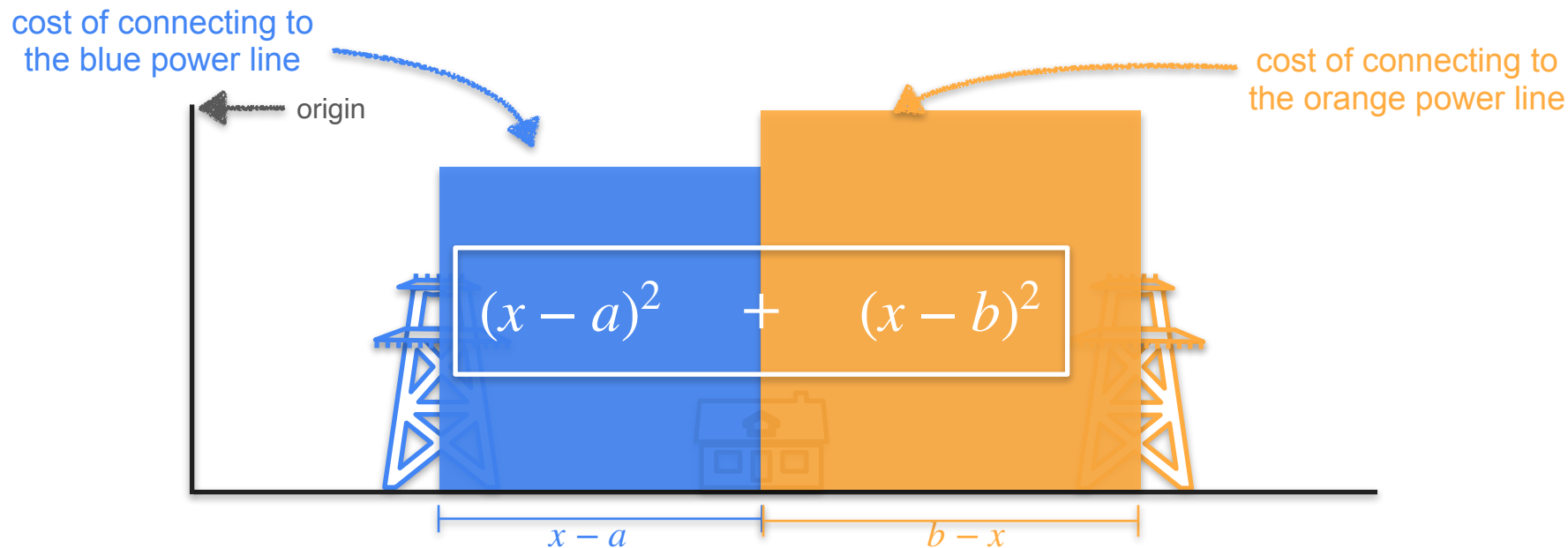
cost of connecting to
the blue power line



Two Power Line Problem - Square Analogy



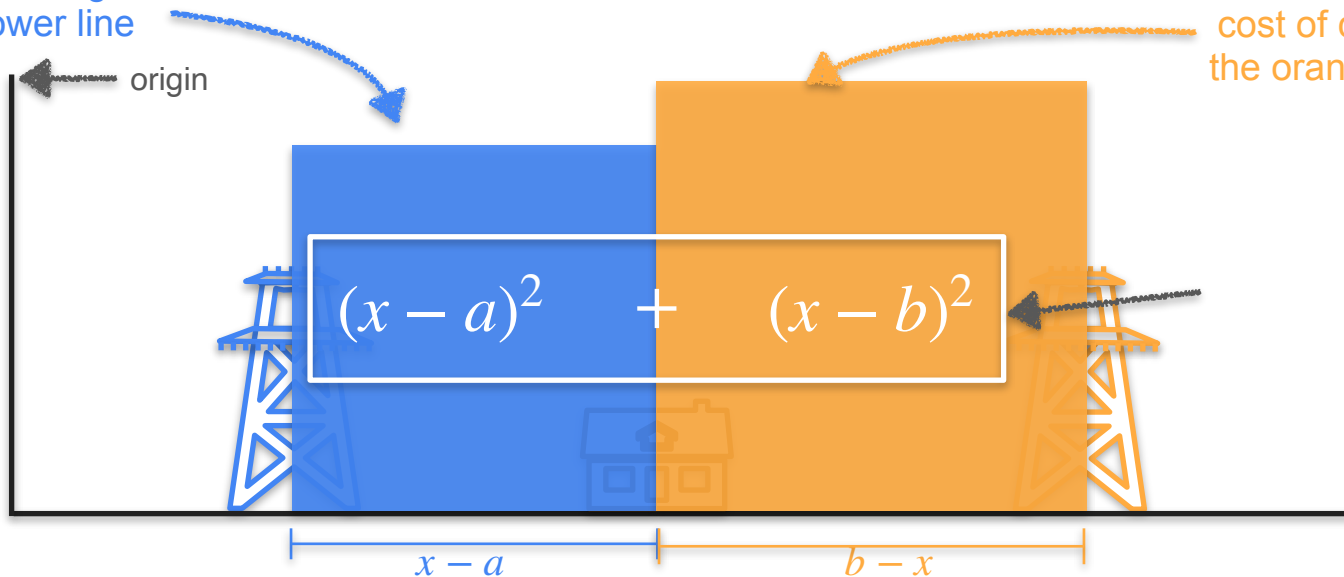
Two Power Line Problem - Square Analogy



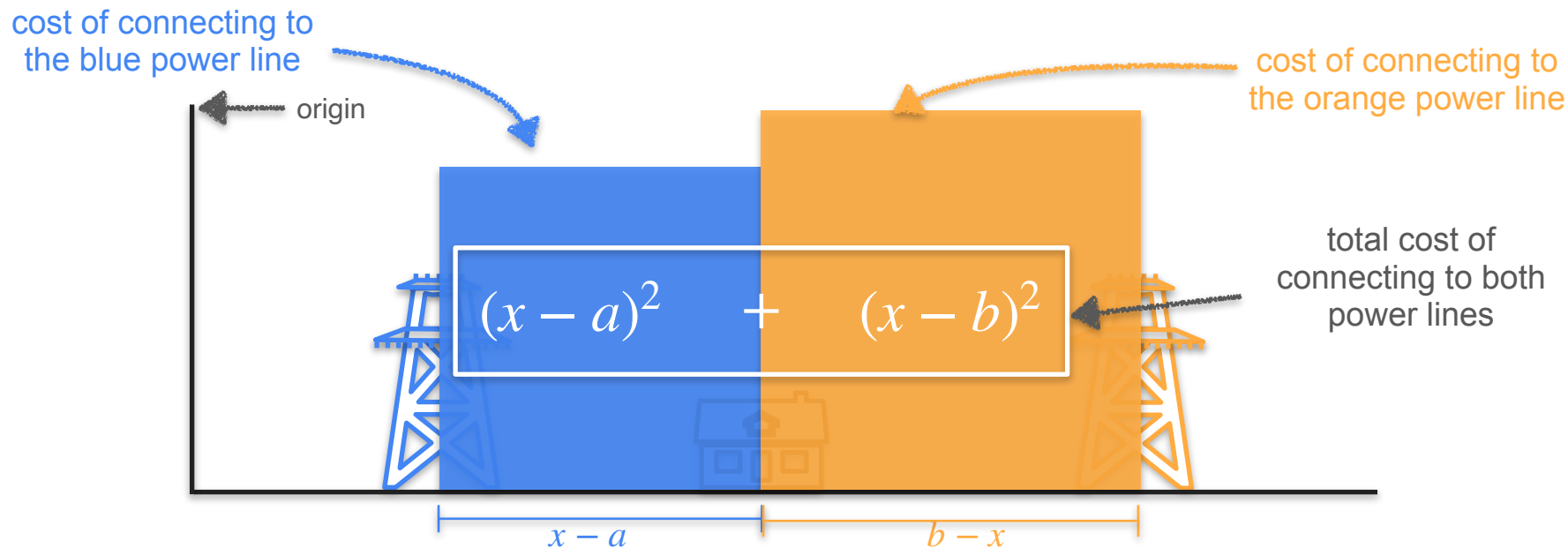
Two Power Line Problem - Square Analogy

cost of connecting to
the blue power line

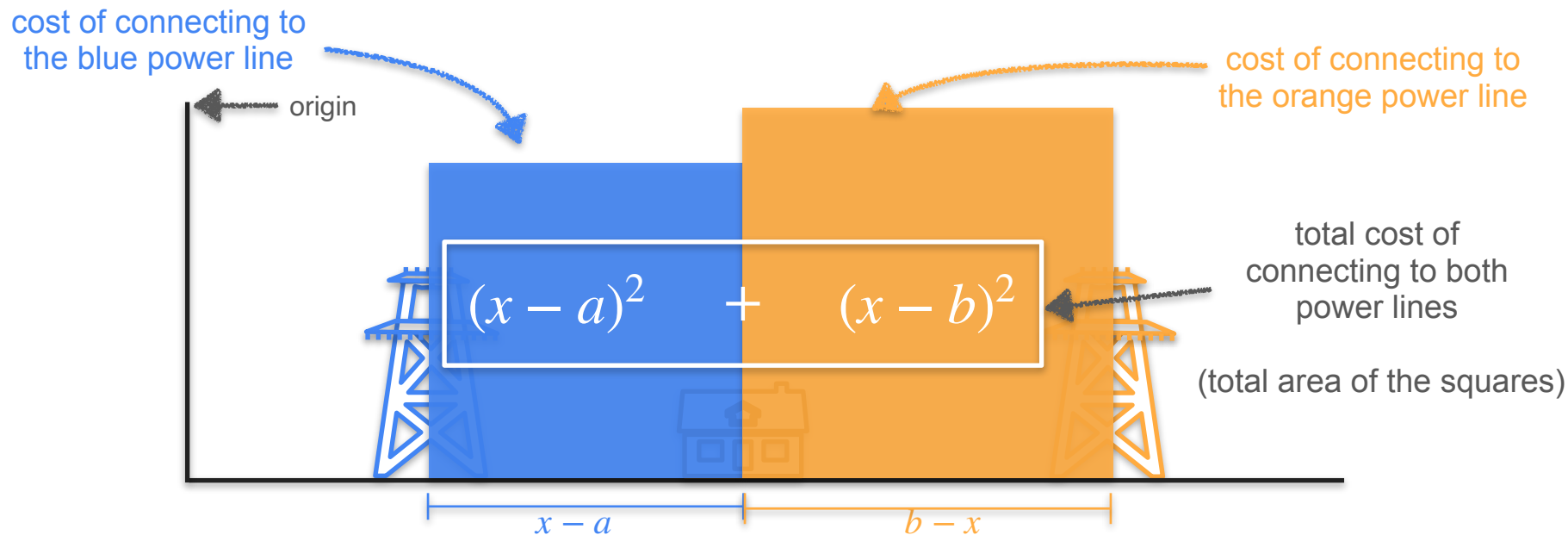
cost of connecting to
the orange power line



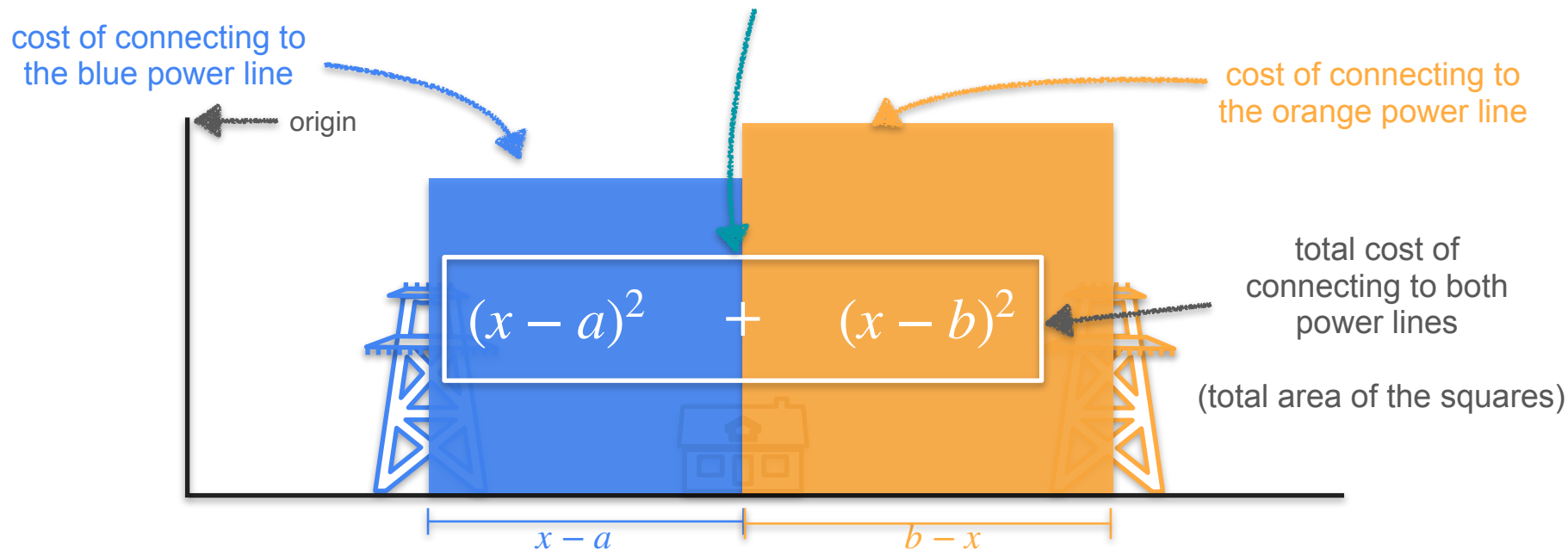
Two Power Line Problem - Square Analogy



Two Power Line Problem - Square Analogy

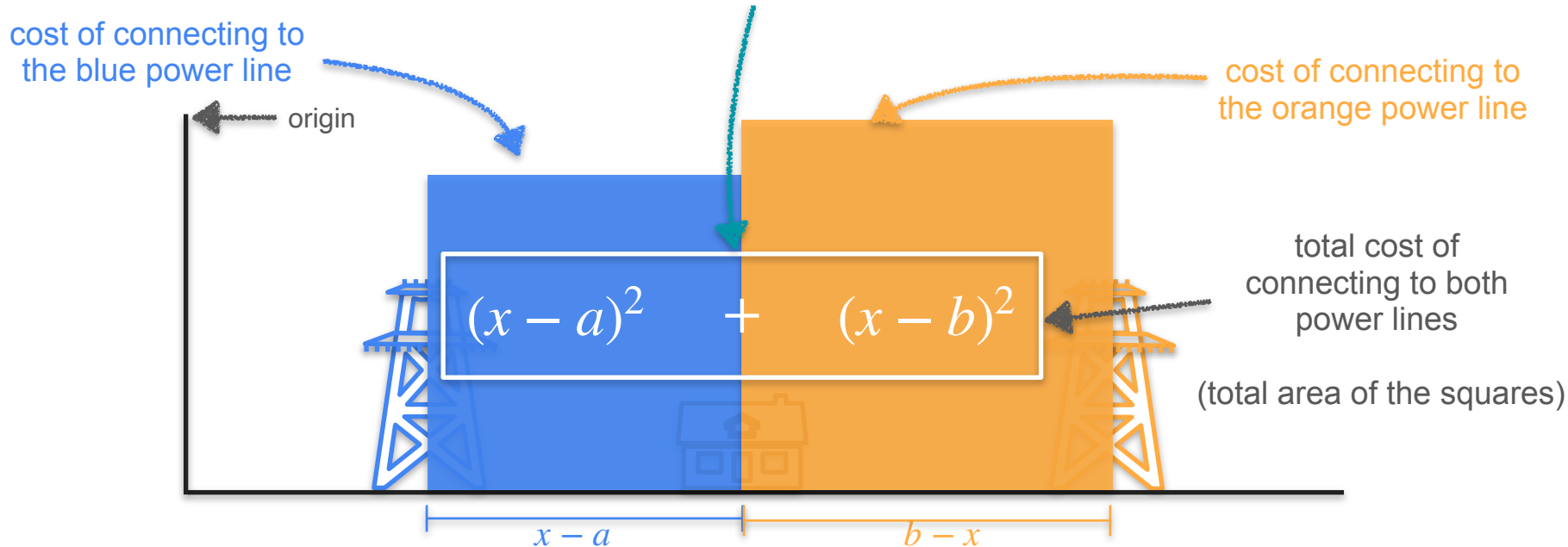


Two Power Line Problem - Square Analogy

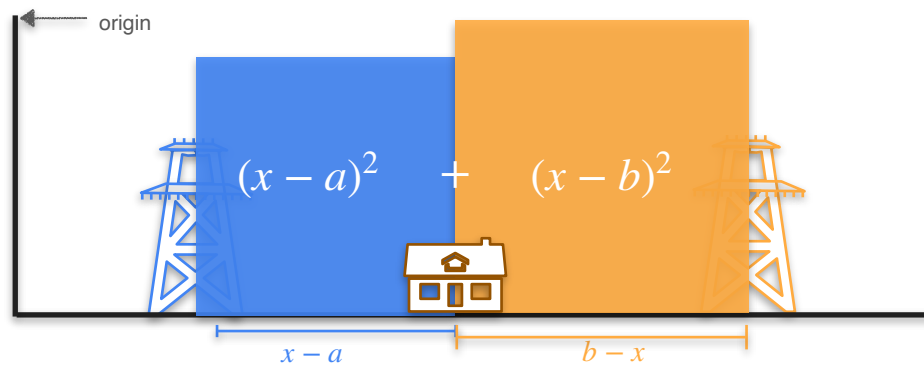


Two Power Line Problem - Square Analogy

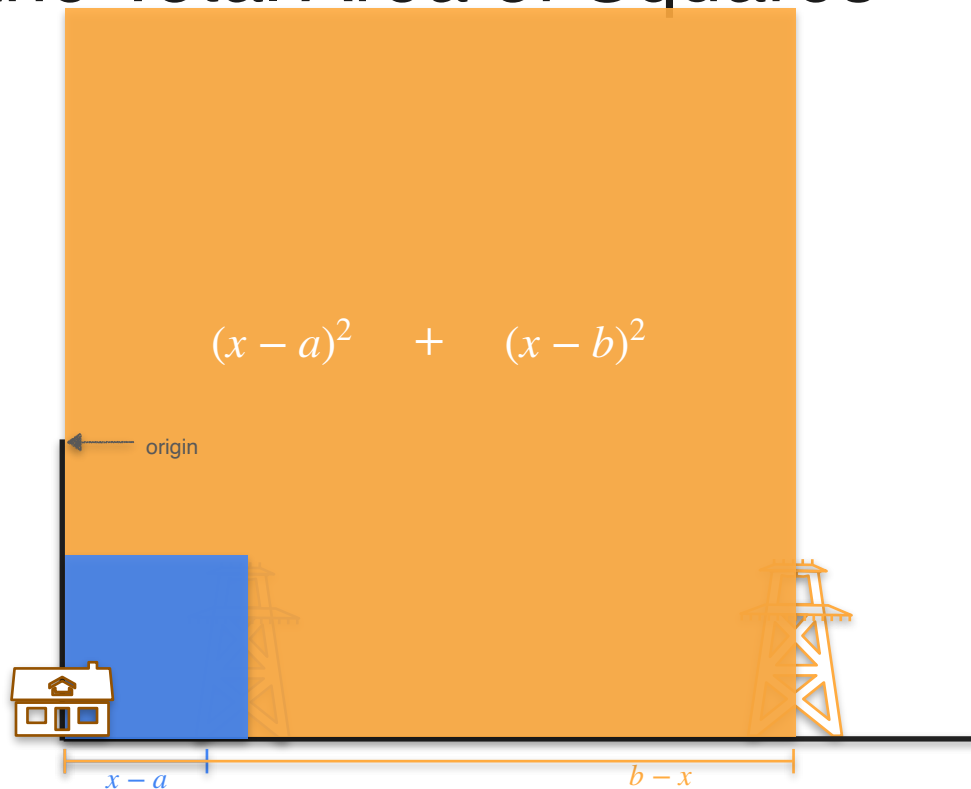
Goal: Minimize the total area of the squares



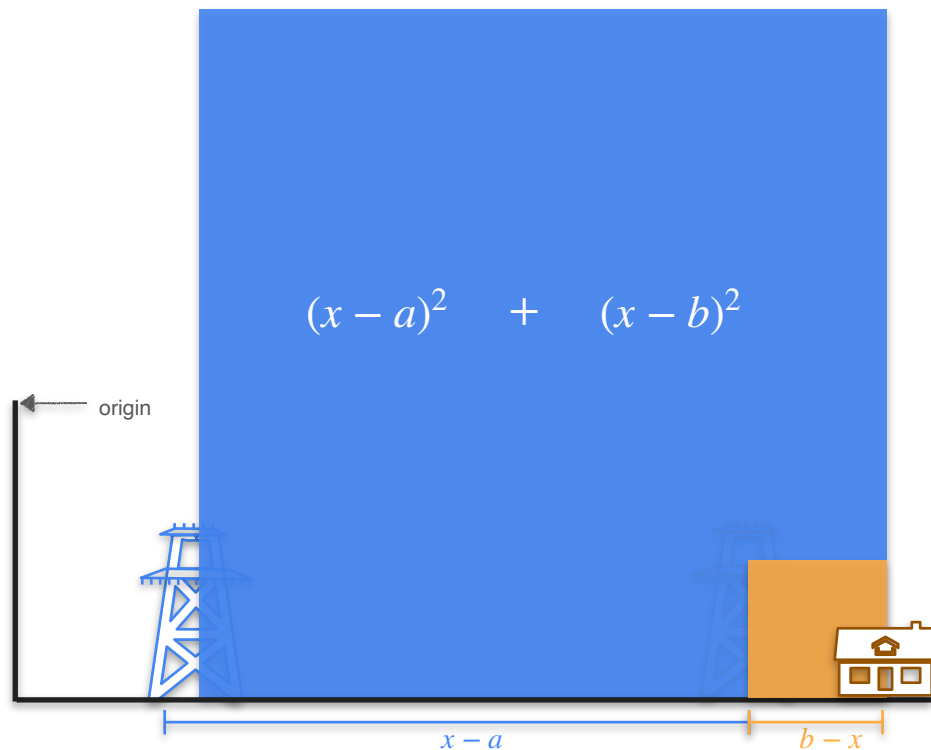
Minimize the Total Area of Squares



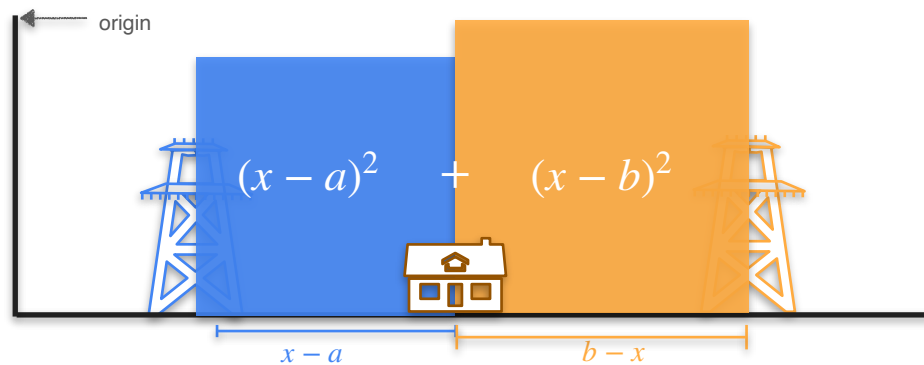
Minimize the Total Area of Squares



Minimize the Total Area of Squares



Minimize the Total Area of Squares



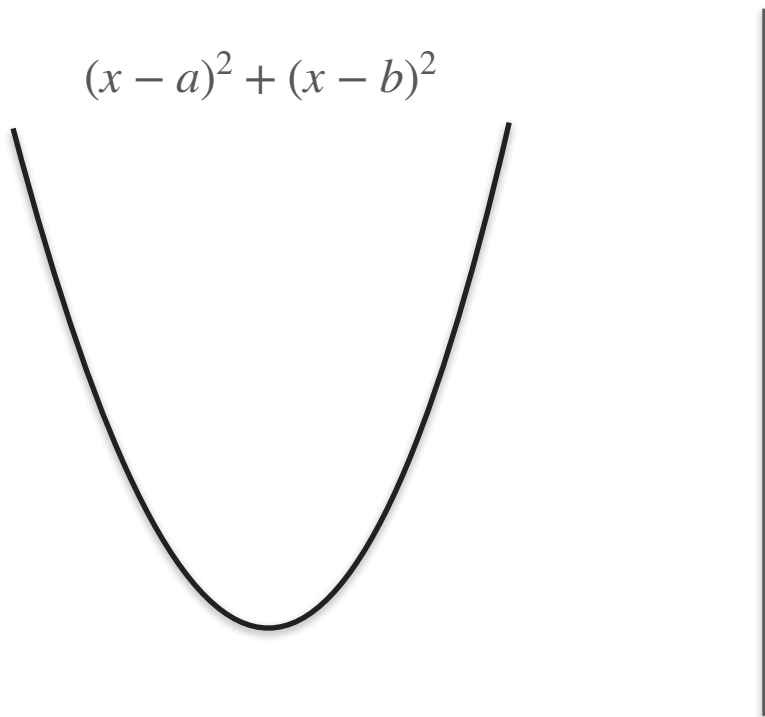
Two Power Line Problem



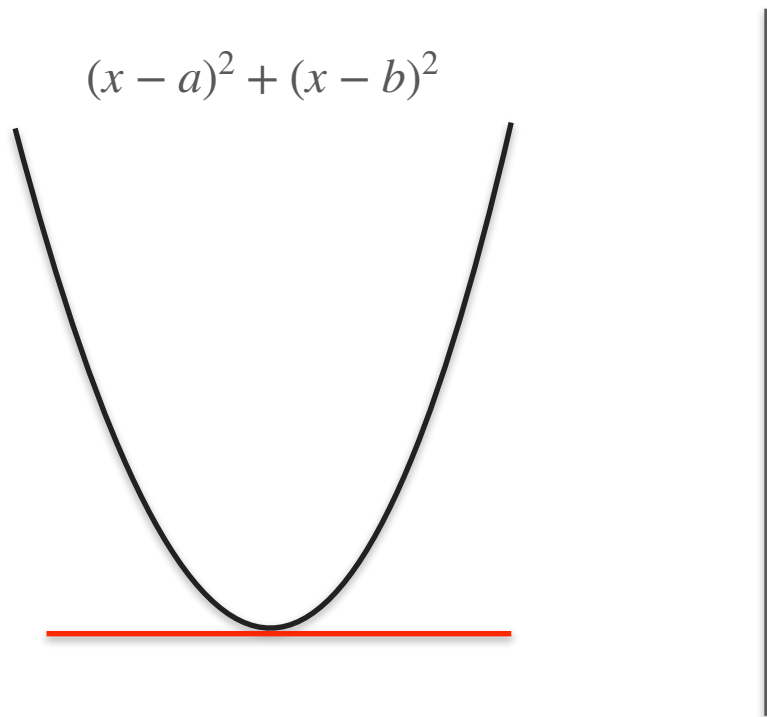
Two Power Line Problem

$$(x - a)^2 + (x - b)^2$$

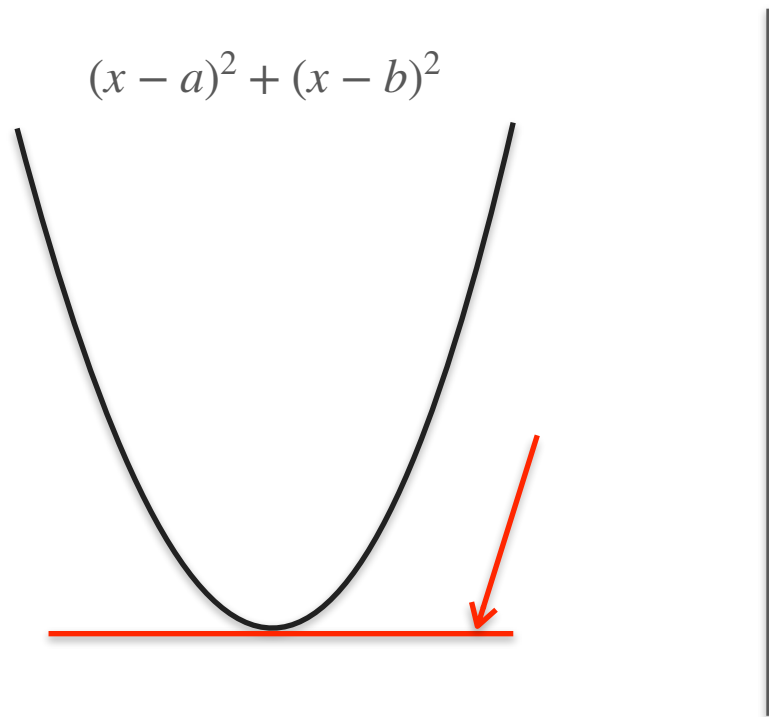
Two Power Line Problem



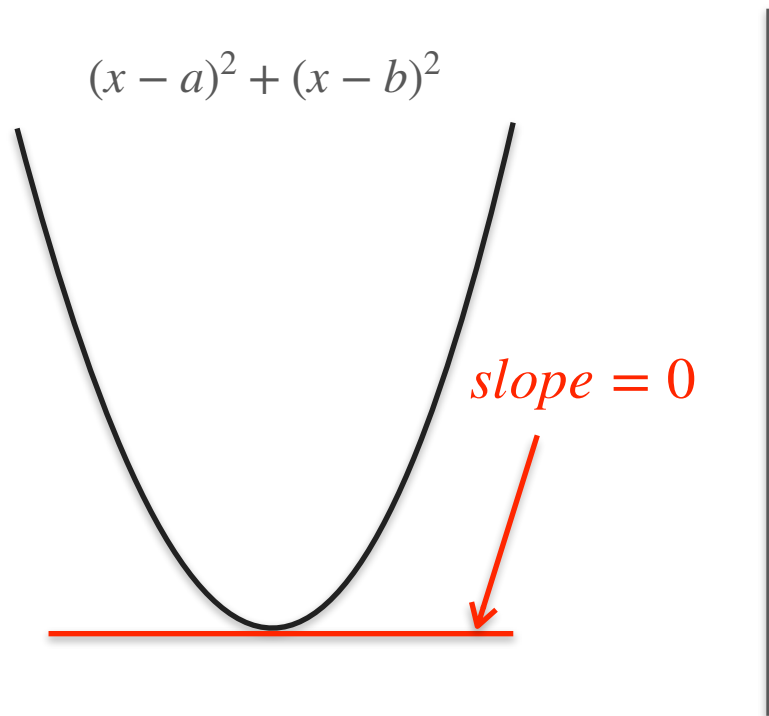
Two Power Line Problem



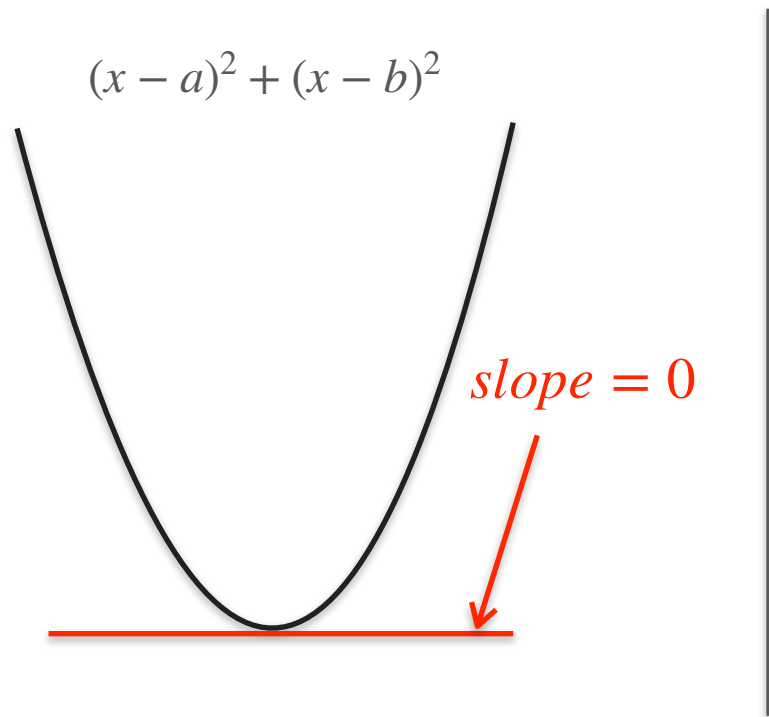
Two Power Line Problem



Two Power Line Problem

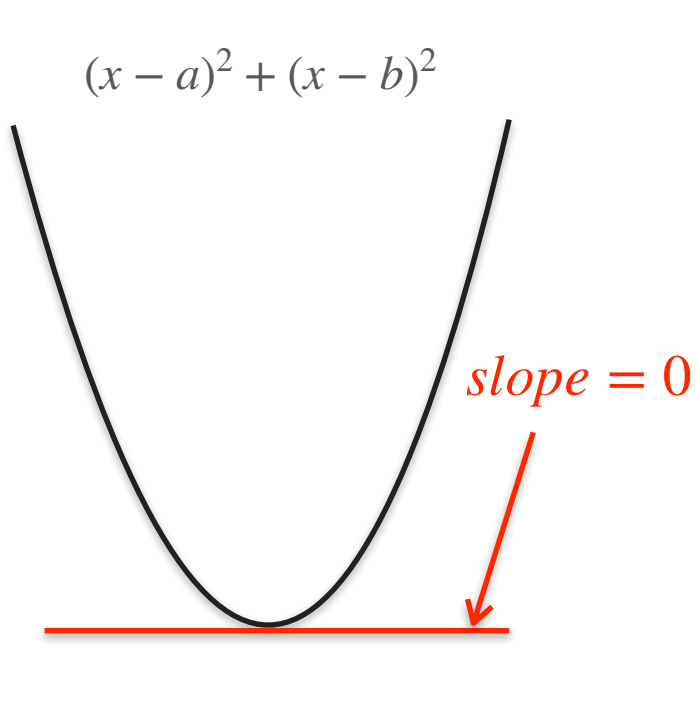


Two Power Line Problem



$$\frac{d}{dx} [(x - a)^2 + (x - b)^2] = 0$$

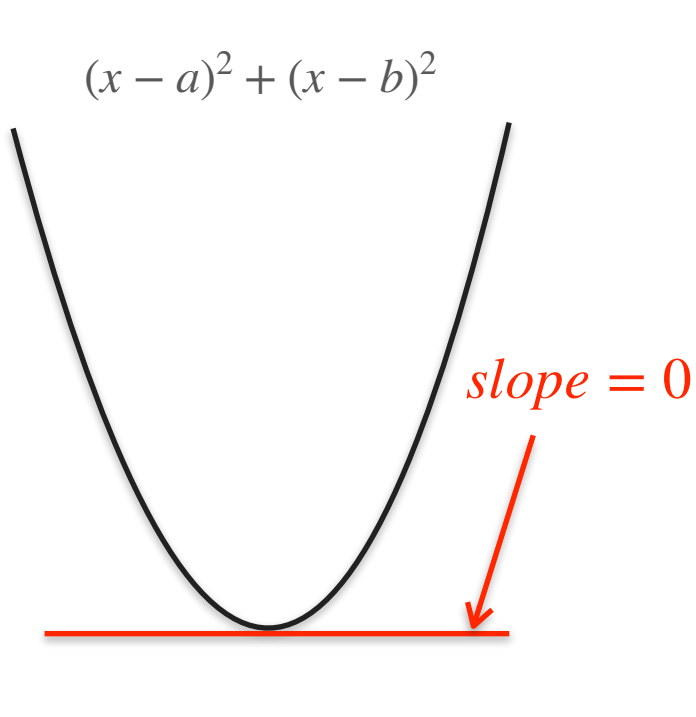
Two Power Line Problem



$$\frac{d}{dx} [(x - a)^2 + (x - b)^2] = 0$$

$$2(x - a) + 2(x - b) = 0$$

Two Power Line Problem

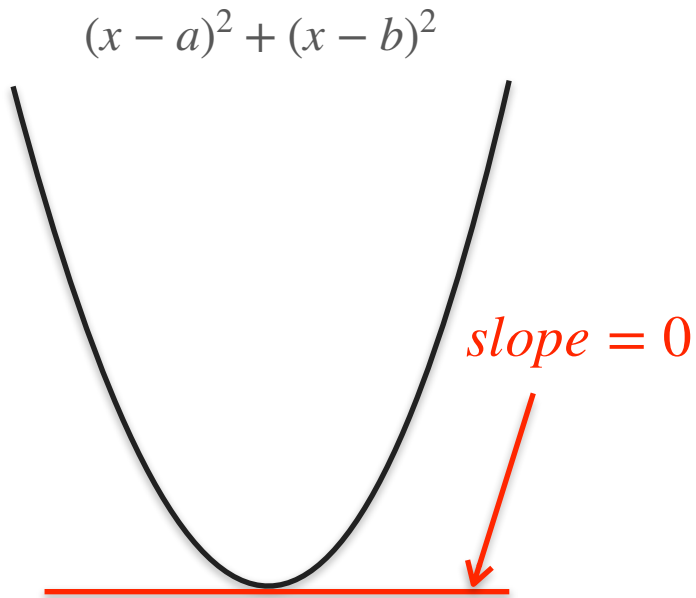


$$\frac{d}{dx} [(x - a)^2 + (x - b)^2] = 0$$

$$2(x - a) + 2(x - b) = 0$$

$$(x - a) + (x - b) = 0$$

Two Power Line Problem



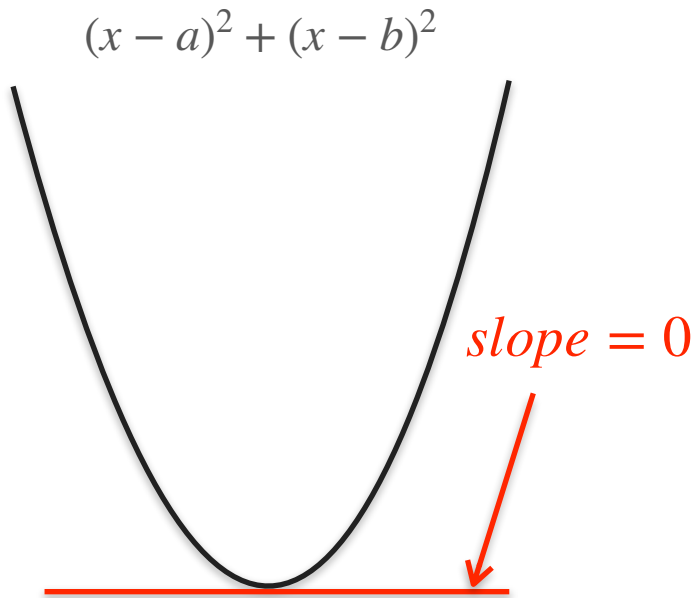
$$\frac{d}{dx} [(x - a)^2 + (x - b)^2] = 0$$

$$2(x - a) + 2(x - b) = 0$$

$$(x - a) + (x - b) = 0$$

$$2x - a - b = 0$$

Two Power Line Problem



$$\frac{d}{dx} [(x - a)^2 + (x - b)^2] = 0$$

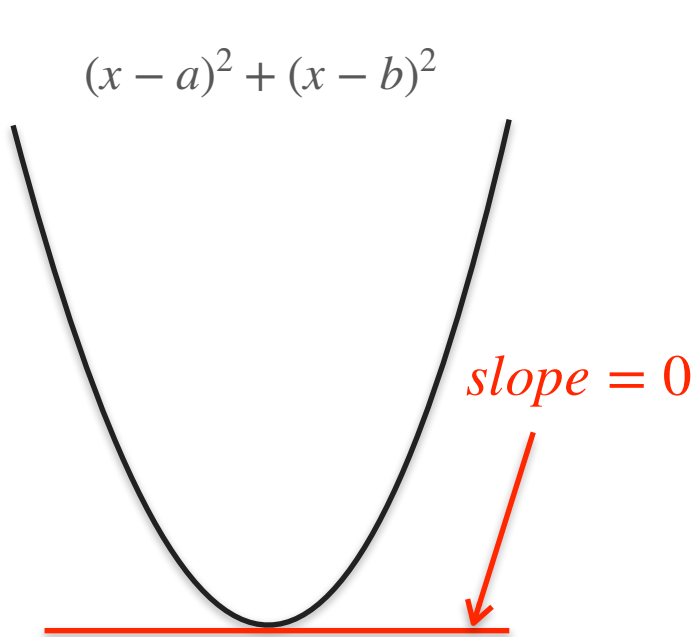
$$2(x - a) + 2(x - b) = 0$$

$$(x - a) + (x - b) = 0$$

$$2x - a - b = 0$$

$$2x = a + b$$

Two Power Line Problem



$$\frac{d}{dx} [(x - a)^2 + (x - b)^2] = 0$$

$$2(x - a) + 2(x - b) = 0$$

$$(x - a) + (x - b) = 0$$

$$2x - a - b = 0$$

$$2x = a + b$$

$$x = \frac{a + b}{2}$$

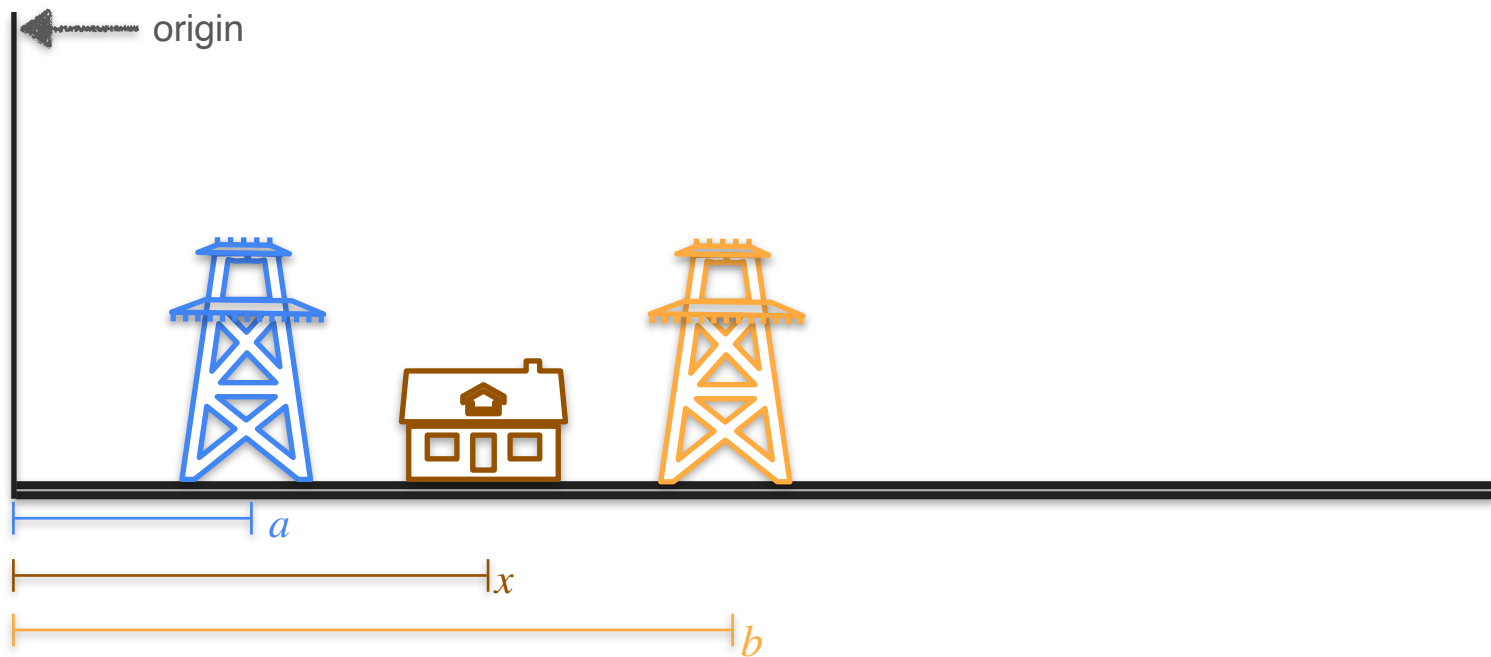


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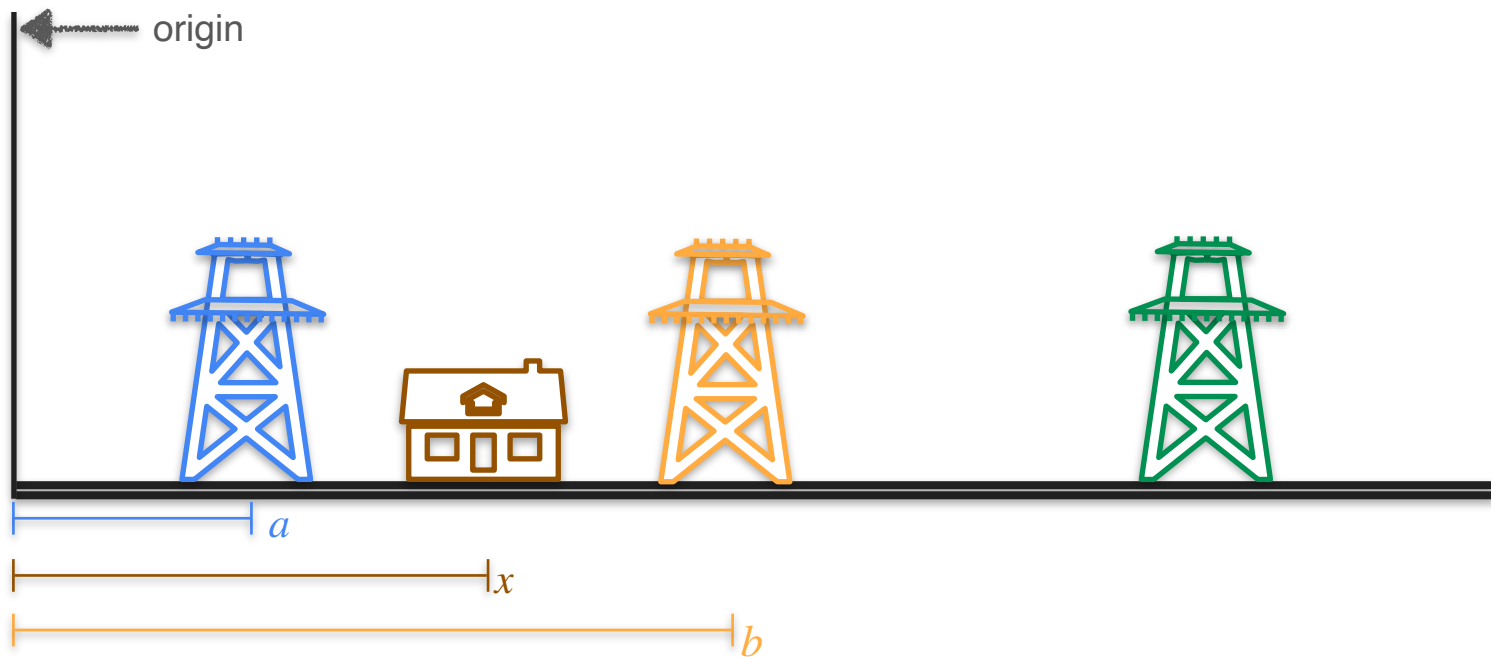
Derivatives and Optimization

**Optimization of squared loss:
The three powerline problem**

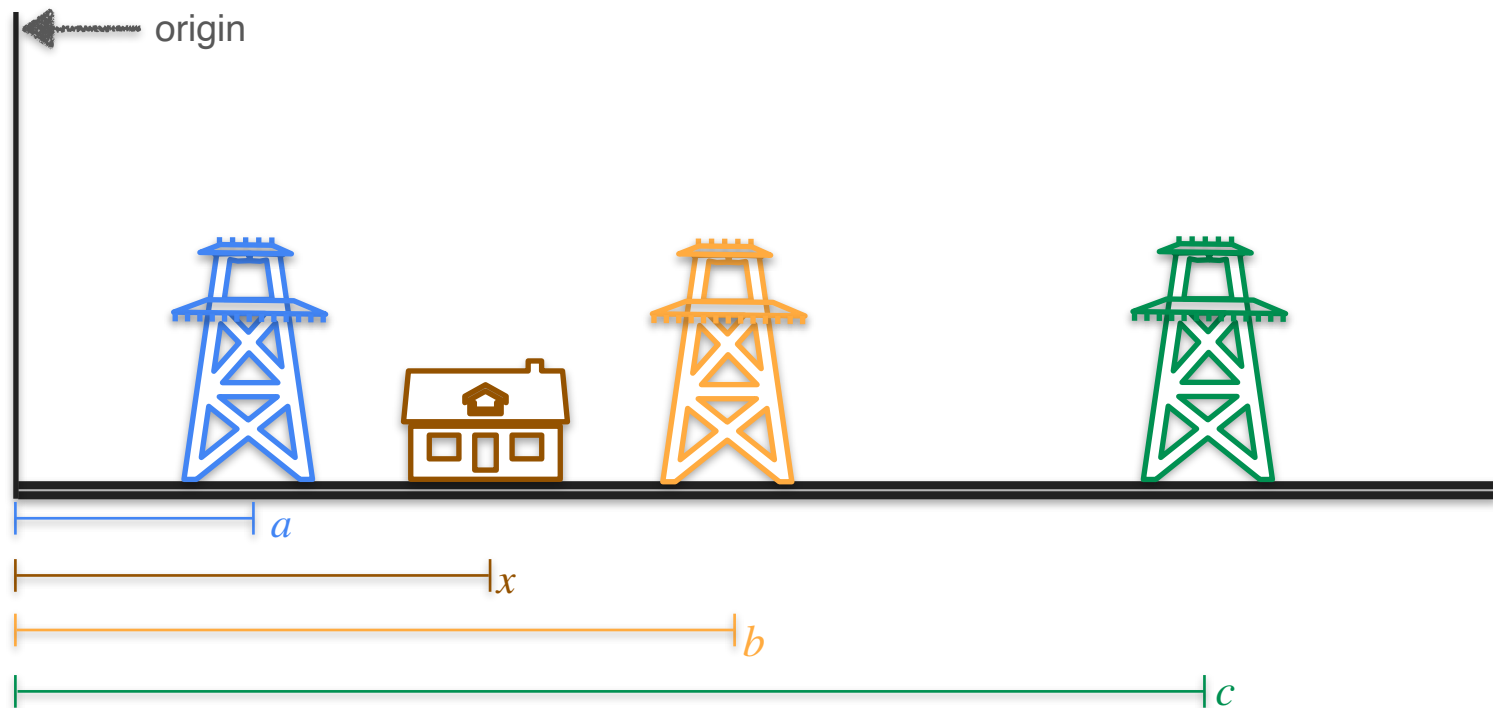
Three Power Line Problem



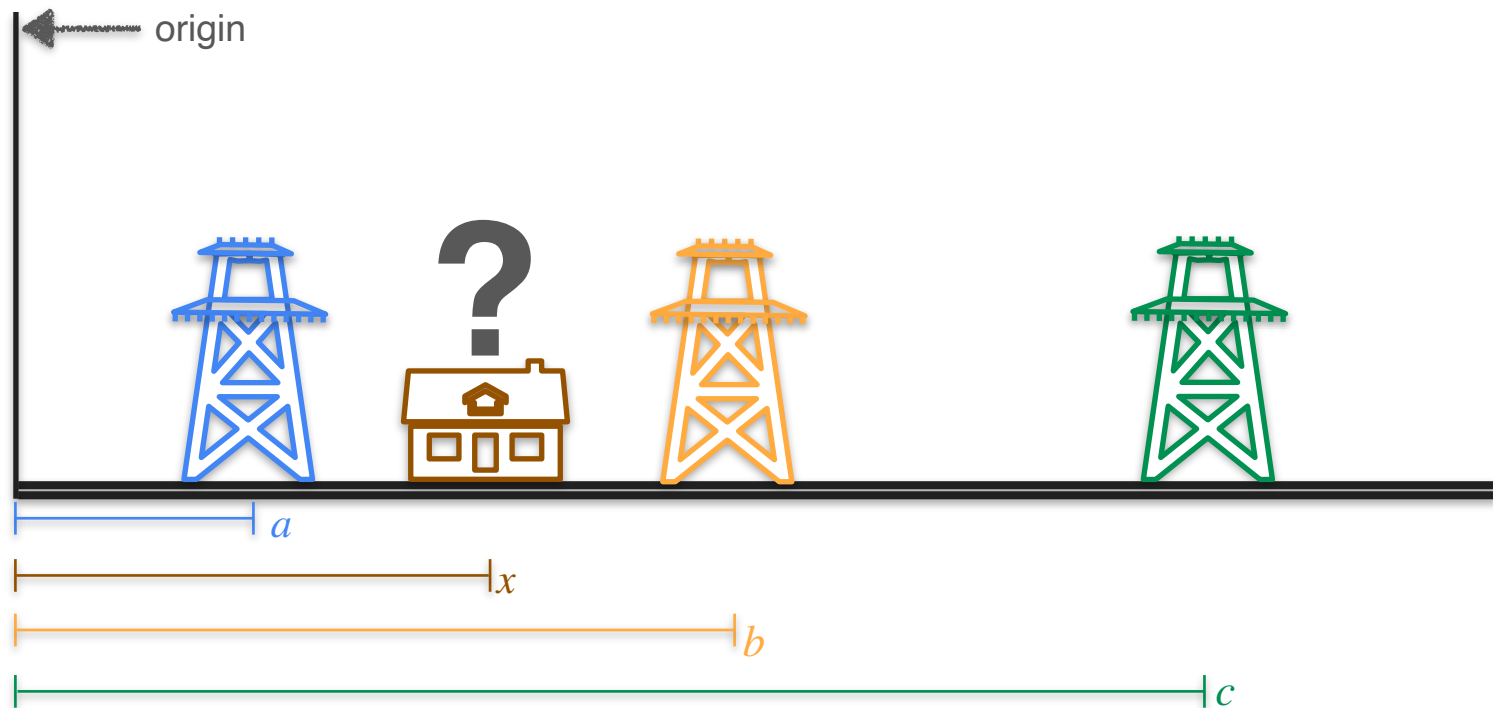
Three Power Line Problem



Three Power Line Problem



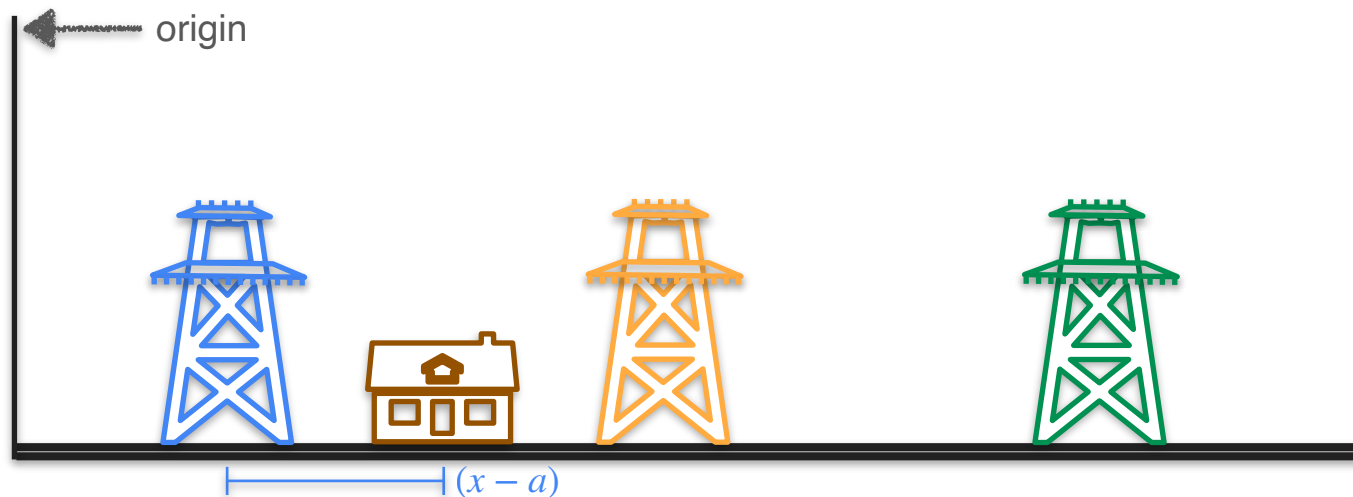
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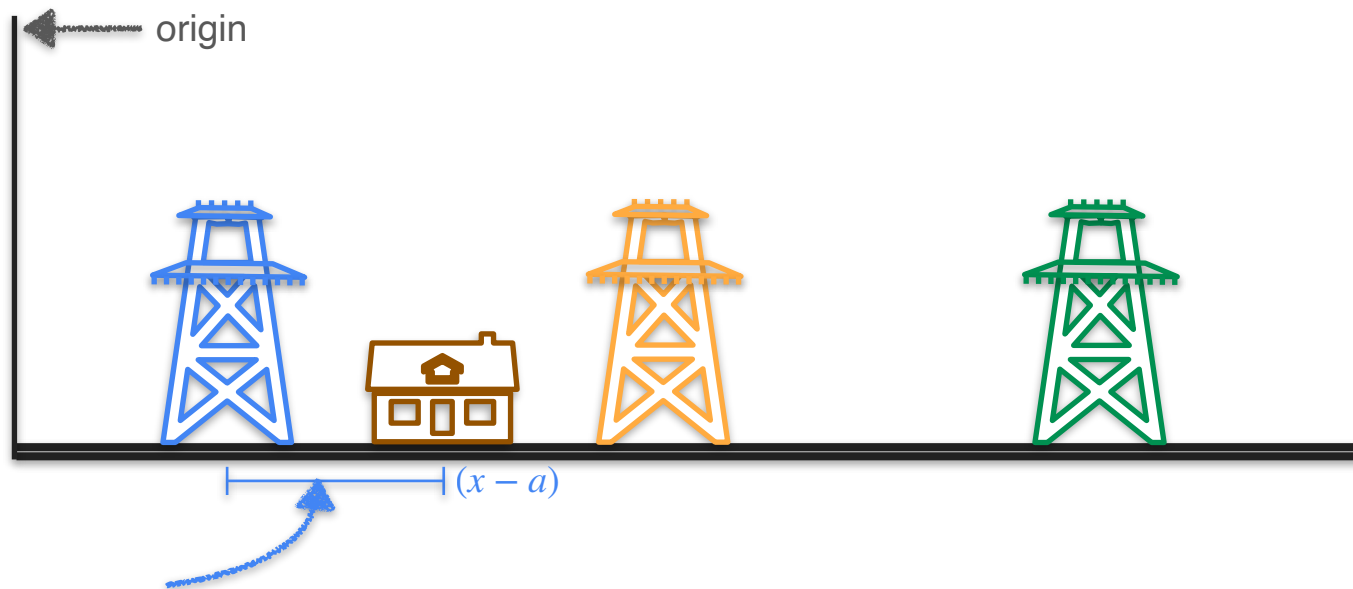
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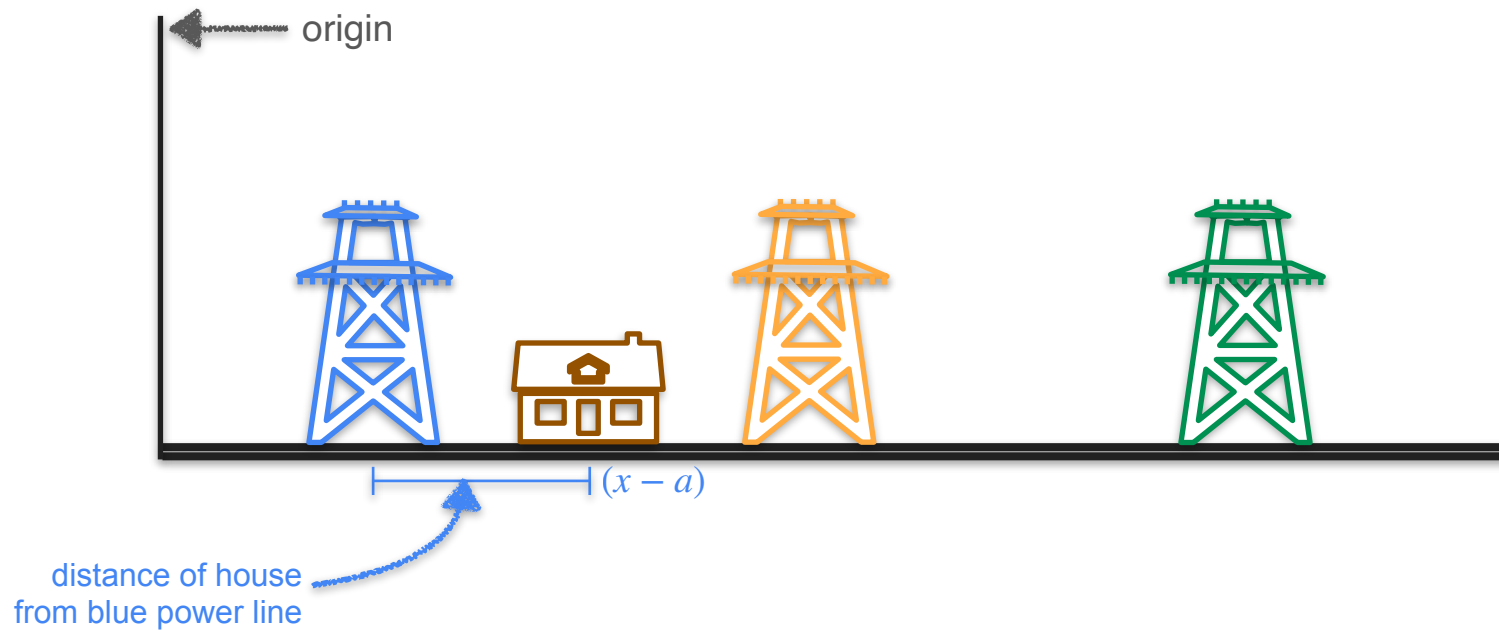
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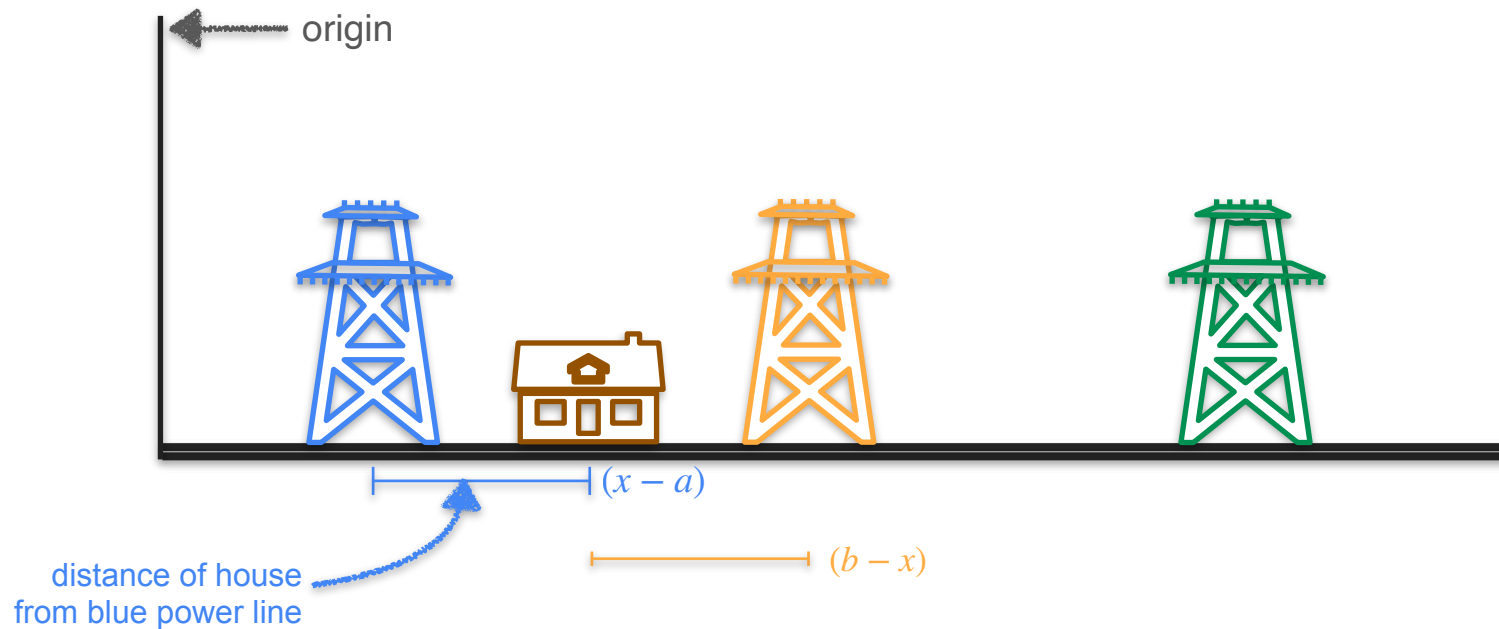
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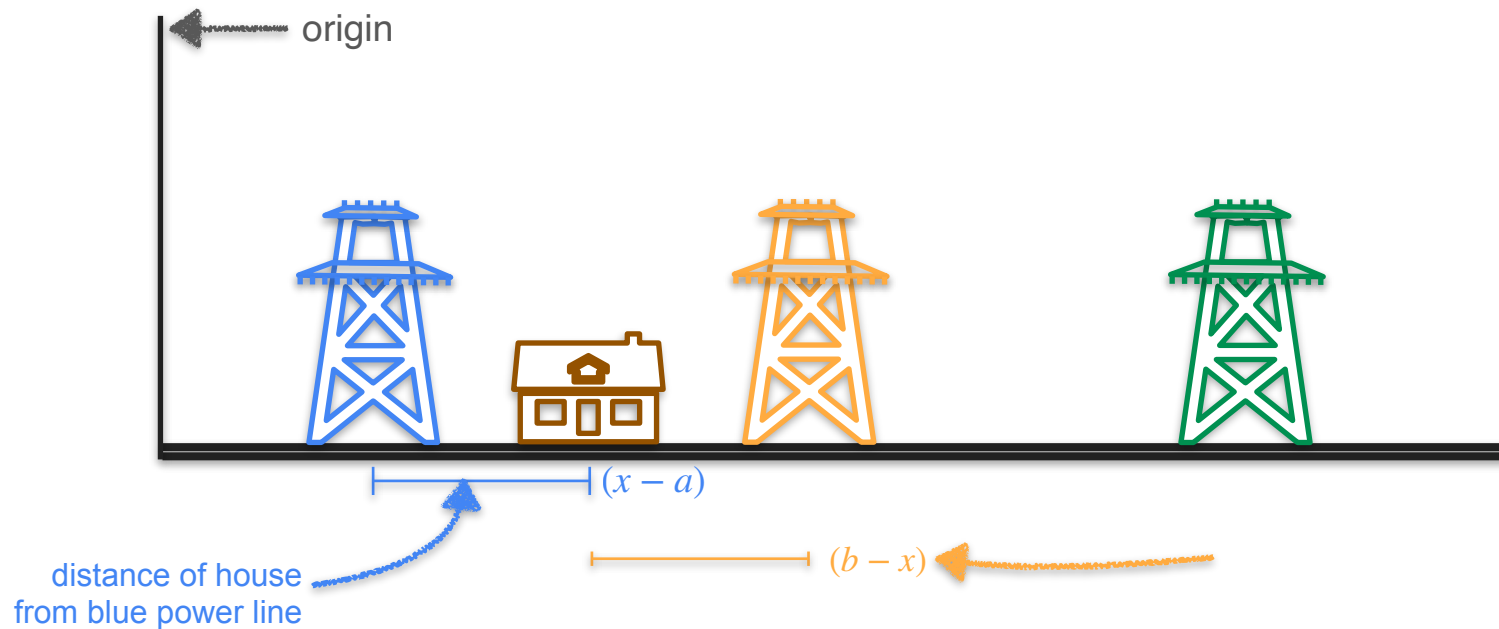
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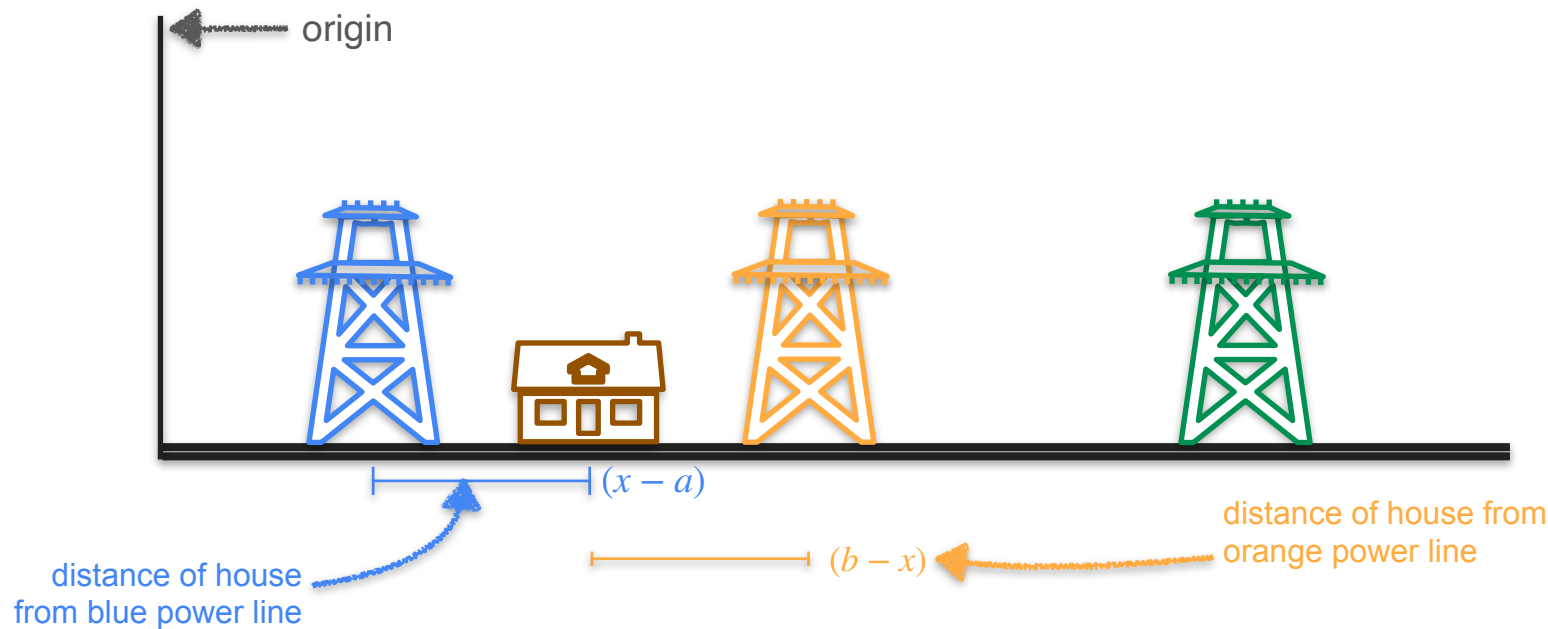
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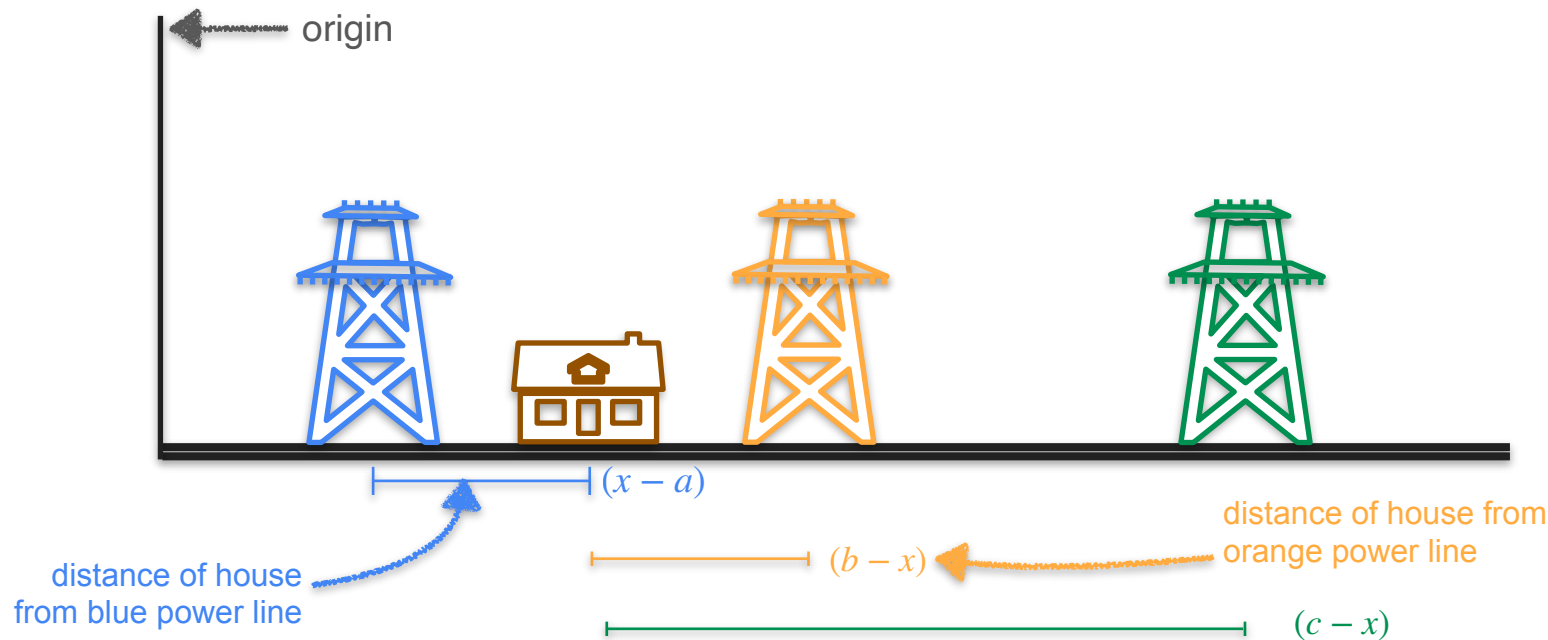
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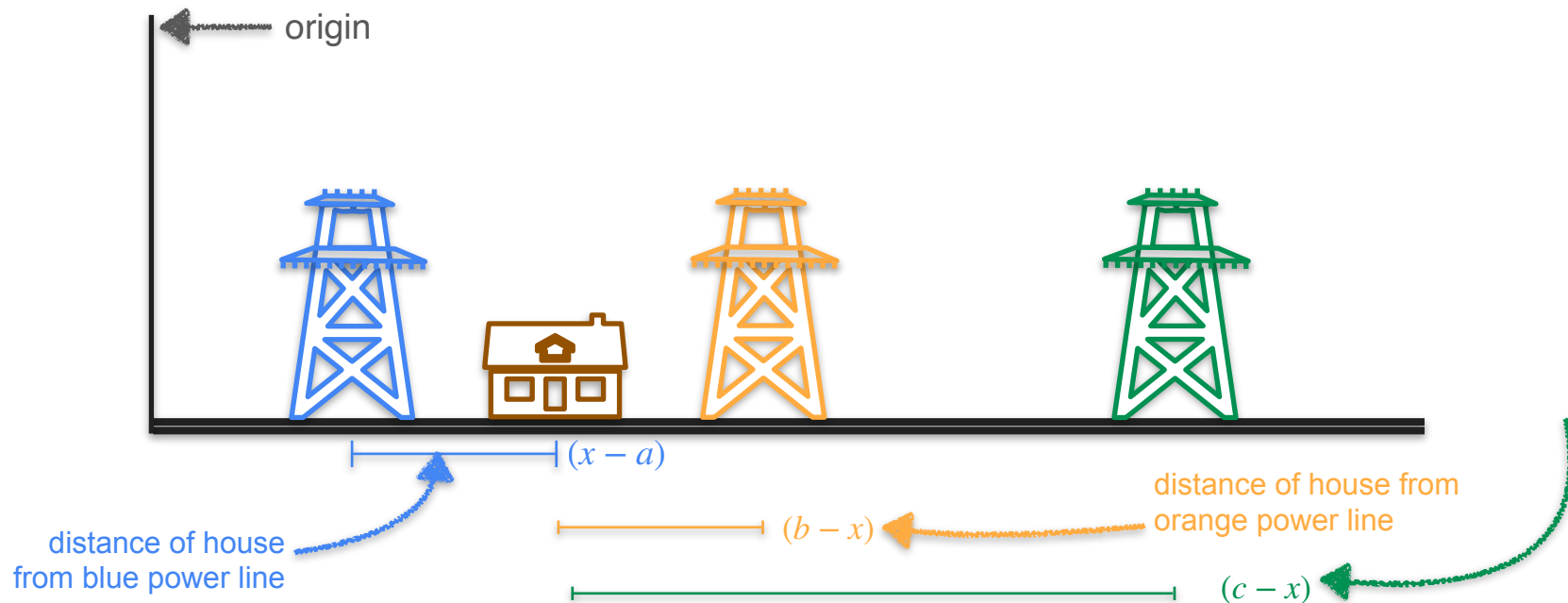
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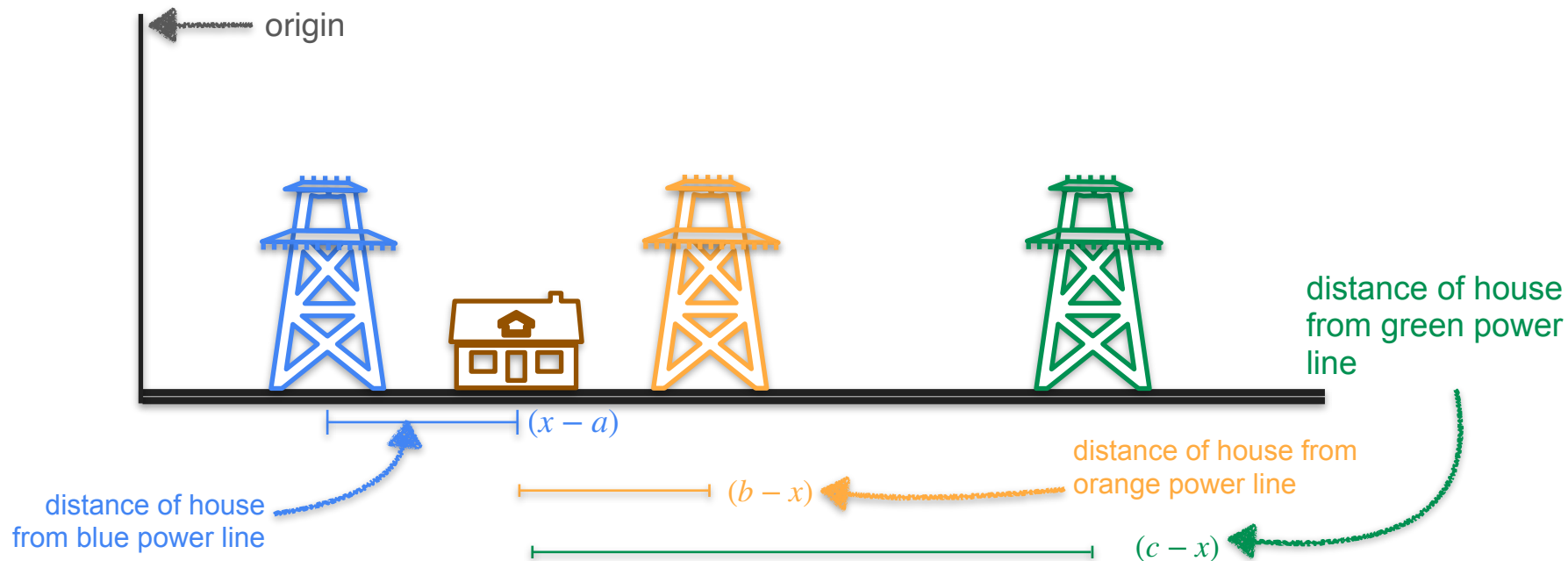
Three Power Line Problem



Three Power Line Problem



Three Power Line Problem



Three Power Line Problem



Three Power Line Problem



Three Power Line Problem



Three Power Line Problem

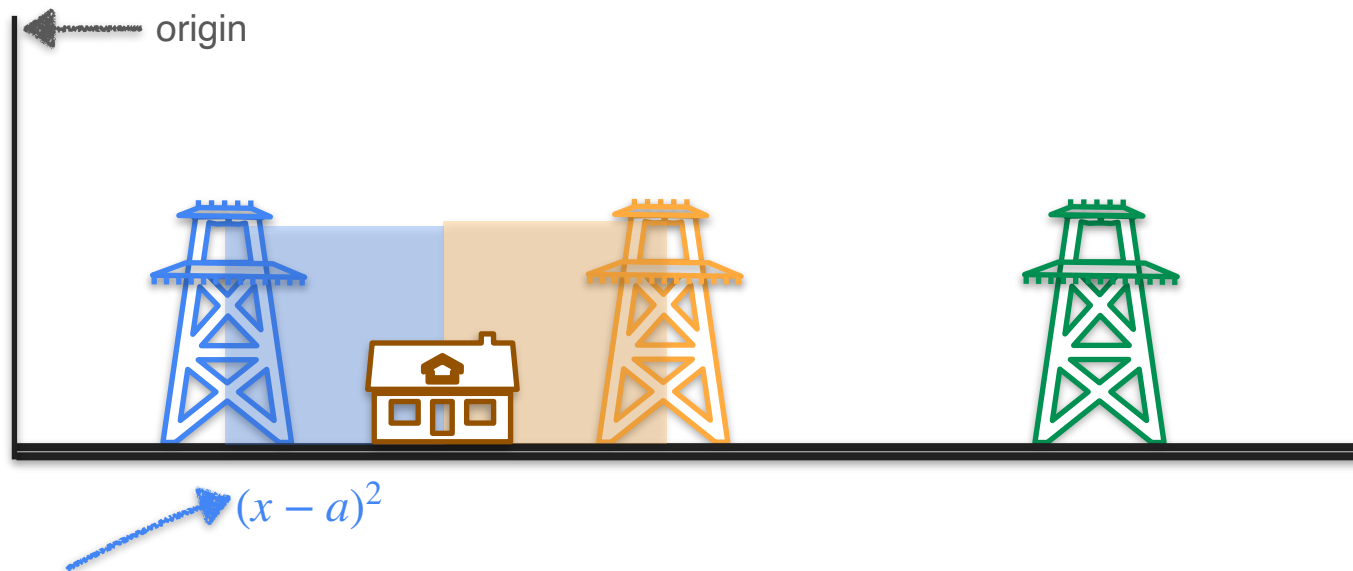


Three Power Line Problem



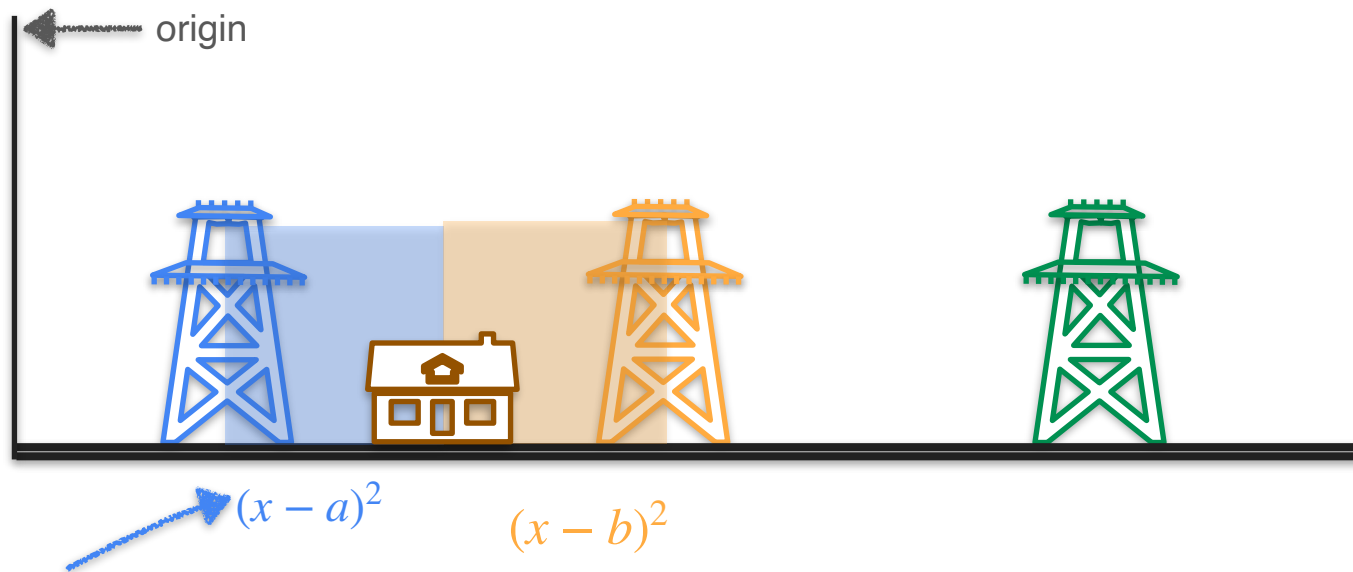
cost of connecting to
the blue power line

Three Power Line Problem



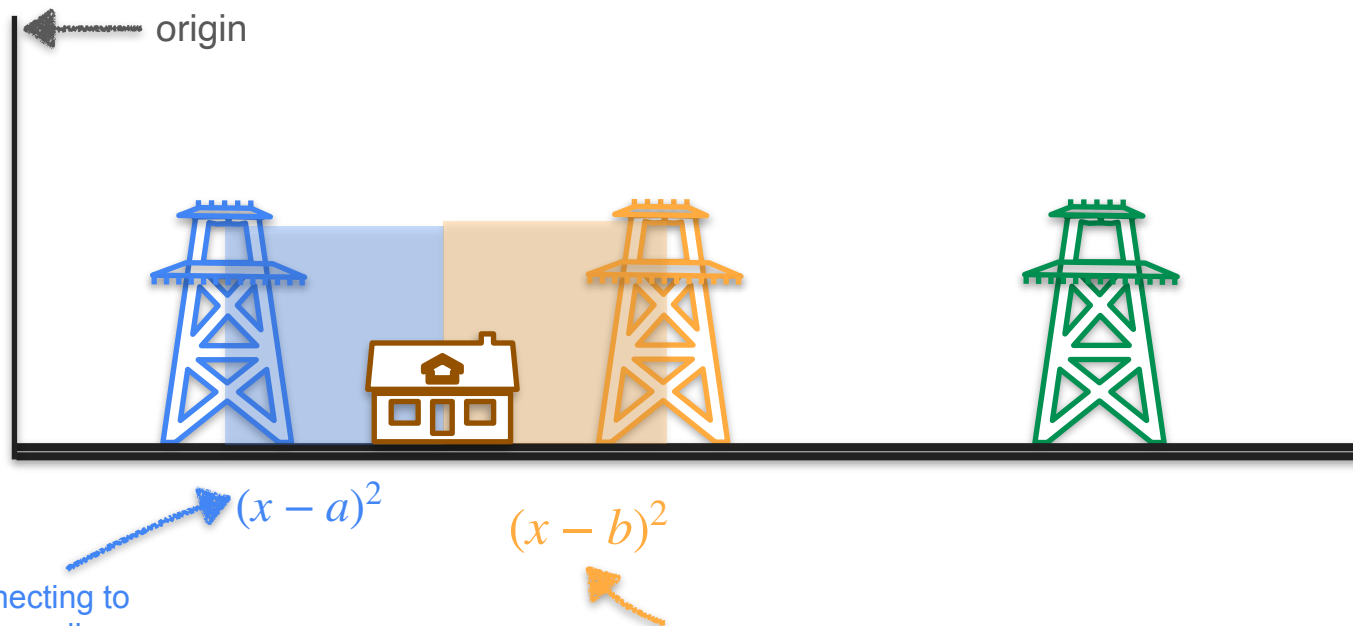
cost of connecting to
the blue power line

Three Power Line Problem



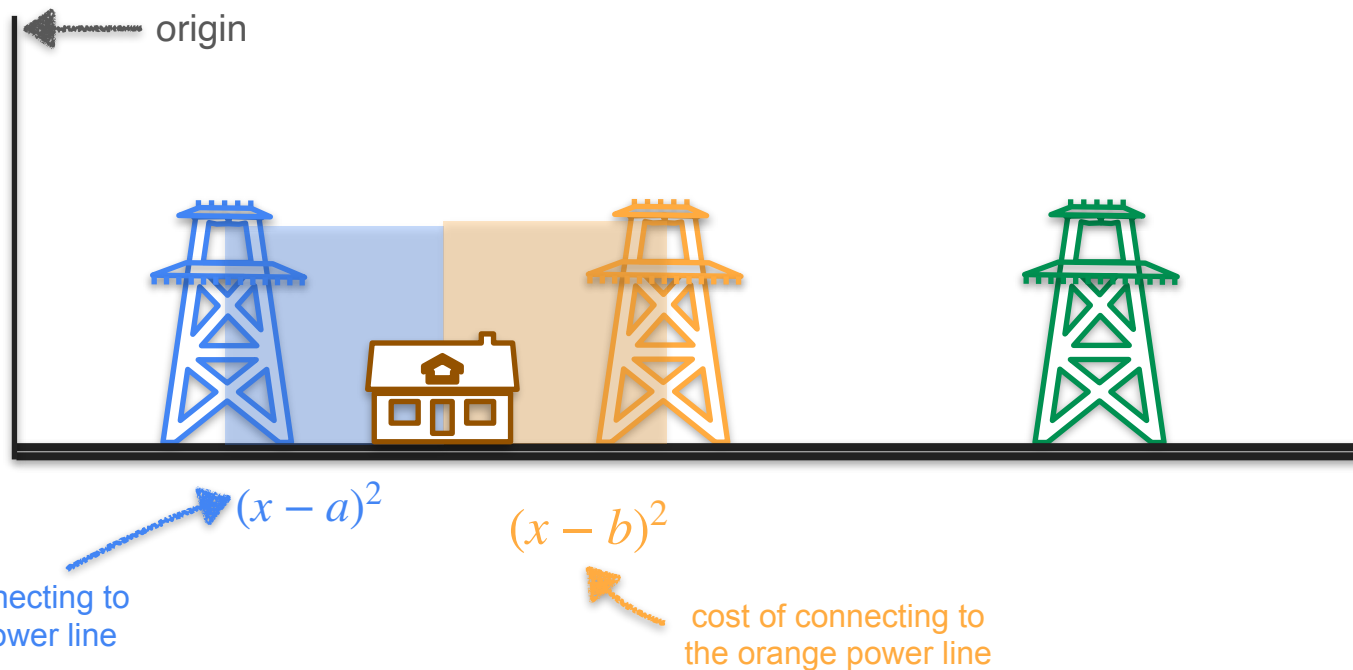
cost of connecting to
the blue power line

Three Power Line Problem

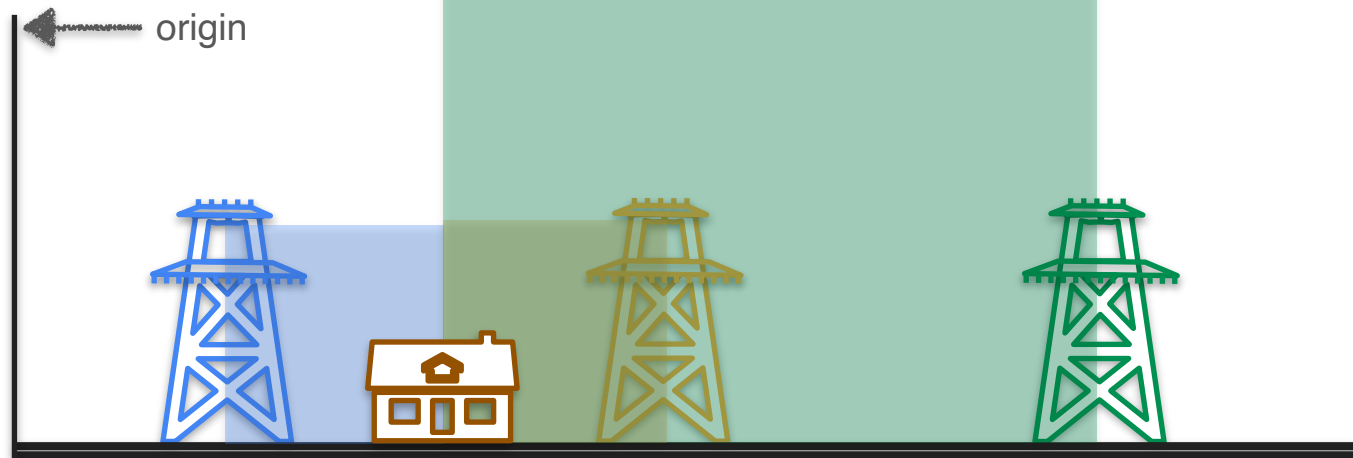


cost of connecting to
the blue power line

Three Power Line Problem



Three Power Line Problem



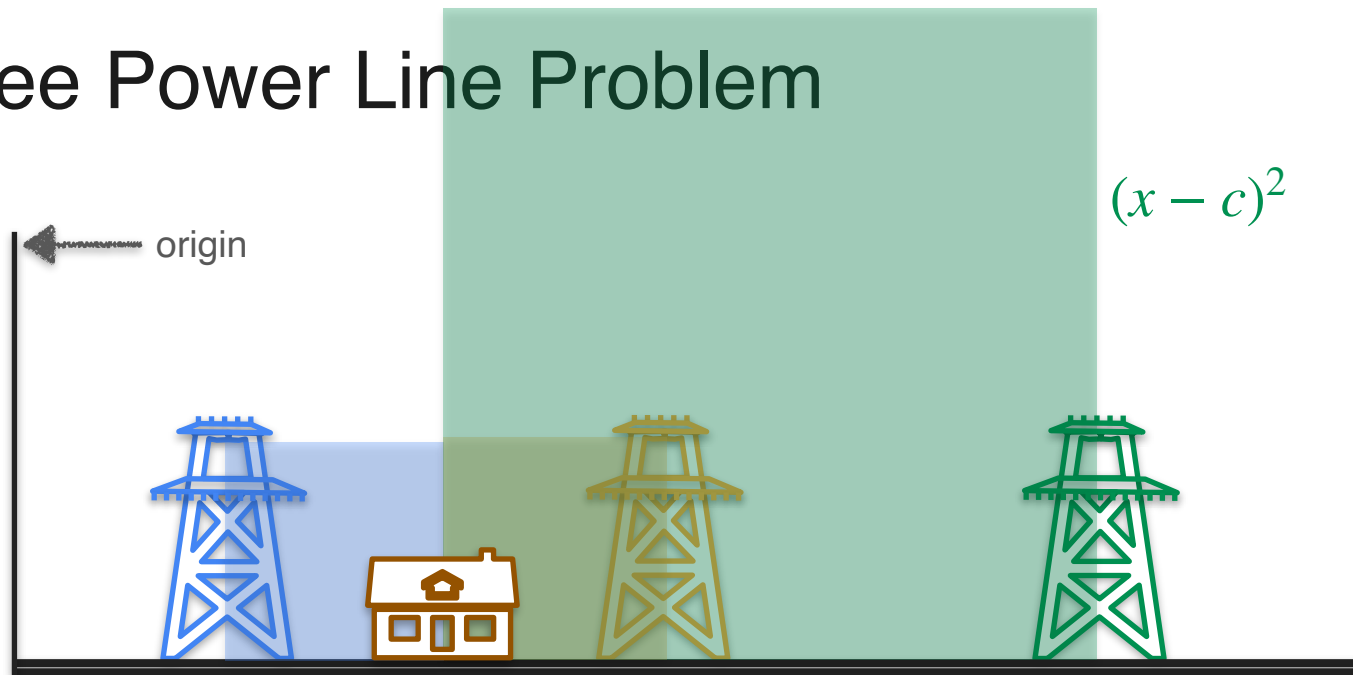
$$(x - a)^2$$

cost of connecting to
the blue power line

$$(x - b)^2$$

cost of connecting to
the orange power line

Three Power Line Problem

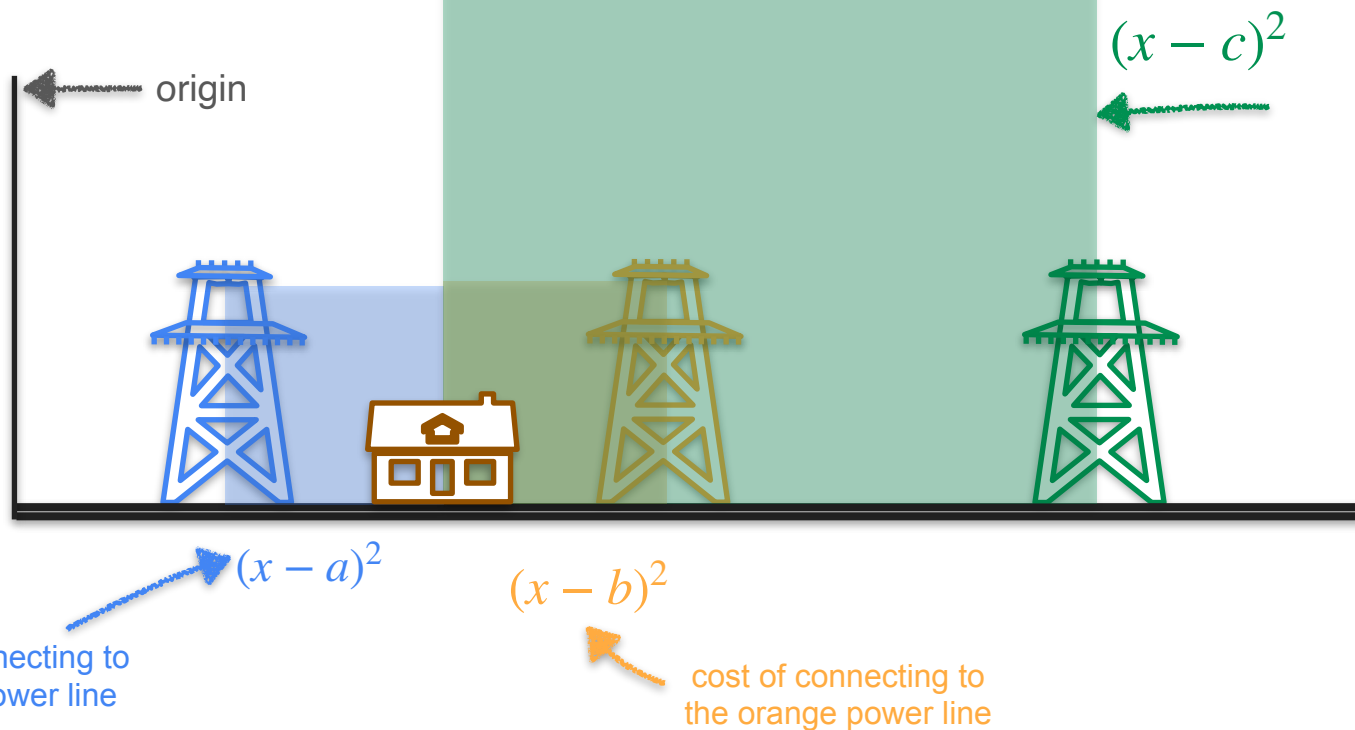


cost of connecting to
the blue power line

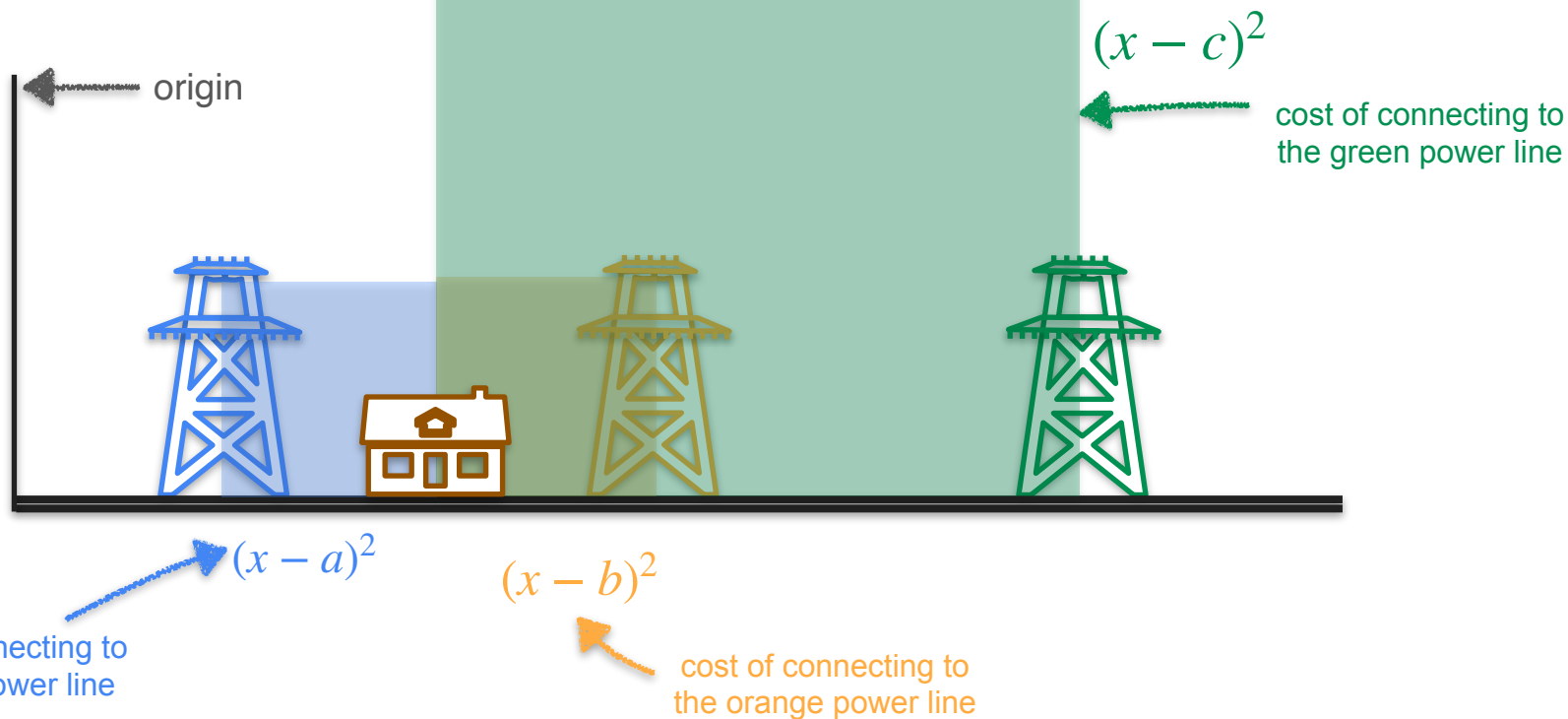
$$(x - b)^2$$

cost of connecting to
the orange power line

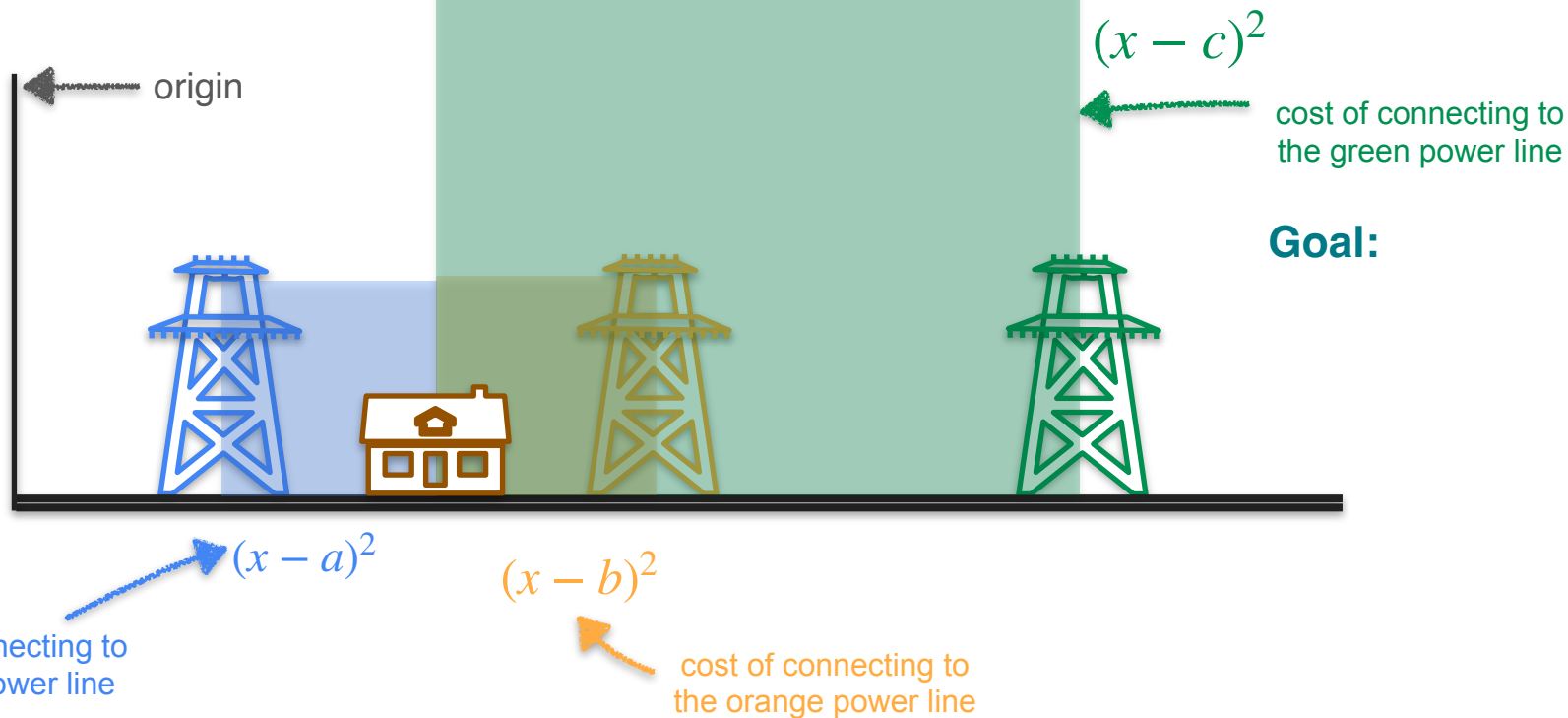
Three Power Line Problem



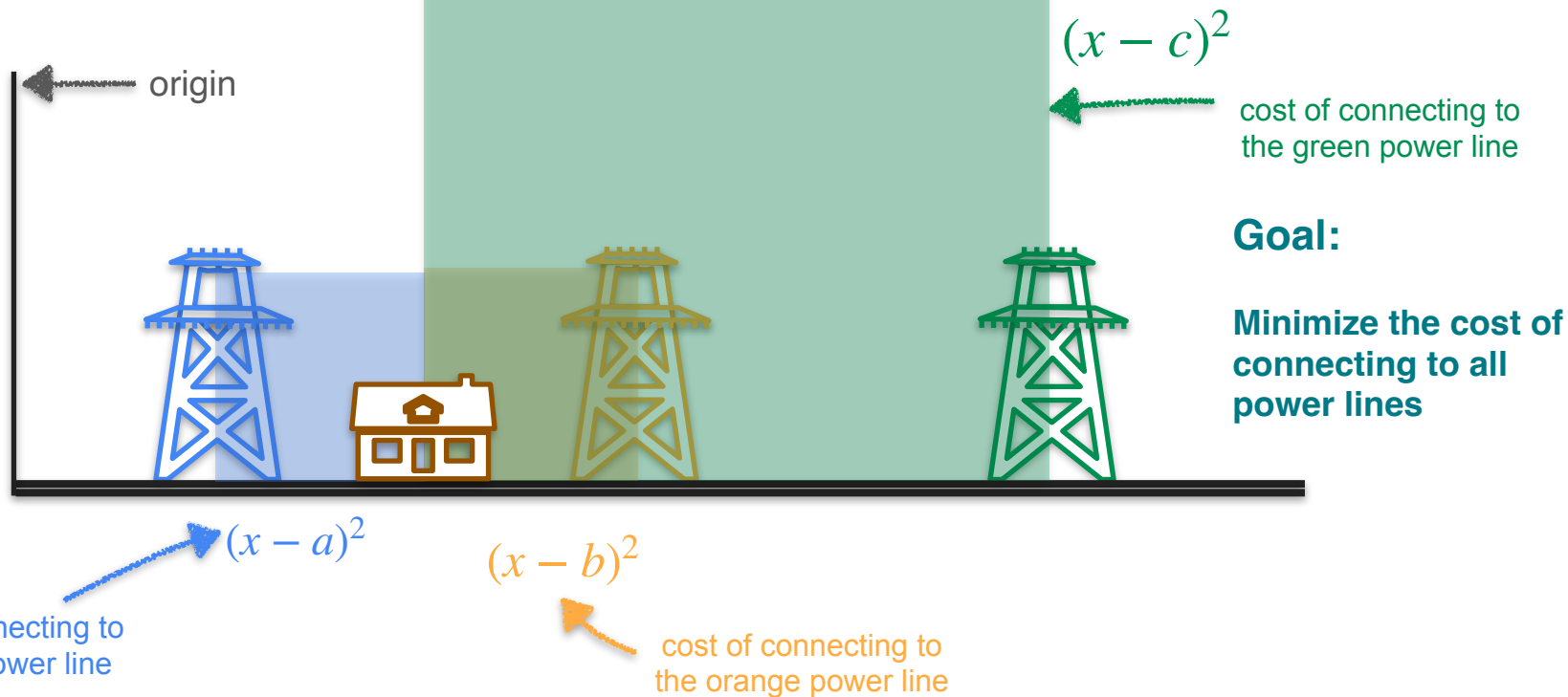
Three Power Line Problem



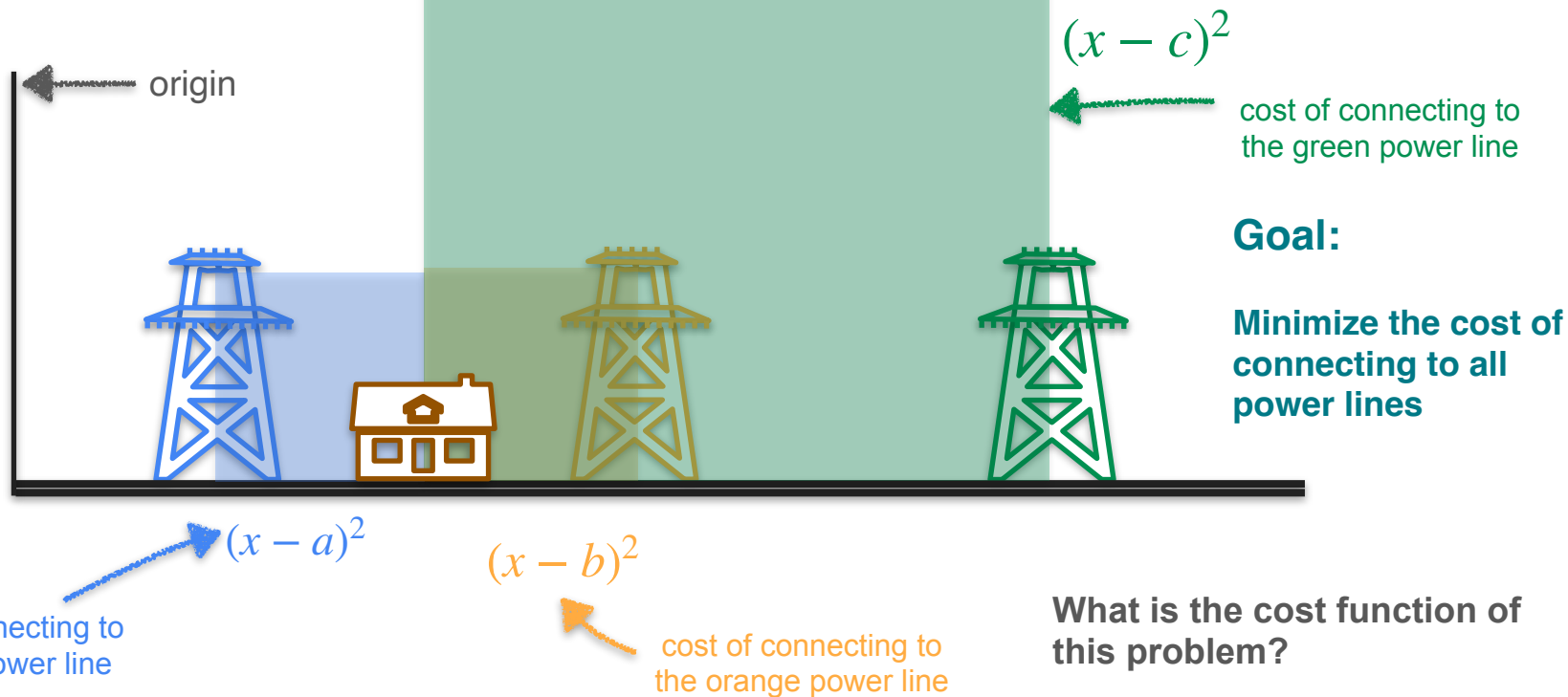
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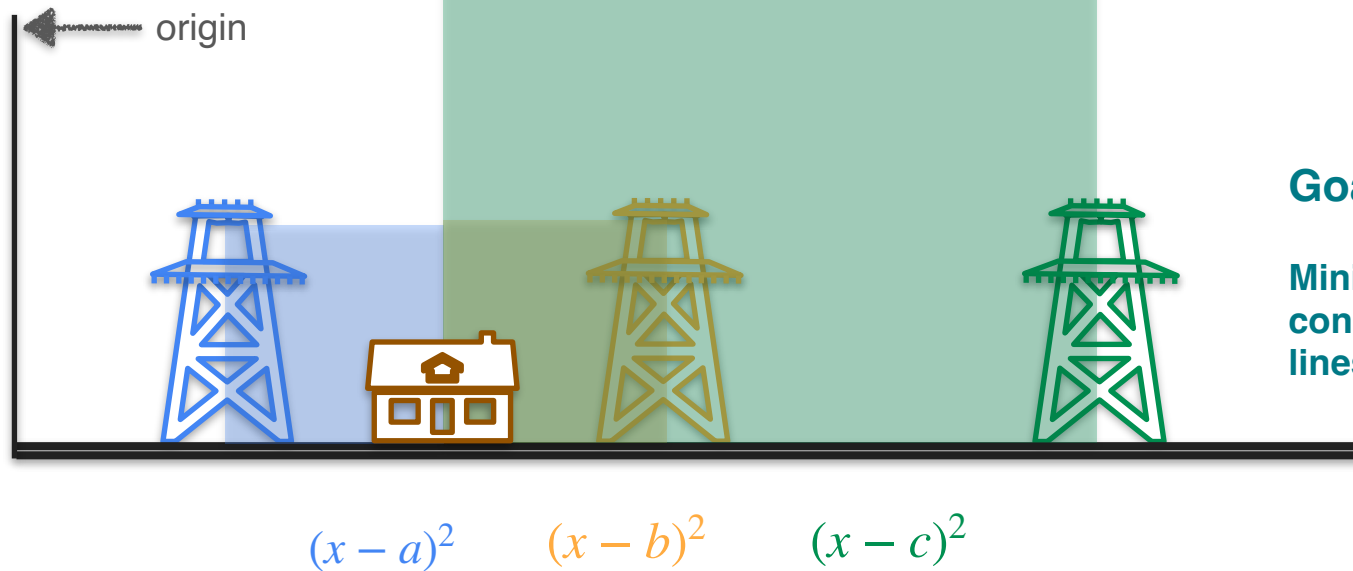
Three Power Line Problem



Three Power Line Problem



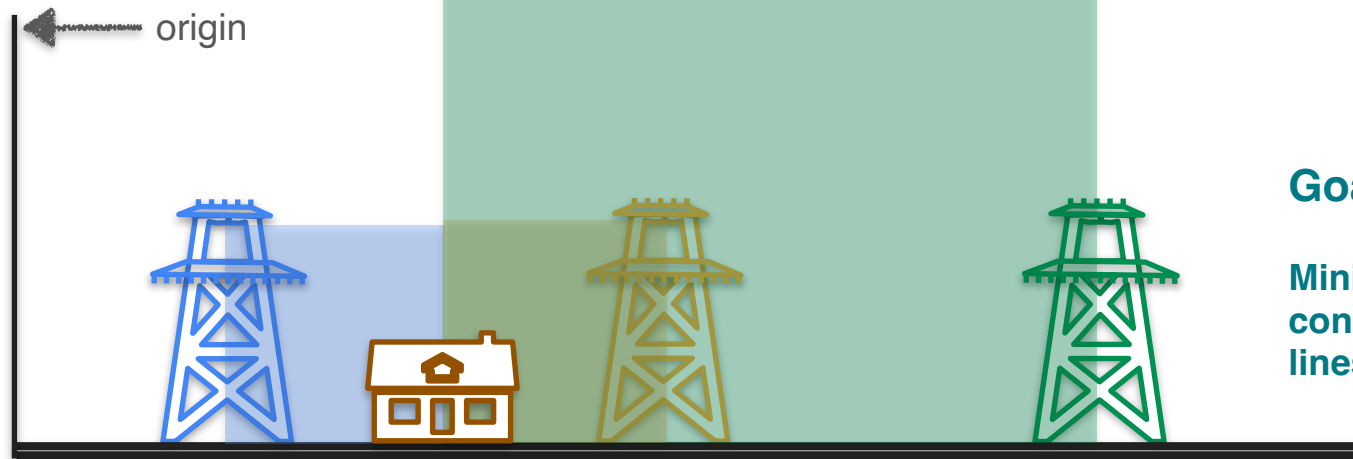
Three Power Line Problem



Goal:

Minimize the cost of
connecting all power
lines

Three Power Line Problem

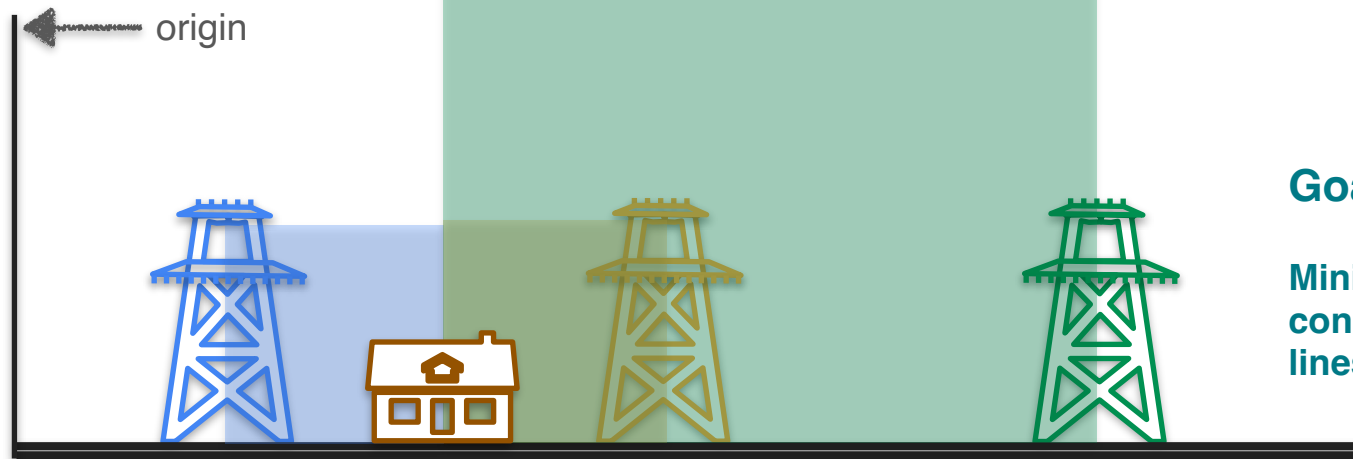


Goal:

**Minimize the cost of
connecting all power
lines**

$$(x - a)^2 + (x - b)^2 + (x - c)^2$$

Three Power Line Problem

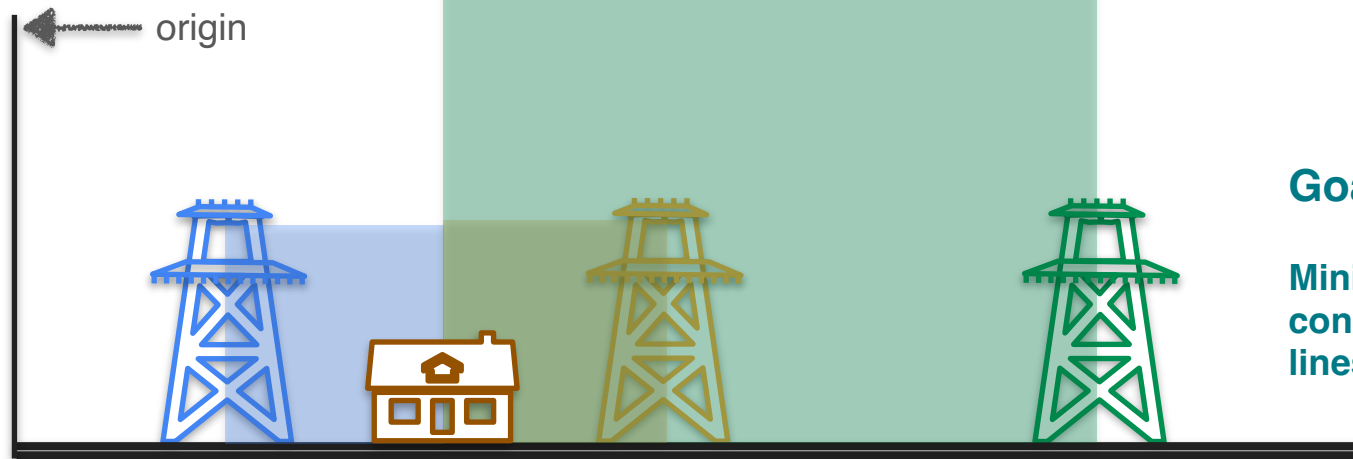


Goal:

Minimize the cost of connecting all power lines

$$\text{Cost function} = (x - a)^2 + (x - b)^2 + (x - c)^2$$

Three Power Line Problem



Goal:

Minimize the cost of connecting all power lines

$$\text{Cost function} = (x - a)^2 + (x - b)^2 + (x - c)^2$$

Total area of squares

Three Power Line Problem



Goal:

Minimize the cost of
connecting all power
lines

$$\text{Cost function} = (x - a)^2 + (x - b)^2 + (x - c)^2$$

Total area of squares

Three Power Line Problem



Goal:

Minimize the cost of connecting all power lines

Cost function = $(x - a)^2 + (x - b)^2 + (x - c)^2$

Total area of squares

Where should you put the house to minimize the cost?

Three Power Line Problem



Goal:

Minimize the cost of connecting all power lines

Cost function = $(x - a)^2 + (x - b)^2 + (x - c)^2$ ← Minimize this cost function

Solution:

Three Power Line Problem



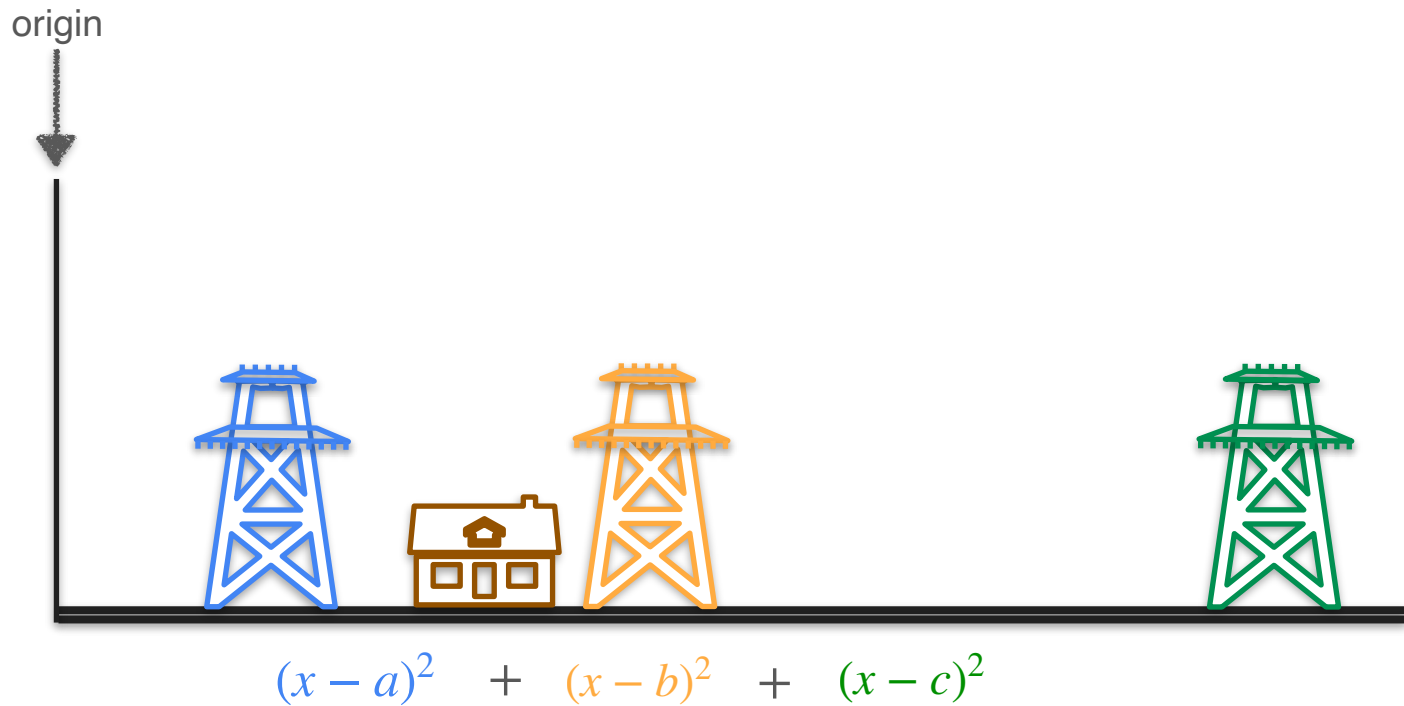
Goal:

Minimize the cost of
connecting all power
lines

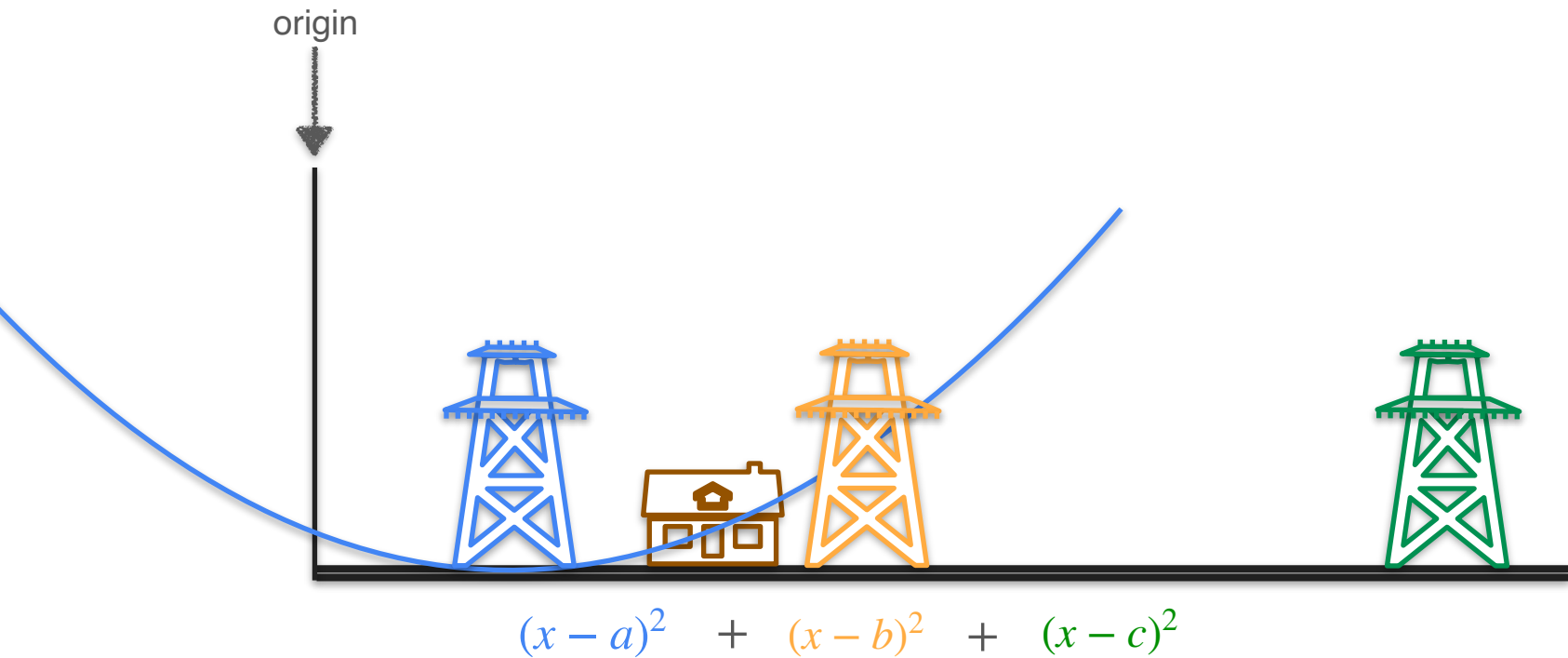
Cost function = $(x - a)^2 + (x - b)^2 + (x - c)^2$ ← Minimize this cost function

Solution: $x = \frac{a + b + c}{3}$

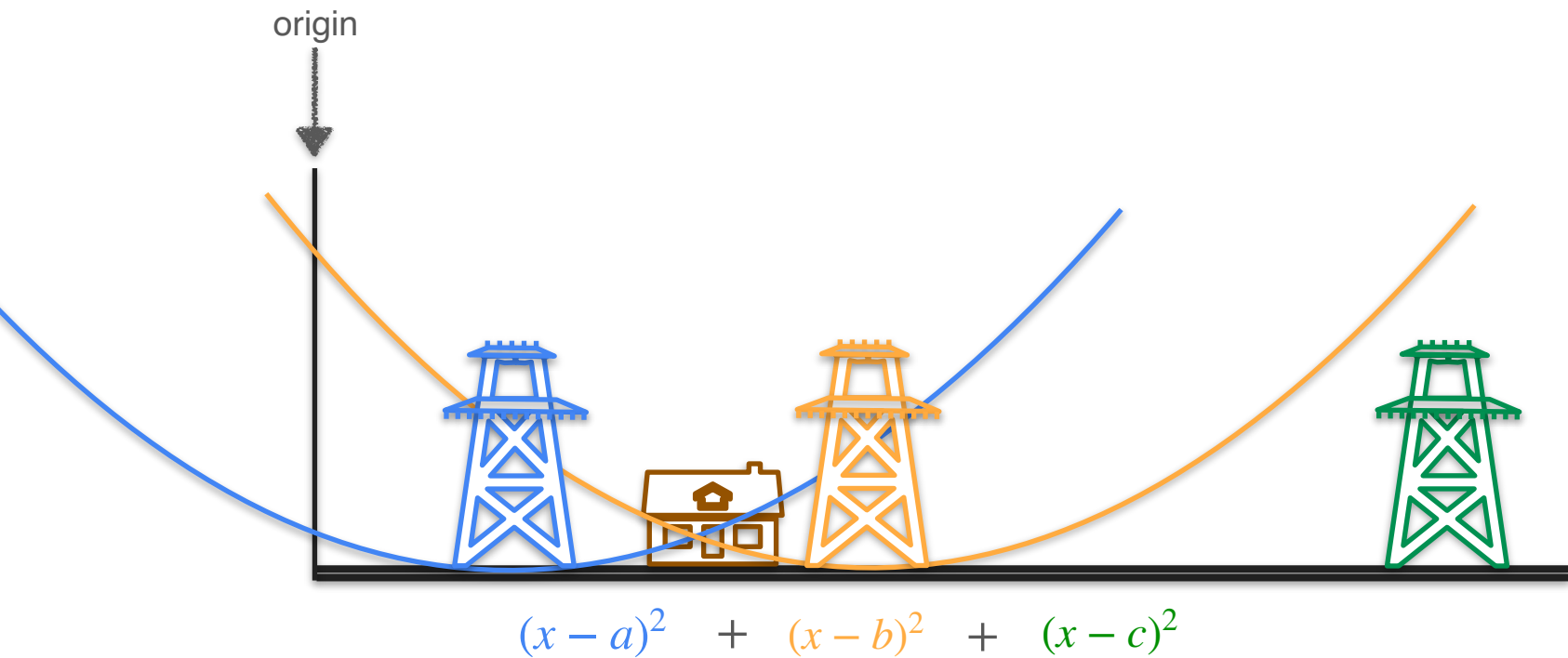
Three Power Line Problem



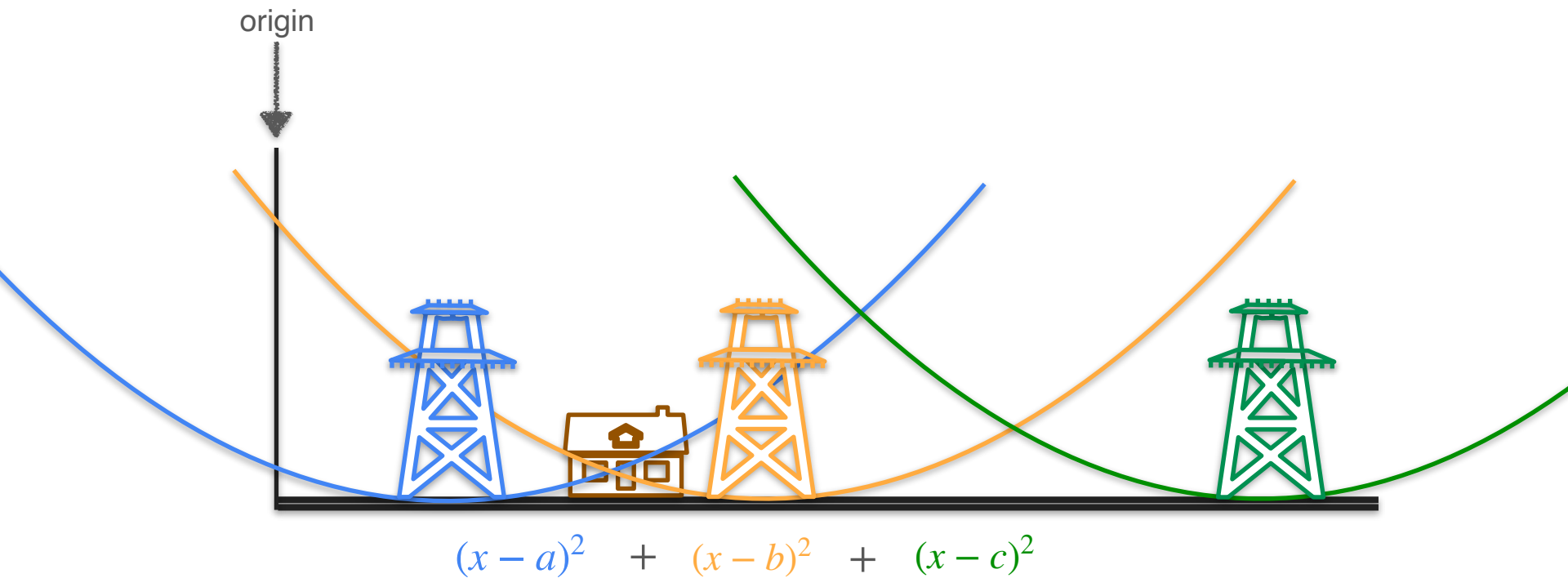
Three Power Line Problem



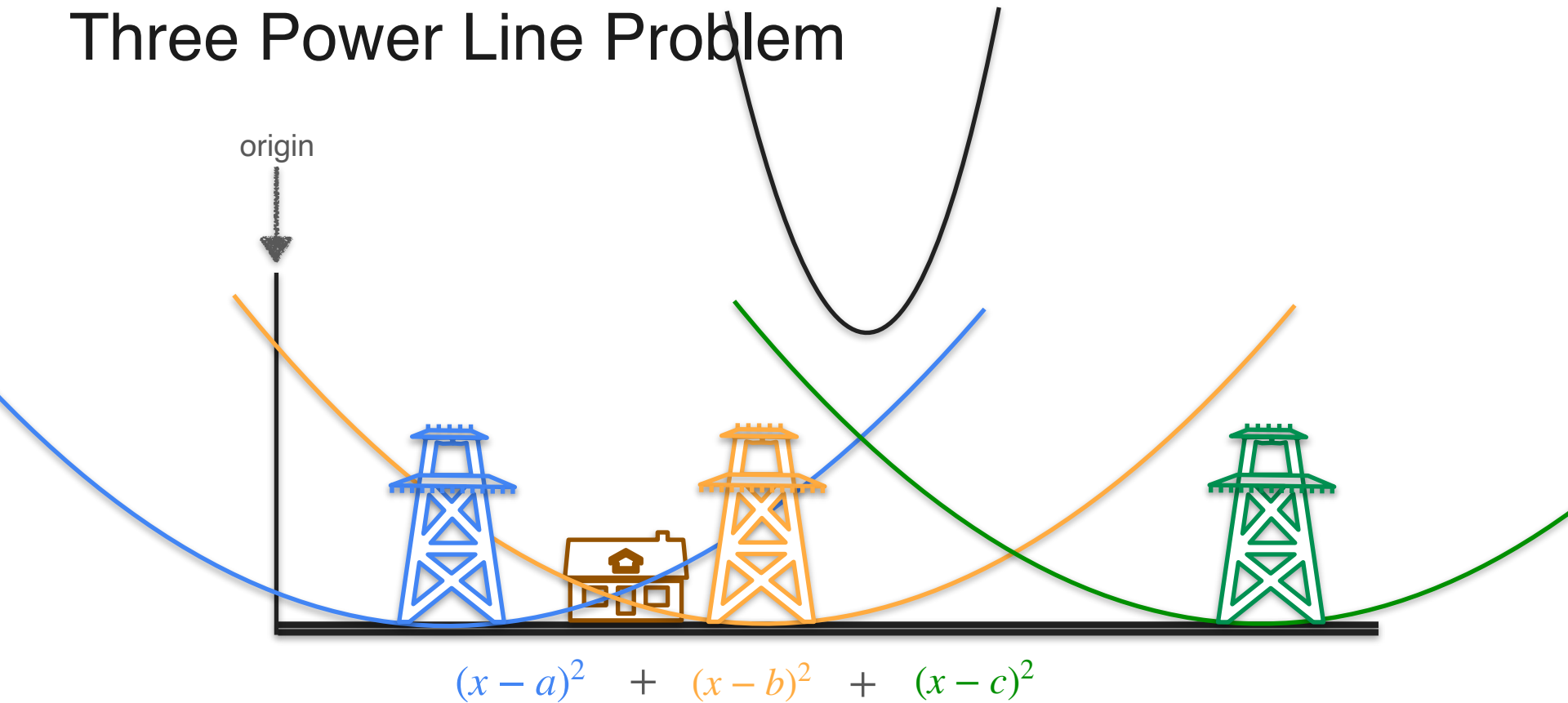
Three Power Line Problem



Three Power Line Problem



Three Power Line Problem



Three Power Line Problem

$$(x - a)^2 + (x - b)^2 + (x - c)^2$$



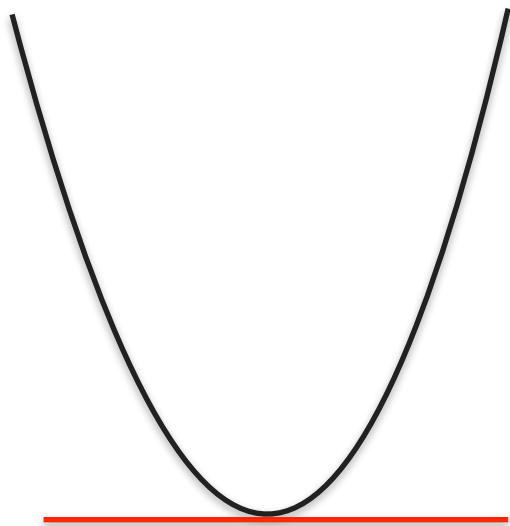
Three Power Line Problem

$$(x - a)^2 + (x - b)^2 + (x - c)^2$$



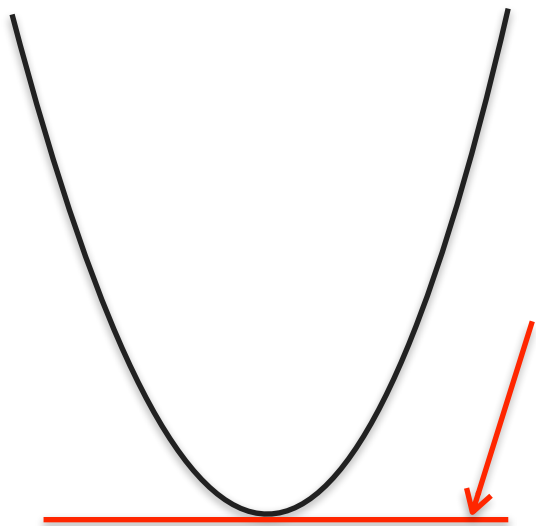
Three Power Line Problem

$$(x - a)^2 + (x - b)^2 + (x - c)^2$$

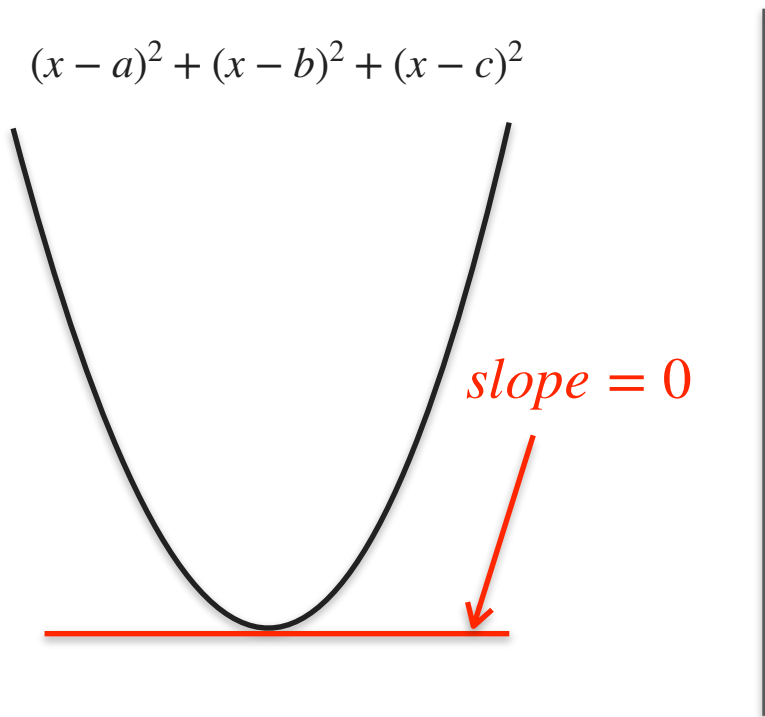


Three Power Line Problem

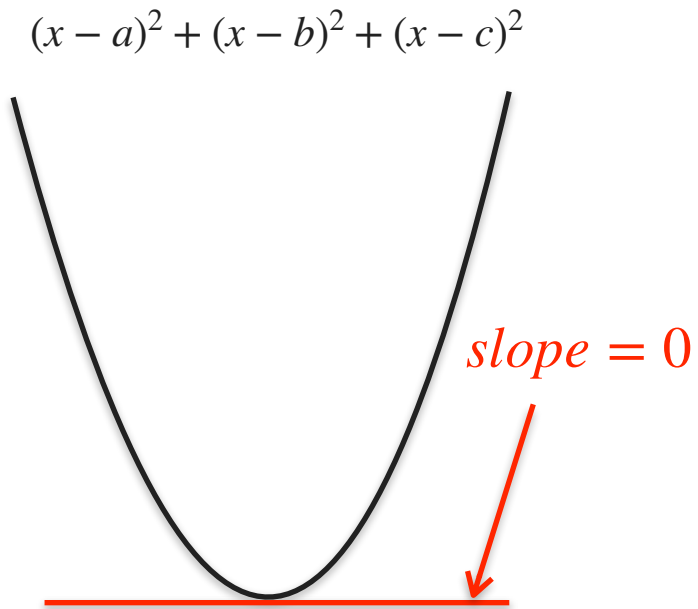
$$(x - a)^2 + (x - b)^2 + (x - c)^2$$



Three Power Line Problem

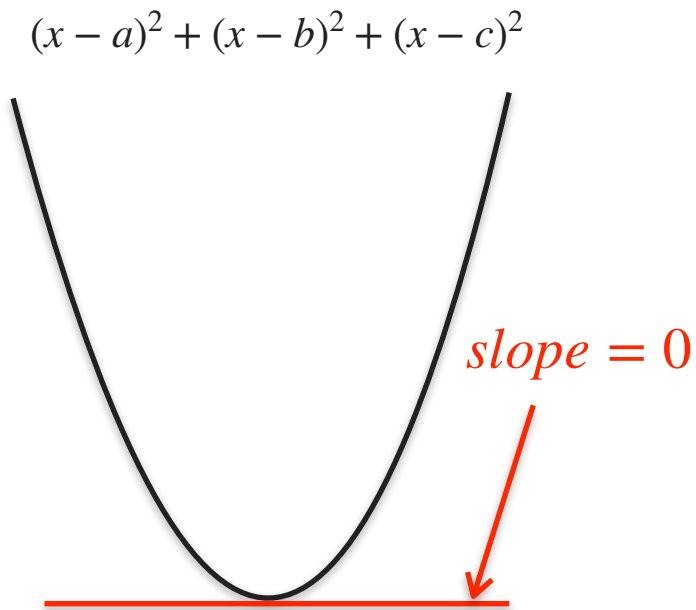


Three Power Line Problem



$$\frac{d}{dx} [(x - a)^2 + (x - b)^2 + (x - c)^2] = 0$$

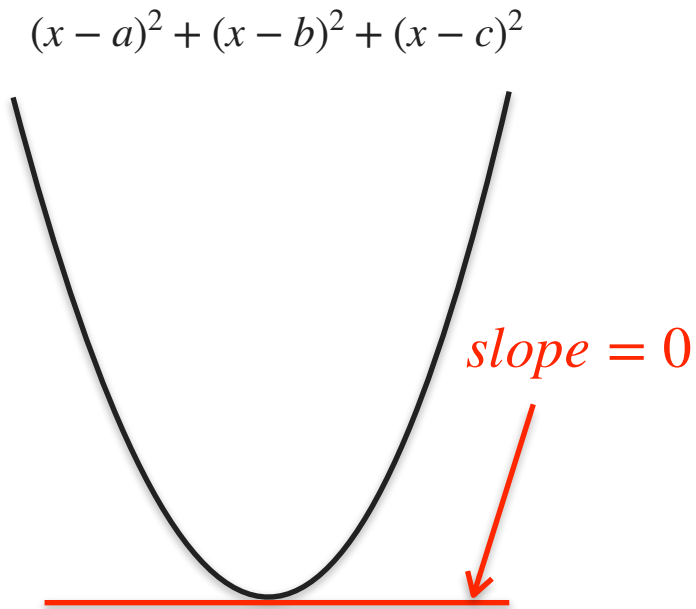
Three Power Line Problem



$$\frac{d}{dx} [(x - a)^2 + (x - b)^2 + (x - c)^2] = 0$$

$$2(x - a) + 2(x - b) + 2(x - c) = 0$$

Three Power Line Problem

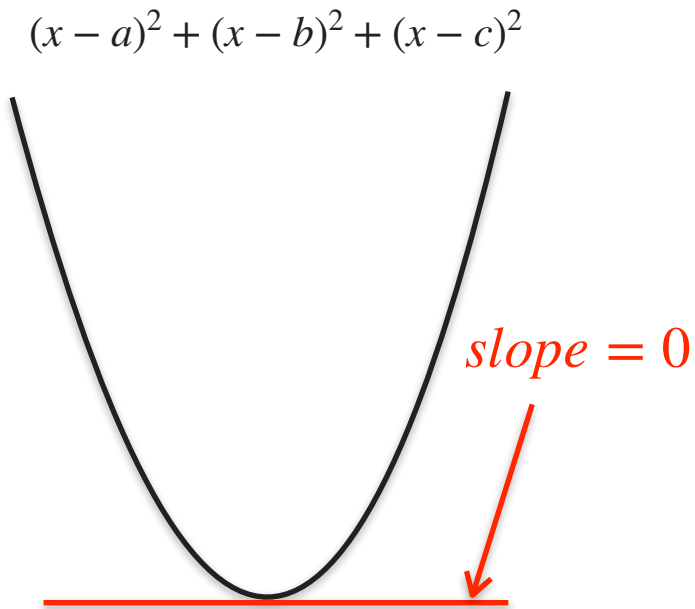


$$\frac{d}{dx} [(x - a)^2 + (x - b)^2 + (x - c)^2] = 0$$

$$2(x - a) + 2(x - b) + 2(x - c) = 0$$

$$(x - a) + (x - b) + (x - c) = 0$$

Three Power Line Problem



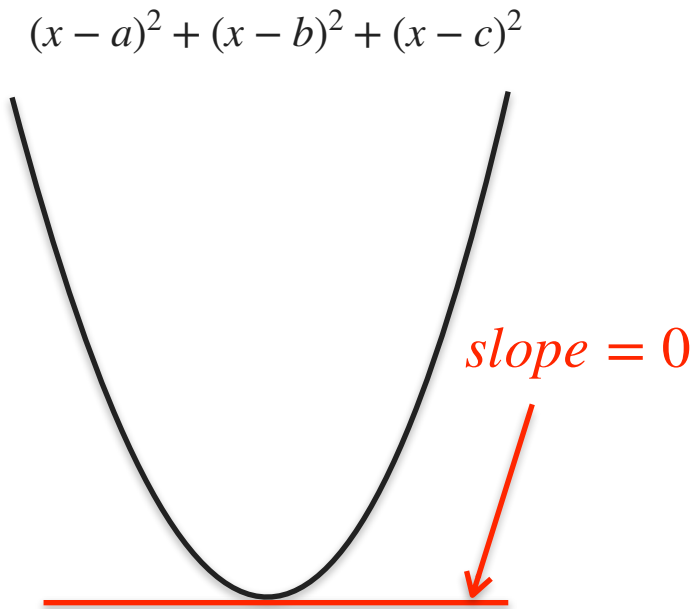
$$\frac{d}{dx} [(x - a)^2 + (x - b)^2 + (x - c)^2] = 0$$

$$2(x - a) + 2(x - b) + 2(x - c) = 0$$

$$(x - a) + (x - b) + (x - c) = 0$$

$$3x - a - b - c = 0$$

Three Power Line Problem



$$\frac{d}{dx} [(x - a)^2 + (x - b)^2 + (x - c)^2] = 0$$

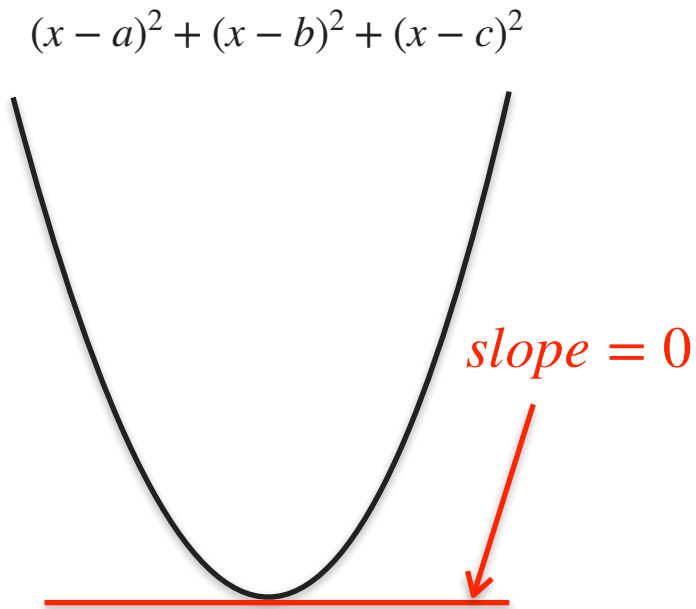
$$2(x - a) + 2(x - b) + 2(x - c) = 0$$

$$(x - a) + (x - b) + (x - c) = 0$$

$$3x - a - b - c = 0$$

$$3x = a + b + c$$

Three Power Line Problem



$$\frac{d}{dx} [(x - a)^2 + (x - b)^2 + (x - c)^2] = 0$$

$$2(x - a) + 2(x - b) + 2(x - c) = 0$$

$$(x - a) + (x - b) + (x - c) = 0$$

$$3x - a - b - c = 0$$

$$3x = a + b + c$$

$$x = \frac{a + b + c}{3}$$

Three Power Line Problem: Square Analogy



Problem: Minimize total area of the squares

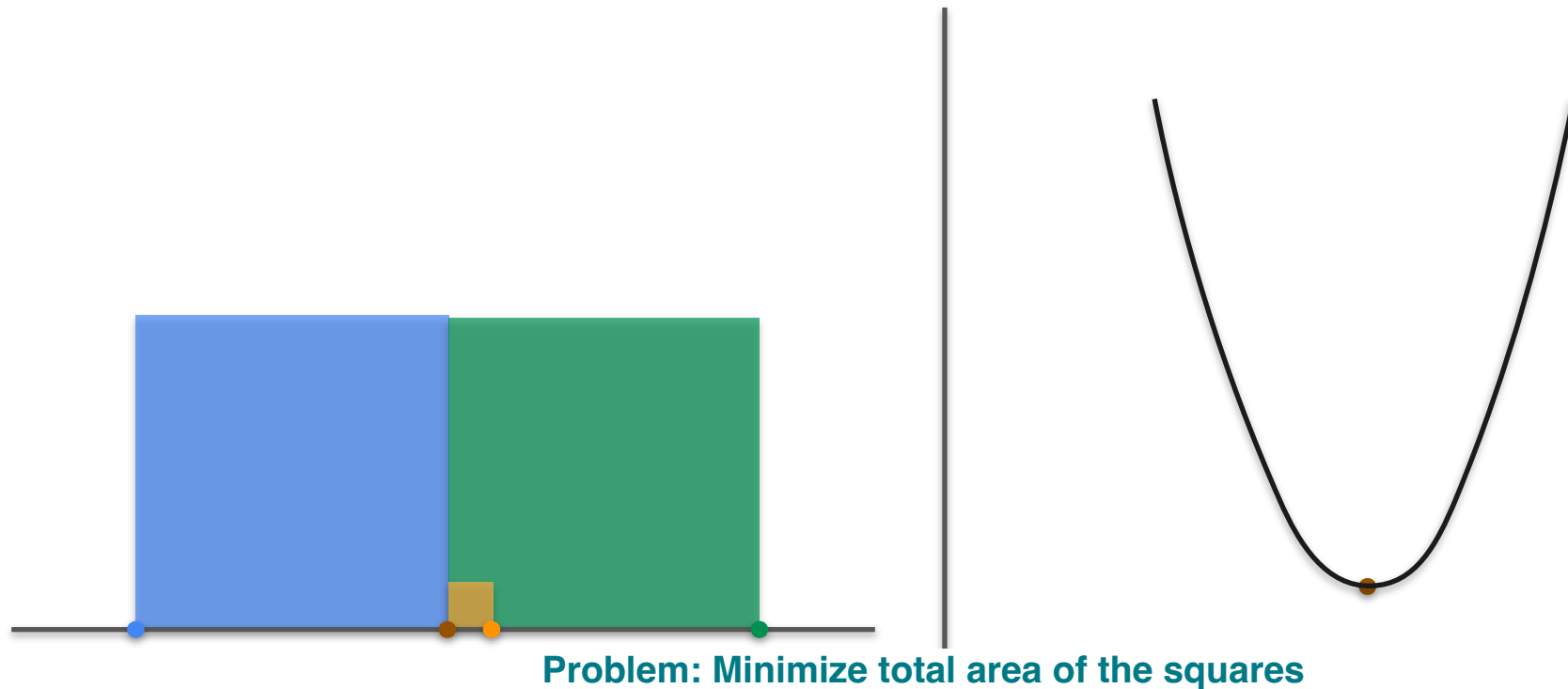
Three Power Line Problem: Square Analogy



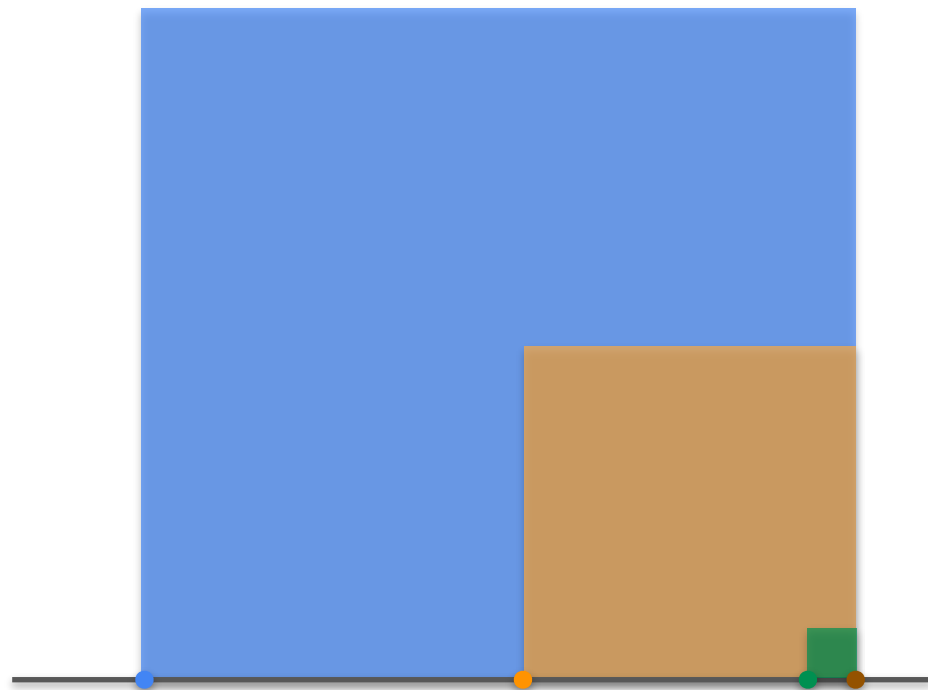
Problem: Minimize total area of the squares



Three Power Line Problem: Square Analogy



Three Power Line Problem: Square Analogy



Problem: Minimize total area of the squares



The Square Loss

$$\text{Minimize } (x - a_1)^2 + (x - a_2)^2 + \cdots + (x - a_n)^2$$

The Square Loss

Minimize $(x - a_1)^2 + (x - a_2)^2 + \dots + (x - a_n)^2$

$$\text{Solution: } x = \frac{a_1 + a_2 + \dots + a_n}{n}$$



DeepLearning.AI

Derivatives and Optimization

Optimization of log-loss Part 1

Coin Toss

Coin Toss



Coin Toss



Coin Toss



Coin 1



70% 30%

Coin Toss



Coin 1



70% 30%

Coin 2



50% 50%

Coin Toss



Coin 1



70% 30%

Coin 2



50% 50%

Coin 3



30% 70%

Quiz

- Which of the three coins would you choose to maximize your chances of winning?

Coin 1



70% 30%

Coin 2



50% 50%

Coin 3



30% 70%

Coins Toss



Coins Toss



Coin 1 $0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.3 \times 0.3 \times 0.3 = 0.7^7 0.3^3$
 $= 0.00222$

Coins Toss



Coin 1 $0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.3 \times 0.3 \times 0.3 = 0.7^7 0.3^3$
 $= 0.00222$

Coin 2 $0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 = 0.5^7 0.5^3$
 $= 0.00097$

Coins Toss



Coin 1 $0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.3 \times 0.3 \times 0.3 = 0.7^7 0.3^3$
 $= 0.00222$

Coin 2 $0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 = 0.5^7 0.5^3$
 $= 0.00097$

Coin 3 $0.3 \times 0.3 \times 0.3 \times 0.3 \times 0.3 \times 0.3 \times 0.3 \times 0.7 \times 0.7 \times 0.7 = 0.3^7 0.7^3$
 $= 0.00008$

Coins Toss



Coin 1 $0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.7 \times 0.3 \times 0.3 \times 0.3 = 0.7^7 0.3^3$
 $= 0.00222$

Coin 2 $0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 = 0.5^7 0.5^3$
 $= 0.00097$

Coin 3 $0.3 \times 0.3 \times 0.3 \times 0.3 \times 0.3 \times 0.3 \times 0.3 \times 0.7 \times 0.7 \times 0.7 = 0.3^7 0.7^3$
 $= 0.00008$

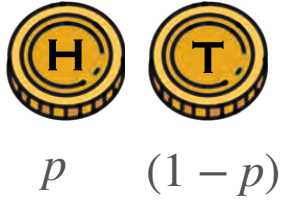
Coin Toss

Coin Toss

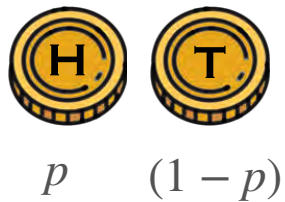


p

Coin Toss



Coin Toss



Chances of winning: $p^7(1 - p)^3 = g(p)$

Coin Toss



p $(1 - p)$

Chances of winning: $p^7(1 - p)^3 = g(p)$

Goal: maximize $g(p)$

Coin Toss

Coin Toss

$$\frac{dg}{dp} = \frac{d}{dp}(p^7(1-p)^3)$$

Coin Toss

Product rule

$$\frac{dg}{dp} = \frac{d}{dp}(p^7(1-p)^3) \stackrel{\downarrow}{=} \frac{d(p^7)}{dp}(1-p)^3 + p^7 \frac{d((1-p)^3)}{dp}$$

Coin Toss

Product rule

$$\begin{aligned}\frac{dg}{dp} &= \frac{d}{dp}(p^7(1-p)^3) \stackrel{\text{Product rule}}{=} \frac{d(p^7)}{dp}(1-p)^3 + p^7 \boxed{\frac{d((1-p)^3)}{dp}} \\ &= 7p^6(1-p)^3 + p^7 \boxed{3(1-p)^2(-1)} \quad \text{Chain rule}\end{aligned}$$

Coin Toss

Product rule

$$\begin{aligned}\frac{dg}{dp} &= \frac{d}{dp}(p^7(1-p)^3) \stackrel{\text{Product rule}}{=} \frac{d(p^7)}{dp}(1-p)^3 + p^7 \boxed{\frac{d((1-p)^3)}{dp}} \\ &= 7p^6(1-p)^3 + p^7 \boxed{3(1-p)^2(-1)} \quad \text{Chain rule} \\ &= p^6(1-p)^2[7(1-p) - 3p]\end{aligned}$$

Coin Toss

Product rule

$$\begin{aligned}\frac{dg}{dp} &= \frac{d}{dp}(p^7(1-p)^3) \stackrel{\text{Product rule}}{=} \frac{d(p^7)}{dp}(1-p)^3 + p^7 \frac{d((1-p)^3)}{dp} \\ &= 7p^6(1-p)^3 + p^7 \boxed{3(1-p)^2(-1)} \quad \text{Chain rule} \\ &= p^6(1-p)^2[7(1-p) - 3p] \\ &= p^6(1-p)^2(7-10p)\end{aligned}$$

Coin Toss

Product rule

$$\begin{aligned}\frac{dg}{dp} &= \frac{d}{dp}(p^7(1-p)^3) \stackrel{\text{Product rule}}{=} \frac{d(p^7)}{dp}(1-p)^3 + p^7 \frac{d((1-p)^3)}{dp} \\ &= 7p^6(1-p)^3 + p^7 \boxed{3(1-p)^2(-1)} \quad \text{Chain rule} \\ &= p^6(1-p)^2[7(1-p) - 3p] \\ &= p^6(1-p)^2(7-10p) = 0\end{aligned}$$

Coin Toss

Product rule

$$\begin{aligned}\frac{dg}{dp} &= \frac{d}{dp}(p^7(1-p)^3) \stackrel{\text{Product rule}}{=} \frac{d(p^7)}{dp}(1-p)^3 + p^7 \boxed{\frac{d((1-p)^3)}{dp}} \\ &= 7p^6(1-p)^3 + p^7 \boxed{3(1-p)^2(-1)} \quad \text{Chain rule} \\ &= p^6(1-p)^2[7(1-p) - 3p] \\ &= \boxed{p^6}(1-p)^2(7-10p) = 0 \\ &\quad \downarrow \\ &\quad p = 0\end{aligned}$$

Coin Toss

Product rule

$$\begin{aligned}\frac{dg}{dp} &= \frac{d}{dp}(p^7(1-p)^3) \stackrel{\text{Product rule}}{=} \frac{d(p^7)}{dp}(1-p)^3 + p^7 \frac{d((1-p)^3)}{dp} \\ &= 7p^6(1-p)^3 + p^7 \boxed{3(1-p)^2(-1)} \quad \text{Chain rule} \\ &= p^6(1-p)^2[7(1-p) - 3p] \\ &= \boxed{p^6} \boxed{(1-p)^2} (7 - 10p) = 0\end{aligned}$$

$p = 0$ $\rightarrow$ $p = 1$

Coin Toss

Product rule

$$\begin{aligned}\frac{dg}{dp} &= \frac{d}{dp}(p^7(1-p)^3) \stackrel{\text{Product rule}}{=} \frac{d(p^7)}{dp}(1-p)^3 + p^7 \frac{d((1-p)^3)}{dp} \\ &= 7p^6(1-p)^3 + p^7 \boxed{3(1-p)^2(-1)} \quad \text{Chain rule} \\ &= p^6(1-p)^2[7(1-p) - 3p] \\ &= \boxed{p^6} \boxed{(1-p)^2} \boxed{(7-10p)} = 0\end{aligned}$$

$p = 0$ $\rightarrow$ $p = 1$ $\rightarrow$ $p = 0.7$

Coin Toss

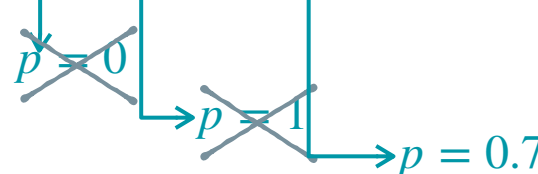
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~~$p \neq 0$~~ $\rightarrow p = 1 \rightarrow p = 0.7$

Coin Toss

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The diagram below the equation shows three boxes: p^6 , $(1-p)^2$, and $(7-10p)$. Arrows from these boxes lead to a decision tree. The first branch is $p=0$, which is crossed out with a large 'X'. The second branch is $p=1$, which is also crossed out with a large 'X'. The third branch is $p=0.7$, which is not crossed out and has an arrow pointing to it.

Coin Toss

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$p^* = 0$ $p = 1$ $p = 0.7$



Coin Toss

Coin Toss

$$\log(g(p))$$

Coin Toss

$$\log(g(p)) = \log(p^7(1-p)^3)$$

Coin Toss

$$\log(g(p)) = \log(p^7(1-p)^3) = \log(p^7) + \log((1-p)^3)$$

Coin Toss

$$\begin{aligned}\log(g(p)) &= \log(p^7(1-p)^3) = \log(p^7) + \log((1-p)^3) \\ &= 7\log(p) + 3\log(1-p)\end{aligned}$$

Coin Toss

$$\begin{aligned}\log(g(p)) &= \log(p^7(1-p)^3) = \log(p^7) + \log((1-p)^3) \\ &= 7\log(p) + 3\log(1-p) = G(p)\end{aligned}$$

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$$\frac{dG(p)}{dp} = \frac{d}{dp}(7\log(p) + 3\log(1-p))$$

Coin Toss

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$$\frac{dG(p)}{dp} = \frac{d}{dp}(7\log(p) + 3\log(1-p)) = 7\frac{1}{p} + 3\frac{1}{1-p}(-1)$$

Coin Toss

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Coin Toss

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$$7(1-p) - 3p = 0$$

Coin Toss

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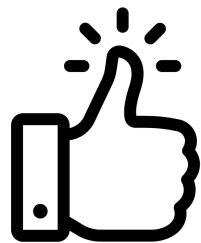
$$7(1-p) - 3p = 0 \quad p = 0.7$$

Coin Toss

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Coin Toss

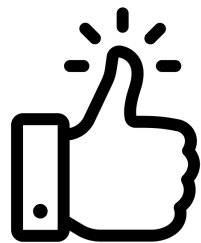
$$\log(g(p)) = \log(p^7(1-p)^3) = \log(p^7) + \log((1-p)^3)$$

$$= 7\log(p) + 3\log(1-p) = G(p)$$

— $G(p)$ is the logloss

$$\begin{aligned}\frac{dG(p)}{dp} &= \frac{d}{dp}(7\log(p) + 3\log(1-p)) = 7\frac{1}{p} + 3\frac{1}{1-p}(-1) \\ &= \frac{7(1-p) - 3p}{p(1-p)} = 0\end{aligned}$$

$$7(1-p) - 3p = 0 \quad p = 0.7$$





DeepLearning.AI

Derivatives and Optimization

Optimization of log-loss Part 2

Relationship With ML

Relationship With ML



Relationship With ML



Data

Relationship With ML



Data



Relationship With ML



Data



p

Relationship With ML



Data

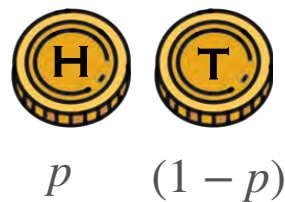


p

Relationship With ML



Data



Relationship With ML



Data



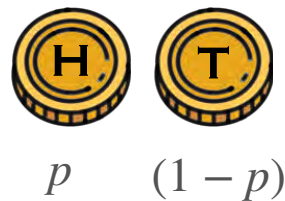
p $(1 - p)$

Model

Relationship With ML



Data



Model

Minimized log-loss

Relationship With ML



Data

Minimized log-loss



p $(1 - p)$

Model

$p = 0.7$

Why the Logarithm?

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1. Derivative of products is hard, derivative of sums is easy

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$$\frac{df}{dp}$$



$$\begin{aligned} & [6p^5](1-p)^2(3-p)^9(p-4)^{13}(10-p)^{500} + \\ & p^6 [2(1-p)](3-p)^9(p-4)^{13}(10-p)^{500}(-1) + \\ & p^6(1-p)^2 [9(3-p)^8](p-4)^{13}(10-p)^{500}(-1) + \\ & p^6(1-p)^2(3-p)^9 [13(p-4)^{12}](10-p)^{500} + \\ & p^6(1-p)^2(3-p)^9(p-4)^{13} [500(10-p)^{499}](-1) \end{aligned}$$

Why the Logarithm?

1. Derivative of products is hard, derivative of sums is easy

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$$\frac{d}{dp} \log(f)$$



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$$\begin{aligned} & \frac{6}{p} + \frac{2}{1-p}(-1) + \frac{9}{3-p}(-1) + \\ & \frac{13}{p-4} + \frac{500}{10-p}(-1) \end{aligned}$$

Why the Logarithm?

1. Derivative of products is hard, derivative of sums is easy

$$f(p) = p^6(1-p)^2(3-p)^9(p-4)^{13}(10-p)^{500}$$

$$\frac{df}{dp}$$



$$\frac{d}{dp} \log(f)$$



Why the Logarithm?

1. Derivative of products is hard, derivative of sums is easy

$$f(p) = p^6(1-p)^2(3-p)^9(p-4)^{13}(10-p)^{500}$$

$$\frac{df}{dp}$$



$$\frac{d}{dp} \log(f)$$



2. Product of lots of tiny things is tiny!



DeepLearning.AI

Derivatives and Optimization

Conclusion